INTEGRATED MATHEMATICAL RESEARCH

Prime Reciprocal Circles and the Riemann Zeta Function

A Unified Energy, Operator, and Sign Theory

Research synthesis prepared for Keio

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Abstract. This manuscript reorganizes the reciprocal-prime-circle investigation into one theory of geometric deficits, centered prime-counting variance, critical Sobolev energy, Mellin singularities, and one-sided scalar sampling. Eleven earlier working notes are consolidated, including the seventh revision of the sign-sparsity analysis; the separate Mittag–Leffler project is excluded.

The synthesis also yields additional deductions. In the divergent regime, the circle deficit is asymptotic to $4\pi^2$ times the centered prime variance. Fixed harmonic deficits have an exact limiting ratio, giving a strict-gap criterion for RH. The previously known total Toeplitz leakage is localized in a single transverse eigenmode, with an explicit finite error estimate. Dilation differences become invertible co-analytic Toeplitz filters under the Laguerre transform, removing the unknown centering constant without removing the zero obstruction. Nonlinear positive-part moments have an exact growth spectrum; all critically damped L^r endpoints with $r > 1$ are finite. When $\Theta > 1/2$, a renormalized limit at $r = 1$ detects whether the supremal zero line is attained. Growing-threshold counts give a quantitative hierarchy of equivalent strip tests.

The proofs retain their quantifiers and distinguish unconditional implications, RH-conditional asymptotics, finite certificates, and an explicitly separated external fixed-strip input. No uniform bound for the original deficit, proof of RH, or historical-priority claim is made.

One unchanged arithmetic target

$$\mathrm{RH} \iff \sup_N D_N < \infty \iff \int_2^\infty \mathcal{A}(x)_+ \frac{dx}{x} < \infty.$$

The equivalences and auxiliary theorems are proved below. The finiteness assertion itself remains unproved.

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1 The synthesis: one object viewed through several structures

The research began with drawing prime-indexed arcs and asking whether the resulting geometry could test primality or constrain zeta zeros. The first proposal, an arc of angle $p\pi$, is degenerate modulo 2π: every odd prime gives a half-turn. Replacing it by a $1/p$-turn arc is a change of construction, not an equivalent transformation. All subsequent arithmetic in this manuscript concerns the reciprocal construction.

The resulting sequence is informative only after its deterministic drift has been accounted for. Write

$$S(x) = \sum_{p \le x} p^{-1}, \qquad E(x) = S(x) - \log\log x - B_{\mathbb{P}}, \tag{1.1}$$

$$P(x) = \pi(x) - \mathrm{Li}_2(x), \qquad \mathrm{Li}_2(x) = \int_2^x \frac{dt}{\log t}, \tag{1.2}$$

$$\mathcal{A}(X) = \int_2^X P(x) x^{-3/2}\, dx, \qquad \Theta = \sup_{\zeta(\rho)=0,\ 0<\Re\rho<1} \Re\rho, \quad \delta = \Theta - \tfrac{1}{2}. \tag{1.3}$$

Here $B_{\mathbb{P}}$ is the prime Mertens constant. Prime-endpoint geometry uses

$$U_n = \exp\{2\pi i(S(p_n) - \log\log p_n)\}, \quad w_n = p_{n+1} - p_{n-1}, \quad W_N = \sum_{n=2}^{N} w_n, \tag{1.4}$$

$$D_N^{(m)} = W_N - \left|\sum_{n=2}^{N} w_n U_n^m\right|, \qquad D_N = D_N^{(1)}. \tag{1.5}$$

The finite statistic D_N requires neither $B_{\mathbb{P}}$ nor a zero of ζ. Its relation to those analytic objects is a theorem, not a property assumed in its definition. For $r > 0$ we also write $\mathscr{M}_r(X) = \int_2^X \mathcal{A}(x)_+^r\, dx/x$, where $y_+ = \max(y, 0)$.

1.1 What the combination adds

The added results are assembled from different mechanisms, rather than from renaming a target. Table 1 separates their mathematical inputs from their precise output. The underlying tools—Taylor expansion, least squares, min–max theory, Laplace transforms, and almost periodicity—are classical. "Added" means derived and written out in this consolidation; it does not assert that no equivalent result exists in the literature.

Table 1: Additional synthesis results and their proof mechanisms.

Result	Structures combined	Exact conclusion
Sharp geometric comparison	Tangent covariance, prime-gap quadrature, and zero-forced lower growth	If RH fails, $D_N \sim 4\pi^2 V(p_{N+1})$, where $V(X) = \min_c \int_2^X (E-c)^2$.
Harmonic ratio dichotomy	Concentration on a circle and the finite-energy criterion	For any fixed integer $m \ge 2$, $\lim D_N^{(m)}/D_N < m^2$ exactly when RH holds; otherwise the limit is m^2.
Single-mode spectral collapse	The tangent expansion of a moment curve and a finite Schur complement estimate	If $D_N \to \infty$, the second eigenvalue of the $(m+1)$-dimensional Toeplitz Gram matrix is asymptotic to $\binom{m+2}{3} D_N$; all later eigenvalues are $o(D_N)$.

Result	Structures combined	Exact conclusion
Constant-free Hardy coordinates	Multiplicative dilation, Laguerre functions, and a singular inner multiplier	On L^2, a dilation difference is represented by $I-qM^*_{\Theta_{L,\alpha}}$, with bounded inverse and sharp norm bounds; its zero-induced Taylor radius is unchanged.
Nonlinear critical spectrum	Positive-part sampling, growth exponents, and weighted L^r estimates	$\limsup \log(1+\mathscr{M}_r(X))/\log X = r\delta$, and $\int_2^\infty \mathcal{A}_+^r x^{-r\delta-1}dx < \infty$ for every $r>1$.
Edge residue and count thresholds	Rectified almost-periodic zero profiles and positive series	When $\delta > 0$, a normalized $r \downarrow 1$ limit detects an attained supremal line; counts above n^λ have a sharp transition at $\lambda = k\delta$.

The strict harmonic criterion deserves care. Every finite ratio is already strictly below m^2 once at least two distinct phases are present. The unresolved issue is whether a positive gap survives the limit. Likewise, critical L^r finiteness for $r>1$ is a proved endpoint result, but its damping depends on the actual zero supremum and it supplies no new numerical bound for Θ.

1.2 The common arithmetic obstruction

Several apparently different targets reduce to the same finiteness statement. With the definitions and endpoint conventions proved later,

$$\begin{aligned}\mathrm{RH} &\iff E \in L^2(2,\infty) \iff \sup_N D_N < \infty \iff \sup_X V(X) < \infty \\ &\iff \sup_X\{R_4(X) + \tfrac{3}{4}\log\log X\} < \infty \iff \mu \in H^{-1}_{1/2}(\mathbb{R}) \\ &\iff \mathcal{A}(X) \le 0 \text{ eventually} \iff \int_2^\infty \mathcal{A}(x)_+ \frac{dx}{x} < \infty. \end{aligned} \tag{1.6}$$

The measure μ is the critically weighted prime measure minus its continuous main term. The correlation R_4 is defined by a finite-cutoff kernel; there is no divergent infinite double sum hidden in the notation. These objects and every arrow in (1.6) receive proofs below. In particular, a Fourier transform of μ is used only after the required distributional or Sobolev membership is established.

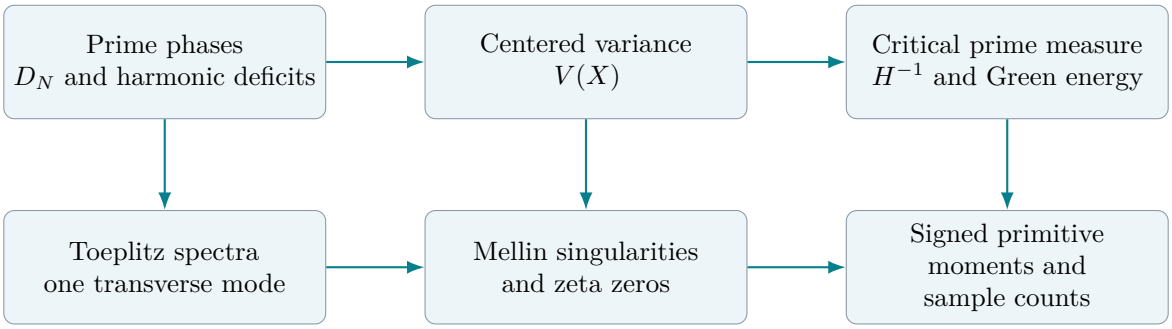


Figure 1: The organization of the synthesis. Arrows denote proved transfers developed in the text, not independent proofs of finite energy. The additional results quantify several of these transfers.

1.3 Status, assumptions, and the meaning of progress

An equivalence is a proved theorem even when both equivalent assertions remain undecided. It does not by itself narrow the set of possible zeta zeros. The concrete progress here consists of exact

constants, error controls, spectral localization, removal of finite-cutoff artifacts, and identification of which quantifiers still require an arithmetic estimate.

The main synthesis results do not use a fixed zero-free strip to the left of 1. Some earlier degree-four arguments did import a stated 2026 preprint result. In this consolidated manuscript those applications are read under the explicit hypothesis

$$(H_b): \qquad \Theta \le b < 7/8, \tag{1.7}$$

or under the directly stated finite eighth-moment hypothesis. The preprint statements and consequences are documented in the polynomial module; their full proofs and formalization are not independently rebuilt here. No such external claim is silently promoted to a result of this manuscript. Purely finite linear algebra and the abstract extremal-polynomial theorems retain their stated hypotheses and do not require (H_b).

Standard inputs are cited where used: Mertens' theorem and the prime number theorem; Euler products and the functional equation; conditional mean-square estimates for the prime-counting error; Stechkin–Popov lower mean estimates; prime-gap estimates; and the stated sieve and large-value theorems. The RH-conditional mean-square estimate is materially stronger for this purpose than merely inserting a pointwise RH bound. The historical studies of integrated prime errors provide relevant context, not a claim of priority for the equivalent formulations recorded here.

1.4 Editorial scope and reproducibility

The eleven source notes are listed in the provenance section. Their common notation and title matter have been consolidated, duplicated foundations are referred to the first proof where possible, and independent proof routes are retained when their different hypotheses matter. The eighty-method transfer ledger is retained as an audit of applicability, not a claim that eighty unrelated theorems have proved RH. The theta-kernel and finite-field analyses are included because they arose in the zeta investigation, even though those auxiliary results have their own scope.

Earlier finite numerical certificates remain explicitly finite. New computations check the synthesis constants on finite model sequences and the algebraic identities on actual primes; they do not test any universal RH assertion. All text, diagrams, tables, and displayed reproduction code needed for this version are contained in this single TEX file. The separate Poisson-coupling manuscript for Mittag–Leffler laws is not included.

Part I

Foundations: reciprocal primes and geometric energy

2 The construction and the scope of the claims

All logarithms are natural. The letter p always denotes a prime, and p_n is the n-th prime, with $p_1 = 2$. Define

$$S(x) = \sum_{p \le x} \frac{1}{p}, \qquad E(x) = S(x) - \log\log x - B_{\mathbb{P}}, \qquad x \ge 2, \tag{2.1}$$

where the exact Meissel–Mertens constant is

$$B_{\mathbb{P}} = \gamma + \sum_{p} \left(\log(1 - p^{-1}) + p^{-1} \right) = 0.261497212847642783755426838608695859 05\ldots. \tag{2.2}$$

The series in (2.2) converges absolutely. Mertens' theorem gives $E(x) = O(1/\log x)$; in particular E is bounded and tends to zero. We use the notation of (2.1) consistently, rather than the several different constants sometimes denoted B_1 in the literature [1, 2].

Let $\mathcal{Z}$ be the multiset of nontrivial zeros of ζ, and set

$$\Theta = \sup_{\rho \in \mathcal{Z}} \Re\rho. \tag{2.3}$$

The functional equation and the location of the zeros give $1/2 \le \Theta \le 1$, and RH is equivalent to $\Theta = 1/2$. We do not assume that the supremum is attained.

2.1 What a complete turn actually means

The reciprocal construction adds $1/p$ of a turn at each prime:

$$C(x) = e^{2\pi \mathrm{i} S(x)}.$$

The lifted number of turns is the real number $S(x)$, not just the point $C(x)$. The number of completed turns is $\lfloor S(x) \rfloor$. Every finite $S(x)$ is an exactly computable rational number.

Proposition 2.1 (No exact closure at a prime endpoint)**.** *For every nonempty finite set of distinct primes, the sum of their reciprocals is not an integer. Consequently $C(p_n) \neq 1$ for every $n \ge 1$.*

Proof. Let Q be the product of the primes in the set and let $A = \sum_{p \mid Q} Q/p$. For each $q \mid Q$, $A \equiv Q/q \not\equiv 0 \pmod{q}$. Thus $\gcd(A, Q) = 1$, and the reduced denominator of A/Q is $Q > 1$. □

If q_m is the first prime with $S(q_m) > m$, then

$$0 < S(q_m) - m < \frac{1}{q_m}. \tag{2.4}$$

A continuously drawn arc passes the integer turn during the final step; its endpoint overshoots it. The fact that q_m is prime is built into this indexing and is not a primality test.

The smooth prediction for the m-th crossing is

$$X_m = \exp\!\big(\exp(m - B_{\mathbb{P}})\big).$$

If one formally replaces the local error by $E(x) \approx 1/(\sqrt{x}\log x)$, then a first-order inversion gives $X_m - q_m \approx \sqrt{X_m}$. This explains why a square-root scale is natural. It does not establish an exact displacement or a uniform one-sided law. Zeta zeros contribute oscillations, and the final prime step is discrete. The theorems below concern the entire error function, not just the sparse sequence of turn crossings.

2.2 The exact centered circle

Introduce the reference motion and the relative phase

$$C_0(x) = e^{2\pi \mathrm{i}(\log\log x + B_{\mathbb{P}})}, \qquad Z(x) = C(x)\overline{C_0(x)} = e^{2\pi \mathrm{i} E(x)}. \tag{2.5}$$

Their chord distance satisfies the exact identity

$$|C(x) - C_0(x)|^2 = |Z(x) - 1|^2 = 4\sin^2(\pi E(x)). \tag{2.6}$$

Because $E(x) \to 0$, this is asymptotic to $4\pi^2 E(x)^2$. The principal argument is not a globally valid replacement for the lift E: for example $E(2) \approx 0.6050 > 1/2$. Only its eventual smallness is needed for the energy comparisons.



Circle: $e^{2\pi \mathrm{i} E}$ Fixed ellipse: $(a\cos 2\pi E, b\sin 2\pi E)$

Figure 2: The same phase on a circle and a fixed nondegenerate ellipse. The squared chord distances have comparable tail integrals when $a, b > 0$. The illustration is schematic, not a plot of prime data.

2.3 Established inputs and deductions in this note

The analytic inputs are Mertens' theorem, the Euler product and functional equation, the conditional mean-square estimate stated in Section 4, and the estimate

$$\pi(x) - \mathrm{li}(x) = O_\Theta\left(\frac{x^\Theta}{\log x}\right), \qquad \frac{1}{2} < \Theta < 1. \tag{2.7}$$

The last estimate is due to Grosswald; the precise form we use is recorded in Zhao's equation (24) [3, 1]. These are cited inputs, not results first proved here. The Mellin converse, discrete comparisons, vector-deficit identities, growth laws, and transform consequences are established below. Their historical priority has not been investigated.

3 Prime zeta decomposition and the Mellin obstruction

For $\Re s > 1$, the prime zeta function is $P(s) = \sum_p p^{-s}$. Stieltjes integration gives

$$P(s) = \int_{2^-}^{\infty} x^{1-s}\,\mathrm{d}S(x) = (s-1)\int_2^\infty S(x)x^{-s}\,\mathrm{d}x. \tag{3.1}$$

The Euler product, followed by absolutely convergent summation, gives

$$\log\zeta(s)=\sum_{k\ge1}\frac{P(ks)}{k},\qquad P(s)=\sum_{k\ge1}\frac{\mu(k)}{k}\log\zeta(ks). \tag{3.2}$$

The logarithms in this half-plane use the Euler-product branch [4, 5]. Each summand $p^{-s}=p^{-\Re s}e^{-\mathrm{i}(\Im s)\log p}$ is a rotating vector, but this observation alone does not continue the prime series to the critical line. The continuation must be justified analytically.

Put

$$F(s)=\int_2^\infty E(x)x^{-s}\,\mathrm{d}x,\qquad c=\log\log 2+B_{\mathbb{P}}.$$

Initially $\Re s>1$. Integrating the elementary part of S in (3.1) gives

$$P(s)=(s-1)F(s)+c\,2^{1-s}+\int_2^\infty\frac{x^{-s}}{\log x}\,\mathrm{d}x. \tag{3.3}$$

Consequently

$$P'(s)=F(s)+(s-1)F'(s)-c(\log 2)2^{1-s}-\frac{2^{1-s}}{s-1}. \tag{3.4}$$

Also,

$$\frac{\zeta'(s)}{\zeta(s)}=P'(s)+\sum_{k\ge2}P'(ks). \tag{3.5}$$

The sum on the right is holomorphic for $\Re s>1/2$, since on every compact sub-half-plane it is dominated by

$$\sum_p\frac{(\log p)p^{-2\sigma}}{1-p^{-\sigma}}<\infty,\qquad \sigma>1/2.$$

Lemma 3.1 (Zero exclusion from Mellin continuation)**.** *Suppose $a\ge1/2$, and the germ F extends holomorphically to $\Re s>a$. Then $\zeta(s)$ has no zeros in that half-plane.*

Proof. Equations (3.4) and (3.5) provide a meromorphic continuation of ζ'/ζ to this half-plane whose only possible pole is the one at $s=1$. A zero of ζ there would give an additional pole of its logarithmic derivative, with positive integral residue. The identity theorem for meromorphic functions rules this out. □

We shall also need the obstruction at $s=1/2$. Define the entire function

$$J_0(s)=c\,2^{1-s}-\gamma-\log\log 2-\sum_{j\ge1}\frac{[-(s-1)\log 2]^j}{j\,j!},$$

and put $R_0(s)=\sum_{k\ge2}P(ks)/k$, holomorphic for $\Re s>1/2$. The exponential-integral expansion in (3.3) gives

$$F(s)=\frac{\log((s-1)\zeta(s))-R_0(s)-J_0(s)}{s-1}. \tag{3.6}$$

The branch is obtained from real $s>1$ on any simply connected zero-free region. At $s=1$, the numerator vanishes because $R_0(1)=\gamma-B_{\mathbb{P}}$, $J_0(1)=B_{\mathbb{P}}-\gamma$, and $(s-1)\zeta(s)$ takes the value 1. Thus $s=1$ is removable.

Lemma 3.2 (Prime-square singularity)**.** *The germ F cannot extend holomorphically to a half-plane $\Re s>a$ with $a<1/2$. Along the real continuation from the right,*

$$F(1/2+\varepsilon)=-\log\varepsilon+O(1),\qquad \varepsilon\downarrow0. \tag{3.7}$$

Proof. The factor $(s-1)\zeta(s)$ has no zero on the real interval $1/2 \le s \le 1$; in particular $\zeta(1/2) < 0$. The $k = 2$ term in R_0 is

$$\tfrac{1}{2}P(1+2\varepsilon) = -\tfrac{1}{2}\log(2\varepsilon) + O(1),$$

whereas its $k \ge 3$ terms remain holomorphic near $s = 1/2$. Equation (3.6) therefore proves (3.7). Equivalently, if P' were regular there, (3.5) would give a residue $-1/2$ from $P'(2s)$, which is impossible for a logarithmic derivative. □

4 The exact quadratic energy criterion

Theorem 4.1 (Scalar, circle, and fixed-ellipse energies)**.** *The following statements are equivalent:*

(i) RH holds.

(ii) $\displaystyle\int_2^\infty E(x)^2\,\mathrm{d}x < \infty.$

(iii) $\displaystyle\int_2^\infty |Z(x)-1|^2\,\mathrm{d}x < \infty.$

(iv) For any fixed $a, b > 0$,

$$\int_2^\infty \Big[(a\cos(2\pi E(x)) - a)^2 + b^2\sin^2(2\pi E(x))\Big]\,\mathrm{d}x < \infty.$$

Proof. Assume (ii). Cauchy–Schwarz shows that F and all its derivatives under the integral sign are holomorphic for $\Re s > 1/2$. For example, on a compact sub-half-plane,

$$\int_2^\infty |E(x)|x^{-\sigma}(\log x)^j\,\mathrm{d}x \le \|E\|_2\left(\int_2^\infty x^{-2\sigma}(\log x)^{2j}\,\mathrm{d}x\right)^{1/2}.$$

Lemma 3.1 excludes zeros to the right of $1/2$; the functional equation excludes those to the left. Hence (ii) implies RH.

For the forward direction we give the analytic argument, rather than infer mean-square convergence from a pointwise RH estimate. The established conditional estimate of Cramer's type is

$$\mathrm{RH} \quad\Longrightarrow\quad \int_X^{2X} (\psi(t)-t)^2\,\mathrm{d}t \ll X^2. \tag{4.1}$$

See Brent–Platt–Trudgian [6] for this mean-square framework. Since $\psi(x) - \vartheta(x) = O(\sqrt{x})$, the same upper bound holds for $R(x) = \vartheta(x) - x$. Set $f(x) = R(x)/(x\log x)$. On dyadic intervals,

$$\int_X^{2X} f(x)^2\,\mathrm{d}x \ll \frac{1}{\log^2 X}.$$

Summing over $X = 2^j$ proves $f \in L^2(2,\infty)$. Partial summation, with the limiting constant in (2.2), gives

$$E(x) = f(x) - \int_x^\infty \frac{f(t)}{t}\left(1 + \frac{1}{\log t}\right)\mathrm{d}t. \tag{4.2}$$

The adjoint Hardy operator $Hg(x) = \int_x^\infty g(t)t^{-1}\,\mathrm{d}t$ satisfies $\|Hg\|_2 \le 2\,\|g\|_2$. For compactly supported g, integration by parts followed by Cauchy–Schwarz proves this inequality; truncation proves the general case. The multiplier in (4.2) is bounded for $t \ge 2$. Therefore $E \in L^2$. This also explains the forward assertion in Zhao's Lemma 9 [1].

By (2.6) and $E(x) \to 0$, the integrands in (ii) and (iii) are comparable on a tail. Initial compact intervals are irrelevant to finiteness. Finally, if $d_{a,b}(t)^2 = a^2(\cos 2\pi t - 1)^2 + b^2\sin^2 2\pi t$, then

$$\min(a^2,b^2)|e^{2\pi i t} - 1|^2 \le d_{a,b}(t)^2 \le \max(a^2,b^2)|e^{2\pi i t} - 1|^2. \tag{4.3}$$

This proves (iii) if and only if (iv). □

Remark 4.2 (Measure and tail rate). The proof also gives, under RH, $\int_X^\infty E(x)^2\,\mathrm{d}x \ll 1/\log X$. The measure $\mathrm{d}x$ matters: under $x = e^t$, it becomes $e^t\,\mathrm{d}t$. Replacing it by $\mathrm{d}t$ without the factor would define a different criterion. Moreover, the common pointwise RH bound $E(x) = O(x^{-1/2}\log x)$ alone gives a divergent majorant for $\int E^2$; the mean-square input is essential.

5 The complete integrability spectrum and geometric losses

Partial summation using $\pi(x) - \mathrm{li}(x)$ gives the exact tail formula

$$E(x) = \frac{\pi(x) - \mathrm{li}(x)}{x} - \int_x^\infty \frac{\pi(t) - \mathrm{li}(t)}{t^2}\,\mathrm{d}t. \tag{5.1}$$

For $1/2 < \Theta < 1$, (2.7) implies

$$E(x) = O_\Theta\left(\frac{x^{\Theta-1}}{\log x}\right). \tag{5.2}$$

Theorem 5.1 (All finite powers, including the endpoint). *For every finite $q > 0$,*

$$\int_2^\infty |E(x)|^q\,\mathrm{d}x < \infty \quad\Longleftrightarrow\quad \Theta \le 1 - \frac{1}{q}. \tag{5.3}$$

In particular no such integral converges for $0 < q < 2$. If $\Theta < 1$, the exact threshold is $q = 1/(1-\Theta)$, and the threshold itself is included. If $\Theta = 1$, no finite q works.

Proof. For $q > 1$, Hölder's inequality makes F holomorphic in $\Re s > 1 - 1/q$ whenever $E \in L^q$. If $q \ge 2$, Lemma 3.1 gives the stated necessary condition. If $1 < q < 2$, this continuation crosses $s = 1/2$, contradicting Lemma 3.2. For $0 < q \le 1$, boundedness of E would imply $E \in L^r$ for any chosen $1 < r < 2$, again a contradiction.

For sufficiency, when $\Theta = 1/2$, Theorem 4.1 and boundedness of E give $E \in L^q$ for all $q \ge 2$. When $1/2 < \Theta < 1$, use (5.2). At the limiting value $q(1-\Theta) = 1$, the majorant is $1/[x(\log x)^q]$, which is integrable because $q > 2$. Larger q are easier. □

Theorem 5.2 (Exact weighted quadratic threshold). *For every real η,*

$$\int_2^\infty x^{1-2\eta}E(x)^2\,\mathrm{d}x < \infty \quad\Longleftrightarrow\quad \eta \ge \Theta. \tag{5.4}$$

Proof. Weighted Cauchy–Schwarz makes F holomorphic in $\Re s > \eta$. If $\eta < 1/2$, Lemma 3.2 gives a contradiction; otherwise Lemma 3.1 requires $\eta \ge \Theta$.

For sufficiency at $\eta = \Theta$, use Theorem 4.1 when $\Theta = 1/2$, (5.2) when $1/2 < \Theta < 1$, and Mertens' $E(x) = O(1/\log x)$ when $\Theta = 1$. The last two cases give the integrable majorant $1/[x(\log x)^2]$. For $\eta > \Theta$ the weight is smaller. □

Corollary 5.3 (Order of contact of a curve). *Suppose a continuous curve $\Gamma : \mathbb{R} \to \mathbb{R}^d$ satisfies $\Gamma(t) - \Gamma(0) = vt^k + O(t^{k+1})$ as $t \to 0$, where k is a positive integer and $v \ne 0$. Then*

$$\int_2^\infty \|\Gamma(E(x)) - \Gamma(0)\|^2\,\mathrm{d}x < \infty \quad\Longleftrightarrow\quad \Theta \le 1 - \frac{1}{2k}. \tag{5.5}$$

Proof. On a tail the integrand is bounded above and below by positive constants times $|E(x)|^{2k}$. Apply Theorem 5.1. □

A fixed nondegenerate ellipse has $k = 1$ and preserves the RH criterion. Projecting onto its horizontal axis gives $(a\cos 2\pi E - a)^2 \asymp E^4$, so the criterion becomes $\Theta \le 3/4$. This is an exact classification, not merely a failure of the original proof.

There is also a continuous transition between these cases. For $\lambda \ge 0$, consider the moving ellipse point

$$\Gamma_\lambda(x) = \big(\cos(2\pi E(x)),\, x^{-\lambda}\sin(2\pi E(x))\big).$$

Its distance from $(1,0)$ has tail square comparable to $E(x)^4 + x^{-2\lambda}E(x)^2$. Theorems 5.1 and 5.2 therefore give

$$\int_2^\infty \|\Gamma_\lambda(x) - (1,0)\|^2 \,\mathrm{d}x < \infty \quad\Longleftrightarrow\quad \Theta \le \min\left(\frac{3}{4}, \frac{1}{2} + \lambda\right). \tag{5.6}$$

Thus pointwise nondegeneracy is insufficient if an axis tends to zero.

Observation	Tail sensitivity	Exact finite-energy condition
Circle / fixed nondegenerate ellipse	E^2	$\Theta = 1/2$
Horizontal projection	E^4	$\Theta \le 3/4$
Curve with contact order k	E^{2k}	$\Theta \le 1 - 1/(2k)$
Axis $b(x) = x^{-\lambda}$	$E^4 + x^{-2\lambda}E^2$	$\Theta \le \min(3/4, 1/2 + \lambda)$

Table 2: How changing the geometry changes the zero-location condition.

6 Exact sampling at prime gaps and integers

Put

$$e_n = E(p_n), \qquad h_n = p_{n+1} - p_n, \qquad b_n = E(p_{n+1}^-) = e_{n+1} - p_{n+1}^{-1}.$$

On the interval $[p_n, p_{n+1})$ there is no prime jump, and

$$E(x) = e_n - (\log\log x - \log\log p_n).$$

Lemma 6.1 (A two-endpoint comparison)**.** *For every $n \ge 1$,*

$$\frac{h_n}{24}(e_n^2 + b_n^2) \le \int_{p_n}^{p_{n+1}} E(x)^2 \,\mathrm{d}x \le h_n(e_n^2 + b_n^2). \tag{6.1}$$

Proof. Write $p = p_n, q = p_{n+1}$ and

$$y = \frac{\log\log x - \log\log p}{\log\log q - \log\log p}.$$

Then $E(x) = (1-y)e_n + yb_n$. The derivative $\mathrm{d}x/\,\mathrm{d}y$ has maximum-to-minimum ratio $q\log q/(p\log p) < 4$ by Bertrand's postulate, while its integral is h_n. It is therefore between $h_n/4$ and $4h_n$. Moreover,

$$\int_0^1 ((1-y)a + yb)^2 \,\mathrm{d}y = \frac{a^2 + ab + b^2}{3} \ge \frac{a^2 + b^2}{6}.$$

This proves the lower bound. For the upper bound, $((1-y)a + yb)^2 \le \max(a^2, b^2) \le a^2 + b^2$ throughout the interval. □

The jump correction has finite weighted square:

$$\sum_{n\ge1} \frac{h_n}{p_{n+1}^2} \le \sum_{n\ge1} \int_{p_n}^{p_{n+1}} \frac{\mathrm{d}x}{x^2} = \frac{1}{2}. \tag{6.2}$$

Consequently replacing b_n by e_{n+1} preserves convergence. For $n \ge 2$, set

$$w_n = h_{n-1} + h_n = p_{n+1} - p_{n-1}. \tag{6.3}$$

Theorem 6.2 (The prime-endpoint criterion)**.** *One has*

$$\mathrm{RH} \iff \sum_{n\geq 2} w_n E(p_n)^2 < \infty \tag{6.4}$$

$$\iff \sum_{n\geq 2} w_n |e^{2\pi \mathrm{i} E(p_n)} - 1|^2 < \infty. \tag{6.5}$$

If $I(X) = \int_2^X E(x)^2 \,\mathrm{d}x$ *and* $A_N = \sum_{n=2}^N w_n e_n^2$*, then*

$$I(p_N) \ll 1 + A_N \ll 1 + I(p_{N+1}). \tag{6.6}$$

Proof. Sum (6.1). The elementary inequalities $(a+b)^2 \leq 2a^2 + 2b^2$ and $a^2 \leq 2(a-b)^2 + 2b^2$, together with (6.2), give both the convergence statement and (6.6). The finite initial endpoint contributes only a constant. Theorem 4.1 and the tail equivalence (2.6) finish the proof. □

The weight w_n records the amount of real x-space represented by a prime endpoint. Dropping it changes the question. Sampling only the primes at which an entire turn is crossed is much sparser and is not covered by this theorem.

Corollary 6.3 (Integer sampling)**.**

$$\mathrm{RH} \quad \iff \quad \sum_{m=2}^{\infty} E(m)^2 < \infty \quad \iff \quad \sum_{m=2}^{\infty} |Z(m) - 1|^2 < \infty.$$

Proof. On $[m, m+1)$, $S(x) = S(m)$ and

$$|E(x) - E(m)| \leq \frac{1}{m \log m}.$$

The difference between E and its integer-step sampling is in L^2. The result follows from Theorem 4.1. □

7 A monotone geometric criterion without an input constant

Define, using no supplied value of $B_{\mathbb{P}}$,

$$U_n = \exp\left(2\pi \mathrm{i}\left[S(p_n) - \log\log p_n\right]\right), \qquad n \geq 2. \tag{7.1}$$

Mertens' theorem implies $U_n \to u_\infty = e^{2\pi \mathrm{i} B_{\mathbb{P}}}$. With the weights in (6.3), put

$$W_N = \sum_{n=2}^{N} w_n = p_N + p_{N+1} - 5, \tag{7.2}$$

$$M_N = \sum_{n=2}^{N} w_n U_n, \qquad m_N = M_N / W_N, \tag{7.3}$$

$$D_N = W_N - |M_N|, \qquad V_N = W_N - \frac{|M_N|^2}{W_N}. \tag{7.4}$$

Here D_N is the sum of the vector lengths minus the length of their resultant. The quantity V_N is their weighted variance.

Lemma 7.1 (Variance and vector deficit)**.** *Let any unit vectors $u_n \to u_\infty$ and positive weights w_n satisfy $W_N \to \infty$. Define M_N, m_N, D_N, V_N as above and let $\mathcal{J}_N = \sum_{n=2}^{N} w_n |u_n - u_\infty|^2$. Then D_N, V_N are nonnegative and nondecreasing, and*

$$V_N = D_N(1 + |M_N|/W_N), \qquad D_N \le V_N \le 2D_N, \tag{7.5}$$

$$V_N = \sum_{n=2}^{N} w_n |u_n - m_N|^2 = \frac{1}{W_N} \sum_{2 \le j < k \le N} w_j w_k |u_j - u_k|^2, \tag{7.6}$$

$$\mathcal{J}_N = V_N + W_N |m_N - u_\infty|^2. \tag{7.7}$$

Moreover,

$$\sup_N D_N < \infty \iff \sup_N V_N < \infty \iff \mathcal{J}_\infty < \infty. \tag{7.8}$$

When these quantities are finite,

$$2D_\infty = V_\infty = \mathcal{J}_\infty. \tag{7.9}$$

Proof. Expanding squares proves (7.5)–(7.7). The triangle inequality gives monotonicity of D_N. If a unit point u of weight w is appended to old quantities W, m, V, direct expansion gives

$$V_{\text{new}} - V = \frac{Ww}{W + w} |u - m|^2 \ge 0. \tag{7.10}$$

Weighted Cesaro convergence gives $m_N \to u_\infty$. If $\mathcal{J}_\infty < \infty$, (7.7) bounds V_N. Conversely, if $V_N \le C$, then for every fixed K

$$\sum_{n=2}^{K} w_n |u_n - m_N|^2 \le C \quad (N \ge K).$$

Let first $N \to \infty$, then $K \to \infty$, to get $\mathcal{J}_\infty \le C$.

To prove the equality of limits, write $A_N = M_N - W_N u_\infty$. When $\mathcal{J}_\infty < \infty$, a finite-prefix/tail split and Cauchy–Schwarz show $A_N = o(\sqrt{W_N})$. Explicitly, for fixed K,

$$\frac{|A_N|}{\sqrt{W_N}} \le \frac{|A_K|}{\sqrt{W_N}} + \left(\sum_{n>K} w_n |u_n - u_\infty|^2 \right)^{1/2}.$$

Now use $\mathcal{J}_N - V_N = |A_N|^2 / W_N \to 0$ and $|M_N|/W_N \to 1$. □

Theorem 7.2 (Constant-free polygon criterion)**.** *For the prime vectors (7.1) and weights (6.3),*

$$\boxed{\text{RH} \iff \sup_{N \ge 2} D_N < \infty \iff \sup_{N \ge 2} V_N < \infty.} \tag{7.11}$$

Under RH,

$$2D_\infty = \sum_{n=2}^{\infty} w_n |e^{2\pi i E(p_n)} - 1|^2. \tag{7.12}$$

Proof. Here $|U_n - u_\infty| = |e^{2\pi i E(p_n)} - 1|$. Apply Lemma 7.1 and Theorem 6.2. □

The finite construction really has no $B_{\mathbb{P}}$ input. Knowledge that the limit exists is used in its proof, but computing D_N or V_N does not require knowing that limit. The pairwise formula (7.6) is also invariant under a common rotation.

There are two convenient alternatives. At integer samples,

$$D_N^{\text{int}} = (N-1) - \left| \sum_{m=2}^{N} e^{2\pi i [S(m) - \log\log m]} \right|$$

is bounded if and only if RH holds, by Corollary 6.3 and Lemma 7.1. The continuous version

$$D(X) = (X-2) - \left| \int_2^X e^{2\pi \mathrm{i}[S(t) - \log\log t]} \, \mathrm{d}t \right|$$

has the same property. Its proof is the integral form of the same variance identities and limiting argument.

7.1 Stable updates

Directly subtracting $W_N - |M_N|$ can lose precision when both terms are large. In addition to (7.10), the deficit has a nonnegative rationalized update. If $r = |M| > 0$,

$$D_{\mathrm{new}} - D = \frac{wr\,|u - M/r|^2}{r + w + |M + wu|}. \tag{7.13}$$

If $M = 0$, the increment is zero. Equation (7.13) follows by multiplying $r + w - |M + wu|$ by its conjugate scalar expression $r + w + |M + wu|$. This is a numerical improvement in evaluating a finite quantity; it does not provide the missing infinite upper bound.

8 Exact growth exponents and the rightmost zeros

The convergence criteria can be sharpened to exact growth laws. All exponents below are limsup exponents. No uniform lower bound at every sufficiently large X is claimed.

Lemma 8.1 (The half-plane barrier)**.** *If F is holomorphic throughout $\Re s > a$, then $a \geq \Theta$.*

Proof. For $a \geq 1/2$, this is Lemma 3.1. For $a < 1/2$, Lemma 3.2 excludes the continuation. □

Theorem 8.2 (The exact moment-growth law)**.** *For every finite $q > 0$, let $I_q(X) = \int_2^X |E(x)|^q \, \mathrm{d}x$. Then*

$$\boxed{\limsup_{X\to\infty} \frac{\log(1 + I_q(X))}{\log X} = \max\{0,\, 1 - q(1-\Theta)\}.} \tag{8.1}$$

In particular,

$$\limsup_{X\to\infty} \frac{\log(1 + I(X))}{\log X} = 2\Theta - 1. \tag{8.2}$$

Proof. The classical bound $\pi(x) - \mathrm{li}(x) = O(x^\Theta \log x)$ for $\Theta < 1$ [1, eq. (23)], together with (5.1), implies

$$E(x) = O_{\Theta,\varepsilon}(x^{\Theta - 1 + \varepsilon}) \quad \text{for every } \varepsilon > 0.$$

This statement also holds for $\Theta = 1$, simply because E is bounded. Integrating its q-th power proves the upper bound in (8.1).

Write $\kappa = 1 - q(1-\Theta)$. If $\kappa \leq 0$, the upper bound and nonnegativity prove the result. If the claimed lower bound failed when $\kappa > 0$, choose $a \geq 0$ strictly between the actual limsup and κ. Then $I_q(X) = O(X^a)$.

For $q > 1$, Hölder's inequality on a dyadic interval gives

$$\int_X^{2X} |E(x)| x^{-\sigma} \, \mathrm{d}x \ll X^{\,a/q + 1 - 1/q - \sigma}.$$

Dyadic summation makes F holomorphic for $\Re s > 1 - 1/q + a/q < \Theta$, contradicting Lemma 8.1. For $q = 1$, the same argument gives holomorphy for $\Re s > a < \Theta$.

For $0 < q < 1$, choose $\varepsilon > 0$ so small that

$$b = a + (\Theta - 1 + \varepsilon)(1 - q) < \Theta.$$

On a dyadic interval,

$$|E(x)| \le O\Big(X^{(\Theta-1+\varepsilon)(1-q)}\Big)\,|E(x)|^q.$$

It follows that $\int_X^{2X} |E(x)|x^{-\sigma}\,\mathrm{d}x \ll X^{b-\sigma}$, so F is holomorphic for $\Re s > b < \Theta$, again impossible. This argument includes $\Theta = 1$; boundedness alone then suffices in the last estimate. □

Corollary 8.3 (Weighted growth and contact order)**.** *For every real η,*

$$\limsup_{X\to\infty} \frac{\log\left(1 + \int_2^X x^{1-2\eta}E(x)^2\,\mathrm{d}x\right)}{\log X} = \max\{0,\, 2(\Theta - \eta)\}. \tag{8.3}$$

A curve of contact order k, as in Corollary 5.3, has energy exponent $\max\{0,\, 2k\Theta - 2k + 1\}$.

Proof. For the weighted upper bound, compare with the finite integral at $\eta = \Theta$ from Theorem 5.2. If a smaller power X^a bounded the truncated weighted energy, dyadic weighted Cauchy–Schwarz would make F holomorphic for $\Re s > \eta + a/2 < \Theta$, a contradiction. The curve statement follows by tail comparison and Theorem 8.2 with $q = 2k$. □

8.1 Recovering the exponent from the moving center

Lemma 8.4 (Growth transfer for a convergent vector sequence)**.** *Under the hypotheses of Lemma 7.1, suppose also $W_{N+1} \le C_0 W_N$ eventually for some fixed C_0. For $0 \le a < 1$,*

$$D_N = O(W_N^a) \iff V_N = O(W_N^a) \iff \mathcal{J}_N = O(W_N^a).$$

These three quantities have the same logarithmic limsup growth exponent, defined as $\limsup_N \log(1+$ *quantity*$)/\log W_N$.

Proof. Only the implication from V_N to $\mathcal{J}_N$ needs work. For $M > N$, variance decomposition on the initial N points gives

$$W_N|m_M - m_N|^2 \le V_M - V_N. \tag{8.4}$$

Starting at any large $N = N_0$, let N_{j+1} be the first index at which W is at least $2W_{N_j}$. Then $2W_{N_j} \le W_{N_{j+1}} < 2C_0 W_{N_j}$. If $V_N = O(W_N^a)$, (8.4) implies

$$|m_{N_{j+1}} - m_{N_j}| \ll W_{N_j}^{(a-1)/2}.$$

The bounds form a summable geometric series since $a < 1$. As $m_{N_j} \to u_\infty$, we get $|m_N - u_\infty| \ll W_N^{(a-1)/2}$. Equation (7.7) now gives $\mathcal{J}_N = O(W_N^a)$.

All three limsup exponents lie in $[0, 1]$, because $\mathcal{J}_N \le 4W_N$. If the exponent for D_N were smaller than that for $\mathcal{J}_N$, choose $a < 1$ strictly between them and apply the proved implication. This is a contradiction, including when the larger exponent is 1. □

For the prime weights, Bertrand's postulate gives

$$p_N \le W_N < 3p_N, \qquad W_{N+1} < 3W_N \quad (N \ge 2).$$

The second inequality follows from $W_N \ge p_{N+1}$ for $N \ge 3$; the case $N = 2$ is $W_3/W_2 = 7/3$. Combining (6.6), the tail chord comparison, and Lemma 8.4 proves the main quantitative result.

Theorem 8.5 (The deficit determines the rightmost-zero boundary)**.** *For D_N defined in (7.4),*

$$\boxed{\Theta = \frac{1}{2}\left(1 + \limsup_{N\to\infty} \frac{\log(1+D_N)}{\log p_N}\right).} \tag{8.5}$$

The same formula holds with V_N in place of D_N.

Proof. Equation (6.6) and $p_{N+1} < 2p_N$ give the same exponent for $I(X)$ and A_N. The chord energy $\mathcal{J}_N$ has that exponent because $E(p_n) \to 0$. Lemma 8.4 transfers it to D_N, V_N, and (8.2) identifies its value. □

In particular, for this specific sequence, even the condition $D_N = O_\varepsilon(p_N^\varepsilon)$ for every $\varepsilon > 0$ already implies RH and hence boundedness of D_N. This implication is an arithmetic consequence of (8.5); it is not a general assertion that every subpower sequence is bounded.

9 Laguerre coefficients, Hardy space, and exact Taylor radius

The same error function admits a countable coefficient criterion. This section also identifies exactly where the coefficient generating function becomes singular.

9.1 A scale-parameter family

For $\alpha > 1/2$, let L_n be the ordinary Laguerre polynomial,

$$L_n(u) = \sum_{k=0}^{n} \binom{n}{k} \frac{(-u)^k}{k!},$$

and define

$$a_n^{(\alpha)} = \sqrt{2\alpha} \int_2^\infty E(x) x^{-\alpha-1/2} L_n(2\alpha \log x)\, \mathrm{d}x. \tag{9.1}$$

Every fixed coefficient converges absolutely without RH: E is bounded, the Laguerre factor is a polynomial in $\log x$, and $\alpha + 1/2 > 1$.

Set

$$f(t) = \begin{cases} 0, & 0 \le t < \log 2, \\ e^{t/2} E(e^t), & t \ge \log 2, \end{cases} \qquad \ell_n^{(\alpha)}(t) = \sqrt{2\alpha}\, e^{-\alpha t} L_n(2\alpha t).$$

The functions $\ell_n^{(\alpha)}$ form an orthonormal basis of $L^2(0,\infty)$. Orthogonality follows from the classical Laguerre weight e^{-u}; completeness also follows from uniqueness of the Laplace transform [7].

Theorem 9.1 (Coefficient and Hardy criteria)**.** *For every fixed $\alpha > 1/2$,*

$$\mathrm{RH} \iff \sum_{n\ge 0} |a_n^{(\alpha)}|^2 < \infty. \tag{9.2}$$

Define the germ at $z = 0$

$$s_\alpha(z) = \frac{1}{2} + \alpha \frac{1+z}{1-z}, \qquad G_\alpha(z) = \frac{\sqrt{2\alpha}}{1-z} F(s_\alpha(z)) = \sum_{n\ge 0} a_n^{(\alpha)} z^n. \tag{9.3}$$

Then

$$\mathrm{RH} \iff G_\alpha \in H^2(\mathbb{D}). \tag{9.4}$$

Here membership means that the germ extends holomorphically to an H^2 function on the whole disk. When these conditions hold,

$$\|G_\alpha\|_{H^2}^2 = \sum_{n\ge 0} |a_n^{(\alpha)}|^2 = \int_2^\infty E(x)^2\, \mathrm{d}x. \tag{9.5}$$

Proof. If RH holds, $f \in L^2(0,\infty)$ by Theorem 4.1, and its Laguerre coefficients are exactly (9.1). Parseval proves (9.5).

For the converse, suppose the coefficients are square summable. There is $h \in L^2(0,\infty)$ with these coefficients. For $g = f - h$, all the integrals $\int_0^\infty g(t)e^{-\alpha t}t^k\,\mathrm{d}t$ vanish: the Laguerre polynomials span the polynomial space. The Laplace transform of g is holomorphic for $\Re w > 1/2$, since $f(t) = O(e^{t/2})$ and $h \in L^2$. All its derivatives vanish at $w = \alpha$, so the identity theorem and Laplace uniqueness give $g = 0$ almost everywhere. Thus $f \in L^2$ and RH follows. This argument avoids assuming Parseval before knowing $f \in L^2$.

The generating identity $\sum_{n\ge0} L_n(u)z^n = (1-z)^{-1}\exp[-uz/(1-z)]$ gives (9.3) initially near zero [7]. For an analytic function on the disk, membership in H^2 is equivalent to square summability of its Taylor coefficients, with

$$\|G\|_{H^2}^2 = \sup_{0<r<1} \frac{1}{2\pi}\int_0^{2\pi} |G(re^{\mathrm{i}\theta})|^2\,\mathrm{d}\theta.$$

This proves the remaining assertions. □

For numerical evaluation at a point in the absolutely convergent half-plane, the finite Laguerre expansion gives

$$a_n^{(\alpha)} = \sqrt{2\alpha}\sum_{k=0}^{n}\binom{n}{k}\frac{(2\alpha)^k}{k!}F^{(k)}\left(\frac12+\alpha\right). \tag{9.6}$$

For real $s > 1$, one may use

$$F(s) = \frac{P(s) - c\,2^{1-s} - \mathrm{E}_1((s-1)\log 2)}{s-1}, \qquad \mathrm{E}_1(y) = \int_y^\infty \frac{e^{-t}}{t}\,\mathrm{d}t. \tag{9.7}$$

9.2 Where the power series stops

If ρ is a zero of multiplicity m_ρ with $\Re\rho > 1/2$, equation (3.6) gives locally

$$F(s) = \frac{m_\rho}{s-1}\log(s-\rho) + \text{a holomorphic function}. \tag{9.8}$$

The terms R_0, J_0 cannot cancel this singularity. Its image in the disk is

$$z_\rho^{(\alpha)} = \frac{\rho - \frac12 - \alpha}{\rho - \frac12 + \alpha}, \qquad |z_\rho^{(\alpha)}| < 1 \iff \Re\rho > \frac12. \tag{9.9}$$

Lemma 3.2 supplies another genuine singularity at $s = 1/2$, corresponding to $z = -1$.

Theorem 9.2 (Exact Taylor radius and root criterion)**.** *The Taylor radius $\mathcal{R}_\alpha$ of G_α is*

$$\boxed{\mathcal{R}_\alpha = \min\left\{1,\ \inf_{\substack{\rho\in\mathcal{Z}\\ \Re\rho>1/2}}\left|\frac{\rho-\frac12-\alpha}{\rho-\frac12+\alpha}\right|\right\}}. \tag{9.10}$$

The infimum of the empty set is $+\infty$. Consequently

$$\limsup_{n\to\infty}|a_n^{(\alpha)}|^{1/n} = \mathcal{R}_\alpha^{-1}, \qquad \text{RH} \iff \limsup_{n\to\infty}|a_n^{(\alpha)}|^{1/n} = 1. \tag{9.11}$$

Proof. On the simply connected image of a disk containing no transformed zero, (3.6) gives a holomorphic logarithm and hence a holomorphic G_α. A transformed zero is a nonremovable singularity by (9.8). If the infimum in (9.10) is less than 1, it is attained: there are finitely many zeros of bounded height, and the transformed moduli tend uniformly to 1 as $|\Im\rho| \to \infty$. If there is no interior transformed zero, G_α is holomorphic in the unit disk, but the prime-square logarithm at $z = -1$ prevents a larger Taylor radius. Cauchy–Hadamard proves (9.11). □

The root criterion is an asymptotic assertion. It cannot be established by inspecting a fixed finite list of coefficients. It is also more delicate numerically than the constant-free deficit because its definition uses the exact lift E.

9.3 Optimal scale for one hypothetical zero

Write a hypothetical off-critical zero as $\rho = 1/2 + \delta + \mathrm{i}\gamma$, with $\delta > 0$. Its image has modulus

$$r_\alpha^2 = \frac{(\alpha-\delta)^2+\gamma^2}{(\alpha+\delta)^2+\gamma^2}. \tag{9.12}$$

Elementary differentiation shows that the best scale for this specified zero is

$$\alpha_* = \sqrt{\delta^2+\gamma^2}, \qquad r_{\min}^{-1} = \frac{\sqrt{\delta^2+\gamma^2}+\delta}{|\gamma|}. \tag{9.13}$$

For large $|\gamma|$ and fixed δ,

$$\log r_1^{-1} \sim \frac{2\delta}{\gamma^2}, \qquad \log r_{\alpha_*}^{-1} \sim \frac{\delta}{|\gamma|}.$$

This optimizes the exponential separation due to one given zero. It is not a running-time theorem for finding such a zero.

There is a complementary limitation. For every fixed n, Laplace's endpoint estimate at $t = \log 2$ gives

$$a_n^{(\alpha)} \sim (-1)^n \frac{2^{n+1}E(2)(\log 2)^n}{n!}\alpha^{n-1/2}2^{-\alpha} \quad (\alpha\to\infty). \tag{9.14}$$

Thus every fixed initial coefficient can be made small by changing the scale, regardless of RH. Under RH, the total squared norm in (9.5) is unchanged; energy moves to higher indices.

9.4 An existing verified height gives a quantitative bound

Platt and Trudgian rigorously verified that all nontrivial zeros with $0 < \Im\rho \le 3\,000\,175\,332\,800$ lie on the critical line [8]. Use the safe rounded value $T = 3\cdot 10^{12}$. For any remaining right-half-plane zero, $0 < \delta < 1/2$ and $|\gamma| > T$. Minimizing (9.12) over those ranges yields

$$\mathcal{R}_\alpha \ge \sqrt{\frac{T^2+(\alpha-\frac12)^2}{T^2+(\alpha+\frac12)^2}}. \tag{9.15}$$

In the normalization $\alpha = 1$,

$$1 \le \limsup_n |a_n^{(1)}|^{1/n} \le \sqrt{\frac{T^2+\frac94}{T^2+\frac14}} < 1 + 1.112\cdot 10^{-25}. \tag{9.16}$$

This is a transfer of a published verified-zero theorem, not a new verification or an assertion about the latest computational record. It quantifies why modest coefficient data cannot resolve the eventual root growth.

9.5 Exact centering error creates a false pole

Theorem 9.3 (Miscentering instability)**.** *Replace $B_{\mathbb{P}}$ by $B_{\mathbb{P}} + \delta$, where $\delta \neq 0$, in the lifted error and coefficients. For every fixed $\alpha > 1/2$, the resulting coefficients satisfy, unconditionally,*

$$\widetilde{a}_n^{(\alpha)} \sim -\frac{\delta\sqrt{2\alpha}}{\alpha-\frac12}\left(-\frac{\alpha+\frac12}{\alpha-\frac12}\right)^n. \tag{9.17}$$

In particular,

$$\widetilde{a}_n^{(1)} \sim -2\sqrt{2}\,\delta\,(-3)^n. \tag{9.18}$$

Proof. Since $\widetilde{E} = E - \delta$,

$$\widetilde{F}(s) = F(s) - \delta\frac{2^{1-s}}{s-1}.$$

Hence

$$\widetilde{G}_\alpha(z) - G_\alpha(z) = -\delta\sqrt{2\alpha}\,\frac{2^{1-s_\alpha(z)}}{(\alpha-\frac{1}{2})+(\alpha+\frac{1}{2})z}. \tag{9.19}$$

This has a simple pole at $z_* = -(\alpha-\frac{1}{2})/(\alpha+\frac{1}{2})$, the image of $s=1$. Its numerator there is $-\delta\sqrt{2\alpha}$. After subtracting

$$-\frac{\delta\sqrt{2\alpha}}{\alpha-\frac{1}{2}}\frac{1}{1-z/z_*},$$

the remainder is holomorphic throughout $|z|<1$. Meanwhile the correctly centered G_α is holomorphic in a disk strictly larger than $|z_*|$: the compact image of $|z|\le|z_*|$ lies in $\Re s \ge 1$, which is zero-free, and $s=1$ is removable by (3.6). The artificial pole therefore dominates the coefficients, proving (9.17). □

A fixed rounding error in $B_{\mathbb{P}}$ eventually overwhelms the true coefficients. In the scale $\alpha = 1$, controlling the centering contribution to fixed absolute tolerance at order n requires the number of accurate decimal digits in $B_{\mathbb{P}}$ to grow at least linearly, with leading rate $n\log_{10}3$. Other numerical errors must also be controlled. The circle itself only needs the phase constant modulo 1; the instability theorem concerns the canonical real lift, where every nonzero δ matters. The deficit construction (7.4) avoids an input value of $B_{\mathbb{P}}$ entirely.

10 A second circle and ellipse: geometry of the zeros

There is another exact geometry, now attached directly to the zeros. For $\rho = \beta + \mathrm{i}\gamma$, define

$$u_\rho = 1 - \frac{1}{\rho}.$$

Then

$$|u_\rho|^2 = \frac{(\beta-1)^2+\gamma^2}{\beta^2+\gamma^2}, \qquad \beta = \frac{1}{2} \iff |u_\rho| = 1. \tag{10.1}$$

The symmetry $\rho \mapsto 1-\rho$ sends u_ρ to u_ρ^{-1}, and conjugation sends it to its complex conjugate. Thus RH is equivalent to every u_ρ lying on the unit circle.

Apply the Joukowski map

$$w = \frac{1}{2}(u + u^{-1}) = 1 - \frac{1}{2\rho(1-\rho)}. \tag{10.2}$$

For $u = re^{\mathrm{i}\theta}$, it gives

$$w = \frac{r+r^{-1}}{2}\cos\theta + \mathrm{i}\,\frac{r-r^{-1}}{2}\sin\theta.$$

The circle $|u| = r$ maps to an ellipse with foci ±1, semimajor axis $(r+r^{-1})/2$, and semiminor axis $|r-r^{-1}|/2$. At a zero, the latter is

$$b_\rho = \frac{|1-2\beta|}{2|\rho|\,|\rho-1|}. \tag{10.3}$$

Hence RH is equivalent to every such spectral ellipse degenerating to the segment $[-1,1]$.

These two ellipse constructions have opposite roles. The physical phase ellipse of Theorem 4.1 must keep nonzero linear sensitivity to E. The spectral ellipse above has zero minor axis precisely when its zero is on the critical line. Confusing the constructions would incorrectly turn a geometric change of variables into a proof.

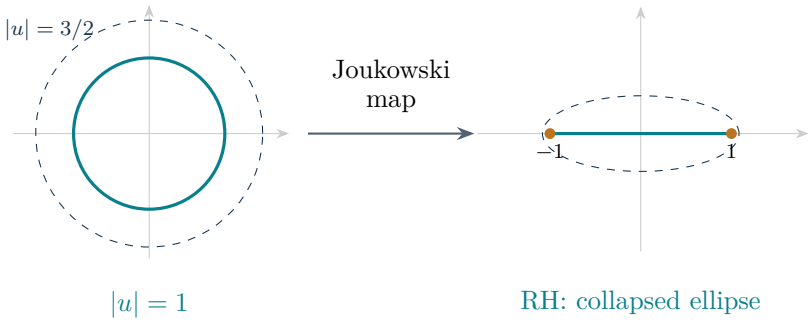


Figure 3: The circle-to-ellipse map for the zeros. This spectral construction is different from applying a fixed ellipse to the prime-reciprocal phase. The dashed ellipse has semiaxes 13/12 and 5/12, corresponding to $|u| = 3/2$.

11 A direct mean-square identity for the remaining bound

The unresolved arithmetic quantity can be isolated without estimating the zeros individually. Put

$$R(x) = \pi(x) - \operatorname{li}(x), \quad f(x) = \frac{R(x)}{x}, \quad A(X) = \int_2^X \frac{R(t)}{t^2}\,\mathrm{d}t, \quad T(X) = \int_2^X \frac{R(t)^2}{t^2}\,\mathrm{d}t.$$

Partial summation and Mertens' theorem give, unconditionally,

$$\begin{aligned} A_\infty &:= \lim_{X\to\infty} A(X) = \log\log 2 + B_{\mathbb{P}} - \frac{\operatorname{li}(2)}{2}, \\ E(x) &= f(x) + A(x) - A_\infty. \end{aligned} \tag{11.1}$$

Numerically, $A_\infty = -0.62759759779276793568\ldots$. Here the prime number theorem gives $f(x) \to 0$; no RH assumption is used in the existence of A_∞.

Proposition 11.1 (Exact optimized variance)**.** *For every $X > 2$,*

$$\min_{c\in\mathbb{R}} \int_2^X (E(x) - c)^2\,\mathrm{d}x = T(X) - \frac{2X}{X-2}A(X)^2. \tag{11.2}$$

Moreover, on putting $H(X) = A_\infty - A(X)$,

$$\int_2^X E(x)^2\,\mathrm{d}x = T(X) + XH(X)^2 - 2A_\infty^2. \tag{11.3}$$

Thus $T(\infty) < \infty$ is equivalent to RH, and when it is finite,

$$T(\infty) = \int_2^\infty E(x)^2\,\mathrm{d}x + 2A_\infty^2, \qquad 2A_\infty^2 = 0.78775748951050582490\ldots. \tag{11.4}$$

Proof. For fixed X, let $H_X(x) = \int_x^X f(t)t^{-1}\,\mathrm{d}t = A(X) - A(x)$. Then $H_X' = -f/x$, $H_X(X) = 0$, and $H_X(2) = A(X)$. Integration by parts gives

$$\int_2^X (f - H_X)^2\,\mathrm{d}x = T(X) - 2A(X)^2, \qquad \int_2^X (f - H_X)\,\mathrm{d}x = 2A(X).$$

Since $E = f - H_X + A(X) - A_\infty$, subtracting the optimal constant subtracts the square of the mean, proving (11.2). Expanding the same identity without subtracting the mean gives (11.3).

If $T(\infty) < \infty$, then

$$XH(X)^2 = X\left(\int_X^\infty \frac{f(t)}{t}\,\mathrm{d}t\right)^2 \le \int_X^\infty f(t)^2\,\mathrm{d}t \longrightarrow 0.$$

Equation (11.3) proves finite E-energy. Conversely that equation and $XH(X)^2 \ge 0$ bound $T(X)$ whenever the E-energy is finite. Theorem 4.1 and passage to the limit finish the proof. □

The subtraction in (11.2) approaches the finite number $2A_\infty^2$. Optimizing the center therefore cannot cancel an unbounded $T(X)$. This identifies the missing global mean-square estimate precisely, rather than attributing it to an uncontrolled choice of center.

12 Compact logarithmic filters preserve the obstruction

The Hardy-space viewpoint suggests averaging the error before measuring its energy. The following result identifies precisely what every fixed compact filter preserves. Signed filters are allowed, and the conclusion does not require their transfer functions to be zero-free.

Extend $E(x)$ by zero for $0 < x < 2$. Let $\mu \neq 0$ be a finite complex Borel measure on $\mathbb{R}$, with compact support contained in $[a, b]$. Define

$$E_\mu(x) = \int_{\mathbb{R}} e^{-u/2} E(xe^{-u})\, \mathrm{d}\mu(u), \qquad x > 0, \tag{12.1}$$

$$M_\mu(z) = \int_{\mathbb{R}} e^{-zu}\, \mathrm{d}\mu(u), \qquad z \in \mathbb{C}, \tag{12.2}$$

$$I_\mu(X) = \int_0^X |E_\mu(x)|^2\, \mathrm{d}x. \tag{12.3}$$

Finite sums of point masses give finite linear combinations of dilations; an integrable density of compact support gives a logarithmic averaging filter. Neither positivity nor normalization of μ is assumed.

Theorem 12.1 (Invariance under compact logarithmic filters)**.** *For every measure μ specified above,*

$$\int_0^\infty |E_\mu(x)|^2\, \mathrm{d}x < \infty \quad \Longleftrightarrow \quad \text{RH}. \tag{12.4}$$

Moreover, the exact energy exponent is unchanged:

$$\boxed{\limsup_{X\to\infty} \frac{\log(1 + I_\mu(X))}{\log X} = 2\Theta - 1.} \tag{12.5}$$

The exponent in (12.5) is a limsup; no lower bound at every sufficiently large X is asserted.

Proof. First, $E_\mu(x) = 0$ when $x < 2e^a$. Compact support makes M_μ entire. It is not identically zero: otherwise all derivatives at zero would vanish, so $\int u^n\, \mathrm{d}\mu(u) = 0$ for every integer $n \geq 0$. Polynomial density in $C([a, b])$, together with finiteness of the total variation of μ, would force $\mu = 0$.

For $\Re s > 1$, define the filtered Mellin transform with the full positive-axis domain:

$$F_\mu(s) = \int_0^\infty E_\mu(x) x^{-s}\, \mathrm{d}x.$$

Since E is bounded and μ has compact support, the relevant integrals are absolutely convergent in this initial half-plane. Fubini's theorem and the substitution $x = e^u y$ give the exact identity

$$F_\mu(s) = M_\mu\left(s - \frac{1}{2}\right) F(s), \qquad \Re s > 1. \tag{12.6}$$

Set $f(t) = e^{t/2} E(e^t)$, using the zero extension, and let

$$g(t) = \int_{\mathbb{R}} f(t - u)\, \mathrm{d}\mu(u).$$

Then $g(t) = e^{t/2}E_\mu(e^t)$, and a change of variables gives

$$\|g\|^2_{L^2(\mathbb{R})} = \int_0^\infty |E_\mu(x)|^2 \,\mathrm{d}x.$$

Under RH, Theorem 4.1 gives $f \in L^2(\mathbb{R})$. Young's inequality for convolution with a finite measure yields

$$\|g\|_2 \le |\mu|(\mathbb{R})\, \|f\|_2 ,$$

which proves the forward implication in (12.4).

Conversely, suppose $E_\mu \in L^2(0,\infty)$. The lower support bound $2e^a > 0$ and Cauchy–Schwarz show that F_μ is holomorphic in $\Re s > 1/2$. Let ρ be a zero of ζ of multiplicity m_ρ, with $\Re\rho > 1/2$. The local continuation of the Mellin identity for F has the form

$$F(s) = \frac{m_\rho}{s-1}\log(s-\rho) + H_\rho(s), \tag{12.7}$$

where H_ρ is holomorphic near ρ and a branch of the logarithm is chosen on a slit neighborhood. Indeed, the prime-power correction is holomorphic for $\Re s > 1/2$, while the local factor $\zeta(s) = (s-\rho)^{m_\rho}h(s)$, $h(\rho) \neq 0$, gives the displayed logarithm.

Continue the right side of (12.6) from its initial half-plane along paths avoiding the isolated zeros. Continuation once around a sufficiently small loop surrounding ρ changes its value by

$$2\pi\mathrm{i}\, m_\rho \frac{M_\mu(s-\frac{1}{2})}{s-1}. \tag{12.8}$$

This is not identically zero on the neighborhood. In particular, a zero of M_μ of any finite order r at $\rho - 1/2$ merely replaces the logarithmic term by a nonzero holomorphic factor times $(s-\rho)^r\log(s-\rho)$; its monodromy remains nonzero. Thus the product in (12.6) cannot agree with the single-valued holomorphic function F_μ throughout $\Re s > 1/2$. There is no such zero ρ, and the functional equation gives RH.

For the growth statement, put $A = |\mu|(\mathbb{R})$, and extend $I(X) = \int_2^X E(x)^2\,\mathrm{d}x$ by zero for $0 < X < 2$. Cauchy–Schwarz with respect to $|\mu|$, followed by Tonelli's theorem, gives

$$\begin{aligned} I_\mu(X) &= \int_{-\infty}^{\log X} |g(t)|^2\,\mathrm{d}t \\ &\le A\int_{\mathbb{R}}\int_{-\infty}^{\log X-u} |f(v)|^2\,\mathrm{d}v\,\mathrm{d}|\mu|(u) \\ &\le A^2 I(e^{-a}X). \end{aligned}$$

Theorem 8.2 therefore implies

$$\Gamma_\mu := \limsup_{X\to\infty} \frac{\log(1+I_\mu(X))}{\log X} \le \gamma := 2\Theta - 1.$$

If $\gamma = 0$, equality follows from $\Gamma_\mu \ge 0$. Otherwise, suppose $\Gamma_\mu < \gamma$, and choose $\Gamma_\mu < \beta < \beta' < \gamma$. Then $I_\mu(X) = O(X^\beta)$, so integration by parts yields

$$\int_0^\infty x^{-\beta'}|E_\mu(x)|^2\,\mathrm{d}x < \infty.$$

With $\eta = (1+\beta')/2 < \Theta$, weighted Cauchy–Schwarz makes F_μ holomorphic in $\Re s > \eta$. By the definition of Θ, there is a zero ρ with $\Re\rho > \eta$. Here $\eta > 1/2$, so (12.7) applies. Its surviving monodromy contradicts holomorphy just as before. This proves (12.5). □

Remark 12.2 (Scope of the obstruction). This theorem concerns fixed, nonzero, compactly supported linear filters of the error. It does not rule out nonlinear constructions, noncompact operators, or new arithmetic inequalities. It shows that a filter in the stated class cannot supply the missing uniform bound merely by cancelling the logarithmic singularities: even a deliberately placed zero of its entire transfer function does not cancel such a branch point.

Part II
Geometric, harmonic, and arithmetic transfers

13 What positivity and Hardy-space transfers require

The positive forms arising from circle variance, Weil's criterion, and finite spectral approximations have different hypotheses. Identifying those differences prevents an automatic positivity statement from being used as an upper bound on arithmetic energy.

13.1 A Gram square is not the Weil zero pairing

Use the multiplicative Fourier convention

$$\widehat{g}(z) = \int_0^\infty g(x) x^{-\mathrm{i}z} \frac{\mathrm{d}x}{x}, \qquad s_\rho = \frac{\rho - \frac{1}{2}}{\mathrm{i}}.$$

For smooth compactly supported test functions, with the usual conditions $\widehat{g}(\pm \mathrm{i}/2) = 0$ removing the pole terms, the zero side of Weil's quadratic form involves

$$\mathcal{Q}(g) = \sum_\rho \widehat{g}(s_\rho)\, \overline{\widehat{g}(\overline{s_\rho})}.$$

Its nonnegativity for every admissible g is the Weil criterion; see [9, Section 4.1]. Under RH, each s_ρ is real and the corresponding summand is an absolute square.

For an off-line pair $s, \overline{s}$, however, put $a = \widehat{g}(s)$ and $b = \widehat{g}(\overline{s})$. Its contribution is

$$2\Re(a\overline{b}) = \begin{pmatrix} \overline{a} & \overline{b} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}.$$

The displayed matrix has eigenvalues 1 and -1. Replacing this pairing by $|a|^2 + |b|^2$ adds exactly $|a - b|^2$, eliminating its negative direction. This block computation distinguishes the two forms; it does not by itself isolate a pair from the full infinite zero sum.

By contrast, a variance such as

$$V_N = \min_{c \in \mathbb{C}} \sum_{n=2}^{N} w_n |U_n - c|^2$$

is nonnegative for every sequence (U_n) and every choice of positive weights. That automatic Gram positivity does not establish Weil positivity, and it gives no uniform upper bound on V_N or D_N. The required arithmetic estimate remains a bound on the size of the specific vector, rather than positivity of its squared norm.

13.2 Noor's density result and the unbounded inverse

Noor considers the functions

$$h_k(z) = \frac{1}{1 - z} \log\left(\frac{1 + z + \cdots + z^{k-1}}{k} \right), \qquad k \geq 2.$$

Their linear span is dense in H^2 if and only if RH. Unconditionally, the span of $(1 - z)h_k$ is dense in H^2, and the span of h_k is dense for the compact-open topology [10, Theorems 8–10].

The missing norm control cannot be recovered by simply dividing by $1 - z$. Indeed, the explicit polynomials

$$p_N(z) = \frac{1 + z + \cdots + z^{N-1}}{\sqrt{N}}$$

satisfy

$$\|p_N\|_{H^2} = 1, \qquad \|(1 - z)p_N\|_{H^2} = \sqrt{\frac{2}{N}} \longrightarrow 0.$$

Thus multiplication by $1 - z$ has no bounded inverse on its range. Convergence after this multiplication, or convergence on every compact subset of the disk, does not supply the boundary norm bound needed for the prime-circle energy.

13.3 The remaining hypotheses in the Connes approximation

Connes' real-zero theorem applies to a lower-bounded selfadjoint operator on a finite interval whose kernel depends on the difference of its arguments. It concludes that the Fourier transform of its lowest eigenfunction has only real zeros, provided that the lowest eigenvalue is simple and isolated and the eigenfunction is even [9, Theorem 6.1].

For the restricted Weil forms, the same paper explicitly leaves two steps to be established: the required simplicity and evenness of the lowest eigenfunction, and sufficiently strong approximation of θ_x by k_λ [9, Section 6.6]. Consequently, its accurate finite approximations do not furnish a uniform bound for the present energy. They identify a substantive spectral route, whose limiting and positivity hypotheses still require proof.

14 Closing the Epstein threshold calculation

This section resolves a concrete claim in the proposed circle–ellipse comparison. The constant $4\pi e^{-\gamma}$ is an exact transition value for an *averaged* Eisenstein series, as already proved by Lagarias–Suzuki [11, 12]. For a fixed rectangular lattice, a positive Bessel correction survives, and the central crossing occurs strictly below that constant. The calculation below establishes the correction, uniqueness, and a certified enclosure; it makes no claim of a new criterion for the Riemann zeta function.

For $k \geq 1$, initially in $\operatorname{Re} s > 1$, set

$$Z_k(s) = \sum_{(m,n)\in\mathbb{Z}^2\setminus\{(0,0)\}} (m^2 + k^2 n^2)^{-s}, \qquad k_0 = 4\pi e^{-\gamma},$$

where γ is Euler's constant. Values outside this half-plane refer to the meromorphic continuation. Let $d(r)$ denote the number of positive divisors of r.

Proposition 14.1 (Exact central value)**.** *For every $k > 0$,*

$$\Phi(k) := Z_k(1/2) = 2\log\frac{k}{k_0} + 8\sum_{r=1}^{\infty} d(r)K_0(2\pi kr). \tag{14.1}$$

In particular, $\Phi(k_0) > 0$.

Proof. The Fourier, or Chowla–Selberg, expansion [11] gives

$$Z_k(s) = 2\zeta(2s) + 2\sqrt{\pi}\, k^{1-2s}\frac{\Gamma(s - 1/2)}{\Gamma(s)}\zeta(2s - 1) + \frac{8\pi^s k^{1/2-s}}{\Gamma(s)}\sum_{r\geq 1} r^{s-1/2}\sigma_{1-2s}(r)K_{s-1/2}(2\pi kr),$$

where $\sigma_u(r) = \sum_{d|r} d^u$. At $s = 1/2 + \varepsilon$, the first two terms are respectively $\varepsilon^{-1} + 2\gamma + O(\varepsilon)$ and $-\varepsilon^{-1} + 2\log(k/(4\pi)) + O(\varepsilon)$. Their poles cancel. The last series converges locally uniformly and becomes the series in (14.1). Every K_0 term is strictly positive. □

Proposition 14.2 (Unique rectangular central crossing)**.** *There is exactly one $k_{\mathrm{rect}} \geq 1$ with $Z_{k_{\mathrm{rect}}}(1/2) = 0$. It satisfies $k_{\mathrm{rect}} < k_0$ and*

$$\boxed{2.96037427750280\,10^{-19} < k_0 - k_{\mathrm{rect}} < 2.96037427750281\,10^{-19}.} \tag{14.2}$$

Thus $k_{\mathrm{rect}} \approx 7.055507955448182762474611110423$, rather than exactly k_0.

Existence and uniqueness. The integral representations of K_0 and K_1, together with $1 + t^2/2 \leq \cosh t \leq e^{t^2/2}$, imply

$$K_0(x) \leq e^{-x}\sqrt{\frac{\pi}{2x}}, \qquad K_1(x) \leq e^{-x}\sqrt{\frac{\pi}{2(x-1)}} \quad (x > 1). \tag{14.3}$$

Here $\log\cosh t \leq t^2/2$ follows by integrating $\tanh t \leq t$. Termwise differentiation in (14.1) gives

$$\Phi'(k) = \frac{2}{k} - 16\pi \sum_{r\geq 1} r\, d(r) K_1(2\pi k r).$$

For fixed $r \geq 1$, the function $ke^{-2\pi kr}/\sqrt{2\pi kr - 1}$ decreases on $k \geq 1$. Consequently, using $d(r) \leq r$ and $q = e^{-2\pi}$,

$$8\pi k \sum_{r\geq 1} r\, d(r) K_1(2\pi k r) \leq B := \frac{8\pi\sqrt{\pi}}{\sqrt{2(2\pi - 1)}} \frac{q(1+q)}{(1-q)^3} < 0.026,$$

$$\Phi'(k) \geq \frac{2(1-B)}{k} > \frac{1.948}{k} > 0.$$

Also, (14.3) yields

$$\Phi(1) \leq -2\log k_0 + \frac{4q}{(1-q)^2} < -3.9001.$$

The Bessel series tends to zero as $k \to \infty$, so $\Phi(k) \to +\infty$. Strict monotonicity proves existence and uniqueness. Since $\Phi(k_0) > 0$, the crossing is below k_0. The numerical enclosure is certified below. □

Corollary 14.3 (A sufficient condition for real off-line zeros)**.** *If $k > k_{\mathrm{rect}}$, then $Z_k(s)$ has a real zero in $(1/2, 1)$ and a reflected real zero in $(0, 1/2)$.*

Proof. We have $Z_k(1/2) > 0$, whereas $\mathrm{Res}_{s=1} Z_k(s) = \pi/k > 0$ implies $Z_k(s) \to -\infty$ as $s \to 1^-$. Continuity supplies the first zero. The functional equation for $\Lambda_k(s) = k^s \pi^{-s} \Gamma(s) Z_k(s)$ is $\Lambda_k(s) = \Lambda_k(1-s)$, giving the reflected zero. □

Remark 14.4. The converse asserted for *all* off-critical zeros would be unjustified: a negative central value alone does not exclude other real or nonreal zeros. The proposition concerns a unique *central crossing*; the corollary gives only the stated sufficient implication.

A finite, rigorous numerical certificate. For $x > 0$ define

$$P_J(x) = \sum_{j=0}^{J} \frac{(-1)^j (2j)!^2}{32^j (j!)^3 x^j}.$$

The exact integral

$$K_0(x) = \frac{e^{-x}}{\sqrt{2x}} \int_0^\infty e^{-u} u^{-1/2} \left(1 + \frac{u}{2x}\right)^{-1/2} \, \mathrm{d}u$$

and Taylor's theorem give, for every integer $M \geq 0$,

$$e^{-x}\sqrt{\frac{\pi}{2x}} P_{2M+1}(x) < K_0(x) < e^{-x}\sqrt{\frac{\pi}{2x}} P_{2M}(x). \tag{14.4}$$

Indeed, the derivatives of $(1+v)^{-1/2}$ alternate in sign for $v \geq 0$, so its Taylor remainder has the required sign; integration then gives the displayed coefficients. These are bounds with signed remainders, not an assumption that an asymptotic series converges. For $q_k = e^{-2\pi k}$, (14.3) also gives

$$0 < \sum_{r \geq 3} d(r) K_0(2\pi k r) \leq \frac{q_k^3 (3 - 2q_k)}{2\sqrt{k}(1 - q_k)^2}. \tag{14.5}$$

Use P_{13} and P_{12} for lower and upper bounds on the first two terms, then add (14.5) only to the upper bound. Outward interval arithmetic gives

$$\begin{aligned} \Phi(k_0 - 2.96037427750280\,10^{-19}) &\in [1.5398, 1.5459]\,10^{-34}, \\ \Phi(k_0 - 2.96037427750281\,10^{-19}) &\in [-1.2949, -1.2888]\,10^{-34}. \end{aligned}$$

These opposite signs and strict monotonicity prove (14.2). The accompanying certificate evaluates only interval constants and elementary functions; it does not rely on a floating-point Bessel routine.

Why averaging gives the exact constant. For a sheared lattice write

$$Z_{x,y}(s) = \sum_{(m,n) \neq (0,0)} \left((m + xn)^2 + y^2 n^2\right)^{-s}.$$

Its central value and horizontal average are

$$\begin{aligned} Z_{x,y}(1/2) &= 2\log(y/k_0) + 8 \sum_{r \geq 1} d(r) \cos(2\pi r x) K_0(2\pi r y), \\ \int_0^1 Z_{x,y}(1/2)\, \mathrm{d}x &= 2\log(y/k_0). \end{aligned}$$

Thus averaging removes precisely the correction retained by the rectangle $x = 0$. In the standard normalization $E^*(x + iy, s) = \frac{1}{2}\pi^{-s}\Gamma(s) y^s Z_{x,y}(s)$, the constant Fourier coefficient is

$$a_0(y, s) = \zeta^*(2s) y^s + \zeta^*(2 - 2s) y^{1-s}, \qquad \zeta^*(w) = \pi^{-w/2}\Gamma(w/2)\zeta(w),$$

and $a_0(y, 1/2) = \sqrt{y}\log(y/k_0)$. Lagarias–Suzuki proved that, for $y \geq 1$, all zeros of a_0 lie on the critical line except for precisely two real zeros when $y > k_0$ [11, 12]. The exact averaged threshold is therefore established prior work. Removing the nonconstant Fourier terms changes the function; this averaging identity supplies no bound on the original prime-circle discrepancy and no proof of RH.

15 Pair energies, harmonic certificates, and the original deficit

The vectors U_n, weights w_n, total mass W_N, and original deficit D_N retain the definitions in (7.1)–(7.4). In particular, neither the vectors nor their weights are changed in this section. For integers $k, m \geq 1$, define

$$M_{k,N} = \sum_{n=2}^{N} w_n U_n^k, \qquad V_N^{(k)} = W_N - \frac{|M_{k,N}|^2}{W_N}, \tag{15.1}$$

$$C_{m,N} = \frac{1}{W_N} \sum_{2 \leq j < k \leq N} w_j w_k |U_j - U_k|^{2m}, \tag{15.2}$$

$$J_{m,N} = \sum_{n=2}^{N} w_n |U_n - u_\infty|^{2m}, \qquad u_\infty = e^{2\pi \mathrm{i} B_{\mathbb{P}}}. \tag{15.3}$$

Every finite $C_{m,N}$ uses only the original prime vectors and weights; it requires no supplied value of $B_{\mathbb{P}}$. The quantity $J_{m,N}$ is used for analysis of its limit. One has $V_N^{(1)} = V_N$ and $C_{1,N} = V_N$. Higher m measure higher powers of pairwise distance, so their boundedness must not be identified with boundedness of the original D_N.

15.1 Exact harmonic identities

Proposition 15.1 (A finite harmonic formula)**.** *For every integer $m \geq 1$ and every $N \geq 2$,*

$$\boxed{C_{m,N} = \sum_{k=1}^{m} (-1)^{k+1} \binom{2m}{m-k} V_N^{(k)}.} \tag{15.4}$$

In particular,

$$\begin{aligned} C_{2,N} &= 4V_N^{(1)} - V_N^{(2)}, \\ C_{3,N} &= 15V_N^{(1)} - 6V_N^{(2)} + V_N^{(3)}, \\ C_{4,N} &= 56V_N^{(1)} - 28V_N^{(2)} + 8V_N^{(3)} - V_N^{(4)} \geq 0. \end{aligned} \tag{15.5}$$

Proof. The binomial expansion gives

$$|1 - e^{\mathrm{i}t}|^{2m} = \binom{2m}{m} + 2 \sum_{k=1}^{m} (-1)^k \binom{2m}{m-k} \cos(kt).$$

Evaluating at $t = 0$ and subtracting rewrites this as

$$|1 - e^{\mathrm{i}t}|^{2m} = 2 \sum_{k=1}^{m} (-1)^{k+1} \binom{2m}{m-k} \{1 - \cos(kt)\}.$$

For $U_j = e^{\mathrm{i}\phi_j}$, expansion of $|M_{k,N}|^2$ gives

$$V_N^{(k)} = \frac{2}{W_N} \sum_{j<\ell} w_j w_\ell \{1 - \cos(k(\phi_j - \phi_\ell))\}.$$

Substitution proves the identity, including its normalization. Nonnegativity follows from the pair definition, even though the harmonic coefficients have alternating signs. □

15.2 A general limit theorem for convergent directions

Theorem 15.2 (Pair moments and the limiting direction)**.** *Let u_n be any unit complex numbers with $u_n \to u_\infty$, and let $w_n > 0$ with $W_N = \sum_{n=2}^{N} w_n \to \infty$. For an integer $m \geq 1$, define $C_{m,N}, J_{m,N}$ as in (15.2)–(15.3), using u_n. Then*

$$0 \leq C_{m,N} \leq 2^{2m-1} J_{m,N}, \tag{15.6}$$

and the extended-real limit satisfies

$$\lim_{N\to\infty} C_{m,N} = J_{m,\infty} := \sum_{n=2}^{\infty} w_n |u_n - u_\infty|^{2m} \ \in [0,\infty]. \tag{15.7}$$

Consequently,

$$\sup_N C_{m,N} < \infty \quad \Longleftrightarrow \quad J_{m,\infty} < \infty. \tag{15.8}$$

No restriction on the relative sizes of the weights is needed here.

Proof. Put $q = 2m$ and $d_n = |u_n - u_\infty|^q$. The triangle inequality and convexity give

$$|u_j - u_k|^q \leq 2^{q-1}(d_j + d_k).$$

As

$$\sum_{j<k} w_j w_k (d_j + d_k) = \sum_j w_j d_j (W_N - w_j) \leq W_N J_{m,N},$$

this proves (15.6).

Fix an index L. Split the pair sum into pairs within $\{2,\dots,L\}$, pairs with one index in this set, and pairs with both indices larger than L. The first part, divided by W_N, tends to zero. For each fixed $j \leq L$, weighted Cesaro convergence gives

$$\frac{1}{W_N} \sum_{k=L+1}^{N} w_k |u_j - u_k|^q \longrightarrow |u_j - u_\infty|^q.$$

Indeed the summand tends to the displayed limit, is bounded, and $(W_N - W_L)/W_N \to 1$. The cross-pair part therefore tends to $J_{m,L}$. All remaining terms are nonnegative, so

$$\liminf_{N\to\infty} C_{m,N} \geq J_{m,L}.$$

If $J_{m,\infty} = \infty$, letting $L \to \infty$ proves divergence of $C_{m,N}$ to infinity.

Suppose instead that $J_{m,\infty} < \infty$. Applying the same triangle inequality to the pairs entirely in the tail gives

$$\frac{1}{W_N} \sum_{L<j<k\leq N} w_j w_k |u_j - u_k|^q \leq 2^{q-1} \frac{W_N - W_L}{W_N} \sum_{n=L+1}^{N} w_n d_n \leq 2^{q-1}(J_{m,\infty} - J_{m,L}).$$

Consequently,

$$J_{m,L} \leq \liminf_N C_{m,N} \leq \limsup_N C_{m,N} \leq J_{m,L} + 2^{q-1}(J_{m,\infty} - J_{m,L}).$$

Letting $L \to \infty$ proves the finite-limit assertion. The equivalence follows either from these limits and (15.6), or directly from the two estimates. □

15.3 Sampling every even moment at prime endpoints

Lemma 15.3 (An elementary interpolation bound)**.** *For every $q > 0$ and $a, b \in \mathbb{R}$,*

$$\frac{|a|^q + |b|^q}{2(q+1)} \le \int_0^1 |(1-y)a + yb|^q \, \mathrm{d}y \le |a|^q + |b|^q. \tag{15.9}$$

Proof. The upper bound follows because the absolute value of a point on the segment is at most $\max(|a|, |b|)$. For the lower bound, set $r = |a|$ and $t = |b|$. The reverse triangle inequality shows that the integral is at least the integral with endpoints $r, -t$. If $r + t > 0$, the latter is

$$\frac{r^{q+1} + t^{q+1}}{(q+1)(r+t)}.$$

The inequality

$$2(r^{q+1} + t^{q+1}) - (r+t)(r^q + t^q) = (r-t)(r^q - t^q) \ge 0$$

proves the claim. The case $r = t = 0$ is immediate. □

Proposition 15.4 (Prime sampling for every $q > 1$)**.** *For fixed $q > 1$, put*

$$I_q(X) = \int_2^X |E(x)|^q \, \mathrm{d}x, \qquad A_{q,N} = \sum_{n=2}^{N} w_n |E(p_n)|^q.$$

Then

$$I_q(p_N) \ll_q 1 + A_{q,N} \ll_q 1 + I_q(p_{N+1}). \tag{15.10}$$

In particular, for $q = 2m$, replacing $A_{q,N}$ by $J_{m,N}$ preserves both these comparisons and the corresponding convergence criterion.

Proof. Use the existing notation $e_n = E(p_n)$, $h_n = p_{n+1} - p_n$, and $b_n = E(p_{n+1}^-) = e_{n+1} - 1/p_{n+1}$. On a prime gap, the change of variable

$$y = \frac{\log\log x - \log\log p_n}{\log\log p_{n+1} - \log\log p_n}$$

gives $E(x) = (1-y)e_n + yb_n$. As in Lemma 6.1, Bertrand's postulate bounds the ratio of the largest and smallest values of $\mathrm{d}x/\,\mathrm{d}y$ by 4. Its integral is h_n, so $\mathrm{d}x/\,\mathrm{d}y \ge h_n/4$. Lemma 15.3 and the pointwise upper bound give

$$\frac{h_n}{8(q+1)}(|e_n|^q + |b_n|^q) \le \int_{p_n}^{p_{n+1}} |E(x)|^q \, \mathrm{d}x \le h_n(|e_n|^q + |b_n|^q). \tag{15.11}$$

The jump correction is summable:

$$\sum_{n \ge 1} \frac{h_n}{p_{n+1}^q} \le \int_2^\infty x^{-q} \, \mathrm{d}x = \frac{2^{1-q}}{q-1} < \infty. \tag{15.12}$$

The inequalities

$$|x+y|^q \le 2^{q-1}(|x|^q + |y|^q)$$

applied in both directions permit replacement of b_n by e_{n+1}. Summing (15.11) through $n = N-1$ proves the left comparison in (15.10); summing through $n = N$ proves the right comparison. The initial endpoint and (15.12) contribute only constants.

Finally,

$$|U_n - u_\infty| = 2|\sin(\pi E(p_n))| \sim 2\pi |E(p_n)|$$

as $n \to \infty$, with the usual interpretation when an endpoint value is zero. Thus the q-th powers are comparable on a tail; the finitely many initial terms are absorbed by the constants. □

Corollary 15.5 (The exact boundedness threshold)**.** *For the original prime vectors and weights and every integer $m \geq 1$,*

$$\boxed{\sup_N C_{m,N} < \infty \quad \Longleftrightarrow \quad \Theta \leq 1 - \frac{1}{2m}.} \tag{15.13}$$

At every finite case the limit is exactly

$$\lim_N C_{m,N} = \sum_{n=2}^{\infty} w_n |e^{2\pi i E(p_n)} - 1|^{2m}.$$

Proof. Combine Theorem 15.2, Proposition 15.4, and the endpoint-inclusive integrability criterion in Theorem 5.1. □

15.4 Moving centers and exact growth

Lemma 15.6 (A moving-center estimate for pair moments)**.** *In the setting of Theorem 15.2, suppose in addition that $W_{N+1} \leq \Lambda W_N$ for some fixed Λ and all sufficiently large N. If $0 \leq a < 1$ and $C_{m,N} = O(W_N^a)$, then*

$$J_{m,N} = O(W_N^a). \tag{15.14}$$

Proof. Write $q = 2m$ and $m_N = W_N^{-1} \sum_{n=2}^{N} w_n u_n$. Convexity and Jensen's inequality give

$$\sum_{n=2}^{N} w_n |u_n - m_N|^q \leq \frac{1}{W_N} \sum_{j,k=2}^{N} w_j w_k |u_j - u_k|^q = 2C_{m,N}. \tag{15.15}$$

If $M > N$, a second application of Jensen on the first N terms yields

$$W_N |m_N - m_M|^q \leq \sum_{n=2}^{N} w_n |u_n - m_M|^q \leq 2C_{m,M}.$$

Starting at a sufficiently large $N_0 = N$, let N_{j+1} be the first index with $W_{N_{j+1}} \geq 2W_{N_j}$. The bounded-overshoot hypothesis implies

$$2W_{N_j} \leq W_{N_{j+1}} \leq 2\Lambda W_{N_j}.$$

Hence

$$|m_{N_{j+1}} - m_{N_j}| \ll W_{N_j}^{(a-1)/q} \leq W_N^{(a-1)/q} 2^{j(a-1)/q}.$$

The series converges because $a < 1$. Weighted Cesaro convergence gives $m_{N_j} \to u_\infty$, so telescoping proves

$$|m_N - u_\infty| \ll W_N^{(a-1)/q}.$$

Finally, (15.15) and the power triangle inequality give

$$J_{m,N} \leq 2^{q-1} \{2C_{m,N} + W_N |m_N - u_\infty|^q\} \ll W_N^a.$$

Finite initial indices cause no difficulty. □

Theorem 15.7 (The pair energy detects its exact zero boundary)**.** *For every integer $m \geq 1$,*

$$\boxed{\limsup_{N \to \infty} \frac{\log(1 + C_{m,N})}{\log p_N} = \max\{0,\, 1 - 2m(1 - \Theta)\}.} \tag{15.16}$$

Proof. Proposition 15.4, Theorem 8.2, and $p_{N+1} < 2p_N$ show that $J_{m,N}$ has the exponent

$$\kappa_m = \max\{0,\, 1 - 2m(1-\Theta)\}.$$

Indeed the two sampling cutoffs are within a fixed factor, so monotonicity of the integral also identifies its exponent along prime cutoffs. The upper bound for $C_{m,N}$ follows from (15.6).

If $\kappa_m = 0$, nonnegativity proves the result. Otherwise $0 < \kappa_m \le 1$. Were the asserted lower bound false, choose a strictly between the actual limsup and κ_m. Then $0 \le a < 1$ and $C_{m,N} = O(p_N^a) = O(W_N^a)$. The formula $W_N = p_N + p_{N+1} - 5$ and Bertrand's postulate give both $W_N \asymp p_N$ and bounded successive ratios. Lemma 15.6 would imply $J_{m,N} = O(p_N^a)$, contradicting its already identified exponent. □

15.5 A conditional four-harmonic bound and a bound for D_N

Corollary 15.8 (Four-harmonic boundedness under (H_b))**.** *Admit the external zero-boundary theorem $\Theta \le b_*$ stated in (91.9) and [13, Theorem 27.1]. Since $b_* < 7/8$,*

$$\sup_N \left(56V_N^{(1)} - 28V_N^{(2)} + 8V_N^{(3)} - V_N^{(4)}\right) < \infty. \tag{15.17}$$

The expression is nonnegative for every N, and its limit is $J_{4,\infty}$. This consequence is conditional on (H_b); it does not separately assume RH. *It retains the original weights and removes the need to supply $B_{\mathbb{P}}$, but it is a higher-order statistic rather than D_N.*

Proof. Theorem 91.1 gives $E \in L^8$. Now apply Proposition 15.4, Theorem 15.2, and (15.5). Alternatively, use Corollary 15.5 directly. □

Proposition 15.9 (A finite Hölder bridge to the original deficit)**.** *For every integer $m \ge 1$ and every $N \ge 2$,*

$$\boxed{D_N \le V_N \le \left(\frac{W_N}{2}\right)^{1-1/m} C_{m,N}^{1/m}.} \tag{15.18}$$

In particular, the right side with $m = 4$ is computable from four harmonic resultants by (15.5). A uniform bound on that fourth pair moment yields $D_N = O(W_N^{3/4})$, not a uniform constant bound.

Proof. Give each unordered pair (j,k) mass $w_j w_k / W_N$. The total mass of this positive measure is

$$\frac{W_N^2 - \sum_{n=2}^N w_n^2}{2W_N} \le \frac{W_N}{2}.$$

Its integrals of $|U_j - U_k|^2$ and $(|U_j - U_k|^2)^m$ are respectively V_N and $C_{m,N}$. Hölder's inequality proves the assertion when $m > 1$. For $m = 1$ it is the identity $C_{1,N} = V_N$; the zero-pair case also satisfies the displayed inequality. □

Remark 15.10 (Nonmonotonicity). Higher pair moments need not be nondecreasing, even though their extended-real limits always exist for convergent directions. With unit weights and the first two points $u_1 = e^{\mathrm{i}\pi/3}, u_2 = e^{-\mathrm{i}\pi/3}$, one has $C_{2,2} = 9/2$. Appending $u_3 = 1$ gives $C_{2,3} = 11/3$. This independent example uses indices starting at 1. It can be extended to a convergent sequence by setting all subsequent points equal to 1.

15.6 Deficits of positive-definite Fourier kernels

Positive-definite kernels supply nonnegative expressions, but a uniform upper budget is a separate requirement. The next result identifies exactly what positive combinations of harmonic variances retain.

Theorem 15.11 (Coercivity of positive harmonic combinations)**.** *Let $a_k \geq 0$ satisfy*

$$0 < A_2 := \sum_{k=1}^{\infty} a_k k^2 < \infty.$$

Define

$$K_{\mathbf{a}}(t) = \sum_{k\geq 1} a_k \cos(kt), \qquad Q_{\mathbf{a}}(t) = K_{\mathbf{a}}(0) - K_{\mathbf{a}}(t), \qquad \mathfrak{C}_N(\mathbf{a}) = \sum_{k\geq 1} a_k V_N^{(k)}.$$

For any convergent unit-vector sequence and positive weights with $W_N \to \infty$, there exist $c > 0, C_0 < \infty$ such that

$$cV_N - C_0 \leq \mathfrak{C}_N(\mathbf{a}) \leq A_2 V_N \qquad \text{for all } N. \tag{15.19}$$

Consequently, for the prime-circle sequence,

$$\sup_N \mathfrak{C}_N(\mathbf{a}) < \infty \quad \Longleftrightarrow \quad \sup_N D_N < \infty \quad \Longleftrightarrow \quad \mathrm{RH}, \tag{15.20}$$

and its logarithmic growth exponent is $2\Theta - 1$.

Proof. The inequality $|z^k - w^k| \leq k|z - w|$ for unit z, w implies $V_N^{(k)} \leq k^2 V_N$, proving the upper bound and convergence of the defining series. The exact pair formula is

$$\mathfrak{C}_N(\mathbf{a}) = \frac{1}{W_N} \sum_{i,j=2}^{N} w_i w_j Q_{\mathbf{a}}(\phi_i - \phi_j), \qquad u_n = e^{\mathrm{i}\phi_n}.$$

For $t \neq 0$ sufficiently close to zero,

$$0 \leq \frac{1 - \cos(kt)}{1 - \cos t} \leq k^2, \qquad \frac{1 - \cos(kt)}{1 - \cos t} \longrightarrow k^2.$$

Dominated convergence in k therefore gives

$$\frac{Q_{\mathbf{a}}(t)}{1 - \cos t} \longrightarrow A_2.$$

Choose a tail of the convergent directions on which all pairwise phase differences lie in an arc where $Q_{\mathbf{a}}(t) \geq c(1 - \cos t)$, with $c = A_2/2$. Let A be the total weight of the omitted finite prefix. In the ordered-pair formula for V_N, pairs with at least one endpoint in that prefix contribute at most $4A$, since $1 - \cos t \leq 2$. All terms of $Q_{\mathbf{a}}$ are nonnegative, so

$$\mathfrak{C}_N(\mathbf{a}) \geq c(V_N - 4A).$$

The same lower inequality is harmless before the end of the prefix. Take $C_0 = 4cA$. The equivalence follows from Theorem 7.2; the exponent follows from (15.19) and Theorem 8.5. □

Remark 15.12 (The exact certificate still required). The kernel $K_{\mathbf{a}}$ is positive definite; $Q_{\mathbf{a}} = K_{\mathbf{a}}(0) - K_{\mathbf{a}}$ is a conditionally negative definite variogram, not a positive-definite kernel. For a nonzero finite family $a_k \geq 0$, a sufficient arithmetic certificate for the original goal is

$$\sum_{k=1}^{m} a_k \{W_N^2 - |M_{k,N}|^2\} \leq C W_N \quad \text{for every } N,$$

with one finite C. The theorem would then give $D_N \le (C + C_0)/c$. Positive definiteness supplies nonnegativity of the expression; it does not supply this upper bound. The theorem does not exclude finding such a certificate using additional arithmetic information. The alternating combination in (15.5) lies outside the nonnegative-coefficient cone and has higher vanishing order at coincident points.

Remark 15.13 (A precise limit of positivity alone). Consider the independent sequence $u_n = e^{\mathrm{i}n^{-1/4}}$, $n \ge 1$, with unit weights. It converges to 1, and $\sum_n |u_n - 1|^8 < \infty$, so its fourth pair moment is bounded. Every finite moment matrix of its positive atomic measure is positive semidefinite. Nevertheless its ordinary deficit satisfies

$$N - \left| \sum_{n=1}^{N} e^{\mathrm{i}n^{-1/4}} \right| \sim \frac{1}{9}\sqrt{N}.$$

Indeed the real and imaginary parts of the sum are respectively

$$N - \sqrt{N} + O(\log N), \qquad \frac{4}{3}N^{3/4} + O(N^{1/4}),$$

and expansion of its modulus gives the stated asymptotic. Thus convergence, moment-matrix positivity, and a bounded fourth pair moment alone do not imply a bounded ordinary deficit. This is not a counterexample to the actual prime-circle assertion.

15.7 An exact Voronoi interpretation and finite bending

Proposition 15.14 (The original weights as Voronoi lengths). *For $n \ge 2$, let*

$$I_n = \left[\frac{p_{n-1} + p_n}{2}, \frac{p_n + p_{n+1}}{2} \right].$$

Then $|I_n| = w_n/2$. Let Γ be the polygonal curve whose derivative equals U_n on the interior of I_n, with $\Gamma(5/2) = 0$. On the union of the first $N - 1$ cells its length is $W_N/2$, and its endpoint is $M_N/2$. Consequently,

$$D_N = 2\{\textit{length of this curve} - \textit{distance between its endpoints}\}. \tag{15.21}$$

Proof. The cells are adjacent and their lengths telescope. Since $|U_n| = 1$, their lengths are also the lengths of the corresponding curve segments. Integrating the tangent gives $\sum_{n=2}^{N} |I_n| U_n = M_N/2$. □

Proposition 15.15 (Unconditional finite bending of an interpolation). *Let $A : [3, \infty) \to \mathbb{R}$ be linear on every prime gap and take the values*

$$A(p_n) = S(p_n) - \log\log p_n \qquad (n \ge 2).$$

Then

$$\int_3^\infty |A'(x)|^2 \, \mathrm{d}x < \infty, \qquad \int_3^\infty \left| \frac{\mathrm{d}}{\mathrm{d}x} e^{2\pi \mathrm{i} A(x)} \right|^2 \mathrm{d}x < \infty. \tag{15.22}$$

These statements are unconditional and concern almost-everywhere derivatives of continuous, locally absolutely continuous functions.

Proof. On a gap with endpoints $p < r$ and length $h = r - p$,

$$\Delta A = \frac{1}{r} - \log\left(\frac{\log r}{\log p}\right), \qquad |\Delta A| \le \frac{1}{r} + \frac{h}{p \log p}.$$

Thus the contribution to the first integral is

$$\frac{|\Delta A|^2}{h} \le \frac{2}{r^2 h} + \frac{2h}{p^2(\log p)^2}.$$

The first series converges by $h \ge 1$. For the second, Bertrand's bound $r < 2p$ gives a uniform comparison of $x^{-2}(\log x)^{-2}$ with its endpoint values. Summing over disjoint gaps therefore compares it with $\int_3^\infty x^{-2}(\log x)^{-2}\,\mathrm{d}x < \infty$. Finally, the squared derivative of the phase interpolation is $4\pi^2|A'(x)|^2$ almost everywhere. □

This continuous interpolation is distinct from the piecewise-constant tangent field in Proposition 15.14. Finite bending does not by itself bound the length excess in (15.21). On an unbounded interval there is no unweighted Poincare inequality of the needed form: $f(x) = x^{-1/4}$ tends to zero and has $f' \in L^2([3,\infty))$, while $f \notin L^2([3,\infty))$. A geometric regularity argument must therefore supply its missing coercive estimate before it can close the original D_N target.

16 Orthogonal polynomials and Toeplitz determinants

The orthogonal-polynomial and determinant methods in items 11–18 of Appendix 118 lead to two different constructions. The first uses the original atomic prime measure and yields a hierarchy of bounded extremal polynomial norms. The second uses finite Laguerre polynomials and gives an exact determinant formulation of the original quadratic problem. Neither construction identifies the prime measure with a random-matrix ensemble.

16.1 A monotone hierarchy on the original prime measure

For an integer $m \ge 1$, define

$$H_{m,N} = \inf_{P(z)=z^m+\sum_{j=0}^{m-1} c_j z^j} \sum_{n=2}^{N} w_n |P(U_n)|^2. \tag{16.1}$$

All coefficients in this finite minimization are computed from the original U_n and w_n; the limiting direction is not an input. These are the monic orthogonal-polynomial norms associated with the positive atomic measure $\sum_{n=2}^{N} w_n \delta_{U_n}$.

Theorem 16.1 (Extremal polynomial hierarchy)**.** *Let $u_n \to u_\infty$ be unit complex numbers, let $w_n > 0$, and suppose $W_N = \sum_{n=2}^{N} w_n \to \infty$. Define $H_{m,N}$ by (16.1) with $U_n = u_n$. Then $H_{m,N}$ is nondecreasing in N, and*

$$\sup_N H_{m,N} < \infty \quad \Longleftrightarrow \quad \sum_{n\ge 2} w_n |u_n - u_\infty|^{2m} < \infty. \tag{16.2}$$

For the prime sequence this is equivalent to $\Theta \le 1 - 1/(2m)$.

Proof. Adding a nonnegative term before taking the infimum proves monotonicity. The polynomial $(z - u_\infty)^m$ proves the implication from right to left. Suppose conversely that all the infima are bounded by C. If the sequence has fewer than m distinct values, convergence forces it to be eventually constant, so the required series is finite. Otherwise fix m distinct values occurring at finite indices. For all sufficiently large N, a minimizing polynomial exists: the quadratic objective is coercive in the lower coefficients because evaluation at those m points has an invertible Vandermonde matrix. Each of its values there is bounded by $\sqrt{C/w_n}$. The same matrix therefore bounds all the lower coefficients uniformly in N. A subsequence converges to a monic degree-m

polynomial P. For each fixed finite prefix, pass to the limit in its energy and then let the prefix grow. This gives $\sum w_n|P(u_n)|^2 \le C$.

If $P(u_\infty) \neq 0$, the summands have a positive lower factor on the tail, contradicting $W_N \to \infty$. Thus $P(z) = (z-u_\infty)^r Q(z)$, where $1 \le r \le m$ and $Q(u_\infty) \neq 0$. On a tail, $|Q(u_n)|$ is bounded below, so the $2r$-moment is finite. Since $|u_n - u_\infty| \le 2$, the $2m$-moment is also finite. The prime assertion follows from the sampled moment criterion in Section 15. □

Put

$$G_{r,N} = \left(\sum_{n=2}^{N} w_n U_n^{j-k} \right)_{0 \le j,k < r}.$$

This is a positive semidefinite Toeplitz Gram matrix. The Schur complement formula for monic least squares gives

$$H_{m,N} = \frac{\det G_{m+1,N}}{\det G_{m,N}} \qquad \text{whenever } G_{m,N} \text{ is invertible.} \tag{16.3}$$

Invertibility means that at least m distinct sampled values have occurred. Before that, a monic polynomial can vanish on the support, but the corresponding ratio may be $0/0$; it must not be used. The orientation of the Gram entries only transposes the more usual moment convention and leaves determinants unchanged. In particular,

$$H_{1,N} = V_N, \qquad D_N = \lambda_{\min}(G_{2,N}), \qquad \det G_{2,N} = W_N V_N.$$

Positive semidefiniteness alone is automatic for any positive measure and does not give an upper bound for its smallest eigenvalue.

Corollary 16.2 (A bounded degree-four extremal norm)**.** *Using the separately identified zero-free theorem in Section 91,*

$$\sup_N H_{4,N} < \infty.$$

Consequently the ratio $\det G_{5,N}/\det G_{4,N}$ *is uniformly bounded over the indices at which its denominator is nonzero. The same assertion for* $H_{1,N}$ *remains equivalent to RH.*

16.2 Dirichlet energy and a double-index Szegő criterion

Fix $\alpha > 1/2$, and let $a_n = a_n^{(\alpha)}$ be the unconditionally defined coefficients in (9.1). For each finite $M \ge 0$, set

$$g_M(z) = \sum_{n=0}^{M} \frac{a_n}{\sqrt{n+1}} z^{n+1}, \qquad L_M(e^{it}) = 2\,\mathrm{Re}\, g_M(e^{it}). \tag{16.4}$$

Only finite polynomials are used here. Assuming in advance that the corresponding infinite series is holomorphic throughout the disk would conflict with the radius obstruction in Theorem 9.2.

Proposition 16.3 (Exact disk and boundary energies)**.** *With dA denoting ordinary planar area,*

$$\mathcal{D}(g_M) := \frac{1}{\pi} \int_{\mathbb{D}} |g_M'(z)|^2 \, \mathrm{d}A(z) = \sum_{n=0}^{M} |a_n|^2, \tag{16.5}$$

$$= \frac{1}{4\pi^2} \int_0^{2\pi} \int_0^{2\pi} \frac{|g_M(e^{it}) - g_M(e^{is})|^2}{|e^{it} - e^{is}|^2} \, \mathrm{d}t \, \mathrm{d}s. \tag{16.6}$$

Thus $\sup_M \mathcal{D}(g_M) < \infty$ *if and only if RH. For every conformal bijection* $\phi : \mathbb{D} \to \Omega$*,*

$$\frac{1}{\pi} \int_\Omega |(g_M \circ \phi^{-1})'(w)|^2 \, \mathrm{d}A(w) = \mathcal{D}(g_M).$$

Proof. Angular orthogonality and radial integration give $\pi^{-1}\int_{\mathbb{D}}|(z^k)'|^2\,\mathrm{d}A = k$. For the boundary formula, simultaneous rotation of t, s eliminates cross terms of different degrees. For a degree k monomial, $(e^{ikt}-e^{iks})/(e^{it}-e^{is})$ is a sum of k orthogonal monomials on the two-torus, so its normalized square integral is k. The coefficient criterion, Theorem 9.1, gives the equivalence. In the change of variables $w=\phi(z)$, the derivative factor and the area Jacobian cancel exactly. □

For a positive symbol w on the unit circle write

$$\widehat{w}(k)=\frac{1}{2\pi}\int_0^{2\pi} w(e^{it})e^{-ikt}\,\mathrm{d}t, \qquad T_L(w)=(\widehat{w}(j-k))_{0\le j,k<L}, \quad L\ge 1.$$

The following is an application of the strong Szegő theorem, in the smooth real-symbol case. A primary proof and the sharp general form are given by Simon [14].

Theorem 16.4 (A two-parameter determinant criterion)**.** *For every finite M,*

$$\lim_{L\to\infty}\log\det T_L(e^{L_M})=\sum_{n=0}^{M}|a_n|^2. \tag{16.7}$$

Moreover the determinants are nondecreasing in L, and hence

$$\boxed{\sup_{M\ge 0,\ L\ge 1}\log\det T_L(e^{L_M})=\sum_{n\ge 0}|a_n|^2.} \tag{16.8}$$

The equality holds in the extended nonnegative real numbers. Finiteness is equivalent to RH; when finite its value is $\int_2^\infty E(x)^2\,\mathrm{d}x$.

Proof. The symbol L_M is a real trigonometric polynomial, its mean is zero, and its positive Fourier coefficients are $\widehat{L}_M(k)=a_{k-1}/\sqrt{k}$, $1\le k\le M+1$. The strong Szegő theorem therefore gives

$$\lim_{L\to\infty}\det T_L(e^{L_M})=\exp\left(\sum_{k=1}^{M+1}k|\widehat{L}_M(k)|^2\right),$$

which is (16.7).

For completeness, monotonicity here follows from a finite extremal argument; it is not an interchange of limits. The ratio of successive Gram determinants is the least norm squared of a monic degree-L polynomial in $L^2(e^{L_M}\,\mathrm{d}t/(2\pi))$. Jensen's inequality and factorization of a monic polynomial $P(z)=\prod(z-r_j)$ give

$$\log\int_0^{2\pi}|P(e^{it})|^2e^{L_M(e^{it})}\frac{\mathrm{d}t}{2\pi}\ge 2\sum_j\log\max(1,|r_j|)\ge 0.$$

Zeros on the circle cause integrable logarithmic singularities and are handled by a limit. Thus each determinant ratio is at least one; also $\det T_1(e^{L_M})\ge 1$. Taking the supremum in L and then in M proves (16.8) without assuming an infinite symbol exists. The coefficient theorem and its Parseval identity finish the proof. □

The theorem transfers methods 16–17 exactly, and connects them with the conformal methods 47 and 68–70. It does not give a common upper bound in M. Each finite symbol has its own finite Szegő constant; the supremum of those constants is the original unknown energy. In particular, compact-open convergence is insufficient: $z^m\to 0$ on compact subsets of $\mathbb{D}$, while $\mathcal{D}(z^m)=m\to\infty$.

16.3 An explicit conversion of an energy bound into the target

Proposition 16.5 (Finite quantitative bridge)**.** *Let $X = p_{N+1}$, and let $\mathcal{V}(X) = \min_{c\in\mathbb{R}} \int_2^X (E(x)-c)^2\,\mathrm{d}x$. For every $N \geq 2$,*

$$D_N \leq 2\pi^2 \left(\sqrt{24\mathcal{V}(p_{N+1})} + \sqrt{\frac{1}{2} - \frac{1}{p_{N+1}}} \right)^2. \tag{16.9}$$

Consequently, a proved bound by C for the supremum in (16.8) would imply the explicit original bound

$$D_N \leq K(C) := 2\pi^2 \left(\sqrt{24C} + \frac{1}{\sqrt{2}} \right)^2 \qquad \text{for all } N. \tag{16.10}$$

Proof. The endpoint inequality in Lemma 6.1 remains true for $E - c$, for every real c. Apply it to gaps $1 \leq n \leq N$, regarding the two endpoints on each gap as coordinates of a weighted Euclidean vector. Replacing the right limit $b_n - c$ by $e_{n+1} - c$ changes that vector's norm by at most

$$\left(\sum_{n=1}^{N} h_n / p_{n+1}^2 \right)^{1/2} \leq \sqrt{1/2 - 1/p_{N+1}}.$$

Minkowski's inequality, followed by omission of the two exterior endpoints, yields

$$\left(\sum_{n=2}^{N} w_n (e_n - c)^2 \right)^{1/2} \leq \sqrt{24 \int_2^{p_{N+1}} (E-c)^2\,\mathrm{d}x} + \sqrt{1/2 - 1/p_{N+1}}.$$

Finally $D_N = \frac{1}{2} \min_{|u|=1} \sum w_n |U_n - u|^2$. Choose $u = e^{2\pi \mathrm{i}(B_{\mathbb{P}}+c)}$ and use $|e^{it} - e^{is}| \leq |t-s|$, then minimize over c. If the determinant supremum is at most C, Theorem 16.4 gives $\int_2^\infty E^2 \leq C$, which proves (16.10). □

17 What the additive circle method transfers

The circle method extracts additive representation counts by Fourier orthogonality. Its success in ternary Goldbach, Waring problems, and systems of forms requires quantitative estimates for the specific exponential sums, together with a positive main term or appropriate local hypotheses. Helfgott's account [15] describes the major-arc, minor-arc and large-sieve inputs in the prime setting.

There is an exact Fourier formulation of our existing quadratic criterion. For $j \geq 1$, set

$$A_j(t) = \sum_{2^j \leq n < 2^{j+1}} E(n) e^{2\pi \mathrm{i} n t}.$$

Finite orthogonality, followed by Corollary 6.3, gives

$$\sum_{j\geq 1} \int_0^1 |A_j(t)|^2\,\mathrm{d}t = \sum_{n=2}^{\infty} E(n)^2 < \infty \quad \Longleftrightarrow \quad \text{RH}. \tag{17.1}$$

For any measurable major-arc region $\mathfrak{M}_j$, with complement $\mathfrak{m}_j$, the norm splits exactly into the sum of its two nonnegative integrals. Thus a summable upper bound for both regions would close the problem. A uniform bound on each dyadic norm would already suffice: the integer-sampling error is square summable, so it would give $I(X) = O(\log X)$, and the exact growth law would force RH.

The published additive existence theorems do not supply those estimates for the coefficients $E(n)$. Products of several exponential sums, or repeated convolution, may suppress exceptional

concentrations; recovering the original quadratic energy requires a quantitative inverse estimate. The higher-contact quantities in Section 15 and the countermodels in Section 18 make that loss explicit.

There is also a useful exact multiscale identity. For two disjoint finite weighted blocks A, B, with means m_A, m_B,

$$V_{A\cup B} = V_A + V_B + \frac{W_A W_B}{W_A + W_B}|m_A - m_B|^2. \tag{17.2}$$

It follows by expanding $W - |M|^2/W$. Iterating over a binary partition expresses the total variance as nonnegative local variance and merger costs. This is a precise transfer of the local-to-global accounting used in items 44, 53–56 and 71. A successful proof must bound the sum of these costs uniformly; boundedness of each individual cost is not enough.

Finally, equidistribution addresses a different limiting measure. Weighted Cesàro convergence gives

$$\frac{1}{W_N}\sum_{n=2}^{N} w_n \delta_{U_n} \implies \delta_{u_\infty}, \qquad \frac{M_N^{(k)}}{W_N} \longrightarrow u_\infty^k.$$

The centered points therefore concentrate at one direction rather than becoming Haar-uniform on the circle. This exact statement imports Weyl's Fourier viewpoint while retaining the correct limiting measure. It supplies $D_N = o(W_N)$, but not $D_N = O(1)$.

18 Arithmetic tests of the prime-circle construction

The elliptic and Diophantine methods in items 21–40 require arithmetic data in addition to a geometric picture: an exact polynomial identity, a controlled family of curves, a suitable Galois representation, or certified Mordell–Weil information. This section applies several of those ideas to the actual prime-reciprocal data. It proves specific obstructions to natural encodings, together with countermodels showing which information cannot alone imply the original bound on D_N.

18.1 Raw torsion and centered phases

Write

$$Q_n = \prod_{j=1}^{n} p_j, \qquad A_n = \sum_{j=1}^{n} Q_n/p_j, \qquad S_n = S(p_n) = A_n/Q_n.$$

The raw point and the point used by the original deficit are

$$C_n = e^{2\pi \mathrm{i} S_n}, \qquad U_n = C_n e^{-2\pi \mathrm{i} \log\log p_n}.$$

The factor distinguishing them is indispensable to the arithmetic audit.

Proposition 18.1 (Exact order and a local obstruction)**.** *The fraction A_n/Q_n is reduced. The point C_n has exact order Q_n, minimal polynomial Φ_{Q_n}, algebraic degree $\varphi(Q_n)$, and absolute logarithmic Weil height zero. Moreover, $v_q(S_n) = -1$ for every $q \mid Q_n$. For each fixed prime ℓ, the sequence (S_n) is not Cauchy in $\mathbb{Q}_\ell$.*

Proof. For $q \mid Q_n$, all summands of A_n except Q_n/q vanish modulo q, and Q_n/q is nonzero modulo q. Hence $\gcd(A_n, Q_n) = 1$, giving the valuation and exact-order assertions. Every conjugate of a root of unity has absolute value one, and it is an algebraic integer and a unit, so its Weil height is zero. For fixed ℓ and all sufficiently large n,

$$|S_{n+1} - S_n|_\ell = |1/p_{n+1}|_\ell = 1.$$

Successive differences therefore do not tend to zero. □

Thus the raw sums have no fixed finite denominator support and no direct ℓ-adic series limit. Formal-group methods require an additional construction; the reciprocal increments do not already converge in a fixed local logarithmic coordinate. None of the preceding arguments establishes algebraicity, torsion, or infinite order for U_n, or for $e^{2\pi \mathrm{i} E(p_n)}$.

Proposition 18.2 (Two barriers to a direct elliptic transport)**.** *Every rational map from a smooth conic to an elliptic curve in characteristic zero is constant. On the other hand, the analytic group-preserving map*

$$S_n \bmod \mathbb{Z} \;\longmapsto\; S_n \bmod (\mathbb{Z} + \tau\mathbb{Z}), \qquad \Im\tau > 0,$$

produces elliptic torsion points of exact order Q_n. If these points are viewed on an elliptic curve defined over a number field, their fields of definition cannot all have bounded degree over $\mathbb{Q}$.

Proof. After extending scalars a smooth conic becomes $\mathbb{P}^1$. A nonconstant map to a genus-one curve would give, by Riemann–Hurwitz, $-2 = \sum_P (e_P - 1) \geq 0$. For the analytic map, rA_n/Q_n belongs to $\mathbb{Z} + \tau\mathbb{Z}$ exactly when it belongs to $\mathbb{Z}$, so its least positive annihilator is Q_n. Merel's uniform torsion theorem bounds the order of a torsion point in terms of the degree of its number field of definition [16]; this is incompatible with $Q_n \to \infty$. □

These statements distinguish a rational coordinate map, an analytic uniformization, and an arithmetic lift. The last requires a rationality or field-degree certificate that the first two do not supply. They do not prohibit other covering constructions with additional hypotheses.

18.2 Identical torsion data with different quadratic energies

Retain exactly the original primes and weights $w_n = p_{n+1} - p_{n-1}$, $n \geq 2$. For a sequence of unit points v_n tending to 1, put

$$\mathcal{A}_q(v) = \sum_{n\geq 2} w_n |v_n - 1|^q, \qquad D_N(v) = W_N - \left| \sum_{n=2}^{N} w_n v_n \right|.$$

These definitions introduce comparison sequences; $D_N(U)$ remains the original target.

Lemma 18.3 (Elementary weighted estimates)**.** *For every fixed $r > 1$, $\sum_{n\geq 2} w_n p_n^{-r} < \infty$. For $0 < r < 1$ the partial sums are $O_r(p_N^{1-r})$, and $\sum_{n=2}^{N} w_n/p_n \asymp \log p_N$. In addition,*

$$\sum_{n\geq 2} \frac{w_n}{p_n^2} \leq \frac{7}{6}. \tag{18.1}$$

Proof. Write $h_n = p_{n+1} - p_n$. Bertrand's postulate and monotonicity of x^{-r} compare the terms $h_{n-1}p_n^{-r}$ and $h_n p_n^{-r}$ with their adjacent integrals, with factors depending only on r. This proves the assertions away from $r = 1$. At $r = 1$,

$$\log \frac{p_{N+1}}{3} \leq \sum_{n=2}^{N} \frac{w_n}{p_n} \leq \log \frac{p_N}{2} + 2 \log \frac{p_{N+1}}{3}.$$

For the sharper square bound, sum

$$\frac{h_{n-1}}{p_n^2} \leq \int_{p_{n-1}}^{p_n} x^{-2}\,\mathrm{d}x, \qquad \frac{h_n}{p_n^2} = \frac{p_{n+1}}{p_n}\left(\frac{1}{p_n} - \frac{1}{p_{n+1}}\right) < 2\left(\frac{1}{p_n} - \frac{1}{p_{n+1}}\right).$$

The resulting bounds are $1/p_1 = 1/2$ and $2/p_2 = 2/3$. □

Theorem 18.4 (An obstruction to inferring the quadratic bound)**.** *There exist two sequences* $v_n^{(0)}, v_n^{(1)} \to 1$ *such that every* $v_n^{(j)}$ *is a primitive* Q_n*-th root of unity and has Weil height zero, but*

$$\sup_N D_N(v^{(0)}) \le \frac{7\pi^2}{3}, \qquad D_N(v^{(1)}) \asymp \log p_N \longrightarrow \infty.$$

For the second sequence, nevertheless,

$$\mathcal{A}_q(v^{(1)}) < \infty \quad \textit{for every real } q > 2, \qquad \mathcal{A}_2(v^{(1)}) = \infty.$$

Proof. Set $v_n^{(0)} = e^{2\pi \mathrm{i}/Q_n}$. Since $Q_n \ge p_n$,

$$D_N(v^{(0)}) \le \frac{1}{2}\sum_{n=2}^{N} w_n |v_n^{(0)} - 1|^2 \le 2\pi^2 \sum_{n\ge 2} \frac{w_n}{Q_n^2} \le \frac{7\pi^2}{3}.$$

For the second sequence choose $a_n \in \{\lfloor\sqrt{p_n}\rfloor, \lfloor\sqrt{p_n}\rfloor + 1\}$ so that $B_n = a_n(Q_n/p_n) + 1$ is not divisible by p_n. There is only one forbidden residue class modulo p_n, so this is possible. For every other $q \mid Q_n$, $B_n \equiv 1 \pmod q$. Thus $\gcd(B_n, Q_n) = 1$, and

$$v_n^{(1)} = e^{2\pi \mathrm{i} B_n/Q_n}, \qquad \theta_n := B_n/Q_n = p_n^{-1/2} + O(p_n^{-1})$$

has the asserted exact order and limit. In particular $|v_n^{(1)} - 1| \asymp p_n^{-1/2}$. Lemma 18.3 proves the moment statements.

It also gives the precise deficit growth. Taylor expansion and the convergent weighted $p_n^{-3/2}$ sum imply

$$\sum_{n=2}^{N} w_n |v_n^{(1)} - 1|^2 = 4\pi^2 \sum_{n=2}^{N} \frac{w_n}{p_n} + O(1) \asymp \log p_N.$$

For $M_N = \sum_{n=2}^{N} w_n v_n^{(1)}$,

$$|M_N - W_N| \ll \sum_{n=2}^{N} w_n p_n^{-1/2} \ll \sqrt{p_N}.$$

Since $W_N \ge p_N$, the moving-center correction $|M_N - W_N|^2/W_N$ is $O(1)$. The variance identity

$$V_N = \sum_{n=2}^{N} w_n |v_n^{(1)} - 1|^2 - \frac{|M_N - W_N|^2}{W_N}, \qquad D_N \le V_N \le 2D_N,$$

therefore gives $D_N(v^{(1)}) \asymp \log p_N$. □

Both comparison points at index n have the same minimal polynomial Φ_{Q_n} as the raw prime point C_n. Even these complete individual algebraic conjugacy data do not force the quadratic bound. Furthermore, for every integer $m \ge 2$,

$$\frac{1}{W_N} \sum_{2\le j<k\le N} w_j w_k |v_j^{(1)} - v_k^{(1)}|^{2m} \le 2^{2m-1} \mathcal{A}_{2m}(v^{(1)}) < \infty$$

uniformly in N. Thus boundedness of all the higher pairwise moments, including the eighth moment, cannot by itself recover boundedness of the quadratic deficit.

Remark 18.5 (Scope of the countermodels). The comparison sequences retain the actual prime-gap weights, primorial orders, zero heights, and convergence. They do not retain the exact reciprocal recurrence and logarithmic centering defining U_n. Theorem 18.4 is consequently no counterexample to the original prime-circle criterion. Any successful inference for that criterion must use additional information specific to those arithmetic relations.

18.3 A natural Frey curve and the surviving conductor

The modular method can be tested on an explicit curve constructed from the reduced fraction $S_n = A_n/Q_n$:

$$\mathcal{E}_n : \quad Y^2 = X(X - A_n)(X + Q_n).$$

This is a nonsingular elliptic curve over $\mathbb{Q}$, and it is modular by the modularity theorem [17].

Proposition 18.6 (Conductor obstruction for this encoding)**.** *The displayed integral model has invariants*

$$\Delta_n = 16A_n^2Q_n^2(A_n + Q_n)^2, \qquad c_{4,n} = 16(A_n^2 + A_nQ_n + Q_n^2). \tag{18.2}$$

At every odd $q \mid Q_n$ it is minimal, has multiplicative reduction, and satisfies $v_q(\Delta_{\min,n}) = 2$. Fix an odd prime ℓ. For $q \mid Q_n$ with $q \neq 2, \ell$, the actual representation on $\mathcal{E}_n[\ell]$ is ramified at q. If that global residual representation is absolutely irreducible, its prime-to-ℓ conductor is divisible by

$$\prod_{\substack{q \mid Q_n \\ q \neq 2,\ell}} q \;\geq\; \frac{Q_n}{2\ell}.$$

Proof. The formulas in (18.2) follow from the Weierstrass invariants. Since Q_n is squarefree and $\gcd(A_n, Q_n) = 1$, an odd $q \mid Q_n$ divides neither A_n nor $A_n + Q_n$, and $c_{4,n} \equiv 16A_n^2 \not\equiv 0 \pmod{q}$. The model is minimal at q, its reduction is multiplicative, and its minimal discriminant has valuation 2. Over an unramified extension, Tate uniformization identifies the tame extension parameter on ℓ-torsion with this valuation modulo ℓ. Since ℓ is odd, it is nonzero, so inertia acts nontrivially. In the absolutely irreducible global case these ramified primes persist in the residual conductor. □

The successful Frey arguments use divisible discriminant valuations to remove the new primes from a residual conductor; see, for example, Darmon–Merel, Lemma 2.1 and its level-lowering application [18]. The valuations in Proposition 18.6 are instead fixed at 2. This natural encoding therefore supplies no collapse to a fixed impossible level. It also does not yet encode the logarithmically centered error.

Remark 18.7 (Residual semisimplification). The unconditional local statement concerns the actual representation on $\mathcal{E}_n[\ell]$. If the global residual representation is reducible, taking its semisimplification can remove extension data. The conductor conclusion used in the usual irreducible level-lowering argument is accordingly stated with an explicit absolute-irreducibility hypothesis. The calculation concerns this particular curve construction, not every possible arithmetic encoding of the original problem.

19 Half-line filters and finite-time heat smoothing

The compact-support hypothesis of Theorem 12.1 can be relaxed in two useful directions. The first permits a finite measure with an arbitrarily long causal tail; the second permits the two-sided Gaussian kernel. Both results concern the particular arithmetic error E, not arbitrary functions with exponential growth. Throughout this section, extend $E(x)$ by zero for $0 < x < 2$, and write

$$f(t) = e^{t/2}E(e^t), \qquad I(X) = \int_0^X |E(x)|^2 \, \mathrm{d}x, \qquad \gamma = 2\Theta - 1.$$

Theorem 19.1 (Finite measures supported on a half-line)**.** *Let $\mu \neq 0$ be a finite complex Borel measure on $\mathbb{R}$, with $\operatorname{supp}\mu \subset [a,\infty)$ for some finite real a. Define*

$$E_\mu(x) = \int_{\mathbb{R}} e^{-u/2} E(xe^{-u})\,\mathrm{d}\mu(u), \qquad I_\mu(X) = \int_0^X |E_\mu(x)|^2\,\mathrm{d}x, \tag{19.1}$$

$$M_\mu(z) = \int_{\mathbb{R}} e^{-zu}\,\mathrm{d}\mu(u), \qquad \Re z > 0. \tag{19.2}$$

Then

$$\int_0^\infty |E_\mu(x)|^2\,\mathrm{d}x < \infty \quad\Longleftrightarrow\quad \text{RH}, \qquad \limsup_{X\to\infty} \frac{\log(1+I_\mu(X))}{\log X} = \gamma. \tag{19.3}$$

Neither positivity nor normalization of μ is required.

Proof. Put $A = |\mu|(\mathbb{R})$. Since E is bounded, E_μ is bounded by $Ae^{-a/2}\sup|E|$, and vanishes for $x < 2e^a$. For every $\sigma > 0$ and integer $n \geq 0$, the function $|u|^n e^{-\sigma u}$ is bounded on $[a,\infty)$. Differentiation under the integral therefore shows that M_μ is holomorphic in $\Re z > 0$.

It cannot vanish identically there. To see this without a compact-support assumption, push μ forward by $u \mapsto v = e^{a-u}$ to a finite complex measure ν on $(0,1]$, and extend ν by zero at $v = 0$. If $M_\mu \equiv 0$, then

$$\int_0^1 v^n\,\mathrm{d}\nu(v) = e^{an} M_\mu(n) = 0 \qquad (n \geq 1).$$

Dominated convergence as real $z \downarrow 0$ also gives $\mu(\mathbb{R}) = 0$. Thus the zeroth moment vanishes as well, and polynomial density in $C([0,1])$ forces $\nu = 0$, hence $\mu = 0$, a contradiction.

For $\Re s > 1$, absolute Fubini is justified by

$$\int_{\mathbb{R}} e^{-(\Re s - 1/2)u}\,\mathrm{d}|\mu|(u) \int_2^\infty |E(y)| y^{-\Re s}\,\mathrm{d}y < \infty.$$

Consequently the Mellin transform on the full positive axis satisfies

$$F_\mu(s) := \int_0^\infty E_\mu(x) x^{-s}\,\mathrm{d}x = M_\mu(s - \tfrac{1}{2}) F(s), \qquad \Re s > 1. \tag{19.4}$$

Writing $g = \mu * f$, one has $g(t) = e^{t/2} E_\mu(e^t)$, and therefore $\|g\|_2^2 = \int_0^\infty |E_\mu|^2$. If RH holds, Theorem 4.1 and Young's inequality give $\|g\|_2 \leq A\|f\|_2 < \infty$.

Conversely, if $E_\mu \in L^2(0,\infty)$, its positive lower support bound and Cauchy–Schwarz make F_μ holomorphic for $\Re s > 1/2$. At any zeta zero ρ of multiplicity m_ρ with $\Re\rho > 1/2$, the local identity (12.7) reads

$$F(s) = \frac{m_\rho}{s-1}\log(s-\rho) + H_\rho(s),$$

with H_ρ holomorphic. Continuation once around ρ changes the right side of (19.4) by

$$2\pi i m_\rho \frac{M_\mu(s-\frac{1}{2})}{s-1}.$$

This is not identically zero on a neighborhood of ρ. A finite-order zero of M_μ at $\rho - 1/2$ still leaves a nonzero multiple of $(s-\rho)^r \log(s-\rho)$. The product cannot be single-valued and holomorphic throughout $\Re s > 1/2$. Thus there is no such ρ, and the functional equation gives RH.

For growth, Cauchy–Schwarz against $|\mu|$ and Tonelli give

$$\begin{aligned} I_\mu(X) &\leq A \int_{\mathbb{R}} \int_{-\infty}^{\log X - u} |f(v)|^2\,\mathrm{d}v\,\mathrm{d}|\mu|(u) \\ &\leq A^2 I(e^{-a}X). \end{aligned}$$

Theorem 8.2 gives the upper limsup bound γ. If $\gamma = 0$, this proves equality. Otherwise, if the filtered limsup were smaller, choose numbers $\beta < \beta' < \gamma$ with β larger than that limsup. Then $I_\mu(X) = O(X^\beta)$, and integration by parts implies

$$\int_0^\infty x^{-\beta'} |E_\mu(x)|^2 \, \mathrm{d}x < \infty.$$

Weighted Cauchy–Schwarz makes F_μ holomorphic for $\Re s > \eta = (1 + \beta')/2 < \Theta$. Here $\eta > 1/2$, and a zero with $\Re\rho > \eta$ exists by the definition of Θ. Its surviving logarithmic monodromy gives the same contradiction. This proves the exact exponent. □

For example, the infinite-tail averaging measure $\mathrm{d}\mu(u) = ce^{-cu}\mathbf{1}_{[0,\infty)}(u)\,\mathrm{d}u$, $c > 0$, has $M_\mu(z) = c/(c + z)$ and satisfies this theorem. The lower support bound, rather than an upper bound on memory length, is what the proof uses.

Theorem 19.2 (Finite-time Gaussian heat smoothing)**.** *For $\tau > 0$, let*

$$h_\tau(u) = \frac{e^{-u^2/(4\tau)}}{\sqrt{4\pi\tau}}, \qquad g_\tau = h_\tau * f, \qquad E_\tau(x) = x^{-1/2} g_\tau(\log x), \qquad I_\tau(X) = \int_0^X |E_\tau(x)|^2 \, \mathrm{d}x. \tag{19.5}$$

For every fixed $\tau > 0$,

$$g_\tau \in L^2(\mathbb{R}) \quad \Longleftrightarrow \quad \mathrm{RH}, \qquad \limsup_{X\to\infty} \frac{\log(1 + I_\tau(X))}{\log X} = \gamma. \tag{19.6}$$

Proof. Boundedness of E gives $|f(v)| \le Ce^{v/2}\mathbf{1}_{[\log 2,\infty)}(v)$, so convolution with the Gaussian is absolutely defined at every t. For $t \le 0$ and $v \ge \log 2 > 0$, one has $(t - v)^2 \ge t^2 + v^2$. Consequently

$$|g_\tau(t)| \le \frac{Ce^{-t^2/(4\tau)}}{\sqrt{4\pi\tau}} \int_{\log 2}^\infty e^{-v^2/(4\tau)+v/2} \, \mathrm{d}v \le C_\tau e^{-t^2/(4\tau)}, \qquad t \le 0. \tag{19.7}$$

In particular, the negative-half-line part of its bilateral Laplace transform is entire.

For $\Re z > 1/2$, absolute Fubini and the Gaussian moment integral give

$$\mathcal{L}g_\tau(z) := \int_{-\infty}^\infty g_\tau(t) e^{-zt} \, \mathrm{d}t = e^{\tau z^2} F(z + \tfrac{1}{2}). \tag{19.8}$$

If RH holds, $f \in L^2$ by Theorem 4.1, and Young's inequality gives $\|g_\tau\|_2 \le \|f\|_2$. Conversely, if $g_\tau \in L^2$, Cauchy–Schwarz makes the positive-half-line part of $\mathcal{L}g_\tau$ holomorphic for $\Re z > 0$; (19.7) handles the negative half-line. The multiplier $e^{\tau z^2}$ is entire and nowhere zero. Hence the logarithmic branch of F at an off-line zero cannot be reconciled with this holomorphic continuation, by the same monodromy computation as in Theorem 19.1. This proves the equivalence.

Changing variables gives $I_\tau(e^T) = \int_{-\infty}^T |g_\tau(t)|^2 \, \mathrm{d}t$. Since h_τ is a probability density, Jensen and Tonelli yield

$$I_\tau(e^T) \le \int_{\mathbb{R}} h_\tau(u) I(e^{T-u}) \, \mathrm{d}u.$$

For each $\varepsilon > 0$, Theorem 8.2 gives $I(Y) \le C_\varepsilon Y^{\gamma+\varepsilon}$ for all $Y > 0$, after enlarging the constant and using $I(Y) = 0$ for $Y < 2$. Therefore

$$I_\tau(X) \le C_\varepsilon e^{\tau(\gamma+\varepsilon)^2} X^{\gamma+\varepsilon}. \tag{19.9}$$

This proves the upper exponent bound, and also equality if $\gamma = 0$.

If the limsup were smaller when $\gamma > 0$, choose $\beta < \beta' < \gamma$, with β larger than that limsup. The bound $I_\tau(X) = O(X^\beta)$ implies weighted square integrability at infinity with weight $x^{-\beta'}$. Near zero, (19.7) gives

$$|E_\tau(x)|^2 \le C_\tau^2 x^{-1} \exp\left(-\frac{(\log x)^2}{2\tau}\right), \qquad 0 < x \le 1,$$

which is integrable against every fixed power of x. Thus

$$\int_{\mathbb{R}} e^{-\beta' t}|g_\tau(t)|^2\,\mathrm{d}t = \int_0^\infty x^{-\beta'}|E_\tau(x)|^2\,\mathrm{d}x < \infty.$$

Cauchy–Schwarz on the positive half-line now extends $\mathcal{L}g_\tau$ holomorphically to $\Re z > \beta'/2$, while its negative-half-line contribution remains entire. A zero ρ with $\Re\rho > (1+\beta')/2$ exists, and its logarithmic monodromy contradicts (19.8). The lower exponent bound follows. □

Remark 19.3 (Which flow is being tested). Theorem 19.2 is the linear heat flow of the logarithmic prime-error function f. It is neither a Ricci flow with surgery, whose nonlinear geometric mechanisms are illustrated by [19], nor the de Bruijn–Newman deformation of Ξ. The result excludes finite-time Gaussian smoothing as a way to remove the present arithmetic obstruction. It does not exclude a new nonlinear arithmetic flow or a new estimate for the unsmoothed data.

20 Finite covers, balancing, and geometric information

The Dirichlet-energy formulation in Section 16.2 retains the arithmetic norm exactly. The following tests clarify what is retained by several other proposed geometric operations.

Proposition 20.1 (Finite ramification does not remove the logarithm)**.** *Let ρ be a zeta zero with $\Re\rho > 1/2$, of multiplicity m_ρ. Let ϕ be nonconstant and holomorphic near w_0, with $\phi(w_0) = \rho$ and local ramification order $e \geq 1$. Then the local continuation of $F \circ \phi$ is not single-valued and holomorphic across w_0. Multiplication by a holomorphic function that is not identically zero does not change this conclusion.*

Proof. Write $\phi(w) - \rho = (w - w_0)^e h(w)$, where h is holomorphic and $h(w_0) \neq 0$. After shrinking the neighborhood, choose a holomorphic logarithm of h. Equation (12.7) gives

$$F(\phi(w)) = \frac{m_\rho}{\phi(w) - 1}\big(e\log(w - w_0) + \log h(w)\big) + H_\rho(\phi(w)).$$

A loop once around w_0 changes this value by

$$\frac{2\pi\mathrm{i}em_\rho}{\phi(w) - 1},$$

which is not identically zero. If a nonzero holomorphic multiplier $B(w)$ is inserted, the change is multiplied by $B(w)$ and is still not identically zero. Its finite-order zeros cannot annihilate the logarithmic branch. □

Finite covers play a central role in Riemann existence and inverse Galois theory over $\mathbb{C}(t)$ [20]. The proposition states a narrower analytic fact: the infinite additive monodromy of this particular logarithm survives every finite local ramification. An infinite universal cover may make the logarithm single-valued, but does not supply a norm bound on the original half-plane. No claim about all nonlocal transforms of F is intended.

Proposition 20.2 (An exact Hobby–Rice split of a finite prefix)**.** *Fix N, positive weights $w_2, \ldots, w_N$, and unit complex numbers $U_2, \ldots, U_N$. Realize each w_n as an interval of length w_n, and put $U(t) = U_n$ on that interval. Write*

$$W = \sum_{n=2}^{N} w_n, \qquad M = \sum_{n=2}^{N} w_n U_n, \qquad D = W - |M|.$$

At most three cuts of $[0,W]$ *suffice to distribute the resulting pieces between two measurable sets* A, B*, with*

$$|A| = |B| = \frac{W}{2}, \qquad \int_A U(t)\,\mathrm{d}t = \int_B U(t)\,\mathrm{d}t = \frac{M}{2}.$$

Consequently, defining $D_A = |A| - |\int_A U|$ *and similarly for* B*, one has*

$$D_A = D_B = \frac{D}{2}, \qquad D_A + D_B = D. \tag{20.1}$$

Proof. The Hobby–Rice theorem, in the form proved by Pinkus [21], applies to the three real integrable functions $1, \Re U, \Im U$ on the nonatomic interval $[0,W]$, after a linear rescaling to $[0,1]$. It gives a function $\sigma(t) \in \{-1,1\}$, with at most three sign changes, such that

$$\int_0^W \sigma(t)\,\mathrm{d}t = 0, \qquad \int_0^W \sigma(t)U(t)\,\mathrm{d}t = 0.$$

Take $A = \{\sigma = 1\}$, $B = \{\sigma = -1\}$, disregarding endpoints of measure zero. The displayed identities give the equal masses and equal complex moments. Formula (20.1) then follows directly from the definitions. □

This is a genuine finite-prefix application, but it may split up to three original weights fractionally. It does not assert an exact partition of indivisible primes or integer masses. Its balanced parts have the same normalized barycenter, so the total deficit is conserved. Other partitions would require their own quantitative analysis. Likewise, multiplying the scalar error by a measurable function of modulus one leaves $\int |E|^2$ unchanged; cancellation of finitely many signed first moments is not an upper bound for that norm.

Proposition 20.3 (Curve geometry without sampling weights is insufficient)**.** *For a positive measure* ν *on the unit circle, define*

$$D(\nu) = \nu(\mathbb{S}^1) - \left|\int_{\mathbb{S}^1} z\,\mathrm{d}\nu(z)\right|.$$

No finite bound depending only on the unparametrized supporting circle can bound $D(\nu)$ *for all such measures.*

Proof. For $L > 0$, take $\mathrm{d}\nu_L(e^{\mathrm{i}\theta}) = L\,\mathrm{d}\theta/(2\pi)$. Its support is the same unit circle for every L, whereas

$$\nu_L(\mathbb{S}^1) = L, \qquad \int_{\mathbb{S}^1} z\,\mathrm{d}\nu_L(z) = 0, \qquad D(\nu_L) = L.$$

Letting $L \to \infty$ proves the assertion. □

This is a test of a proposed transfer, not a counterexample involving the actual primes. It shows why a curve invariant that forgets the sampling weights cannot by itself produce the desired constant. A successful geometric argument must retain those arithmetic weights and establish a uniform upper estimate for their error energy.

21 Circle, Fourier, Toeplitz, and additive-number-theory transfers

The word "circle" has different meanings in the proposed methods. In additive number theory it is the Fourier parameter in a coefficient extraction integral; in the attached paper it is the range of a prime-reciprocal phase. Toeplitz methods use yet another circle, the support of a spectral measure. A useful transfer therefore requires an explicit operator or identity, rather than the shared picture

alone. This section supplies such identities. None proves a new unconditional zero-free region. The results below are additional deductions relative to the attached note; no claim of historical novelty is made.

Write

$$S(x)=\sum_{p\le x}\frac{1}{p},\quad E(x)=S(x)-\log\log x-B_{\mathbb{P}},\quad U(x)=\exp\bigl(2\pi i[S(x)-\log\log x]\bigr),\quad u_\infty=e^{2\pi iB_{\mathbb{P}}}.$$

Mertens' theorem gives $E(x)\to 0$ and $U(x)\to u_\infty$. We use the attached paper's established quadratic criterion

$$\mathrm{RH}\iff E\in L^2(2,\infty)\iff U-u_\infty\in L^2(2,\infty). \tag{21.1}$$

The arithmetic input needed for the forward implication is the conditional mean-square prime-number-theorem estimate, not merely a pointwise RH error bound.

21.1 A constant-free multiplicative-window criterion

For a fixed $\lambda>1$ define

$$A_\lambda(x)=\sum_{x<p\le\lambda x}\frac{1}{p}-\log\left(\frac{\log(\lambda x)}{\log x}\right)=E(\lambda x)-E(x). \tag{21.2}$$

This is a reciprocal-prime discrepancy on a multiplicative interval. It requires neither the Meissel–Mertens constant nor a fitted limiting phase. Its circle version satisfies the exact identity

$$|U(\lambda x)-U(x)|^2=4\sin^2(\pi A_\lambda(x)). \tag{21.3}$$

Lemma 21.1 (Dilation inversion with quantitative constants)**.** *Let $\beta>-1$, let $f:[2,\infty)\to\mathbb{C}$ be measurable, locally square integrable, and satisfy $f(x)\to 0$. Set $g(x)=f(x)-f(\lambda x)$ and*

$$\|f\|_\beta^2=\int_2^\infty x^\beta|f(x)|^2\,dx,\qquad c=\lambda^{-(\beta+1)/2}<1.$$

Then f has finite β-norm if and only if g does. When finite,

$$(1-c)\|f\|_\beta\le\|g\|_\beta\le(1+c)\|f\|_\beta. \tag{21.4}$$

Proof. For $T_\lambda f(x)=f(\lambda x)$, substitution gives

$$\|T_\lambda f\|_\beta^2=\lambda^{-\beta-1}\int_{2\lambda}^\infty t^\beta|f(t)|^2\,dt\le c^2\|f\|_\beta^2.$$

Thus if f has finite norm, both inequalities follow from the triangle and reverse triangle inequalities. Conversely, if g has finite norm, then $\sum_{j\ge0}T_\lambda^j g$ converges in this weighted L^2 space and its norm is at most $\|g\|_\beta/(1-c)$. Its partial sums are

$$\sum_{j=0}^{J-1}g(\lambda^j x)=f(x)-f(\lambda^J x)\longrightarrow f(x).$$

An almost-everywhere convergent subsequence of the L^2 limit identifies that limit with f. Alternatively Fatou's lemma applied to these partial sums proves directly that $\|f\|_\beta\le\|g\|_\beta/(1-c)$. □

Theorem 21.2 (Multiplicative windows, scalar and circle forms)**.** *For every fixed $\lambda>1$, each of the following is equivalent to RH:*

1. $\int_2^\infty A_\lambda(x)^2\,dx < \infty$;
2. $\int_2^\infty |U(\lambda x) - U(x)|^2\,dx < \infty$;
3. $\sum_{m=2}^{\infty} A_\lambda(m)^2 < \infty$;
4. $\sum_{m=2}^{\infty} 4\sin^2(\pi A_\lambda(m)) < \infty$.

In the scalar continuous criterion the two norms satisfy

$$(1-\lambda^{-1/2})\|E\|_2 \le \|A_\lambda\|_2 \le (1+\lambda^{-1/2})\|E\|_2. \tag{21.5}$$

The same constants compare $\|U-u_\infty\|_2$ *with* $\|U(\lambda\cdot)-U\|_2$.

Proof. Apply Lemma 21.1, with $\beta = 0$, first to E and then to $U - u_\infty$. Equation (21.1) proves the continuous claims and the norm bounds. Since $A_\lambda(x) \to 0$, $4\sin^2(\pi A_\lambda(x)) \sim 4\pi^2 A_\lambda(x)^2$ on a tail.

It remains to justify integer sampling; no prime-gap assumption is needed. For $x \in [m, m+1)$, $S(x) = S(m)$ and

$$|E(x) - E(m)| \le \frac{1}{m\log m}.$$

The interval $[\lambda m, \lambda(m+1)]$ contains at most $\lceil\lambda\rceil + 1$ integers, hence at most that many primes. Every reciprocal-prime jump in it is at most $1/(\lambda m)$. The smooth logarithmic change is at most $1/[m\log(\lambda m)]$. Consequently

$$\sup_{m\le x<m+1} |A_\lambda(x) - A_\lambda(m)| \le \frac{\lceil\lambda\rceil+1}{\lambda m} + \frac{1}{m\log(\lambda m)} + \frac{1}{m\log m} \ll_\lambda \frac{1}{m}. \tag{21.6}$$

The difference from the integer-step sampling is therefore in L^2. This proves (i)⇔(iii). Finally the tail comparison of sine and its argument proves (iii)⇔(iv). □

Why this transfer preserves the arithmetic obstruction. For $F(s) = \int_2^\infty E(x)x^{-s}\,dx$ and $H_\lambda(s) = \int_2^\infty A_\lambda(x)x^{-s}\,dx$, initially $\Re s > 1$,

$$H_\lambda(s) = (\lambda^{s-1} - 1)F(s) - \lambda^{s-1}\int_2^{2\lambda} E(t)t^{-s}\,dt. \tag{21.7}$$

The finite integral is entire in s. The multiplier $\lambda^{s-1}-1$ has zeros only on $\Re s = 1$, at $1 + 2\pi i k/\log\lambda$. It therefore does not cancel a zero-induced singularity strictly inside the critical strip. The new criterion is an exact arithmetic reformulation, not a way of removing unwanted zeros by a filter.

Weighted transfer. For every $\eta < 1$, Lemma 21.1 with $\beta = 1 - 2\eta$ gives the exact finiteness equivalence

$$\int_2^\infty x^{1-2\eta}A_\lambda(x)^2\,dx < \infty \quad\Longleftrightarrow\quad \int_2^\infty x^{1-2\eta}E(x)^2\,dx < \infty, \tag{21.8}$$

with norm-comparison constants $1 \pm \lambda^{\eta-1}$. The same statement holds for the circle difference and centered circle. Thus any proved weighted threshold for the original error transfers without alteration. In particular, accepting the attached note's endpoint theorem, the exact threshold is $\eta \ge \Theta$. For $\eta \ge 1$ finiteness follows directly from $A_\lambda(x) = O_\lambda(1/\log x)$, so no endpoint inversion at $\eta = 1$ is being assumed.

Proposition 21.3 (Exact window-energy growth exponent)**.** *Let* $J_\lambda(X) = \int_2^X A_\lambda(x)^2\,dx$*. Then*

$$\limsup_{X\to\infty} \frac{\log(1 + J_\lambda(X))}{\log X} = 2\Theta - 1. \tag{21.9}$$

The circle-difference integral has the same exponent.

Proof. The upper bound follows from

$$J_\lambda(X) \le 2\int_2^X E(x)^2\,dx + \frac{2}{\lambda}\int_2^{\lambda X} E(x)^2\,dx$$

and the source note's exact quadratic moment-growth law. If $\Theta = 1/2$, this upper bound and nonnegativity already suffice. Otherwise, suppose the left-hand exponent were smaller than $2\Theta - 1$. Choose $a \ge 0$ strictly between the two exponents, and then choose $(1 + a)/2 < \eta < \Theta \le 1$. We would have $J_\lambda(X) = O(X^a)$. Dyadic summation then gives

$$\int_2^\infty x^{1-2\eta} A_\lambda(x)^2\,dx < \infty,$$

because $a + 1 - 2\eta < 0$. By (21.8) the same weighted integral for E is finite. Weighted Cauchy–Schwarz would continue its Mellin transform to $\Re s > \eta$, excluding every zeta zero to the right of η, contrary to $\eta < \Theta$. This lower-bound argument uses only the necessary part of the weighted criterion, not its delicate endpoint sufficiency. The circle claim follows by the eventual comparison in (21.3). □

21.2 An exact Toeplitz and spectral criterion

Let $u_n = U(p_n)$, $w_n = p_{n+1} - p_{n-1}$ for $n \ge 2$, and $W_N = \sum_{n=2}^N w_n$. Define the probability measure and its moments by

$$\mu_N = \frac{1}{W_N}\sum_{n=2}^N w_n \delta_{u_n}, \qquad m_{k,N} = \int z^k\,d\mu_N(z), \qquad V_N^{(k)} = W_N(1 - |m_{k,N}|^2).$$

Thus $V_N^{(1)}$ is exactly the original weighted variance V_N. For a fixed integer $r \ge 1$, set

$$T_{r,N} = (m_{j-k,N})_{0\le j,k\le r}, \quad \mathcal{T}_{r,N} = W_N e_2(T_{r,N}), \quad \mathcal{L}_{r,N} = W_N(r + 1 - \lambda_{\max}(T_{r,N})),$$

where $e_2(T)$ is the sum of all principal 2×2 minors, equivalently the second elementary symmetric polynomial of its eigenvalues. All these quantities are invariant under a common rotation of the u_n and require no centering constant.

Theorem 21.4 (Toeplitz impurity and top-eigenvalue criteria)**.** *For every* N *and fixed* $r \ge 1$*,* $T_{r,N}$ *is positive semidefinite,* $\operatorname{tr} T_{r,N} = r + 1$*, and*

$$\mathcal{T}_{r,N} = \sum_{k=1}^r (r + 1 - k)V_N^{(k)}. \tag{21.10}$$

Furthermore,

$$rV_N \le \mathcal{T}_{r,N} \le \frac{r(r+1)^2(r+2)}{12} V_N, \tag{21.11}$$

$$\frac{r+1}{2}\,\mathcal{L}_{r,N} \le \mathcal{T}_{r,N} \le (r+1)\mathcal{L}_{r,N}. \tag{21.12}$$

The sequence $\mathcal{T}_{r,N}$ *is nondecreasing. For the prime vectors,*

$$\mathrm{RH} \iff \sup_N \mathcal{T}_{r,N} < \infty \iff \sup_N \mathcal{L}_{r,N} < \infty. \tag{21.13}$$

Both diagnostics have limsup growth exponent $2\Theta - 1$ *when the normalizing variable is* p_N*.*

Proof. For $a = (a_0, \dots, a_r)$,

$$a^* T_{r,N} a = \int \left| \sum_{j=0}^{r} \overline{a_j} z^j \right|^2 d\mu_N(z) \geq 0.$$

The diagonal entries equal one. A principal minor indexed by $j < k$ is $1 - |m_{k-j,N}|^2$, and there are $r + 1 - k$ pairs separated by k. This proves (21.10).

The exact pairwise variance formula is

$$V_N^{(k)} = \frac{1}{W_N} \sum_{2 \leq j < \ell \leq N} w_j w_\ell |u_j^k - u_\ell^k|^2.$$

For unit complex numbers, factorization gives $|u^k - v^k| \leq k|u - v|$. Hence $0 \leq V_N^{(k)} \leq k^2 V_N$. Retaining the $k = 1$ summand proves the lower bound in (21.11). The upper bound follows from

$$\sum_{k=1}^{r} (r + 1 - k)k^2 = \frac{r(r+1)^2(r+2)}{12}.$$

For each k, the ordinary weighted-variance update formula is

$$V_{\text{new}}^{(k)} - V^{(k)} = \frac{Ww}{W + w} |u^k - m_k|^2 \geq 0.$$

Thus (21.10) also proves monotonicity.

Put $L = r + 1$ and let the nonnegative eigenvalues have largest member Λ. Set $\delta = L - \Lambda$. Since $\sum \lambda_j^2 \leq \Lambda \sum \lambda_j = \Lambda L$,

$$e_2(T) = \frac{L^2 - \sum \lambda_j^2}{2} \geq \frac{L\delta}{2}.$$

Writing the other eigenvalues as a list with sum δ also gives

$$e_2(T) = \Lambda\delta + e_2(\text{other eigenvalues}) \leq (L - \delta)\delta + \frac{\delta^2}{2} \leq L\delta.$$

Multiplication by W_N proves (21.12). Finally, the original paper proves RH equivalent to bounded V_N and proves that its growth exponent is $2\Theta - 1$ (since $D_N \leq V_N \leq 2D_N$). The fixed positive comparison constants transfer both statements. □

Why the full determinant is the wrong diagnostic. For $r \geq 2$, take $\mu_W = (\delta_1 + \delta_{\exp(iW^{-a})})/2$, where $0 < a < 1/2$. Then μ_W converges to a point mass, and $T_r(\mu_W)$ has rank at most two, so $\det T_r(\mu_W) = 0$. Nevertheless

$$W(1 - |m_1|^2) = W \sin^2(W^{-a}/2) \sim \tfrac{1}{4} W^{1-2a} \longrightarrow \infty.$$

These are test measures, not the prime measures. They prove that positivity, concentration, and a small full determinant alone cannot supply a general bound for the required variance. The sum of second minors in (21.10) avoids this rank-loss problem.

A genuine random-matrix integral, but not a prime estimate. The Toeplitz moment determinant obeys the exact Vandermonde identity

$$\det T_{r,N} = \frac{1}{(r+1)!} \int \cdots \int \prod_{0 \leq j < k \leq r} |z_j - z_k|^2 \prod_{j=0}^{r} d\mu_N(z_j). \tag{21.14}$$

To prove it, write the product as the squared determinant of $(z_j^k)_{0 \leq j,k \leq r}$, expand both determinants, and integrate term by term. Each permutation pair contributes a product of moments; reindexing gives $(r + 1)!$ copies of the Toeplitz determinant. This is a valid Dyson/random-matrix bridge. It does not make the prime phases random, and the preceding two-atom example explains why controlling this full determinant is insufficient.

21.3 Two rigorously identifiable failed shortcuts

Proposition 21.5 (Fixed additive differences are already square summable)**.** *For every fixed positive integer h,*

$$\sum_{m=2}^{\infty} |U(m+h) - U(m)|^2 < \infty$$

unconditionally. More generally, if $a_0, \dots, a_h$ are fixed complex numbers with $\sum_{j=0}^{h} a_j = 0$, then

$$\sum_{m=2}^{\infty} \left| \sum_{j=0}^{h} a_j U(m+j) \right|^2 < \infty.$$

Proof. For $\phi_m = S(m) - \log\log m$,

$$|\phi_{m+1} - \phi_m| \le \frac{\mathbf{1}_{\{m+1 \text{ prime}\}}}{m+1} + \frac{1}{m \log m}.$$

The squares of this upper bound are summable. Since $|e^{2\pi i a} - e^{2\pi i b}| \le 2\pi|a-b|$, the first differences of $U(m)$ are in ℓ^2. A fixed h-step difference is a sum of h translated first differences. For the final assertion, use $\sum_j a_j U(m+j) = \sum_j a_j [U(m+j) - U(m)]$ and the finite triangle inequality in ℓ^2. □

This is not the unbounded-scale window criterion of Theorem 21.2. A model makes the loss explicit: for $u_m = e^{im^{-a}}$, $0 < a \le 1/2$, the first differences are in ℓ^2 while $\sum_m |u_m - 1|^2 = \infty$. Thus a compact-curve Fourier or isoperimetric estimate controlling only local motion cannot be used as a substitute for the required global centered energy.

Proposition 21.6 (Szegő smoothing is not uniform at a point mass)**.** *Let $0 < t < 1$ and*

$$f_t(\theta) = \frac{1 - t^2}{|1 - te^{i\theta}|^2}$$

be the Poisson smoothing of a unit point mass. With normalized circle measure $d\theta/(2\pi)$, write c_k for the Fourier coefficients of $\log f_t$. Then

$$c_0 = \log(1 - t^2), \qquad c_k = \frac{t^{|k|}}{|k|} \ (k \neq 0), \qquad \sum_{k \ge 1} k|c_k|^2 = -\log(1 - t^2) \longrightarrow \infty. \tag{21.15}$$

For the $n \times n$ Toeplitz matrix of f_t,

$$\det(t^{|j-k|})_{0 \le j,k < n} = (1 - t^2)^{n-1}. \tag{21.16}$$

Proof. Expand $-\log(1 - te^{i\theta}) = \sum_{k \ge 1} t^k e^{ik\theta}/k$ and add its conjugate. The coefficient formula and logarithmic series prove (21.15). The Poisson kernel itself has Fourier moments $t^{|k|}$. Starting from the last row and proceeding upward, subtract t times row $j-1$ from row j. The resulting matrix is upper triangular with diagonal $1, 1 - t^2, \dots, 1 - t^2$, proving (21.16). □

Even a point mass with exactly zero geometric deficit has a diverging Szegő logarithmic constant under this regularization. Therefore a fixed-symbol asymptotic cannot simply be passed through the joint prime-cutoff and smoothing limits. Uniform control in those parameters would have to be separately proved.

Proposition 21.7 (Infinite turning is not real-space equidistribution)**.** *Let $C(x) = e^{2\pi i S(x)}$. Then*

$$\frac{1}{X}\int_2^X C(x)\,dx = C(X)(1+o(1)), \qquad X \to \infty. \tag{21.17}$$

In particular its first Fourier average has modulus tending to one, not zero, so the circle motion is not Haar-equidistributed under uniform real-x averaging.

Proof. Fix $0 < \delta < 1$. Uniformly for $\delta X \le x \le X$,

$$S(x) - S(X) = \log\log x - \log\log X + E(x) - E(X) = o(1).$$

Indeed the first difference is $O_\delta(1/\log X)$ and the second tends uniformly to zero by Mertens' theorem. Thus $C(x)/C(X) = 1 + o(1)$ on this interval. The complementary interval has length at most δX and the integrand has modulus one. Take $X \to \infty$, then $\delta \downarrow 0$. □

22 An overlapping-window local criterion

This section gives an extension not stated in the supplied manuscript. It is a new deduction relative to that manuscript, not a claim of historical novelty, and does not establish an unconditional bound needed to prove RH.

For integers $m \ge 2$, put

$$u_m = \exp\bigl(2\pi i[S(m) - \log\log m]\bigr).$$

No value of the Meissel–Mertens constant is needed to compute these vectors. For $k \ge 1$, define the local deficit

$$d_k = 3\cdot 2^k - \left|\sum_{m=2^k}^{2^{k+2}-1} u_m\right|.$$

Thus the windows overlap and their endpoint ratio is four. Write

$$L_K = \sum_{k=1}^{K-1} d_k, \qquad D_K^* = (2^{K+1}-2) - \left|\sum_{m=2}^{2^{K+1}-1} u_m\right| \quad (K \ge 2).$$

Here D_K^* is the original integer-sampled deficit at a dyadic cutoff.

Theorem 22.1 (Finite local-to-global comparison)**.** *For every sequence of complex unit vectors u_m, without any arithmetic or convergence assumption,*

$$\frac{D_K^*}{18+12\sqrt{2}} \le L_K \le 2D_K^*.$$

Proof. Partition the full range into blocks $B_k = \{2^k, \dots, 2^{k+1}-1\}$ of size $a_k = 2^k$, and write

$$b_k = \frac{1}{a_k}\sum_{m\in B_k} u_m, \qquad H_k = \sum_{m\in B_k} |u_m - b_k|^2.$$

Let Q_k denote the variance on $B_k \cup B_{k+1}$, with its own mean. The variance decomposition gives exactly

$$Q_k = H_k + H_{k+1} + \frac{2}{3}a_k|b_{k+1} - b_k|^2.$$

For a set of W unit vectors whose sum has modulus r, its deficit is $D = W - r$ and its variance is $V = W - r^2/W = D(1 + r/W)$. Consequently $d_k \le Q_k \le 2d_k$.

Put $C_H = (1-2^{-1/2})^{-2} = 6+4\sqrt{2}$. The elementary weighted Hardy inequality

$$\sum_{k=1}^{K} a_k |b_k - b_K|^2 \le C_H \sum_{k=1}^{K-1} a_k |b_{k+1} - b_k|^2$$

follows on setting $x_j = 2^{j/2}(b_{j+1} - b_j)$: the vector on the left after square roots is the backward convolution $\sum_{j\ge k} 2^{-(j-k)/2} x_j$. Each backward shift is an ℓ^2 contraction, so its norm is at most $\sum_{r\ge 0} 2^{-r/2} = (1-2^{-1/2})^{-1}$. All sequences here are finite, extended by zero where needed.

The global variance V_K^* minimizes the squared error over all complex centers, hence

$$\begin{aligned} D_K^* \le V_K^* &\le \sum_{m=2}^{2^{K+1}-1} |u_m - b_K|^2 \\ &= \sum_{k=1}^{K} H_k + \sum_{k=1}^{K} a_k |b_k - b_K|^2 \\ &\le \frac{3}{2} C_H \sum_{k=1}^{K-1} Q_k \le 3 C_H L_K. \end{aligned}$$

Here the penultimate inequality uses $\sum Q_k \ge \sum H_k + (2/3)\sum a_k |b_{k+1} - b_k|^2$.

For the opposite inequality, let v be the unit vector in the direction of the full resultant, or any unit vector if that resultant is zero. For each local window, $d_k \le \frac{1}{2}\sum_{m\in B_k \cup B_{k+1}} |u_m - v|^2$. Every integer occurs in at most two windows, so

$$L_K \le \sum_{m=2}^{2^{K+1}-1} |u_m - v|^2 = 2 D_K^*.$$

This proves both bounds. □

Theorem 22.2 (Constant-free bounded-local-deficit criterion)**.** *For the prime-reciprocal vectors above, the following are equivalent:*

$$\text{RH}, \qquad \sum_{k\ge 1} d_k < \infty, \qquad \sup_{k\ge 1} d_k < \infty.$$

Proof. RH implies boundedness of the original integer-sampled deficit by the supplied note's energy, integer-sampling, and variance lemmas. The finite comparison then proves $\sum d_k < \infty$. Summability of the nonnegative d_k implies their boundedness.

It remains to prove that bounded local deficits imply RH. If $d_k \le A$ for all k, the finite theorem gives $D_K^* = O(K)$. Hence $V_K^* = O(K)$ too. Let $W_K = 2^{K+1} - 2$ and let c_K be the mean of u_m over the full prefix of this size. By Mertens' theorem, $u_m \to u_\infty = e^{2\pi i B_{\mathbb{P}}}$, and therefore $c_K \to u_\infty$. Variance decomposition when one prefix is enlarged gives

$$W_K |c_{K+1} - c_K|^2 \le V_{K+1}^* - V_K^* \le O(K+1).$$

Telescoping to the known limit yields

$$|c_K - u_\infty| \le \sum_{j\ge K} O(\sqrt{j+1}\, 2^{-j/2}) = O(\sqrt{K+1}\, 2^{-K/2}).$$

The fixed-center variance identity now implies

$$\sum_{m=2}^{2^{K+1}-1} |u_m - u_\infty|^2 = V_K^* + W_K |c_K - u_\infty|^2 = O(K+1).$$

Since $E(m) \to 0$ and $|u_m - u_\infty|^2 = 4\sin^2(\pi E(m)) \asymp E(m)^2$ on a tail, we obtain $\sum_{m \le X} E(m)^2 = O(\log X)$. On $[m, m+1)$, $|E(x) - E(m)| \le 1/(m \log m)$; this error is square-integrable. Thus $\int_2^X E(x)^2\,dx = O(\log X)$ as well.

Finally, for every $\sigma > 1/2$, Cauchy–Schwarz on dyadic intervals gives

$$\int_{2^j}^{2^{j+1}} |E(x)| x^{-\sigma}\,dx \ll \sqrt{j+1}\, 2^{j(1/2-\sigma)}.$$

The same estimates with powers of $\log x$ justify all complex derivatives. Consequently $F(s) = \int_2^\infty E(x) x^{-s}\,dx$ is holomorphic on $\Re s > 1/2$. The supplied note's Euler-product Mellin obstruction excludes any zeta zero there; the functional equation gives RH. □

The equivalence between bounded individual d_k and summability is an arithmetic consequence, not a theorem about arbitrary convergent unit-vector sequences. No bound $\sup_k d_k < \infty$ has been proved here.

Why overlap cannot simply be omitted. For the generic sequence $u_m = e^{i/k}$ on each block B_k, the vectors converge to 1, and every individual nonoverlapping block has deficit zero. Nevertheless,

$$\sum_{m \ge 2} |u_m - 1|^2 = \sum_{k \ge 1} 2^k |e^{i/k} - 1|^2 = \infty.$$

So zero deficit in each disjoint block is insufficient. The overlapping windows supply the missing comparison between block means.

Computable local pairwise form. If $I_k = B_k \cup B_{k+1}$, then

$$Q_k = \frac{1}{3 \cdot 2^k} \sum_{\substack{m<n \\ m,n \in I_k}} |u_m - u_n|^2, \qquad d_k \le Q_k \le 2d_k.$$

Thus the same criterion has a positive pairwise form depending only on prime sums and logarithms, with no supplied centering constant. Computing via the resultant avoids the quadratic cost of this formula.

23 What the geometric methods actually close

The geometric attempts yield unconditional local regularity, an exact multiscale separation, and a global harmonic comparison. They do not prove boundedness of the prime deficit. The identities below are elementary consequences of variance decomposition, Bertrand's postulate, and trigonometry. Their historical novelty is not asserted. In particular, the multiscale identity is a Haar/ANOVA identity, not a new general orthogonality theorem.

23.1 Finite nearest-neighbor turning energy

Extend the manuscript's notation to the first prime by setting

$$U_n = \exp\left(2\pi i \left[\sum_{r \le n} \frac{1}{p_r} - \log\log p_n\right]\right), \qquad n \ge 1.$$

Proposition 23.1 (Unconditional local turning bound)**.** *Without assuming RH,*

$$\sum_{n \ge 1} |U_{n+1} - U_n|^2 < 12\pi^2.$$

Proof. Write $a_n = \log p_n$ and $\delta_n = \log a_{n+1} - \log a_n$. Cauchy–Schwarz and Bertrand's postulate give

$$\delta_n^2 = \left(\int_{a_n}^{a_{n+1}} \frac{dt}{t}\right)^2 \le (a_{n+1} - a_n) \int_{a_n}^{a_{n+1}} \frac{dt}{t^2} < (\log 2)\left(\frac{1}{a_n} - \frac{1}{a_{n+1}}\right).$$

Summing yields $\sum_n \delta_n^2 \le 1$. The exact phase increment, measured in turns, is $p_{n+1}^{-1} - \delta_n$. Hence

$$|U_{n+1} - U_n|^2 \le 4\pi^2 (p_{n+1}^{-1} - \delta_n)^2 \le 8\pi^2 (p_{n+1}^{-2} + \delta_n^2).$$

Finally, $\sum_{n\ge 1} p_{n+1}^{-2} \le \sum_{m\ge 3} m^{-2} < 1/2$. □

This result controls an unweighted nearest-neighbor quantity. The manuscript's RH criterion concerns accumulated displacement with prime-gap weights. Interchanging these two energies would be an error.

23.2 Every fixed dyadic scale is unconditionally finite

Use integer samples, which the manuscript proves equivalent to the prime-endpoint criterion, and set

$$z_r = \exp\left(2\pi i \left[\sum_{p \le r+2} \frac{1}{p} - \log\log(r+2)\right]\right), \quad r \ge 0.$$

For $j \ge 1$ and $k \ge 0$, let $L = 2^j$, and let $m_{j,k}^-$ and $m_{j,k}^+$ be the arithmetic means on the first and second halves of the block $kL \le r < (k+1)L$. Define the nonnegative Haar merge cost

$$H_{j,k} = \frac{L}{4} |m_{j,k}^- - m_{j,k}^+|^2.$$

No Meissel–Mertens constant enters these definitions.

Theorem 23.2 (Exact scale separation)**.** *For $Q = 2^J$,*

$$Q - \frac{1}{Q}\left|\sum_{r=0}^{Q-1} z_r\right|^2 = \sum_{j=1}^{J} \sum_{k=0}^{2^{J-j}-1} H_{j,k}.$$

Moreover, with $K = 2\pi(1 + 1/\log 2)$,

$$\sum_{k\ge 0} H_{j,k} \le 2^j \left(1 + \frac{K^2\pi^2}{24}\right) < \infty \qquad \textit{for every fixed } j.$$

Using the manuscript's integer-sampling theorem,

$$\text{RH} \iff \sum_{j\ge 1} \sum_{k\ge 0} H_{j,k} < \infty.$$

Proof. For two equal-size blocks of $L/2$ points, variance decomposition gives

$$V_{\text{merged}} = V_{\text{left}} + V_{\text{right}} + \frac{L}{4} |m_{\text{left}} - m_{\text{right}}|^2.$$

Iterate down to singletons, whose variance is zero, to obtain the finite identity.

For the scale bound, two integer sample locations s, t in a block starting at $a = 2 + kL$ satisfy

$$|S(s) - S(t)| \le L/a, \qquad |\log\log s - \log\log t| \le L/(a \log a).$$

Thus any two unit vectors in that block are at distance at most $\min\{2, KL/a\}$. The same bound holds between the two means, by averaging pairwise differences. For $k \geq 1$ this gives $H_{j,k} \leq K^2 L/(4k^2)$; for $k = 0$, use $H_{j,0} \leq L$. Summing $\sum_{k\geq 1} k^{-2} = \pi^2/6$ proves the estimate.

The finite sums in the first identity increase with J and exhaust all pairs (j, k). The variance and vector deficit are within a factor of two, as in the manuscript. The integer deficit is monotone, so bounding its dyadic subsequence is equivalent to bounding all its partial sums. The integer-sampling criterion finishes the proof. □

This theorem closes each fixed spatial scale. It also isolates the missing statement: a summable bound across the scale index j. The proved upper bound grows like 2^j, and therefore does not supply that statement. It would be misleading to count the infinitely many fixed-scale conclusions as a proof of their infinite sum.

23.3 Global harmonic certificates

For arbitrary unit complex numbers $u_1, \ldots, u_N$ and positive weights $w_1, \ldots, w_N$, write $W = \sum_n w_n$ and define

$$V^{(k)} = W - \frac{1}{W}\left|\sum_n w_n u_n^k\right|^2.$$

Proposition 23.3 (Two harmonics retain all pairwise dispersion)**.** *For every finite configuration, without a convergence assumption,*

$$\frac{3}{4}V^{(1)} \leq V^{(2)} + V^{(3)} \leq 13V^{(1)}.$$

More generally, let F be a nonempty finite set of positive integers and let $c_k > 0$ for $k \in F$. Then constants $0 < A \leq B < \infty$ satisfying

$$AV^{(1)} \leq \sum_{k\in F} c_k V^{(k)} \leq BV^{(1)}$$

for every finite weighted configuration exist if and only if $\gcd(F) = 1$.

Proof. The pairwise variance formula gives

$$V^{(k)} = \frac{1}{W}\sum_{r<s} w_r w_s |u_r^k - u_s^k|^2.$$

For $z, w \in \mathbb{S}^1$ and $c = \Re(z\overline{w})$,

$$|z^2 - w^2|^2 + |z^3 - w^3|^2 = |z - w|^2(4c^2 + 6c + 3).$$

On $[-1, 1]$, the polynomial on the right ranges from $3/4$ to 13. This proves the first assertion, including at $z = w$ by continuity.

For the general assertion, the relevant ratio is

$$R(\theta) = \frac{\sum_{k\in F} c_k(1 - \cos k\theta)}{1 - \cos\theta}.$$

Its continuous value at 0 is $\sum c_k k^2 > 0$. When $\gcd(F) = 1$, its numerator has no other zero on the circle. Compactness gives positive finite extrema. If $d = \gcd(F) > 1$, use two positive-weight points 1 and $e^{2\pi i/d}$: every $V^{(k)}$ on the proposed right side vanishes but $V^{(1)} > 0$. □

Consequently the prime criterion can equally be tested using the sum of second- and third-harmonic variances. This finite-configuration comparison is stronger than a merely eventual local comparison, but it gives no new upper bound for the actual prime configuration. It explains exactly when a finite positive Fourier-kernel substitution loses information through aliasing.

23.4 A sharp obstruction to purely geometric concentration arguments

Proposition 23.4 (Finite total turning does not bound the deficit)**.** *Let* $v_m = \exp(i/\log(m+1))$ *and* $\mathcal{D}_N = N - |\sum_{m=1}^{N} v_m|$*. Then* $v_m \to 1$*,* $|v_m - 1| = O(1/\log m)$*, and*

$$\sum_{m\geq 1} |v_{m+1} - v_m| \leq \frac{1}{\log 2} < \infty.$$

Nevertheless, for all $N \geq 16$*,*

$$\mathcal{D}_N \geq \frac{(\log(4/3))^2}{32\pi^2} \frac{N}{\log^4(N+1)} \longrightarrow \infty.$$

Proof. The angles $\theta_m = 1/\log(m+1)$ decrease to zero, so the total turning bound follows from $|e^{ia} - e^{ib}| \leq |a-b|$ and telescoping. Choose

$$A = \{\lceil N/4\rceil, \dots, \lfloor N/2\rfloor\}, \quad B = \{\lceil 3N/4\rceil, \dots, N\}.$$

Each set has at least $N/8$ elements. For $m \in A, n \in B$,

$$\theta_m - \theta_n \geq \frac{\log((3N/4+1)/(N/2+1))}{\log(N/2+1)\log(3N/4+1)} \geq \frac{\log(4/3)}{\log^2(N+1)}.$$

All angle differences are smaller than π, so $|v_m - v_n| \geq 2|\theta_m - \theta_n|/\pi$. Keeping only these cross pairs in $\mathcal{V}_N = N^{-1}\sum_{m<n} |v_m - v_n|^2$ gives

$$\mathcal{V}_N \geq \frac{(\log(4/3))^2}{16\pi^2} \frac{N}{\log^4(N+1)}.$$

Finally $\mathcal{V}_N \leq 2\mathcal{D}_N$. □

This is not a counterexample involving the prime phase identity and is not a counterexample to RH. It refutes an attempted general implication from convergence, a Mertens-size envelope, or even finite total turning to bounded deficit. It also explains why the local result in Proposition 23.1 cannot finish the problem.

23.5 Exact merge costs and Voronoi weights

Two additional identities clarify the geometric transfers. For any partition into blocks B, put $M_B = \sum_{n\in B} w_n u_n$ and $d_B = \sum_{n\in B} w_n - |M_B|$. Then

$$D = \sum_B d_B + \left(\sum_B |M_B| - \left|\sum_B M_B\right|\right).$$

Both terms are nonnegative. Applying this recursively gives an exact tree decomposition. No uniform bound is obtained by discarding positive merge costs.

For the prime weights, the one-dimensional Voronoi cell

$$I_n = \left[\frac{p_{n-1}+p_n}{2}, \frac{p_n + p_{n+1}}{2}\right)$$

has length $w_n/2$. Let $v(x) = U_n$ on I_n and $\Omega_N = \bigcup_{n=2}^{N} I_n$. Then exactly

$$D_N = 2\left(|\Omega_N| - \left|\int_{\Omega_N} v(x)\,dx\right|\right).$$

This gives the correct geometric volume element. Replacing the cells by equal weights would change the arithmetic question.

24 Arithmetic obstructions to elliptic transfers

In this section only, write

$$S_N = \sum_{j=1}^{N} \frac{1}{p_j} = \frac{A_N}{Q_N}$$

in lowest terms, with $Q_N > 0$. These are the *uncentered* reciprocal sums. Their raw circle endpoints are $C_N = \exp(2\pi i S_N)$, whereas the manuscript's centered endpoints are $U_N = \exp(2\pi i(S_N - \log\log p_N))$. The cyclotomic assertions below concern C_N, not automatically U_N. No assertion about the transcendence of U_N is needed. The conclusions close specified algebraic transfer attempts; they do not prove the required upper bound on D_N. No historical novelty is asserted.

Theorem 24.1 (Exact algebraic degree of the raw endpoints)**.** *For every $N \geq 1$,*

$$Q_N = \prod_{j=1}^{N} p_j, \qquad \gcd(A_N, Q_N) = 1, \qquad [\mathbb{Q}(C_N) : \mathbb{Q}] = \varphi(Q_N) = \prod_{j=1}^{N} (p_j - 1).$$

Let K be a fixed number field and let $f \in K(T)$ be a nonconstant rational function of degree d. Whenever $f(C_N)$ is finite,

$$[K(f(C_N)) : K] \;\geq\; \frac{\varphi(Q_N)}{d[K : \mathbb{Q}]}.$$

Consequently neither the raw endpoints, nor their images under any fixed nonconstant rational coordinate function, have uniformly bounded algebraic degree over a fixed number field.

Proof. Put $Q = \prod_{j=1}^{N} p_j$ and $A = \sum_{j=1}^{N} Q/p_j$. For each $p_j \mid Q$, $A \equiv Q/p_j \not\equiv 0 \pmod{p_j}$. Thus A/Q is already reduced, and C_N is a primitive Q-th root of unity. Cyclotomic irreducibility gives its degree $\varphi(Q)$ [22].

Write $f = P/R$, where $P, R \in K[T]$ are coprime and $\max(\deg P, \deg R) = d$. Set $\alpha = C_N$ and $\beta = f(\alpha)$. The polynomial $P(T) - \beta R(T)$ is nonzero, has degree at most d, and annihilates α over $K(\beta)$. Hence

$$[K(\alpha) : K(\beta)] \leq d, \qquad [K(\alpha) : K] \geq \frac{[\mathbb{Q}(\alpha) : \mathbb{Q}]}{[K : \mathbb{Q}]}.$$

The tower law gives the claimed lower bound. A rational function has only finitely many poles, and the C_N have pairwise distinct orders, so only finitely many indices can be excluded as poles. □

This does not prevent arithmetic on a fixed algebraic variety with points in growing number fields: every C_N already belongs to the fixed multiplicative group. It excludes only the stated fixed-field, bounded-degree compression.

Theorem 24.2 (Squarefree denominators obstruct direct elliptic lifts)**.** *If $a_2, a_4, a_6 \in \mathbb{Z}$, then for no $N \geq 1$ and no $y \in \mathbb{Q}$ does*

$$y^2 = S_N^3 + a_2 S_N^2 + a_4 S_N + a_6.$$

More generally, fix $a \in \mathbb{Q} \setminus \{0\}$, $b \in \mathbb{Q}$, and $a_2, a_4, a_6 \in \mathbb{Q}$. For every sufficiently large N, the number $x_N = aS_N + b$ is not the x-coordinate of a rational point on

$$y^2 = x^3 + a_2 x^2 + a_4 x + a_6.$$

No nonsingularity assumption is needed; in particular the assertions apply to elliptic Weierstrass models.

Proof. Choose a prime $p \mid Q_N$. The preceding theorem gives $v_p(S_N) = -1$. For integral coefficients, the four terms on the right of the first equation have valuations -3, at least -2, at least -1, and at least 0, respectively. A zero term is understood to have infinite valuation. Since the minimum is unique, the right side has valuation -3. This is impossible for a nonzero rational square, whose valuation is even.

For the second assertion, fix an odd prime p such that $v_p(a) = 0$ and b, a_2, a_4, a_6 are p-adic integers. Only finitely many primes fail these conditions. Once $p \le p_N$, we have $v_p(S_N) = -1$, and therefore $v_p(aS_N + b) = -1$. The same uniquely minimal valuation -3 again contradicts rational squareness. □

Completing the square treats rational Weierstrass equations with xy or y terms at a prime outside the finite set of coefficient denominators and the prime 2. The obstruction is to this *fixed affine coordinate* encoding. It does not exclude nonlinear choices such as $x = S_N^2$, covers, growing number fields, or curves that vary with N. Those require further arguments. For example, putting $x = S_N$ on $y^2 = x^3 - 157^2 x$ is impossible for every N, not just for the finitely many prefixes checked computationally.

Theorem 24.3 (Growing bad reduction in a canonical Frey encoding)**.** *For each N, the curve*

$$\mathcal{E}_N : \qquad y^2 = x(x - A_N)(x + Q_N)$$

is elliptic. Its displayed integral Weierstrass model has

$$c_4 = 16(A_N^2 + A_N Q_N + Q_N^2), \qquad \Delta = 16 A_N^2 Q_N^2 (A_N + Q_N)^2.$$

At every odd prime $p \mid Q_N$, this model is minimal and has multiplicative reduction of type I_2. In particular,

$$\frac{Q_N}{2} \;\mid\; \operatorname{cond}(\mathcal{E}_N).$$

Proof. The roots $0, A_N, -Q_N$ are distinct because $A_N, Q_N > 0$. The Weierstrass coefficients are $a_2 = Q_N - A_N$, $a_4 = -A_N Q_N$, and $a_1 = a_3 = a_6 = 0$. The standard formulas

$$b_2 = 4a_2, \quad b_4 = 2a_4, \quad b_6 = 0, \quad b_8 = -a_4^2, \quad c_4 = b_2^2 - 24 b_4, \quad \Delta = -b_2^2 b_8 - 8 b_4^3$$

give the asserted invariants by direct substitution.

Let p be odd and divide Q_N. Squarefreeness gives $v_p(Q_N) = 1$, and coprimality gives $p \nmid A_N(A_N + Q_N)$. Hence

$$v_p(\Delta) = 2, \qquad v_p(c_4) = 0.$$

The standard multiplicative-reduction criterion says that an integral model with c_4 a unit and discriminant of positive valuation is minimal and of type $I_{v_p(\Delta)}$; a multiplicative prime has conductor exponent one [23]. Applying this criterion and multiplying the distinct odd prime factors proves the result. □

This closes the attempt to obtain a bounded-conductor family simply by inspecting these original Frey curves. It is not an impossibility theorem for level lowering or for alternative Frey encodings. In particular, conductor growth of the curves alone does not rule out a more subtle argument with residual representations. The present construction does not control the centered error or D_N.

Proposition 24.4 (A conic does not rationally parametrize an elliptic curve)**.** *Every rational map over $\mathbb{C}$ from the smooth projective completion of $x^2 + y^2 = 1$ to a smooth projective elliptic curve is constant.*

Proof. The conic is isomorphic to $\mathbb{P}^1$ over $\mathbb{C}$. A rational map from a smooth projective curve to a projective curve extends across its missing points. If the resulting map to an elliptic curve were nonconstant, it would have a positive degree d, and Riemann–Hurwitz would imply

$$-2 = d(2 \cdot 1 - 2) + \sum_P (e_P - 1) = \sum_P (e_P - 1) \geq 0,$$

a contradiction [24]. □

Analytic uniformization, covers of the conic, correspondences, and index-dependent constructions are not excluded. A circle or a nondegenerate plane ellipse is genus zero; changing between them does not create the elliptic-curve arithmetic used by the modular method.

Exact checks. An independent rational-arithmetic program verified the reduced denominators, cyclotomic-degree formula, local valuations, Frey invariants, and the two consecutive-cube identities used below for the first 100 prime prefixes, through $p_{100} = 541$. The Frey invariants and cube identities were also checked symbolically. These calculations test the formulas; the preceding arguments, rather than the finite tests, prove their general validity.

25 Topological and conformal transfers

A transfer attempt here means specifying a target construction, checking its necessary hypotheses, and recording its consequence or missing estimate. It does not mean that every possible use of the source theorem has been exhausted. Four rigorous outcomes follow: a quantitative obstruction to trace-only arguments, unitary classification of disk-chart changes, a finite winding certificate, and persistence of logarithmic monodromy under finite covers. None gives the missing uniform arithmetic bound. The finite certificate uses classical complex analysis; no claim of historical novelty is made for it or its elementary zero-free example.

25.1 Same geometric trace, different deficit growth

For $a > 0$ set

$$u_a(x) = e^{ix^{-a}}, \quad W(X) = X - 1, \quad M_a(X) = \int_1^X u_a(x)\,dx, \quad D_a(X) = W(X) - |M_a(X)|.$$

The measure dx is part of this continuous vector-deficit construction.

Theorem 25.1 (A trace-only obstruction)**.** *Every u_a has exactly the same oriented circular-arc trace, from e^i towards 1, and the same total curvature 1. Nevertheless,*

$$D_a(X) \sim \frac{a^2 X^{1-2a}}{2(1-2a)(1-a)^2} \quad (0 < a < \tfrac{1}{2}), \qquad D_{1/2}(X) \sim \tfrac{1}{2}\log X,$$

whereas for $a > 1/2$,

$$D_a(X) \longrightarrow \int_1^\infty [1 - \cos(x^{-a})]\,dx < \infty.$$

Thus no invariant of the unparametrized arc alone distinguishes bounded from unbounded sampling-weighted deficit.

Proof. The variable $\theta = x^{-a}$ decreases bijectively from 1 to 0, so all oriented traces coincide. A unit circular arc of length 1 has total curvature 1. Put

$$A = \int_1^X [1 - \cos(x^{-a})]\,dx, \qquad B = \int_1^X \sin(x^{-a})\,dx.$$

Then $M_a = W - A + iB$, $A = o(W)$, $B = o(W)$, and exactly

$$D_a = \frac{2WA - A^2 - B^2}{W + |M_a|}. \tag{25.1}$$

If $0 < a < 1/2$, the small-angle expansions imply $A \sim X^{1-2a}/[2(1-2a)]$ and $B \sim X^{1-a}/(1-a)$. Also $A^2/W = o(X^{1-2a})$ and $W + |M_a| \sim 2X$. Substitution gives coefficient

$$\frac{1}{2(1-2a)} - \frac{1}{2(1-a)^2} = \frac{a^2}{2(1-2a)(1-a)^2} > 0.$$

For $a = 1/2$, the fourth-order cosine remainder is integrable, giving $A = \frac{1}{2}\log X + O(1)$, while $B \sim 2\sqrt{X}$. Hence $B^2/W \to 4$ and $A^2/W \to 0$, proving the logarithmic assertion. For $a > 1/2$, A converges. The inequality $|\sin(x^{-a})| \le x^{-a}$ gives $B^2/W \to 0$: the respective bounds for $1/2 < a < 1$, $a = 1$, and $a > 1$ are $O(X^{1-2a})$, $O((\log X)^2/X)$, and $O(X^{-1})$. Equation (25.1) now gives the finite limit. □

For example, $a = 1/4$ gives $D_a(X) \sim X^{1/2}/9$, $a = 1/2$ gives $(\log X)/2$, and $a = 1$ gives a finite limit. Appending the same complementary circular arc makes all images the same smooth closed unit circle. Its winding number, knot type, total curvature, and any defined reparametrization-invariant curve energy are identical. This closure does not change the deficit measured on the original arc. These are not counterexamples about primes: they disprove an inference based only on shared trace geometry, with the arithmetic measure omitted.

25.2 Unitary classification of Hardy chart changes

Use the source manuscript's notation

$$s_\alpha(z) = \tfrac{1}{2} + \alpha\frac{1+z}{1-z}, \quad G_\alpha(z) = \frac{\sqrt{2\alpha}}{1-z}F(s_\alpha(z)), \quad \alpha > \tfrac{1}{2}.$$

Proposition 25.2. *For $b \in \mathbb{D}$, set*

$$\psi_b(z) = \frac{z+b}{1+\bar{b}z}, \qquad (U_bG)(z) = \frac{\sqrt{1-|b|^2}}{1+\bar{b}z}G(\psi_b(z)).$$

This is a unitary operator on $H^2(\mathbb{D})$, with inverse U_{-b}. For $\alpha, \beta > 1/2$ and $b = (\beta-\alpha)/(\beta+\alpha)$, $G_\beta = U_bG_\alpha$ as continued germs, and globally whenever either extends to the whole disk. No interior logarithmic branch point is removed by this change of chart.

Proof. On $|z| = 1$ one has $|\psi_b'(z)| = (1-|b|^2)/|1+\bar{b}z|^2$. Changing angular variable gives $\|U_bG\|_{H^2} = \|G\|_{H^2}$ first for polynomials, then by density for every H^2 function. Direct substitution gives inverse U_{-b}. For the displayed real b,

$$\alpha\frac{1+\psi_b(z)}{1-\psi_b(z)} = \beta\frac{1+z}{1-z}, \qquad \frac{\sqrt{1-b^2}\sqrt{2\alpha}}{(1+bz)(1-\psi_b(z))} = \frac{\sqrt{2\beta}}{1-z}.$$

These identities prove the scale relation, initially via the real continuation with $s > 1$, and then by analytic continuation. A biholomorphism has nonzero derivative, and the prefactor is nonzero throughout the disk. They transport an interior logarithmic branch point to another interior point rather than canceling it. □

Conformal coordinates can improve numerical conditioning, but not change whether the Hardy norm is finite. Moreover, local uniform convergence is insufficient for boundary-energy control: $1 + z + \cdots + z^n$ converges locally to $(1-z)^{-1}$ in the disk, whereas its squared H^2 norm is $n+1$ and the limit is not in H^2.

25.3 Finite winding certificates

Let $H(s) = (s-1)\zeta(s)$, with removable value $H(1) = 1$, and let

$$H_M(s) = (s-1)\sum_{n=1}^{M} n^{-s} + M^{1-s} \qquad (M \in \mathbb{N}).$$

Euler summation, initially for $\Re s > 1$ and then continued, gives for $\sigma = \Re s > 0$

$$H(s) - H_M(s) = -s(s-1)\int_M^\infty \{x\}x^{-s-1}\,dx, \qquad |H - H_M| \le \frac{|s(s-1)|}{\sigma}M^{-\sigma}. \tag{25.2}$$

The integral converges locally uniformly absolutely in this half-plane.

Theorem 25.3 (A finite zero-count certificate)**.** *Let Γ be a positively oriented simple polygon with its interior in $\Re s > 1/2$. Subdivide its edges at s_j with maximum segment length Δ. Choose*

$$\sigma_0 = \min_\Gamma \Re s > \tfrac{1}{2}, \quad A \ge \max_\Gamma |s-1|, \quad C \ge \max_\Gamma |s(s-1)|,$$

$$B_M = \frac{C}{\sigma_0}M^{-\sigma_0}, \qquad L_M = \sum_{n=1}^{M}(1 + A\log n)n^{-\sigma_0} + (\log M)M^{1-\sigma_0}.$$

If $\min_j |H_M(s_j)| > B_M + L_M\Delta$, the number of zeta zeros inside Γ, counted with multiplicity, equals the winding number about 0 of the polygon with vertices $H_M(s_j)$ in their original order. Winding number 0 certifies the entire enclosed region zero-free.

Proof. Differentiation gives $|H_M'| \le L_M$ along every boundary segment. Hence its image starting at s_j stays within $L_M\Delta$ of $H_M(s_j)$, and $|H_M| > B_M$ on the entire boundary. Equation (25.2) and Rouché's theorem give equal zero counts for H and H_M. Each image arc and its straight chord lie in the same disk of radius $L_M\Delta$ centered at $H_M(s_j)$; this disk avoids 0. Linear interpolation therefore replaces every image arc by its chord without changing winding number. The argument principle finishes the proof. The removable value $H(1) = 1$ means there is no extraneous zero at 1. □

These are finite, explicit certification targets. An implementation must use rigorous enclosures of vertex values and winding decisions; a floating-point plot is insufficient. No unreported high-height computation is claimed here.

There is an exact elementary application: $H_1(s) = s$, and (25.2) proves $H(s) \ne 0$ whenever

$$\sigma > \tfrac{1}{2}, \qquad |s-1| < \sigma, \quad \text{equivalently} \quad (\Im s)^2 < 2\sigma - 1.$$

In particular the entire rectangle $3/4 \le \Re s \le 5/4$, $|\Im s| \le 1/4$ is certified zero-free. This is a genuine finite-domain application of method 65, albeit an elementary region already covered by much stronger known results. Uniform control at every height is not proved.

25.4 Finite covers do not eliminate the zero obstruction

The source manuscript's analytic identity is

$$(s-1)F(s) + R_0(s) + J_0(s) = \log H(s), \qquad R_0(s) = \sum_{k\ge 2} P(ks)/k.$$

Here R_0 and J_0 are single-valued and holomorphic in $\Re s > 1/2$. Continuation of the logarithm around a loop avoiding zeros changes its value by $2\pi i$ times its zero-count winding. At a fixed base point $s_* \ne 1$, the corresponding increment of F is this number divided by $s_* - 1$. A

simply connected subdomain avoiding $s = 1$ and containing a defining base point $s_* > 1$ admits a holomorphic continuation of F exactly when H is zero-free there. If the domain includes 1, removability also requires the normalized branch $\log H(1) = 0$; this is automatic when the domain contains the real segment joining 1 to s_*. For a multiply connected zero-free domain, a single-valued logarithm additionally requires zero winding on every closed loop.

Proposition 25.4. *If ρ is a zero of multiplicity $m \geq 1$ with $\Re\rho > 1/2$, no finite branched holomorphic cover of a neighborhood of ρ makes the pullback of F single-valued and holomorphic at a point above ρ.*

Proof. Locally $F(s) = m\log(s-\rho)/(s-1) + h(s)$ with h holomorphic. A cover of local degree $k \geq 1$ has $s = \rho + ct^k(1+O(t))$, $c \neq 0$. Pullback replaces $\log(s-\rho)$ by $k\log t + \log c + \log(1+O(t))$. Its leading logarithmic coefficient $km/(\rho-1)$ is nonzero. Equivalently one circuit changes the value by $2\pi ikm/(s(t)-1) \neq 0$. The branch singularity persists. □

The universal cover of a punctured domain can support a logarithm, but this is not a holomorphic function on the original disk. Finite algebraic covering theory does not remove this analytic obstruction.

Part III
Centered prime variance and the correlation target

26 The unchanged target and the result

Let p_n be the nth prime, let log denote the natural logarithm, and put

$$S(x) = \sum_{p \le x} \frac{1}{p}, \qquad E(x) = S(x) - \log\log x - B_{\mathbb{P}}, \tag{26.1}$$

$$U_n = e^{2\pi i(S(p_n) - \log\log p_n)}, \qquad w_n = p_{n+1} - p_{n-1} \quad (n \ge 2). \tag{26.2}$$

Here $B_{\mathbb{P}} = \lim_{x\to\infty}(S(x) - \log\log x)$ is the prime Mertens constant. Define

$$W_N = \sum_{n=2}^{N} w_n = p_N + p_{N+1} - 5, \qquad M_N = \sum_{n=2}^{N} w_n U_n, \qquad \boxed{D_N = W_N - |M_N|.} \tag{26.3}$$

The triangle inequality gives $0 \le D_N \le W_N$ and $D_{N+1} \ge D_N$. The original target is the existence of a finite constant K such that $D_N \le K$ for every $N \ge 2$.

Write

$$\Theta = \sup\{\Re\rho : \zeta(\rho) = 0,\ 0 < \Re\rho < 1\}. \tag{26.4}$$

The functional equation and the classical zero-free boundary give $1/2 \le \Theta \le 1$ and RH $\iff \Theta = 1/2$. The main result of this addendum is

$$\boxed{\lim_{N\to\infty} \frac{\log(1 + D_N)}{\log p_N} = 2\Theta - 1.} \tag{26.5}$$

In particular, when $\Theta > 1/2$, for every $\varepsilon > 0$,

$$D_N \gg_\varepsilon p_N^{2\Theta - 1 - \varepsilon} \quad \text{for every sufficiently large } N. \tag{26.6}$$

This is stronger than a statement about the upper limit or about a selected subsequence. It also applies to the best center in the definition of D_N, rather than only to a fixed-center energy. Here $A \ll B$ or $A \gg B$ means a bound by a positive constant multiple in the indicated direction; constants may depend on displayed parameters. They are not claimed to be effective.

Exact status. The proofs below establish (26.5) and its consequences. They do not establish a new zero-free half-plane, boundedness of D_N, or the Riemann hypothesis. The main analytic lower bound is an imported 1996 theorem, explicitly credited below. This note strengthens the growth statement in the preceding project draft; it does not assert that this transfer or its consequences have no antecedents in the literature. The original constants, phases, and weights in (26.2)–(26.3) are retained throughout.

27 A lower bound that survives every bounded recentering

For E and Θ as defined above, let

$$\mathrm{Li}_2(x) = \int_2^x \frac{dt}{\log t}, \qquad R_\pi(x) = \pi(x) - \mathrm{Li}_2(x),$$

$$b_0 = \log\log 2 + B_{\mathbb{P}}, \qquad C_\pi = R_\pi(2) - 2E(2) = 2b_0.$$

The normalization of the logarithmic integral differs from the usual principal-value normalization by a constant only. Such a constant does not affect the large-interval lower bounds below.

Established input. Stechkin and Popov [25, Theorem 12, equation (15.2)] prove that, if $\Theta > 1/2$, then for every $\delta > 0$,

$$\int_X^{2X} |R_\pi(t)|\,dt \gg_\delta X^{1+\Theta-\delta} \qquad (X \geq X_\delta).$$

This is a bound for every sufficiently large X, not an Omega statement along a subsequence. Their theorem includes $\Theta = 1$. The constants and the starting point are generally ineffective. If $\Theta > 1/2$ is attained by the real part of a zero, equation (16.2) gives the stronger lower bound

$$\int_X^{2X} |R_\pi(t)|\,dt \gg \frac{X^{1+\Theta}}{\log X}.$$

These are imported theorems, not conclusions newly proved here.

Lemma 27.1 (Exact inverse partial summation)**.** *For every real c and every $x \geq 2$,*

$$R_\pi(x) = x(E(x) - c) - \int_2^x (E(t) - c)\,dt + C_\pi + 2c.$$

Consequently, for $|c| \leq C$ and $X \geq 2$,

$$\int_X^{2X} |R_\pi(t)|\,dt \leq 3X \int_2^{2X} |E(t) - c|\,dt + X(|C_\pi| + 2C).$$

Proof. Abel summation gives

$$xS(x) - \int_2^x S(t)\,dt = \pi(x), \qquad S(x) = \sum_{p \leq x} p^{-1}.$$

Integration by parts gives

$$x \log\log x - \int_2^x \log\log t\,dt = 2\log\log 2 + \mathrm{Li}_2(x).$$

Subtracting proves the formula at $c = 0$. Replacing E by $E - c$ changes $xE(x) - \int_2^x E$ by $-2c$, yielding the claimed formula for every c. Integrating its absolute value over $[X, 2X]$, using $t \leq 2X$, proves the inequality. □

Theorem 27.2 (Uniform centered lower bounds)**.** *Suppose $\Theta > 1/2$. For every $C > 0$ and every $\varepsilon > 0$, there are positive constants c_1, c_2 and X_0 such that, for all $Y \geq X_0$ and every $|c| \leq C$,*

$$\int_2^Y |E(t) - c|\,dt \geq c_1 Y^{\Theta-\varepsilon}, \qquad \int_2^Y |E(t) - c|^2\,dt \geq c_2 Y^{2\Theta-1-\varepsilon}.$$

The constants do not depend on c, even when c is chosen separately for each Y. If $\Theta > 1/2$ is attained, the bounds strengthen to

$$\int_2^Y |E(t) - c|\,dt \gg_C \frac{Y^\Theta}{\log Y}, \qquad \int_2^Y |E(t) - c|^2\,dt \gg_C \frac{Y^{2\Theta-1}}{(\log Y)^2}.$$

Proof. Apply the established lower bound with $0 < \delta < \min(\Theta, \varepsilon/2)$ to Lemma 27.1. The term $X(|C_\pi| + 2C)$ is negligible compared with $X^{1+\Theta-\delta}$, uniformly for $|c| \leq C$. Set $Y = 2X$. This proves the first assertion, with a smaller epsilon if necessary. Cauchy–Schwarz gives

$$\int_2^Y |E - c|^2\,dt \geq \frac{1}{Y-2}\left(\int_2^Y |E - c|\,dt\right)^2,$$

and proves the second assertion after renaming epsilon. The attained case uses equation (16.2) of Stechkin–Popov in the identical argument. □

Relation to Zhao's Remark 4. Taking $c = 0$ supplies the all-Y first- and second-moment lower bounds discussed, but not proved, in Zhao's Remark 4 [1]. The uniformity in the moving center is needed for the original circle deficit. Merely proving the uncentered lower bound does not by itself justify passage to that deficit.

28 Transfer to the original discrete circle deficit

Use the original definitions (26.2)–(26.3), and write $a = e^{2\pi i B_{\mathbb{P}}}$ and $E_n = E(p_n)$. Thus $U_n = ae^{2\pi i E_n}$. The original phases and weights are unchanged.

Lemma 28.1 (Sampling with a center-independent error)**.** *For every fixed $C > 0$, uniformly for $|c| \le C$ and $N \ge 3$,*

$$\int_2^{p_N} |E(x) - c|^2 \, dx \le 2 \sum_{n=2}^{N} w_n |E_n - c|^2 + O_C(1).$$

Proof. On each gap $[p_n, p_{n+1})$, the function $E(x) - c$ decreases between $E_n - c$ and $E_{n+1} - c - 1/p_{n+1}$. Hence, if $h_n = p_{n+1} - p_n$,

$$\int_{p_n}^{p_{n+1}} |E(x) - c|^2 \, dx \le h_n \left(|E_n - c|^2 + 2|E_{n+1} - c|^2 + \frac{2}{p_{n+1}^2} \right).$$

Sum from $n = 2$ to $N - 1$, use $w_n = h_{n-1} + h_n$, and note that

$$\sum_{n \ge 2} \frac{h_n}{p_{n+1}^2} \le \sum_{n \ge 2} \int_{p_n}^{p_{n+1}} \frac{dt}{t^2} = \frac{1}{3}.$$

The remaining interval $[2, 3]$ contributes $O_C(1)$. No estimate for prime-gap sizes is needed. □

Theorem 28.2 (Every-scale lower bound for the original deficit)**.** *If $\Theta > 1/2$, then for every $\varepsilon > 0$,*

$$D_N \gg_\varepsilon p_N^{2\Theta - 1 - \varepsilon} \qquad \text{for every sufficiently large } N.$$

If $\Theta > 1/2$ is attained, then

$$D_N \gg \frac{p_N^{2\Theta - 1}}{(\log p_N)^2}.$$

Proof. Mertens' theorem gives $U_n \to a$; positive weighted averaging gives $M_N / W_N \to a$. Therefore, for large N,

$$b_N = \frac{M_N}{|M_N|} = ae^{2\pi i c_N}$$

with real $c_N \to 0$, choosing the argument near zero. Exactly,

$$\sum_{n=2}^{N} w_n |U_n - b_N|^2 = 2D_N.$$

Fix n_0 so that $|E_n| < 1/8$ for $n \ge n_0$, and take N large enough that $|c_N| < 1/8$. Since $2|\sin(\pi t)| \ge 4|t|$ for $|t| \le 1/2$,

$$|U_n - b_N| \ge 4|E_n - c_N| \qquad (n_0 \le n \le N).$$

The earlier indices contribute a bounded amount, uniformly in N. Consequently Lemma 28.1 yields

$$\int_2^{p_N} |E(x) - c_N|^2 \, dx \le \frac{1}{4} D_N + O(1).$$

The uniformity in c in Theorem 27.2 now gives the result. An additive $O(1)$ is absorbed because one may first choose $\varepsilon < 2\Theta - 1$; larger epsilon assertions follow by weakening the resulting bound. □

29 The RH endpoint without a prime-gap hypothesis

The following proof is included to make the endpoint used in the corollaries independent of the earlier manuscripts.

Proposition 29.1 (RH gives bounded original deficit)**.** *Assume RH. Then*

$$\int_2^\infty E(x)^2\,dx<\infty, \qquad \sum_{n=2}^{\infty} w_n E_n^2<\infty, \qquad \sup_N D_N<\infty.$$

Proof. Write

$$\theta(x)=\sum_{p\le x}\log p, \qquad \psi(x)=\sum_{p^k\le x}\log p, \qquad R(x)=\theta(x)-x.$$

Cramér's classical RH mean-square theorem, also stated and sharpened by Brent, Platt, and Trudgian [6, equation (1), Theorem 1], gives

$$\int_X^{2X}(\psi(t)-t)^2\,dt\ll X^2.$$

The elementary bound $\psi(x)-\theta(x)\ll\sqrt{x}$ therefore implies $\int_X^{2X}R(t)^2\,dt\ll X^2$. Summing over dyadic intervals yields

$$f(x):=\frac{R(x)}{x\log x}\in L^2([2,\infty)), \qquad \int_{2^k}^{2^{k+1}}|f(x)|^2\,dx\ll k^{-2}.$$

Exact partial summation, with the Mertens constant fixed by $E(x)\to 0$, gives

$$E(x)=f(x)-\int_x^\infty f(t)\left(1+\frac{1}{\log t}\right)\frac{dt}{t}.$$

The dual Hardy inequality

$$\int_2^\infty\left|\int_x^\infty\frac{g(t)}{t}\,dt\right|^2dx\le 4\int_2^\infty|g(t)|^2dt$$

then proves $E\in L^2$. For completeness, the inequality follows for truncated integrals by integration by parts and Cauchy–Schwarz; the lower-boundary term is nonpositive. Passing to the limit gives the stated inequality.

The transfer from the integral to the actual prime endpoints has an elementary proof which does not require RH or a prime-gap moment estimate. On a gap $[p,q)=[p_n,p_{n+1})$, put

$$h=q-p, \qquad v=\log\log x-\log\log p, \qquad L=\log\log q-\log\log p,$$
$$A=E(p), \qquad B=E(q)-1/q=A-L.$$

Then $E(x)=A-v$ and $dx/dv=x\log x$. Bertrand's inequality implies

$$\frac{\max_{[p,q]}x\log x}{\min_{[p,q]}x\log x}\le 4,$$

so that

$$\int_p^q E(x)^2\,dx\ge\frac{h}{4L}\int_0^L(A-v)^2\,dv=\frac{h}{12}(A^2+AB+B^2)\ge\frac{h}{24}(A^2+B^2).$$

Since $E(q)=B+1/q$, it follows that

$$h(E(p)^2+E(q)^2)\le 48\int_p^q E(x)^2\,dx+\frac{2h}{q^2}.$$

Summing over consecutive gaps, using $\sum_{n\geq 2} h_n/p_{n+1}^2 \leq 1/3$, gives

$$\sum_{n=2}^{\infty} w_n E_n^2 \leq O(1) + 48\int_3^{\infty} E(x)^2\,dx < \infty.$$

Finally, choosing the fixed center a gives

$$D_N \leq \frac{1}{2}\sum_{n=2}^{N} w_n|U_n - a|^2 \leq 2\pi^2 \sum_{n=2}^{N} w_n E_n^2.$$

This proves the proposition without any additional prime-gap estimate. The RH assumption enters through Cramér's mean-square bound; it has not been removed. □

30 An actual growth limit and sparse tests

Corollary 30.1 (A full growth limit)**.** *For the unchanged deficit,*

$$\lim_{N\to\infty} \frac{\log(1+D_N)}{\log p_N} = 2\Theta - 1.$$

In particular, the upper limit and the lower limit agree.

Proof. For $1/2 < \Theta < 1$, the classical bound $R_\pi(x) = O_\Theta(x^\Theta/\log x)$, recorded in [1, equation (24)], implies by partial summation

$$E(x) = \frac{R_\pi(x)}{x} - \int_x^{\infty} \frac{R_\pi(t)}{t^2}\,dt = O_\Theta(x^{\Theta-1}/\log x).$$

The displayed identity follows by taking the limit at infinity in Abel summation; the additive normalization constant in Li_2 cancels from its right-hand side. This estimate and $D_N \leq 2\pi^2\sum_{n=2}^{N} w_n E_n^2$ give

$$D_N = O_\Theta\left(\frac{p_N^{2\Theta-1}}{(\log p_N)^2}\right).$$

To justify this weighted summation, apply Bertrand's inequality $p_{n+1} \leq 2p_n$ to the decreasing function $x^{2\Theta-2}/(\log x)^2$. The discrete weighted sum is bounded by a constant multiple of its integral up to p_{N+1}. That integral is $O_\Theta(p_N^{2\Theta-1}/(\log p_N)^2)$. For $\Theta = 1$, Mertens' unconditional estimate $E(x) = O(1/\log x)$ gives the same upper bound. Theorem 28.2 gives the matching lower exponent. For $\Theta = 1/2$, functional-equation symmetry gives RH, and Proposition 29.1 gives bounded D_N. □

Corollary 30.2 (Attained right boundary)**.** *If $\Theta > 1/2$ is the real part of a nontrivial zero, then*

$$D_N \asymp \frac{p_N^{2\Theta-1}}{(\log p_N)^2}.$$

The comparison constants are not claimed to be effective.

Corollary 30.3 (Subsequence upper bounds suffice)**.** *The Riemann hypothesis is equivalent to each of the following:*

1. *$D_N \leq \log p_N$ for infinitely many N.*
2. *There is an unbounded sequence N_j with $\log(1+D_{N_j})/\log p_{N_j} \to 0$.*
3. *For every $\varepsilon > 0$, there are arbitrarily large N with $D_N \leq p_N^{\varepsilon}$.*

Proof. If RH holds, Proposition 29.1 makes D_N bounded, so all three conditions hold. If $\Theta > 1/2$, Theorem 28.2 gives an eventual positive-power lower bound, excluding the first condition. Corollary 30.1 excludes the second. For the third, choose an exponent strictly between zero and $2\Theta - 1$; the all-large-N lower bound with a smaller loss excludes infinitely many such indices. □

What remains unproved. These results do not establish even the sparse upper bound $D_N \le \log p_N$ infinitely often, nor the requested uniform upper bound $D_N \le K$ for all N. They show that the growth caused by an off-critical zero cannot be hidden at arbitrarily late indices by choosing the best center. The decisive lower estimate is inherited from Stechkin and Popov and transferred by the exact inverse Abel identity; it is not a newly proved zero-free region.

31 A finite-scale inequality and the next sufficient target

The preceding proof gives a useful quantitative inequality without any hypothesis about zeros. For the real numbers $c_N \to 0$ constructed in the proof of Theorem 28.2,

$$\int_2^{p_N} |E(x) - c_N|^2 \, dx \le \frac{1}{4} D_N + C \tag{31.1}$$

with a fixed constant C. Put $X = p_N/2$ in Lemma 27.1, apply Cauchy–Schwarz, and use (31.1). Increasing the constant to cover the finitely many small admissible indices gives, for $N \ge 3$,

$$\boxed{\int_{p_N/2}^{p_N} |\pi(t) - \mathrm{Li}_2(t)| \, dt \ll p_N^{3/2} \sqrt{1 + D_N}.} \tag{31.2}$$

Thus a small deficit forces a small prime-counting error averaged over the most recent multiplicative interval. If $\Theta > 1/2$, the Stechkin–Popov theorem prohibits this smallness beyond a positive power scale, at every sufficiently large index. This is the precise arithmetic information that rules out arbitrarily late dips in the growth exponent of D_N.

Corollary 30.3 gives the following sufficient target for the original problem:

$$\boxed{\forall N_0 \; \exists N \ge N_0 : \quad D_N \le \log p_N.} \tag{31.3}$$

If (31.3) were proved unconditionally, the preceding theorems would imply RH and then the original bound $\sup_N D_N < \infty$. Conversely, RH implies (31.3) for every sufficiently large N. No assertion of (31.3) is proved here. Rewriting the target with fewer upper-bound obligations does not itself supply the missing upper estimate.

More generally, the same equivalence holds with $\log p_N$ replaced by $h(p_N)$ whenever $h(x) \to \infty$ and $\log h(x)/\log x \to 0$. Indeed boundedness of D_N implies that upper bound eventually, whereas any positive power of p_N ultimately exceeds $h(p_N)$. The specific choice in (31.3) keeps the next target concrete.

Why finite computation does not settle it. The lower-bound constants and starting points depend on information about possible zeros to the right of $1/2$; an effective universal starting point is not produced. Consequently a finite computed run of D_N below $\log p_N$ cannot verify (31.3) or exclude an eventual positive growth exponent. Nothing in the proof turns an observed finite pattern into a claim about infinitely many indices.

32 What the reciprocal recurrence alone cannot force

Theorem 32.1 (An integer sequence with a logarithmically divergent deficit)**.** *Fix $c > 0$ and any finite prefix of the ordinary primes. There exists a strictly increasing sequence of positive integers $q_1, q_2, \ldots$ that agrees with this prefix, has $q_1 = 2, q_2 = 3$, and satisfies*

$$q_n \sim n \log n, \qquad q_{n+1} - q_n \sim \log q_n.$$

With its exact reciprocal sum and its own Mertens constant,

$$S_q(x) = \sum_{q_n \le x} \frac{1}{q_n}, \quad B_q = \lim_{x\to\infty} \{S_q(x) - \log\log x\}, \quad E_q(x) = S_q(x) - \log\log x - B_q,$$

one has

$$E_q(x) = \frac{c}{\sqrt{x}} + O(x^{-1}).$$

Define, with precisely the same form of phase and weights as in the prime-circle construction,

$$U_n^{(q)} = e^{2\pi i(S_q(q_n) - \log\log q_n)}, \qquad w_n^{(q)} = q_{n+1} - q_{n-1},$$
$$D_N^{(q)} = \sum_{n=2}^{N} w_n^{(q)} - \left| \sum_{n=2}^{N} w_n^{(q)} U_n^{(q)} \right|.$$

Then

$$\boxed{D_N^{(q)} = 4\pi^2 c^2 \log q_N + O(1) \longrightarrow \infty.}$$

Nevertheless every higher centered moment converges:

$$\sum_{n\ge 2} w_n^{(q)} |U_n^{(q)} - e^{2\pi i B_q}|^r < \infty \qquad (r > 2).$$

The weighted nearest-neighbor turning energy also converges, and its tail is $O(X^{-1})$ when restricted to $q_n \ge X$.

Proof. Choose an integer T beyond the prescribed prefix and large enough that

$$a(x) = \frac{1}{\log x} - \frac{c}{2\sqrt{x}}$$

is positive and less than $1/2$ on $[T, \infty)$. Retain every ordinary prime at most T. Put

$$F(x) = \int_T^x a(t)\, dt = \mathrm{Li}(x) - \mathrm{Li}(T) - c(\sqrt{x} - \sqrt{T}).$$

For each integer $j \ge 1$, let $x_j > T$ be the unique solution of $F(x_j) = j$, and append $\lceil x_j \rceil$ to the retained prefix. Since $a < 1/2$, consecutive x_j are separated by more than two, so the appended integers are strictly increasing and exceed T. The resulting counting function $Q(x)$ satisfies, for $x \ge T$,

$$Q(x) = \pi(T) + \lfloor F(\lfloor x \rfloor) \rfloor.$$

Writing $A(x) = \pi(T) + F(x)$, we have

$$R(x) := Q(x) - A(x) = O(1), \qquad R(T) = 0.$$

In particular,

$$Q(x) = \mathrm{Li}(x) - c\sqrt{x} + O(1),$$

where the fixed T-dependent constant is absorbed in the error. This proves $q_n \sim n \log n$. The mean-value theorem applied to $F(x_{j+1}) - F(x_j) = 1$, together with $a(x) \sim 1/\log x$, gives $x_{j+1} - x_j \sim \log x_j$; rounding changes a gap by at most one. Thus $q_{n+1} - q_n \sim \log q_n$.

Stieltjes partial summation gives

$$\int_{(T,x]} \frac{1}{t}\, dR(t) = \frac{R(x)}{x} + \int_T^x \frac{R(t)}{t^2}\, dt.$$

On the other hand,

$$\int_T^x \frac{a(t)}{t}\,dt = \log\log x - \log\log T + \frac{c}{\sqrt{x}} - \frac{c}{\sqrt{T}}.$$

The boundedness of R proves that B_q exists and yields exactly

$$E_q(x) = \frac{c}{\sqrt{x}} + \frac{R(x)}{x} - \int_x^\infty \frac{R(t)}{t^2}\,dt = \frac{c}{\sqrt{x}} + O(x^{-1}).$$

The reciprocal recurrence is therefore not being replaced by arbitrarily selected phases: it holds by the construction itself.

We record the elementary weighted summation estimates required for the deficit. If $h_n = q_{n+1} - q_n = O(\log q_n)$, then, for every $s > 0$,

$$\sum_{n=2}^{N} (h_{n-1} + h_n) q_n^{-s} = 2\int_{q_2}^{q_N} t^{-s}\,dt + C_s + o(1)$$

for a finite constant C_s. To see this, express the left side as the trapezoidal sum over the interior gaps plus two boundary terms. On a gap $[q_n, q_{n+1}]$ the difference from twice the integral is $O_s(h_n^2 q_n^{-s-1})$. This is summable, because $q_n \sim n\log n$ and $h_n = O(\log q_n)$; the final boundary term is $O(q_N^{-s}\log q_N) = o(1)$. In particular,

$$\sum_{n=2}^{N} w_n^{(q)} q_n^{-1} = 2\log q_N + O(1), \qquad \sum_{n=2}^{N} w_n^{(q)} q_n^{-1/2} = 4\sqrt{q_N} + O(1),$$

and the corresponding infinite sums converge whenever $s > 1$.

Rotate all the vectors by $e^{-2\pi i B_q}$, which leaves the deficit unchanged. Set $k = 2\pi c$ and $\theta_n = 2\pi E_q(q_n) = k q_n^{-1/2} + O(q_n^{-1})$. The weighted sum of θ_n^4 is finite and the weighted sum of $|\theta_n|$ through q_N is $O(\sqrt{q_N})$. Moreover

$$\sum_{n=2}^{N} w_n^{(q)} \theta_n^2 = 2k^2 \log q_N + O(1).$$

Thus, if $W_N^{(q)} = q_N + q_{N+1} - 5 \sim 2q_N$,

$$\Re \sum_{n=2}^{N} w_n^{(q)} e^{i\theta_n} = W_N^{(q)} - k^2 \log q_N + O(1), \qquad \left|\Im \sum_{n=2}^{N} w_n^{(q)} e^{i\theta_n}\right| = O(\sqrt{q_N}).$$

The real part is positive and asymptotic to $W_N^{(q)}$. For a complex number z with positive real part, $|z| - \Re z = (\Im z)^2/(|z| + \Re z)$. Consequently the difference between the modulus and the displayed real part is $O(1)$, giving $D_N^{(q)} = k^2 \log q_N + O(1)$ as claimed.

Since $|e^{i\theta_n} - 1| \asymp q_n^{-1/2}$ on a tail, each centered moment of order $r > 2$ converges by the preceding summation estimate. Finally, the exact phase increment in turns is

$$\frac{1}{q_{n+1}} - \log\log q_{n+1} + \log\log q_n.$$

Because $h_n = O(\log q_n)$, its absolute value is $O(q_n^{-1})$. Thus the weighted squared turning is bounded by a constant times $\sum w_n^{(q)} q_n^{-2}$, and its tail is $O(X^{-1})$. □

Remark 32.2 (Exact scope of the obstruction). The numbers q_n following the fixed prefix are not asserted to be prime. Therefore this is not a counterexample to the prime-circle criterion or to RH. It preserves the reciprocal recurrence, logarithmic centering, integer denominators, a prescribed

finite prime prefix, prime-number-theorem scale, and the exact form of the Voronoi weights. It does not preserve the Euler product and analytic identities of the ordinary Riemann zeta function.

The example therefore excludes only an inference from the recurrence and the listed geometric properties alone. For example, if $0 < c < 1/(2\pi)$, then $D_N^{(q)} \le \log q_N$ eventually, despite $D_N^{(q)} \to \infty$. This does not contradict Corollary 30.3: that corollary is specifically about the ordinary primes and uses their connection with the zeros of ζ.

33 An exact identity for the optimal center

Put

$$C_\pi = 2(\log\log 2 + B_{\mathbb{P}}), \qquad H(x) = \int_2^x E(t)\,dt.$$

Abel summation, including the endpoint at 2, gives

$$P(x) - C_\pi = xE(x) - H(x). \tag{33.1}$$

For example, it follows directly by subtracting $x\log\log x - \int_2^x \log\log t\,dt = 2\log\log 2 + \mathrm{Li}_2(x)$ from $xS(x) - \int_2^x S(t)\,dt = \pi(x)$.

Theorem 33.1 (Finite variance identity)**.** *For every $X > 2$,*

$$V(X) = \min_{c\in\mathbb{R}} \int_2^X (E(t)-c)^2\,dt \tag{33.2}$$

$$= \int_2^X \frac{P(t)^2}{t^2}\,dt - \frac{2X}{X-2}\left(\int_2^X \frac{P(t)}{t^2}\,dt\right)^2. \tag{33.3}$$

The equality is exact at every finite X and requires no hypothesis about zeros of ζ.

Proof. Fix $c \in \mathbb{R}$ and write $F(x) = H(x) - c(x-2)$. Then $F' = E - c$ almost everywhere and

$$P(x) - C_\pi - 2c = xF'(x) - F(x).$$

Since

$$\left(F' - \frac{F}{x}\right)^2 = (F')^2 - \left(\frac{F^2}{x}\right)',$$

integration gives the exact identity

$$\int_2^X (E-c)^2\,dx = \int_2^X \frac{(P - C_\pi - 2c)^2}{x^2}\,dx + \frac{(H(X) - c(X-2))^2}{X}. \tag{33.4}$$

The primitive F is absolutely continuous, so the jumps of E introduce no omitted boundary terms.

The left-hand side is minimized at $\bar{E}_X = H(X)/(X-2)$, where the last term vanishes. Moreover,

$$\int_2^X \frac{P(t) - C_\pi}{t^2}\,dt = \frac{H(X)}{X}, \qquad \int_2^X \frac{dt}{t^2} = \frac{X-2}{2X}.$$

Consequently the weighted mean of P is $C_\pi + 2\bar{E}_X$. It is exactly the minimizing constant in the first integral on the right. This proves (33.2). The usual weighted least-squares formula then gives (33.3). □

Corollary 33.2 (The amount removed by centering is bounded)**.** *Let $b_0 = \log\log 2 + B_{\mathbb{P}}$ and $Q(X) = \int_2^X P(t)^2 t^{-2}\,dt$. Unconditionally,*

$$V(X) = Q(X) - 2b_0^2 + o(1) \qquad (X \to \infty). \tag{33.5}$$

Proof. Partial summation also gives

$$E(X) = \frac{P(X)}{X} + \int_2^X \frac{P(t)}{t^2}\,dt - b_0.$$

The ordinary prime number theorem and Mertens' theorem imply $P(X)/X \to 0$ and $E(X) \to 0$. Thus the integral on the right tends to b_0, and (33.3) proves the assertion. □

The definition of V is unchanged if $P = \pi - \mathrm{Li}_2$ is replaced by $\pi - \mathrm{li}$: the two versions differ by a constant, which the minimization absorbs. Neither the Mertens constant nor any future prime needs to be numerically evaluated to compute this finite-interval variance.

34 Summing the quadrature errors at the actual primes

We use two established unconditional inputs: the quantitative prime number theorem and Peck's mean-square prime-gap estimate,

$$P(x) \ll_A \frac{x}{(\log x)^A} \quad (A > 0), \qquad \sum_{p_n \le Y} h_n^2 \ll_\eta Y^{5/4+\eta}, \quad h_n = p_{n+1} - p_n. \tag{34.1}$$

For the latter, see Maynard's reproduction [26, Theorem 3.1], which explicitly credits Peck. These are imported results. In particular, no new prime-gap estimate is claimed. The prime number theorem and partial summation imply $E(x) = O((\log x)^{-2})$.

Lemma 34.1. *Let $X = p_{N+1}$, $c \in \mathbb{R}$, and define*

$$T_N(c) = \sum_{n=1}^{N} h_n\big((E(p_n) - c)^2 + (E(p_{n+1}) - c)^2\big).$$

There is a constant C such that, for every $N \ge 2$ and every real c,

$$\left| T_N(c) - 2\int_2^X (E(t) - c)^2\,dt \right| \le C(1 + |c|) + 2|c| \log(X/2). \tag{34.2}$$

In particular, the error is $O(1)$ along any choice $c = c_X$ for which $|c_X| \log X = O(1)$.

Proof. The prime number theorem and partial summation give $E(x) = O((\log x)^{-2})$, and E is bounded on $[2, \infty)$. On a single gap $[p, q]$, where q is the prime following p, put $h = q - p$, and extend the left-hand branch to its right endpoint:

$$F(t) = E(p) - c - \log\log t + \log\log p, \qquad p \le t \le q.$$

Except at the right endpoint this is $E(t) - c$. Its endpoint value is $F(q) = E(q) - c - 1/q$. For $G = F^2$,

$$G''(t) = \frac{2}{t^2 \log^2 t} + \frac{2F(t)(\log t + 1)}{t^2 \log^2 t}.$$

Thus the trapezoidal-rule error on this smooth branch is at most

$$\left| h\big(G(p) + G(q)\big) - 2\int_p^q G(t)\,dt \right| \ll (1 + |c|)\frac{h^3}{p^2 \log p}.$$

The replacement of $F(q)$ by the actual prime value adds exactly

$$h\left(\frac{2(E(q) - c)}{q} - \frac{1}{q^2}\right).$$

Its absolute value is at most

$$\frac{2h|E(q)|}{q}+\frac{2|c|h}{q}+\frac{h}{q^2}.$$

All terms except the one multiplied by $|c|$ are summable. Indeed, decreasing-integrand comparisons give

$$\sum_n \frac{h_n|E(p_{n+1})|}{p_{n+1}}<\infty, \qquad \sum_n \frac{h_n}{p_{n+1}^2}<\infty, \qquad \sum_{n=1}^{N}\frac{h_n}{p_{n+1}}\le\int_2^X\frac{dt}{t}=\log(X/2).$$

For the smooth error, monotonicity of finite-dimensional ℓ^r norms and Peck's estimate give

$$\sum_{Y\le p_n<2Y} h_n^3 \le \left(\sum_{p_n\le 2Y} h_n^2\right)^{3/2} \ll_\eta Y^{15/8+3\eta/2}.$$

Choosing $0<\eta<1/12$ and summing over dyadic Y proves

$$\sum_n \frac{h_n^3}{p_n^2\log p_n}<\infty.$$

Adding the estimates for the disjoint gaps proves (34.2). □

35 The direct upper bound

Theorem 35.1. *There is a finite constant C, independent of N, such that*

$$\boxed{D_N\le 4\pi^2V(p_{N+1})+C \qquad (N\ge 2).} \tag{35.1}$$

The primes, reciprocal phases, neighboring-prime weights, and deficit are those of (126.3).

Proof. For every real c, choose the unit vector $z=e^{2\pi i(B_{\mathbb{P}}+c)}$. The variational formula for the modulus and $1-\cos u\le u^2/2$ give

$$D_N\le W_N-\Re(\bar z M_N)\le 2\pi^2\sum_{n=2}^{N} w_n(E(p_n)-c)^2. \tag{35.2}$$

For $X=p_{N+1}$, the exact endpoint relation is

$$T_N(c)=\sum_{n=2}^{N} w_n(E(p_n)-c)^2+h_1(E(2)-c)^2+h_N(E(X)-c)^2.$$

Both exterior terms are nonnegative. Thus no unproved estimate on the final prime gap is needed when taking an upper bound.

Choose $c=\bar E_X=(X-2)^{-1}\int_2^X E(t)\,dt$. Splitting the integral at $\sqrt{X}$ and using boundedness of E on the first part and $E(t)=O((\log t)^{-2})$ on the second shows

$$\bar E_X=O((\log X)^{-2}), \qquad |\bar E_X|\log X=O(1).$$

Lemma 34.1 now gives $T_N(\bar E_X)=2V(X)+O(1)$. Combining this with (35.2) proves (35.1). □

The proof can retain the last endpoint term, giving

$$D_N\le 4\pi^2V(X)-2\pi^2h_N(E(X)-\bar E_X)^2+C, \qquad X=p_{N+1}. \tag{35.3}$$

No bound showing that this single endpoint correction controls the accumulated variance is obtained.

Where this enters the original target. For example, if one could prove, for one fixed $0 < \varepsilon < 1/(4\pi^2)$, that

$$V(p_{N+1}) \le \left(\frac{1}{4\pi^2} - \varepsilon\right) \log p_N$$

at infinitely many indices, then the constant C in (35.1) would be absorbed by $4\pi^2\varepsilon \log p_N$, proving (126.4). This observation specifies the estimate needed to use the upper bound; it does not prove that estimate or replace the original target.

36 Testing the available analytic inputs

36.1 An unconditional prime number theorem estimate

The classical Vinogradov–Korobov form of the error term, recorded for example in [25, §1, equation (2)], and partial summation imply that, for some fixed $a > 0$,

$$|P(x)| \ll xe^{-aL(x)}, \qquad L(x) = \frac{(\log x)^{3/5}}{(\log\log x)^{1/5}} \quad (x \text{ sufficiently large}). \tag{36.1}$$

Fixed logarithmic factors, if present in the input estimate, can be absorbed by decreasing a. Explicit versions of prime number theorem estimates are also given by Johnston and Yang [27]. We make no claim that the displayed bound is the strongest available in November 2026.

Theorem 35.1 and $V(X) \le Q(X)$ give

$$D_N \ll 1 + \int_{x_0}^{p_{N+1}} e^{-2aL(t)}\, dt.$$

Since $xL'(x) \to 0$ and $L(x) = o(\log x)$, differentiation of $xe^{-2aL(x)}$, or l'Hôpital's rule, gives

$$\int_{x_0}^{X} e^{-2aL(t)}\, dt \sim Xe^{-2aL(X)}.$$

Therefore this unconditional input yields

$$D_N \ll p_{N+1}e^{-2aL(p_{N+1})}. \tag{36.2}$$

The numerical coefficient a has not been optimized. The positive function on the right, divided by $\log p_N$, tends to infinity: its logarithm is $\log X - 2aL(X) - \log\log X + o(1) \to \infty$. Thus choosing a subsequence of this proved upper envelope does not establish (126.4). This is a statement about the available upper bound, not a lower estimate for the actual D_N.

Likewise, importing a fixed zero-free boundary $B > 1/2$ into the classical estimate $P(x) = O_B(x^B/\log x)$ would give a bound of order $p_N^{2B-1}/(\log p_N)^2$. For any fixed $B > 1/2$, its ratio to $\log p_N$ still tends to infinity. A fixed positive power improvement alone does not finish this method.

36.2 The tempting mean-square input assumes RH

The bound

$$\int_X^{2X} (\psi(t) - t)^2\, dt \ll X^2$$

is explicitly conditional on RH in [6, equation (1), Theorem 1]. The same paper's unconditional estimate is a lower bound. Using the upper estimate here without its hypothesis would make the attempted proof circular. We therefore do not use it in Theorems 33.1 and 35.1.

36.3 Averaging residue classes can remove the needed component

Let $\psi(x;q,a) = \sum_{n\le x,\ n\equiv a \pmod q} \Lambda(n)$ and $\psi_q(x) = \sum_{n\le x,\ (n,q)=1} \Lambda(n)$. An exact variance decomposition is

$$\sum_{(a,q)=1} \left|\psi(x;q,a) - \frac{x}{\varphi(q)}\right|^2 = \sum_{(a,q)=1} \left|\psi(x;q,a) - \frac{\psi_q(x)}{\varphi(q)}\right|^2 + \frac{|\psi_q(x)-x|^2}{\varphi(q)}. \tag{36.3}$$

The mixed term vanishes because the centered class errors sum to zero. At $q=1$, the first term on the right is identically zero and the last term is $|\psi(x)-x|^2$. Thus a theorem about dispersion around the actual residue-class mean does not control the total prime-counting error needed in this argument. A theorem including that total error must be used with its stated modulus range and all its remainder terms intact.

37 The unchanged target and its definitions

Let p_N be the Nth ordinary prime. All logarithms are natural. Set

$$P(x) = \pi(x) - \mathrm{Li}_2(x), \qquad \mathrm{Li}_2(x) = \int_2^x \frac{dt}{\log t}, \qquad E(x) = \sum_{p\le x} \frac{1}{p} - \log\log x - B_{\mathbb{P}},$$

where $B_{\mathbb{P}}$ is the prime Mertens constant, and define

$$V(X) = \min_{a\in\mathbb{R}} \int_2^X \frac{(P(t)-a)^2}{t^2}\, dt. \tag{37.1}$$

The sole open target is

$$\boxed{\forall N_0\ \exists N \ge N_0 : \quad V(p_{N+1}) \le \frac{\log p_N}{8\pi^2}.} \tag{T}$$

This is an unnormalized cumulative squared error. A small value of P at an endpoint, or a small derivative of V there, does not by itself bound the error already accumulated. The exact identity

$$V(X) = \min_{c\in\mathbb{R}} \int_2^X (E(t)-c)^2\, dt$$

is proved as the case $a=2$ of Lemma 38.1 below. Both P and E are right-continuous at primes; single endpoints do not affect their Lebesgue integrals.

Under RH, the established square-integrability estimate for E implies that V is bounded, so (T) holds eventually [1, Lemma 9]. Appendix 129 proves the converse directly from the sparse bound. Neither implication proves (T) without RH. The results below apply unconditionally unless a hypothesis is explicitly stated.

38 The local contribution can be completely bounded

38.1 The exact variance identity on every subinterval

For $2 \le a < b$, write

$$v(a,b) = \min_{c\in\mathbb{R}} \int_a^b (E(t)-c)^2\, dt, \qquad \mu_{a,b} = \frac{1}{b-a}\int_a^b E(t)\, dt,$$

$$\omega_{a,b} = \int_a^b \frac{dt}{t^2} = \frac{b-a}{ab}, \qquad r_{a,b} = \frac{1}{\omega_{a,b}} \int_a^b \frac{P(t)}{t^2}\, dt.$$

Lemma 38.1 (Exact local identity)**.** *For every $2 \le a < b$,*

$$v(a,b) = \min_{d\in\mathbb{R}} \int_a^b \frac{(P(t)-d)^2}{t^2}\,dt, \qquad r_{a,b} = P(a) - aE(a) + a\mu_{a,b}.$$

These identities include endpoints at primes and involve no assumption about the zeros of ζ.

Proof. Stieltjes partial summation gives, for $a \le t \le b$,

$$P(t) - C_a = tE(t) - H_a(t), \qquad C_a = P(a) - aE(a), \qquad H_a(t) = \int_a^t E(u)\,du.$$

Here P and E are right-continuous. For a real c, put $F(t) = H_a(t) - c(t-a)$. Then F is absolutely continuous, $F(a) = 0$, $F' = E - c$ almost everywhere, and

$$P(t) - C_a - ac = tF'(t) - F(t).$$

The elementary identity

$$\left(F' - \frac{F}{t}\right)^2 = (F')^2 - \left(\frac{F^2}{t}\right)'$$

therefore gives

$$\int_a^b (E-c)^2\,dt = \int_a^b \frac{(P - C_a - ac)^2}{t^2}\,dt + \frac{(H_a(b) - c(b-a))^2}{b}.$$

The left side is minimized at $c = \mu_{a,b}$, where the last term vanishes. Also

$$\int_a^b \frac{P(t) - C_a}{t^2}\,dt = \frac{H_a(b)}{b}.$$

Dividing by $\omega_{a,b}$ shows that the corresponding constant $C_a + a\mu_{a,b}$ is the weighted mean of P, and hence minimizes the first integral on the right. This proves the lemma. □

38.2 The unconditional sieve input and its exact scope

We use only the following classical consequence of the Brun–Titchmarsh theorem:

$$\pi(x+h) - \pi(x) < \frac{2h}{\log h} \qquad (x > 0,\ h > 1). \tag{38.1}$$

This is equation (1.12) of Montgomery and Vaughan's original paper [28, Theorem 2, p.121]. It is the modulus-one case of their Brun–Titchmarsh inequality, valid for every $h > 1$. In particular, no assertion that a prime exists between consecutive squares is needed.

Lemma 38.2 (A finite-interval upper bound)**.** *If $a \ge 2$, $h > 1$, and $h \le a^2$, then*

$$v(a, a+h) \le \frac{h^3}{a(a+h)\log^2 h}.$$

Proof. For $a \le u < v \le a + h$,

$$P(v) - P(u) = (\pi(v) - \pi(u)) - \int_u^v \frac{dt}{\log t}.$$

Both terms in parentheses are nonnegative. The first is at most $2h/\log h$ by (38.1), applied to the full interval, and the second is at most $h/\log a$. Since $h \le a^2$, $h/\log a \le 2h/\log h$. Therefore

$$\operatorname{osc}_{[a,a+h]} P \le \frac{2h}{\log h}.$$

For any positive finite measure of mass ω, a real function of range at most R has minimum centered quadratic integral at most $\omega R^2/4$: choose the midpoint of its range as a trial center. Apply this with measure $t^{-2}dt$, mass $h/(a(a+h))$, and then use Lemma 38.1. □

Theorem 38.3 (Summable local variance on square blocks)**.** *For $m \geq 2$, put*

$$I_m = [m^2, (m+1)^2], \qquad v_m = v(m^2, (m+1)^2).$$

Then, unconditionally,

$$0 \leq v_m \leq \frac{(2m+1)^3}{m^2(m+1)^2 \log^2(2m+1)} \leq \frac{125}{8m \log^2 m}. \tag{38.2}$$

In particular,

$$L_\infty := v(2,4) + \sum_{m=2}^{\infty} v_m < \infty.$$

The following completely explicit, though loose, bound is valid:

$$L_\infty \leq \frac{\log^2 2}{2} + \frac{125}{8}\left(\frac{1}{2\log^2 2} + \frac{1}{\log 2}\right) < 40. \tag{38.3}$$

Proof. Take $a = m^2$, $h = 2m+1$ in Lemma 38.2. Its length conditions hold for $m \geq 2$. The second inequality in (38.2) follows from $2m+1 \leq (5/2)m$, $m^2(m+1)^2 \geq m^4$, and $\log(2m+1) \geq \log m$.

On $[2,4]$, the total increase of the reciprocal-prime sum away from its left endpoint is $1/3$, whereas the total increase of $\log\log t$ is $\log 2$. Thus the oscillation of E on this interval is at most $\log 2$, and

$$v(2,4) \leq \frac{2}{4}\log^2 2.$$

Finally, the decreasing-integrand estimate

$$\sum_{m=2}^{\infty} \frac{1}{m \log^2 m} \leq \frac{1}{2\log^2 2} + \int_2^\infty \frac{dt}{t \log^2 t} = \frac{1}{2\log^2 2} + \frac{1}{\log 2}$$

proves (38.3). For example, the elementary bounds $0.69 < \log 2 < 0.70$ already prove that its displayed right side is less than 40. □

38.3 Every endpoint, including the final incomplete block

For $X > 2$, partition $[2, X]$ into the nonempty intervals

$$[2, \min(4, X)], \qquad [m^2, \min((m+1)^2, X)] \quad (m \geq 2,\ m^2 < X).$$

Call this finite partition $\mathcal{P}_X$; endpoints of adjacent intervals have measure zero and are immaterial. For $I = [a, b]$ in this partition, write

$$\ell_I = b - a, \quad \omega_I = \frac{1}{a} - \frac{1}{b}, \quad \mu_I = \mu_{a,b}, \quad r_I = r_{a,b}, \quad v_I = v(a,b), \qquad L(X) = \sum_{I \in \mathcal{P}_X} v_I.$$

Theorem 38.4 (Exact decomposition with a uniformly bounded local part)**.** *Let*

$$\overline{E}_X = \frac{1}{X-2}\int_2^X E(t)\,dt, \qquad a_X = \frac{\int_2^X P(t)t^{-2}\,dt}{\int_2^X t^{-2}\,dt}.$$

Then, for every $X > 2$,

$$V(X) = L(X) + \sum_{I \in \mathcal{P}_X} \ell_I(\mu_I - \overline{E}_X)^2, \tag{38.4}$$

$$= L(X) + \sum_{I \in \mathcal{P}_X} \omega_I(r_I - a_X)^2. \tag{38.5}$$

The same $L(X)$ occurs in both formulas, exactly. Moreover,

$$0 \le L(X) \le L_\infty < 40, \qquad L(X) \uparrow L_\infty,$$

and, if $M = \lfloor \sqrt{X} \rfloor \ge 3$, then

$$0 \le L_\infty - L(X) \le \frac{125}{8\log(M-1)} = O\Big(\frac{1}{\log X}\Big). \tag{38.6}$$

Proof. On each interval, expand $E - \overline{E}_X = (E - \mu_I) + (\mu_I - \overline{E}_X)$. The cross term integrates to zero. Summing gives (38.4). The identical calculation for P with measure $t^{-2}dt$, together with Lemma 38.1, gives (38.5).

Minimum centered variance is monotone when its interval is enlarged: if $a < b < c$, then

$$\min_z \int_a^b (E-z)^2 \le \int_a^b (E-\mu_{a,c})^2 \le \min_z \int_a^c (E-z)^2.$$

Thus the incomplete final block contributes no more than the corresponding full block, and its contribution increases as its right endpoint increases. This proves $L(X) \uparrow L_\infty$. For $M = \lfloor \sqrt{X} \rfloor$, all possible omitted energy belongs to blocks with index at least M. Hence

$$L_\infty - L(X) \le \frac{125}{8} \sum_{m=M}^{\infty} \frac{1}{m\log^2 m} \le \frac{125}{8} \int_{M-1}^{\infty} \frac{dt}{t\log^2 t},$$

which proves (38.6). A block of length zero at an exact square is simply omitted. □

38.4 The exact nonnegative cost of joining blocks

For every $2 < a < b$, the ordinary merge identity for variance gives

$$V(b) - V(a) = v(a,b) + \frac{(a-2)(b-a)}{b-2}\left(\mu_{a,b} - \overline{E}_a\right)^2. \tag{38.7}$$

For clarity, this follows by decomposing $[2,b]$ into $[2,a]$ and $[a,b]$, then minimizing the sum

$$V(a) + (a-2)(\overline{E}_a - c)^2 + v(a,b) + (b-a)(\mu_{a,b} - c)^2$$

over c. Thus, apart from the first block, the variance is the sum of the local terms $v(a,b)$ and explicitly nonnegative costs for the separation between a new block mean and the mean over all preceding points.

There is the completely equivalent weighted formula

$$V(b) - V(a) = v(a,b) + \frac{\Omega_a \omega_{a,b}}{\Omega_a + \omega_{a,b}}(r_{a,b} - a_a)^2, \qquad \Omega_a = \frac{1}{2} - \frac{1}{a}.$$

Indeed,

$$r_{a,b} - a_a = a(\mu_{a,b} - \overline{E}_a),$$

so its second term is exactly the second term in (38.7), not merely comparable with it.

The arithmetic expression for the displacement of a block mean from its own initial value is

$$\mu_{a,b} - E(a) = \frac{1}{b-a}\left[\sum_{a<p\le b} \frac{b-p}{p} - \int_a^b \frac{b-t}{t\log t}\,dt\right].$$

The sieve bound controls this local expression. It does not bound the additional term $E(a) - \overline{E}_a$ in

$$\mu_{a,b} - \overline{E}_a = (\mu_{a,b} - E(a)) + (E(a) - \overline{E}_a).$$

This is the precise long-range arithmetic term left uncontrolled by the local argument.

38.5 A general summability criterion

Let $a_{j+1} = a_j + h_j$, where $a_0 \geq 2$, $1 < h_j \leq a_j$, and the intervals cover $[a_0, \infty)$. If

$$\sum_j \frac{h_j^3}{a_j(a_j + h_j)\log^2 h_j} < \infty, \tag{38.8}$$

then the sum of the local variances over this partition is finite, by Lemma 38.2. This includes every last incomplete block as above.

For a convenient integral version, suppose $\log h_j \geq \kappa \log a_j$ for some fixed $\kappa > 0$, and let $H(t) = h_j$ on $[a_j, a_{j+1})$. Then

$$\sum_j v(a_j, a_{j+1}) \leq \kappa^{-2} \sum_j \frac{h_j^3}{a_j^2 \log^2 a_j} \leq 16\kappa^{-2} \int_{a_0}^{\infty} \frac{H(t)^2}{t^2 \log^2 t}\, dt.$$

The last comparison uses $a_j \leq t \leq 2a_j$ and $\log(2a_j) \leq 2\log a_j$. Consequently, after any finite initial part, one may choose successive lengths of the following orders and still obtain finite total local variance:

$$h(a) = \sqrt{a}\,(\log a)^{\alpha} \quad (\alpha < 1/2), \qquad h(a) = \frac{\sqrt{a \log a}}{(\log\log a)^{1/2+\varepsilon}} \quad (\varepsilon > 0).$$

The relevant comparison integrals are, respectively,

$$\int^{\infty} \frac{dt}{t(\log t)^{2-2\alpha}}, \qquad \int^{\infty} \frac{dt}{t \log t (\log\log t)^{1+2\varepsilon}},$$

which converge. The functions specifying these lengths vary by a bounded factor on $[a, 2a]$, so replacing the step function H by the displayed length functions is legitimate.

Status of the original target. For $X = p_{N+1}$, all formulas above concern the actual variance in the required inequality

$$V(p_{N+1}) \leq \frac{\log p_N}{8\pi^2} \quad \text{for infinitely many } N.$$

The local component $L(X)$ has now been bounded uniformly, with an explicit constant and a vanishing tail. No upper bound with the required logarithmic budget has been proved for the nonnegative between-block component in (38.4)–(38.5), or equivalently for the sum of the second terms in (38.7). The original target remains unproved. Historical priority is not claimed.

39 Exact evolution of the original optimized energy

For $X > 2$, define

$$m(X) = \frac{1}{X-2}\int_2^X E(t)\, dt, \qquad r(X) = E(X) - m(X), \qquad a(X) = \frac{\int_2^X P(t)t^{-2}\, dt}{\int_2^X t^{-2}\, dt}.$$

These definitions give the same original V, and introduce no substitute objective. The exact variance identity yields

$$r(X) = \frac{P(X) - a(X)}{X}.$$

All the following assertions are unconditional.

Lemma 39.1 (Evolution and jumps)**.** *The function V is continuous and locally absolutely continuous on $(2,\infty)$. Away from primes,*

$$V'(X) = r(X)^2, \qquad m'(X) = \frac{r(X)}{X-2}, \qquad r'(X) = -\frac{1}{X\log X} - \frac{r(X)}{X-2}.$$

At a prime $p > 2$, m is continuous and

$$r(p) = r(p^-) + \frac{1}{p}, \qquad V'_+(p) = r(p)^2, \qquad V'_-(p) = r(p^-)^2.$$

Moreover $V(2+) = 0$, $r(2+) = 0$, and

$$\frac{d}{dX}\big((X-2)r(X)\big) = -\frac{X-2}{X\log X} < 0$$

between consecutive primes.

Proof. Write

$$V(X) = \int_2^X E(t)^2\,dt - \frac{1}{X-2}\left(\int_2^X E(t)\,dt\right)^2.$$

Differentiation on a prime-free interval proves the first two formulas. On that interval $E'(X) = -1/(X\log X)$, which proves the third. A prime contributes a jump of $1/p$ to E, while an integral and hence m remain continuous. One-sided differentiation proves the formulas at a prime. Continuity of E from the right at 2 gives the stated limits. The integrating-factor formula follows immediately. □

40 An unconditional sparse result and its limitation

Littlewood's theorem gives unbounded oscillations of $P(x)$ in both signs; see Pintz [29, p.255, (1.3)–(1.4)]. Changing the normalization of the logarithmic integral adds a constant and does not affect this fact. The unconditional PNT and Mertens' theorem give

$$a(X) \longrightarrow 2(\log\log 2 + B_{\mathbb{P}}).$$

Theorem 40.1 (Infinitely many actual primes with small energy slope)**.** *There are infinitely many primes q such that*

$$-\frac{1}{q} \le r(q^-) \le 0 < r(q) \le \frac{1}{q}.$$

In particular, at these actual primes,

$$0 \le V'_-(q) \le q^{-2}, \qquad 0 < V'_+(q) \le q^{-2}, \qquad 0 \le (q-2)r(q)^2 < q^{-1}.$$

For every $H > 0$ at any such q, one also has

$$0 \le V(q+H) - V(q) \le \frac{H\big(1+(1+1/\log q)H\big)^2}{(q-2)^2}.$$

For example,

$$V(q+\sqrt{q}) - V(q) = O(q^{-1/2})$$

along this infinite prime subsequence.

Proof. Since $a(X)$ stays bounded and $P(X)$ has unbounded oscillations in both signs, $r(X) = (P(X) - a(X))/X$ takes both signs at arbitrarily large arguments. On a prime-free interval, $(X-2)r(X)$ strictly decreases. Therefore a passage from nonpositive to positive values can only occur at a prime, where the jump of r is exactly $1/p$. There must be infinitely many such passages. At each one, $r(q^-) \le 0 < r(q^-) + 1/q$, proving the bounds.

For the local estimate put $u(x) = (x-2)r(x)$. For $q \le x \le q + H$, the exact evolution is

$$u(x) = u(q) + \sum_{q<p\le x} \left(1 - \frac{2}{p}\right) - \int_q^x \frac{1 - 2/t}{\log t}\, dt.$$

Here $0 < u(q) < 1$. There are at most H integers and hence at most H primes in $(q, q+H]$, while the integral is at most $H/\log q$. Thus

$$|r(x)| \le \frac{1 + (1 + 1/\log q)H}{q - 2}.$$

Integrate $V' = r^2$ over the interval. No prime-gap theorem or probabilistic assumption enters this step. □

This theorem controls the *increase* of the original energy near an unbounded sequence of actual primes. It does not bound the energy accumulated before those primes. In particular it does not prove $V(q) \le \log p_{k-1}/(8\pi^2)$ when $q = p_k$.

41 The exact accumulated-energy balance

The following separates the small diagonal term from the correlation that is not controlled by the preceding theorem. Define, for $X > 2$,

$$J(X) = \sum_{2<p\le X} \frac{p-2}{p^2},$$

and define the entirely deterministic quantity

$$C(X) = \sum_{2<p\le X} \left(1 - \frac{2}{p}\right) r(p^-) - \int_2^X \frac{1 - 2/t}{\log t}\, r(t)\, dt.$$

The left limit in the prime sum is essential. The prime 2 is excluded: it is the initial endpoint, not an interior jump. If X is itself prime, its jump is included in both sums, and $r(X)$ below denotes the post-jump value.

Proposition 41.1 (Energy balance with all jump terms retained)**.** *For every $X > 2$,*

$$\boxed{V(X) + (X-2)r(X)^2 = J(X) + 2C(X).}$$

Furthermore,

$$J(X) = \log\log X + B_{\mathbb{P}} - 2\sum_p \frac{1}{p^2} + o(1), \qquad C(X) \ge -\tfrac{1}{2} J(X).$$

At the infinitely many primes from Theorem 40.1,

$$V(q) = J(q) + 2C(q) - \varepsilon_q, \qquad 0 \le \varepsilon_q < 1/q.$$

Proof. Let $B(X) = (X-2)r(X)^2$. Between primes,

$$B'(X) = -r(X)^2 - 2\frac{X-2}{X\log X}r(X).$$

At a prime $p > 2$,

$$B(p) - B(p^-) = 2\left(1 - \frac{2}{p}\right)r(p^-) + \frac{p-2}{p^2}.$$

Integrate the continuous part and sum these jumps, starting at $2+$, where B and V both tend to zero. Since $\int_2^X r(t)^2 dt = V(X)$, the exact identity follows. The formula for J follows from

$$J(X) = \sum_{p\le X}\frac{1}{p} - 2\sum_{p\le X}\frac{1}{p^2};$$

the term at $p = 2$ is zero. Apply Mertens' theorem and absolute convergence of $\sum_p p^{-2}$. The lower bound for C follows from nonnegativity of V and B. At the crossing primes, use $0 \le B(q) < 1/q$. □

Where this attack stops. The diagonal contribution $J(q)$ is only $O(\log\log q)$, and the terminal boundary energy can be made $O(1/q)$ along actual primes, without RH. What remains is an upper bound for the deterministic correlation $C(q)$ on those primes. The proof provides a lower bound for C, but no upper bound strong enough to force the requested logarithmic estimate for V. Replacing C by zero or treating it as a mean-zero martingale term would introduce an unjustified random-prime assumption. No such replacement is made here. Consequently the user's requested inequality remains unproved.

42 A certified exclusion of every index below 10^{25}

The following is a *finite* computer-assisted result for the exact threshold in (T). It is not a verification of infinitely many indices and does not imply that a later successful index exists.

Theorem 42.1 (Certified finite exclusion)**.** *For every index N with $p_N \le 10^{25}$,*

$$V(p_{N+1}) > \frac{\log p_N}{8\pi^2}.$$

Proof. First, V is nondecreasing: for $2 < X < Y$ and every real a,

$$\int_2^Y \frac{(P-a)^2}{t^2}\,dt \ge \int_2^X \frac{(P-a)^2}{t^2}\,dt \ge V(X).$$

Taking the minimum on the left proves the assertion.

The interval-arithmetic computation documented below and reproduced in Appendix 121 gives the following outward enclosures. They are error-bounded intervals, not decimal point estimates.

X	Certified enclosing interval for $V(X)$	Lower bound used
3	[0.0171114, 0.0171669]	1/60
5	[0.0451688, 0.0453412]	1/25
23	[0.1757966, 0.1761344]	17/100
200000	[0.7405992, 0.7413277]	37/50

The same interval arithmetic verifies

$$\frac{\log 3}{8\pi^2} < \frac{1}{60}, \quad \frac{\log 19}{8\pi^2} < \frac{1}{25}, \quad \frac{\log 200000}{8\pi^2} < \frac{17}{100}, \quad \frac{25\log 10}{8\pi^2} < \frac{37}{50}.$$

The four overlapping ranges of p_N are

$$[2,3], \qquad [3,19], \qquad [19,200000], \qquad [200000,10^{25}].$$

In these ranges the next prime is at least 3, 5, 23, and 200000, respectively. Monotonicity of V and of the logarithm now proves the desired strict inequality in each range. □

42.1 Analytic error bounds underlying the certificate

Put $F(x) = \sum_{p\le x} p^{-1} - 1/4 - \log\log x$. It differs from E by a constant, so

$$V(X) = I_2(X) - \frac{I_1(X)^2}{X-2}, \qquad I_j(X) = \int_2^X F(t)^j\,dt \quad (j=1,2).$$

Partition at every prime up to 200000, including the final interval $[199999, 200000]$. On a gap $[a,b)$ the prime sum is constant. Extend that smooth branch to b without its prime jump. Writing f for this branch, one has

$$f'(x) = -\frac{1}{x\log x}, \qquad f''(x) = \frac{\log x + 1}{x^2\log^2 x}.$$

The second expression is decreasing for $x \ge 2$. If $M = \max(|f(a)|, |f(b)|)$, monotonicity of f gives

$$|(f^2)''(x)| \le \frac{2}{a^2\log^2 a} + \frac{2M(\log a + 1)}{a^2\log^2 a} \qquad (a \le x \le b).$$

For k equal trapezoid panels, the integral error is bounded by $(b-a)^3 \sup|g''|/(12k^2)$ for either $g = f$ or $g = f^2$. All these bounds are accumulated with interval arithmetic at 50 decimal digits. The sieve uses exact integer arithmetic; only rational operations and logarithms enter the integrals. The final $I_2 - I_1^2/(X-2)$ is evaluated as an interval, including all accumulated quadrature errors. A floating-point square root is used solely to choose a panel count, not to evaluate a bound.

The calculation requires the primes only through 200000; there are 17984 of them. Its exclusion up to 10^{25} comes from monotonicity and the slowly increasing threshold, not from enumerating primes up to 10^{25}. No zeta-zero computation, probabilistic model, or assumption of RH enters this certificate.

43 The original target and the result of this investigation

Let p_N be the N-th ordinary prime. All logarithms are natural. Use the Mertens constant $B_{\mathbb{P}}$ and the exact functions

$$E(x) = \sum_{p\le x}\frac{1}{p} - \log\log x - B_{\mathbb{P}}, \qquad P(x) = \pi(x) - \mathrm{Li}_2(x), \qquad \mathrm{Li}_2(x) = \int_2^x \frac{dt}{\log t}.$$

They are right-continuous at primes. Mertens' theorem gives $E(x) \to 0$, unconditionally. The original variance is

$$V(X) = \min_{c\in\mathbb{R}} \int_2^X (E(t)-c)^2\,dt = \min_{a\in\mathbb{R}} \int_2^X (P(t)-a)^2\frac{dt}{t^2}. \tag{43.1}$$

The equality of these two optimized quantities, and the bounded centering correction used below, are proved in Appendix 125. The unchanged target is

$$\mathsf{B}: \quad \forall N_0\ \exists N \geq N_0: \quad V(p_{N+1}) \leq \frac{\log p_N}{8\pi^2}. \tag{43.2}$$

The integral measure, endpoint, numerical constant, and quantifiers in (43.2) are not modified.

Theorem 43.1 (An exact window reformulation)**.** *Define*

$$g_4(x) = \log \frac{\log(4x)}{\log x} - \sum_{x<p\leq 4x} \frac{1}{p}, \qquad H_4(X) = \int_2^X g_4(x)^2\, dx.$$

Then the following are equivalent:

$$\boxed{\mathsf{B} \iff \mathrm{RH} \iff E \in L^2([2,\infty), dx) \iff H_4(\infty) < \infty.}$$

Whenever these equivalent conditions hold,

$$\frac{1}{4}\int_2^\infty E(x)^2 dx \leq H_4(\infty) \leq \frac{9}{4}\int_2^\infty E(x)^2 dx. \tag{43.3}$$

In particular, if $H_4(\infty) = K < \infty$, then $V(X) \leq 4K$ for every $X > 2$, and the original inequality holds for every index with

$$\log p_N \geq 32\pi^2 K. \tag{43.4}$$

This theorem proves equivalence and the stated implications; it does not assert that the equivalent conditions hold.

The arithmetic content remains in the square-integrability of the explicit signed prime-window discrepancy g_4. The constant $B_{\mathbb{P}}$ cancels from that discrepancy. An exact kernel expansion below turns its bounded energy into a one-sided compensated-correlation estimate. Further sections test whether local sieve bounds, multiplicative convolution, or Hardy factorization provide that estimate.

44 Prime reciprocals meet a strict dilation contraction

The two structures combined here are the exact reciprocal-prime recurrence and a Hilbert-space dilation. No random-prime model is used.

Lemma 44.1 (Inverse dilation without an integrability assumption)**.** *Let $Uf(x) = f(4x)$ on $L^2([2,\infty), dx)$. Then $\|U\| = 1/2$ and*

$$\tfrac{1}{2}\|f\|_2 \leq \|(I-U)f\|_2 \leq \tfrac{3}{2}\|f\|_2.$$

If instead f is initially only locally square-integrable, $f(x) \to 0$, and $h(x) = f(x) - f(4x)$ belongs to L^2, then $f \in L^2$ and

$$f = \sum_{k\geq 0} U^k h \quad \text{in } L^2, \qquad \|f\|_2 \leq 2\|h\|_2.$$

Proof. A change of variables gives

$$\|Uf\|_2^2 = \frac{1}{4}\int_8^\infty |f(t)|^2 dt \leq \tfrac{1}{4}\|f\|_2^2.$$

Equality is attained by functions supported in $[8,\infty)$. The two-sided estimate is the triangle inequality. For the converse, $\sum_{k=0}^K U^k h$ has norm at most $\sum_{k=0}^K 2^{-k}\|h\|_2 \leq 2\|h\|_2$. It equals $f(x) - f(4^{K+1}x)$ pointwise, and therefore tends to $f(x)$. Fatou's lemma yields $\|f\|_2 \leq 2\|h\|_2$. The geometric series also converges in L^2; a convergent subsequence identifies its limit with f. Thus square-integrability of f has not been assumed in proving the reverse implication. □

The exact prime identity is

$$E(x) - E(4x) = \log \frac{\log(4x)}{\log x} - \sum_{x<p\le 4x} \frac{1}{p} = g_4(x).$$

Mertens' limit and Lemma 44.1 prove the last equivalence in Theorem 43.1 and (43.3). Moreover, using $c = 0$ in (43.1) gives $V(X) \le \|E\|_2^2 \le 4K$, proving (43.4). Appendix 125 proves $\mathsf{B} \Rightarrow$ RH. The established zeta factorization and the finite-norm estimates in Section 48 prove RH $\Rightarrow E \in L^2$ without using the target. These implications complete the proof of Theorem 43.1.

44.1 Why a finite endpoint does not close the estimate

Put $F(X) = \int_2^X E(x)^2 dx$. Expanding the square gives the exact identity

$$H_4(X) = F(X) + \tfrac{1}{4}\{F(4X) - F(8)\} - 2\int_2^X E(x)E(4x)dx.$$

Consequently,

$$F(X) \le 2H_4(X) + \tfrac{1}{2}\{F(4X) - F(8)\}. \tag{44.1}$$

The coefficient $1/2$ multiplies the future energy $F(4X)$, not the current energy $F(X)$. Moving that term to the left as though the arguments agreed would be invalid. The valid infinite-horizon inverse requires the arithmetic premise $H_4(\infty) < \infty$, which is still unproved.

45 A finite-cutoff kernel for balanced reciprocal-prime windows

For $X > 2$, let the finite signed measure on $(2, 4X]$ be

$$\eta(dt) = \sum_{2<p\le 4X} \frac{\delta_p(dt)}{p} - \frac{dt}{t\log t}.$$

Put

$$g_4(x) = E(x) - E(4x) = -\eta((x, 4x]) \quad (2 \le x \le X), \qquad H_4(X) = \int_2^X g_4(x)^2\, dx.$$

In particular,

$$g_4(x) = \log\Big(\frac{\log(4x)}{\log x}\Big) - \sum_{x<p\le 4x} \frac{1}{p}$$

contains no unknown additive Mertens constant.

Proposition 45.1 (Exact banded kernel)**.** *Define*

$$K_X(u, v) = \big[\min(X, u, v) - \max(2, u/4, v/4)\big]_+.$$

Then

$$H_4(X) = \iint_{(2,4X]^2} K_X(u, v)\, \eta(du)\eta(dv).$$

The kernel is nonnegative and positive semidefinite. Moreover,

$$K_X(u, v) = 0 \quad \textit{unless} \quad \frac{1}{4} < \frac{u}{v} < 4.$$

At either boundary ratio $u/v = 1/4$ or $u/v = 4$, the kernel is zero.

Proof. For fixed u, the condition $x < u \le 4x$ is $u/4 \le x < u$. Therefore the Lebesgue measure of the common set of allowed $x \in [2, X]$ for u and v is exactly $K_X(u, v)$. Differences at interval endpoints have Lebesgue measure zero.

Since η has finite total variation and $0 \le K_X \le X - 2$, Fubini's theorem applies absolutely with respect to its total variation measures and proves the identity. The Gram representation

$$K_X(u, v) = \int_2^X \mathbf{1}_{x<u\le 4x}\mathbf{1}_{x<v\le 4x}\, dx$$

proves positive semidefiniteness. If, for example, $v \ge 4u$, the lower endpoint $v/4$ is at least the upper endpoint u, so the overlap length is zero. The other direction is symmetric. The prime 2 contributes to no window, because the lower endpoint is open and $x \ge 2$. Equivalently $K_X(2, v) = 0$; there is no omitted endpoint atom. □

Proposition 45.2 (The atomic diagonal)**.** *Let*

$$D_4(X) = \sum_{2<p\le 4X} \frac{K_X(p, p)}{p^2}.$$

Unconditionally,

$$D_4(X) = \frac{3}{4}\log\log X + C_D + o(1) = \frac{3}{4}\log\log X + O(1),$$

where the finite constant can be written

$$C_D = \frac{3}{4}B_{\mathbb{P}} - \frac{3}{8} - \frac{5}{36} - \frac{3}{100} - \frac{1}{196}.$$

Proof. For $X \ge 8$, the exact decomposition is

$$D_4(X) = \sum_{p=3,5,7} \frac{p-2}{p^2} + \frac{3}{4}\sum_{8<p\le X} \frac{1}{p} + \sum_{X<p\le 4X} \frac{X - p/4}{p^2}.$$

The last sum is nonnegative and $O(1/\log X)$. For example, each summand is at most $1/X$, while Brun–Titchmarsh [30] on $(X, 4X]$ gives at most $6X/\log(3X)$ primes. Thus it is bounded above by $6/\log(3X)$. Alternatively, Chebyshev's bound and partial summation give the same order.

Mertens' theorem $\sum_{p\le X} p^{-1} = \log\log X + B_{\mathbb{P}} + o(1)$ proves the asserted asymptotic. The constant follows by subtracting $(3/4)\sum_{p\le 7} 1/p$ from the displayed small prime contribution. A prime at $p = X$ belongs only to the middle sum, and an atom at $p = 4X$, if present, has coefficient zero; both boundaries are therefore covered. □

At every finite X, expansion of the signed measure gives

$$H_4(X) = D_4(X) + R_4(X),$$

where, writing $d\nu(t) = dt/(t\log t)$ on $(2, 4X]$,

$$R_4(X) = \sum_{\substack{2<p,q\le 4X\\ p\ne q}} \frac{K_X(p, q)}{pq} - 2\sum_{2<p\le 4X} \frac{1}{p}\int_{(2,4X]} K_X(p, t)\, d\nu(t) + \iint_{(2,4X]^2} K_X(s, t)\, d\nu(s) d\nu(t).$$

The atomic off-diagonal sum and the last term are nonnegative; the mixed term has a negative sign. Positive semidefiniteness of K_X proves $H_4(X) \ge 0$, not $R_4(X) \ge 0$. If $H_4(\infty) < \infty$, then in fact

$$R_4(X) = -\frac{3}{4}\log\log X + \big(H_4(\infty) - C_D\big) + o(1).$$

Thus the compensated remainder must cancel the divergent diagonal in that case. No such cancellation is proved here. All expansions are made at finite X; no exchange of separately divergent infinite terms is used.

Corollary 45.3 (An induced one-sided correlation criterion)**.** *With the explicit finite-cutoff remainder R_4 above, the original target* B *is equivalent to*

$$\boxed{\exists C, X_0\ \forall X \geq X_0 : \quad R_4(X) \leq -\tfrac{3}{4}\log\log X + C.} \tag{45.1}$$

It is also equivalent to the same inequality, with one fixed constant C, along any unbounded sequence of endpoints $X_j \to \infty$. Equivalently, $R_4(X) + \frac{3}{4}\log\log X$ has a finite limit. None of these equivalent arithmetic assertions is proved here.

Proof. Write $D_4(X) = \frac{3}{4}\log\log X + C_D + \varepsilon(X)$, where $\varepsilon(X) \to 0$. Then

$$H_4(X) = R_4(X) + \tfrac{3}{4}\log\log X + C_D + \varepsilon(X).$$

An upper bound of the stated form along $X_j \to \infty$ makes $H_4(X_j)$ uniformly bounded. Since H_4 is nondecreasing, it is then bounded at every endpoint. Conversely, if $H_4(\infty) < \infty$, then

$$R_4(X) + \tfrac{3}{4}\log\log X \longrightarrow H_4(\infty) - C_D.$$

Apply Theorem 43.1. The automatic lower bound $R_4(X) \geq -D_4(X)$ supplies no missing upper bound. □

This is an exact reformulation using the original ordinary primes. It reveals a signed correlation budget involving only comparable scales: $K_X(u, v) = 0$ when their ratio is outside $(1/4, 4)$. The cancellation of the diagonal is essential. Treating the prime atoms as independent, or treating every term of the signed-measure expansion as positive, would not prove (45.1).

46 A positive Selberg kernel is not a positive quadratic form

Let Λ be the ordinary von Mangoldt function, and for compactly supported real f on $[2, \infty)$, extended by zero below 2, define

$$(\mathcal{S}f)(x) = \frac{1}{\log x}\sum_{n \leq x}\frac{\Lambda(n)}{n} f(x/n), \qquad x \geq 2.$$

Every coefficient of this actual arithmetic operator is nonnegative. Nevertheless it is neither a bounded operator on the relevant $L^2(dx)$ space nor an accretive quadratic form.

Proposition 46.1 (An explicit failure of the proposed coercivity)**.** *For $X > 8$, put*

$$a(x) = \mathbf{1}_{[2,4]}(x), \qquad b_X(x) = X^{-1/2}\mathbf{1}_{[X,2X]}(x), \qquad f_X = a - b_X.$$

Then, unconditionally,

$$\|f_X\|_2^2 = 3, \qquad \int_2^\infty f_X(x)(\mathcal{S}f_X)(x)\,dx \sim -\frac{(\log 2)\sqrt{X}}{\log X}.$$

In particular no constant c can satisfy $\langle f, \mathcal{S}f\rangle \geq -c\|f\|_2^2$ for all such compactly supported real functions.

Proof. Partial summation and the ordinary prime number theorem imply, uniformly as $x \to \infty$,

$$\sum_{x/4 \le n \le x/2} \frac{\Lambda(n)}{n} = \log 2 + o(1).$$

Endpoint terms are $O((\log x)/x)$ and do not affect the limit. Thus

$$(\mathcal{S}a)(x) = \frac{\log 2 + o(1)}{\log x}.$$

This already proves $\mathcal{S}a \notin L^2$. Since every nonzero term has $n \ge 2$, $\mathcal{S}a = 0$ almost everywhere on $[2, 4]$, and $\mathcal{S}b_X = 0$ almost everywhere on $[2, 2X]$. Consequently the two self-pairings and the opposite cross-pairing vanish exactly, leaving

$$\langle f_X, \mathcal{S}f_X \rangle = -\langle b_X, \mathcal{S}a \rangle = -X^{-1/2} \int_X^{2X} \frac{\log 2 + o(1)}{\log x}\, dx \sim -\frac{(\log 2)\sqrt{X}}{\log X}.$$

All pairings are finite because the test functions have compact support. No assertion that $\mathcal{S}f_X \in L^2$ is used. □

This does not say that every use of Selberg's identity must fail, or that the actual prime error behaves like these test functions. It rules out the specific inference "nonnegative convolution coefficients imply a coercive quadratic form" on the space required for the target. Any successful use of that identity must supply further arithmetic restrictions or a genuinely different form.

47 Where the Selberg–Möbius remainder enters

The exact multiplicative identity is

$$\Lambda_2(n) := \Lambda(n) \log n + (\Lambda * \Lambda)(n) = (\mu * (\log)^2)(n).$$

Selberg's asymptotic formula [31] gives

$$\sum_{n \le x} \Lambda_2(n) = 2x \log x + O(x).$$

For completeness, the prime-power version follows directly from Selberg's original prime-only formula. The additional single-sum terms are $O(\sqrt{x} \log^2 x) = O(x)$, by Chebyshev's estimate. The additional convolution terms are bounded by

$$2 \sum_{p^k \le x,\ k \ge 2} (\log p) \psi(x/p^k) \ll x \sum_{p, k \ge 2} \frac{\log p}{p^k} \ll x.$$

Writing $\psi(x) = x + R(x)$, partial summation and $\sum_{n \le x} \Lambda(n)/n = \log x + O(1)$ yield

$$R(x) \log x + \sum_{n \le x} \Lambda(n) R(x/n) = O(x).$$

For $e(x) = R(x)/x$, extended with its actual values on $[1, 2)$, this becomes

$$e(x) + \frac{1}{\log x} \sum_{n \le x} \frac{\Lambda(n)}{n} e(x/n) = O(1/\log x).$$

This is a genuine equation using ordinary multiplicativity. It does not give an $L^2(dx)$ forcing bound: the envelope $1/\log x$ has square integral of order $X/\log^2 X$. In addition, the proposed quadratic coercivity for its operator fails by the preceding proposition. The exact remainder may contain more information than this envelope; no cancellation estimate for that remainder sufficient for the target has been proved in this attempt.

The failure survives the target's logarithmic normalization. Put $u(x) = (\psi(x) - x)/(x \log x)$ for $x \geq 2$. The terms with $1 \leq x/n < 2$ in the preceding equation have bounded total weight and can be included in its remainder. Dividing by $\log x$ gives

$$u(x) + (\mathcal{T}u)(x) = O(\log^{-2} x), \qquad (\mathcal{T}f)(x) = \frac{1}{\log^2 x} \sum_{n \leq x/2} \frac{\Lambda(n)}{n} \log(x/n) f(x/n).$$

The coefficients are again nonnegative. For the same $a = \mathbf{1}_{[2,4]}$, weighted partial summation gives

$$(\mathcal{T}a)(x) = \frac{\frac{3}{2}(\log 2)^2 + o(1)}{\log^2 x},$$

because $\int_2^4 (\log t) dt/t = \frac{3}{2}(\log 2)^2$. The support argument in the proposition is unchanged, so

$$\langle f_X, \mathcal{T} f_X \rangle \sim -\frac{\frac{3}{2}(\log 2)^2 \sqrt{X}}{\log^2 X} \longrightarrow -\infty.$$

Thus the counterexample does not depend on dropping a logarithmic weight. Also the available forcing envelope still has divergent square integral of order $X/\log^4 X$.

For precision, u and the original E have the same square-integrability status, without assuming RH. First, Chebyshev's estimate gives $\psi(x) - \vartheta(x) = O(\sqrt{x})$, so replacing ψ by ϑ changes u by an L^2 function. If $F(x) = P(x)/x$, exact partial summation gives

$$\frac{\vartheta(x) - x}{x \log x} = F(x) - \frac{1}{x \log x} \int_2^x F(t)\, dt - \frac{2}{x \log x}.$$

On a tail $[A, \infty)$ with $\log A > 2$, the integral operator with lower endpoint A has L^2 norm at most $2/\log A < 1$, by Hardy's inequality. The finite initial segment contributes only a constant multiple of $(x \log x)^{-1}$, which belongs to L^2. This proves the forward implication. For the reverse implication the Neumann inverse constructs an L^2 solution; uniqueness among locally integrable solutions follows by differentiating the homogeneous integral equation with initial value zero. Hence this normalized ϑ error lies in L^2 exactly when F does.

Finally, the project's exact variance identity gives $\int_2^X F^2 = V(X) + O(1)$. Boundedness of V, together with $E(x) \to 0$, is equivalent to $E \in L^2$: the means $(X-2)^{-1} \int_2^X E$ tend to zero, and each fixed initial interval can then be tested against these minimizing centers. This supplies the asserted identification with the original energy and prevents a change of target in the operator audit.

The Mellin integrating factor makes the unresolved issue explicit. For $\Re s > 0$, write

$$\widehat{e}(s) = \int_1^\infty e(x) x^{-s-1}\, dx.$$

Absolute convergence and the ordinary bound $e(x) = O(1)$ justify differentiating and exchanging the convolution sum. If $\mathcal{B}e = e(x) \log x + \sum_{n \leq x} \Lambda(n) e(x/n)/n$, then

$$\widehat{\mathcal{B}e}(s) = -\widehat{e}'(s) - \frac{\zeta'(s+1)}{\zeta(s+1)} \widehat{e}(s) = -\frac{(\zeta(s+1)\widehat{e}(s))'}{\zeta(s+1)}.$$

Hence formal inversion introduces $1/\zeta(s+1)$. Extending the resulting control from $\Re s > 0$ to the critical $L^2(dx)$ boundary $\Re s = -1/2$ requires handling exactly the zeros with $\Re(s+1) > 1/2$. Omitting this denominator, or assuming a uniform bound for its inverse, would hide the unresolved zero exclusion in the proposed contraction.

Products versus ratios. There is another precise distinction between the algebraic symmetry and the desired positive energy. For $\Re z > 1$, let $A(z) = \sum_n \Lambda(n) n^{-z} = -\zeta'(z)/\zeta(z)$. Then

$$A(z)^2 - A'(z) = \frac{\zeta''(z)}{\zeta(z)} = \sum_n \frac{\Lambda_2(n)}{n^z}.$$

For $z = \sigma + it$, the double series in $A(z)^2$ has phases $e^{-it\log(mn)}$, whereas the positive energy $|A(z)|^2$ has phases $e^{-it\log(m/n)}$. Selberg's product convolution is therefore not the ratio correlation that appears in a Fourier quadratic energy. Replacing one with the other without an additional identity or estimate is not justified by multiplicativity.

The Dirichlet hyperbola identity does extract a genuine square:

$$C(Y) := \sum_{mn \le Y} \Lambda(m)\Lambda(n) = 2 \sum_{m \le \sqrt{Y}} \Lambda(m)\psi(Y/m) - \psi(\sqrt{Y})^2.$$

This follows because every pair has at least one coordinate at most $\sqrt{Y}$, and the doubly counted square contributes $\psi(\sqrt{Y})^2$. Thus a product convolution can be connected exactly to a positive square. However, at $Y = X^2$ the $O(Y)$ remainder in the stated Selberg formula becomes $O(X^2)$. Without an additional cancellation estimate for that remainder and for the remaining mixed sum, this connection does not bound the square of a prime error at the required scale. No such estimate is supplied by the identity alone.

Sources and status. Selberg's original prime-only form appears as equations (2.8), (2.9), and (2.12), printed pp. 309–310, in Atle Selberg, *An elementary proof of the prime-number theorem*, Annals of Mathematics **50** (1949), 305–313, DOI 10.2307/1969455. The von Mangoldt form above is the usual prime-power version; its convolution identity was written explicitly to expose every operator used here. The original paper was inspected at https://www.math.lsu.edu/~mahlburg/teaching/handouts/2014-7230/Selberg-ElemPNT1949.pdf.

The dilation estimate is a valid bridge from an actual arithmetic difference to the original energy. This attempt does not obtain the required upper bound for that difference. The Selberg–Möbius collision fails at two explicit steps: the standard remainder is too large at the target scale, and the nonnegative kernel is not a positive coercive form. No unconditional proof of the original infinite-endpoint inequality, or of RH, results from these calculations.

48 Separating a controlled Hardy part from the actual zero defect

This section combines the ordinary-prime Mellin transform with the known Blaschke factorization of zeta. All prime-power and outer-factor terms are bounded unconditionally in the critical Hardy norm. The remaining inner factor is not controlled by this argument.

Put

$$P(x) = \pi(x) - \mathrm{Li}_2(x), \qquad \mathrm{Li}_2(x) = \int_2^x \frac{dt}{\log t}, \qquad F_P(s) = \int_2^\infty P(x) x^{-s-1}\, dx \quad (\Re s > 1).$$

Write $\mathcal{P}(s) = \sum_p p^{-s}$ for the prime zeta function in its half-plane of absolute convergence, and define

$$Q_{\mathrm{pp}}(s) = \sum_{k \ge 2} \frac{\mathcal{P}(ks)}{k}, \qquad L(s) = \int_2^\infty \frac{x^{-s}}{\log x}\, dx \quad (\Re s > 1).$$

The first series is normally convergent on $\Re s > 1/2$. Abel summation and the Euler product give the exact identity

$$sF_P(s) = \mathcal{P}(s) - L(s) = \log \zeta(s) - Q_{\mathrm{pp}}(s) - L(s), \qquad \Re s > 1. \tag{48.1}$$

Here and below the logarithm of zeta in this half-plane is the Euler-product branch, real on the real interval $(1,\infty)$.

We use the normalization

$$\|H\|_{\mathcal{H}}^2 = \sup_{\sigma>1/2} \frac{1}{2\pi}\int_{\mathbb{R}} |H(\sigma+it)|^2\,dt, \qquad \mathcal{H} = H^2(\Re s > 1/2).$$

The Mellin version of the Paley–Wiener theorem identifies this space isometrically with $L^2([1,\infty), x^{-2}dx)$:

$$H(s) = \int_1^\infty h(x)x^{-s-1}\,dx, \qquad \|H\|_{\mathcal{H}}^2 = \int_1^\infty |h(x)|^2 \frac{dx}{x^2}.$$

Indeed, after $x = e^u$, this is the ordinary Fourier–Laplace isometry for $e^{-u/2}h(e^u)$, supported on $u \ge 0$.

48.1 Three contributions whose critical norms are finite

Lemma 48.1 (The prime-power correction has finite critical energy)**.** *Let*

$$R_{\mathrm{pp}}(x) = \sum_{k\ge 2} \frac{1}{k}\pi(x^{1/k}) \qquad (x \ge 1).$$

Then, unconditionally,

$$\frac{Q_{\mathrm{pp}}(s)}{s} = \int_1^\infty R_{\mathrm{pp}}(x)x^{-s-1}\,dx \in \mathcal{H}.$$

If C_π is any Chebyshev constant such that $\pi(y) \le C_\pi y/\log y$ for all $y \ge 2$, then

$$\int_1^\infty R_{\mathrm{pp}}(x)^2 \frac{dx}{x^2} \le 2C_\pi^2 \left\{ \frac{1}{\log 4} + \frac{3\,4^{-1/3}}{(\log 2)^2} \right\}. \tag{48.2}$$

Proof. The integral identity follows first by summing the atoms at p^k for $\Re s > 1/2$. The function is zero for $x < 4$. For $x \ge 4$, Chebyshev's bound and $k \le \log x/\log 2$ give

$$R_{\mathrm{pp}}(x) \le \frac{C_\pi}{\log x} \sum_{2\le k\le \log x/\log 2} x^{1/k} \le C_\pi \left\{ \frac{\sqrt{x}}{\log x} + \frac{x^{1/3}}{\log 2} \right\}.$$

Squaring with $(a+b)^2 \le 2a^2 + 2b^2$ and integrating proves (48.2). The isometry then proves the Hardy assertion. □

Lemma 48.2 (A deterministic main-term correction)**.** *On $\Re s > 1$, set*

$$k(s) = \log\frac{s}{s-1} - L(s).$$

The function $k(s)/s$ extends to $\mathcal{H}$. Its inverse Mellin transform is the real function

$$v(x) = \begin{cases} \displaystyle\int_1^x \frac{t-1}{t\log t}\,dt, & 1 \le x \le 2, \\ \displaystyle v(2) - \log\frac{\log x}{\log 2}, & x \ge 2, \end{cases}$$

and

$$\int_1^\infty v(x)^2 \frac{dx}{x^2} \le 1 + \frac{1}{\log 2} + \frac{1}{(\log 2)^2}. \tag{48.3}$$

Proof. Frullani's identity, initially for $\Re s > 1$, gives

$$\log \frac{s}{s-1} = \int_1^\infty \frac{x-1}{\log x} x^{-s-1}\, dx.$$

Subtracting $L(s)$ leaves $\int_1^\infty v'(x)x^{-s}\,dx$. Integration by parts therefore gives $k(s)/s = \int_1^\infty v(x)x^{-s-1}dx$. Since $\log t \geq (t-1)/t$, one has $0 \leq v(x) \leq 1$ on $[1,2]$. For $x \geq 2$,

$$|v(x)| \leq 1 + \log \frac{\log x}{\log 2} \leq \frac{\log x}{\log 2}.$$

Integration of these bounds gives (48.3). □

For the third contribution we use the established factorization

$$\frac{s-1}{s}\zeta(s) = \zeta_B(s)B(s), \qquad \Re s > 1/2. \tag{48.4}$$

Here B is the Blaschke product of the actual zeros of zeta in $\Re s > 1/2$, with multiplicity, normalized by $B(1) > 0$. The zero-free factor ζ_B is specified by

$$\ell(s) := \log \zeta_B(s) = \frac{2}{\pi}\left(s - \frac{1}{2}\right) \int_0^\infty \frac{\log|\zeta(\frac{1}{2}+it)|}{t^2 + (s-\frac{1}{2})^2}\, dt. \tag{48.5}$$

We invoke the full formula (48.4), not merely a generic inner–outer factorization: there is no additional omitted singular inner factor. A primary source is M. Kunik, *Logarithmic Fourier integrals for the Riemann Zeta Function*, equations (1.2)–(1.6) [32], https://arxiv.org/pdf/0804.4829.

Put

$$M_2 = \frac{1}{2\pi}\int_{\mathbb{R}} \frac{\log^2|\zeta(\frac{1}{2}+it)|}{1/4+t^2}\, dt, \quad \Omega_\zeta = \frac{1}{2\pi}\int_{\mathbb{R}} \frac{\log|\zeta(\frac{1}{2}+it)|}{1/4+t^2}\, dt.$$

The same factorization gives

$$\Omega_\zeta = \ell(1) = -\log B(1) = \sum_{\Re\rho > 1/2} \log\left|\frac{\rho}{1-\rho}\right| \geq 0. \tag{48.6}$$

Lemma 48.3 (Unconditional control of the outer logarithm)**.** *Unconditionally, $M_2 < \infty$, and*

$$\frac{\ell(s)}{s} \in \mathcal{H}, \qquad \left\|\frac{\ell(s)}{s}\right\|_{\mathcal{H}}^2 = 2M_2 - \Omega_\zeta^2. \tag{48.7}$$

Proof. The standard local formula

$$\frac{\zeta'}{\zeta}(\sigma+it) = \sum_{|\gamma - t|<1} \frac{1}{\sigma+it-\rho} + O(\log t), \qquad -1 \leq \sigma \leq 2,$$

together with the bound $O(\log T)$ for the number of zeros whose ordinates lie in any fixed-length interval near T, is unconditional. For example, K. Kedlaya gives these statements as Lemmas 9.4 and 9.6 of his analytic-number-theory notes [33], https://kskedlaya.org/ant/chap-von-mangoldt.html. For $t \geq 3$ away from zero ordinates, integrate horizontally from $2+it$ to $1/2+it$ and take real parts. Since $\log|\zeta(2+it)| = O(1)$, this yields

$$\log|\zeta(\tfrac{1}{2}+it)| = \sum_{|\gamma-t|<1} \log|\tfrac{1}{2}+it-\rho| + O(\log t).$$

Uniformly for $0 < \Re\rho < 1$,

$$\int_T^{T+1} \mathbf{1}_{\{|t-\gamma|<1\}} \big|\log|\tfrac{1}{2}+it-\rho|\big|^2\,dt \ll 1:$$

the positive part is bounded, and the negative part is dominated by $|\log|t-\gamma||$. Cauchy–Schwarz across the $O(\log T)$ relevant zeros consequently gives

$$\int_T^{T+1} \log^2|\zeta(\tfrac{1}{2}+it)|\,dt \ll \log^2 T.$$

The compact part has only finitely many logarithmic singularities. Summing the displayed estimate with weight T^{-2}, and using conjugation symmetry for negative ordinates, proves $M_2 < \infty$.

Under $z = 1 - 1/s$, the boundary $s = 1/2 + it$ has $d\theta = |dt|/(1/4+t^2)$. Formula (48.5) is the analytic Schwarz extension of its real boundary data $\log|\zeta(1/2+it)|$, with real center value $\ell(1) = \Omega_\zeta$. More explicitly, put $\widetilde{\ell}(z) = \ell(1/(1-z))$. The disk Fourier coefficients give $\|\widetilde{\ell}\|^2_{H^2(\mathbb{D})} = 2M_2 - \Omega_\zeta^2$. The conformal Hardy isometry sends $\widetilde{\ell}(z)$ to $\widetilde{\ell}(1-1/s)/s = \ell(s)/s$. This proves (48.7). □

48.2 The exact remainder and its finite-endpoint energy

Proposition 48.4 (An unconditional finite-energy separation)**.** *Define, on $\Re s > 1/2$,*

$$F_{\mathrm{good}}(s) = \frac{\ell(s) + k(s) - Q_{\mathrm{pp}}(s)}{s},$$

where the continuation of $k(s)/s$ is the Mellin transform of v. Then $F_{\mathrm{good}} \in \mathcal{H}$, and

$$F_P(s) = F_{\mathrm{good}}(s) + \frac{\log B(s)}{s}, \qquad \Re s > 1. \tag{48.8}$$

The logarithm of B in (48.8) is the branch real on $(1,\infty)$. This branch is well-defined there; it is not asserted to continue holomorphically through an interior zero of B.

There is a real function $g \in L^2([1,\infty), x^{-2}dx)$ with Mellin transform F_{good}. Extend P by zero on $[1,2)$, call the extension P_0, and put $d = P_0 - g$. If

$$Q(X) = \int_2^X P(x)^2 \frac{dx}{x^2}, \qquad W(X) = \int_1^X d(x)^2 \frac{dx}{x^2},$$

then for every $X \geq 2$,

$$|\sqrt{Q(X)} - \sqrt{W(X)}| \leq C_*, \tag{48.9}$$

where one unconditional finite choice is

$$C_* = \sqrt{2M_2 - \Omega_\zeta^2} + \sqrt{1 + \frac{1}{\log 2} + \frac{1}{(\log 2)^2}} + C_\pi \sqrt{2\left\{\frac{1}{\log 4} + \frac{3\,4^{-1/3}}{(\log 2)^2}\right\}}.$$

Moreover $\log B(s)/s$ is the Mellin transform of this actual remainder d on $\Re s > 1$.

Proof. All three terms defining F_{good} belong to $\mathcal{H}$ by the preceding lemmas. Factorization (48.4) yields

$$\log\zeta(s) = \ell(s) + \log B(s) + \log\frac{s}{s-1} \qquad (\Re s > 1),$$

with no additive multiple of $2\pi i$, because all branches are real for real $s > 1$. Substitution in (48.1) proves (48.8). The Mellin isometry produces g; conjugation symmetry and uniqueness make it real almost everywhere. The weighted reverse triangle inequality on $[1, X]$ proves (48.9). All Mellin integrals for the difference converge absolutely on $\Re s > 1$, by Cauchy–Schwarz for g and elementary growth for P. Their difference is $\log B(s)/s$. □

What this does and does not settle. The separation retains the original primes. It proves that the prime-power correction, the deterministic main term, and the outer logarithm contribute only bounded critical energy. Since

$$V(X) = Q(X) - \frac{2X}{X-2}A_X^2, \qquad A_X = \int_2^X P(t)t^{-2}dt \longrightarrow \log\log 2 + B_{\mathbb{P}},$$

the bounded scalar correction in V does not resolve the remaining energy either. No logarithmic subsequence upper bound for $W(X)$, or for $V(X)$, has been proved here. The unknown factor B is the Blaschke product of the actual off-line zeta zeros. It cannot be dropped merely because $|B(1/2+it)| = 1$ almost everywhere.

Indeed, differentiating the exact identity on $\Re s > 1$ gives the meromorphic continuation

$$(sF_P(s))' = \ell'(s) + \frac{B'(s)}{B(s)} + \frac{1}{s} + \frac{2^{1-s}-1}{s-1} - \sum_{k\geq 2}\mathcal{P}'(ks), \qquad \Re s > 1/2.$$

Every term except B'/B is holomorphic in this half-plane; the apparent singularity at $s = 1$ is removable. At an off-line zero, B'/B has the positive integer residue given by its multiplicity. The controlled terms therefore cannot cancel that interior singularity.

48.3 A concrete causality check

An elementary Laplace example shows precisely why boundary square-integrability cannot replace the missing Hardy condition. For $a > 0$, let

$$f_+(u) = e^{au}\mathbf{1}_{[0,\infty)}(u).$$

Its Laplace transform is $1/(s-a)$, initially only for $\Re s > a$. Analytic continuation to the imaginary axis gives

$$\int_{\mathbb{R}} \left|\frac{1}{it-a}\right|^2 dt = \frac{\pi}{a} < \infty.$$

Nevertheless $f_+ \notin L^2$. With Fourier convention $\widehat{f}(t) = \int_{\mathbb{R}} f(u)e^{-itu}du$, the actual inverse Fourier transform of that boundary function is

$$f_-(u) = -e^{au}\mathbf{1}_{(-\infty,0)}(u), \qquad \widehat{f}_-(t) = \frac{1}{it-a}.$$

The pole changes the causal support. A finite boundary norm without analyticity in the intervening half-plane cannot justify replacing the original causal function by this inverse transform. In the prime problem, (48.8) records that missing interior information explicitly in $\log B/s$.

Corollary 48.5 (The conditional finite-energy implication)**.** *If RH holds, then* $\int_2^\infty P(x)^2x^{-2}\,dx < \infty$ *and* $\int_2^\infty E(x)^2\,dx < \infty$.

Proof. Under RH the actual Blaschke product is $B \equiv 1$. Proposition 48.4 therefore gives $F_P = F_{\text{good}}$ on $\Re s > 1$, where $F_{\text{good}} \in H^2(\Re s > 1/2)$ is the Mellin transform of $g \in L^2([1,\infty), x^{-2}dx)$. For any fixed $\sigma > 1$, both $P_0(e^u)e^{-\sigma u}$ and $g(e^u)e^{-\sigma u}$ belong to $L^1([0,\infty))$: the former by $P(x) = O(x)$, and the latter by Cauchy–Schwarz. Equality of their Fourier transforms, obtained by setting $s = \sigma + it$, implies $P_0 = g$ almost everywhere. Thus $Q(\infty) < \infty$. The exact variance identity gives $V(X) \leq Q(X)$. Unconditionally $E(x) \to 0$, so its averages $m_X = (X-2)^{-1}\int_2^X E(t)\,dt$ tend to zero. For fixed $R \geq 2$ and all $X \geq R$, $\int_2^R (E - m_X)^2 \leq V(X) \leq Q(\infty)$. First let $X \to \infty$, then $R \to \infty$, to obtain $\int_2^\infty E^2 \leq Q(\infty) < \infty$. This proves an implication from RH, not an unconditional proof of RH or of the original logarithmic target. □

49 Local prime energy and a discrete spectral obstruction

Retain

$$E(x) = \sum_{p \le x} p^{-1} - \log\log x - B_{\mathbb{P}}, \qquad V(X) = \min_c \int_2^X (E(t) - c)^2 \, dt.$$

For $m \ge 2$, put

$$I_m = [m^2, (m+1)^2], \qquad w_m = 2m + 1, \qquad \mu_m = \frac{1}{w_m} \int_{I_m} E(t) \, dt.$$

The identity

$$\int_{I_m} E(t)^2 dt = \int_{I_m} (E(t) - \mu_m)^2 dt + w_m \mu_m^2$$

separates within-block fluctuations and block means. The sieve bound below gives $\operatorname{osc}_{I_m} E \ll 1/(m \log m)$, so the sum of within-block variances is finite. Hence $E \in L^2$ if and only if $\sum_{m \ge 2} w_m \mu_m^2 < \infty$. The finite initial interval $[2, 4]$ has no effect. The question here is whether local arithmetic control of adjacent means yields that global sum.

49.1 An unconditional Dirichlet estimate for the actual primes

We use the classical inequality

$$\pi(x + h) - \pi(x) < \frac{2h}{\log h} \qquad (x > 0, \ h > 1),$$

namely Montgomery–Vaughan, *The large sieve*, Mathematika **20** (1973), 119–134, Theorem 2, equation (1.12), printed p. 121 [30]. The original author-hosted PDF is https://personal.science.psu.edu/rcv4/personal/Publications/large_sieve.pdf.

Proposition 49.1 (The actual neighboring block energy is finite)**.** *Unconditionally,*

$$\mu_m \longrightarrow 0, \qquad \sum_{m \ge 2} (2m + 1) |\mu_{m+1} - \mu_m|^2 < \infty.$$

More precisely,

$$|\mu_{m+1} - \mu_m| \ll \frac{1}{m \log m}, \qquad \sum_{m \ge M} (2m + 1) |\mu_{m+1} - \mu_m|^2 \ll \frac{1}{\log M} \quad (M \ge 3).$$

Proof. The union $I_m \cup I_{m+1}$ has left endpoint $a = m^2$ and length $h = 4m + 4$. For any $a \le u < v \le a + h$,

$$E(v) - E(u) = \sum_{u < p \le v} \frac{1}{p} - \int_u^v \frac{dt}{t \log t}.$$

The two terms being subtracted are nonnegative. The first is at most $2h/(a \log h)$, and the second is at most $h/(a \log a)$. Since $h \le a^2$ for $m \ge 2$, the first bound is at least the second. Thus

$$\operatorname{osc}_{I_m \cup I_{m+1}} E \le \frac{2h}{a \log h} \ll \frac{1}{m \log m}.$$

The two means lie between the same infimum and supremum. The stated energy and tail bounds follow from $\sum_{m \ge M} 1/(m \log^2 m) \ll 1/\log M$. Finally, Mertens' theorem gives $E(x) \to 0$, hence $\mu_m \to 0$. □

49.2 The exact reciprocal recurrence still contains a primitive

Define the genuine prime-block discrepancy

$$d_m = \sum_{m^2<p\le(m+1)^2} \frac{1}{p} - \log\left(\frac{\log((m+1)^2)}{\log(m^2)}\right),$$

and the within-block displacement

$$A_m = \frac{1}{w_m}\left[\sum_{m^2<p\le(m+1)^2} \frac{(m+1)^2-p}{p} - \int_{m^2}^{(m+1)^2} \frac{(m+1)^2-t}{t\log t}\,dt\right].$$

Fubini's theorem and the reciprocal-prime recurrence give

$$\mu_m = E(m^2) + A_m = -\sum_{k=m}^{\infty} d_k + A_m. \tag{49.1}$$

The tail sum is an ordinary telescoping limit, justified by $E((K+1)^2) \to 0$. The same sieve oscillation bound shows

$$|A_m| \ll \frac{1}{m\log m}, \qquad \sum_{m\ge2} w_m|A_m|^2 < \infty.$$

Thus exact reciprocity changes the global block-mean problem into a tail-sum operator on the signed arithmetic discrepancies d_m, up to a square-summable error. It does not remove that tail-sum operator. Absolute local bounds on d_m do not supply the needed cancellation of its tails.

49.3 There are infinitely many independent modes of arbitrarily small energy

Let

$$\mathcal{H}_w = \ell^2(\{2,3,\dots\}, w), \qquad \|f\|_{\mathcal{H}_w}^2 = \sum_{m\ge2} w_m|f_m|^2, \qquad Q(f) = \sum_{m\ge2} w_m|f_{m+1}-f_m|^2.$$

Since $Q(f) \le 4\|f\|_{\mathcal{H}_w}^2$, this quadratic form defines a bounded nonnegative operator $\mathcal{A}$. Its formula on finitely supported sequences is

$$(\mathcal{A}f)_2 = f_2 - f_3, \qquad (\mathcal{A}f)_m = \frac{w_m(f_m-f_{m+1}) + w_{m-1}(f_m-f_{m-1})}{w_m} \quad (m>2).$$

The associated bilinear form gives $\|\mathcal{A}f\|_{\mathcal{H}_w} \le 2\sqrt{Q(f)}$.

Theorem 49.2 (No finite-dimensional correction produces coercivity)**.** *There exist finitely supported, mutually orthonormal real sequences $z^{(j)} \in \mathcal{H}_w$, with supports tending to infinity, such that*

$$\sum_m w_m z_m^{(j)} = 0, \qquad Q(z^{(j)}) \longrightarrow 0, \qquad \|\mathcal{A}z^{(j)}\|_{\mathcal{H}_w} \longrightarrow 0.$$

More strongly, for every finite-dimensional subspace $F \subset \mathcal{H}_w$, every prescribed finite initial segment, and every $\varepsilon > 0$, there is a finitely supported z vanishing on that segment such that

$$\|z\|_{\mathcal{H}_w} = 1, \qquad z \perp F, \qquad \sum_m w_m z_m = 0, \qquad Q(z) < \varepsilon.$$

Consequently no estimate of the form

$$\|f\|_{\mathcal{H}_w}^2 \le CQ(f) + C\|\Pi_F f\|_{\mathcal{H}_w}^2$$

can hold for all finitely supported f, for any finite C and finite-dimensional F. Removing finitely many moments or modes cannot repair this lack of coercivity.

Proof. Choose a nonconstant real function $\phi \in C^2(\mathbb{R})$ supported on $[1,2]$; for example $\phi(t) = (t-1)^3(2-t)^3$ there and 0 outside. For an integer $M \geq 3$, put

$$g_m = \phi(m/M), \qquad b_m^{(M)} = \frac{M(g_{m+1} - g_m)}{2m+1}.$$

The support is contained in $[M, 2M-1]$, and telescoping gives $\sum_m w_m b_m^{(M)} = 0$. Uniform differentiation estimates give

$$|b_m^{(M)}| \ll M^{-1}, \qquad |b_{m+1}^{(M)} - b_m^{(M)}| \ll M^{-2}.$$

Also, by a Riemann-sum limit,

$$\|b^{(M)}\|_{\mathcal{H}_w}^2 = M^2 \sum_m \frac{(g_{m+1} - g_m)^2}{2m+1} \longrightarrow \int_1^2 \frac{|\phi'(t)|^2}{2t}\, dt > 0.$$

After normalizing to $z^{(M)}$, the same pointwise bounds hold, and $Q(z^{(M)}) \ll M^{-2}$. Taking $M = 4^j$ makes supports and the adjacent edges disjoint. The normalized sequences are orthonormal, and the operator estimate proves the assertion about $\mathcal{A}$.

If $\dim F = d$, take $d+1$ such packets sufficiently far out that each has energy less than ε. Their span has a unit vector orthogonal to F. Both its norm and its energy are sums of the corresponding coefficient-weighted quantities, because supports and edges are disjoint. Its energy is therefore less than ε, and its weighted mean remains exactly zero. The same dimension argument accommodates any additional finite list of linear moment constraints on finitely supported sequences. Taking $\varepsilon < 1/C$ contradicts the proposed coercive inequality. □

A direct limitation for logarithmic budgets. The obstruction is not confined to a uniform norm inequality. Let $M_j = 4^j$, and normalize the packets in the proof. Then

$$f = \sum_j \sqrt{M_j}\, z^{(M_j)}$$

is a well-defined real sequence, since the supports are disjoint. It satisfies

$$f_m = O(m^{-1/2}), \qquad f_{m+1} - f_m = O(m^{-3/2}), \qquad Q(f) < \infty.$$

Every completed packet has weighted mean zero and contributes M_j to the squared norm. Hence the finite-prefix variances

$$B_R(f) = \min_c \sum_{m=2}^{R} w_m (f_m - c)^2$$

satisfy $B_R(f) \gg R$ for all sufficiently large R: at $R = 2M_j - 1$ the variance is $\sum_{i \leq j} M_i$, and between these indices monotonicity of minimum variance and $M_{j+1} = 4M_j$ give the bound. Thus $B_R(f)/\log R \to \infty$. In particular, even decay faster than every inverse power of $\log m$, finite local Dirichlet energy, and exact zero weighted means for each distant packet do not imply recurrent logarithmic budgets.

This is a functional counterexample. It is not asserted to be the block means of the ordinary primes or of an exact reciprocal prime sum. Theorem 49.2 excludes a proposed generic coercive step; it does not disprove the original arithmetic target.

49.4 The sharp power-weighted Hardy inequality that does hold

Theorem 49.3 (Critical weighted Hardy inequality)**.** *For every finitely supported sequence f,*

$$\sum_{m\geq 2}(2m+1)|f_m|^2 \leq 2\sum_{m\geq 2}(m+1)^3|f_{m+1}-f_m|^2. \tag{49.2}$$

It also holds if $f_m \to 0$ and the right side is finite. The constant 2 is optimal. For every $\sigma > 0$, replacing the weight $(m+1)^3$ by $(m+1)^{3-\sigma}$ makes any analogous uniform inequality false, even after any finite-dimensional correction.

Proof. Write $S = \sum_{m\geq 2}(2m+1)|f_m|^2$. Summation by parts gives

$$S + 4|f_2|^2 = \sum_{m\geq 2}(m+1)^2(|f_m|^2 - |f_{m+1}|^2).$$

Cauchy–Schwarz bounds the right side by

$$\left(\sum_{m\geq 2}(m+1)^3|f_m - f_{m+1}|^2\right)^{1/2}\left(\sum_{m\geq 2}(m+1)|f_m + f_{m+1}|^2\right)^{1/2}.$$

The second squared factor is at most $2S$, since

$$2\sum_{m\geq 2}(m+1)(|f_m|^2 + |f_{m+1}|^2) = 2S - 4|f_2|^2.$$

Dropping $4|f_2|^2$ proves (49.2).

For a sequence tending to zero, apply the finite-support result to $f_m - f_{M+1}$ for $2 \leq m \leq M$, extended by zero for $m > M$, and use Fatou's lemma as $M \to \infty$. For optimality, take $f_m = m^{-1-\delta}$ with $\delta \downarrow 0$. Its two sums are respectively

$$S = \delta^{-1} + O(1), \qquad \sum_{m\geq 2}(m+1)^3|f_{m+1} - f_m|^2 = \frac{(1+\delta)^2}{2\delta} + O(1),$$

so their ratio tends to 2.

Finally, the packets $z^{(M)}$ used above satisfy

$$\sum_m (m+1)^{3-\sigma}|z^{(M)}_{m+1} - z^{(M)}_m|^2 \ll M^{-\sigma}.$$

The finite-dimensional argument in Theorem 49.2 therefore applies unchanged. □

The actual-prime estimate in Proposition 49.1 controls the weight $2m+1$, not the critical cubic weight. Replacing the former by the latter is not a valid inference. No finite cubic-weighted estimate for the actual prime block means has been proved here, and that stronger hypothesis is not used as a premise or silently substituted for the original target.

49.5 A dilation has a gap, but requires a new arithmetic input

The preceding obstruction is specific to transferring a nearest-neighbor estimate. It does not preclude a different functional bridge. Define Tf on $\mathcal{H}_w$ by $(Tf)_m = f_{2m}$. Then

$$\|Tf\|^2_{\mathcal{H}_w} = \sum_{m\geq 2} w_m|f_{2m}|^2 \leq \frac{5}{9}\sum_{m\geq 2} w_{2m}|f_{2m}|^2 \leq \frac{5}{9}\|f\|^2_{\mathcal{H}_w}.$$

Thus $I - T$ is invertible on $\mathcal{H}_w$, with

$$\|(I-T)^{-1}\| \le (1-\sqrt{5/9})^{-1}.$$

Moreover, the converse applies without assuming $\mu \in \mathcal{H}_w$: if $\mu_m \to 0$ and $d_m^* = \mu_m - \mu_{2m} \in \mathcal{H}_w$, then

$$\sum_{k=0}^{K} T^k d^* = (\mu_m - \mu_{2^{K+1}m})_{m\ge 2}.$$

The left side converges in $\mathcal{H}_w$, while each coordinate of the right side tends to μ_m. Hence $\mu \in \mathcal{H}_w$. This completely proves the functional implication.

It does not prove the arithmetic premise $\sum_m w_m|\mu_m - \mu_{2m}|^2 < \infty$. The sieve estimate on the enlarged interval only gives $|\mu_m - \mu_{2m}| \ll 1/\log m$; a classical PNT estimate gives arbitrarily high inverse powers of $\log m$, but neither bound makes that weighted sum convergent. Dilation removes the operator obstruction while placing an unproved requirement on correlations between widely separated prime blocks.

Conclusion and scope. The combination establishes an actual-prime adjacent-block Dirichlet estimate, an exact reciprocal forcing factorization, an infinite-dimensional obstruction to finite-rank coercivity, and a sharp critical Hardy inequality. A separate dilation bridge is rigorously valid, including its converse without a circular square-integrability assumption. None of these results supplies the missing long-range arithmetic estimate. The exact original claim

$$V(p_{N+1}) \le \frac{\log p_N}{8\pi^2} \quad \text{for infinitely many } N$$

therefore remains unproved. No historical-priority claim is made.

50 The unchanged correlation target

All primes are ordinary primes and all logarithms are natural. Put

$$P(x) = \pi(x) - \mathrm{Li}_2(x), \qquad \mathrm{Li}_2(x) = \int_2^x \frac{dt}{\log t}, \qquad E(x) = \sum_{p\le x} \frac{1}{p} - \log\log x - B_{\mathbb{P}}.$$

Here $B_{\mathbb{P}}$ is the prime Mertens constant, and Mertens' theorem gives $E(x) \to 0$. Functions are right-continuous at primes. Define

$$g_4(x) = E(x) - E(4x) = \log\frac{\log(4x)}{\log x} - \sum_{x<p\le 4x} \frac{1}{p}, \qquad H_4(X) = \int_2^X g_4(x)^2 dx.$$

The finite-cutoff kernel and its atomic diagonal are

$$K_X(u,v) = [\min(X,u,v) - \max(2,u/4,v/4)]_+, \qquad D_4(X) = \sum_{2<p\le 4X} \frac{K_X(p,p)}{p^2}.$$

The same compensated correlation as in the original target is

$$R_4(X) = \sum_{\substack{2<p,q\le 4X\\ p\ne q}} \frac{K_X(p,q)}{pq} - 2\sum_{2<p\le 4X} \frac{1}{p}\int_2^{4X} \frac{K_X(p,t)}{t\log t}\,dt + \int_2^{4X}\int_2^{4X} \frac{K_X(u,v)}{uv\log u\log v}\,du\,dv.$$

Thus no new sensor or replacement objective is being substituted. The target is

$$\mathsf{B}: \qquad \sup_{X\ge 8}\left(R_4(X)+\frac{3}{4}\log\log X\right)<\infty. \tag{50.1}$$

The restriction to $X \ge 8$ only removes a compact interval. All finite-cutoff expressions are locally bounded.

Lemma 50.1 (The compensated sum is the original window energy)**.** *For every finite $X > 2$, $H_4(X) = D_4(X) + R_4(X)$, and*

$$D_4(X)=\frac{3}{4}\log\log X + C_D + o(1), \qquad C_D=\frac{3}{4}B_{\mathbb{P}}-\frac{3}{8}-\frac{5}{36}-\frac{3}{100}-\frac{1}{196}.$$

Consequently B *is equivalent to $H_4(\infty) < \infty$.*

Proof. On $(2, 4X]$, let $\eta(dt) = \sum_{2<p\le 4X} p^{-1}\delta_p(dt) - dt/(t\log t)$. The Gram identity

$$K_X(u,v)=\int_2^X \mathbf{1}_{x<u\le 4x}\mathbf{1}_{x<v\le 4x}\,dx$$

and finite-total-variation Fubini give $H_4(X) = \iint K_X\, d\eta\, d\eta = D_4(X) + R_4(X)$. For $X \ge 8$, the diagonal is exactly

$$D_4(X)=\sum_{p=3,5,7}\frac{p-2}{p^2}+\frac{3}{4}\sum_{8<p\le X}\frac{1}{p}+\sum_{X<p\le 4X}\frac{X-p/4}{p^2}.$$

The last sum is $O(1/\log X)$ by Chebyshev's bound, or by Brun–Titchmarsh [30]. Mertens' theorem gives the stated asymptotic and constant. Since H_4 is nonnegative and nondecreasing, the boundedness equivalence follows. No separately divergent infinite double sum is exchanged. □

Lemma 50.2 (Exact equality of the two centered variances)**.** *For every real $X > 2$,*

$$V(X):=\min_{a\in\mathbb{R}}\int_2^X\frac{(P(t)-a)^2}{t^2}\,dt=\min_{c\in\mathbb{R}}\int_2^X (E(t)-c)^2\,dt. \tag{50.2}$$

Put

$$Q(X)=\int_2^X\frac{P(t)^2}{t^2}\,dt, \qquad A(X)=\int_2^X\frac{P(t)}{t^2}\,dt, \qquad b_0=\log\log 2+B_{\mathbb{P}}.$$

Then

$$Q(X)-V(X)=\frac{2X}{X-2}A(X)^2, \qquad A(X)=b_0+E(X)-\frac{P(X)}{X}\longrightarrow b_0. \tag{50.3}$$

In particular $Q(X) = V(X) + O(1)$, uniformly for $X \ge 3$.

Proof. Let $J(t) = \int_2^t E(u)\,du$. Stieltjes partial summation gives

$$P(t)=tE(t)-J(t)+2b_0. \tag{50.4}$$

Indeed, both sides have Stieltjes differential $t\,dE(t)$ and agree at $t = 2$, where $P(2) = 1$ and $E(2) = 1/2 - b_0$. For a real c, set $J_c(t) = J(t) - c(t-2)$. This function is absolutely continuous, $J_c(2) = 0$, and

$$\frac{P(t)-2b_0-2c}{t}=J_c'(t)-\frac{J_c(t)}{t} \quad \text{almost everywhere.}$$

The identity $(J_c' - J_c/t)^2 = (J_c')^2 - (J_c^2/t)'$ therefore proves

$$\int_2^X (E-c)^2dt=\int_2^X\frac{(P-2b_0-2c)^2}{t^2}\,dt+\frac{(J(X)-c(X-2))^2}{X}.$$

The left side is minimized at $c_X = J(X)/(X-2)$, where the last term vanishes. Integrating (50.4) after division by t^2 gives

$$A(X) = 2b_0\left(\frac{1}{2} - \frac{1}{X}\right) + \frac{J(X)}{X}.$$

Thus the corresponding $a_X = 2b_0 + 2c_X$ is also the weighted mean of P for the measure $t^{-2}dt$, and minimizes the other integral. This proves (50.2).

Expanding that weighted variance, whose total mass is $(X-2)/(2X)$, gives the first identity in (50.3). Combining the displayed formula for $A(X)$ with (50.4) gives its second identity. Mertens' theorem gives $E(X) \to 0$, and Chebyshev's estimate $\pi(X) = O(X/\log X)$, together with the elementary analogous bound for $\mathrm{Li}_2(X)$, gives $P(X)/X \to 0$. The bounded-correction conclusion follows. □

Proposition 50.3 (Finite energy retains the target exactly)**.** *The following assertions are equivalent:*

$$\mathsf{B}, \qquad H_4(\infty) < \infty, \qquad E \in L^2([2,\infty), dx), \qquad \sup_{X>2} V(X) < \infty, \qquad Q(\infty) < \infty.$$

Proof. On $L^2(dx)$, $Uf(x) = f(4x)$ has norm $1/2$. Thus $E \in L^2$ implies $g_4 = (I-U)E \in L^2$. Conversely, if $g_4 \in L^2$, then

$$\sum_{k=0}^{M} g_4(4^k x) = E(x) - E(4^{M+1}x) \to E(x), \qquad \left\| \sum_{k=0}^{M} g_4(4^k x) \right\|_2 \le 2\|g_4\|_2.$$

Mertens' limit and Fatou's lemma give $E \in L^2$, without assuming it in advance. If V is bounded, its minimizing centers $c_X = (X-2)^{-1}\int_2^X E$ tend to zero. For any fixed Y, pass to the limit in $\int_2^Y (E - c_X)^2 \le V(X)$, then let $Y \to \infty$. This proves the reverse implication from bounded variance. The other direction uses the trial center zero. Finally, Lemma 50.2 gives $Q = V + O(1)$, and Lemma 50.1 supplies the correlation equivalence. □

The original preceding endpoint target is also retained:

$$V(p_{N+1}) \le \frac{\log p_N}{8\pi^2} \quad \text{for infinitely many } N. \tag{50.5}$$

Theorem 52.1 and Appendix 51 show that (50.5), B, and RH are equivalent. Indeed, (50.5) gives $Q(p_{N+1}) = O(\log p_{N+1})$ along an unbounded prime subsequence. Plancherel then implies the fixed-band subpower condition of that theorem. Conversely, finite $Q(\infty)$ makes V bounded, so (50.5) holds at every sufficiently large prime endpoint. None of these equivalent assertions is assumed as an unconditional premise.

51 The conditional finite-energy implication used above

Lemma 51.1. *If RH holds, then* $Q(\infty) = \int_2^\infty P(x)^2 x^{-2} dx < \infty$.

Proof. Here is a compact derivation recording all required terms. For $\Re s > 1$, define

$$F(s) = \int_2^\infty P(x) x^{-s-1} dx, \quad L_0(s) = \int_2^\infty \frac{x^{-s}}{\log x} dx, \quad Q_{\mathrm{pp}}(s) = \sum_{k\ge 2} \frac{\mathcal{P}(ks)}{k}.$$

Abel summation and the Euler product give

$$sF(s) = \log \zeta(s) - Q_{\mathrm{pp}}(s) - L_0(s).$$

The logarithm here is only the Euler-product branch on $\Re s > 1$. The established exact factorization [32, (1.2)–(1.6)] is

$$\frac{s-1}{s}\zeta(s) = e^{\ell(s)}B(s), \qquad \ell(s) = \frac{2}{\pi}(s-\tfrac{1}{2})\int_0^\infty \frac{\log|\zeta(1/2+it)|}{t^2+(s-1/2)^2}dt, \quad \Re s > 1/2.$$

It includes the full Blaschke factor of the off-line zeros and has no omitted singular inner factor. Under RH, $B = 1$. On $\Re s > 1$ we consequently have

$$F(s) = \frac{\ell(s) + k_0(s) - Q_{\mathrm{pp}}(s)}{s}, \qquad k_0(s) = \log\frac{s}{s-1} - L_0(s).$$

Each term on the right has finite norm in

$$\mathcal{H} = H^2(\Re s > 1/2), \qquad \|H\|_{\mathcal{H}}^2 = \sup_{\sigma>1/2}\frac{1}{2\pi}\int_{\mathbb{R}}|H(\sigma+it)|^2dt.$$

For completeness, the inverse Mellin transform of $Q_{\mathrm{pp}}(s)/s$ is

$$R_{\mathrm{pp}}(x) = \sum_{k\geq 2}\frac{\pi(x^{1/k})}{k} \ll \frac{\sqrt{x}}{\log x} + x^{1/3} \quad (x \geq 4),$$

and it vanishes on $[1, 4)$. Chebyshev's estimate proves this bound, and its square is integrable against $x^{-2}dx$. The inverse transform of $k_0(s)/s$ is

$$v(x) = \begin{cases} \displaystyle\int_1^x \frac{t-1}{t\log t}dt, & 1 \leq x \leq 2, \\ \displaystyle v(2) - \log\frac{\log x}{\log 2}, & x \geq 2. \end{cases}$$

Frullani's identity and integration by parts prove the formula; $v \in L^2([1,\infty), x^{-2}dx)$ is immediate.

Finally,

$$M_2 = \frac{1}{2\pi}\int_{\mathbb{R}}\frac{\log^2|\zeta(1/2+it)|}{1/4+t^2}dt < \infty$$

unconditionally. Indeed, the local formula for ζ'/ζ and the $O(\log T)$ count of zeros in a unit interval [34, Lemmas 9.4 and 9.6] give, by horizontal integration and Cauchy–Schwarz,

$$\int_T^{T+1}\log^2|\zeta(1/2+it)|dt \ll \log^2 T.$$

In detail, the logarithm is a sum of $O(\log T)$ terms $\log|1/2+it-\rho|$, plus $O(\log T)$. Each term has uniformly bounded squared integral on a unit interval: its negative part is bounded by $|\log|t-\Im\rho||$, and its positive part is bounded. Compact ranges contain only finitely many such logarithmic singularities. The weight T^{-2} proves convergence. The disk map $z = 1 - 1/s$ has boundary measure $d\theta = |dt|/(1/4+t^2)$. The analytic Schwarz extension of these real data has disk squared norm $2M_2 - |\ell(1)|^2$. The conformal Hardy isometry therefore gives $\ell(s)/s \in \mathcal{H}$.

The Mellin–Paley–Wiener isometry now produces a real $g \in L^2([1,\infty), x^{-2}dx)$ with the displayed transform. It agrees with the transform of P_0, where $P_0 = 0$ on $[1, 2)$ and $P_0 = P$ otherwise. For any fixed $\sigma > 1$, both $e^{-\sigma u}g(e^u)$ and $e^{-\sigma u}P_0(e^u)$ are integrable on $[0,\infty)$. Fourier uniqueness identifies them almost everywhere. Hence $g = P_0$ and $Q(\infty) < \infty$. This is a conditional implication from RH, not a proof of RH. □

Part IV
Critical spectral energy and frequency localization

52 Independent sparse control on each fixed frequency band

For $X > 2$, define the entire function

$$F_X(s) = \int_2^X P(x)x^{-s-1}\,dx, \qquad I_T(X) = \int_{-T}^{T} |F_X(\tfrac{1}{2}+it)|^2\,dt \quad (T>0).$$

The finite endpoint X, the band radius T, and the frequency variable t are independent parameters.

Theorem 52.1 (A fixed-band subpower criterion)**.** *The following three statements are equivalent:*

1. *RH holds.*
2. *For every finite $T > 0$, $\liminf_{X\to\infty} I_T(X) < \infty$.*
3. *For every finite $T > 0$,*
$$\liminf_{X\to\infty} \frac{\log(1+I_T(X))}{\log X} = 0. \tag{52.1}$$

Each band may have its own unbounded sequence of endpoints. No common endpoint sequence, no uniform bound in T, and no growth bound for the constants in assertion 2 are required. The theorem remains valid if endpoints are restricted throughout to any prescribed unbounded subset of $(2,\infty)$, including the ordinary prime endpoints $X = p_{N+1}$.

We first isolate the analytic estimate used in the proof.

Lemma 52.2 (Small growth inside a sufficiently tall strip rectangle)**.** *Let $B > 1$, and let K be a compact subset of $\{1/2 < \Re s < B\}$. For every $\varepsilon > 0$, there is a finite T, depending on B, K, ε, with the following property. If $X_j \to \infty$ and*

$$\log(1 + I_T(X_j)) = o(\log X_j),$$

then

$$\sup_{s\in K} |F_{X_j}(s)| \ll_{B,K,T,\varepsilon} X_j^{\varepsilon} \tag{52.2}$$

for all sufficiently large j. If K is a fixed neighborhood of another compact set K_0, the same bound holds on K_0 for $G_{X_j}(s) = (sF_{X_j}(s))'$.

Proof. Only the elementary bound $P(x) = O(x)$ is needed here. Set $u_X(s) = \log(1 + |F_X(s)|)$, a nonnegative subharmonic function, and consider

$$\mathcal{R}_T = \{1/2 < \Re s < B,\ |\Im s| < T\}.$$

On its right edge, $u_X \le C_B$, independently of X. On either horizontal edge, the uniform crude bound $F_X(s) = O(\sqrt{X})$, valid for $\Re s \ge 1/2$, gives

$$u_X(s) \le \tfrac{1}{2}\log X + C_0.$$

Here is an explicit bound for the horizontal harmonic measure. Write $W = B - 1/2$, $m = (B + 1/2)/2$, $\kappa = \pi/(2W)$, and

$$v_T(s) = \sqrt{2}\cos\bigl(\kappa(\Re s - m)\bigr)\frac{\cosh(\kappa \Im s)}{\cosh(\kappa T)}.$$

This function is nonnegative and harmonic on the rectangle; it is at least one on both horizontal edges. Also $\sup_K v_T \to 0$ as $T \to \infty$.

For the left-edge data, use its Poisson extension in the half-plane $\Re s > 1/2$, setting the data to zero outside $[-T, T]$:

$$U_X(s) = \frac{1}{\pi}\int_{-T}^{T} \frac{\Re s - 1/2}{(\Re s - 1/2)^2 + (\Im s - t)^2} u_X(\tfrac{1}{2} + it)\, dt.$$

This harmonic function is nonnegative. If $\delta = \min_{s\in K}(\Re s - 1/2) > 0$, then

$$\sup_K U_X \le \frac{1}{\pi\delta}\int_{-T}^{T} u_X(\tfrac{1}{2} + it)\, dt.$$

Concavity of $\log(1 + y)$, followed by Cauchy–Schwarz, gives

$$\int_{-T}^{T} u_X(\tfrac{1}{2} + it)\, dt \le 2T \log\left(1 + \sqrt{\frac{I_T(X)}{2T}}\right) \ll_T 1 + \log(1 + I_T(X)).$$

Thus along the chosen sequence $\sup_K U_{X_j} = o(\log X_j)$.

The subharmonic maximum principle, applied with the harmonic majorant $U_X + C_B + (\frac{1}{2}\log X + C_0)v_T$, yields

$$\sup_K u_{X_j} \le o(\log X_j) + C_B + (\tfrac{1}{2}\log X_j + C_0)\sup_K v_T.$$

The two corner points cause no difficulty: their harmonic measure is zero, or they may be removed and recovered by a limit. Choose T large enough that $\sup_K v_T < \varepsilon$. The right side is at most $\varepsilon \log X_j + O(1)$ for sufficiently large j, which proves (52.2). Cauchy's derivative estimate on a fixed neighborhood of K_0 gives the assertion for G_{X_j}. □

Proof of Theorem 52.1. Lemma 51.1 in Appendix 51 gives RH $\Rightarrow Q(\infty) < \infty$. More explicitly, under RH the Blaschke factor is one; the outer logarithm, prime-power correction, and deterministic main-term correction all have finite critical Hardy norm, and uniqueness of their Mellin transform identifies the inverse with P. Plancherel for the finite transform then gives

$$\int_{\mathbb{R}} |F_X(\tfrac{1}{2} + it)|^2\, dt = 2\pi\int_2^X \frac{P(x)^2}{x^2}\, dx \le 2\pi Q(\infty).$$

This proves assertion 2, even with a uniform bound in X, T. Assertion 2 implies assertion 3 immediately.

Suppose assertion 3 holds, and assume for contradiction that $\zeta(\rho) = 0$ with $\Re\rho > 1/2$. Initially for $\Re s > 1$, the Euler product and Abel summation give

$$G(s) = \left(s\int_2^\infty P(x)x^{-s-1}dx\right)' = \frac{\zeta'(s)}{\zeta(s)} - \sum_{k\ge 2}\mathcal{P}'(ks) + \frac{2^{1-s}}{s-1}. \tag{52.3}$$

Here $\mathcal{P}(s) = \sum_p p^{-s}$ initially for $\Re s > 1$. For every $\sigma_0 > 1/2$,

$$\sum_{k\ge 2}\sum_p (\log p)p^{-k\sigma_0} = \sum_p \frac{(\log p)p^{-2\sigma_0}}{1 - p^{-\sigma_0}} < \infty,$$

so the prime-power sum is normally convergent on $\Re s > 1/2$. The principal parts at $s = 1$ cancel. Consequently G is meromorphic throughout that half-plane, with a simple pole at each zeta zero there, whose residue is the positive integer multiplicity of the zero.

Choose a fixed closed rectangle $\mathcal{R}$ containing ρ in its interior, with left edge $\Re s = \alpha$, where $1/2 < \alpha < \Re\rho$, and right edge $\Re s = b > 1$. Choose its left and horizontal edges to avoid all zeta zeros. Let $\rho_1, \dots, \rho_m$ be its distinct interior zeros and set

$$\mathcal{C}(s) = \prod_{r=1}^{m}(s - \rho_r), \qquad H_X(s) = \mathcal{C}(s)(G_X(s) - G(s)).$$

Thus H_X is holomorphic on a neighborhood of the closed rectangle. If $\rho = \rho_r$ has multiplicity m_ρ, then, for every X,

$$H_X(\rho) = -m_\rho \prod_{\ell \neq r}(\rho - \rho_\ell) \neq 0. \tag{52.4}$$

Let $\omega \in (0, 1)$ be the harmonic measure of the right edge of $\mathcal{R}$, evaluated at ρ. Fix

$$0 < \varepsilon < \frac{\omega(b-1)}{1-\omega}.$$

Choose $B > b$ and a compact neighborhood K of $\mathcal{R}$ inside $\{1/2 < \Re s < B\}$. Lemma 52.2, with a slightly larger compact neighborhood if necessary for Cauchy's estimate, specifies one sufficiently large but finite band radius T. Assertion 3 supplies a sequence $X_j \to \infty$ for this particular T such that $\log(1 + I_T(X_j)) = o(\log X_j)$. The lemma gives $G_{X_j} = O(X_j^{\varepsilon})$ on the boundary of $\mathcal{R}$. Since G has no poles on that boundary,

$$|H_{X_j}(s)| \ll X_j^{\varepsilon} \quad (s \in \partial\mathcal{R}).$$

On its right edge the defining tail integral gives the strictly stronger estimate

$$G_{X_j}(s) - G(s) = -\int_{X_j}^{\infty} P(x)x^{-s-1}(1 - s\log x)\,dx = O(X_j^{1-b}\log X_j).$$

The two-constants theorem, or the subharmonic maximum principle for $\log|H_{X_j}|$, therefore gives

$$\log|H_{X_j}(\rho)| \leq \left[-\omega(b-1) + (1-\omega)\varepsilon\right]\log X_j + \omega\log\log X_j + O(1) \longrightarrow -\infty.$$

This contradicts the fixed nonzero value (52.4). Thus no zero has real part greater than $1/2$; the functional equation proves RH. Only one band-dependent endpoint sequence was used for each hypothetical zero. All steps hold on any prescribed unbounded endpoint set, proving the final assertion. □

What has not been proved. This theorem weakens the quantifiers of a sufficient analytic premise while preserving the original RH-equivalent target. It does not establish (52.1) for the actual primes. Finite computation on any collection of bands and endpoints cannot establish the displayed limiting assertion. No claim of historical priority is attached to this criterion.

Corollary 52.3 (A fixed finite band would detect an off-line zero)**.** *If RH is false, there are finite $T_0 > 0$, $\delta > 0$, and X_0 such that*

$$I_{T_0}(X) \geq X^{\delta} - 1 \qquad (X \geq X_0).$$

Proof. The nonnegative ratios in (52.1) cannot have liminf zero for every T if RH is false. Choose a band with positive liminf and take δ strictly below that liminf. The stated eventual lower bound follows from the definition of liminf. An infinite liminf causes no difficulty. □

Thus a failure of the target cannot be accounted for solely by energy moving into ever higher frequency bands. This is a conditional detection statement, not a proof that the fixed-band upper bounds hold. It is sufficient to prove, for every $T, \varepsilon, X_0 > 0$, the existence of $X \geq X_0$ with $I_T(X) \leq X^{\varepsilon}$; a finite collection of tests does not discharge these quantifiers.

53 The critical prime measure without an assumed Fourier transform

Let $t_0 = \log 2$, $L = \log 4$, and extend all the following measures to the full real line. We include the prime 2:

$$\mu(dt) = \sum_p p^{-1/2}\delta_{\log p}(dt) - \mathbf{1}_{[t_0,\infty)}(t)\frac{e^{t/2}}{t}\,dt. \tag{53.1}$$

This is initially only a locally finite signed measure, hence a distribution in $\mathcal{D}'(\mathbb{R})$. Temperedness is not assumed. Adding the prime 2 changes the version with $p > 2$ by one compactly supported atom and does not change any window $(x, 4x]$ for $x \geq 2$.

Define the compactly supported kernel

$$k(t) = e^{t/2}\mathbf{1}_{[-L,0]}(t), \qquad h = -k * \mu.$$

Convolution with this compactly supported kernel is well-defined as a distribution even before μ is known to be tempered. It is represented by the locally bounded function

$$h(t) = -\int_{(t,t+L]} e^{-(u-t)/2}\mu(du).$$

Endpoint conventions here do not affect its Lebesgue class. For $t \geq t_0$, it satisfies the exact identity

$$h(t) = e^{t/2}g_4(e^t), \qquad g_4(x) = \log\frac{\log(4x)}{\log x} - \sum_{x<p\leq 4x}\frac{1}{p}. \tag{53.2}$$

The function h vanishes below $t_0 - L$, and its energy on $[t_0 - L, t_0]$ is a finite fixed boundary contribution. Consequently $h \in L^2(\mathbb{R})$ is equivalent to $H_4(\infty) = \int_2^\infty g_4(x)^2 dx < \infty$.

For a tempered distribution ν, write

$$\|\nu\|^2_{-1,1/2} = \frac{1}{2\pi}\int_{\mathbb{R}}\frac{|\widehat{\nu}(\xi)|^2}{1/4+\xi^2}\,d\xi, \qquad \widehat{f}(\xi) = \int_{\mathbb{R}} f(t)e^{-i\xi t}\,dt,$$

when its Fourier transform is a function and the displayed integral is finite. The resulting Hilbert space is denoted $H^{-1}_{1/2}(\mathbb{R})$. Its dual norm on test functions is $\|\varphi\|^2_{1,1/2} = \int(|\varphi'|^2 + |\varphi|^2/4)$.

Proposition 53.1 (The original prime-counting energy is the exact norm)**.** *For the actual measure* (53.1)*,*

$$\mu \in H^{-1}_{1/2}(\mathbb{R}) \quad\Longleftrightarrow\quad Q(\infty) := \int_2^\infty \frac{P(x)^2}{x^2}\,dx < \infty,$$

and, whenever finite,

$$\|\mu\|^2_{-1,1/2} = Q(\infty), \qquad P(x) = \pi(x) - \int_2^x \frac{du}{\log u}. \tag{53.3}$$

Proof. The locally integrable function

$$c(t) = e^{-t/2}P(e^t)\mathbf{1}_{[t_0,\infty)}(t)$$

satisfies the distributional identity

$$(D + \tfrac{1}{2})c = \mu. \tag{53.4}$$

In particular the jump at t_0 is $e^{-t_0/2}P(2) = 2^{-1/2}$, exactly the prime-2 atom. If $Q(\infty) < \infty$, then $c \in L^2$, and Plancherel applied to (53.4) proves (53.3).

Conversely, if $\mu \in H^{-1}_{1/2}$, its Fourier multiplier $(1/2 + i\xi)^{-1}$ defines a unique L^2 solution c_0 of $(D + 1/2)c_0 = \mu$, with $\|c_0\|_2 = \|\mu\|_{-1,1/2}$. The difference $c - c_0$, considered in $\mathcal{D}'$, solves the

homogeneous first-order equation and is therefore $Ae^{-t/2}$ for a constant A. On $(-\infty, t_0)$, one has $c = 0$; square-integrability of c_0 there forces $A = 0$. Thus $c = c_0$, proving the converse and the exact norm. No Fourier transform of the original untempered distribution was taken before its Sobolev membership was proved. □

Proposition 53.2 (The window multiplier and the missing growth condition)**.** *For the actual primes,*

$$h \in L^2(\mathbb{R}) \quad \Longleftrightarrow \quad \mu \in H^{-1}_{1/2}(\mathbb{R}).$$

When these quantities are finite,

$$\frac{1}{2}\|\mu\|_{-1,1/2} \le \|h\|_2 \le \frac{3}{2}\|\mu\|_{-1,1/2}. \tag{53.5}$$

The reverse implication does not assume that μ is tempered.

Proof. The Fourier transform of the kernel is

$$\widehat{k}(\xi) = \frac{1 - \frac{1}{2}e^{i\xi L}}{1/2 - i\xi}, \qquad \frac{1/2}{\sqrt{1/4 + \xi^2}} \le |\widehat{k}(\xi)| \le \frac{3/2}{\sqrt{1/4 + \xi^2}}.$$

Thus $\mu \in H^{-1}_{1/2}$ implies the desired assertion and both estimates by Plancherel.

To prove the converse without circular use of Fourier analysis, let $S\nu = \nu(\cdot + L)$ denote translation. Differentiating the compact kernel in $\mathcal{D}'$ gives

$$k' = \tfrac{1}{2}k + \tfrac{1}{2}\delta_{-L} - \delta_0, \qquad (I - \tfrac{1}{2}S)\mu = (D - \tfrac{1}{2})h.$$

If $h \in L^2$, put $q = (D - 1/2)h \in H^{-1}_{1/2}$. Translation is an isometry on this space, so $\sum_{j\ge 0} 2^{-j}S^j q$ converges there, with norm at most $2\|h\|_2$. The finite identity in $\mathcal{D}'$ is

$$\mu = \sum_{j=0}^{N} 2^{-j}S^j q + 2^{-N-1}S^{N+1}\mu.$$

For every fixed compact interval I, Chebyshev's bound gives

$$|\mu|(I + nL) \ll_I \frac{2^n}{n+1} \qquad (n \longrightarrow \infty).$$

Indeed, the prime weights are $O_I(2^{-n})$, while the number of relevant primes is $O_I(4^n/(n+1))$; direct integration gives the same bound for the continuous part. It follows that $2^{-n}S^n\mu \to 0$ in $\mathcal{D}'$. The convergent Sobolev series therefore equals the original μ, proving its membership and completing the proof. □

Why a real-axis multiplier alone is insufficient. Without the preceding growth argument, the converse would be false for general locally finite measures. For example, a complex density proportional to $e^{(1/2+2\pi i/L)t}\mathbf{1}_{[0,\infty)}(t)$ has a compactly supported window output, because its complete length-L weighted window integral vanishes. The density itself is not tempered. It is excluded in the actual prime argument by the $O_I(2^n/(n+1))$ growth bound.

54 A finite Fourier test with controlled smooth cutoffs

Choose once and for all $\chi \in C^\infty(\mathbb{R})$ with $0 \le \chi \le 1$, $\chi(t) = 1$ for $t \le 0$, and $\chi(t) = 0$ for $t \ge 1$. For $T > t_0$, set

$$\chi_T(t) = \chi(t - T), \qquad \mu_T = \chi_T\mu, \qquad Y = e^{T+1}.$$

These are compactly supported finite signed measures. Their Fourier transforms are ordinary finite expressions:

$$\widehat{\mu}_T(\xi) = \sum_{p \le Y} \frac{\chi(\log p - T)}{p^{1/2+i\xi}} - \int_{t_0}^{T+1} \chi(u-T) \frac{e^{(1/2-i\xi)u}}{u}\, du. \tag{54.1}$$

Define the nonnegative number

$$\mathcal{N}(T) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{|\widehat{\mu}_T(\xi)|^2}{1/4+\xi^2}\, d\xi.$$

It is finite for each T, without any hypothesis.

Proposition 54.1 (The exact cutoff criterion)**.** *The following statements are equivalent:*

$$\mu \in H^{-1}_{1/2}, \qquad \sup_{T>t_0} \mathcal{N}(T) < \infty, \qquad \liminf_{T\to\infty} \mathcal{N}(T) < \infty.$$

For every finite T, one also has the absolutely justified kernel identity

$$\mathcal{N}(T) = \iint e^{-|u-v|/2} \mu_T(du) \mu_T(dv). \tag{54.2}$$

Proof. Let $M = \|\chi'\|_\infty$. On $H^1_{1/2}$, the product rule gives the uniform multiplier estimate

$$\|\chi_T \varphi\|_{1,1/2} \le (1+2M) \|\varphi\|_{1,1/2}.$$

By duality it holds on $H^{-1}_{1/2}$ as well. This proves that Sobolev membership implies the uniform bound. Conversely, a bounded subsequence μ_{T_j}, with $T_j \to \infty$, has a weakly convergent subsequence in the Hilbert space $H^{-1}_{1/2}$. Its distributional limit is necessarily μ, because the cutoffs equal one on each fixed compact set for all sufficiently large j. Thus μ belongs to that space. Finally,

$$\frac{1}{2\pi} \int_{\mathbb{R}} \frac{e^{i\xi(u-v)}}{1/4+\xi^2}\, d\xi = e^{-|u-v|/2}.$$

Since μ_T has finite total variation and the Fourier weight is integrable, Fubini proves (54.2). □

Every Fourier integral above is either attached to a finite measure or follows a proved Sobolev membership statement. Neither this criterion nor its proof supplies the missing uniform upper bound. The unconditional high-frequency estimate below controls one part of $\mathcal{N}(T)$; the complementary growing frequency band must still be bounded using arithmetic information that has not been established.

55 Exact finite-cutoff Green energy with neutralized endpoints

For $Y > 2$, define the finite signed measure

$$\nu_Y = \sum_{p \le Y} p^{-1/2} \delta_{\log p} - \mathbf{1}_{[\log 2, \log Y]}(s) \frac{e^{s/2}}{s}\, ds.$$

The prime 2 is included. For a real number a, put

$$\nu_{Y,a} = \nu_Y - \frac{a}{\sqrt{2}} \delta_{\log 2} - \frac{P(Y)-a}{\sqrt{Y}} \delta_{\log Y}.$$

Proposition 55.1 (Exact centered resolvent identity)**.** *For every $Y > 2$ and every real a,*

$$\|\nu_{Y,a}\|^2_{-1,1/2} = \int_2^Y \frac{(P(t)-a)^2}{t^2}\,dt.$$

Consequently the original variance is exactly

$$\boxed{V(Y) = \min_{a\in\mathbb{R}} \|\nu_{Y,a}\|^2_{-1,1/2}.}$$

For the uncorrected measure,

$$\|\nu_Y\|^2_{-1,1/2} = \int_2^Y \frac{P(t)^2}{t^2}\,dt + \frac{P(Y)^2}{Y}.$$

The right endpoint term cannot be discarded or assumed bounded.

Proof. Let

$$c_{Y,a}(s) = e^{-s/2}(P(e^s)-a)\mathbf{1}_{[\log 2,\log Y]}(s).$$

On the interior of the interval, $(D+1/2)c_{Y,a} = e^{-s/2}dP(e^s)$. The latter has prime masses $p^{-1/2}$ and continuous part $-e^{s/2}ds/s$. At the left endpoint the jump is $(1-a)/\sqrt{2}$, because $P(2)=1$. At the right endpoint, the interior prime atom, if Y is prime, combines with $-(P(Y)-a)/\sqrt{Y}$ to give exactly the jump to zero. Thus

$$(D+\tfrac{1}{2})c_{Y,a} = \nu_{Y,a}$$

as distributions, with all endpoint atoms accounted for. The same Fourier multiplier calculation as above gives

$$\|\nu_{Y,a}\|^2_{-1,1/2} = \|c_{Y,a}\|_2^2 = \int_2^Y (P(t)-a)^2 t^{-2}\,dt.$$

For ν_Y without endpoint correction, the causal solution instead continues as $P(Y)e^{-s/2}$ for $s > \log Y$. Its extra squared norm is $P(Y)^2/Y$. □

Equivalently, for every finite signed measure σ,

$$\|\sigma\|^2_{-1,1/2} = \iint e^{-|u-v|/2}\,\sigma(du)\sigma(dv),$$

because the Fourier transform of $e^{-|s|/2}$ is $(\xi^2+1/4)^{-1}$. Finite total variation justifies the Fourier and measure interchanges. This is a positive Laplace-resolvent kernel with exponential decay in the log ratio. For example, the raw atomic diagonal for ν_Y is $\sum_{p\le Y} 1/p = \log\log Y + B_{\mathbb{P}} + o(1)$. The cancellations with continuous and endpoint terms remain essential; positivity of the complete norm does not bound it.

56 A sieve improvement of the high-frequency estimate

The following estimate concerns the ordinary primes themselves. It uses the same smooth cutoff as above, includes the prime 2, and takes Fourier transforms only of finite measures. In this section R denotes a frequency threshold, whereas T denotes the logarithmic cutoff parameter and $Y = e^{T+1}$.

Lemma 56.1 (Reciprocal nearest-neighbour spacing)**.** *For $Y \ge 8$, let*

$$\delta_p^{(Y)} = \min_{\substack{q\le Y \text{ prime}\\ q\ne p}} |\log p - \log q|, \qquad \mathcal{E}(Y) = \sum_{p\le Y} \frac{1}{p\delta_p^{(Y)}}.$$

There is an absolute constant such that

$$\mathcal{E}(Y) \ll \frac{Y\log\log Y}{(\log Y)^2}. \tag{56.1}$$

Proof. Write d_p for the smaller of the two adjacent prime gaps at p, and use the one adjacent gap at $p = 2$. Including primes beyond Y can only decrease the nearest logarithmic distance. If $p_- < p < p_+$ are adjacent primes, then

$$p \log(p/p_-) \geq p - p_-, \qquad p \log(p_+/p) \geq \tfrac{1}{2}(p_+ - p),$$

where the second inequality uses Bertrand's bound $p_+ < 2p$. Consequently

$$\frac{1}{p\delta_p^{(Y)}} \leq \frac{2}{d_p}.$$

Let $G = \lfloor \log Y \rfloor$ for sufficiently large Y. The contribution from $d_p > G$ is at most $2\pi(Y)/G$. Every prime with $d_p \leq G$ can be charged to an adjacent pair at distance d_p, and each unordered pair is charged at most twice. Its smaller prime is at most Y, even if its larger prime exceeds Y. Thus, on putting

$$N_Y(d) = \#\{n \in \mathbb{N} : n \leq Y,\ n \text{ and } n + d \text{ are prime}\},$$

we have

$$\mathcal{E}(Y) \leq \frac{2\pi(Y)}{G} + 4 \sum_{1 \leq d \leq G} \frac{N_Y(d)}{d}. \tag{56.2}$$

For even $d \leq G$, the uniform upper-bound sieve gives

$$N_Y(d) \ll \mathfrak{S}(d) \frac{Y}{(\log Y)^2}, \qquad \mathfrak{S}(d) = 2C_2 \prod_{\substack{\ell \mid d \\ \ell > 2 \text{ prime}}} \frac{\ell - 1}{\ell - 2}, \quad C_2 = \prod_{\ell > 2} \left(1 - \frac{1}{(\ell - 1)^2}\right). \tag{56.3}$$

The implied constant is independent of d and Y. One precise source is [35, Lemma 2.3], applied with $k = 2$ and shifts $0, d$; that lemma permits distinct shifts throughout $[-Y, Y]$, uniformly, and records the upper factor $2!2^2(1 + O(\log\log Y/\log Y))$. We apply it only to the admissible even shifts. For odd d, one of the two primes must be 2, so $N_Y(d) \leq 1$.

The required weighted singular-series average has an elementary proof. If $a(r) = \prod_{\ell \mid r} (\ell - 2)^{-1}$ for odd squarefree r, then

$$\prod_{\substack{\ell \mid d \\ \ell > 2}} \frac{\ell - 1}{\ell - 2} = \sum_{\substack{r \mid d \\ r \text{ odd squarefree}}} a(r).$$

Dropping the parity restriction on d for an upper bound gives

$$\begin{aligned} \sum_{\substack{d \leq G \\ 2 \mid d}} \frac{\mathfrak{S}(d)}{d} &\leq 2C_2 \sum_{\substack{r \leq G \\ r \text{ odd squarefree}}} \frac{a(r)}{r} \sum_{m \leq G/r} \frac{1}{m} \\ &\leq 2C_2(1 + \log G) \prod_{\ell > 2} \left(1 + \frac{1}{\ell(\ell - 2)}\right) \ll \log(2G). \end{aligned}$$

The Euler product converges absolutely. Combining this with (56.2), (56.3), and $\pi(Y) \ll Y/\log Y$, we obtain

$$\mathcal{E}(Y) \ll \frac{Y}{G \log Y} + \frac{Y \log(2G)}{(\log Y)^2} + \log(2G).$$

Choosing the indicated G proves (56.1) for large Y; enlarging the constant covers $8 \leq Y$ below that range. □

Theorem 56.2 (An unconditional shrinking high-frequency tail)**.** *Let $\chi \in C^\infty(\mathbb{R})$ be fixed, with $0 \le \chi \le 1$, $\chi = 1$ on $(-\infty, 0]$, and $\chi = 0$ on $[1, \infty)$. For $Y = e^{T+1} \ge 8$, define*

$$\begin{aligned} M_T(\xi) &= A_T(\xi) - B_T(\xi), \\ A_T(\xi) &= \sum_{p \le Y} \frac{\chi(\log p - T)}{p^{1/2+i\xi}}, \\ B_T(\xi) &= \int_{\log 2}^{T+1} \chi(u - T) \frac{e^{(1/2-i\xi)u}}{u}\, du. \end{aligned}$$

Then, uniformly for $R \ge 1$,

$$\int_{|\xi| \ge R} \frac{|M_T(\xi)|^2}{1/4 + \xi^2}\, d\xi \ll_\chi \frac{1 + \log\log Y}{R} + \frac{Y \log\log Y}{R^2 (\log Y)^2} + \frac{Y}{R^3 (\log Y)^2}. \tag{56.4}$$

In particular, for every fixed $\alpha < 1$, taking $R = \sqrt{Y}/(\log Y)^\alpha$ makes the left side tend to zero as $T \to \infty$. More sharply, the explicit choice

$$R(Y) = \frac{\sqrt{Y} \log\log Y}{\log Y} \tag{56.5}$$

gives the bound $O_\chi(1/\log\log Y)$. Any threshold

$$R(Y) = \frac{\sqrt{Y \log\log Y}}{\log Y}\, w(Y), \qquad w(Y) \longrightarrow \infty,$$

also makes the tail tend to zero.

Proof. Montgomery–Vaughan's weighted Hilbert inequality [36, Corollary 2, equation (1.9)] implies, for distinct real λ_r and $\delta_r = \min_{s \ne r} |\lambda_r - \lambda_s|$,

$$\int_b^{b+H} \left| \sum_r a_r e^{i\lambda_r \xi} \right|^2 d\xi \le H \sum_r |a_r|^2 + 3\pi \sum_r \frac{|a_r|^2}{\delta_r}.$$

The shifted-interval version follows by multiplying each coefficient by $e^{i\lambda_r b}$. Apply it with $\lambda_p = -\log p$ and $a_p = \chi(\log p - T)/\sqrt{p}$. Since $|\chi| \le 1$, for $S(Y) = \sum_{p \le Y} 1/p$ this gives

$$\int_H^{2H} |A_T(\xi)|^2 d\xi \le H S(Y) + 3\pi \mathcal{E}(Y).$$

The coefficients are real, so the integrand is even. Splitting the two tails into dyadic intervals and using $(1/4 + \xi^2)^{-1} \le (2^j R)^{-2}$ on the j-th interval, we obtain the explicit intermediate inequality

$$\int_{|\xi| \ge R} \frac{|A_T(\xi)|^2}{1/4 + \xi^2}\, d\xi \le \frac{4 S(Y)}{R} + \frac{8\pi \mathcal{E}(Y)}{R^2}. \tag{56.6}$$

Chebyshev's estimate and partial summation yield $S(Y) \ll 1 + \log\log Y$; Lemma 56.1 bounds the other term.

For the continuous part put $F(u) = e^{u/2}/u$. On $[\log 2, T+1]$, F has at most one turning point, at $u = 2$. Its total variation is at most $F(\log 2) + F(T+1)$. Since χ' is supported in $[0, 1]$, integration by parts in B_T therefore gives

$$|B_T(\xi)| \le \frac{C_\chi}{|\xi|} \frac{\sqrt{Y}}{\log Y} \qquad (\xi \ne 0).$$

Here the fixed lower endpoint and the length-one transition interval are absorbed into C_χ; this is uniform for $Y \geq 8$. Consequently

$$\int_{|\xi|\geq R} \frac{|B_T(\xi)|^2}{1/4+\xi^2}\,d\xi \ll_\chi \frac{Y}{R^3(\log Y)^2}. \tag{56.7}$$

The elementary inequality $|A_T - B_T|^2 \leq 2|A_T|^2 + 2|B_T|^2$ now proves (56.4). With the specified R, its three terms are respectively

$$O\Big(\frac{(\log Y)^\alpha(1+\log\log Y)}{\sqrt{Y}}\Big), \qquad O\Big(\frac{\log\log Y}{(\log Y)^{2-2\alpha}}\Big), \qquad O\Big(\frac{(\log Y)^{3\alpha-2}}{\sqrt{Y}}\Big),$$

and all tend to zero for fixed $\alpha < 1$. For (56.5), the three terms in (56.4) are respectively $O(\log Y/\sqrt{Y})$, $O(1/\log\log Y)$, and $O(\log Y/(\sqrt{Y}(\log\log Y)^3))$. For the last threshold in the statement they are respectively $O(\log Y\sqrt{\log\log Y}/(\sqrt{Y}\,w(Y)))$, $O(w(Y)^{-2})$, and $O(\log Y/(\sqrt{Y}(\log\log Y)^{3/2}w(Y)^3))$, so all again tend to zero. □

Sharp cutoffs. The same bound (56.4), with an absolute implied constant, holds for

$$M_Y^{\mathrm{sharp}}(\xi) = \sum_{p\leq Y} p^{-1/2-i\xi} - \int_2^Y \frac{x^{-1/2-i\xi}}{\log x}\,dx.$$

The atomic proof is unchanged. Integration by parts for the continuous part now includes its two endpoint values, which are both absorbed by $O(\sqrt{Y}/(|\xi|\log Y))$. This observation is a finite-cutoff estimate only; it does not assert that sharp multiplication is uniformly bounded on $H^{-1}_{1/2}$.

What has, and has not, been bounded. The polynomial support bound in the classical mean-value theorem alone would replace the second term of (56.4) by $Y/(R^2\log Y)$. The actual-prime spacing argument gains $\log Y/\log\log Y$ in that term. Both statements are unconditional applications of established sieve and Hilbert-inequality theorems; no historical novelty claim is made. The controlled band moves to infinity with Y. The complementary integral

$$\int_{|\xi|\leq\sqrt{Y}/(\log Y)^\alpha} \frac{|M_T(\xi)|^2}{1/4+\xi^2}\,d\xi$$

has not been bounded uniformly. The high-frequency estimate therefore does not prove $\sup_T \mathcal{N}(T) < \infty$, the boundedness of $R_4(X) + (3/4)\log\log X$, or RH.

57 Transfer of the high-frequency bound to actual prime endpoints

Define the actual upward-crossing prime set

$$\mathcal{Q}_\uparrow = \{q \text{ prime} : P(q^-) \leq 0 < P(q)\}.$$

Littlewood's sign oscillation, in the form recorded by Pintz [37], shows that this set is unbounded. Indeed, P decreases between primes and jumps upward by one at each prime. Its unbounded oscillations of both signs are unchanged by the constant normalization of Li_2. Every passage from a nonpositive to a positive value must therefore occur at a prime, and

$$0 < P(q) \leq 1 \qquad (q \in \mathcal{Q}_\uparrow). \tag{57.1}$$

Let a_Y be the optimal center in the weighted variance, and use the exactly corrected finite measure ν_{Y,a_Y} of Proposition 55.1. Lemma 50.2 gives

$$a_Y = \frac{2Y}{Y-2}A(Y) \longrightarrow 2b_0.$$

At crossing primes the correction from ν_q consists of two atoms whose total variation is bounded independently of q:

$$\frac{|a_q|}{\sqrt{2}} + \frac{|P(q)-a_q|}{\sqrt{q}} = O(1).$$

Corollary 57.1 (An exact low-band remainder at infinitely many primes)**.** *For sufficiently large $q \in \mathcal{Q}_\uparrow$, put*

$$R(q) = \frac{\sqrt{q}\log\log q}{\log q}, \qquad \mathcal{L}(q) = \frac{1}{2\pi}\int_{|\xi|\le R(q)} \frac{|\widehat{\nu}_{q,a_q}(\xi)|^2}{1/4+\xi^2}\,d\xi.$$

Unconditionally,

$$\boxed{V(q) = \mathcal{L}(q) + \varepsilon(q), \qquad 0 \le \varepsilon(q) \ll \frac{1}{\log\log q}.} \tag{57.2}$$

Consequently the original target B *is equivalent to*

$$\liminf_{\substack{q\to\infty \\ q\in\mathcal{Q}_\uparrow}} \mathcal{L}(q) < \infty.$$

The boundedness of $\mathcal{L}(q)$ is not proved here.

Proof. The Fourier transform of a finite measure is bounded by its total variation. The two correction atoms therefore have weighted high-frequency squared norm $O(1/R(q))$. The sharp-cutoff version of Theorem 56.2, combined with $|a+b|^2 \le 2|a|^2 + 2|b|^2$, bounds the corrected tail by $O(1/\log\log q)$. Proposition 55.1 identifies the complete corrected norm with $V(q)$, proving (57.2) with a nonnegative error. If $\mathcal{L}$ is bounded along an unbounded crossing subsequence, then so is V. Minimum variance is nondecreasing in its upper endpoint, so V is then bounded everywhere. Proposition 50.3 gives B. The converse follows at once from the same proposition and $\mathcal{L}(q) \le V(q)$. □

This corollary addresses the original optimized variance, including its endpoint terms. No assumption that $P(Y)^2/Y$ is bounded at arbitrary endpoints is made. The crossing-prime choice is used exactly where that boundary control is needed.

58 One fixed frequency band at power-dense endpoints

The preceding criterion allows arbitrarily sparse endpoints but asks for every fixed frequency band. A complementary criterion uses just one band, provided the chosen endpoints do not have fixed power-size gaps. The new analytic input is polynomial approximation on short logarithmic blocks. No Riemann hypothesis is used in this input.

Lemma 58.1 (Polynomial transfer from an interval to a compact set)**.** *For every fixed $T, R > 0$, there is $C_{T,R} > 1$ such that every complex polynomial p of degree at most n satisfies*

$$\sup_{|z|\le R} |p(z)| \le C_{T,R}^{n+1} \|p\|_{L^2([-T,T])}.$$

Proof. Expand p in the orthonormal Legendre basis

$$q_k(t) = \sqrt{\frac{2k+1}{2T}} P_k(t/T), \qquad 0 \le k \le n.$$

Each expansion coefficient has modulus at most $\|p\|_2$. The generating function

$$\sum_{k=0}^{\infty} P_k(w) v^k = (1 - 2wv + v^2)^{-1/2}$$

and Cauchy's coefficient estimate, on a sufficiently small fixed circle $|v| = r_{T,R} > 0$, give $\sup_{|z|\le R} |P_k(z/T)| \le C_0 r_{T,R}^{-k}$. Cauchy–Schwarz on the finite expansion gives the stated exponential bound, after enlarging $C_{T,R}$ to absorb the polynomial factors in n. The constant is independent of the coefficients and degree. □

Lemma 58.2 (Subexponential propagation through logarithmic blocks)**.** *Let f be locally integrable on $[u_0, \infty)$, and suppose $|f(u)| \le Ce^{au}$ there for some $a \ge 0$. Let $U_j \uparrow \infty$ satisfy*

$$\frac{U_{j+1} - U_j}{U_j} \longrightarrow 0.$$

Define

$$A(U, t) = \int_{u_0}^{U} f(u) e^{-itu}\, du.$$

Fix $T > 0$ and $c \ge 0$. If

$$\limsup_{j\to\infty} \frac{\log(1 + \|A(U_j, \cdot)\|_{L^2([-T,T])})}{U_j} \le c, \tag{58.1}$$

then the initially absolutely convergent Laplace transform $\int_{u_0}^{\infty} f(u)e^{-zu}\, du$, $\Re z > a$, has a holomorphic continuation to $\Re z > c$.

Proof. First thin the endpoint sequence greedily: retain the first endpoint at least one unit beyond the previously retained endpoint. Its new gaps $L_j = U_{j+1} - U_j$ satisfy

$$1 \le L_j = o(U_j), \qquad U_j \ge j + O(1).$$

Indeed, minimality gives a new gap at most one plus one original gap whose lower endpoint is less than $U_j + 1$. The original relative gap hypothesis makes that additional gap $o(U_j)$. The growth hypothesis is retained on this subsequence.

Put

$$B_j(w) = \int_0^{L_j} f(U_j + v) e^{-iwv}\, dv.$$

For real t,

$$B_j(t) = e^{itU_j} (A(U_{j+1}, t) - A(U_j, t)).$$

Consequently

$$\limsup_{j} U_j^{-1} \log(1 + \|B_j\|_{L^2([-T,T])}) \le c. \tag{58.2}$$

The total mass in the block obeys, for a fixed $C_0 > 0$,

$$M_j := \int_0^{L_j} |f(U_j + v)|\, dv \le e^{C_0 U_j} \tag{58.3}$$

for all sufficiently large j. This follows from $M_j \le CL_j e^{a(U_j + L_j)}$, $L_j = o(U_j)$, and $\log L_j = o(U_j)$.

Here are explicit approximation parameters. Let

$$d_j = \log \frac{U_j}{1+L_j} \longrightarrow \infty, \qquad n_j = \left\lceil \frac{4(C_0+2)U_j}{d_j} \right\rceil.$$

Only sufficiently large indices are needed, so $d_j > 0$. Then $n_j = o(U_j)$. Let p_j be the Taylor polynomial of B_j at zero of degree n_j. For every fixed $R_* > 0$, Taylor's formula for the exponential and (58.3) give

$$\sup_{|w|\le R_*} |B_j(w) - p_j(w)| \le M_j e^{R_* L_j} \frac{(R_* L_j)^{n_j+1}}{(n_j+1)!}.$$

Using $m! \ge (m/e)^m$, the logarithm of the right side is at most

$$C_0 U_j + R_* L_j - (n_j+1) \log \frac{n_j+1}{e R_* L_j}.$$

Since

$$\log \frac{n_j+1}{e R_* L_j} \ge d_j - \log d_j + O_{C_0,R_*}(1) \ge \frac{1}{2} d_j$$

eventually, this logarithm is at most $-U_j$ for all sufficiently large j. Thus

$$\sup_{|w|\le R_*} |B_j(w) - p_j(w)| \le e^{-U_j}. \tag{58.4}$$

The same degree sequence works for every fixed R_*; the starting index may depend on R_*.

Apply Lemma 58.1 with $R_* \ge \max(T, R)$. Equations (58.2) and (58.4) give

$$\sup_{|w|\le R} |B_j(w)| \le C_{T,R}^{n_j+1} \left(\|B_j\|_{L^2([-T,T])} + \sqrt{2T}\, e^{-U_j} \right) + e^{-U_j}.$$

Because $n_j = o(U_j)$, it follows that

$$\limsup_j U_j^{-1} \log \left(1 + \sup_{|w|\le R} |B_j(w)| \right) \le c \qquad (R > 0). \tag{58.5}$$

For $\Re z > c$, define

$$\mathcal{F}(z) = \int_{u_0}^{U_1} f(u) e^{-zu}\, du + \sum_{j\ge 1} e^{-zU_j} B_j(-iz). \tag{58.6}$$

On a compact subset with $\Re z \ge c + \eta$, (58.5) bounds the summands by $e^{-\eta U_j/2}$ eventually. These bounds are summable because $U_j \ge j + O(1)$. Hence the series converges normally there and defines a holomorphic function. For $\Re z > \max(a, c)$, absolute convergence of the original integral identifies (58.6) with it. This proves the continuation. It does not assert absolute convergence of the ungrouped integral on the larger half-plane. □

Remark 58.3 (A quantitative transfer cost). The preceding proof gives a concrete norm inequality, not just an identity of coordinates. Its degree cost is

$$n_j = O\left(\frac{U_j}{\log(U_j/(1+L_j))} \right) = o(U_j).$$

For bounded logarithmic gaps L_j, the loss in transferring a block from one real frequency interval to any fixed complex compact set is at most $\exp(O_{T,R}(U_j/\log U_j))$, apart from an exponentially small approximation error. This is subpower in $X_j = e^{U_j}$. It does not by itself bound the local input norm in (58.2) for the ordinary primes.

Theorem 58.4 (A single-band criterion with power-dense endpoints)**.** *Let $X_j \uparrow \infty$ be any prescribed sequence satisfying*

$$\frac{\log X_{j+1}}{\log X_j} \longrightarrow 1. \tag{58.7}$$

Fix any one $T > 0$. Then RH is equivalent to

$$\limsup_{j\to\infty} \frac{\log(1 + I_T(X_j))}{\log X_j} = 0. \tag{58.8}$$

It is also equivalent to the existence of one such T and one such endpoint sequence satisfying (58.8)*. The original compensated-correlation target is consequently equivalent to these conditions as well.*

Proof. Under RH, the previously proved critical-energy result gives $I_T(X) \le 2\pi Q(\infty) < \infty$ for every T, X. This proves the forward implication for every prescribed sequence.

Conversely, set

$$f(u) = e^{-u/2}P(e^u), \qquad u \ge u_0 = \log 2, \qquad U_j = \log X_j.$$

Chebyshev's elementary bound, or simply $P(x) = O(x)$, gives $|f(u)| = O(e^{u/2})$. Moreover

$$A(U_j, t) = F_{X_j}(1/2 + it), \qquad \|A(U_j, \cdot)\|_2^2 = I_T(X_j).$$

Condition (58.8) therefore supplies Lemma 58.2 with $c = 0$. It follows that

$$F(s) = \int_2^\infty P(x)x^{-s-1}\,dx \qquad (\Re s > 1)$$

has a holomorphic continuation to $\Re s > 1/2$. Consequently $(sF(s))'$ is holomorphic throughout that half-plane. But initially on $\Re s > 1$,

$$(sF(s))' = \frac{\zeta'(s)}{\zeta(s)} - \sum_{k\ge2} \mathcal{P}'(ks) + \frac{2^{1-s}}{s-1}, \qquad \mathcal{P}(s) = \sum_p p^{-s}.$$

The prime-power series is normally convergent on $\Re s > 1/2$, and the principal parts at $s = 1$ cancel. At every zero ρ of ζ with $\Re\rho > 1/2$, the right side has a simple pole with nonzero residue equal to the multiplicity of ρ. Meromorphic uniqueness makes such a pole incompatible with the holomorphic continuation of $(sF(s))'$. There are therefore no such zeros. The functional equation yields RH. □

Corollary 58.5 (One band witnesses every off-line zero on a power mesh)**.** *For every fixed $T > 0$, every sequence satisfying* (58.7)*, and every zeta zero ρ with $\beta = \Re\rho > 1/2$,*

$$\limsup_{j\to\infty} \frac{\log(1 + I_T(X_j))}{\log X_j} \ge 2\beta - 1.$$

Thus, if $\Theta = \sup_\rho \Re\rho$, the same limsup is at least $2\Theta - 1$. In particular, for every $0 < \delta < 2\beta - 1$, one has $I_T(X_j) > X_j^\delta$ for infinitely many j.

Proof. If the displayed limsup were less than $2\beta - 1$, choose $c \ge 0$ strictly between half that limsup and $\beta - 1/2$. Lemma 58.2 would continue $F(s)$ holomorphically to $\Re s > 1/2 + c$. This half-plane contains ρ, and the identical nonzero-pole argument gives a contradiction. Taking the supremum over zeros proves the assertion with Θ. The final statement follows from the definition of limsup, using an intermediate exponent between δ and $2\beta - 1$ to absorb the harmless additive one. □

Examples and the remaining arithmetic input. The prescribed endpoint sequence may be the ordinary primes, $X_j = 2^j$, or $X_j = \exp(j^2)$. For primes, Bertrand's inequality $p_{j+1} < 2p_j$ gives (58.7); no sharp prime-gap theorem is required. With $T = 1$ and $X_j = 2^j$, the exact new criterion becomes

$$\forall \varepsilon > 0 \quad \exists j_0 \quad \forall j \geq j_0 : \qquad I_1(2^j) \leq 2^{\varepsilon j}.$$

This is a one-band condition with all sufficiently late sampled endpoints. It must not be confused with a liminf at an arbitrary sparse endpoint set: the latter is not proved sufficient for one band. No unconditional proof of this actual-prime input is supplied here.

59 Removing the nearest-spacing logarithm by a positive Fourier kernel

Put

$$A_Y(t) = \sum_{p \leq Y} a_p e^{-it \log p}, \qquad |a_p| \leq p^{-1/2}.$$

The coefficients may be complex. This includes the sharp prime sum and any fixed smooth cutoff with coefficients bounded in modulus by $p^{-1/2}$.

Lemma 59.1 (An unweighted singular-series mean)**.** *Let*

$$\mathfrak{S}(d) = 2C_2 \prod_{\substack{\ell \mid d \\ \ell > 2}} \frac{\ell - 1}{\ell - 2} \quad (d \text{ even}), \qquad C_2 = \prod_{\ell > 2} \left(1 - \frac{1}{(\ell - 1)^2}\right).$$

For every real $D \geq 1$,

$$\sum_{\substack{1 \leq d \leq D \\ 2 \mid d}} \mathfrak{S}(d) \leq D.$$

Proof. For odd squarefree r, set $b(r) = \prod_{\ell \mid r} (\ell - 2)^{-1}$. Expanding the finite product gives

$$\sum_{\substack{d \leq D \\ 2 \mid d}} \mathfrak{S}(d) = 2C_2 \sum_{\substack{r \geq 1 \\ r \text{ odd squarefree}}} b(r) \left\lfloor \frac{D}{2r} \right\rfloor \leq C_2 D \prod_{\ell > 2} \left(1 + \frac{1}{\ell(\ell - 2)}\right) = D.$$

All summands are nonnegative, the product converges absolutely, and its factors are precisely the reciprocals of the factors in C_2. □

Proposition 59.2 (Uniform mean value without nearest-gap loss)**.** *Uniformly for $Y \geq 8$, $H \geq 1$, and $b \in \mathbb{R}$,*

$$\int_b^{b+H} |A_Y(t)|^2 dt \ll H \sum_{p \leq Y} \frac{1}{p} + \frac{Y}{(\log Y)^2}. \tag{59.1}$$

The implied constant is absolute and independent of the coefficients.

Proof. For $1 \leq H < 8$, Chebyshev's upper bound and partial summation give

$$\int_b^{b+H} |A_Y(t)|^2 dt \leq H \left(\sum_{p \leq Y} p^{-1/2}\right)^2 \ll \frac{Y}{(\log Y)^2}.$$

It remains to treat $H \geq 8$.

Let $c = b + H/2$ and

$$W_H(t) = \left(\frac{\sin((t - c)/H)}{(t - c)/H}\right)^2.$$

At the removable singularity use the continuous value one. On $[b, b+H]$, $W_H(t) \geq c_0 := (\sin(1/2)/(1/2))^2 > 0$. The elementary Fourier identity is

$$\int_{\mathbb{R}} W_H(t) e^{-itv} dt = \pi H e^{-icv} \left(1 - \frac{H|v|}{2}\right)_+ .$$

Expanding the finite sum and taking the absolute value of each off-diagonal term therefore yields

$$\int_b^{b+H} |A_Y(t)|^2 dt \leq \frac{\pi H}{c_0} \sum_{\substack{p,q \leq Y \\ |\log(p/q)| \leq 2/H}} \frac{1}{\sqrt{pq}}. \tag{59.2}$$

The diagonal contributes $(\pi H/c_0) \sum_{p \leq Y} 1/p$.

For the off-diagonal, take $p < q$, write $d = q - p$, and group the smaller prime in dyadic intervals $Z \leq p < 2Z$, where $Z = 2^j$, $j \geq 1$. Since $2/H \leq 1/4$,

$$d \leq p(e^{2/H} - 1) \leq \frac{4p}{H} < \frac{8Z}{H} \leq Z.$$

No pair containing the prime 2 can occur here, since $\log(3/2) > 1/4$. Thus all differences are even. With

$$N_{2Z}(d) = \#\{n \leq 2Z : n, n+d \text{ are prime}\},$$

the uniform upper-bound sieve gives

$$N_{2Z}(d) \ll \mathfrak{S}(d) \frac{Z}{(\log Z)^2} \qquad (2 \mid d,\ d \leq Z).$$

This is an immediate specialization of the uniform two-prime sieve [35, Lemma 2.3]; increasing the absolute constant covers the finitely many small Z. Hence Lemma 59.1 implies

$$\begin{aligned} \sum_{\substack{Z \leq p < 2Z,\ p < q \leq Y \\ \log(q/p) \leq 2/H}} \frac{1}{\sqrt{pq}} &\leq \frac{1}{Z} \sum_{\substack{d \leq 8Z/H \\ 2 \mid d}} N_{2Z}(d) \\ &\ll \frac{Z}{H(\log Z)^2}. \end{aligned}$$

If $8Z/H < 1$, the sum is empty and the same estimate is valid. Finally,

$$\sum_{\substack{Z = 2^j \\ Z \leq Y}} \frac{Z}{(\log Z)^2} \ll \frac{Y}{(\log Y)^2}.$$

For example, split at $\sqrt{Y}$: the lower part is $O(\sqrt{Y})$, while the upper part is bounded by a geometric sum divided by $(\frac{1}{2}\log Y)^2$. Inserting these bounds in (59.2), and counting both orders of each pair, proves (59.1). □

Theorem 59.3 (Improved unconditional high-frequency tail)**.** *For the sharp actual-prime discrepancy*

$$M_Y(t) = \sum_{p \leq Y} p^{-1/2-it} - \int_2^Y \frac{x^{-1/2-it}}{\log x}\, dx,$$

one has, uniformly for $Y \geq 8$ and $R \geq 1$,

$$\boxed{\int_{|t| \geq R} \frac{|M_Y(t)|^2}{1/4 + t^2} dt \ll \frac{1 + \log\log Y}{R} + \frac{Y}{R^2 (\log Y)^2} + \frac{Y}{R^3 (\log Y)^2}.} \tag{59.3}$$

The same estimate holds, with a cutoff-dependent constant, for the fixed smooth logarithmic cutoffs of the preceding sections.

Proof. Apply Proposition 59.2 to each dyadic interval $[2^jR, 2^{j+1}R]$ and its negative counterpart. On such an interval, $(1/4+t^2)^{-1} \le (2^jR)^{-2}$. The geometric sums give

$$\int_{|t|\ge R} \frac{|A_Y(t)|^2}{1/4+t^2}dt \ll \frac{\sum_{p\le Y} 1/p}{R} + \frac{Y}{R^2(\log Y)^2}.$$

For the continuous part, integration by parts in

$$B_Y(t) = \int_{\log 2}^{\log Y} \frac{e^{(1/2-it)u}}{u}\,du$$

and the total variation of $e^{u/2}/u$ give

$$|B_Y(t)| \ll \frac{\sqrt{Y}}{|t|\log Y} \quad (t\neq 0), \qquad \int_{|t|\ge R} \frac{|B_Y(t)|^2}{1/4+t^2}dt \ll \frac{Y}{R^3(\log Y)^2}.$$

The lower endpoint is absorbed in an absolute constant for $Y \ge 8$. The prime reciprocal sum is $O(1+\log\log Y)$. Combining the atomic and continuous estimates via $|A_Y - B_Y|^2 \le 2|A_Y|^2 + 2|B_Y|^2$ proves the theorem. For a fixed smooth cutoff, Proposition 59.2 still applies, and the extra derivative of the cutoff in the integration by parts has bounded support and a fixed supremum norm. □

Corollary 59.4 (A smaller remaining frequency band)**.** *Put*

$$R_0(Y) = \frac{\sqrt{Y}}{\log Y}.$$

The tail in (59.3) is bounded uniformly as $Y\to\infty$ when $R = R_0(Y)$. If

$$R(Y) = R_0(Y)w(Y), \qquad w(Y)\longrightarrow\infty,$$

the tail tends to zero. In particular,

$$R_1(Y) = \frac{\sqrt{Y\log\log Y}}{\log Y}$$

gives a tail of order $O(1/\log\log Y)$.

Proof. The three terms for $R = R_0 w$ are respectively

$$O\left(\frac{\log Y(1+\log\log Y)}{\sqrt{Y}\,w}\right), \qquad O(w^{-2}), \qquad O\left(\frac{\log Y}{\sqrt{Y}\,w^3}\right).$$

Take $w = 1$, any $w\to\infty$, or $w = \sqrt{\log\log Y}$, as appropriate. □

At the actual upward-crossing primes q, the optimal-center endpoint correction in the preceding sections has uniformly bounded total variation. Its additional weighted tail is therefore $O(1/R)$. Consequently, with

$$\mathcal{L}_0(q) = \frac{1}{2\pi}\int_{|t|\le R_0(q)} \frac{|\widehat{\nu}_{q,a_q}(t)|^2}{1/4+t^2}\,dt,$$

one has the exact decomposition

$$V(q) = \mathcal{L}_0(q) + \varepsilon_0(q), \qquad 0 \le \varepsilon_0(q) \ll 1.$$

Thus the unchanged original boundedness target is equivalent to $\liminf_{q\to\infty,\ q\in\mathcal{Q}_\uparrow} \mathcal{L}_0(q) < \infty$. Using R_1 instead gives a vanishing error $O(1/\log\log q)$. The argument proving the equivalence uses only monotonicity of V, the exact endpoint identity, and the proved tail bound.

Scope. The earlier nearest-spacing argument had the term $Y \log\log Y/(R^2 \log^2 Y)$. The positive Fourier kernel removes this logarithmic loss by counting all close pairs without the reciprocal-gap weight. This is an unconditional improvement using established sieve tools; no historical-priority claim is made. It does not bound any fixed frequency band by Y^{ε}. At a fixed R, the right side still allows order $Y/\log^2 Y$, and the required actual-prime cancellation remains unproved.

60 A centered short-interval transfer below the square-root cutoff

Put $\Omega(Y) = (\log Y/\log\log Y)^{1/3}$ for Y sufficiently large. The unconditional input used here is Montgomery–Vaughan, *Multiplicative Number Theory III*, Theorem 28.6 [38]: there are absolute $A, c_0 > 0$ such that

$$\int_0^Z |\psi(x+h) - \psi(x) - h|^2 dx \ll Zh^2 \left((Z^{1/6}/h)^{2/5} (\log Z)^A + e^{-c_0 \Omega(Z)} \right), \qquad h \geq 1. \tag{60.1}$$

This is unconditional; the substantially stronger RH-conditional Selberg estimate is not used.

Theorem 60.1. *There is an absolute constant $c > 0$ such that, with*

$$R_*(Y) = \sqrt{Y}\, e^{-c\Omega(Y)}, \qquad M_Y(t) = \sum_{p \leq Y} p^{-1/2-it} - \int_2^Y \frac{x^{-1/2-it}}{\log x}\, dx,$$

one has unconditionally

$$\int_{|t| \geq R_*(Y)} \frac{|M_Y(t)|^2}{1/4 + t^2}\, dt \ll e^{-c\Omega(Y)}. \tag{60.2}$$

The constants are not evaluated numerically here.

Proof. We first record an averaged-window version of Gallagher's inequality, valid for every finite complex measure ν on the real line:

$$\int_{-H}^{H} |\widehat{\nu}(t)|^2 dt \ll H^3 \int_{1/(4H)}^{1/(2H)} \int_{\mathbb{R}} |\nu((u, u+\delta])|^2 du\, d\delta. \tag{60.3}$$

Indeed Plancherel expresses the inner integral as $(2\pi)^{-1} \int |\widehat{\nu}(t)|^2 |(e^{it\delta} - 1)/t|^2 dt$. For $|t| \leq H$ and the displayed range of δ, the last multiplier is bounded below by a constant times H^{-2}. Integrating in δ proves (60.3).

Let $Z_0 = Y^{3/4}$ and $S = Y^{2/3}$. The already proved elementary tail bound, applied at Z_0 and at Y, shows that the contribution of M_{Z_0} above $R_*(Y)$, and that of M_Y above S, are $O(Y^{-\eta} e^{O(\Omega(Y))} (\log Y)^A)$ for some $\eta > 0$. Thus it suffices to treat the measure on $(Z_0, Y]$ and frequencies $R_*(Y) \leq |t| \leq S$.

Temporarily include prime powers and put $w(x) = x^{-1/2}/\log x$:

$$\nu(du) = [w(x)(d\psi(x) - dx)]_{x = e^u, Z_0 < x \leq Y}.$$

The additional prime-power measure has total variation

$$\sum_{k \geq 2,\ p^k \leq Y} \frac{p^{-k/2}}{k} \ll 1 + \log\log Y,$$

so its weighted Fourier tail is $O((1 + \log\log Y)^2/R_*(Y))$, which is negligible.

Consider H between $R_*(Y)$ and $2S$ and $\delta \in [1/(4H), 1/(2H)]$. For windows fully inside $(\log Z_0, \log Y]$, set $x = e^u$ and $h = x(e^{\delta} - 1) \asymp x/H$. On a dyadic block $x \asymp Z$, where $Z \gg Z_0$, $h \asymp Z/H \gg 1$. Freeze the smooth weight at x:

$$\nu((u, u+\delta]) = w(x)(\psi(x+h) - \psi(x) - h) + \mathcal{E}(x, h).$$

Since $|w'(v)| \ll Z^{-3/2}/\log Z$ and the trivial bound $\psi(x+h)-\psi(x) \ll (h+1)\log Z$ holds,

$$|\mathcal{E}(x,h)| \ll h^2/Z^{3/2}.$$

The change of variables satisfies $du = dx/x$ and $d\delta = dh/(x+h)$. In (60.3), the contribution of the main term on this block is at most

$$\frac{H^3}{Z^3\log^2 Z}\int_{c_1 Z/H}^{c_2 Z/H}\int_0^{2Z}|\psi(x+h)-\psi(x)-h|^2 dx\,dh \ll Z^{2/3}H^{2/5}(\log Y)^A + Ze^{-c_1'\Omega(Y)}.$$

Here (60.1) was used with endpoint $2Z$; throughout $Z \gg Y^{3/4}$, its exponential term is at most $e^{-c_1'\Omega(Y)}$ after decreasing the fixed constant. The frozen-weight error contributes $O(Z/H^2)$. Summing the geometric blocks costs no factor growing with Y.

Windows intersecting either endpoint of $(\log Z_0, \log Y]$ have total u-length $O(H^{-1})$. At an endpoint of size Z, the same trivial prime-power bound gives window mass $O(\sqrt{Z}/H)$. Their contribution to (60.3) is $O(Z/H)$. Consequently

$$\int_{-H}^{H}|\widehat{\nu}(t)|^2 dt \ll Y^{2/3}H^{2/5}(\log Y)^A + Ye^{-c_1'\Omega(Y)} + Y/H + Y/H^2. \tag{60.4}$$

Dyadically cover $R \le |t| \le S$, bound each annulus by its containing symmetric band, and divide by t^2. This gives

$$\int_{R\le|t|\le S}\frac{|\widehat{\nu}(t)|^2}{1/4+t^2}dt \ll \frac{Y^{2/3}(\log Y)^A}{R^{8/5}} + \frac{Ye^{-c_1'\Omega(Y)}}{R^2} + \frac{Y}{R^3} + \frac{Y}{R^4}. \tag{60.5}$$

Choose $0 < c < c_1'/4$ in $R = R_*(Y)$. The first term is $Y^{-2/15}e^{(8c/5)\Omega(Y)}(\log Y)^A$; the second is $e^{-(c_1'-2c)\Omega(Y)}$; the other terms have negative powers of Y times subpower factors. The discarded small-scale, prime-power, and far-frequency terms are also bounded by $e^{-c\Omega(Y)}$ for all sufficiently large Y, after decreasing c if needed. This proves the theorem. □

Define

$$F_{Y,a}(t) = \int_2^Y (P(x)-a)x^{-3/2-it}\,dx.$$

The exact endpoint identity is $(1/2+it)F_{Y,a}(t) = \widehat{\nu}_{Y,a}(t)$, so the weighted measure norm and the centered-transform norm agree.

At the actual upward-crossing primes q, the exact endpoint correction from the preceding sections has uniformly bounded total variation. Its weighted tail is $O(1/R_*(q))$. Hence, writing

$$\mathcal{L}_*(q) = \frac{1}{2\pi}\int_{|t|\le R_*(q)}|F_{q,a_q}(t)|^2 dt,$$

the original variance satisfies unconditionally

$$V(q) = \mathcal{L}_*(q) + \varepsilon_*(q), \qquad 0 \le \varepsilon_*(q) \ll e^{-c\Omega(q)}.$$

The boundedness of $\mathcal{L}_*$ remains unproved.

61 Exceptional frequency sets and a logarithmic criterion

For the same entire finite transform F_X, put

$$J_T(X) = \int_{-T}^{T}\log(1+|F_X(1/2+it)|)\,dt,$$

and, for $\varepsilon > 0$, define

$$m_{T,\varepsilon}(X) = \big|\{t \in [-T, T] : |F_X(1/2 + it)| > X^{\varepsilon}\}\big|.$$

The bars on a set denote Lebesgue measure. This criterion allows large values on small exceptional frequency sets.

Proposition 61.1 (An explicit exceptional-set tradeoff)**.** *For each fixed $T, \varepsilon > 0$, as $X \to \infty$,*

$$m_{T,\varepsilon}(X) \le \frac{J_T(X)}{\varepsilon \log X}, \tag{61.1}$$

$$\frac{J_T(X)}{\log X} \le 2T\varepsilon + \frac{1}{2} m_{T,\varepsilon}(X) + O_T(1/\log X). \tag{61.2}$$

Consequently, on any sequence $X_j \to \infty$,

$$J_T(X_j) = o(\log X_j) \quad \Longleftrightarrow \quad m_{T,\varepsilon}(X_j) \to 0 \text{ for every } \varepsilon > 0.$$

Proof. On the exceptional set, $\log(1 + |F_X|) > \varepsilon \log X$, giving (61.1). The elementary bound $P(x) = O(x)$ implies, uniformly for real t,

$$|F_X(1/2 + it)| \le C\sqrt{X}, \qquad \log(1 + |F_X|) \le \tfrac{1}{2} \log X + O(1).$$

Off the exceptional set the stronger bound $\log(1 + |F_X|) \le \varepsilon \log X + \log 2$ holds. Integrating separately proves (61.2). For the last assertion, use (61.1) in one direction; in the other, take a limsup in (61.2) and let $\varepsilon \downarrow 0$. □

Theorem 61.2 (An induced criterion allowing frequency spikes)**.** *Each of the following is equivalent to RH and to the unchanged compensated R_4 bound:*

1. *For every fixed $T > 0$,*
$$\liminf_{X \to \infty} \frac{J_T(X)}{\log X} = 0.$$
2. *For every $T > 0$, there is a sequence $X_j \to \infty$ such that, for every $\varepsilon > 0$, $m_{T,\varepsilon}(X_j) \to 0$.*
3. *For every $T, \varepsilon, \eta > 0$ and every X_0, there exists $X \ge X_0$ with*
$$m_{T,\varepsilon}(X) < \eta.$$

The endpoints may be restricted throughout to any prescribed unbounded set, including the ordinary primes. Sequences may depend on T. These statements do not require the L^2 energy of the exceptional set to be small.

Proof. Under RH, the critical square norm is finite, so $I_T(X)$ is uniformly bounded. Jensen and Cauchy–Schwarz give

$$J_T(X) \le 2T \log\left(1 + \sqrt{I_T(X)/(2T)}\right),$$

which proves assertion 1. Proposition 61.1 shows the equivalence of assertions 1 and 2. Assertion 2 implies assertion 3. Conversely, for fixed T, choose successive endpoints in assertion 3 with $\varepsilon = \eta = 1/j$ and with $X_j \to \infty$. Equation (61.2) gives assertion 1.

For the remaining implication, inspect the harmonic-majorant proof of the fixed-band criterion. Its left-edge Poisson extension satisfies, for a compact set at distance $\delta > 0$ from $\Re s = 1/2$,

$$\sup_K U_X \le \frac{J_T(X)}{\pi\delta}.$$

The rest of that proof uses only this upper bound and the crude uniform bound $|F_X| \ll \sqrt{X}$, not the full L^2 norm on the band. Assertion 1 thus supplies the same $o(\log X_j)$ left-edge contribution. A sufficiently tall rectangle makes its horizontal harmonic measure small, giving $F_{X_j} = O(X_j^{\epsilon})$ on each required compact set. Clearing the finitely many poles of ζ'/ζ, the right-edge convergence $O(X_j^{1-b}\log X_j)$ then contradicts the fixed nonzero residue of any hypothetical zero with $\Re\rho > 1/2$, exactly as in the existing compact-band proof. Hence RH. Only the chosen unbounded endpoint sequence was used. □

Scope. The tradeoff is an unconditional analytic estimate, and the equivalence follows from the proved harmonic-measure argument. The required exceptional-set decay for the actual primes is not established. Thus this is a weaker observation requirement, not an unconditional completion of the original upper bound.

62 Removing the sharp-endpoint artifact at every cutoff

Keep

$$P(x) = \pi(x) - \mathrm{Li}_2(x), \qquad Q(Y) = \int_2^Y P(x)^2 x^{-2}\,dx, \qquad F_Y(s) = \int_2^Y P(x)x^{-s-1}\,dx.$$

Write $u_0 = \log 2$ and

$$\Omega(Y) = \left(\frac{\log Y}{\log\log Y}\right)^{1/3}$$

for sufficiently large Y. The already proved unconditional raw-measure estimate supplies fixed $c_0, C > 0$ such that

$$r(Y) = \sqrt{Y}\,e^{-c_0\Omega(Y)}, \qquad \int_{|t|\ge r(Y)} \frac{|M_Y(t)|^2}{1/4+t^2}\,dt \le Ce^{-c_0\Omega(Y)}, \tag{62.1}$$

where

$$M_Y(t) = \sum_{p\le Y} p^{-1/2-it} - \int_2^Y \frac{x^{-1/2-it}}{\log x}\,dx.$$

Its arithmetic input is the unconditional short-interval mean square from Montgomery–Vaughan [38, Theorem 28.6], transferred by the averaged Gallagher argument in Section 60.

62.1 The exact artifact in a sharp primitive

For $R \ge 1$, put

$$T_Y(R) = \frac{1}{2\pi}\int_{|t|>R} |F_Y(1/2+it)|^2 dt, \qquad U_Y(R) = \frac{1}{2\pi}\int_{|t|>R} \frac{|M_Y(t)|^2}{1/4+t^2}\,dt,$$

and

$$b(R) = \frac{1}{2\pi}\int_{|t|>R} \frac{dt}{1/4+t^2} = 1 - \frac{2}{\pi}\arctan(2R) \sim \frac{1}{\pi R}.$$

Proposition 62.1 (Exact sharp-endpoint comparison)**.** *For every real $Y > 2$ and $R \ge 1$,*

$$\left|\sqrt{T_Y(R)} - \frac{|P(Y)|}{\sqrt{Y}}\sqrt{b(R)}\right| \le \sqrt{U_Y(R)}. \tag{62.2}$$

In particular, a uniform bound $T_Y(r(Y)) = O(1)$ at all sufficiently large sharp endpoints would imply the additional pointwise estimate

$$|P(Y)| \ll Y^{3/4}e^{-c_0\Omega(Y)/2}. \tag{62.3}$$

No assertion that (62.3) holds is made here.

Proof. Finite Stieltjes partial summation, including the prime 2, gives

$$(1/2+it)F_Y(1/2+it) = M_Y(t) - P(Y)Y^{-1/2-it}.$$

It holds whether or not Y is a prime. Apply the reverse triangle inequality in the Hilbert space $L^2(\{|t|>R\}, dt/(2\pi))$. The endpoint term has norm $|P(Y)|Y^{-1/2}\sqrt{b(R)}$, proving (62.2). At $R = r(Y)$, (62.1) bounds $U_Y(R)$ and $b(R) \asymp R^{-1}$. If $T_Y(R)$ is uniformly bounded, the same comparison gives $|P(Y)| \ll \sqrt{YR}$, which is (62.3). □

Thus the raw-measure tail bound cannot simply be declared a bound for the sharp primitive at arbitrary endpoints. The obstruction is an explicitly identified boundary term. We now remove that term by smoothing the primitive itself, preserving the original target through a two-sided energy comparison.

62.2 A separated-frequency multiplication estimate

Use the Fourier convention $\widehat{f}(t) = \int_{\mathbb{R}} f(u)e^{-itu}\,du$. For $f \in L^2(\mathbb{R})$, let $\Pi_{>R}$ denote the Fourier projection onto $\{|t|>R\}$.

Lemma 62.2. *Suppose $0 \le \phi \le 1$, $\phi \in C_c^\infty(\mathbb{R})$, and $\|\phi^{(N+1)}\|_1 \le C_N$, where $N \ge 1$ is an integer. For $R \ge 1$ and $f \in L^2(\mathbb{R})$,*

$$\|\Pi_{>R}(\phi f)\|_2 \le \|\Pi_{>R/2} f\|_2 + C'_N R^{-N}\|f\|_2. \tag{62.4}$$

Here C'_N depends only on N and C_N, not on the length of the support of ϕ.

Proof. Split $f = f_{\text{high}} + f_{\text{low}}$, with Fourier supports $|t| > R/2$ and $|t| \le R/2$, respectively. Multiplication by ϕ has L^2 norm at most one, so

$$\|\Pi_{>R}(\phi f_{\text{high}})\|_2 \le \|f_{\text{high}}\|_2.$$

Integration by parts $N+1$ times gives

$$\int_{|v|>R/2} |\widehat{\phi}(v)|\,dv \le C''_N R^{-N}.$$

For an output frequency $|t|>R$ and an input frequency $|s| \le R/2$, one has $|t-s| > R/2$. The convolution identity $\widehat{\phi f_{\text{low}}} = (2\pi)^{-1}\widehat{\phi} * \widehat{f}_{\text{low}}$, Young's inequality, and Plancherel therefore give

$$\|\Pi_{>R}(\phi f_{\text{low}})\|_2 \le C'_N R^{-N}\|f_{\text{low}}\|_2.$$

The triangle inequality completes the proof. Notice that no bound for the full L^1 norm of $\widehat{\phi}$ was used. □

62.3 Smooth primitive tails at every real endpoint

Choose a fixed nonincreasing $\chi \in C^\infty(\mathbb{R})$ with $0 \le \chi \le 1$, $\chi(v) = 1$ for $v \le 0$, and $\chi(v) = 0$ for $v \ge 1$. Define

$$h_Y(u) = \chi(u - \log Y)e^{-u/2}P(e^u)\mathbf{1}_{[u_0,\infty)}(u),$$

and its ordinary finite transform

$$\mathcal{F}_{\chi,Y}(t) = \widehat{h}_Y(t) = \int_2^{eY} \chi\Big(\log\frac{x}{Y}\Big)P(x)x^{-3/2-it}\,dx. \tag{62.5}$$

The smooth cutoff is applied to the primitive, not to the critical prime measure. These are distinct operations.

Theorem 62.3 (An unconditional tail at every endpoint)**.** *Put*

$$R_\chi(Y) = 2r(eY) = 2\sqrt{eY}\, e^{-c_0\Omega(eY)}.$$

For all sufficiently large real Y, unconditionally,

$$\frac{1}{2\pi}\int_{|t|>R_\chi(Y)} |\mathcal{F}_{\chi,Y}(t)|^2 dt \ll_\chi e^{-c_0\Omega(eY)}. \tag{62.6}$$

No sign-crossing selection and no bound for $P(Y)^2/Y$ is needed. Equivalently, after decreasing a positive constant, one may use a threshold of the form $\sqrt{Y}e^{-c\Omega(Y)}$ and a remainder $O_\chi(e^{-c\Omega(Y)})$. This keeps the previous asymptotic cutoff scale; the improvement is the uniform validity for smooth primitive energies at every endpoint.

Proof. For a finite $Z > 2$, introduce the causal primitive

$$c_Z(u) = \begin{cases} 0, & u < u_0, \\ e^{-u/2}P(e^u), & u_0 \le u \le \log Z, \\ P(Z)e^{-u/2}, & u > \log Z. \end{cases}$$

Its value at a single endpoint is immaterial for the integrals. The distributional identity

$$(D + 1/2)c_Z = \sum_{p\le Z} p^{-1/2}\delta_{\log p} - \mathbf{1}_{[\log 2,\log Z]}(u)\frac{e^{u/2}}{u}\, du$$

has no extra right-endpoint atom. At a prime endpoint the jump of P is precisely the prime atom already displayed. Consequently

$$\widehat{c}_Z(t) = \frac{M_Z(t)}{1/2 + it}, \qquad \|c_Z\|_2^2 = Q(Z) + \frac{P(Z)^2}{Z} \ll \frac{Z}{\log^2 Z}. \tag{62.7}$$

The last inequality is only Chebyshev's bound $|P(x)| \ll x/\log x$, integrated over a finite interval; it does not assume a uniform bound for Q.

Choose a fixed smooth η, with $0 \le \eta \le 1$, $\eta = 0$ on $(-\infty, u_0 - 1]$ and $\eta = 1$ on $[u_0, \infty)$. Set

$$Z = eY, \qquad \phi_Y(u) = \eta(u)\chi(u - \log Y).$$

Then $h_Y = \phi_Y c_Z$. The left and right transition intervals are disjoint for sufficiently large Y, and hence, for every integer $m \ge 1$,

$$\|\phi_Y^{(m)}\|_1 \le \|\eta^{(m)}\|_1 + \|\chi^{(m)}\|_1,$$

uniformly in Y. In particular Lemma 62.2 applies with $N = 2$ and a constant independent of the expanding plateau length.

Take $R = 2r(Z)$, square (62.4), and use (62.1) and (62.7). This gives

$$\begin{aligned} \|\Pi_{>R}h_Y\|_2^2 &\ll_\chi e^{-c_0\Omega(Z)} + R^{-4}\frac{Z}{\log^2 Z} \\ &\ll_\chi e^{-c_0\Omega(Z)} + \frac{e^{4c_0\Omega(Z)}}{Z\log^2 Z} \ll_\chi e^{-c_0\Omega(Z)}. \end{aligned}$$

The final comparison uses $\Omega(Z) = o(\log Z)$. Plancherel proves (62.6). For the alternative threshold, take any fixed sufficiently small $0 < c < c_0$. Eventually $\sqrt{Y}e^{-c\Omega(Y)} \ge 2r(eY)$, and $e^{-c_0\Omega(eY)} \le e^{-c\Omega(Y)}$. □

Corollary 62.4 (Exact equivalence after smoothing the primitive)**.** *Define*

$$Q_\chi(Y) = \|h_Y\|_2^2, \qquad \mathcal{L}_\chi(Y) = \frac{1}{2\pi}\int_{|t|\le R_\chi(Y)} |\mathcal{F}_{\chi,Y}(t)|^2 dt.$$

Then

$$Q(Y) \le Q_\chi(Y) \le Q(eY), \qquad Q_\chi(Y) = \mathcal{L}_\chi(Y) + \epsilon_\chi(Y), \qquad 0 \le \epsilon_\chi(Y) \ll_\chi e^{-c_0\Omega(eY)}. \tag{62.8}$$

For every prescribed unbounded set $\mathcal{S} \subset (2,\infty)$,

$$Q(\infty) < \infty \iff \liminf_{\substack{Y\to\infty \\ Y\in\mathcal{S}}} \mathcal{L}_\chi(Y) < \infty. \tag{62.9}$$

This is consequently equivalent to the unchanged original target that $R_4(X) + (3/4)\log\log X$ is bounded above.

Proof. The cutoff is one below Y, between zero and one on $[Y, eY]$, and zero above eY; this proves the two energy inequalities. Plancherel and Theorem 62.3 give the exact decomposition. Finite $Q(\infty)$ bounds every $\mathcal{L}_\chi(Y)$. Conversely, bounded values along an unbounded subsequence in $\mathcal{S}$ bound $Q(Y)$ there by (62.8). Since Q is nondecreasing, this bounds Q everywhere. The previously proved original-target equivalence then applies without changing its constants or primes. □

62.4 The same comparison for the original centered variance

One can retain the optimization over centers as well. Put

$$V_\chi(Y) = \min_{a\in\mathbb{R}} \int_2^{eY} \chi\Big(\log\frac{x}{Y}\Big)^2 \frac{(P(x)-a)^2}{x^2}\,dx,$$

and let $a_{\chi,Y}$ be its unique minimizing center.

Corollary 62.5 (Every-endpoint centered energy)**.** *The centers $a_{\chi,Y}$ are bounded as $Y\to\infty$. If*

$$\mathcal{F}_{\chi,Y,a}(t) = \int_2^{eY} \chi\Big(\log\frac{x}{Y}\Big)(P(x)-a)x^{-3/2-it}\,dx,$$

and

$$\mathcal{L}_{\chi,V}(Y) = \frac{1}{2\pi}\int_{|t|\le R_\chi(Y)} |\mathcal{F}_{\chi,Y,a_{\chi,Y}}(t)|^2 dt,$$

then, for every sufficiently large real Y,

$$V(Y) \le V_\chi(Y) \le V(eY), \qquad V_\chi(Y) = \mathcal{L}_{\chi,V}(Y) + \epsilon_{\chi,V}(Y), \qquad 0 \le \epsilon_{\chi,V}(Y) \ll_\chi e^{-c_0\Omega(eY)}. \tag{62.10}$$

In particular, the original target is equivalent to bounded $\mathcal{L}_{\chi,V}(Y)$ along an unbounded subsequence of any prescribed unbounded endpoint set.

Proof. The total weight $\int \chi^2 x^{-2}dx$ tends to $1/2$ and is at least $1/4$ for $Y \ge 4$. The primitive

$$J_2(x) = \int_2^x P(t)t^{-2}\,dt$$

is bounded unconditionally by the exact Mertens identity proved earlier. Integration by parts writes the numerator of the weighted mean $a_{\chi,Y}$ as

$$\int_0^1 -\frac{d}{dv}(\chi(v)^2)\,J_2(Ye^v)\,dv.$$

Since χ is nonincreasing, its weight is nonnegative and has total mass one. This proves $a_{\chi,Y} = O(1)$.

The transform subtracted when changing the center from zero to $a_{\chi,Y}$ is the transform of

$$a_{\chi,Y} g_Y(u), \qquad g_Y(u) = e^{-u/2}\chi(u - \log Y)\mathbf{1}_{[u_0,\infty)}(u).$$

Its distributional derivative has uniformly bounded total variation, including the fixed jump at u_0. Hence $|\widehat{g}_Y(t)| \ll_\chi |t|^{-1}$ for $t \neq 0$, and its squared Fourier tail is $O_\chi(R^{-1})$. At $R = R_\chi(Y)$, this is negligible compared with $e^{-c_0\Omega(eY)}$. Combining with Theorem 62.3 proves the claimed centered remainder.

For every fixed a, its weighted energy lies between the corresponding integrals on $[2, Y]$ and $[2, eY]$. Taking minima gives $V(Y) \le V_\chi(Y) \le V(eY)$. The last assertion follows from these inequalities, the vanishing remainder, and monotonicity of V. □

Scope of the refinement. The smoothing acts on an actual prime-discrepancy primitive, and (62.8) and (62.10) prove that its boundedness target has not changed. The high-frequency remainder is now proved at every endpoint, without requiring $P(q^-) \le 0 < P(q)$. The cutoff still grows like $Y^{1/2-o(1)}$; the remaining low-band energy has not been bounded. In particular, neither eventual negativity of $\mathcal{A}(X) = \int_2^X P(x)x^{-3/2}dx$ nor the original compensated-correlation upper bound is established here. All imported arithmetic input is identified, and no claim of historical priority is made.

63 One-band propagation from a positive-measure frequency set

The earlier one-band criterion can also tolerate gaps in the observed frequency interval and can use a logarithmic average rather than an L^2 norm. The following elementary Remez-type inequality makes the required propagation explicit.

Lemma 63.1 (Polynomial control from a measurable set)**.** *Let $E \subset [-T, T]$ have Lebesgue measure at least $m > 0$, and let p be a polynomial of degree at most $n \ge 1$. If $|p(t)| \le M$ for almost every $t \in E$, then*

$$\sup_{|z|\le R} |p(z)| \le M \left(\frac{2e(R+T)}{m}\right)^n. \tag{63.1}$$

Proof. Discard a null subset on which the stated bound fails. Choose $n+1$ quantile points $t_0 < \cdots < t_n$ of the remaining set, equally spaced in cumulative Lebesgue measure. If the actual measure is $m' \ge m$, the cumulative measure is 1-Lipschitz, so the points can be taken in its essential closure with

$$t_j - t_i \ge \frac{(j-i)m'}{n} \ge \frac{(j-i)m}{n} \quad (i < j).$$

The bound $|p(t_j)| \le M$ follows by continuity. The endpoint quantiles mean the essential infimum and supremum of the set. Lagrange interpolation, $|z - t_k| \le R + T$, and the separation give

$$|p(z)| \le M \left(\frac{n(R+T)}{m}\right)^n \sum_{j=0}^{n} \frac{1}{j!(n-j)!} = M \left(\frac{n(R+T)}{m}\right)^n \frac{2^n}{n!}.$$

Since $n! \ge (n/e)^n$, this proves (63.1). The degree-zero case holds directly without this formula. □

Theorem 63.2 (A logarithmic criterion on any one positive-measure set)**.** *Fix any bounded measurable set $E \subset \mathbb{R}$ of positive measure, and any prescribed $X_j \uparrow \infty$ satisfying $\log X_{j+1}/\log X_j \to 1$. Put*

$$J_E(X) = \int_E \log\left(1 + |F_X(1/2 + it)|\right)\, dt.$$

Then

$$\mathrm{RH} \iff \frac{J_E(X_j)}{\log X_j} \longrightarrow 0. \tag{63.2}$$

The arithmetic condition on the right is not proved here.

Proof. Under RH, Plancherel and the critical-energy bound give a uniform upper bound for $\int_E |F_X(1/2+it)|^2 dt$. Jensen's inequality and Cauchy–Schwarz then bound $J_E(X)$ uniformly in X.

For the converse, write $U_j = \log X_j$, and use the same greedy thinning, blocks B_j, and Taylor approximants p_j as in Lemma 58.2. Thinning preserves the assumed limit. Choose T so that $E \subset [-T, T]$, and put $m = |E| > 0$. On the real axis, the relation between a block and its two endpoint transforms yields

$$L_j^* := \int_E \log(1 + |B_j(t)|) dt \le J_E(X_j) + J_E(X_{j+1}) = o(U_j).$$

This uses $1 + |a - b| \le (1 + |a|)(1 + |b|)$, and is independent of the phases multiplying the block.

If $L_j^* = 0$, then B_j vanishes on a set of positive measure and hence identically. Otherwise Markov's inequality gives a measurable $E_j \subset E$ of measure at least $m/2$ on which

$$\log(1 + |B_j(t)|) \le \frac{2L_j^*}{m} = o(U_j).$$

The previously proved Taylor approximation has degree $n_j = o(U_j)$ and error at most e^{-U_j} on every fixed compact set. Thus on E_j, the polynomial p_j is bounded by $\exp(o(U_j))$. Lemma 63.1, whose constant depends on T, R, m but not on the shape of E_j, shows

$$\sup_{|z| \le R} |B_j(z)| \le \exp(o(U_j)) \qquad \text{for every fixed } R > 0.$$

The same normally convergent block series as in Lemma 58.2 continues $F(s)$ holomorphically to $\Re s > 1/2$. The nonzero-pole argument for $(sF(s))'$ then proves RH. □

Corollary 63.3 (A vanishing exceptional-frequency formulation)**.** *With the same E and endpoint sequence, RH is equivalent to*

$$\left| \left\{ t \in E : |F_{X_j}(1/2+it)| > \log X_j \right\} \right| \longrightarrow 0.$$

Proof. RH gives a uniform $L^2(E)$ bound, so the exceptional measure is $O((\log X_j)^{-2})$ by Chebyshev's inequality. Conversely, the unconditional estimate $P(x) = O(x)$ implies $|F_X(1/2+it)| \ll \sqrt{X}$, uniformly for real t. On the good set the logarithmic integrand is at most $\log(1 + \log X_j) = o(\log X_j)$. On the exceptional set it is at most $\frac{1}{2}\log X_j + O(1)$. A vanishing exceptional measure therefore implies $J_E(X_j) = o(\log X_j)$, and Theorem 63.2 applies. □

Theorem 63.4 (A fixed majority of frequencies suffices)**.** *Fix a bounded measurable set E of measure $m > 0$, a number $\theta \in (1/2, 1)$, and a prescribed sequence $X_j \uparrow \infty$ with $\log X_{j+1}/\log X_j \to 1$. Define*

$$G_j = \{ t \in E : |F_{X_j}(1/2+it)| \le \log X_j \}.$$

Then

$$\mathrm{RH} \iff |G_j| \ge \theta m \quad \textit{for every sufficiently large } j. \tag{63.3}$$

The good sets may change completely with j, and no bound is required on the complementary frequencies. In particular one may fix $E = [-1, 1]$, $\theta = 3/4$, and $X_j = 2^j$, obtaining

$$\mathrm{RH} \iff |\{ t \in [-1,1] : |F_{2^j}(1/2+it)| \le j \log 2 \}| \ge \frac{3}{2} \quad \textit{for every sufficiently large } j.$$

This majority condition for the actual primes remains unproved.

Proof. Under RH, the uniform $L^2(E)$ bound gives $m - |G_j| = O((\log X_j)^{-2})$, so every fixed $\theta < 1$ is eventually admissible.

Conversely, greedily thin the endpoint sequence as in Lemma 58.2, so that $U_j = \log X_j$ and its new gaps obey $1 \le U_{j+1} - U_j = o(U_j)$. The majority hypothesis survives thinning. For each pair of retained consecutive endpoints,

$$|G_j \cap G_{j+1}| \ge (2\theta - 1)m > 0.$$

On this intersection, the block relation gives

$$|B_j(t)| = |F_{X_{j+1}}(1/2 + it) - F_{X_j}(1/2 + it)| \le U_{j+1} + U_j = O(U_j).$$

The block Taylor polynomial has degree $n_j = o(U_j)$ and error at most e^{-U_j} on every fixed compact set. Choose T with $E \subset [-T, T]$. Lemma 63.1, applied to $G_j \cap G_{j+1}$, therefore bounds the polynomial on each fixed complex compact by

$$O(U_j) \left(\frac{2e(R+T)}{(2\theta - 1)m} \right)^{n_j} = \exp(o(U_j)).$$

Adding the Taylor error gives the same bound for B_j. The normally convergent block continuation and nonzero-pole argument prove RH exactly as before. No common good-frequency set for all endpoints is needed. □

Status of the reductions. The interpolation inequalities and the polynomial propagation estimate are unconditional. The eventual quartic-grid signs and the one-set logarithmic or exceptional-frequency conditions are unproved inputs, each equivalent to the original target. These results do not establish RH, and no historical-priority claim is made for the general Landau, interpolation, or Remez principles.

64 A wider local projection with unconditional critical forcing regularity

Put

$$G(x) = \sum_{p \le x} \frac{1}{p} - \log\log x, \qquad E(x) = G(x) - B_{\mathbb{P}}, \qquad P(x) = \pi(x) - \mathrm{Li}_2(x).$$

For every integer $m \ge 16$, define

$$a_m = \frac{m^2 \log m}{(\log\log m)^2}, \qquad h_m = a_{m+1} - a_m, \qquad I_m = [a_m, a_{m+1}).$$

Use also the initial interval $I_0 = [2, a_{16})$. On each cell let $q_I = |I|^{-1} \int_I G(t)\,dt$, let QG be the function equal to q_I on I, and define

$$r(x) = G(x) - QG(x).$$

This residual is unchanged if G is replaced everywhere by E; no Mertens constant needs to be evaluated.

Theorem 64.1 (A near-square-root logarithmic enlargement)**.** *The cells are strictly increasing, cover $[2, \infty)$, and $1 < h_m < a_m$. Moreover,*

$$h_m \sim \frac{2m \log m}{(\log\log m)^2} \sim \sqrt{2}\, \frac{\sqrt{a_m \log a_m}}{\log\log a_m}.$$

Unconditionally,

$$r \in L^2([2, \infty), dx), \qquad \int_{a_M}^{\infty} |r(x)|^2\,dx \ll \frac{1}{\log\log M} \qquad (M \ge 16).$$

Proof. For real $t \geq 16$, write $L = \log t$ and $\ell = \log L$. The defining function $a(t) = t^2 L/\ell^2$ satisfies

$$\frac{a'(t)}{a(t)} = \frac{1}{t}\left(2 + \frac{1}{L} - \frac{2}{L\ell}\right).$$

Since $L > e$ and $\ell > 1$, the parenthesis lies strictly between $3/2$ and $5/2$. Thus a is increasing and

$$1 < \frac{a(m+1)}{a(m)} \leq \left(1 + \frac{1}{m}\right)^{5/2} \leq (17/16)^{5/2} < 2.$$

Also $a(t) \geq t^2$: the function $L/(\log L)^2$ has minimum $e^2/4 > 1$ for $L > 1$. Hence

$$h_m = \int_m^{m+1} a'(t)\, dt \geq \frac{3a(m)}{2(m+1)} > 1,$$

while the preceding ratio bound gives $h_m < a_m$. Differentiation and the mean-value theorem give the first asymptotic. The second follows from $\log a_m \sim 2 \log m$ and $\log \log a_m \sim \log \log m$.

Use the unconditional Brun–Titchmarsh estimate

$$\pi(a+h) - \pi(a) < 2h/\log h \qquad (a > 0,\ h > 1).$$

For $a \leq u < v \leq a + h$, with $1 < h \leq a$,

$$G(v) - G(u) = \sum_{u < p \leq v} \frac{1}{p} - \int_u^v \frac{dt}{t \log t}.$$

The two nonnegative terms on the right are at most $2h/(a \log h)$ and $h/(a \log a)$, respectively. Therefore

$$\operatorname{osc}_{[a,a+h]} G \leq \frac{2h}{a \log h}.$$

The variance on a set of measure h is at most one quarter of h times the squared range, by choosing the midpoint of the range as a trial center. Consequently

$$\int_{I_m} |r(x)|^2\, dx \leq \frac{h_m^3}{a_m^2 \log^2 h_m} \ll \frac{1}{m \log m (\log \log m)^2}.$$

The initial interval contributes a finite amount. Finally,

$$\int_M^\infty \frac{dt}{t \log t (\log \log t)^2} = \frac{1}{\log \log M}$$

proves convergence and the stated tail bound, with a fixed constant absorbing the finitely many initial cells. □

Input and scope. The sieve inequality is Montgomery–Vaughan, *The large sieve*, Mathematika **20** (1973), 119–134, Theorem 2, equation (1.12), printed p. 121 [30]: https://personal.science.psu.edu/rcv4/personal/Publications/large_sieve.pdf. No prime-in-every-cell assertion, RH, or global zero-free hypothesis is used. The theorem bounds a residual after local centering; it does not bound the energy of the cell means. The same computation for $a_m = m^2 \log m/(\log \log m)^{2\beta}$, $\beta > 1/2$, gives a tail $O((\log \log M)^{1-2\beta})$, after a fixed sufficiently large initial index. The explicit choice $\beta = 1$, $m \geq 16$ avoids any issue with small logarithms.

64.1 The fine forcing is genuinely a negative Sobolev distribution

Use the Fourier convention $\widehat{f}(\xi) = \int_{\mathbb{R}} e^{-i\xi s} f(s)\, ds$, with the usual L^2 extension. Write

$$\|\tau\|^2_{-1,1/2} := \frac{1}{2\pi} \int_{\mathbb{R}} \frac{|\widehat{\tau}(\xi)|^2}{\xi^2 + 1/4}\, d\xi$$

when this expression is finite; it is an equivalent negative Sobolev norm of order one. Set $s_0 = \log 2$ and define the zero extension

$$F(s) = \begin{cases} e^{s/2} r(e^s), & s \geq s_0, \\ 0, & s < s_0. \end{cases}$$

Theorem 64.2 (Unconditional regularity of the fine critical forcing)**.** *The distribution*

$$\nu_{\mathrm{fine}} = (D - \tfrac{1}{2})F, \qquad D = d/ds,$$

is tempered and has finite negative Sobolev norm. In fact,

$$\boxed{\|\nu_{\mathrm{fine}}\|^2_{-1,1/2} = \|F\|_2^2 = \int_2^\infty |r(x)|^2\, dx < \infty.}$$

For

$$\nu_{\mathrm{fine},M} = (D - \tfrac{1}{2})(F\, \mathbf{1}_{[\log a_M, \infty)}),$$

including its cutoff boundary atom, one likewise has

$$\|\nu_{\mathrm{fine},M}\|^2_{-1,1/2} = \int_{a_M}^\infty |r(x)|^2\, dx \ll \frac{1}{\log\log M}.$$

Proof. The change of variable $x = e^s$ gives $\|F\|_2^2 = \int_2^\infty |r(x)|^2 dx$, which is finite by Theorem 64.1. An L^2 function and its distributional derivative are tempered. Moreover,

$$\widehat{\nu_{\mathrm{fine}}}(\xi) = (i\xi - \tfrac{1}{2})\widehat{F}(\xi).$$

Plancherel's theorem and $|i\xi - \frac{1}{2}|^2 = \xi^2 + 1/4$ give the exact norm identity. The same proof applies to the cutoff L^2 function and proves the tail statement. □

There is also an exact locally finite measure description. Enumerate the cell boundaries as

$$b_0 = 2, \quad b_1 = a_{16}, \quad b_2 = a_{17}, \quad \ldots,$$

and let q_j be the mean of G on $[b_j, b_{j+1})$. The full critical prime discrepancy is well-defined as a locally finite signed measure

$$\nu = \sum_{p \geq 2} p^{-1/2} \delta_{\log p} - \mathbf{1}_{[s_0,\infty)}(s) \frac{e^{s/2}}{s}\, ds.$$

Without assuming that this full measure is tempered, one has the exact identity in distributions on compactly supported tests:

$$\boxed{\nu = \nu_{\mathrm{fine}} + \sqrt{2}\,(q_0 + \log\log 2)\delta_{s_0} + \sum_{j \geq 1} \sqrt{b_j}\,(q_j - q_{j-1})\delta_{\log b_j}.} \tag{64.1}$$

Indeed, applying $D - 1/2$ to the zero extension of $e^{s/2}G(e^s)$ produces $\nu - \sqrt{2}\log\log 2\, \delta_{s_0}$. The term $-\sqrt{2}\log\log 2$ is the difference between its initial value $\sqrt{2}G(2) = 1/\sqrt{2} - \sqrt{2}\log\log 2$ and the prime-2 atom $1/\sqrt{2}$. Applying the same operator to the locally constant cell projection produces

its initial atom and the displayed boundary jumps. Subtraction proves (64.1), including any coincident prime and cell boundary.

Thus the fine component has been proved to lie in H^{-1}, and the uncontrolled component is an explicit atomic comb at the cell boundaries. All coefficients are independent of the additive Mertens constant. The assertion is local distributionally; it does not assert that the separate infinite prime and continuous sums converge absolutely against arbitrary Schwartz functions. Temperedness of the fine component has been proved from its L^2 primitive, rather than assumed.

Part V
Signed primitives, positive kernels, and sparse sampling

65 A scalar one-sided criterion and the exact prime-power drift

Keep the normalization $P(x) = \pi(x) - \mathrm{Li}_2(x)$, where $\mathrm{Li}_2(x) = \int_2^x dt/\log t$, and define

$$\mathcal{A}(X) = \int_2^X \frac{P(x)}{x^{3/2}}\,dx.$$

This is a signed linear statistic, not the nonnegative variance. The change of target below is justified by a two-sided theorem. One-sided weighted integral criteria and Landau's positivity method have substantial precedents; in particular, see Johnston [39, Theorem 1.4]. Our statements retain the Li_2 normalization and make the critical $\log\log X$ compensation and its constant explicit. No new general positivity principle or historical priority is claimed.

Write $\mathcal{P}(s) = \sum_p p^{-s}$ for $\Re s > 1$, and put

$$C_{\mathrm{pp}} = B_{\mathbb{P}} - \log 2 + 2\sum_{k\ge 3} \frac{\mathcal{P}(k/2)}{k}, \tag{65.1}$$

$$C_* = 2\log(-\zeta(1/2)) + 2\,\mathrm{Ei}\Big(\frac{\log 2}{2}\Big) - C_{\mathrm{pp}}. \tag{65.2}$$

The logarithm is real because $\zeta(1/2) < 0$. The series is absolutely convergent. Numerical evaluation gives $C_* \approx -0.06142989385861936738$; the exact formula, not this decimal, defines the constant.

Proposition 65.1 (The entire prime-power drift is unconditional)**.** *Let*

$$J(x) = \sum_{k\ge 1} \frac{1}{k}\pi(x^{1/k}), \qquad \mathcal{B}_J(X) = \int_2^X \frac{J(x) - \mathrm{Li}_2(x)}{x^{3/2}}\,dx.$$

Then, without RH,

$$\mathcal{A}(X) + \log\log X = \mathcal{B}_J(X) - C_{\mathrm{pp}} + O(1/\log X). \tag{65.3}$$

More precisely, the square contribution is exactly

$$\frac{1}{2}\int_2^X \frac{\pi(\sqrt{x})}{x^{3/2}}\,dx = \sum_{p\le\sqrt{X}} \frac{1}{p} - \frac{\pi(\sqrt{X})}{\sqrt{X}}, \tag{65.4}$$

and the contribution of powers of order at least three converges to $2\sum_{k\ge 3}\mathcal{P}(k/2)/k$, with remainder $O(X^{-1/6}/\log X)$.

Proof. For each finite cutoff, all relevant sums are finite and $J - \pi = \sum_{k\ge 2}\pi(x^{1/k})/k$. Substitution $u = \sqrt{x}$ and finite partial summation prove (65.4). Mertens' estimate $\sum_{p\le y} p^{-1} = \log\log y + B_{\mathbb{P}} + O(1/\log y)$, together with Chebyshev's bound, makes this

$$\log\log X + B_{\mathbb{P}} - \log 2 + O(1/\log X).$$

For a primary exposition of the elementary remainder in Mertens' theorem, see [40].

For each $k \ge 3$, Tonelli's theorem gives

$$\frac{1}{k}\int_2^\infty \frac{\pi(x^{1/k})}{x^{3/2}}\,dx = \frac{2}{k}\sum_p p^{-k/2} = \frac{2}{k}\mathcal{P}(k/2).$$

The sum over $k \geq 3$ converges because

$$\sum_{k\geq 3} \frac{\mathcal{P}(k/2)}{k} \leq \sum_{p} \sum_{k\geq 3} p^{-k/2} = \sum_{p} \frac{p^{-3/2}}{1-p^{-1/2}} < \infty.$$

For large x, Chebyshev's bound for $k = 3$ and the trivial bound for $4 \leq k \leq \log x/\log 2$ yield

$$\sum_{k\geq 3} \frac{1}{k}\pi(x^{1/k}) \ll \frac{x^{1/3}}{\log x} + x^{1/4}\log\log x \ll \frac{x^{1/3}}{\log x}.$$

Integrating its product with $x^{-3/2}$ from X to infinity proves the asserted tail bound. Combining the square and higher-power terms proves (65.3). □

Lemma 65.2 (Landau positivity for a Mellin integral)**.** *Let $b : [2,\infty) \to [0,\infty)$ be locally integrable, and suppose $L(z) = \int_2^\infty b(x)x^{-z-1}dx$ converges in some right half-plane. If its abscissa of convergence σ_c is finite, then the real point $z = \sigma_c$ is a singularity of L.*

Proof. Suppose instead that L is analytic in a disk of radius $r > 0$ about σ_c. Set $a = \sigma_c + r/4$, $b_1 = \sigma_c - r/4$. The Taylor series about a converges at b_1, and differentiation in the original convergence half-plane gives

$$(-1)^n L^{(n)}(a) = \int_2^\infty b(x)(\log x)^n x^{-a-1}\,dx \geq 0.$$

All terms in the Taylor series evaluated at b_1 are nonnegative. Tonelli's theorem therefore gives

$$L(b_1) = \sum_{n\geq 0} \frac{(a-b_1)^n}{n!} \int_2^\infty b(x)(\log x)^n x^{-a-1}\,dx = \int_2^\infty b(x)x^{-b_1-1}\,dx < \infty.$$

This contradicts $b_1 < \sigma_c$. □

Proposition 65.3 (A one-sided subpower bound already excludes off-line zeros)**.** *Suppose that for every $\varepsilon > 0$ there is a finite C_ε such that*

$$\mathcal{A}(X) \leq C_\varepsilon X^\varepsilon \quad \text{for all sufficiently large } X. \tag{65.5}$$

Then RH holds. The same conclusion follows if the inequality is replaced by $\mathcal{A}(X) \geq -C_\varepsilon X^\varepsilon$.

Proof. Fix $0 < \varepsilon < 1/2$, enlarge $C_\varepsilon > 0$ to cover the initial compact interval, and put $b_\varepsilon(X) = C_\varepsilon X^\varepsilon - \mathcal{A}(X) \geq 0$. Chebyshev's bound gives $\mathcal{A}(X) = O(\sqrt{X}/\log X)$, so its Mellin integral converges initially for $\Re z > 1/2$. If

$$F(s) = \int_2^\infty P(x)x^{-s-1}\,dx \qquad (\Re s > 1),$$

integration by parts, using $\mathcal{A}(2) = 0$, gives

$$L_\varepsilon(z) := \int_2^\infty b_\varepsilon(x)x^{-z-1}\,dx = \frac{C_\varepsilon 2^{\varepsilon-z}}{z-\varepsilon} - \frac{F(1/2+z)}{z}. \tag{65.6}$$

We need only continuation along the positive real axis at this point; no zero-free complex half-plane is assumed. For $\Re s > 1$, the Euler product gives

$$sF(s) = H(s) - \sum_{k\geq 2} \frac{\mathcal{P}(ks)}{k}, \tag{65.7}$$

where, writing $u_0 = \log 2$,

$$H(s) = \log((s-1)\zeta(s)) + \gamma + \log u_0 + \sum_{n\geq 1} \frac{[-(s-1)u_0]^n}{n\, n!}.$$

This follows by expanding $\int_2^\infty x^{-s}/\log x\, dx = E_1((s-1)u_0)$. The function $(s-1)\zeta(s)$ is positive and nonzero on the real interval $s > 1/2$, including its removable value at $s = 1$. Indeed, $\zeta(s) < 0$ for $0 < s < 1$ by the alternating eta series. Thus H, with its real logarithm continued from $s > 1$, is analytic in a neighborhood of each point of this real interval. The prime-power sum is normally convergent on $\Re s > 1/2$. Consequently $F(1/2 + z)$ has analytic continuation along every real $z > 0$.

If the convergence abscissa of L_ε were greater than ε, equation (65.6) and this real-axis continuation would contradict Lemma 65.2. Hence that abscissa is at most ε, and the nonnegative integral defines a holomorphic function for $\Re z > \varepsilon$. Solving (65.6) for F gives a holomorphic continuation to $\Re s > 1/2 + \varepsilon$.

In that half-plane the derivative of $sF(s)$ must agree with the meromorphic function

$$\frac{\zeta'(s)}{\zeta(s)} - \sum_{k\geq 2} \mathcal{P}'(ks) + \frac{2^{1-s}}{s-1}.$$

Its pole at 1 cancels, but a nontrivial zeta zero has a nonzero residue equal to its multiplicity. Hence there can be no zero with real part exceeding $1/2+\varepsilon$. Letting $\varepsilon \downarrow 0$ and using the functional equation proves RH. For the lower-bound variant, use $b_\varepsilon = C_\varepsilon X^\varepsilon + \mathcal{A}$ and change the sign of the second term in (65.6); the argument is identical. □

Theorem 65.4 (The compensated scalar limit under RH)**.** *If RH holds, then*

$$\mathcal{A}(X) + \log\log X \longrightarrow C_*. \tag{65.8}$$

No convergence rate is asserted here.

Proof. The previously established implication RH $\Rightarrow Q(\infty) < \infty$ gives

$$f(u) = e^{-u/2} P(e^u) \mathbf{1}_{[u_0,\infty)}(u) \in L^2(\mathbb{R}).$$

Its Laplace transform is $F(1/2 + z)$ for $\Re z > 0$, by uniqueness from the original domain $\Re z > 1/2$. This use of the critical Hardy norm is conditional on RH; it is not an unconditional assertion of temperedness.

We first calculate the local singularity. The constant identity

$$B_{\mathbb{P}} = \gamma - \sum_{j\geq 2} \frac{\mathcal{P}(j)}{j}$$

and the Euler product give, locally as $z \to 0$ with $\Re z > 0$,

$$\mathcal{P}(1 + 2z) = -\log(2z) + B_{\mathbb{P}} - \gamma + O(z).$$

In (65.7), separate its $k = 2$ term. The remainder is analytic near $s = 1/2$, since $\zeta(1/2) \neq 0$. On the real interval $1/2 < s < 1$,

$$H(s) = \log(-\zeta(s)) + \mathrm{Ei}((1-s)u_0).$$

It follows, with the principal logarithm on $\Re z > 0$, that

$$F(1/2 + z) = \log z + (C_* + \gamma) + O(z \log z). \tag{65.9}$$

This also verifies every constant in (65.2).

Now put

$$m(u) = -\frac{\mathbf{1}_{[u_0,\infty)}(u)}{u}, \qquad \eta(u) = e^{-(u-u_0)}\mathbf{1}_{[u_0,\infty)}(u), \qquad c = C_* - \log u_0.$$

Both m and η belong to L^2, and $\int \eta = 1$. Their Laplace transforms are $-E_1(zu_0)$ and $e^{-zu_0}/(1+z)$, respectively. Let $r = f - m - c\eta$. From (65.9) and $E_1(zu_0) = -\gamma - \log(zu_0) + O(z)$, its boundary Fourier transform satisfies

$$\widehat{r}(\xi) = O(|\xi| \log(1/|\xi|)) \quad (\xi \to 0).$$

Boundary values here agree almost everywhere with the L^2 Fourier transform: exponential damping converges in L^2, while the displayed local analytic formula gives the pointwise boundary value away from zero.

Therefore $\widehat{r}(\xi)/(i\xi) \in L^1(\mathbb{R})$. Near zero this follows from integrability of $\log(1/|\xi|)$; for $|\xi| \geq 1$, it follows from Cauchy–Schwarz applied to $\widehat{r} \in L^2$ and $1/\xi \in L^2$. Let

$$v(U) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{\widehat{r}(\xi)}{i\xi} e^{i\xi U}\, d\xi.$$

By the Riemann–Lebesgue lemma, v is continuous and tends to zero as $U \to \pm\infty$. In distributions, $v' = r$. Since r vanishes below u_0, v is constant there, and its limit at $-\infty$ makes that constant zero. Hence $v(U) = \int_{u_0}^{U} r(u)du$.

For $U = \log X \geq u_0$, we obtain

$$\mathcal{A}(e^U) = -\log U + \log u_0 + c(1 - e^{-(U-u_0)}) + v(U).$$

Letting $U \to \infty$ gives $\mathcal{A}(X) + \log\log X \to \log u_0 + c = C_*$, as claimed. □

Corollary 65.5 (Exact induced scalar criteria)**.** *Each of the following is equivalent to RH and hence to the unchanged bound on $R_4(X) + (3/4)\log\log X$:*

1. *$\mathcal{A}(X) + \log\log X$ is bounded above;*
2. *$\mathcal{A}(X) \leq 0$ for every sufficiently large X;*
3. *the one-sided subpower condition (65.5);*
4. *$\mathcal{A}(X) + \log\log X \to C_*$;*
5. *$\mathcal{B}_J(X)$ is bounded above.*

Proof. Theorem 65.4 proves all the stated bounds under RH, including eventual negativity. Each of assertions 1, 2, and 4 implies the one-sided subpower upper bound; Proposition 65.3 then gives RH. Assertion 5 is equivalent to assertion 1 by Proposition 65.1. □

Corollary 65.6 (Unconditional negative visits and conditional oscillation)**.** *Unconditionally, $\mathcal{A}(X) < 0$ for arbitrarily large X. If ζ has a zero with real part $\beta > 1/2$, then for every $0 < \delta < \beta - 1/2$,*

$$\limsup_{X\to\infty} \frac{\mathcal{A}(X)}{X^\delta} = +\infty, \qquad \liminf_{X\to\infty} \frac{\mathcal{A}(X)}{X^\delta} = -\infty.$$

Proof. If $\mathcal{A} \geq 0$ eventually, the lower-bound version of Proposition 65.3 would imply RH, whereas Theorem 65.4 would force $\mathcal{A} \to -\infty$. Thus eventual nonnegativity is impossible. For the second assertion, the proof of Proposition 65.3 with the single exponent $\varepsilon = \delta$ excludes a zero with real part greater than $1/2 + \delta$, if either one-sided eventual bound with that exponent holds. Neither can therefore hold. □

Corollary 65.7 (Sparse scalar recurrence holds unconditionally)**.** *Put* $\mathcal{S}(X) = \mathcal{A}(X) + \log\log X$. *Unconditionally,*

$$\liminf_{X\to\infty} |\mathcal{S}(X) - C_*| = 0, \qquad \liminf_{n\to\infty} |\mathcal{S}(n) - C_*| = 0.$$

If RH is false, $\mathcal{S}(X) = C_*$ *at arbitrarily large real values of* X.

Proof. If RH is true, Theorem 65.4 gives convergence. If RH is false, Corollary 65.6 gives $\limsup \mathcal{S} = +\infty$ and $\liminf \mathcal{S} = -\infty$, because a positive power dominates $\log\log X$. Continuity and the intermediate value theorem give arbitrarily late exact crossings of C_*. For the integer version, Chebyshev's bound gives, uniformly on $[n, n+1]$,

$$|\mathcal{S}(x) - \mathcal{S}(n)| \le \int_n^{n+1} \left(\frac{|P(t)|}{t^{3/2}} + \frac{1}{t\log t} \right) dt \ll \frac{1}{\sqrt{n}\log n} \longrightarrow 0.$$

Thus real crossings can be approximated by integer endpoints. □

This recurrence is not a substitute for eventual one-sided control: it holds in both the RH and non-RH alternatives. Weakening the scalar criterion to arbitrarily late close visits therefore loses its equivalence to the original target.

The unresolved arithmetic step. There is an exact finite prime-sum formula,

$$\mathcal{A}(X) = 2\sum_{p\le X} \frac{1}{\sqrt{p}} - \frac{2P(X)}{\sqrt{X}} - 2\,\mathrm{Ei}\Big(\frac{\log X}{2}\Big) + 2\,\mathrm{Ei}\Big(\frac{\log 2}{2}\Big).$$

The preceding arguments do not prove that this signed quantity is eventually nonpositive, or that its compensated version is bounded above. The prime-power drift has been removed unconditionally, but an upper bound for the remaining $\mathcal{B}_J$ has not been obtained. In particular, the critical choice $x^{-3/2}$ cannot be replaced by the easier weight x^{-2}: the latter belongs to a different, unconditionally controlled problem as shown in [39, Theorem 1.3].

66 Quartic sampling of the signed primitive

Keep

$$P(x) = \pi(x) - \mathrm{Li}_2(x), \qquad \mathcal{A}(X) = \int_2^X P(x)x^{-3/2}\,dx, \qquad S(X) = \mathcal{A}(X) + \log\log X.$$

The new input in this section is an unconditional interpolation inequality for the actual prime-counting staircase. Its proof uses the positive sign of each jump, not an assumption about cancellation or about the location of zeta zeros.

Proposition 66.1 (One-sided curvature control)**.** *One may take* $a_0 = 17$ *and* $K_0 = 59/20$ *in the following estimate. For every* $b > a \ge a_0$, *every* $x \in [a, b]$, *and* $\theta = (x-a)/(b-a)$,

$$\mathcal{A}(x) \le (1-\theta)\mathcal{A}(a) + \theta\mathcal{A}(b) + \frac{K_0(x-a)(b-x)}{2a^{3/2}\log a}. \tag{66.1}$$

In particular,

$$\sup_{a\le x\le b} \mathcal{A}(x) \le \max\{\mathcal{A}(a), \mathcal{A}(b)\} + \frac{K_0(b-a)^2}{8a^{3/2}\log a}. \tag{66.2}$$

Both estimates are unconditional and include intervals containing arbitrarily many primes.

Proof. The first derivative $\mathcal{A}'(x) = P(x)x^{-3/2}$ is locally of bounded variation. Its distributional derivative on $(2,\infty)$ is the signed measure

$$\mathcal{A}'' = \sum_p p^{-3/2}\delta_p - \left(\frac{x^{-3/2}}{\log x} + \frac{3}{2}P(x)x^{-5/2}\right)dx. \tag{66.3}$$

Since $\mathrm{Li}_2(x) \geq 0$, we have $P(x) \leq \pi(x)$. The explicit upper bound $\pi(x) < (13/10)x/\log x$ for $x \geq 17$, recorded in [39, Lemma 2.2] from Rosser–Schoenfeld, therefore gives

$$\frac{x^{-3/2}}{\log x} + \frac{3}{2}P(x)x^{-5/2} \leq \frac{59}{20}\frac{x^{-3/2}}{\log x} \leq \frac{K_0}{a^{3/2}\log a} \qquad (a \leq x \leq b).$$

The atoms in (66.3) are nonnegative. Consequently

$$\mathcal{A}'' \geq -\frac{K_0}{a^{3/2}\log a}\,dx \quad \text{on } (a,b).$$

Adding $K_0x^2/(2a^{3/2}\log a)$ thus makes $\mathcal{A}$ convex on $[a,b]$. This standard distributional form of convexity also follows by mollifying on inner subintervals and then taking a limit, using continuity of $\mathcal{A}$. The convex chord inequality gives (66.1), since $(1-\theta)a^2 + \theta b^2 - x^2 = (x-a)(b-x)$. Finally $(x-a)(b-x) \leq (b-a)^2/4$, proving (66.2). Atoms at either endpoint do not affect the chord inequality, since $\mathcal{A}$ itself is continuous. □

Theorem 66.2 (A general scalar sampling criterion)**.** *Let $a_j \uparrow \infty$ be any prescribed sequence, let $h_j = a_{j+1} - a_j$, and suppose*

$$\frac{\log\left(1 + h_j^2/(a_j^{3/2}\log a_j)\right)}{\log a_j} \longrightarrow 0. \tag{66.4}$$

Then the following are equivalent:

1. *RH;*
2. *$\mathcal{A}(a_j) \leq 0$ for every sufficiently large j;*
3. *for every $\varepsilon > 0$, there is a finite C_ε such that $\mathcal{A}(a_j) \leq C_\varepsilon a_j^{\varepsilon}$ for every sufficiently large j.*

These statements are also equivalent to the original bound on $R_4(X) + (3/4)\log\log X$.

Proof. The previously proved conditional asymptotic $\mathcal{A}(X) = -\log\log X + C_* + o(1)$ shows that RH implies assertion 2, which implies assertion 3.

Assume assertion 3. Condition (66.4) says exactly that its nonnegative defect is bounded by every positive power of a_j, up to an exponent-dependent constant. In particular, taking any fixed exponent less than $1/2$ shows that $h_j/a_j \to 0$, so $a_{j+1}/a_j \to 1$. For $x \in [a_j, a_{j+1}]$, apply (66.2). For every fixed $\varepsilon > 0$, use assertion 3 and the mesh bound with, for example, exponent $\varepsilon/2$. Both endpoint values and the interpolation defect are then at most $C'_\varepsilon x^{\varepsilon}$. Thus

$$\mathcal{A}(x) \leq C'_\varepsilon x^{\varepsilon} \quad \text{for all sufficiently large } x.$$

The one-sided Landau criterion already proved in Proposition 65.3 gives RH. No negativity of $\mathcal{A}$ between sampled points has been assumed: the interpolation error is explicitly retained. □

Corollary 66.3 (A quartic-grid sign test)**.** *One has the exact equivalence*

$$\boxed{\text{RH} \iff \exists n_0\ \forall n \geq n_0 : \quad \mathcal{A}(n^4) \leq 0.} \tag{66.5}$$

Unconditionally, for $n^4 \leq x \leq (n+1)^4$,

$$\mathcal{A}(x) \leq \max\{\mathcal{A}(n^4), \mathcal{A}((n+1)^4)\} + O(1/\log x). \tag{66.6}$$

Proof. Here $h_n = (n+1)^4 - n^4 = 4n^3(1 + O(1/n))$. Hence

$$\frac{h_n^2}{(n^4)^{3/2}\log(n^4)} = \frac{16 + O(1/n)}{\log(n^4)}.$$

Proposition 66.1 proves (66.6), and Theorem 66.2 gives the equivalence. □

For a finite interval, the same estimate is an explicit certificate: if $n \geq 3$, $h_n = (n+1)^4 - n^4$, and both sampled endpoint values are at most

$$-\frac{59h_n^2}{160n^6\log(n^4)},$$

then $\mathcal{A}(x) \leq 0$ throughout $[n^4, (n+1)^4]$. This finite certificate does not assert that the endpoint inequalities persist for every sufficiently large n.

Corollary 66.4 (The first interpolation bound on a coarser mesh)**.** *Fix a real $k > 4$. If $\mathcal{A}(n^k) \leq 0$ for every sufficiently large integer n, then $\zeta(s)$ has no zeros in*

$$\Re s > 1 - \frac{2}{k}.$$

This is an implication from an unproved arithmetic hypothesis, not an unconditional zero-free strip.

Proof. The interpolation defect is

$$O\left(\frac{x^{1/2-2/k}}{\log x}\right) \quad (n^k \leq x \leq (n+1)^k).$$

Thus the assumed sampled sign gives the one-sided bound $\mathcal{A}(x) \ll x^d_k$, with $d_k = 1/2 - 2/k \in (0, 1/2)$. The proof of Proposition 65.3, applied with this single exponent d_k, excludes zeros with real part greater than $1/2 + d_k = 1 - 2/k$. □

This records only the direct curvature estimate. Corollary 82.4 later upgrades the same eventual-sign hypothesis to RH for every fixed $k > 0$.

An explicit arithmetic form. At each sampled point the signed primitive has the finite formula

$$\mathcal{A}(n^4) = 2\sum_{p \leq n^4} \frac{1}{\sqrt{p}} - \frac{2P(n^4)}{n^2} - 2\,\mathrm{Ei}(2\log n) + 2\,\mathrm{Ei}\left(\frac{\log 2}{2}\right).$$

Theorem 66.2 does not prove the required sign of this expression. It proves that no additional continuum estimate is needed once those eventual sampled signs are known. A finite number of sample checks cannot establish (66.5).

66.1 Two-sided interpolation and exact sampled limits

The positive-jump argument is enough for the upper criterion. Brun–Titchmarsh also bounds the total size of the jumps and gives a two-sided approximation. We use the unconditional bound $\pi(a+h) - \pi(a) \ll h/\log h$ for $h \geq 2$, as in Montgomery–Vaughan [30].

Proposition 66.5 (A two-sided interpolation error)**.** *Let a be sufficiently large and $2 \le h = b - a \le a$. For $x \in [a, b]$, let $\ell_{a,b}(x)$ be the affine interpolant of $\mathcal{A}(a), \mathcal{A}(b)$. Then*

$$\sup_{a \le x \le b} |\mathcal{A}(x) - \ell_{a,b}(x)| \ll \frac{h^2}{a^{3/2} \log h}. \tag{66.7}$$

Consequently, writing ℓ_n^S for the affine interpolant of $S(n^4), S((n+1)^4)$,

$$\sup_{n^4 \le x \le (n+1)^4} |S(x) - \ell_n^S(x)| \ll \frac{1}{\log n}.$$

In particular, as equalities of extended real numbers,

$$\limsup_{X \to \infty} S(X) = \limsup_{n \to \infty} S(n^4),$$
$$\liminf_{X \to \infty} S(X) = \liminf_{n \to \infty} S(n^4).$$

Thus $S(X) \to C$ if and only if $S(n^4) \to C$, for every finite real C. The same limsup and liminf identities hold for $\mathcal{A}$ itself.

Proof. Chebyshev's bound and the elementary bound for Li_2 give $|P(x)| \ll x/\log x$. The total variation of the continuous part of (66.3) on $[a, b]$ is therefore at most $Ch/(a^{3/2} \log a)$. Brun–Titchmarsh bounds the atomic part by

$$\sum_{a < p \le b} p^{-3/2} \le a^{-3/2}(\pi(b) - \pi(a)) \ll \frac{h}{a^{3/2} \log h}.$$

Since $h \le a$, the continuous part obeys the same upper bound. Thus the signed measure $\mathcal{A}''$, restricted to the open interval (a, b), has total variation at most $Ch/(a^{3/2} \log h)$.

The Dirichlet Green kernel of $-d^2/dx^2$ on $[a, b]$ is

$$G_{a,b}(x, t) = \frac{(\min(x, t) - a)(b - \max(x, t))}{b - a}, \qquad 0 \le G_{a,b}(x, t) \le \frac{h}{4}.$$

Distributional integration with the prescribed endpoint values gives

$$\mathcal{A}(x) - \ell_{a,b}(x) = -\int_{(a,b)} G_{a,b}(x, t)\, d\mathcal{A}'(t).$$

The total-variation estimate proves (66.7).

For the quartic grid, $h \asymp n^3$, $a = n^4$, and $\log h \asymp \log n$, so this error is $O(1/\log n)$. The variation of $\log \log x$ across this interval is $O(h/(a \log a)) = O(1/(n \log n))$, so adding this compensation preserves the claimed interpolation error for S. Every affine interpolant lies between its two endpoint values. Taking upper and lower limits, and using that the error tends to zero, gives the two identities. The reverse inequalities follow by restricting the continuous variable to the sampled points. □

67 The first scalar correction and a lower-barrier criterion

The earlier conditional scalar limit can be sharpened. This section uses the classical integrated explicit formula, not an unconditional bound for the original target. Put $U = \log X$ and retain $\mathcal{S}(X) = \mathcal{A}(X) + \log \log X$ and the exact constant C_*.

Theorem 67.1. *Assume RH. With zeros counted with multiplicity, define*

$$Z(U) = \sum_{\rho=1/2+i\gamma} \frac{e^{i\gamma U}}{\rho\, i\gamma}.$$

This is a real-valued, uniformly and absolutely convergent series on the real line. Then

$$\mathcal{S}(X) = C_* + \frac{2 - Z(\log X)}{\log X} + O\Big(\frac{1}{\log^2 X}\Big). \tag{67.1}$$

Moreover $|Z(U)| < 0.04621$ *for every real* U*. Consequently, under RH,*

$$\frac{1.95}{\log X} \le \mathcal{S}(X) - C_* \le \frac{2.05}{\log X} \qquad \textit{for every sufficiently large } X. \tag{67.2}$$

No numerical starting point is asserted.

Proof. Let $u_0 = \log 2$, $D(x) = \psi(x) - x$, and $f_\psi(u) = e^{-u/2}D(e^u)$ for $u \ge u_0$. The integrated explicit formula for ψ (see, for example, [39, Lemma 2.6]) gives in distributions on (u_0, ∞)

$$f_\psi(u) = -Z'(u) + r(u), \qquad r(u) = -e^{-u/2}\log(2\pi) - \tfrac{1}{2}e^{-u/2}\log(1 - e^{-2u}). \tag{67.3}$$

One rigorous route to this identity is to start with the absolutely convergent zero series for $\int_1^x \psi(t)dt$, differentiate in distributions, and change variables. Under RH the series defining Z is uniformly convergent since $\sum_\rho |\rho\gamma|^{-1} < \infty$; its distributional derivative is exactly the zero term after this change of variables. Thus no termwise differentiation of a conditionally convergent pointwise series is required.

The function r is $O(e^{-u/2})$ on $[u_0, \infty)$. Both Z and its primitive

$$Z_1(u) = \sum_\rho \frac{e^{i\gamma u}}{\rho(i\gamma)^2}$$

are bounded, by absolute uniform convergence. In particular f_ψ has a bounded primitive on $[u_0, \infty)$. Integration by parts in (67.3) proves, for each fixed integer $j \ge 1$,

$$\begin{aligned} T_j(U) &:= \int_U^\infty \frac{f_\psi(v)}{v^j}\, dv \\ &= \frac{Z(U)}{U^j} - j\int_U^\infty \frac{Z(v)}{v^{j+1}}\, dv + \int_U^\infty \frac{r(v)}{v^j}\, dv = \frac{Z(U)}{U^j} + O(U^{-j-1}). \end{aligned} \tag{67.4}$$

For the last estimate integrate the Z integral once more using the bounded primitive Z_1. These improper integrals converge; absolute integrability of $f_\psi(v)/v$ is not assumed.

Finite partial summation gives, exactly,

$$J(x) - \mathrm{Li}_2(x) = \frac{D(x)}{\log x} + \int_2^x \frac{D(t)}{t\log^2 t}\, dt + \frac{2}{\log 2}.$$

Integrating against $x^{-3/2}$ and using finite Fubini gives

$$\begin{aligned} \mathcal{B}_J(e^U) = &\int_{u_0}^U \frac{f_\psi(v)}{v}\, dv + 2\int_{u_0}^U \frac{f_\psi(v)}{v^2}\, dv \\ &- 2e^{-U/2}\int_{u_0}^U \frac{f_\psi(v)e^{v/2}}{v^2}\, dv + \frac{4}{\log 2}(2^{-1/2} - e^{-U/2}). \end{aligned} \tag{67.5}$$

The bounded primitive of f_ψ gives

$$e^{-U/2}\int_{u_0}^U \frac{f_\psi(v)e^{v/2}}{v^2}\, dv = O(U^{-2})$$

by integration by parts; the fixed initial segment is exponentially small. Hence (67.4) and (67.5) yield

$$\mathcal{B}_J(e^U) = \mathcal{B}_J(\infty) - \frac{Z(U)}{U} + O(U^{-2}). \tag{67.6}$$

The previously proved scalar limit identifies $\mathcal{B}_J(\infty) = C_* + C_{\mathrm{pp}}$.

We also need one more term of the unconditional prime-square drift. The classical prime number theorem with an error smaller than $y/\log^4 y$, followed by partial summation, gives

$$\sum_{p\le y}\frac{1}{p} = \log\log y + B_{\mathbb{P}} + O(\log^{-2} y), \qquad \frac{\pi(y)}{y} = \frac{1}{\log y} + O(\log^{-2} y).$$

Apply these at $y = \sqrt{X}$ in the exact square identity (65.4). The already bounded contribution of higher powers has tail $O(X^{-1/6}/\log X)$. It follows that, unconditionally,

$$\mathcal{S}(e^U) = \mathcal{B}_J(e^U) - C_{\mathrm{pp}} + \frac{2}{U} + O(U^{-2}). \tag{67.7}$$

Equations (67.6) and (67.7) prove (67.1).

Finally, Brent–Platt–Trudgian [41, Corollary 1] rigorously bound the full sum over positive ordinates by

$$\sum_{\gamma>0}\gamma^{-2} = 0.023104993115418970788933810 4 + \vartheta\, 5\cdot 10^{-28}, \qquad |\vartheta|\le 1.$$

This is a bound for the infinite sum with a proved tail, not a truncation being treated as complete. Under RH, $|\rho| > |\gamma|$, so $|Z(U)| \le \sum_\rho |\rho\gamma|^{-1} < 2\sum_{\gamma>0}\gamma^{-2} < 0.04621$. The strict margins between 2 ± 0.04621 and $1.95, 2.05$ absorb the $O(U^{-2})$ error for sufficiently large U. □

Corollary 67.2 (A fixed lower barrier is also an exact target)**.** *The unchanged original target, RH, and the assertion*

$$\mathcal{A}(X) + \log\log X \ge C_* \qquad \textit{for every sufficiently large } X$$

are equivalent.

Proof. RH implies the stronger positive margin (67.2). Conversely the displayed lower bound gives, for each $\varepsilon > 0$, an eventual bound $\mathcal{A}(X) \ge -C_\varepsilon X^\varepsilon$. The lower-bound version of Proposition 65.3 therefore proves RH. □

The lower barrier has not been proved unconditionally. In particular, the bound on the coefficient sum does not control factors $X^{\rho-1/2}$ when an off-line zero is allowed. Dropping those factors' growth would assume the very assertion being tested. The advance in this section is the proved conditional remainder and its positive margin, together with a fully justified equivalent lower-barrier target.

68 Positive kernels, eventual moments, and an exact disk obstruction

Keep

$$\mathcal{A}(X) = \int_2^X \frac{\pi(x) - \mathrm{Li}_2(x)}{x^{3/2}}dx, \qquad u_0 = \log 2, \qquad a(u) = -\mathcal{A}(e^u) \quad (u \ge u_0).$$

The previous scalar theorem proves that eventual positivity of a is equivalent to the unchanged RH-equivalent target. The Laplace transform, initially for $\Re z > 1/2$, is

$$H(z) = \int_{u_0}^{\infty} a(u)e^{-zu}du = -\frac{F(1/2+z)}{z} = \frac{E_1((z-1/2)u_0) - \mathcal{P}(z+1/2)}{z(z+1/2)}. \tag{68.1}$$

Here $F(s) = \int_2^{\infty} P(x)x^{-s-1}dx$ initially for $\Re s > 1$, and $\mathcal{P}(s) = \sum_p p^{-s}$ for $\Re s > 1$. Chebyshev's estimate gives $a(u) = O(e^{u/2}/u)$, justifying absolute convergence and all the derivatives used below.

Proposition 68.1 (The raw complete-monotonicity attempt fails)**.** *The function H is negative for all real $z > 1/2 + 2/\log 2$, in particular for every integer $z \geq 4$. It therefore cannot be completely monotone on $(1/2, \infty)$, whether or not RH holds. More precisely,*

$$H(z) \sim -\frac{2^{-z-1/2}}{z(z+1/2)} \qquad (z \to +\infty).$$

Proof. For $t > 0$, $E_1(t) < e^{-t}/t$. Set $s = z + 1/2$. If $(s-1)\log 2 > 2$, then

$$E_1((s-1)\log 2) < \frac{2^{1-s}}{(s-1)\log 2} < 2^{-s} \leq \mathcal{P}(s).$$

Equation (68.1) gives the sign. Also $\mathcal{P}(s) = 2^{-s}(1 + o(1))$, while the displayed bound for E_1 is $O(2^{-s}/s)$, proving the asymptotic. The obstruction comes from the initial interval where $P(2) = 1$ and $\mathcal{A}$ initially increases; it is not an off-line-zero obstruction. □

For a real base $m > 1/2$ and an integer $n \geq 0$, define

$$M_n(m) = (-1)^n H^{(n)}(m) = \int_{u_0}^{\infty} u^n a(u)e^{-mu}du. \tag{68.2}$$

The useful positivity question is eventual positivity as $n \to \infty$ at each base, rather than complete monotonicity of the unmodified transform.

Gamma averaging and disk geometry. Extend a by zero on $[0, u_0)$. If U has the gamma distribution with shape $n+1$ and rate m, then

$$\frac{m^{n+1}}{n!}M_n(m) = \mathbb{E}[a(U)].$$

For the untruncated test mode $e^{(\delta+i\gamma)u}$ on $[0,\infty)$, with $\delta < m$, its exact response is

$$\mathbb{E}[e^{(\delta+i\gamma)U}] = \left(\frac{m}{m-\delta-i\gamma}\right)^{n+1}.$$

The modulus of this gain exceeds one precisely when $\delta^2 + \gamma^2 < 2m\delta$, the disk condition used below. This is a probability interpretation of the same geometry. The proof does not assume that the actual prime error has a convergent expansion into such modes.

Proposition 68.2 (An exact positive-kernel arithmetic test)**.** *Let*

$$\begin{aligned} K_{n,m}(v) &= 2\int_v^{\infty} u^n e^{-mu}(e^{-v/2} - e^{-u/2})du \\ &= 2e^{-v/2}m^{-n-1}\Gamma(n+1, mv) - 2(m+1/2)^{-n-1}\Gamma(n+1,(m+1/2)v). \end{aligned} \tag{68.3}$$

Then $K_{n,m} > 0$, $K'_{n,m} < 0$, and $K''_{n,m} > 0$ for $v > 0$. Unconditionally,

$$M_n(m) = \int_{u_0}^{\infty} \frac{e^v}{v}K_{n,m}(v)dv - \sum_p K_{n,m}(\log p). \tag{68.4}$$

Thus eventual moment positivity is exactly an eventual weighted prime-sum upper bound for explicit positive, decreasing, convex kernels.

Proof. The finite identity for $\mathcal{A}$ can be written

$$-\mathcal{A}(e^u) = 2\int_{u_0}^{u}(e^{-v/2} - e^{-u/2})\frac{e^v}{v}dv - 2\sum_{p\le e^u}(p^{-1/2} - e^{-u/2}).$$

Multiply by $u^n e^{-mu}$ and integrate. Since $m > 1/2$, both the positive continuous term and the positive prime term are separately integrable. Tonelli therefore proves (68.4). Formula (68.3) gives strict positivity, and direct differentiation, whose boundary contribution vanishes, gives

$$K'_{n,m}(v) = -e^{-v/2}\int_v^\infty u^n e^{-mu}du < 0,$$

$$K''_{n,m}(v) = \tfrac{1}{2}e^{-v/2}\int_v^\infty u^n e^{-mu}du + v^n e^{-(m+1/2)v} > 0.$$

□

An equivalent formula uses only absolutely convergent prime sums and elementary incomplete gamma functions. Put

$$D_c(m) = \int_{u_0}^\infty u^{c-1}e^{-(m-1/2)u}du - \sum_p (\log p)^c p^{-m-1/2}.$$

For $c = 0$ the integral is $E_1((m-1/2)u_0)$; for $c \ge 1$ it equals $(m-1/2)^{-c}\Gamma(c, (m-1/2)u_0)$. Leibniz's rule gives

$$\frac{M_n(m)}{n!} = \sum_{a+b+c=n}\frac{D_c(m)}{c!\,m^{a+1}(m+1/2)^{b+1}}. \tag{68.5}$$

Neither positivity of the kernel nor positivity of the Euler-product coefficients determines the sign of this continuous-minus-prime difference.

Lemma 68.3 (Pringsheim's principle with finitely many exceptional coefficients)**.** *Suppose $G(w) = \sum_{n\ge0} c_n w^n$ has finite positive radius of convergence R, and $c_n \ge 0$ for every sufficiently large n. Then $w = R$ is a singularity of G.*

Proof. Subtract the finite exceptional polynomial. If the resulting series were analytic near R, choose $b < R$ close enough to R that its Taylor series about b converges at some $R + \delta$, $\delta > 0$. All derivatives at $b > 0$ are nonnegative. Expanding them from the original nonnegative series and applying Tonelli to the Taylor series at $R + \delta$ shows that $\sum c_n(R+\delta)^n$ converges, a contradiction. □

Let

$$\mathcal{D}_m = \{z \in \mathbb{C} : |z - m| < m\}.$$

This disk lies in $\Re z > 0$, and the point $z = 0$ lies on its boundary. A zeta zero ρ is represented in this coordinate by $z_\rho = \rho - 1/2$.

Theorem 68.4 (An exact disk criterion for eventual moment positivity)**.** *For every fixed real $m > 1/2$, the following are equivalent:*

1. *$M_n(m) \ge 0$ for all sufficiently large integers n;*
2. *ζ has no zero ρ with $\rho - 1/2 \in \mathcal{D}_m$.*

If these conditions hold, then

$$\frac{M_n(m)}{n!} = m^{-n-1}\left(\log n - \log m - C_* + O_m(1/n)\right). \tag{68.6}$$

Zeros on the boundary of $\mathcal{D}_m$, if any, do not alter the leading term; their contributions are absorbed by $O_m(1/n)$.

Proof. Expand $G_m(w) = H(m - w)$ at $w = 0$. Its coefficient of w^n is $M_n(m)/n!$. The local expansion already proved at the critical point is

$$H(z) = -\frac{\log z + (C_* + \gamma)}{z} + O(\log z) \qquad (z \to 0,\ \Re z > 0). \tag{68.7}$$

Consequently $w = m$ is a nonremovable singularity of G_m, and its Taylor radius is finite and at most m. On the other hand, H has analytic continuation along every positive real z, as established in the scalar Landau proof. If the coefficients are eventually nonnegative, Lemma 68.3 therefore forces the Taylor radius to be m: a smaller radius would give a singularity at a positive real $z = m - R$. The Taylor series then defines a holomorphic function throughout $\mathcal{D}_m$. A zeta zero there would give a logarithmic singularity of H, since the prime-power correction is analytic for $\Re z > 0$ and the factors z, $z + 1/2$ are nonzero there. This proves assertion 2.

Conversely, assume assertion 2. There are only finitely many zeta zeros $\lambda_j = \rho_j - 1/2$ on the boundary of $\mathcal{D}_m$, counted below with multiplicities r_j. None equals zero. Write $q_j = m - \lambda_j$, so $|q_j| = m$ and $q_j \neq m$. Choose $R > m$, sufficiently close to m, such that the following hold in $|z - m| < R$: there are no other zeros of $\zeta(1/2 + z)$; the function $2z\zeta(1 + 2z)$ has no zeros; and $\Re(1/2 + z) > 1/3$. Such an R exists by discreteness and compactness. For the second condition, zeros of $\zeta(1 + 2z)$ have strictly negative real part and hence lie strictly outside the closed disk $|z - m| \leq m$; its pole at zero is canceled by $2z$.

For $|w| < m$, with all logarithmic branches continued from $w = 0$, the Euler-product identity takes the form

$$(z + 1/2)F(z + 1/2) = \tfrac{1}{2}\log z + B(z) + \sum_j r_j \log(1 - w/q_j), \qquad w = m - z, \tag{68.8}$$

where B extends holomorphically to the enlarged disk. The identity of logarithmic branches is asserted on $|w| < m$, not through the boundary logarithmic singularities. To see this directly, factor the displayed boundary zeros out of $(s - 1)\zeta(s)$. Its remaining factor has a holomorphic logarithm there. In the square correction use $\mathcal{P}(2s) = \log\zeta(2s) - \sum_{k\geq 2}\mathcal{P}(2ks)/k$, with the latter sum normally convergent, and factor its pole at $s = 1/2$. The terms $\mathcal{P}(ks)$, $k \geq 3$, are normally convergent on this slightly enlarged disk. These steps prove (68.8) without assuming any statement about zeros outside that disk.

Subtract from each boundary logarithm its value at $w = m$:

$$E_j(w) = \log(1 - w/q_j) - \log(1 - m/q_j).$$

The branch at the second point is the radial continuation from $|w| < m$, and $E_j(m) = 0$. The local constant in (68.7) fixes the remaining constant term. Using $-1/[z(z + 1/2)] = -2/z + 2/(z + 1/2)$, one obtains

$$G_m(w) = -\frac{\log(m - w) + (C_* + \gamma)}{m - w} + \frac{\log(m - w)}{m + 1/2 - w} + C(w) - \sum_j \frac{r_j E_j(w)}{(m - w)(m + 1/2 - w)}, \tag{68.9}$$

with C holomorphic for $|w| < R$, after decreasing R if necessary so that $R < m + 1/2$.

The coefficient of the first term in (68.9) is

$$m^{-n-1}(H_n - \log m - C_* - \gamma), \qquad H_n = \sum_{k=1}^{n} \frac{1}{k}.$$

The second term has coefficient $O_m(m^{-n}/n)$, by expanding the logarithm and summing the geometric convolution with ratio $m/(m + 1/2) < 1$. The analytic term contributes $O(R_0^{-n})$ for any fixed $m < R_0 < R$.

For each boundary logarithm, put $r = m/q_j$, so $|r| = 1$, $r \neq 1$. Then

$$[w^n]\frac{E_j(w)}{m-w} = m^{-n-1}\left(-\log(1-r) - \sum_{k=1}^{n}\frac{r^k}{k}\right) = O_j(m^{-n}/n),$$

because the tail of $\sum r^k/k$ is $O_j(1/n)$ by summation by parts. The coefficient of $E_j(w)/(m+1/2-w)$ is again $O_{m,j}(m^{-n}/n)$ by the same geometric convolution. Hence the last term in (68.9) contributes this order. Finally, $H_n = \log n + \gamma + O(1/n)$ gives (68.6), which is positive for all sufficiently large n. □

Corollary 68.5 (A discrete positivity criterion for the unchanged target)**.** *The original compensated R_4 bound is equivalent to*

$$\forall m \in \{1,2,3,\dots\} \quad \exists N_m \quad \forall n \geq N_m : \qquad M_n(m) \geq 0. \tag{68.10}$$

The threshold N_m may depend on m.

Proof. The disks $\mathcal{D}_m$, for positive integer m, exhaust $\Re z > 0$: a point z belongs to $\mathcal{D}_m$ whenever $m > |z|^2/(2\Re z)$. Theorem 68.4 therefore identifies (68.10) with absence of zeta zeros in $\Re s > 1/2$, which is RH by the functional equation. □

Corollary 68.6 (How an off-line zero defeats the moment test)**.** *If $\rho = 1/2 + \delta + i\gamma$ is a zero with $\delta > 0$, then every integer*

$$m > \frac{\delta^2+\gamma^2}{2\delta}$$

has infinitely many n with $M_n(m) < 0$ and infinitely many with $M_n(m) > 0$. If

$$r_m = \min\left(m, \inf_{\rho}|m-(\rho-1/2)|\right) < m,$$

then, more quantitatively,

$$\limsup_{n\to\infty}\left|\frac{M_n(m)}{n!}\right|^{1/n} = \frac{1}{r_m}.$$

Proof. The condition on m puts the zero inside $\mathcal{D}_m$. Eventual nonnegativity is impossible by Theorem 68.4. Eventual nonpositivity would allow application of Lemma 68.3 to $-G_m$, giving the same impossible zero-free disk. This proves both sign assertions. The singularities of G_m of modulus less than m are precisely the nonremovable logarithmic singularities produced by those zeta zeros; the prime-power correction is holomorphic there. The Cauchy–Hadamard formula gives the stated growth rate. □

Proposition 68.7 (A finite collection of bases closed analytically)**.** *For every integer $1 \leq m \leq 85$, the asymptotic (68.6), and hence eventual positivity of $M_n(m)$, holds unconditionally. This is a consequence of a known first Li-coefficient identity, not a new global zero-free estimate.*

Proof. The standard first Li identity [42, (1.1), (1.3)] is

$$\lambda_1 = \sum_{\rho}\Re\frac{1}{\rho} = 1 + \frac{\gamma}{2} - \frac{1}{2}\log(4\pi) = 0.023095708966\dots.$$

The series of real parts converges absolutely, and each term is positive since $0 < \Re\rho < 1$. Suppose that a zero lies in the closed disk $|(\rho-1/2)-m| \leq m$. Then $\rho = 1/2+\delta+i\gamma_\rho$, with $0 < \delta < 1/2$, and $\gamma_\rho^2 \leq 2m\delta - \delta^2$. Its functional-equation and conjugation quartet contributes at least

$$\frac{2(1/2+\delta)}{1/4+(2m+1)\delta} + \frac{2(1/2-\delta)}{1/4+(2m-1)\delta} \geq \frac{2}{m+3/4}$$

to λ_1. Both terms in the preceding sum are decreasing in $\delta \in (0, 1/2)$, so the last bound follows by taking $\delta \uparrow 1/2$. But for $m \le 85$,

$$\lambda_1 < \frac{8}{343} = \frac{2}{85 + 3/4} \le \frac{2}{m + 3/4}.$$

For example, the elementary bounds $\gamma < 0.5773$ and $\log(4\pi) > 2.531$ already give $\lambda_1 < 0.02315 < 8/343$. The contradiction proves that the closed disk is zero-free. Theorem 68.4 supplies the asserted infinite-order positivity at each of these bases. □

What remains. Neither (68.10) for all integer bases nor eventual $\mathcal{A}(X) \le 0$ has been proved. The positive kernels in (68.4) expose the missing inequality as a signed comparison between the actual prime sum and its continuous reference. Positivity of the separate summands does not give that comparison. The analytically closed bases $1 \le m \le 85$ exclude only a finite collection of disks; the remaining disks still exhaust the full unresolved half-plane.

69 Positive transfer between Gamma bases and its sharp direction

Keep $a(u) = -\mathcal{A}(e^u)$ for $u \ge \log 2$, extended by zero below $\log 2$, and put

$$g_n(m) = \frac{m^{n+1}}{n!} M_n(m), \qquad m > \frac{1}{2}, \quad n \ge 0.$$

The elementary estimate $|a(u)| \ll e^{u/2}$ is enough for all interchanges in this section. No hypothesis on zeta zeros is used.

Theorem 69.1 (Exact positive transfer)**.** *Let $1/2 < b < m$ and $r = b/m$. Then*

$$M_n(b) = \sum_{k=0}^{\infty} \frac{(m-b)^k}{k!} M_{n+k}(m), \tag{69.1}$$

$$g_n(b) = \sum_{k=0}^{\infty} \binom{n+k}{k} r^{n+1} (1-r)^k g_{n+k}(m). \tag{69.2}$$

Both series converge absolutely. The coefficients in (69.2) are nonnegative and sum to one. Consequently, if $M_j(m) \ge 0$ for every integer $j \ge N$, then

$$M_n(b) \ge 0 \qquad \text{for every real } b \in (1/2, m] \text{ and every integer } n \ge N.$$

The same threshold N works throughout this whole interval of bases.

Proof. Expand $e^{(m-b)u}$ inside the defining integral for $M_n(b)$. Absolute interchange follows from

$$\sum_{k \ge 0} \frac{(m-b)^k}{k!} \int_0^{\infty} u^{n+k} |a(u)| e^{-mu}\, du = \int_0^{\infty} u^n |a(u)| e^{-bu}\, du < \infty.$$

Multiplication by $b^{n+1}/n!$ gives (69.2). The binomial generating function gives

$$\sum_{k \ge 0} \binom{n+k}{k} r^{n+1} (1-r)^k = 1.$$

For $n \ge N$ all the sampled orders $n+k$ satisfy the assumed sign, which proves the final assertion. □

Remark 69.2 (A probability interpretation). If K has the negative-binomial law

$$\Pr(K=k)=\binom{n+k}{k}r^{n+1}(1-r)^k,$$

then $g_n(b)=\mathbb{E}[g_{n+K}(m)]$. Equivalently, a Gamma random variable of shape $n+1$ and rate b is a positive mixture of Gamma random variables of shapes $n+K+1$ and rate m. Its larger mean is obtained by increasing the shape in this mixture.

Corollary 69.3 (A uniform continuous range of bases)**.** *There exists an integer N such that*

$$M_n(b)>0 \qquad \left(\frac{1}{2}<b\le 85, \quad n\ge N\right).$$

Proof. The previously established zero-disk argument gives $M_n(85)>0$ for all sufficiently large n. Apply Theorem 69.1, using this one threshold. □

The corollary strengthens the quantifiers over the bases in a fixed bounded interval, but it is a consequence of the already excluded finite collection of zero disks. It is not a new global zero-free region and does not imply eventual positivity at any larger base.

The exact differential equation. Differentiation under the integral gives

$$\frac{\partial}{\partial m}M_n(m)=-M_{n+1}(m), \qquad \frac{\partial}{\partial \log m}g_n(m)=(n+1)\big(g_n(m)-g_{n+1}(m)\big).$$

Thus, on writing $b=me^{-\tau}$, the normalized moments evolve by

$$\frac{\partial}{\partial \tau}g_n(me^{-\tau})=(n+1)\big(g_{n+1}(me^{-\tau})-g_n(me^{-\tau})\big).$$

This is the positive pure-birth evolution whose transition weights are those in (69.2). Increasing the base means running this evolution in the opposite direction; positivity is not preserved by that inverse operation.

Proposition 69.4 (A countermodel to an upward positivity bootstrap)**.** *Fix $c>1/2$ and $C>0$. There is a smooth real function a_c on $[0,\infty)$ such that $|a_c(u)|\ll e^{u/2}$ and $a_c''(u)\le Ce^{u/2}$, with the following properties. Define its normalized Gamma moments by*

$$q_n(m)=\frac{m^{n+1}}{n!}\int_0^\infty u^n a_c(u)e^{-mu}\,du.$$

For every $1/2<m\le c$ and every $n\ge 0$, $q_n(m)>0$. For every $m>c$, the sequence $q_n(m)$ has infinitely many positive terms and infinitely many negative terms.

Proof. Take

$$\delta=\frac{1}{4}, \qquad \gamma=\sqrt{\frac{c}{2}-\frac{1}{16}}, \qquad 0<\varepsilon\le\min\left(\frac{1}{2},\frac{2C}{c}\right),$$

and set

$$a_c(u)=1-\varepsilon e^{\delta u}\cos(\gamma u).$$

Since $\delta^2+\gamma^2=c/2$, direct differentiation gives

$$|a_c''(u)|\le \varepsilon\frac{c}{2}e^{u/4}\le Ce^{u/2}.$$

Exact integration gives

$$q_n(m)=1-\varepsilon\,\mathrm{Re}\left(\frac{m}{m-\delta-i\gamma}\right)^{n+1}.$$

Write the displayed ratio as $Re^{i\theta}$. Here $0 < \theta < \pi/2$, and

$$R \le 1 \quad \Longleftrightarrow \quad m \le c.$$

For $m \le c$ we therefore have $q_n(m) \ge 1 - \varepsilon > 0$. For $m > c$ we have $R > 1$. The sequence $\cos((n+1)\theta)$ is bounded away from zero on the positive side infinitely often, and on the negative side infinitely often. For a rational rotation this follows by inspecting one period; for an irrational rotation it follows from density on the circle. The factor R^{n+1} then proves both sign assertions. □

This countermodel concerns a proposed general implication between Gamma moments. It is not the prime error function, is not a counterexample to the desired prime-sum inequality, and says nothing against RH. It shows that positive transfer, positivity at every order for a bounded range of bases, and a common upper-curvature bound cannot by themselves supply the missing passage to all bases. A successful passage must use some further property of the actual prime error, beyond the features shared by this model.

70 Consecutive moment blocks and recurrent sign obstructions

The next refinement combines the exact singularity structure of the prime Laplace transform with finite Fourier block averages and a second-moment inequality. It strengthens the previously proved infinite sign oscillation to bounded gaps, with an explicit bound in terms of the nearest singularities. The argument uses established singularity analysis [43] and elementary Fourier estimates; no historical novelty of this general block-sign principle is asserted.

Keep H, $M_n(m)$, and $\mathcal{D}_m$ from Section 68. In this section $m > 1/2$ is fixed. For a set of nonnegative integers, "bounded gaps eventually" means that some $L \ge 1$ and $N \ge 0$ have the following property: every integer interval $[n, n+L-1]$, $n \ge N$, meets that set.

Lemma 70.1 (Coefficients of a logarithm times an analytic factor)**.** *Let $q \ne 0$, $|q| = r$, and let C be analytic in a disk of radius strictly greater than r. Then*

$$[w^n]\{-C(w)\log(1 - w/q)\} = \frac{C(q)q^{-n}}{n} + O(r^{-n}/n^2).$$

The logarithm denotes its Taylor branch at zero.

Proof. Write $C(w) = \sum_{k\ge 0} c_k w^k$. For some $s > r$, $|c_k| \ll s^{-k}$. The coefficient on the left is

$$q^{-n} \sum_{k=0}^{n-1} \frac{c_k q^k}{n-k}.$$

Subtract $q^{-n}C(q)/n$. For $k \le n/2$ use

$$\left|\frac{1}{n-k} - \frac{1}{n}\right| \le \frac{2k}{n^2}$$

and the convergence of $\sum k|c_k q^k|$. The remaining terms are $O((r/s)^{n/2})$, including the tail of $C(q)$. This is $O(n^{-2})$ and proves the assertion. □

Lemma 70.2 (An explicit two-sign bound from Fourier block averages)**.** *Suppose*

$$T(n) = \sum_{j=1}^{J} c_j \omega_j^n, \qquad n \in \mathbb{Z},$$

is real for every integer n, the numbers ω_j are distinct, $|\omega_j| = 1$, no $\omega_j = 1$, and each $c_j \neq 0$. Put

$$B = \sum_j |c_j|, \qquad q = \sum_j |c_j|^2,$$

$$A_1 = \sum_j \frac{2|c_j|}{|1-\omega_j|}, \qquad A_2 = \sum_{j\neq k} \frac{2|c_j c_k|}{|1-\omega_j\overline{\omega_k}|}.$$

Set $\eta = q/(8B)$, and choose an integer

$$L \geq \max\left\{1, \frac{8BA_1}{q}, \frac{2A_2}{q}\right\}. \tag{70.1}$$

Every block of L consecutive integers contains an index where $T(n) > \eta$ and an index where $T(n) < -\eta$.

Proof. For a block starting at any integer h, the finite geometric sum gives

$$\left|\frac{1}{L}\sum_{n=h}^{h+L-1} T(n)\right| \leq \frac{A_1}{L} \leq \frac{q}{8B}.$$

Since T is real, $T(n)^2 = |T(n)|^2$. Expanding the square and applying the same geometric-sum bound to the off-diagonal terms gives

$$\frac{1}{L}\sum_{n=h}^{h+L-1} T(n)^2 \geq q - \frac{A_2}{L} \geq \frac{q}{2}.$$

Also $|T(n)| \leq B$ and $0 < \eta \leq B/8$, because $q \leq B^2$. If every value in the block were at least $-\eta$, then

$$(T(n)+\eta)(B - T(n)) \geq 0, \qquad T(n)^2 \leq (B-\eta)T(n) + B\eta.$$

Averaging the last inequality yields

$$\frac{1}{L}\sum T(n)^2 \leq (B-\eta)\frac{q}{8B} + B\eta \leq \frac{q}{4},$$

contradicting the lower bound $q/2$. Applying the same argument to $-T$ rules out all values being at most η. Thus both strict signs occur with the indicated margins. □

Theorem 70.3 (An unconditional dichotomy for the moment signs)**.** *For every fixed real $m > 1/2$, exactly one of the following alternatives holds.*

1. *The disk $\mathcal{D}_m$ contains no shifted zeta zero $\rho - 1/2$. In this case $M_n(m) > 0$ for all sufficiently large n, with the asymptotic in (68.6).*
2. *The disk $\mathcal{D}_m$ contains a shifted zeta zero. In this case there are $L_m \geq 1$ and N_m such that every L_m consecutive indices beyond N_m contain both a strictly positive and a strictly negative value of $M_n(m)$.*

In particular, unconditionally, the strictly positive moments have bounded gaps eventually at every fixed base $m > 1/2$.

Proof. The first alternative is Theorem 68.4. Suppose the second alternative holds. Put

$$r = \min_\rho |m - (\rho - 1/2)| < m,$$

where only nontrivial zeta zeros enter the minimum. There are finitely many distinct zeros ρ_j on this nearest circle; write their multiplicities as b_j and set

$$z_j = \rho_j - 1/2, \qquad q_j = m - z_j = re^{i\theta_j}, \qquad c_j = \frac{b_j}{z_j(z_j + 1/2)}.$$

All c_j are nonzero. The q_j are distinct and nonreal, since the zeta function has no real nontrivial zeros. In particular no θ_j is zero modulo 2π.

Choose r_1 strictly between r and m, sufficiently close to r that the closed disk $|w| \le r_1$ contains no further shifted zeros in the coordinate $z = m - w$. Factoring the nearest zeros out of $(s-1)\zeta(s)$, with $s = 1/2 + m - w$, and canceling the pole at $s = 1$ against the E_1 term in (68.1), gives

$$H(m - w) = G(w) - \frac{1}{(m - w)(m + 1/2 - w)} \sum_j b_j \log(1 - w/q_j), \tag{70.2}$$

where G is holomorphic on a neighborhood of $|w| \le r_1$. This equality initially means the Taylor branches at zero. The prime-power correction is holomorphic there because $\Re s > 1/2$ throughout the disk. The rational prefactor is analytic for $|w| < m$, so no other singularity has been introduced.

Lemma 70.1, together with a Cauchy bound on the coefficients of G, proves

$$\frac{M_n(m)}{n!} = \frac{1}{n} \sum_j c_j q_j^{-n} + O_m(r^{-n}/n^2). \tag{70.3}$$

The conjugation symmetry of the zeros makes

$$T_m(n) = \sum_j c_j e^{-in\theta_j}$$

real. Thus

$$nr^n \frac{M_n(m)}{n!} = T_m(n) + O_m(1/n).$$

Apply Lemma 70.2, with $\omega_j = e^{-i\theta_j}$, and then take N_m so large that the error is at most $\eta/2$. The normalizing factor is positive. In every sufficiently late block of the explicit length in (70.1), the normalized moments therefore take a value greater than $\eta/2$ and a value less than $-\eta/2$. This proves the second alternative. The block bound depends on the nearest zeros; it is not a uniform numerical bound independent of unknown zero locations. □

Corollary 70.4 (Arbitrarily long positive blocks are an exact criterion)**.** *For a fixed $m > 1/2$, the following are equivalent:*

1. *$\mathcal{D}_m$ contains no shifted zeta zero;*
2. *$M_n(m) > 0$ for all sufficiently large n;*
3. *for every integer $L \ge 1$, there exists $n \ge 0$ such that $M_{n+j}(m) > 0$ for $0 \le j < L$;*
4. *for every integer $L \ge 1$, there exists $n \ge 0$ such that $M_{n+j}(m) \ge 0$ for $0 \le j < L$.*

Therefore the unchanged compensated R_4 target is equivalent to the block-existence statement

$$\forall m \in \mathbb{N} \quad \forall L \in \mathbb{N} \quad \exists n \ge 0 \quad \forall j \in \{0, \dots, L-1\} : \qquad M_{n+j}(m) > 0. \tag{70.4}$$

The start n can equivalently be required to exceed any prescribed lower bound.

Proof. The first implication follows from the disk theorem, and the next two implications are immediate. If the disk contains a shifted zero, Theorem 70.3 gives a negative moment in every L_m-block beyond N_m. Any block of length $N_m + L_m$ contains one of those negative moments. This contradicts the fourth condition, proving the equivalence. Positive-integer disks exhaust the right half-plane, so Corollary 68.5 gives (70.4). The same argument, or eventual positivity in the zero-free case, gives the final assertion. □

Corollary 70.5 (A zero-density version)**.** *For each fixed $m > 1/2$, the disk $\mathcal{D}_m$ is zero-free if and only if*

$$\liminf_{X\to\infty} \frac{1}{X}\#\{0 \le n < X : M_n(m) < 0\} = 0.$$

Consequently imposing this condition at every positive integer m is another exact formulation of the original target. If the disk is not zero-free, the liminf is at least $1/L_m > 0$, for a constant supplied by Theorem 70.3.

Proof. In the zero-free case only finitely many moments are negative. Otherwise, partition the indices beyond N_m into disjoint blocks of length L_m. Each contains a negative moment, giving the stated positive lower density. □

Proposition 70.6 (A quantitative lower density of each sign)**.** *Suppose $\mathcal{D}_m$ contains a shifted zero, and let J, c_j, $B = \sum_j |c_j|$, and $q = \sum_j |c_j|^2$ refer to its nearest singularity circle as above. Then both the strictly positive and the strictly negative moment sets have lower asymptotic density at least*

$$\frac{q}{2B^2} \ge \frac{1}{2J}. \tag{70.5}$$

For every integer $m \ge 1$, the negative density is consequently at least the entirely explicit quantity

$$\frac{1}{2\lambda_1(m+3/4)}, \qquad \lambda_1 = 1 + \frac{\gamma}{2} - \frac{1}{2}\log(4\pi), \tag{70.6}$$

whenever $\mathcal{D}_m$ contains a shifted zero.

Proof. Finite geometric sums give the Cesàro limits

$$\frac{1}{N}\sum_{n=1}^{N} T_m(n) \longrightarrow 0, \qquad \frac{1}{N}\sum_{n=1}^{N} T_m(n)^2 \longrightarrow q.$$

Since $|T_m| \le B$, the inequality $T_m^2 \le B|T_m|$ gives

$$\liminf_{N\to\infty} \frac{1}{N}\sum_{n=1}^{N} T_m(n)^- \ge \frac{q}{2B},$$

and the same holds for T_m^+. The normalized moment $S_n = nr^n M_n(m)/n!$ satisfies $S_n = T_m(n) + O_m(1/n)$. When $M_n(m) \ge 0$, one has $T_m(n)^- = O_m(1/n)$; at other indices $T_m(n)^- \le B$. Summing this statement and using $\sum_{n\le N} 1/n = O(\log N)$ proves the lower negative density $q/(2B^2)$. The proof for strictly positive moments is identical. Cauchy–Schwarz gives $B^2 \le Jq$, proving (70.5).

The J nearest zeros occur in conjugate pairs, all in $\Re s > 1/2$. Their $J/2$ functional-equation quartets are disjoint. The quartet computation in the proof of Proposition 68.7 shows that each contributes at least $2/(m + 3/4)$ to the positive sum defining λ_1. Ignoring multiplicities only decreases this contribution. Thus

$$\lambda_1 \ge \frac{J}{m+3/4}, \qquad J \le \lambda_1(m+3/4).$$

Combining this bound with (70.5) proves (70.6). □

In particular, another exact sufficient-and-necessary target is

$$\forall m \in \{86, 87, \ldots\} : \quad \liminf_{N\to\infty} \frac{1}{N}\#\{0 \le n < N : M_n(m) < 0\} < \frac{1}{2\lambda_1(m + 3/4)}.$$

The bases $1 \le m \le 85$ are already treated by Proposition 68.7. If RH holds, all the displayed lower densities are zero. If RH fails, a shifted zero enters some positive-integer disk and the displayed strict inequality fails there. No such density upper bound for all bases is asserted here.

The exact remaining distinction. The theorem proves unconditionally that at least one positive moment occurs in every sufficiently late block of some fixed length. The unresolved target requires blocks of arbitrarily large length in which *every* moment is positive. These are different quantifier patterns. Neither the latter block condition for every integer base, nor zero density of negative moments for every base, is proved here. The result shows rigorously that a false-RH obstruction cannot be an arbitrarily isolated sequence of negative moment defects; once a zero enters a fixed disk, negative defects recur with bounded gaps.

71 A monotone Poisson budget for the negative moment defects

The moment kernel also has an exact Markov interpretation. Define, for $m > 1/2$,

$$d_n(m) = \frac{m^n M_n(m)}{n!} = \int_0^\infty a(u)e^{-mu}\frac{(mu)^n}{n!}\,du, \qquad \mathfrak{D}(m) = \sum_{n\ge 0} d_n(m)^- \in [0, \infty],$$

where $y^- = \max\{-y, 0\}$. Thus $d_n(m)$ uses the Poisson probability of n with mean mu. It differs by a factor m from the normalized Gamma expectation $m^{n+1}M_n(m)/n!$; this factor is essential for the conservation law below. Every individual integral is absolutely convergent, since $a(u) = O(e^{u/2}/u)$.

Theorem 71.1 (Thinning contraction and exact recovery)**.** *Unconditionally,*

$$0 \le \mathfrak{D}(m_1) \le \mathfrak{D}(m_2) \le \int_0^\infty a(u)^-\,du \qquad (1/2 < m_1 < m_2),$$

with extended-real values allowed. Moreover,

$$\lim_{m\to\infty} \mathfrak{D}(m) = \int_0^\infty a(u)^-\,du = \int_2^\infty \frac{\mathcal{A}(x)^+}{x}\,dx. \tag{71.1}$$

Proof. Put $r = m_1/m_2 \in (0, 1)$. Poisson thinning gives

$$e^{-m_1u}\frac{(m_1u)^n}{n!} = \sum_{k=n}^\infty \binom{k}{n} r^n(1-r)^{k-n}e^{-m_2u}\frac{(m_2u)^k}{k!}.$$

Integrating against $a(u)$ is justified absolutely for each fixed n: replacing a by $|a|$ makes the right side equal the finite corresponding integral at m_1 by Tonelli. Consequently

$$d_n(m_1) = \sum_{k\ge n} B_{n,k}d_k(m_2), \qquad B_{n,k} = \binom{k}{n} r^n(1-r)^{k-n}.$$

The entries are nonnegative and $\sum_{n=0}^k B_{n,k} = 1$. Taking negative parts and then summing, again by Tonelli for nonnegative terms, yields

$$\mathfrak{D}(m_1) \le \sum_n \sum_{k\ge n} B_{n,k}d_k(m_2)^- = \mathfrak{D}(m_2).$$

Similarly, $d_n(m)^-$ is at most the integral of a^- against its Poisson kernel. Summing the probabilities over n gives $\mathfrak{D}(m) \le \int a^-$.

For the reverse limiting inequality, let ϕ be a continuous function of compact support on $[0,\infty)$, with $0 \le \phi \le 1$. Choose $B > 0$ with $\operatorname{supp}\phi \subseteq [0,B]$. Since only finitely many terms occur,

$$\mathfrak{D}(m) \ge -\sum_{n\ge 0} \phi(n/m) d_n(m) = -\int_0^\infty a(u)\, Q_m\phi(u)\, du,$$

where

$$Q_m\phi(u) = \mathbb{E}\left[\phi(\mathrm{Pois}(mu)/m)\right].$$

For fixed u, the mean of $\mathrm{Pois}(mu)/m$ is u and its variance is u/m. Therefore $Q_m\phi(u) \to \phi(u)$.

To justify the integral limit despite the infinite positive background of a, put $c = 1 - e^{-1} > 0$. Exponential Markov inequality gives

$$0 \le Q_m\phi(u) \le \mathbb{P}\{\mathrm{Pois}(mu) \le mB\} \le \exp(mB - mcu).$$

For $u \ge 2B/c$ this is at most $e^{-mcu/2}$. Taking $m \ge 4/c$ gives the integrable dominating tail $|a(u)|e^{-2u}$; on the remaining finite interval use $Q_m\phi \le 1$. Dominated convergence now proves

$$\liminf_{m\to\infty} \mathfrak{D}(m) \ge -\int_0^\infty a(u)\phi(u)\, du.$$

Taking the supremum over such compactly supported test functions gives $\int a^-$, also when this integral is infinite. For example, approximate the indicator of the negative set on successive compact intervals by continuous functions between zero and one. This proves the first equality in (71.1). The last equality is the change of variables $x = e^u$, since $a(u)^- = \mathcal{A}(e^u)^+$. □

Corollary 71.2 (Finiteness of one budget detects exactly one disk)**.** *For each fixed $m > 1/2$,*

$$\mathfrak{D}(m) < \infty \quad\Longleftrightarrow\quad \mathcal{D}_m \text{ contains no shifted zeta zero.}$$

Proof. In the zero-free case, Theorem 68.4 makes all sufficiently late $M_n(m)$ strictly positive. Only finitely many nonzero terms remain in $\mathfrak{D}(m)$. If the disk contains a zero, use the radius $r < m$ and the fixed negative margin in the proof of Theorem 70.3. At infinitely many indices (indeed, with bounded gaps),

$$d_n(m)^- \ge \frac{\eta}{2n}\left(\frac{m}{r}\right)^n.$$

These terms do not tend to zero; in fact they grow without bound along that subsequence. Hence the budget is infinite. □

Corollary 71.3 (An exact one-sided integrability criterion)**.** *The following statements are equivalent to RH and to the unchanged compensated R_4 bound:*

1. $\displaystyle\int_2^\infty \mathcal{A}(x)^+\, dx/x < \infty$;
2. $\displaystyle\sup_{m>1/2} \mathfrak{D}(m) < \infty$;
3. $\mathfrak{D}(m) < \infty$ *for every positive integer m;*
4. $\mathfrak{D}(m_j) < \infty$ *along some unbounded sequence of positive real bases m_j.*

The bound in item 2, when finite, is exactly the integral in item 1. It is enough in item 2 to take positive integer bases, or any prescribed unbounded sequence of bases.

Proof. The first two statements are equivalent by Theorem 71.1, and imply the third and fourth. If the fourth holds, monotonicity makes the budget finite at every fixed base, hence every positive-integer disk is zero-free by Corollary 71.2. Those disks exhaust $\Re z > 0$, so RH follows. Conversely, under RH the previously proved scalar asymptotic gives $a(u) = \log u - C_* + o(1)$ and thus $a(u) > 0$ eventually. Its negative part is supported in a compact interval and is integrable, proving item 1. The equality and the final assertion follow from monotonicity and exact recovery. □

What the contraction does and does not prove. Thinning transports control from a larger base to every smaller base, and the transport preserves total probability. It does not supply a bound at larger bases from bounds at smaller ones. The theorem proves the monotonicity and identifies the exact limiting budget unconditionally; it does not prove that this budget is finite. Thus the original problem is now an exact finite-negative-area question, with its positivity-preserving transport law fully established.

72 A single moving Gamma diagonal

The fixed-base condition has the quantifier order $\forall m\, \exists N_m\, \forall n \geq N_m$. It cannot be restricted to a diagonal by a formal exchange of quantifiers. We prove a uniform localization estimate for the actual prime-counting primitive, and use that estimate to obtain a genuinely single-sequence equivalent condition.

Put $u_0 = \log 2$, and extend

$$a(u) = -\mathcal{A}(e^u) \quad (u \geq u_0), \qquad a(u) = 0 \quad (0 \leq u < u_0).$$

For real $m > 1/2$, retain

$$M_n(m) = \int_0^\infty u^n a(u) e^{-mu}\, du.$$

For integers $N \geq 3$, set

$$m_N = \left\lceil \frac{N}{\log N} \right\rceil, \qquad \mathfrak{G}_N = \frac{m_N^N}{(N-1)!} M_{N-1}(m_N). \tag{72.1}$$

Thus $\mathfrak{G}_N = \mathbb{E}[a(U_N)]$, where U_N has Gamma shape N and rate m_N. Both indices $N-1, m_N$ belong to the original integer-indexed moment family.

72.1 An explicit global one-sided estimate

Proposition 72.1 (Prime jumps and Gamma Jensen)**.** *Let $C_0 = 1/2 + 1/\log 2$. If U has Gamma shape $k > 0$ and rate $m > 1/2$, and $\mu = k/m$, then*

$$\mathbb{E}[a(U)] \leq a(\mu) + 4C_0 \left[\left(1 - \frac{1}{2m}\right)^{-k} - e^{\mu/2} \right]. \tag{72.2}$$

This inequality is unconditional.

Proof. On $[0, \infty)$, the distributional second derivative is

$$a''(u) = \mathbf{1}_{(u_0,\infty)}(u) \left(\tfrac{1}{2} e^{-u/2} P(e^u) + \frac{e^{u/2}}{u} \right) du - \sum_p p^{-1/2} \delta_{\log p}.$$

The atom at $\log 2$ is included: the extension is continuous, while its right derivative jumps from zero to $-1/\sqrt{2}$. Since $P(x) \leq \pi(x) \leq x$, one has $a'' \leq C_0 e^{u/2} du$. Hence $a(u) - 4C_0 e^{u/2}$ is concave. The elementary bound $|a(u)| \ll e^{u/2}$ and $m > 1/2$ justify Jensen's inequality and all expectations. Finally $\mathbb{E}[e^{U/2}] = (1 - 1/(2m))^{-k}$. □

For the exact-center real rate $m = N/\log N$, the bracket in (72.2) is $O((\log N)^2/\sqrt{N})$. This alone supplies a one-sided localization sufficient for the RH criterion below. The next estimate is two-sided and improves one logarithm by the sieve input.

72.2 Two-sided localization with Brun–Titchmarsh

Lemma 72.2 (A local Taylor remainder)**.** *For sufficiently large $x = e^{\mu}$, choose the right derivative $b = a'_+(\mu)$. Uniformly for $|h| \le 1/2$,*

$$|a(\mu + h) - a(\mu) - bh| \ll \frac{\sqrt{x}}{\log x} h^2 + \frac{1}{x}. \tag{72.3}$$

Proof. The elementary bounds $|P(x)| \ll x/\log x$ show that the continuous part of a'', on a logarithmic interval of radius $r \le 1/2$ about μ, has total variation $O(r\sqrt{x}/\log x)$. The atomic part has total variation bounded by

$$Cx^{-1/2}\#\{p : xe^{-r} \le p \le xe^{r}\}.$$

If $x^{-3/4} \le r \le 1/2$, the physical interval has length $\asymp xr$, and $\log(xr) \ge (\log x)/4$. Brun–Titchmarsh [30], with at most one endpoint prime added, bounds the prime count by $Cxr/\log x$. Thus the total variation of a'' on this interval is $O(r\sqrt{x}/\log x)$.

If $0 \le r < x^{-3/4}$, counting integers instead gives total variation $O(\sqrt{x}\, r + x^{-1/2})$. The Taylor remainder has absolute value at most r times the corresponding total variation. In this second range it is $O(\sqrt{x}\, r^2 + x^{-1/2} r) = O(x^{-1})$. The distributional Taylor formula uses the right derivative at μ; when the increment is negative, it includes any atom at μ. The bounds just given include that atom and therefore also apply when x itself is prime. Combining the two ranges proves (72.3). □

Theorem 72.3 (Unconditional diagonal localization)**.** *For $m_N, \mathfrak{G}_N$ in (72.1),*

$$\boxed{\mathfrak{G}_N = -\mathcal{A}(N) + O\Big(\frac{\log N}{\sqrt{N}}\Big).} \tag{72.4}$$

The implied constant is absolute, and no RH assumption is used. More generally, for each fixed $c \ge 1$, put

$$m_{N,c} = \left\lceil \frac{N}{c \log N} \right\rceil, \qquad \mathfrak{G}_{N,c} = \frac{m_{N,c}^N}{(N-1)!} M_{N-1}(m_{N,c}).$$

Then

$$\mathfrak{G}_{N,c} = -\mathcal{A}(N^c) + O_c\Big(N^{c/2-1} \log N\Big). \tag{72.5}$$

Proof. Write $m = m_{N,c}$, $\mu = N/m$, and $x = e^{\mu}$. Then

$$\mu = c \log N + O_c((\log N)^2/N), \qquad x \asymp_c N^c.$$

For $U \sim \mathrm{Gamma}(N, m)$, write $h = U - \mu$. The exact identities $\mathbb{E}h = 0$ and $\mathbb{E}h^2 = \mu^2/N$ cancel the linear Taylor term. Applying Lemma 72.2 on $|h| \le 1/2$ gives a contribution at most

$$C \frac{\sqrt{x}}{\mu} \frac{\mu^2}{N} + \frac{C}{x} \ll_c N^{c/2-1} \log N.$$

Here $c \ge 1$ ensures that x^{-1} is absorbed by the last quantity.

For completeness, the omitted tails do not require a growth assumption. The Gamma Chernoff bounds give, for sufficiently large N,

$$\mathbb{P}(|U - \mu| > 1/2) \le 2\exp(-c_1 N/\mu^2),$$

with an absolute $c_1 > 0$. Also

$$\mathbb{E}e^U = (1 - 1/m)^{-N} = \exp(\mu + O_c(\mu^2/N)),$$

where $m > 1$ eventually. Since $|a(v)| \ll e^{v/2}$ and $|a'_+(\mu)| \ll e^{\mu/2}/\mu$, Cauchy–Schwarz bounds the expected absolute Taylor remainder on the tails by $O_c(N^{c/2}\exp(-c_2 N/(\log N)^2))$, after reducing $c_2 > 0$. It is smaller than every fixed negative power of N. This also justifies using the exact, untruncated mean cancellation.

We have proved $\mathbb{E}a(U) = a(\mu) + O_c(N^{c/2-1}\log N)$. Finally, throughout the interval between μ and $c\log N$, $|a'(v)| \ll_c N^{c/2}/\log N$. The rounding displacement of size $O_c((\log N)^2/N)$ therefore changes a by at most $O_c(N^{c/2-1}\log N)$. This proves (72.5); $c = 1$ gives (72.4). □

72.3 A single-sequence equivalent condition

Theorem 72.4 (One moving diagonal is equivalent to RH)**.** *For each fixed $1 \le c \le 2$,*

$$\mathrm{RH} \iff \exists N_0\ \forall N \ge N_0 :\ \mathfrak{G}_{N,c} \ge 0. \tag{72.6}$$

In particular, one may fix $c = 1$ and the integer rates $m_N = \lceil N/\log N\rceil$. Moreover, writing $y^- = \max(0, -y)$,

$$\mathrm{RH} \iff \limsup_{N\to\infty} \frac{\log(1 + \mathfrak{G}_N^-)}{\log N} = 0. \tag{72.7}$$

Proof. Assume eventual nonnegativity in (72.6). The unconditional localization estimate gives

$$\mathcal{A}(N^c) \le C_c N^{c/2-1}\log N \le C_c \log N.$$

Between consecutive points $N^c, (N+1)^c$, the elementary bound $|\mathcal{A}'(x)| \ll 1/(\sqrt{x}\log x)$ gives a variation $O_c(N^{c/2-1}/\log N)$. Thus $\mathcal{A}(X) \le C_c \log X$ for all sufficiently large real X. The previously proved one-sided subpower criterion for $\mathcal{A}$ implies RH. For $c = 1$ one obtains the stronger actual intermediate conclusion $\mathcal{A}(X) \le C\log X/\sqrt{X}$.

Conversely, under RH the already established scalar expansion gives $a(u) = \log u - C_* + O(1/u)$ for sufficiently large u. For $U \sim \mathrm{Gamma}(N, m_{N,c})$, its mean $\mu \sim c\log N$ tends to infinity and its variance μ^2/N tends to zero. The initial region has exponentially small probability, and a is bounded there. In particular, $\mathbb{E}a(U) > 0$ eventually. More quantitatively, the standard Gamma identities $\mathbb{E}\log U = \psi(N) - \log m_{N,c}$, $\mathbb{E}U^{-1} = m_{N,c}/(N-1)$, and the same Chernoff bound show

$$\mathfrak{G}_{N,c} = \log(c\log N) - C_* + O_c(1/\log N).$$

This proves the reverse implication without assuming any unproved uniformity in the fixed-base thresholds N_m.

For (72.7), subpower growth of the negative part, combined with (72.4), gives $\mathcal{A}(N) \le N^{o(1)}$. Interpolation between consecutive integers gives the same one-sided subpower bound for all real X, and hence RH. Under RH the diagonal is eventually positive, so its negative part eventually vanishes. □

Exact remaining arithmetic statement. The positive normalizing factor in (72.1) does not affect signs. Using the earlier kernel identity, the remaining single-sequence assertion is precisely

$$\exists N_0\ \forall N \ge N_0 : \qquad \sum_p K_{N-1,m_N}(\log p) \le \int_2^\infty \frac{K_{N-1,m_N}(\log x)}{\log x}\,dx, \qquad m_N = \left\lceil \frac{N}{\log N}\right\rceil.$$

This inequality is not proved here. The unconditional advances are the explicit Jensen bound and the two-sided localization estimate; the new diagonal formulation is proved equivalent in both directions. The earlier finitely many fixed bases do not settle the diagonal, because $m_N \to \infty$.

Why the diagonal does not follow by changing quantifiers. A generic array can be positive eventually in each fixed column and negative along this diagonal: for example, take $b_{n,m} = 1$ when $n \geq e^m$, and $b_{n,m} = -1$ otherwise. For each fixed m, $b_{n,m}$ is eventually positive, whereas $b_{N-1,m_N} = -1$ for all sufficiently large N. The equivalence above therefore uses the actual arithmetic localization and the RH scalar asymptotic, not a formal implication between quantifier patterns.

72.4 A finite logarithmic mass of negative diagonal values

The diagonal sign condition has a useful relaxation. It permits infinitely many negative values, provided their total mass on the logarithmic scale is finite.

Theorem 72.5 (Negative-mass criterion)**.** *With $\mathfrak{G}_N$ as in (72.1),*

$$\mathrm{RH} \iff \sum_{N\geq 3} \frac{\mathfrak{G}_N^-}{N} < \infty, \qquad y^- := \max(0,-y). \tag{72.8}$$

Equivalently,

$$\mathrm{RH} \iff \int_2^\infty \mathcal{A}(x)^+ \frac{dx}{x} < \infty, \qquad y^+ := \max(0,y). \tag{72.9}$$

These are equivalent criteria, not asserted convergence results.

Proof. The map $y \mapsto y^+$ is 1-Lipschitz. Therefore Theorem 72.3 gives

$$|\mathfrak{G}_N^- - \mathcal{A}(N)^+| \ll \frac{\log N}{\sqrt{N}}.$$

The errors are summable after multiplication by $1/N$. On $[N, N+1]$, $|\mathcal{A}(x) - \mathcal{A}(N)| \ll 1/(\sqrt{N}\log N)$. Also $|\mathcal{A}(N)| \ll \sqrt{N}$, so replacing $\int_N^{N+1} dx/x$ by $1/N$ has a summable error. Consequently the series in (72.8) converges if and only if the integral in (72.9) converges.

Under RH, $\mathcal{A}(x) = -\log\log x + C_* + o(1)$, so $\mathcal{A}(x)^+$ vanishes eventually. This proves one implication.

For the converse, write $b(u) = \mathcal{A}(e^u)$ for $u \geq u_0 = \log 2$. The assumed convergence says $\int_{u_0}^\infty b(u)^+ du < \infty$. Thus

$$L_+(z) = \int_{u_0}^\infty b(u)^+ e^{-zu}\, du$$

is holomorphic for $\Re z > 0$. Let $\sigma_- \leq 1/2$ be the abscissa of convergence of the Laplace transform $L_-(z)$ of the nonnegative function b^-. Initially, for $\Re z > 1/2$, Fubini gives

$$L_+(z) - L_-(z) = \int_{u_0}^\infty b(u) e^{-zu}\, du = \frac{F(1/2+z)}{z}.$$

The real-axis cancellation already used in the scalar Landau criterion continues $F(1/2+z)/z$ holomorphically through a neighborhood of each positive real z. In detail, the pole at $s = 1$ cancels between the prime reference and the logarithmic integral; $\zeta(s)$ has no real zero for $s > 1/2$, and the prime-power corrections are holomorphic there.

If $\sigma_- > 0$, the identity $L_- = L_+ - F(1/2+z)/z$, continued along the positive real axis, would make L_- holomorphic through its real convergence abscissa σ_-. Landau's positivity theorem for Laplace transforms excludes this. Therefore $\sigma_- \leq 0$, so both L_+ and L_- are holomorphic on $\Re z > 0$. It follows that $F(s)$ is holomorphic on $\Re s > 1/2$. The earlier identity

$$(sF(s))' = \frac{\zeta'}{\zeta}(s) - \sum_{k\geq 2} \mathcal{P}'(ks) + \frac{2^{1-s}}{s-1}$$

then excludes every zeta zero with real part greater than $1/2$: the correction is holomorphic there, apart from its canceled singularity at $s = 1$, while an off-line zero would contribute a nonzero residue. Functional-equation symmetry completes RH. □

73 A summable positive correction between consecutive primes

The remaining integral is

$$\mathcal{I}(X) = \int_2^X \mathcal{A}(x)^+ \frac{dx}{x}, \qquad y^+ = \max(y, 0).$$

Its finiteness is equivalent to the unchanged target by the preceding positive-area criterion. This section separates a positive, summable geometric correction from a series determined entirely by values at consecutive prime endpoints. The arithmetic input is the established Baker–Harman–Pintz prime-gap theorem, not an improved prime-gap claim.

Lemma 73.1 (Concavity in the reciprocal-square-root coordinate)**.** *Let $p < q$ be consecutive primes, put $v = q^{-1/2}$, $w = p^{-1/2}$, and define $F(y) = \mathcal{A}(y^{-2})$ on $[v, w]$. Then, on the open interval,*

$$F'(y) = -2P(y^{-2}), \qquad F''(y) = -\frac{4y^{-3}}{\log(y^{-2})} < 0.$$

Let $L_{p,q}(x)$ be the interpolation of $\mathcal{A}(p)$ and $\mathcal{A}(q)$ which is affine in $x^{-1/2}$:

$$L_{p,q}(x) = \frac{x^{-1/2} - q^{-1/2}}{p^{-1/2} - q^{-1/2}}\mathcal{A}(p) + \frac{p^{-1/2} - x^{-1/2}}{p^{-1/2} - q^{-1/2}}\mathcal{A}(q).$$

For every $p \le x \le q$,

$$0 \le \mathcal{A}(x) - L_{p,q}(x) \le \frac{2q^{3/2}}{\log p}(x^{-1/2} - q^{-1/2})(p^{-1/2} - x^{-1/2}). \tag{73.1}$$

Proof. On (p, q) the prime count is a constant integer j, so $P'(x) = -1/\log x$. The chain rule gives the two derivatives. The function F is continuous at both endpoints; its one-sided derivatives are sufficient for interpolation on the closed interval. Concavity gives the lower bound in (73.1). Moreover

$$-\frac{4q^{3/2}}{\log p} \le F''(y) \le 0.$$

Put $K = 4q^{3/2}/\log p$ and let ℓ be the affine chord of F. The function

$$F(y) - \ell(y) - \frac{K}{2}(y - v)(w - y)$$

has nonnegative second derivative and zero endpoint values, hence is nonpositive. This proves the upper bound. □

The correction in (73.1) does not depend on the unknown size of P inside the gap. In fact the correction itself, before taking positive parts, is a deterministic function of p, q. For $y \in [v, w]$, write

$$W(y) = 2y\,\mathrm{Li}_2(y^{-2}) - 2\,\mathrm{Ei}(-\log y).$$

Direct integration gives $F(y) = c - 2jy + W(y)$ for a constant c. The constant and linear terms disappear on subtracting the chord, so $F - \ell$ is exactly W minus its own endpoint chord.

Proposition 73.2 (An explicit positive area error on one prime gap)**.** *For consecutive primes $p < q$, let*

$$B(p, q) = \int_p^q L_{p,q}(x)^+ \frac{dx}{x}, \qquad E(p, q) = \int_p^q (\mathcal{A}(x)^+ - L_{p,q}(x)^+)\frac{dx}{x}.$$

Then

$$0 \le E(p, q) \le \frac{2q^2}{3\log p}(p^{-1/2} - q^{-1/2})^3 \le \frac{(q - p)^3}{3p^{5/2}\log p}. \tag{73.2}$$

Proof. The positive-part map is increasing and 1-Lipschitz. Therefore $0 \le \mathcal{A}^+ - L^+_{p,q} \le \mathcal{A} - L_{p,q}$. Change variables to $y = x^{-1/2}$, so $|dx|/x = 2\,dy/y$. Using the preceding parabolic bound and $1/y \le \sqrt{q}$ gives

$$E(p,q) \le \frac{4q^2}{\log p}\int_v^w (y-v)(w-y)dy = \frac{2q^2}{3\log p}(w-v)^3.$$

Since

$$w - v = \frac{q-p}{\sqrt{p}\sqrt{q}(\sqrt{p}+\sqrt{q})} \le \frac{q-p}{2p^{3/2}},$$

and Bertrand's inequality gives $q < 2p$, the final bound follows. □

Each $B(p,q)$ has an elementary closed expression in its two endpoint values. With $v = q^{-1/2}$ and $w = p^{-1/2}$, put

$$\beta = \frac{\mathcal{A}(p) - \mathcal{A}(q)}{w - v}, \qquad \alpha = \frac{w\mathcal{A}(q) - v\mathcal{A}(p)}{w - v}.$$

Then

$$B(p,q) = 2\int_v^w [\alpha + \beta y]_+ \frac{dy}{y}. \tag{73.3}$$

If the positive set of $\alpha + \beta y$ in $[v,w]$ is the nonempty interval $[c,d]$ up to endpoints, its value is

$$2\{\alpha\log(d/c) + \beta(d-c)\};$$

if the positive set is empty, the value is zero. The case $\beta = 0$ has the evident constant-function interpretation. No infinite prime sum is needed to evaluate an endpoint, because

$$\mathcal{A}(X) = 2\sum_{p\le X} p^{-1/2} - \frac{2P(X)}{\sqrt{X}} - 2\,\mathrm{Ei}\left(\frac{\log X}{2}\right) + 2\,\mathrm{Ei}\left(\frac{\log 2}{2}\right). \tag{73.4}$$

This identity follows by finite summation and integration by parts. At a prime endpoint, its new prime's two contributions cancel, as required by continuity of $\mathcal{A}$.

Theorem 73.3 (All hidden between-prime area is summable)**.** *Write p_n for the consecutive primes, with $p_1 = 2$. Unconditionally,*

$$\sum_{n\ge 1} E(p_n, p_{n+1}) < \infty.$$

More precisely, for sufficiently large prime endpoints $P = p_M$,

$$\sum_{n\ge M} E(p_n, p_{n+1}) \ll \frac{P^{-9/20}}{\log P}. \tag{73.5}$$

Consequently, with

$$\mathcal{B}_M = \sum_{n<M} B(p_n, p_{n+1}), \qquad \mathcal{E}_M = \sum_{n<M} E(p_n, p_{n+1}),$$

the exact finite identity is

$$\mathcal{I}(p_M) = \mathcal{B}_M + \mathcal{E}_M, \qquad 0 \le \mathcal{E}_M \uparrow \mathcal{E}_\infty < \infty.$$

In particular, the unchanged RH-equivalent target is equivalent to

$$\boxed{\sum_{n\ge 1} B(p_n, p_{n+1}) < \infty.} \tag{73.6}$$

The convergence in (73.6) itself is not proved here.

Proof. Baker–Harman–Pintz [44, Theorem 1] implies $p_{n+1}-p_n \ll p_n^{\theta}$ with $\theta = 21/40$. For example, apply their backwards-interval theorem at $x = p_n + 2p_n^{\theta}$; for large p_n its interval begins strictly above p_n. Put $g_n = p_{n+1} - p_n$. The preceding bound gives

$$\sum_{n\geq M} E(p_n, p_{n+1}) \ll \sum_{n\geq M} \frac{g_n^3}{p_n^{5/2}\log p_n} \ll \sum_{n\geq M} \frac{g_n p_n^{-29/20}}{\log p_n}.$$

On each partition interval $[p_n, p_{n+1}]$, Bertrand's inequality makes the last summand at most an absolute constant times $\int_{p_n}^{p_{n+1}} x^{-29/20}\,dx/\log x$. Summing the partition intervals yields

$$\sum_{n\geq M} E(p_n, p_{n+1}) \ll \int_P^{\infty} \frac{x^{-29/20}}{\log x}dx \leq \frac{20}{9}\frac{P^{-9/20}}{\log P},$$

up to the already fixed implied constant. The finitely many initial gaps contribute a finite amount. The finite identity is additive over the prime-gap partition. Since its correction has finite total mass, convergence of the prime-endpoint series is equivalent to finiteness of the original positive-area integral. □

Corollary 73.4 (Positive humps with nonpositive prime endpoints)**.** *Unconditionally, the total logarithmic area contributed by gaps with $\mathcal{A}(p_n) \leq 0$ and $\mathcal{A}(p_{n+1}) \leq 0$ is finite. Its tail satisfies (73.5). More generally, the area where $L_{p_n,p_{n+1}} \leq 0$ but $\mathcal{A} > 0$ has finite total mass with the same tail bound.*

Proof. The chord is nonpositive when both endpoint values are nonpositive. On every portion where the chord is nonpositive, the positive part of $\mathcal{A}$ is bounded by the nonnegative error integrand in Proposition 73.2. Apply the theorem. □

Scope. This closes the summability of the geometric positive humps that can be hidden between prime endpoints. It does not control the positive endpoint contributions $B(p_n, p_{n+1})$ themselves. The prime-gap exponent is imported from an established theorem; no new prime-gap bound or historical-priority claim is made.

74 A sparse endpoint series with a summable continuum error

The preceding prime-gap decomposition has a nonnegative correction. A different combination, Brun–Titchmarsh with a deterministic sparse mesh, gives a signed correction whose absolute value is summable. Each new summand uses only two finite prime sums. The convergence of the endpoint series remains the arithmetic input to be proved.

Theorem 74.1 (An area criterion for a general mesh)**.** *Let $x_n \uparrow \infty$, let $h_n = x_{n+1} - x_n$, and suppose that, for all sufficiently large n,*

$$2 \leq h_n \leq x_n, \qquad \sum_n \frac{h_n^3}{x_n^{5/2}\log h_n} < \infty.$$

Write ℓ_n for the affine interpolant, in the variable x, of $\mathcal{A}(x_n)$ and $\mathcal{A}(x_{n+1})$, and put

$$C_n = \int_{x_n}^{x_{n+1}} \ell_n(x)^+ \frac{dx}{x} \geq 0.$$

Then, for any sufficiently large starting index n_0,

$$\int_{x_{n_0}}^{x_M} \mathcal{A}(x)^+ \frac{dx}{x} = \sum_{n=n_0}^{M-1} C_n + R_M, \tag{74.1}$$

where R_M converges to a finite real number R_∞ and

$$|R_\infty - R_M| \ll \sum_{n\ge M} \frac{h_n^3}{x_n^{5/2}\log h_n}. \tag{74.2}$$

In particular, the unchanged target B *is equivalent to* $\sum_{n\ge n_0} C_n < \infty$.

Proof. Proposition 66.5, which is unconditional, gives

$$\sup_{x_n\le x\le x_{n+1}} |\mathcal{A}(x) - \ell_n(x)| \ll \frac{h_n^2}{x_n^{3/2}\log h_n}.$$

The positive-part map is 1-Lipschitz. Since $\log(x_{n+1}/x_n) \le h_n/x_n$, it follows that

$$\int_{x_n}^{x_{n+1}} |\mathcal{A}(x)^+ - \ell_n(x)^+| \frac{dx}{x} \ll \frac{h_n^3}{x_n^{5/2}\log h_n}. \tag{74.3}$$

The series of signed cell errors therefore converges absolutely, which proves both identities and the tail estimate. A compact initial interval has finite integral. The equivalence with B is the already proved finite-positive-area criterion. □

Corollary 74.2 (A log-corrected quartic mesh)**.** *Fix $\alpha > 0$ and take*

$$x_n = \frac{n^4}{(\log n)^\alpha}$$

for all sufficiently large n. Theorem 74.1 applies, and

$$|R_\infty - R_M| \ll_\alpha (\log M)^{-\alpha/2}.$$

In particular, the fixed choice $\alpha = 2$ gives the exact induced criterion

$$\boxed{\mathsf{B} \iff \sum_{n\ge n_0} C_n < \infty, \qquad x_n = \frac{n^4}{(\log n)^2},} \tag{74.4}$$

with an unconditional continuum-error tail $O(1/\log M)$.

Proof. Differentiating $f(t) = t^4/(\log t)^\alpha$ gives

$$f'(t) = \frac{t^3}{(\log t)^\alpha}\left(4 - \frac{\alpha}{\log t}\right).$$

For large n, the mean value theorem yields $h_n \asymp_\alpha n^3/(\log n)^\alpha$ and $\log h_n \asymp \log n$, with $2 \le h_n \le x_n$. Consequently

$$\frac{h_n^3}{x_n^{5/2}\log h_n} \ll_\alpha \frac{1}{n(\log n)^{1+\alpha/2}}.$$

The integral test gives $\sum_{n\ge M} 1/[n(\log n)^{1+\alpha/2}] \ll_\alpha (\log M)^{-\alpha/2}$. □

An elementary formula for each nonnegative summand. Let $A_n = \mathcal{A}(x_n)$, and set

$$\beta_n = \frac{A_{n+1} - A_n}{h_n}, \qquad \alpha_n = \frac{x_{n+1}A_n - x_nA_{n+1}}{h_n}.$$

Then $\ell_n(x) = \alpha_n + \beta_n x$. Its positive set in the cell is either empty or an interval, up to endpoints. If the nonempty interval has endpoints $c_n < d_n$, then exactly

$$C_n = \alpha_n \log\frac{d_n}{c_n} + \beta_n(d_n - c_n). \tag{74.5}$$

If the positive set is empty, $C_n = 0$. For $\beta_n \ne 0$ its only possible interior endpoint is $-\alpha_n/\beta_n$; the constant case is immediate. The endpoint identity in (73.4) expresses each A_n through the finite list of primes not exceeding x_n and standard special functions. There is no improper integral or unknown zeta zero inside the definition of one C_n.

The exact gain and its limitation. For the unmodified grid $x_n = n^4$, the same argument gives only a cell-error majorant of order $1/(n \log n)$, which is not summable. A logarithmic increase in sampling density makes the total error summable. This observation concerns the present upper bound; it is not a proof that the actual error diverges on the unmodified grid, nor a claim of an optimal sampling threshold. The summable estimate also bounds the total area on all sets where $\ell_n \le 0$ but $\mathcal{A} > 0$. No separate continuum obstruction remains after the endpoint-area series is specified. Its convergence in (74.4) has not been proved. A finite computation of the summands cannot establish it.

75 Abel regularization and its exact singularity barrier

Retain a, H, and $M_n(m)$, with $m > 1/2$, and put

$$d_n(m) = \frac{m^n M_n(m)}{n!}, \qquad \mathfrak{D}(m,t) = \sum_{n\ge 0} t^n d_n(m)^-, \qquad t \ge 0.$$

The sum is interpreted in $[0,\infty]$. At $t = 1$ it is the previous Poisson negative mass $\mathfrak{D}(m)$. Its associated signed generating function is

$$F_m(w) = \sum_{n\ge 0} d_n(m) w^n = H(m(1-w)),$$

initially near zero. Below, equality with H outside its original Laplace half-plane means analytic continuation, not an unjustified integral evaluation.

Lemma 75.1 (The mean of the negative part of a finite oscillation)**.** *Let $T(n) = \sum_{j=1}^{J} c_j \omega_j^n$ be real, with distinct $|\omega_j| = 1$, no $\omega_j = 1$, and nonzero c_j. Write $B = \sum_j |c_j|$ and $q = \sum_j |c_j|^2$. The Cesàro means of T^+, T^-, and $|T|$ exist, and*

$$\mu := \operatorname{mean}(T^-) = \operatorname{mean}(T^+) = \tfrac{1}{2} \operatorname{mean} |T| \ge \frac{q}{2B} > 0.$$

Moreover, for $0 < v < 1$,

$$\sum_{n\ge 1} \frac{v^n T(n)^-}{n} = \mu \log \frac{1}{1-v} + o\Big(\log \frac{1}{1-v}\Big) \qquad (v \uparrow 1).$$

Proof. Every polynomial in $T(n)$ is a finite sum of unit-modulus exponentials, whose Cesàro mean exists by finite geometric sums. Uniform polynomial approximation of the continuous functions y^+, y^-, and $|y|$ on $[-B,B]$ therefore proves existence of their means. The mean of T is zero and that of T^2 is q. Since $T^2 \le B|T|$, the asserted lower bound follows.

For completeness, any bounded real sequence b_n with mean μ satisfies the displayed Abelian assertion. Put $S_N = \sum_{n\le N}(b_n - \mu) = o(N)$ and apply summation by parts to $\sum v^n (b_n - \mu)/n$. For every $\varepsilon > 0$, the part beyond a fixed index has absolute value at most $\varepsilon \log(1/(1-v))$; the initial part is $O_\varepsilon(1)$. Letting $v \uparrow 1$ and then $\varepsilon \downarrow 0$ proves the result. Apply it to $b_n = T(n)^-$. □

Theorem 75.2 (The precise Abel deficit dichotomy)**.** *Fix $m > 1/2$.*

1. *If the open disk $\mathcal{D}_m$ contains no shifted zeta zero, then $\mathfrak{D}(m,t)$ is a polynomial in t with nonnegative coefficients.*
2. *Otherwise, let*

$$r = \min_{\rho} |m - (\rho - 1/2)| < m, \qquad R_m = \frac{r}{m} < 1,$$

and let T_m be the nearest-singularity oscillation in (70.3). Put $\mu_m = \operatorname{mean}(T_m^-) > 0$. Then

$$\mathfrak{D}(m,t) = \mu_m \log \frac{1}{1 - t/R_m} + o_m\!\left(\log \frac{1}{1 - t/R_m}\right) \qquad (t \uparrow R_m). \tag{75.1}$$

The sum is finite for $0 \le t < R_m$ and infinite for every $t \ge R_m$.

In particular, for every $0 < t < 1$,

$$\mathfrak{D}(m,t) < \infty \quad \Longleftrightarrow \quad \{\rho : \zeta(\rho) = 0,\ 0 < \Re\rho < 1, |(\rho - 1/2) - m| \le mt\} = \varnothing. \tag{75.2}$$

At $t = 1$, the correct condition is the previous open-*disk criterion, so the closed-boundary wording in (75.2) must not be extended to $t = 1$.*

Proof. In the zero-free case, the earlier disk theorem makes $d_n(m)$ strictly positive for all sufficiently large n. The negative-part series therefore has only finitely many nonzero coefficients.

In the other case, the proved nearest-logarithm expansion gives

$$d_n(m) = \frac{R_m^{-n}}{n}(T_m(n) + O_m(1/n)).$$

The negative-part map is 1-Lipschitz. Consequently

$$d_n(m)^- = \frac{R_m^{-n}}{n} T_m(n)^- + O_m(R_m^{-n}/n^2).$$

After multiplication by t^n, the error has a uniformly bounded sum for $0 \le t \le R_m$. Lemma 75.1 proves (75.1). Monotone convergence gives $\mathfrak{D}(m, R_m) = \infty$, and nonnegative coefficients give divergence for every larger t. Absolute convergence below R_m follows directly from the same coefficient bound. For $0 < t < 1$, a shifted zero in the smaller closed disk is equivalent to $r \le mt$; the two cases just proved give (75.2). □

Where the signed power series is absolutely convergent, there is also the exact total-variation identity

$$\mathfrak{D}(m,t) = \frac{1}{2}\left(\sum_{n\ge 0} t^n |d_n(m)| - H(m(1-t))\right).$$

At a bad radius $R_m < 1$, the last term remains analytic on the positive real axis, while the absolute-value and negative-part series diverge logarithmically. Thus analytic continuation of the signed transform by itself cannot control this positive mass.

75.1 Tilting the source and a uniformly controlled region

Theorem 75.3 (A tilted Poisson recovery law)**.** *For fixed $c \ge 0$ and $m > \max(c, 1/2)$, put*

$$\mathfrak{D}_c(m) = \mathfrak{D}\left(m, 1 - \frac{c}{m}\right), \qquad L_-(c) = \int_0^\infty a(u)^- e^{-cu}\, du.$$

Then, with extended-real values allowed,

$$\mathfrak{D}_c(m) \text{ is nondecreasing in } m, \qquad \lim_{m\to\infty} \mathfrak{D}_c(m) = L_-(c).$$

Unconditionally, there is a finite constant C such that

$$\mathfrak{D}(m,t) \le C \quad \text{whenever} \quad 0 \le t < 1, \quad m > 1/2, \quad m(1-t) \ge 1/2. \tag{75.3}$$

Proof. Set $b = m - c > 0$ and $a_c(u) = a(u)e^{-cu}$. The exact identity

$$\left(1 - \frac{c}{m}\right)^n d_n(m) = \int_0^\infty a_c(u) e^{-bu} \frac{(bu)^n}{n!}\, du$$

identifies $\mathfrak{D}_c(m)$ with the Poisson negative mass of the tilted source at rate b. Each row integral is absolutely convergent because $b + c = m > 1/2$. The proof of Theorem 71.1 applies unchanged: thinning proves monotonicity in b, and compact-test recovery proves convergence to $\int a_c^- = L_-(c)$. Its dominating tail remains valid because the tilt only decreases the growth of a.

For the uniform bound, the classical quantitative prime number theorem gives $P(x) = O_K(x/\log^K x)$ for each fixed K; taking $K = 2$ yields

$$\int_2^\infty \frac{|P(t)|}{t^2}\, dt < \infty.$$

Absolute Tonelli gives

$$L_-(1/2) \le \int_2^\infty \frac{|\mathcal{A}(x)|}{x^{3/2}}\, dx \le \int_2^\infty \frac{|P(t)|}{t^{3/2}} \int_t^\infty x^{-3/2}\, dx\, dt = 2\int_2^\infty \frac{|P(t)|}{t^2}\, dt =: C < \infty.$$

The kernel inequality $\mathfrak{D}(m,t) \le L_-(m(1-t))$ follows directly by summing the Poisson probabilities against a^-. Since $L_-(c) \le L_-(1/2)$ for $c \ge 1/2$, this proves (75.3), including $t = 0$ directly. □

What is and is not closed. The entire closed regularization region $m(1-t) \ge 1/2$ is controlled by one unconditional finite bound. Extending that bound uniformly as $c = m(1-t) \downarrow 0$ would give $L_-(0) < \infty$ by monotone convergence, and hence the unchanged RH-equivalent target. This extension has not been proved. The exact logarithmic barrier identifies where a putative off-line zero forces the positive deficit series to diverge, even though the signed transform has analytic continuation along the real axis. No historical novelty of Abelian summation or Poisson tilting is claimed.

76 Exact sign-boundary flux and a globally bounded tilted budget

The thinning inequality can be differentiated, but the resulting boundary terms matter. This section gives the exact differentiation law and closes a genuinely global, exponentially tilted version of the negative-mass problem. The exponent of the tilt is retained explicitly throughout.

Write $I_n(m) = \mathbf{1}_{\{d_n(m)<0\}}$, and define

$$D_r(m) := \mathfrak{D}(m,r) = \sum_{n\ge 0} r^n d_n(m)^-, \qquad 0 \le r < 1.$$

The convention $r^0 = 1$ is used also at $r = 0$. For $m(1-r) > 1/2$ the series and its first derivatives are locally absolutely convergent. Indeed, summing the absolute Poisson majorants yields

$$\sum_{n\ge 0} r^n |d_n(m)| \le \int_0^\infty |a(u)| e^{-m(1-r)u}\, du,$$

and first derivatives introduce only one further power of u.

Proposition 76.1 (Kato transport law)**.** *For every fixed $n \ge 0$ and $m > 1/2$,*

$$m d_n'(m) = n d_n(m) - (n+1) d_{n+1}(m). \tag{76.1}$$

For almost every (m,r) in the region $m(1-r) > 1/2$,

$$m\partial_m D_r(m) + (1-r)\partial_r D_r(m) = \Phi_r(m) \ge 0, \tag{76.2}$$

where

$$\Phi_r(m) = \sum_{k\geq 1} kr^{k-1} d_k(m)\big(I_{k-1}(m) - I_k(m)\big). \tag{76.3}$$

Every summand in (76.3) *is nonnegative.*

Proof. Differentiate $m^n M_n(m)/n!$ and use $M_n'(m) = -M_{n+1}(m)$. For a locally absolutely continuous real function f, the chain rule $(f^-)' = -\mathbf{1}_{\{f<0\}} f'$ holds almost everywhere. This remains valid on its zero set, where $f' = 0$ almost everywhere. Applying this rule termwise, using the preceding absolute majorants, gives

$$m\partial_m D_r = \sum_{k\geq 1} kr^{k-1} I_{k-1} d_k - \sum_{k\geq 1} kr^k I_k d_k,$$
$$\partial_r D_r = -\sum_{k\geq 1} kr^{k-1} I_k d_k.$$

Their indicated combination is (76.2). If d_{k-1} and d_k have the same strict sign, the summand vanishes. If $d_{k-1} < 0 \leq d_k$, it is $kr^{k-1} d_k \geq 0$. If $d_{k-1} \geq 0 > d_k$, it is $-kr^{k-1} d_k > 0$. Cases with $d_k = 0$ contribute zero. □

There is a precise boundary obstruction to applying the same calculation directly to an undamped infinite sum. At every finite cutoff K,

$$m\frac{d}{dm}\sum_{n=0}^{K} d_n(m)^- = \sum_{k=1}^{K} k d_k(m)(I_{k-1}(m) - I_k(m)) + (K+1) d_{K+1}(m) I_K(m) \tag{76.4}$$

almost everywhere. The last term can have either sign. It cannot be discarded, and no vanishing of it as $K \to \infty$ is asserted.

The characteristic invariant of (76.2) is

$$c = m(1 - r).$$

The following theorem includes the boundary value $c = 1/2$, where the crude bound $a(u) = O(e^{u/2}/u)$ alone does not give integrability.

Theorem 76.2 (Exact characteristic flux and an explicit bound)**.** *Unconditionally, for every* $c \geq 1/2$ *define*

$$I(c) = \int_0^\infty a(u)^- e^{-cu}\, du = \int_2^\infty \mathcal{A}(x)^+ x^{-c-1}\, dx, \qquad H(c) = \int_0^\infty a(u) e^{-cu}\, du.$$

These integrals are finite. The function

$$B_c(m) = D_{1-c/m}(m), \qquad m > c,$$

is locally absolutely continuous and nondecreasing, with

$$\lim_{m\downarrow c} B_c(m) = H(c)^-, \qquad \lim_{m\to\infty} B_c(m) = I(c). \tag{76.5}$$

Its exact total sign-boundary flux is

$$\int_c^\infty \frac{\Phi_{1-c/m}(m)}{m}\, dm = I(c) - H(c)^-. \tag{76.6}$$

Moreover,

$$\sup_{m>1/2} \sum_{n\geq 0} \left(1 - \frac{1}{2m}\right)^n d_n(m)^- = I(1/2) < 180. \tag{76.7}$$

In particular, the total flux in (76.6) *is less than* 180 *at* $c = 1/2$.

Proof. Put $b_c(u) = a(u)e^{-cu}$ and $\lambda = m - c$. Direct multiplication of Poisson probabilities gives the exact identity

$$\left(1 - \frac{c}{m}\right)^n d_n(m) = e_n(\lambda) := \int_0^\infty b_c(u) e^{-\lambda u} \frac{(\lambda u)^n}{n!}\, du. \tag{76.8}$$

Thus $B_c(m) = \sum_n e_n(m-c)^-$ is the ordinary Poisson negative budget of the tilted signed density b_c.

We first check endpoint integrability using a published unconditional prime number theorem estimate. Johnston and Yang [27, Corollary 1.3] prove, for every $x \geq 2$,

$$|\pi(x) - \mathrm{li}(x)| \leq 9.59 x (\log x)^{0.515} \exp(-0.8274\sqrt{\log x}). \tag{76.9}$$

Since $\mathrm{Li}_2(x) = \mathrm{li}(x) - \mathrm{li}(2)$, this bounds $|P(x)|$ by the right side plus $|\mathrm{li}(2)|$. Tonelli's theorem gives, for $c > 0$,

$$\int_0^\infty |a(u)| e^{-cu}\, du \leq \frac{1}{c} \int_2^\infty |P(t)| t^{-c-3/2}\, dt. \tag{76.10}$$

At $c = 1/2$ the last integral converges by (76.9). The same argument with polynomial weights shows

$$\int_0^\infty u^j |b_c(u)|\, du < \infty \qquad (c \geq 1/2, \quad j = 0, 1, 2, \ldots). \tag{76.11}$$

For completeness, after changing variables $x = e^u$, the inner integral introduced by Tonelli is $\int_t^\infty (\log x)^j x^{-3/2}\, dx \ll_j t^{-1/2}(1 + (\log t)^j)$; the remaining integral is dominated by a polynomial in $\log t$ times $e^{-0.8274\sqrt{\log t}} dt/t$.

For an integrable b_c, thinning proves monotonicity of its Poisson negative budget, while the approximate-identity argument of Theorem 71.1 gives its limit $\int b_c^- = I(c)$. Alternatively, dominated convergence on compactly supported test functions suffices, with $|b_c|$ itself as the dominating function. As $\lambda \downarrow 0$, $e_0(\lambda) \to \int b_c = H(c)$ and

$$\sum_{n \geq 1} |e_n(\lambda)| \leq \int_0^\infty |b_c(u)| (1 - e^{-\lambda u})\, du \longrightarrow 0.$$

This proves (76.5).

To justify the flux law even at $c = 1/2$, differentiate the kernels in (76.8) directly. With the convention $p_{-1}(v) = 0$ and $p_n(v) = e^{-v} v^n / n!$,

$$\partial_\lambda p_n(\lambda u) = u\{p_{n-1}(\lambda u) - p_n(\lambda u)\}, \qquad \sum_{n \geq 0} |e_n'(\lambda)| \leq 2 \int_0^\infty u |b_c(u)|\, du < \infty.$$

Consequently the signed sequence is locally absolutely continuous in ℓ^1, and the negative-part sum is locally absolutely continuous. The preceding Kato calculation gives, almost everywhere,

$$\frac{d}{dm} B_c(m) = \frac{1}{m - c} \sum_{k \geq 1} k e_k(m - c)(I_{k-1}(m) - I_k(m)) = \frac{1}{m} \Phi_{1 - c/m}(m).$$

Here the signs agree because the weights in (76.8) are positive for $m > c$. Integrating this nonnegative derivative and using the endpoint limits proves (76.6).

For the explicit constant, write $\alpha = 0.8274$. At $c = 1/2$, (76.10) and (76.9) give

$$\begin{aligned} I(1/2) &\leq 19.18 \int_2^\infty (\log t)^{0.515} e^{-\alpha\sqrt{\log t}} \frac{dt}{t} + |\mathrm{li}(2)| \\ &= 38.36 \alpha^{-3.03} \Gamma(3.03, \alpha\sqrt{\log 2}) + |\mathrm{li}(2)|, \end{aligned}$$

where $\Gamma(s,y)=\int_y^\infty v^{s-1}e^{-v}\,dv$. The following deliberately loose estimates certify the stated integer bound without a numerical quadrature:

$$\Gamma(3.03,y)\le\Gamma(3.03)\le\Gamma(3)^{0.97}\Gamma(4)^{0.03}=2\cdot3^{0.03}<2.07,\qquad \alpha^{-3.03}<0.82^{-4}<2.213.$$

The middle inequality is the log-convexity of the Gamma integral, an immediate consequence of Hölder's inequality. Finally, $|\operatorname{li}(2)|<2$: put $t=\log 2\in(1/2,1)$ in

$$\operatorname{li}(2)=\gamma+\log t+\sum_{k\ge1}\frac{t^k}{k\,k!},$$

use $0<\gamma<1$, $-1<\log t<0$, and $0<\sum t^k/(k\,k!)\le e^t-1=1$. Therefore

$$I(1/2)<38.36\cdot2.213\cdot2.07+2=177.7237076<180.$$

The exact equality in (76.7) follows from monotonicity and (76.5). □

Why the tilt has not disappeared. Although $1-c/m\to1$ as $m\to\infty$, the relevant index of a Poisson kernel with mean mu is $n\approx mu$. Consequently $(1-c/m)^n\to e^{-cu}$ at that scale. More decisively, (76.8) already identifies the retained factor exactly. Thus the bound 180 controls $\int\mathcal{A}(x)^+x^{-3/2}\,dx$, whereas the unchanged RH-equivalent target is $\int\mathcal{A}(x)^+dx/x<\infty$. The latter corresponds to $c=0$. Passing from the proved characteristic $c=1/2$ to $c=0$ requires additional cancellation beyond the absolute prime-error estimate (76.9); no such bound is proved here. The sign-boundary flux formula is exact at every scale where it is used, and the explicit global tilted bound is unconditional, but neither is a proof of the original compensated R_4 bound.

77 Short-interval cancellation removes a larger class of excursions

Keep $P(x)=\pi(x)-\operatorname{Li}_2(x)$ and $\mathcal{A}(X)=\int_2^X P(t)t^{-3/2}\,dt$. The previous interpolation estimates used upper bounds on the total number of primes in a short interval. Here an asymptotic formula for that count supplies actual cancellation between the discrete and continuous terms. This closes the total positive area of an explicitly larger family of excursions and permits a mesh sparser than the pure quartic mesh. It does not establish convergence of the remaining positive endpoint series.

Write

$$\varepsilon_{\mathrm{si}}(x)=\exp\bigl(-(\log x)^{1/4}\bigr).$$

The external input is the following unconditional theorem of Guth–Maynard [45, Corollary 1.3]: for each fixed $\varepsilon>0$, uniformly for $x^{17/30+\varepsilon}\le h\le x^{0.99}$,

$$\pi(x+h)-\pi(x)=\frac{h}{\log x}+O_\varepsilon\bigl(h\varepsilon_{\mathrm{si}}(x)\bigr).\tag{77.1}$$

We use only the range $h\ge x^{2/3}$, by fixing $\varepsilon=1/10$. Thus no limiting use of the lower endpoint $17/30$ is involved. Real rather than integer endpoints cause no difficulty: rounding changes the count by $O(1)$, which is absorbed by $h\varepsilon_{\mathrm{si}}(x)$ in this range.

Lemma 77.1 (A uniform interpolation estimate with cancellation)**.** *For a sufficiently large and $0<h\le a^{0.99}$, let $\ell_{a,h}$ be the affine interpolant of $\mathcal{A}(a),\mathcal{A}(a+h)$ in the variable x. Then*

$$\sup_{a\le x\le a+h}\bigl|\mathcal{A}'_+(x)-\mathcal{A}'_+(a)\bigr|\ll ha^{-3/2}\varepsilon_{\mathrm{si}}(a)+\frac{a^{-5/6}}{\log a},\tag{77.2}$$

$$\sup_{a\le x\le a+h}\bigl|\mathcal{A}(x)-\ell_{a,h}(x)\bigr|\ll h^2a^{-3/2}\varepsilon_{\mathrm{si}}(a)+\frac{ha^{-5/6}}{\log a}.\tag{77.3}$$

The constants are uniform in a, h. In particular,

$$\int_a^{a+h} \left|\mathcal{A}(x)^+ - \ell_{a,h}(x)^+\right| \frac{dx}{x} \ll h^3 a^{-5/2}\varepsilon_{\mathrm{si}}(a) + \frac{h^2 a^{-11/6}}{\log a}. \tag{77.4}$$

Here $\mathcal{A}'_+(x) = P(x)x^{-3/2}$ uses the right derivative at a prime.

Proof. Put $L = \log a$ and

$$R_a(t) = P(a+t) - P(a), \qquad 0 \le t \le h.$$

For $a^{2/3} \le t \le h$, (77.1) gives

$$R_a(t) = O(t\varepsilon_{\mathrm{si}}(a)) + \frac{t}{\log a} - \int_a^{a+t} \frac{dv}{\log v}.$$

The last difference is at most $t^2/(2aL^2)$ in absolute value. Since $t \le a^{0.99}$, this is $O(t\varepsilon_{\mathrm{si}}(a))$ for sufficiently large a. For $t < a^{2/3}$, monotonicity of π and Brun–Titchmarsh on the containing interval of length $a^{2/3}$ give

$$|R_a(t)| \ll \frac{a^{2/3}}{\log a}.$$

Consequently, throughout $0 \le t \le h$,

$$|R_a(t)| \ll h\varepsilon_{\mathrm{si}}(a) + \frac{a^{2/3}}{\log a}. \tag{77.5}$$

The ordinary quantitative prime number theorem, for example the Johnston–Yang bound already used in this paper, gives $|P(a)| \ll a\varepsilon_{\mathrm{si}}(a)$. Therefore

$$\mathcal{A}'_+(a+t) - \mathcal{A}'_+(a) = \frac{R_a(t)}{(a+t)^{3/2}} + P(a)\big((a+t)^{-3/2} - a^{-3/2}\big),$$
$$\left|P(a)\big((a+t)^{-3/2} - a^{-3/2}\big)\right| \ll ha^{-3/2}\varepsilon_{\mathrm{si}}(a).$$

This proves (77.2). The chord slope is the average of $\mathcal{A}'$ on the interval. Subtracting this average and integrating proves (77.3). The positive-part map is 1-Lipschitz, and $\int_a^{a+h} dx/x \le h/a$, proving (77.4). □

77.1 All positive excursions below a larger duration threshold

A positive excursion means a connected component (a, b) of $\{x > 2 : \mathcal{A}(x) > 0\}$; a priori $b = \infty$ is allowed.

Lemma 77.2 (All positive components are finite)**.** *Unconditionally, every positive excursion has finite endpoints $2 \le a < b < \infty$ and $\mathcal{A}(a) = \mathcal{A}(b) = 0$. For every fixed T, only finitely many excursions have left endpoint $a < T$.*

Proof. If an unbounded positive component existed, then $\mathcal{A}(X) \ge 0$ for all sufficiently large X. The lower-bound version of Proposition 65.3 would imply RH. Theorem 65.4 would then give $\mathcal{A}(X) = -\log\log X + C_* + o(1) \to -\infty$, contradicting that component. This contradiction excludes unbounded positive components without assuming RH.

On each prime gap, P is strictly decreasing. Hence $\mathcal{A}' = Px^{-3/2}$ has at most one zero, and $\mathcal{A}$ has at most two zeros on that gap. Every compact interval meets finitely many prime gaps, so its zero set is finite. Thus only finitely many component left endpoints can lie below a fixed T. Continuity gives zero values at the finite component endpoints; the possible initial endpoint $a = 2$ also satisfies $\mathcal{A}(2) = 0$. □

Write $\mathcal{E}$ for this family of bounded components. The logarithmic positive area of one component is

$$J(a,b)=\int_a^b \mathcal{A}(x)\frac{dx}{x}.$$

Each $J(a,b)$ is finite, since its interval is compact.

Theorem 77.3 (A summable family of positive excursions)**.** *Fix $0\le\kappa<1/2$, and put*

$$H_\kappa(a)=a^{3/4}\exp\bigl(\kappa(\log a)^{1/4}\bigr).$$

Unconditionally,

$$\sum_{\substack{(a,b)\ \text{positive excursion}\\ b-a\le H_\kappa(a)}} J(a,b)<\infty. \tag{77.6}$$

More precisely, with $c=1-2\kappa>0$, the sub-sum over left endpoints $a\ge T$ is

$$\ll_\kappa (1+(\log T)^{3/4})e^{-c(\log T)^{1/4}}+\frac{T^{-1/12}e^{\kappa(\log T)^{1/4}}}{\log T}\qquad (T\to\infty). \tag{77.7}$$

In particular one may take the fixed value $\kappa=1/4$.

Proof. Write $h=b-a$. For a sufficiently large, $H_\kappa(a)\le a^{0.99}$, so the preceding lemma applies. Both endpoint values vanish, making $\ell_{a,h}=0$. Consequently

$$J(a,b)\ll h^3a^{-5/2}\varepsilon_{\mathrm{si}}(a)+\frac{h^2a^{-11/6}}{\log a}.$$

Use $h^2\le H_\kappa(a)^2$ in the first term and $h\le H_\kappa(a)$ in the second term to obtain

$$J(a,b)\ll h\left\{\frac{e^{-c(\log a)^{1/4}}}{a}+\frac{a^{-13/12}e^{\kappa(\log a)^{1/4}}}{\log a}\right\}. \tag{77.8}$$

Both weights in braces are eventually decreasing and are comparable at all points of $[a,2a]$. Since $h\le a^{0.99}\le a$, the right side of (77.8) is bounded by a constant times the integral of those weights over $[a,b]$. Disjointness of the excursions now gives an upper bound by

$$\int_T^\infty e^{-c(\log x)^{1/4}}\frac{dx}{x}+\int_T^\infty\frac{x^{-13/12}e^{\kappa(\log x)^{1/4}}}{\log x}\,dx.$$

The first integral is exactly $4c^{-4}\Gamma(4,c(\log T)^{1/4})$. The second is $O_\kappa(T^{-1/12}e^{\kappa(\log T)^{1/4}}/\log T)$: after $u=\log x$, the exponent $-u/12+\kappa u^{1/4}$ has derivative at most $-1/24$ for sufficiently large u. The elementary formula $\Gamma(4,z)=e^{-z}(z^3+3z^2+6z+6)$ proves (77.7). The finitely many excursions with left endpoint in a fixed initial compact interval contribute a finite amount. □

Corollary 77.4 (Only longer excursions can carry the unresolved mass)**.** *For any fixed $0\le\kappa<1/2$, the unchanged positive-area condition has the exact equivalence*

$$\begin{aligned} I(0)&:=\int_2^\infty \mathcal{A}(x)^+\frac{dx}{x},\\ I(0)<\infty\quad&\Longleftrightarrow\quad \sum_{\substack{(a,b)\in\mathcal{E}\\ b-a>H_\kappa(a)}} J(a,b)<\infty. \end{aligned} \tag{77.9}$$

The same equivalence holds if the sum on the right is restricted to $a\ge T$, for any fixed sufficiently large T. The number of these longer excursions with left endpoints in $[X,2X]$ is at most

$$1+X^{1/4}\exp\bigl(-\kappa(\log X)^{1/4}\bigr)$$

for X sufficiently large.

Proof. Fix T large enough for Theorem 77.3. Tonelli's theorem for the disjoint positive components gives the exact identity in $[0,\infty]$

$$I(0) = \sum_{\substack{(a,b)\in\mathcal{E}\\ a<T}} J(a,b) + \sum_{\substack{(a,b)\in\mathcal{E}\\ a\ge T,\ b-a\le H_\kappa(a)}} J(a,b) + \sum_{\substack{(a,b)\in\mathcal{E}\\ a\ge T,\ b-a>H_\kappa(a)}} J(a,b).$$

The first sum is finite by Lemma 77.2; this includes any component starting below T and ending far above it. The second sum is finite by Theorem 77.3. Therefore finiteness of $I(0)$ is exactly finiteness of the third sum. Adding back the finitely many initial long components proves (77.9). For the count, all but at most one excursion starting in $[X,2X]$ end before $2X$. Their disjoint lengths sum to at most X, and each such length exceeds $H_\kappa(X)$ because H_κ is increasing. □

The corollary does not bound the areas of the longer excursions. It removes a specified family by a proved summable estimate; the remaining arithmetic condition is still unproved. In particular, large duration is not itself a substitute for an upper bound on height or area.

77.2 A mesh sparser than every logarithmically coarser quartic mesh

Define, for integers $n \ge 2$,

$$y_n = n^4 \exp\big((\log n)^{1/4}\big), \qquad h_n = y_{n+1} - y_n.$$

Let $\widetilde{\ell}_n$ be the affine interpolant of $\mathcal{A}(y_n), \mathcal{A}(y_{n+1})$, and set

$$\widetilde{C}_n = \int_{y_n}^{y_{n+1}} \widetilde{\ell}_n(x)^+ \frac{dx}{x}.$$

Theorem 77.5 (Summable interpolation on a coarser mesh)**.** *Unconditionally, for n_0 sufficiently large,*

$$\sum_{n\ge n_0} \int_{y_n}^{y_{n+1}} \left|\mathcal{A}(x)^+ - \widetilde{\ell}_n(x)^+\right| \frac{dx}{x} < \infty.$$

Writing $\delta = \sqrt{2} - 1/2 > 0$, the tail starting at $n = N$ is

$$\ll (1 + (\log N)^{3/4}) e^{-\delta(\log N)^{1/4}}. \tag{77.10}$$

Consequently there is a convergent real sequence $\widetilde{R}_M$ such that

$$\int_{y_{n_0}}^{y_M} \mathcal{A}(x)^+ \frac{dx}{x} = \sum_{n=n_0}^{M-1} \widetilde{C}_n + \widetilde{R}_M.$$

The unchanged target is therefore equivalent to $\sum_{n\ge n_0} \widetilde{C}_n < \infty$.

Proof. We have $h_n \asymp y_n/n$ and $h_n \le y_n^{0.99}$ for sufficiently large n. Apply (77.4). If $v = \log n$, then $\log y_n = 4v + v^{1/4} \ge 4v$, whence

$$h_n^3 y_n^{-5/2} \varepsilon_{\mathrm{si}}(y_n) \ll \frac{1}{n} \exp\big(-(\sqrt{2} - 1/2) v^{1/4}\big),$$
$$\frac{h_n^2 y_n^{-11/6}}{\log y_n} \ll \frac{n^{-4/3} e^{v^{1/4}/6}}{\log n}.$$

Both are summable. Integrating the first majorant gives $4\delta^{-4}\Gamma(4, \delta(\log N)^{1/4})$ up to an initial summand. The tail of the second majorant is $O(N^{-1/3} e^{(\log N)^{1/4}/6}/\log N)$, which is smaller than (77.10). Summing the signed cell errors gives the convergent remainder. The equivalence uses the already established original positive-area criterion. □

More generally, the same proof applies to $y_n = n^4 \exp(\lambda(\log n)^{1/4})$ for any fixed $0 < \lambda < 2\sqrt{2}$: replace δ by $\sqrt{2} - \lambda/2 > 0$. In particular the pure quartic mesh $y_n = n^4$, and every mesh $n^4(\log n)^B$ with fixed real B, also have a summable signed-chord area error. The short-interval estimate is the additional arithmetic input that removes the logarithmic divergence of the earlier Brun–Titchmarsh-only error bound. No historical priority for the external short-interval theorem, or for this style of interpolation argument, is claimed.

78 Weighted curvature and a one-node positive-mass criterion

The signed-chord estimate from the preceding section does not by itself control the chord of the two positive endpoint values: taking a positive part can create a corner at a zero. A weighted variation argument controls exactly these additional corners. Applied after polygonal interpolation, it works on the coarser mesh already constructed.

Lemma 78.1 (A weighted positive-part variation inequality)**.** *Let f be continuous on $[T, \infty)$, with a locally bounded-variation first derivative. Assume that locally it is piecewise smooth with finitely many isolated zeros and zero intervals; this includes the applications below. Let $w > 0$ be continuously differentiable. If*

$$w(x)|f'(x)| \longrightarrow 0, \qquad \int_T^\infty |w'(x)f'(x)|\,dx < \infty, \qquad \int_{(T,\infty)} w\,d(D^2 f)^- < \infty,$$

then

$$\int_{(T,\infty)} w\,d|D^2(f^+)| \le w(T)|f'_+(T)| + \int_T^\infty |w'f'|\,dx + 2\int_{(T,\infty)} w\,d(D^2 f)^-. \tag{78.1}$$

At a nonsmooth point, the first displayed limit is interpreted for both one-sided derivatives, and $f'_+(T)$ denotes the right derivative.

Proof. Put $g = f^+$. On each region where $f > 0$, its derivative is f'; where $f < 0$, it is zero. At an isolated zero with left and right derivatives s_-, s_+, the corresponding derivative jump of g is $\max(s_+, 0) - \min(s_-, 0) \ge 0$. The endpoints of a zero interval have the same one-sided interpretation. At a point where $f > 0$, the derivative jump of g is the original jump; at a point where $f < 0$ it is zero. These statements also follow by smooth convex approximation to the positive-part function. Consequently

$$(D^2 g)^- \le (D^2 f)^-, \qquad |g'| \le |f'|.$$

For a finite upper endpoint R which is not a break point, weighted Stieltjes integration by parts gives

$$\int_{(T,R)} w\,dDg' = w(R)g'(R) - w(T)g'_+(T) - \int_T^R w'g'\,dx.$$

Add twice the weighted negative variation to obtain the weighted total variation. Bound the resulting terms in absolute value and let $R \to \infty$ through non-break points. The assumed endpoint limit and monotone convergence prove (78.1). □

Proposition 78.2 (Summable weighted curvature of the actual positive part)**.** *Unconditionally, with $g(x) = \mathcal{A}(x)^+$,*

$$\int_{(T,\infty)} \frac{\sqrt{x}}{\log x}\,d|D^2 g|(x) \ll \frac{1}{\log T} \qquad (T \longrightarrow \infty). \tag{78.2}$$

In particular, the non-prime zero crossings satisfy

$$\sum_{\substack{\xi > T,\ \mathcal{A}(\xi)=0 \\ \xi \text{ mnot prime}}} \frac{\sqrt{\xi}}{\log \xi} |\mathcal{A}'(\xi)| \ll \frac{1}{\log T}.$$

Tangencies with zero derivative contribute zero.

Proof. There are only finitely many zeros of $\mathcal{A}$ on a compact interval. Indeed, between consecutive primes the reciprocal-root coordinate makes $\mathcal{A}$ strictly concave, so there are at most two zeros on each such interval. The curvature identity is

$$D^2\mathcal{A} = \sum_p p^{-3/2}\delta_p - k(x)dx, \qquad k(x) = \frac{x^{-3/2}}{\log x} + \frac{3}{2}P(x)x^{-5/2}.$$

The ordinary Chebyshev bound gives $|\mathcal{A}'(x)| \ll x^{-1/2}/\log x$ and $k(x)^+ \ll x^{-3/2}/\log x$. In particular, $(D^2\mathcal{A})^- \le k(x)^+dx$. For $w_0(x) = \sqrt{x}/\log x$, one has

$$w_0|\mathcal{A}'| \ll (\log x)^{-2} \to 0, \qquad |w_0'\mathcal{A}'| + w_0k^+ \ll \frac{1}{x(\log x)^2}.$$

Lemma 78.1 proves the asserted tail bound. At a non-prime zero ξ, the atom of D^2g is exactly $|\mathcal{A}'(\xi)|$, proving the second assertion. □

For the coarser grid, retain

$$y_n = n^4e^{(\log n)^{1/4}}, \qquad h_n = y_{n+1} - y_n, \qquad v_n = \mathcal{A}(y_n)^+.$$

Let L be the continuous, piecewise affine interpolant of the signed values $\mathcal{A}(y_n)$, and let

$$s_n = \frac{\mathcal{A}(y_{n+1}) - \mathcal{A}(y_n)}{h_n}$$

be its slope on the nth cell. Define the larger weight

$$w(x) = \sqrt{x}\exp\left(\tfrac{1}{2}(\log x)^{1/4}\right).$$

The preceding proposition alone does not permit this larger weight. The short-interval prime number theorem controls the slope changes of L and supplies the additional input.

Lemma 78.3 (Weighted variation after polygonal interpolation)**.** *Unconditionally,*

$$\int_{(T,\infty)} w(x)\, d|D^2(L^+)|(x) \ll (1 + (\log T)^{3/4})e^{-(\log T)^{1/4}/2} \tag{78.3}$$

for sufficiently large grid endpoints $T = y_N$. All zero-crossing atoms of the polygonal function are included.

Proof. Neighboring cell lengths are comparable, and $h_n \asymp y_n^{3/4}\exp((\log n)^{1/4}/4)$. The derivative-oscillation estimate (77.2), applied across two adjacent cells, therefore gives

$$|s_n - s_{n-1}| \ll h_n y_n^{-3/2}e^{-(\log y_n)^{1/4}} + \frac{y_n^{-5/6}}{\log y_n}. \tag{78.4}$$

The slopes are averages of $\mathcal{A}'$, so their difference is bounded by its oscillation on the union. The possible prime jumps are included in that oscillation estimate.

Since $D^2L = \sum_n (s_n - s_{n-1})\delta_{y_n}$, multiplying (78.4) by $w(y_n)$ and comparing sums over the cell partition with integrals bounds its weighted total variation beyond T by a constant times

$$\int_T^\infty e^{-(\log x)^{1/4}/2}\frac{dx}{x} + \int_T^\infty \frac{x^{-13/12}e^{(\log x)^{1/4}/2}}{\log x}\,dx. \tag{78.5}$$

For the second term we used $h_n \gg y_n^{3/4}$. Functions in this comparison vary by bounded factors on each cell, because $h_n/y_n \asymp 1/n \to 0$. A finite shift of the initial index only changes the implied constant.

The first integral in (78.5) is $4\int_{(\log T)^{1/4}}^{\infty} t^3 e^{-t/2}dt$, and hence is $O((1+(\log T)^{3/4})e^{-(\log T)^{1/4}/2})$. The second is smaller than this bound for large T.

The global prime number theorem (76.9) also gives, on each sufficiently late cell,

$$|L'(x)| \ll x^{-1/2}(\log x)^{0.515}e^{-0.8\sqrt{\log x}}.$$

The harmless reduction from 0.8274 to 0.8 allows a uniform bound over adjacent endpoints. Thus $w|L'| \to 0$ and

$$\int_T^{\infty} |w'L'|\,dx \ll \int_{\log T}^{\infty} u^{0.515}e^{-0.8\sqrt{u}+u^{1/4}/2}\,du,$$

which is finite and eventually smaller than the asserted tail. The endpoint term $w(T)|s_N|$ satisfies the same bound. Apply Lemma 78.1 to L, whose local zero set consists of finitely many points and possibly zero intervals. This proves (78.3). □

Theorem 78.4 (One-node quadrature with an absolutely summable error)**.** *Let q_n be the affine interpolant of v_n and v_{n+1} on $[y_n, y_{n+1}]$. Unconditionally,*

$$\sum_{n\ge N}\int_{y_n}^{y_{n+1}} |\mathcal{A}(x)^+ - q_n(x)|\frac{dx}{x} \ll (1+(\log N)^{3/4})e^{-(\log N)^{1/4}/\sqrt{2}}. \tag{78.6}$$

Consequently the unchanged target has the exact criterion

$$\boxed{\mathsf{B} \iff \sum_{n\ge 3}\frac{\mathcal{A}\left(n^4 e^{(\log n)^{1/4}}\right)^+}{n} < \infty.} \tag{78.7}$$

The series convergence in (78.7) is not proved.

Proof. On each cell, the chord of L^+ is exactly q_n. The Dirichlet Green-kernel bound therefore gives

$$\int_{y_n}^{y_{n+1}} |L(x)^+ - q_n(x)|\frac{dx}{x} \le \frac{h_n^2}{4y_n}|D^2(L^+)|((y_n, y_{n+1})).$$

The corner at a zero inside the cell is retained in the measure on the right. Atoms at the grid endpoints have zero Green kernel. Furthermore,

$$\frac{h_n^2}{y_n} \ll \sqrt{y_n}\exp\left(\tfrac{1}{2}(\log y_n)^{1/4}\right),$$

because $(\log y_n)^{1/4} \sim \sqrt{2}(\log n)^{1/4}$ and $1/2 < 1/\sqrt{2}$ in the comparison of the extra exponents. More explicitly, $h_n^2/y_n \asymp \sqrt{y_n}e^{(\log n)^{1/4}/2}$, whereas the right-hand weight is $\sqrt{y_n}e^{(\log y_n)^{1/4}/2}$. The weight w is increasing. Summing and applying Lemma 78.3 bounds the total error between L^+ and q_n.

Theorem 77.5 already bounds the absolute error between $\mathcal{A}^+$ and L^+, with the faster exponential coefficient $\sqrt{2}-1/2$ in $(\log N)^{1/4}$. The triangle inequality proves (78.6), since $(\log y_N)^{1/4} \ge \sqrt{2}(\log N)^{1/4}$.

For the exact discrete weights, put

$$\alpha_n = \frac{y_{n+1}}{h_n}\log\frac{y_{n+1}}{y_n} - 1, \qquad \beta_n = 1 - \frac{y_n}{h_n}\log\frac{y_{n+1}}{y_n}.$$

Both numbers are positive and direct integration gives

$$\int_{y_n}^{y_{n+1}} q_n(x)\frac{dx}{x} = \alpha_n v_n + \beta_n v_{n+1}. \tag{78.8}$$

Since $h_n/y_n \sim 4/n$, Taylor expansion gives $\alpha_n \sim 2/n$, $\beta_n \sim 2/n$, and hence $W_n := \beta_{n-1}+\alpha_n \sim 4/n$. Nonnegativity allows exact regrouping, with infinite values permitted:

$$\sum_{n\ge n_0} \int_{y_n}^{y_{n+1}} q_n(x)\frac{dx}{x} = \alpha_{n_0} v_{n_0} + \sum_{n>n_0} W_n v_n.$$

The absolute error in (78.6) makes this sum finite exactly when $I(0)$ is finite. The positive weights $W_n \asymp 1/n$ give (78.7); the finite initial terms permit the fixed starting index 3. □

A finite identity, not merely a convergence comparison. For every $M > n_0$,

$$\int_{y_{n_0}}^{y_M} \mathcal{A}(x)^+ \frac{dx}{x} = \alpha_{n_0} v_{n_0} + \sum_{n=n_0+1}^{M-1} W_n v_n + \beta_{M-1} v_M + R_M,$$

where R_M has a finite limit and its tail obeys (78.6). The terminal term is kept; it need not tend to zero without additional information. Each v_n is determined by the finite prime formula (73.4). Thus a single endpoint value per node replaces the earlier two-endpoint positive-area summand, while both the signed interpolation error and the extra zero-crossing corners have been summed unconditionally.

79 Critical damping, rectified zero modes, and a summable remainder

Retain $u_0 = \log 2$, $b(u) = \mathcal{A}(e^u)$, $a = -b$, and the real-axis continuation of $H(z) = -F(1/2+z)/z$. Write

$$I(c) = \int_{u_0}^{\infty} b(u)^+ e^{-cu}\, du, \qquad \mathcal{I}_c(X) = \int_2^X \mathcal{A}(x)^+ x^{-c-1}\, dx.$$

The value $c = 0$ is the unchanged target. No bound for $I(0)$ is assumed in this section.

Proposition 79.1 (Positive damping removes the one-sided ambiguity)**.** *For every $c > 0$, the following are equivalent:*

$$I(c) < \infty, \qquad \int_{u_0}^{\infty} |a(u)| e^{-cu}\, du < \infty.$$

Whenever they hold,

$$\int_{u_0}^{\infty} |a(u)| e^{-cu}\, du = H(c) + 2I(c), \qquad |H(c+it)| \le H(c) + 2I(c) \quad (t \in \mathbb{R}). \tag{79.1}$$

Moreover, ζ then has no nontrivial zero in the closed half-plane $\Re s \ge 1/2 + c$.

Proof. Only the first implication needs proof. Set $L_\pm(z) = \int b(u)^\pm e^{-zu} du$, initially where the integrals converge. The hypothesis makes L_+ holomorphic on $\Re z > c$ and finite at c. Let $\sigma_- \le 1/2$ be the convergence abscissa of the nonnegative transform L_-. The identity $L_+ - L_- = -H$, initially valid for $\Re z > 1/2$, and the already proved holomorphic continuation of H through every positive real point rule out $\sigma_- > c$ by Landau positivity. Consequently $L_-(c+\varepsilon) = L_+(c+\varepsilon) + H(c+\varepsilon)$ for $\varepsilon > 0$. As $\varepsilon \downarrow 0$, monotone convergence and the continuity of H at c show that $L_-(c)$ is finite. This proves the mass identity. Absolute convergence gives the Fourier bound and continuity on the closed half-plane.

Its holomorphic interior cannot contain a shifted zeta zero: the existing logarithmic-derivative identity for $(sF(s))'$ would have a nonzero residue there. At a boundary zero $\rho = 1/2 + c + i\gamma$ of multiplicity m_ρ, the local branch is

$$H(z) = -\frac{m_\rho}{z(z+1/2)} \log(z - (\rho - 1/2)) + h_\rho(z),$$

with h_ρ holomorphic locally. Approaching horizontally from the right makes this unbounded, contradicting (79.1). The pole of ζ at 1 is canceled in H and is not a nontrivial zero. □

Put

$$\Theta = \sup_{\zeta(\rho)=0,\ 0<\Re\rho<1} \Re\rho, \qquad \delta = \Theta - \tfrac{1}{2} \geq 0.$$

The remaining theorems distinguish $\delta > 0$ from $\delta = 0$; the latter is RH and has already been proved to imply eventual negativity of $\mathcal{A}$. Sums below run over distinct zero locations, with multiplicities m_ρ included explicitly.

Theorem 79.2 (An absolutely integrable critical remainder)**.** *Suppose $\delta > 0$. Define*

$$c_\rho = \frac{m_\rho}{\rho(\rho - 1/2)}, \qquad Z(u) = \sum_\rho c_\rho e^{(\rho-1/2)u}, \qquad B = \sum_\rho |c_\rho| < \infty.$$

The series is real-valued and converges absolutely, uniformly on every compact real u-interval. Unconditionally, under the stated case distinction $\delta > 0$, the remainder

$$R(u) = \mathcal{A}(e^u) + \frac{Z(u)}{u}$$

satisfies

$$R(u) = O_\delta\left(1 + \log u + \frac{e^{\delta u}}{u^2}\right), \qquad K_\delta := \int_{u_0}^{\infty} |R(u)| e^{-\delta u}\, du < \infty. \tag{79.2}$$

In particular, for every $X \geq 2$,

$$\left| \mathcal{I}_\delta(X) - \int_{u_0}^{\log X} \left[-\sum_\rho c_\rho e^{-(\Theta-\Re\rho)u} e^{i\Im\rho u} \right]_+ \frac{du}{u} \right| \leq K_\delta. \tag{79.3}$$

This closes the total critical mass of the lower-order remainder; it does not assert convergence of the rectified zero integral.

Proof. There is no real nontrivial zero, and the classical zero count gives $\sum_\rho m_\rho/(1 + |\Im\rho|^2) < \infty$. Thus $B < \infty$. Since $\Re\rho - 1/2 \leq \delta$, one has $|Z(u)| \leq Be^{\delta u}$ for $u \geq u_0$. The integrated explicit formula used earlier in the note remains valid without RH, with this definition of Z:

$$f_\psi(u) := e^{-u/2}(\psi(e^u) - e^u) = -Z'(u) + r(u), \qquad r(u) = O(e^{-u/2}),$$

as distributions on (u_0, ∞). Indeed, the absolutely convergent series for the integrated explicit formula can first be differentiated in distributions; all subsequent integrations here are over finite intervals.

The already established finite partial-summation identity is

$$\begin{aligned} \mathcal{B}_J(e^U) = &\int_{u_0}^{U} \frac{f_\psi(v)}{v}\, dv + 2\int_{u_0}^{U} \frac{f_\psi(v)}{v^2}\, dv \\ &- 2e^{-U/2} \int_{u_0}^{U} \frac{f_\psi(v) e^{v/2}}{v^2}\, dv + \frac{4}{\log 2}(2^{-1/2} - e^{-U/2}). \end{aligned}$$

Substitute $f_\psi = -Z' + r$ and integrate by parts. The terms $-2Z(U)/U^2$ and $+2Z(U)/U^2$ cancel. Apart from bounded terms and exponentially decaying endpoint terms, the result is

$$\begin{aligned} \mathcal{B}_J(e^U) + \frac{Z(U)}{U} = &-\int_{u_0}^{U} \frac{Z(v)}{v^2}\, dv - 4\int_{u_0}^{U} \frac{Z(v)}{v^3}\, dv \\ &- e^{-U/2} \int_{u_0}^{U} \frac{Z(v) e^{v/2}}{v^2}\, dv + 4e^{-U/2} \int_{u_0}^{U} \frac{Z(v) e^{v/2}}{v^3}\, dv + O(1). \end{aligned}$$

For $\eta > 0$ and $j \geq 2$, $\int_{u_0}^{U} e^{\eta v} v^{-j} dv = O_\eta(1 + e^{\eta U}/U^j)$. Using the exact prime-power drift $\mathcal{A}(e^U) = \mathcal{B}_J(e^U) - \log U - C_{\mathrm{pp}} + O(1/U)$ proves the pointwise bound in (79.2).

For integrability one can also avoid this asymptotic estimate. Tonelli gives, for $j \geq 2$,

$$\int_{u_0}^{\infty} e^{-\delta U} \int_{u_0}^{U} \frac{|Z(v)|}{v^j} dv\, dU \leq \frac{B}{\delta} \int_{u_0}^{\infty} \frac{dv}{v^j} < \infty,$$

and, for the exponentially weighted terms, the same calculation gives $B/(\delta + 1/2)$ in place of B/δ. The bounded terms and $\log U$ are integrable against $e^{-\delta U} dU$. This proves $K_\delta < \infty$. Finally the positive-part map is 1-Lipschitz; applying it to $\mathcal{A}(e^u) = -Z(u)/u + R(u)$ proves (79.3). □

Theorem 79.3 (The supremal line determines the leading critical mass)**.** *Continue to suppose $\delta > 0$, and let*

$$T(u) = -\sum_{\Re\rho=\Theta} c_\rho e^{i\Im\rho u}, \qquad \mu = \operatorname{mean}(T^+) = \tfrac{1}{2} \operatorname{mean}|T|.$$

An empty sum is zero. Then

$$u e^{-\delta u} \mathcal{A}(e^u) = T(u) + o(1), \tag{79.4}$$

and

$$\mathcal{I}_\delta(X) = \mu \log\log X + o(\log\log X), \tag{79.5}$$

$$I(\delta + h) = \mu \log(1/h) + o(\log(1/h)) \qquad (h \downarrow 0). \tag{79.6}$$

The constant μ is positive if and only if there is a zero with real part exactly Θ. More explicitly, when that line is nonempty, put

$$B_* = \sum_{\Re\rho=\Theta} |c_\rho|, \qquad q_* = \sum_{\Re\rho=\Theta} |c_\rho|^2.$$

Then

$$0 < \frac{q_*}{2B_*} \leq \mu \leq \frac{B_*}{2}. \tag{79.7}$$

If the supremum is not attained, the conclusions give $\mathcal{I}_\delta(X) = o(\log\log X)$ and $I(\delta + h) = o(\log(1/h))$; they do not imply $I(\delta) < \infty$.

Proof. Absolute summability and dominated convergence give

$$\left| e^{-\delta u} Z(u) - \sum_{\Re\rho=\Theta} c_\rho e^{i\Im\rho u} \right| \leq \sum_{\Re\rho<\Theta} |c_\rho| e^{-(\Theta - \Re\rho)u} \longrightarrow 0.$$

No separation between the largest and the other real parts is needed. Multiplying (79.2) by $ue^{-\delta u}$ proves (79.4).

The absolutely convergent trigonometric series T is real and uniformly almost periodic, with mean zero and mean square q_*. The continuous positive-part function also has a mean: approximate it uniformly on the bounded range of T by polynomials and then approximate T uniformly by finite trigonometric sums. Consequently $\operatorname{mean} T^+ = \operatorname{mean} T^- = \frac{1}{2} \operatorname{mean}|T|$. Since $|T| \leq B_*$, $T^2 \leq B_*|T|$, which gives (79.7).

Taking positive parts in (79.4) gives $e^{-\delta u}\mathcal{A}(e^u)^+ = (T(u)^+ + o(1))/u$. If a bounded function has Cesàro mean μ, partial summation gives its harmonic integral $\mu \log U + o(\log U)$; its exponentially damped harmonic integral is $\mu \log(1/h) + o(\log(1/h))$. These facts follow directly by writing its primitive as $\mu U + o(U)$ and integrating by parts. The $o(1)/u$ remainder has the same respective $o(\log U)$ and $o(\log(1/h))$ bounds. This proves both asymptotic formulas. All $I(\delta + h)$ are finite by Theorem 79.2 and $|Z(u)| \leq Be^{\delta u}$. □

Corollary 79.4 (Persistent growth when the supremum is attained)**.** *For $\delta > 0$ define*

$$K(u) = \int_0^\infty e^{-\delta t} T(u-t)^+ dt.$$

Then

$$\mathcal{I}_0(X) = \frac{X^\delta}{\log X}\{K(\log X) + o(1)\}.$$

If the supremal line contains a zero, K is uniformly almost periodic and bounded above and below by positive constants. In that case

$$\mathcal{I}_0(X) \asymp \frac{X^\delta}{\log X} \qquad (X \longrightarrow \infty).$$

If the supremum is not attained, $K = 0$ and the displayed profile instead says $\mathcal{I}_0(X) = o(X^\delta/\log X)$.

Proof. Use (79.4) in the integral defining $\mathcal{I}_0(e^U)$, multiply by $Ue^{-\delta U}$, and put $t = U - u$. On each bounded t-interval the factor $U/(U-t)$ tends uniformly to one and the scalar remainder tends uniformly to zero. The portion $u \geq U/2$ is dominated by a constant times $e^{-\delta t}$; the portion $u < U/2$ tends to zero exponentially, using the bound $\mathcal{A}(e^u)^+ \ll_\delta 1 + e^{\delta u}/u$. This proves the profile by a finite-interval and tail argument. Convolution with an integrable kernel preserves uniform almost periodicity. If T is nonzero, its mean is zero and its positive part is positive on some interval. Relatively dense almost periods supply translates of such an interval with a fixed positive height in every sufficiently long interval. Integrating over the most recent such interval proves a positive uniform lower bound for K; the upper bound is B_*/δ. □

Proposition 79.5 (A sufficient logarithmic separation condition)**.** *Suppose $\delta > 0$, the supremum Θ is not attained, and*

$$\sum_\rho |c_\rho| \log^+ \frac{1}{\Theta - \Re\rho} < \infty. \tag{79.8}$$

Then $I(\delta) < \infty$. More generally, for every $c > \delta$,

$$I(c) \leq K_\delta + \sum_\rho |c_\rho| E_1\big((c + 1/2 - \Re\rho)u_0\big), \qquad E_1(y) = \int_y^\infty e^{-v}\frac{dv}{v}. \tag{79.9}$$

The same inequality holds at $c = \delta$ if its right side is finite.

Proof. By positive-part Lipschitzness and Tonelli,

$$I(c) \leq \int |R(u)| e^{-cu} du + \sum_\rho |c_\rho| \int_{u_0}^\infty e^{-(c+1/2-\Re\rho)u}\frac{du}{u}.$$

For $c \geq \delta$ the first term is at most K_δ. The integral in each summand is precisely the stated E_1 value. For $0 < y \leq 1$, $E_1(y) \leq \log(1/y) + E_1(1)$; for $y \geq 1$, $E_1(y) \leq E_1(1)$. Absolute summability of the c_ρ and (79.8) therefore make the endpoint sum finite. No assertion that (79.8) holds at the unknown Θ is made here. □

Corollary 79.6 (The exact growth exponent of the undamped area)**.** *Unconditionally,*

$$\limsup_{X\to\infty} \frac{\log(1 + \mathcal{I}_0(X))}{\log X} = \Theta - \tfrac{1}{2}.$$

If RH is false, the abscissa of convergence of $I(c)$ is exactly $\delta > 0$, with convergence at $c = \delta$ left to the preceding endpoint distinction. If RH holds, $\mathcal{A}^+$ is compactly supported and $I(c)$ is finite for every real c; its abscissa is then $-\infty$, not zero.

Proof. In the case $\delta > 0$, Theorem 79.2 gives convergence for $c > \delta$. For $0 < c < \delta$, convergence would contradict Proposition 79.1, since the definition of Θ supplies a zero with real part greater than $1/2 + c$. At $c \le 0$, convergence would imply $I(0) < \infty$ and hence RH by the criterion already proved. Thus the abscissa is δ.

For the growth formula, put $J(U) = \mathcal{I}_0(e^U)$. Convergence at $c > \delta$ gives $J(U) \le e^{cU} I(c)$. If instead $\limsup U^{-1} \log(1 + J(U)) < \delta$, choose a positive $c < \delta$ above that limsup; Stieltjes integration by parts then makes $\int e^{-cu} dJ(u) = I(c)$ finite, a contradiction. Under RH, $J(U)$ is eventually constant, so the stated limsup is zero. □

What has and has not been closed. The finite critical mass of the integrated explicit-formula remainder is proved, as are the exact leading critical mass, the undamped profile when the supremum is attained, and the logarithmic separation sufficient condition. The unknown arithmetic content is the value of Θ and, at an unattained positive supremum, the approach of zero real parts to that supremum. These results do not prove $\Theta = 1/2$ or the convergence of $I(0)$. They use established explicit-formula and almost-periodic principles; no historical priority or nonduplication claim is attached to them.

80 Instability of zero-tilt control under integer staircase perturbations

The positive-jump geometry and the tilted flux estimates do not, by themselves, force a finite zero-tilt budget. This can be proved within a much narrower class than arbitrary smooth counterexamples: integer counting staircases with unit jumps, logarithmic gaps, the same reference integral, and the same $-\log\log x$ drift. The result identifies a limitation of those inputs, not a counterexample to RH. The additional staircases constructed below are not the actual primes.

For a nondecreasing integer counting function N, define

$$P_N(x) = N(x) - \mathrm{Li}_2(x), \qquad \mathcal{A}_N(x) = \int_2^x P_N(t) t^{-3/2}\, dt, \qquad a_N(u) = -\mathcal{A}_N(e^u),$$

with $a_N = 0$ below $\log 2$. Let

$$I_N(c) = \int_2^\infty \mathcal{A}_N(x)^+ x^{-c-1}\, dx$$

in the extended sense. Write $D_N(m, r)$ for the corresponding Poisson negative budget.

Theorem 80.1 (A prescribed tilt barrier with arbitrarily small critical discrepancy)**.** *Fix $0 < \delta < 1/2$, $\gamma > 0$, and $\eta > 0$. For every finite cutoff X_0 and every $\varepsilon > 0$, there is a counting function N with the following properties.*

1. *Its jumps occur at distinct integers and have size one, and $N(x) = \pi(x)$ for $2 \le x \le X_0$.*
2. *As $x \to \infty$, $N(x) \sim \mathrm{Li}_2(x)$. Its sufficiently late consecutive jump locations have gaps $O(\log x)$. Uniformly for sufficiently large a and $2 \le h \le a$,*

$$N(a + h) - N(a) \ll h/\log h.$$

It also satisfies the following uniform short-interval prime-number-theorem estimate:

$$N(x + h) - N(x) = \frac{h}{\log x} + O\Big(h \exp\big(-(\log x)^{1/4}\big)\Big) \quad (x^{2/3} \le h \le x^{0.99}). \tag{80.1}$$

The error is uniform in h, for all sufficiently large x; its constant and starting point may depend on the constructed model. Indeed, the same assertion holds on every fixed range $x^{a_0} \le h \le x^{b_0}$ with $0 < a_0 < b_0 < 1$.

3. *It has the exact between-jump geometry used in the earlier interpolation arguments. In particular, on each gap, with* $y = x^{-1/2}$,

$$\frac{d^2}{dy^2}\mathcal{A}_N(y^{-2}) = -\frac{4y^{-3}}{\log(y^{-2})} < 0.$$

The distribution $\mathcal{A}_N''$ *has positive atoms of size* $q^{-3/2}$ *at its jump locations* q.

4. *The entire previously controlled tilted region is close to that of the actual primes:*

$$\int_0^\infty |a_N(u) - a(u)|e^{-u/2}\,du < \varepsilon, \tag{80.2}$$

and consequently

$$|D_N(m,r) - D(m,r)| < \varepsilon \quad (m > 1/2,\ 0 \le r < 1,\ m(1-r) \ge 1/2). \tag{80.3}$$

5. *Nevertheless, for* $c \ge 0$,

$$I_N(c) < \infty \quad \Longleftrightarrow \quad c > \delta, \tag{80.4}$$

and the divergence has the precise leading term

$$I_N(c) = \frac{\eta}{\pi}\log\frac{1}{c-\delta} + O(1) \qquad (c \downarrow \delta). \tag{80.5}$$

At the critical exponent itself,

$$\int_2^X \mathcal{A}_N(x)^+ x^{-\delta-1}\,dx = \frac{\eta}{\pi}\log\log X + O(1). \tag{80.6}$$

In addition, N *can be chosen so that* $I_N(1/2) < 180$. *All constants in the* $O(1)$ *terms may depend on the constructed staircase and on* δ, γ, η.

Proof. Put $\beta = 1/2 + \delta < 1$, and prescribe the primitive

$$g(x) = -\log\log x + \eta\frac{x^\delta\cos(\gamma\log x)}{\log x}, \qquad E(x) = x^{3/2}g'(x).$$

Explicitly,

$$E(x) = -\frac{\sqrt{x}}{\log x} + \eta\Re\left\{x^{\beta+i\gamma}\left(\frac{\delta+i\gamma}{\log x} - \frac{1}{(\log x)^2}\right)\right\}. \tag{80.7}$$

Hence $E'(x) = O_{\delta,\gamma,\eta}(x^{\beta-1}/\log x)$. For a sufficiently large integer $T \ge X_0$, define, for $x \ge T$,

$$F_T(x) = \pi(T) + \mathrm{Li}_2(x) - \mathrm{Li}_2(T) + E(x) - E(T).$$

Choose T so large that throughout $x \ge T$,

$$\frac{1}{2\log x} \le F_T'(x) \le \frac{3}{2\log x} < 1. \tag{80.8}$$

Now take

$$N(x) = \begin{cases} \pi(x), & 2 \le x \le T, \\ \lfloor F_T(\lfloor x\rfloor)\rfloor, & x > T. \end{cases}$$

The definitions agree at T, since $F_T(T) = \pi(T)$. For consecutive integers $n-1, n \ge T$, the increase of F_T lies strictly between zero and one. Thus the integer differences of its floors are either zero or one. This proves the unit-jump and prefix assertions. Furthermore, for $x \ge T$,

$$P_N(x) = E(x) + C_T + R_T(x), \qquad C_T = P(T) - E(T), \qquad -2 < R_T(x) \le 0. \tag{80.9}$$

The bounded rounding error follows from (80.8). As $E(x) = O(x^{\beta}/\log x)$ with $\beta < 1$, this gives $N(x) \sim \mathrm{Li}_2(x)$.

The lower bound in (80.8) implies that an interval of length $12\log x$, with x sufficiently large, increases F_T by more than two. Rounding cannot erase all of that increase, so there is a jump in every such interval. The upper bound gives

$$N(a+h) - N(a) \le \frac{3h}{2\log a} + 2 \ll \frac{h}{\log h} \qquad (2 \le h \le a).$$

These prove the gap and local-count assertions. To verify the stronger short-interval assertion directly, subtract (80.9) at $x+h$ and x. For $1 \le h \le x$, the mean value theorem gives

$$\left| N(x+h) - N(x) - \frac{h}{\log x} \right| \ll_{\delta,\gamma,\eta} \frac{h^2}{x(\log x)^2} + \frac{hx^{\delta-1/2}}{\log x} + 1. \tag{80.10}$$

Here the first term is the variation of $1/\log t$, the second comes from E', and the bounded rounding discrepancy contributes the last term. On $x^{a_0} \le h \le x^{b_0}$, division by h bounds the right side by a constant times

$$\frac{x^{b_0-1}}{(\log x)^2} + \frac{x^{\delta-1/2}}{\log x} + x^{-a_0}.$$

Since $a_0 > 0$, $1 - b_0 > 0$, and $1/2 - \delta > 0$, each term is $O(\exp(-(\log x)^{1/4}))$. This proves (80.1), including its claimed uniformity.

Distributional differentiation gives exactly

$$\mathcal{A}_N'' = \sum_q q^{-3/2}\delta_q - \left(\frac{x^{-3/2}}{\log x} + \frac{3}{2} P_N(x) x^{-5/2} \right) dx.$$

On a gap $P_N'(x) = -1/\log x$, yielding the displayed reciprocal-square-root curvature by the chain rule. Since $N(x) \sim x/\log x$, the earlier sufficiently-large one-sided curvature bound holds with the same admissible constant $59/20$.

We next quantify the critical discrepancy. The primitives agree up to T, and Tonelli gives

$$\begin{aligned} \int_0^{\infty} |a_N - a| e^{-u/2}\, du &= \int_T^{\infty} |\mathcal{A}_N(x) - \mathcal{A}(x)| x^{-3/2}\, dx \\ &\le 2\int_T^{\infty} \frac{|N(t) - \pi(t)|}{t^2}\, dt \\ &\le 2\left\{ \frac{|P(T)| + |E(T)| + 2}{T} + \int_T^{\infty} \frac{|E(t)| + |P(t)|}{t^2}\, dt \right\}. \end{aligned} \tag{80.11}$$

The last expression tends to zero: $P(T)/T \to 0$ by the prime number theorem, the absolute P-tail is integrable by the already used Johnston–Yang estimate, and $\beta < 1$ handles E. Thus T can be increased until (80.2) holds. For the Poisson kernels, the negative-part map is 1-Lipschitz, so

$$\begin{aligned} |D_N(m,r) - D(m,r)| &\le \sum_{n \ge 0} r^n |d_n^N(m) - d_n(m)| \\ &\le \int_0^{\infty} |a_N(u) - a(u)| e^{-m(1-r)u}\, du \le \int_0^{\infty} |a_N - a| e^{-u/2}\, du. \end{aligned}$$

All sums here converge absolutely in the stated region. This proves (80.3). The same Lipschitz argument gives $|I_N(1/2) - I(1/2)| < \varepsilon$. The preceding explicit proof actually gave $I(1/2) < 177.7237076$. Requiring the critical discrepancy also to be less than one therefore ensures $I_N(1/2) < 180$.

Finally, integrate (80.9). The identity $E(x)x^{-3/2} = g'(x)$ and absolute convergence of the rounding integral give a finite constant K_T such that

$$\mathcal{A}_N(x) = -\log\log x + K_T + \eta\frac{x^\delta\cos(\gamma\log x)}{\log x} + O_T(x^{-1/2}). \tag{80.12}$$

For example, the error bound follows by writing the constant C_T contribution as $2C_T(T^{-1/2}-x^{-1/2})$ and using $|\int_x^\infty R_T(t)t^{-3/2}dt| \le 4x^{-1/2}$.

Write $U_0 = \log T$ and $v = c - \delta$. For $c \ge \delta/2$, the 1-Lipschitz property of positive part and (80.12) imply

$$I_N(c) = \eta\int_{U_0}^\infty e^{-vu}\frac{\cos(\gamma u)^+}{u}\,du + O(1) \qquad (c > \delta), \tag{80.13}$$

with a remainder bounded uniformly for c near δ. The periodic function $\cos(\gamma u)^+$ has mean $1/\pi$; its mean-zero part has a bounded primitive. Integration by parts against e^{-vu}/u therefore gives, uniformly for $v > 0$ near zero,

$$\int_{U_0}^\infty e^{-vu}\frac{\cos(\gamma u)^+}{u}\,du = \frac{1}{\pi}E_1(vU_0) + O(1) = \frac{1}{\pi}\log\frac{1}{v} + O(1).$$

This proves (80.5). The identical bounded-primitive argument at $v = 0$ and a finite upper endpoint proves (80.6). Thus $I_N(\delta) = \infty$; monotonicity in c gives divergence for $c < \delta$. For $c > \delta$, (80.12) gives absolute integrability directly. This proves (80.4). □

A fully explicit discrepancy estimate. The existence statement above has a numerical majorant. Put $L = \log T$, $\alpha = 0.8274$, and $C_E = \eta(\sqrt{\delta^2+\gamma^2}+1)$. For $T \ge e$,

$$|E(T)|/T + \int_T^\infty |E(t)|t^{-2}\,dt \le \frac{3}{\sqrt{T}L} + C_E\left(1 + \frac{1}{1/2-\delta}\right)\frac{T^{\delta-1/2}}{L}.$$

Using Johnston–Yang [27, Corollary 1.3] and $|\operatorname{li}(2)| < 2$ in (80.11), the critical discrepancy is at most

$$2\Big\{9.59L^{0.515}e^{-\alpha\sqrt{L}} + 19.18\alpha^{-3.03}\Gamma(3.03,\alpha\sqrt{L}) + \frac{6}{T} + \frac{3}{\sqrt{T}L} + C_E\left(1 + \frac{1}{1/2-\delta}\right)\frac{T^{\delta-1/2}}{L}\Big\}.$$

Every displayed term tends to zero. No numerical verification of the prime error beyond the finite prefix is required to bound this tail.

The real-axis continuation also survives. The comparison model is not excluded merely by requiring continuation of the signed Laplace transform through positive real points. In fact, (80.12) gives, initially for $\Re z > \delta$,

$$H_N(z) = \frac{e^{-zU_0}(\log U_0 - K_T) + E_1(zU_0)}{z} - \frac{\eta}{2}\left\{E_1((z-\delta+i\gamma)U_0) + E_1((z-\delta-i\gamma)U_0)\right\} + Q_T(z),$$

where $U_0 = \log T$ and Q_T is holomorphic on $\Re z > -1/2$. Indeed, Q_T is the finite-prefix transform plus the transform of the $O_T(e^{-u/2})$ remainder. The displayed reference term follows by integrating $\log u$ by parts. Using the principal continuation of E_1, the arguments $z - \delta \pm i\gamma$ avoid its cut when z is positive real. Thus H_N continues holomorphically through every such real point. Near zero it even has the familiar leading form

$$H_N(z) = -\frac{\log z + \gamma_E + K_T}{z} + O(1) \qquad (z \to 0,\ \Re z > 0),$$

where γ_E is Euler's constant; the local logarithm is the principal one. These are statements about the comparison transform, not identities with the actual zeta function.

The flux implication and its exact limitation. For each constructed staircase and $c > \delta$, all polynomial moments of $a_N e^{-cu}$ are absolutely integrable. Therefore the earlier Kato argument applies. In this paragraph its model moment family is naturally defined for every $m > \delta$, since $a_N(u) = O_T(1 + e^{\delta u}/u)$; thus the characteristic starting at $m = c$ lies entirely inside the model's convergence domain even when $c < 1/2$. Its total characteristic flux is $I_N(c) - H_N(c)^-$, where $H_N(c) = \int a_N e^{-cu}$. The signed transform $H_N(c)$ has a finite limit as $c \downarrow \delta$: the oscillatory integral $\int e^{-(c-\delta)u} \cos(\gamma u) du/u$ converges by Dirichlet's test, and the remaining terms are absolutely integrable there. Hence this total flux also equals

$$\frac{\eta}{\pi} \log \frac{1}{c - \delta} + O(1).$$

The model thus retains all the geometry and every tilted budget in the controlled region to arbitrary accuracy while developing a logarithmic barrier at a freely prescribed lower tilt. It also retains the short-interval estimate (80.1) and the positive-real-axis continuation of its signed transform. A transfer from these inputs alone to $c = 0$ is therefore impossible. This does not rule out a proof for the actual primes: such a proof must use additional arithmetic structure not shared by these counting staircases. In particular, the actual zeta function and its functional equation have not been assigned to the comparison model.

81 Mean-square zero estimates and the duration of positive excursions

This section combines the classical mean-square explicit-formula argument with disjoint-interval geometry. It uses the actual supremum $\Theta = \sup_\rho \Re\rho$ and $\delta = \Theta - 1/2 \in [0, 1/2]$; neither RH nor attainment of the supremum is assumed. The external mean-square argument is the one in Montgomery–Vaughan [46, Theorem 13.5 and its proof]. We record explicitly why its upper bound extends from the line $1/2$ to the actual Θ.

Lemma 81.1 (A mean-square derivative bound using the actual zero supremum)**.** *For every sufficiently large X,*

$$\int_X^{2X} |\psi(x) - x|^2 \, dx \ll X^{2\Theta+1}, \tag{81.1}$$

$$\int_X^{2X} |P(x)|^2 \, dx \ll \frac{X^{2\Theta+1}}{\log^2 X}, \tag{81.2}$$

$$\int_X^{2X} |\mathcal{A}'(x)|^2 \, dx \ll \frac{X^{2\delta-1}}{\log^2 X}. \tag{81.3}$$

The derivative in the last integral is the almost-everywhere derivative of the locally absolutely continuous function $\mathcal{A}$.

Proof. The truncated explicit formula [46, Theorem 12.5 and proof of Theorem 13.5], with truncation height $T = X$, gives for almost every $x \in [X, 2X]$

$$\psi(x) - x = -\sum_{|\gamma| \le X} \frac{x^\rho}{\rho} + R_X(x), \qquad \int_X^{2X} |R_X(x)|^2 dx \ll X \log^4 X.$$

The remainder estimate is the unconditional part of that classical mean-square argument; RH is used there only later to replace β by $1/2$ in the zero sum. Exceptional prime-power points have measure zero and cause no problem for the square integral. The usual zero count in unit intervals is $O(\log(2 + |t|))$. It implies

$$\sum_{\rho,\rho'} \frac{1}{|\rho\rho'|(1 + |\gamma - \gamma'|)} < \infty, \tag{81.4}$$

where zero multiplicities are included in the sums. Indeed, for fixed $\gamma \geq 1$, split γ' into $|\gamma'| \leq \gamma/2$, $\gamma/2 < \gamma' < 3\gamma/2$, $\gamma' \geq 3\gamma/2$, and $\gamma' \leq -\gamma/2$. Summation over unit intervals bounds the inner sum by $O(\log^2(2+\gamma)/\gamma)$. The remaining outer sum is bounded by a constant times $\sum_{n\geq 1} \log^3(2+n)/n^2$. Negative ordinates follow by conjugation, and the finitely many low ordinates are harmless.

For $\rho = \beta + i\gamma$ and $\rho' = \beta' + i\gamma'$, direct integration gives

$$\left| \int_X^{2X} x^{\rho+\overline{\rho'}}\, dx \right| \ll \frac{X^{2\Theta+1}}{1+|\gamma-\gamma'|}.$$

Here $1 \leq \beta + \beta' + 1 \leq 3$, so the denominator bound is uniform. Expanding the finite zero sum, using (81.4), and bounding the error square by $O(X \log^4 X)$ proves (81.1), since $\Theta \geq 1/2$.

Chebyshev's elementary bound gives $\psi(x) - \vartheta(x) = O(\sqrt{x})$. Consequently $D_\vartheta(x) = \vartheta(x) - x$ satisfies the same mean-square bound. Finite partial summation gives exactly

$$P(x) = \frac{D_\vartheta(x)}{\log x} + \int_2^x \frac{D_\vartheta(t)}{t \log^2 t}\, dt + \frac{2}{\log 2}.$$

On a dyadic interval $[Y, 2Y]$, Cauchy–Schwarz bounds the absolute integral of its middle integrand by

$$\left(\int_Y^{2Y} |D_\vartheta(t)|^2 dt \right)^{1/2} \left(\int_Y^{2Y} \frac{dt}{t^2 \log^4 t} \right)^{1/2} \ll \frac{Y^\Theta}{\log^2 Y}.$$

Summing the geometric dyadic scales yields $\int_2^x D_\vartheta(t)/(t \log^2 t)\, dt = O(x^\Theta/\log^2 x)$. This proves (81.2). Finally, $\mathcal{A}'(x) = P(x)x^{-3/2}$ almost everywhere gives (81.3). □

For a positive component $E = (a, b)$ of $\mathcal{A}$, write $h(E) = b - a$ and

$$J_c(E) = \int_a^b \mathcal{A}(x) x^{-c-1}\, dx \qquad (c \geq 0).$$

Every such component is bounded, and only finitely many start in a fixed bounded interval, by Lemma 77.2.

Lemma 81.2 (Disjoint excursions convert square energy into area)**.** *Let $\mathscr{E}$ be any family of positive components with left endpoints in $[Y, 2Y]$, lengths at most H, and $H \leq 2Y$. Then, for every fixed $c \geq 0$,*

$$\sum_{E \in \mathscr{E}} J_c(E) \ll_c \frac{H Y^{\delta-c-1}}{\log Y}. \tag{81.5}$$

Proof. For $E = (a, b)$, the two endpoint values are zero. Integration by parts with the midpoint $m = (a + b)/2$ and Cauchy–Schwarz give

$$\int_a^b \mathcal{A}(x)\, dx = -\int_a^b (x - m)\mathcal{A}'(x)\, dx \leq \frac{h(E)^{3/2}}{\sqrt{12}} \left(\int_E |\mathcal{A}'(x)|^2 dx \right)^{1/2}.$$

All the components lie in $[Y, 4Y]$ and are disjoint. Thus $\sum_E h(E) \leq 3Y$ and $\sum_E h(E)^3 \leq 3H^2 Y$. Since $x^{-c-1} \leq Y^{-c-1}$, a second application of Cauchy–Schwarz and Lemma 81.1 give

$$\begin{aligned} \sum_E J_c(E) &\leq \frac{Y^{-c-1}}{\sqrt{12}} \left(\sum_E h(E)^3 \right)^{1/2} \left(\int_Y^{4Y} |\mathcal{A}'(x)|^2 dx \right)^{1/2} \\ &\ll \frac{H Y^{\delta-c-1}}{\log Y}. \end{aligned}$$

This argument includes components crossing $2Y$; none is lost at the dyadic boundary. □

Theorem 81.3 (A Dini family with finite critical positive mass)**.** *Let $L : [T_0, \infty) \to [1, \infty)$ be nondecreasing and suppose*

$$\int_{T_0}^{\infty} \frac{dx}{x \log x \, L(x)} < \infty. \tag{81.6}$$

Then, without assuming attainment of Θ,

$$\sum_{\substack{E=(a,b) \text{ positive component} \\ a \geq T_0,\ h(E) \leq a/L(a)}} J_\delta(E) < \infty. \tag{81.7}$$

More quantitatively, the contribution of such components with $a \geq T$ is

$$\ll \frac{1}{\log T \, L(T)} + \int_T^{\infty} \frac{dx}{x \log x \, L(x)}. \tag{81.8}$$

For every fixed $\varepsilon > 0$, this applies in particular to

$$h(E) \leq \frac{a}{(\log \log a)^{1+\varepsilon}},$$

and the corresponding tail is $O_\varepsilon((\log \log T)^{-\varepsilon})$.

Proof. For left endpoints in $[Y, 2Y]$, monotonicity gives $h(E) \leq 2Y/L(Y) \leq 2Y$. Lemma 81.2 with $c = \delta$ bounds their total by $O(1/(\log Y \, L(Y)))$. Sum over $Y = 2^j T$. The summand is nonincreasing in j, and comparison with the integral in (81.6) proves both assertions. For the example, substitute $u = \log \log x$ in that integral. □

Remark 81.4. The same proof gives finite *undamped* mass for the Θ-dependent family $h(E) \leq a^{1-\delta}/L(a)$ under (81.6). This is a parameterized consequence, not a proof that the unknown δ is zero. The previous Guth–Maynard excursion theorem remains useful: its unconditional undamped threshold is larger in parts of the possible range of δ.

81.1 Every fixed power saving in duration gives an exact criterion

Fix $0 < q < 1$. Let $\mathscr{E}_{\leq q}$ consist of positive components $E = (a, b)$ with $h(E) \leq a^q$, and let $\mathscr{E}_{>q}$ be its complement. Define truncated masses

$$S_q(X) = \sum_{E \in \mathscr{E}_{\leq q}} \int_{E \cap [2,X]} \mathcal{A}(x) \frac{dx}{x}, \qquad L_q(X) = \sum_{E \in \mathscr{E}_{>q}} \int_{E \cap [2,X]} \mathcal{A}(x) \frac{dx}{x}.$$

These definitions count only the portion up to X, even when a component extends farther. Hence $\mathcal{I}_0(X) = S_q(X) + L_q(X)$ exactly.

Theorem 81.5 (Sublinear durations have a strict growth gap)**.** *Put $\alpha = \delta + q - 1$. Then*

$$S_q(X) \ll_{q,\delta} \begin{cases} 1, & \alpha < 0, \\ 1 + \log \log X, & \alpha = 0, \\ 1 + X^\alpha / \log X, & \alpha > 0. \end{cases} \tag{81.9}$$

In the first case the tail mass of full components starting at $a \geq T$ is $O_{q,\delta}(T^\alpha / \log T)$. If RH is false, so that $\delta > 0$, then

$$\limsup_{X \to \infty} \frac{\log(1 + L_q(X))}{\log X} = \delta. \tag{81.10}$$

Thus all of the full positive growth exponent is already present among components longer than a^q, for every fixed $q < 1$. No maximal zero is assumed to exist.

Proof. Components with left endpoints in $[Y, 2Y]$ have $h(E) \le (2Y)^q$. Lemma 81.2, with $c = 0$, bounds their full mass by $O_{q,\delta}(Y^{\delta+q-1}/\log Y)$. The components starting in a fixed initial interval have finite total mass. Sum the remaining dyadic bounds: if $\alpha < 0$ this is a convergent geometric tail; if $\alpha = 0$ it is a harmonic sum in the dyadic index; if $\alpha > 0$ it is bounded by its last geometric scales. This proves (81.9) and the tail assertion. Counting the full mass of components with $a \le X$ only increases the truncated $S_q(X)$, so a component crossing X does not affect the upper bound.

By Corollary 79.6, the growth exponent of $\mathcal{I}_0(X)$ is δ. The bound just proved gives a growth exponent for S_q at most $\max(0, \delta + q - 1) < \delta$ when $\delta > 0$. Since S_q and L_q are nonnegative and add exactly to $\mathcal{I}_0$, the exponent of their sum is the maximum of their individual limsup exponents. The exponent of L_q must therefore equal δ, proving (81.10). □

Corollary 81.6 (A pure duration criterion for the unchanged target)**.** *For any fixed $q \in (0, 1)$, each of the following is equivalent to RH, and hence to the unchanged assertion* B*:*

1. *Only finitely many positive components (a, b) satisfy $b - a > a^q$.*
2. *All sufficiently late positive components satisfy $b - a \le a^q$.*
3. $\displaystyle\sum_{E \in \mathscr{E}_{>q}} J_0(E) < \infty.$

In particular, one may fix $q = 99/100$ once and for all.

Proof. Under RH, the previously proved asymptotic $\mathcal{A}(X) = -\log\log X + C_* + O(1/\log X)$ makes $\mathcal{A}$ eventually negative, so there are only finitely many positive components in total. Conversely, if RH is false, Theorem 81.5 gives $\delta > 0$ as the growth exponent of L_q. Thus L_q is unbounded and its total mass is infinite. Since every individual component has finite endpoints and finite mass, infinitely many components must belong to $\mathscr{E}_{>q}$. This contradicts any of the three conditions. □

The remaining unproved assertion. The corollary proves an exact duration-only reformulation; it does not prove the eventual bound $b - a \le a^{99/100}$ for actual positive components. The Dini theorem closes a much larger family at the unknown critical damping $c = \delta$, not automatically at the undamped target $c = 0$. The distinction is necessary. No historical-priority claim is made for these consequences of the classical mean-square method.

82 Every fixed polynomial mesh retains the full obstruction

The summable interpolation theorems above give additive control of the positive mass on a particular quartic-scale mesh. A different argument retains the exact criterion on every fixed polynomial mesh, even when the total interpolation error is not known to be finite. The crucial point is that the growth exponent controlling that error is determined by the same zeta zeros as the positive mass itself. Their exponents differ by a fixed amount supplied by the mesh. No conclusion about the convergence of the actual sampled series is assumed.

Write

$$\Theta = \sup_{\rho} \Re\rho, \qquad \delta = \Theta - \tfrac{1}{2}, \qquad \mathcal{I}_0(X) = \int_2^X \mathcal{A}(x)^+ \frac{dx}{x}.$$

Here and below the supremum is over nontrivial zeta zeros, with no assumption that it is attained.

Lemma 82.1 (A derivative bound parameterized by the zero strip)**.** *Let $1/2 \le B \le 1$, and suppose all nontrivial zeta zeros have real part at most B. Then*

$$P(x) \ll x^B \log x, \qquad |\mathcal{A}'_+(x)| \ll x^{B-3/2} \log x \quad (x \ge 3). \tag{82.1}$$

In particular, taking $B = \Theta$ gives

$$|\mathcal{A}'_+(x)| \ll x^{\delta-1} \log x.$$

Proof. For completeness, use the classical truncated explicit formula in the form of Tao [47, Theorem 21]. With its fixed parameter $1/4$ and truncation height $T = x$, it gives

$$\psi(x) - x = -\sum_{\substack{\Re\rho \geq 1/4 \\ |\Im\rho| \leq x}} \frac{x^\rho}{\rho} + O(x^{1/4} \log^2 x + \log^2 x).$$

The standard zero count $N(T) = O(T \log(T+2))$, or the unit-interval zero count in the same source, implies $\sum_{|\Im\rho| \leq x} |\rho|^{-1} \ll \log^2 x$. The assumed strip therefore yields

$$\psi(x) - x \ll x^B \log^2 x$$

in absolute value. Higher prime powers contribute $\psi(x) - \vartheta(x) = O(\sqrt{x} \log^2 x)$, absorbed because $B \geq 1/2$. Partial summation in

$$\pi(x) = \frac{\vartheta(x)}{\log x} + \int_2^x \frac{\vartheta(t)}{t \log^2 t}\, dt$$

then gives $P(x) = O(x^B \log x)$; the constant arising from the Li_2 normalization is harmless. Finally, $\mathcal{A}'_+(x) = P(x)x^{-3/2}$. □

Let x_n be a strictly increasing sequence tending to infinity, with $x_{n_0} \geq 3$. Suppose that for some fixed $\eta > 0$ and $C < \infty$,

$$h_n := x_{n+1} - x_n \leq C x_n^{1-\eta} \qquad (n \geq n_0). \tag{82.2}$$

After removing finitely many nodes we may suppose $x_{n+1} \leq 2x_n$. Define

$$v_n = \mathcal{A}(x_n)^+, \qquad \lambda_n = \log(x_{n+1}/x_n) > 0.$$

For $X \geq x_{n_0}$, let m be the unique integer with $x_m \leq X < x_{m+1}$, and retain the partial terminal cell:

$$S_{\mathcal{X}}(X) = \sum_{n=n_0}^{m-1} \lambda_n v_n + v_m \log(X/x_m). \tag{82.3}$$

The terminal term is zero when $X = x_m$. This nonnegative function is continuous and nondecreasing in X.

Theorem 82.2 (Growth-preserving sampling on power-dense meshes)**.** *For a mesh satisfying (82.2), the exact identity*

$$\int_{x_{n_0}}^X \mathcal{A}(x)^+ \frac{dx}{x} = S_{\mathcal{X}}(X) + E_{\mathcal{X}}(X)$$

has a remainder whose total absolute cell errors on $[Y, 2Y]$ are

$$\ll \frac{Y^{\delta-\eta}}{\log Y}. \tag{82.4}$$

The estimate includes cells crossing a dyadic boundary, by enlarging the containing block by a fixed factor. Consequently,

$$|E_{\mathcal{X}}(X)| \ll \begin{cases} 1, & \delta < \eta, \\ 1 + \log\log X, & \delta = \eta, \\ 1 + X^{\delta-\eta}/\log X, & \delta > \eta. \end{cases} \tag{82.5}$$

When $\delta < \eta$, the remainder converges, with total absolute cell error after X bounded by $O(X^{\delta-\eta}/\log X)$. For every such mesh, unconditionally,

$$\limsup_{X\to\infty} \frac{\log(1+S_{\mathcal{X}}(X))}{\log X} = \delta. \tag{82.6}$$

In particular the following are equivalent:

1. *RH;*
2. *$\sum_{n\ge n_0} \lambda_n \mathcal{A}(x_n)^+ < \infty$;*
3. *$\mathcal{A}(x_n) \le 0$ for all sufficiently large n;*
4. *for every $\varepsilon > 0$, $\mathcal{A}(x_n) \le C_\varepsilon x_n^\varepsilon$ for all sufficiently large n.*

These equivalences preserve the unchanged target B *through the previously proved equivalence* $\mathsf{B} \iff \mathrm{RH}$.

Proof. The positive-part map is 1-Lipschitz. Absolute continuity of $\mathcal{A}$ therefore gives, for every full or partial cell $[x_n, z] \subseteq [x_n, x_{n+1}]$,

$$\left| \int_{x_n}^{z} \mathcal{A}(x)^+ \frac{dx}{x} - v_n \log(z/x_n) \right| \le \lambda_n \int_{x_n}^{x_{n+1}} |\mathcal{A}'(t)|\, dt.$$

On cells with $x_n \asymp Y$, condition (82.2) gives $\lambda_n \ll Y^{-\eta}$. Their interiors are disjoint. Summing and using Cauchy–Schwarz and Lemma 81.1 on a fixed number of dyadic blocks yields

$$\sum_{\text{cells in the block}} |\text{cell error}| \ll Y^{-\eta} \int_{Y/2}^{4Y} |\mathcal{A}'(t)|\, dt \ll Y^{\delta-\eta}/\log Y.$$

This proves (82.4). Geometric summation over dyadic blocks proves (82.5), including the convergent-tail assertion when $\delta < \eta$.

If $\delta = 0$, then RH holds by functional-equation symmetry, and the already proved asymptotic $\mathcal{A}(x) = -\log\log x + C_* + O(1/\log x)$ implies eventual negativity. The sampled mass is then eventually constant, proving (82.6) in this case.

Suppose $\delta > 0$, and put $e = \max\{0, \delta - \eta\} < \delta$. For every $\varepsilon > 0$, the error is $O_\varepsilon(X^{e+\varepsilon})$. The exact continuum growth law in the preceding section gives

$$\limsup_{X\to\infty} \frac{\log(1+\mathcal{I}_0(X))}{\log X} = \delta.$$

The finite initial integral does not affect this exponent. The triangle inequality first bounds the limsup for the sampled mass by δ. Conversely choose $0 < \varepsilon < (\delta - e)/3$. Along an unbounded sequence where $\mathcal{I}_0(X) \ge X^{\delta-\varepsilon}$, the error is smaller by a positive power of X, so $S_{\mathcal{X}}(X) \ge \frac{1}{2} X^{\delta-\varepsilon}$ for sufficiently large members of this sequence. Letting $\varepsilon \downarrow 0$ proves (82.6).

RH gives (ii), (iii), and (iv) by eventual negativity. If (ii) holds, the sampled mass is bounded, so its growth exponent is zero and (82.6) gives $\delta = 0$. Condition (iii) implies (ii). Finally, if (iv) holds, then for every $\varepsilon > 0$,

$$S_{\mathcal{X}}(X) \le O_\varepsilon(1) + C_\varepsilon \sum_{n<m} \lambda_n x_n^\varepsilon + C_\varepsilon x_m^\varepsilon \log(X/x_m) \ll_\varepsilon 1 + X^\varepsilon,$$

using $x_n^\varepsilon \le x^\varepsilon$ on each cell and integrating $x^\varepsilon\, dx/x$. Again (82.6) implies $\delta = 0$. □

Remark 82.3 (The power saving also follows from the pointwise bound). The mean-square estimate sharpens only the logarithmic factors. Lemma 82.1 alone bounds the block error by $O(Y^{\delta-\eta}\log Y)$, giving cumulative bounds $O(1)$, $O(1+(\log X)^2)$, and $O(1+X^{\delta-\eta}\log X)$ in the three respective cases. Those weaker bounds still have exponent $\max\{0,\delta-\eta\}<\delta$ when $\delta>0$, and therefore independently prove the same sampling equivalences and exact growth exponents. The finite strip argument below uses this pointwise input directly.

Corollary 82.4 (Every fixed polynomial mesh). *For every fixed real $k>0$,*

$$\boxed{\mathsf{B} \iff \mathrm{RH} \iff \sum_{n\ge n_0}\frac{\mathcal{A}(n^k)^+}{n}<\infty \iff \mathcal{A}(n^k)\le 0 \text{ eventually.}} \tag{82.7}$$

Here n_0 is any fixed integer with $n_0^k\ge 3$. The weighted positive partial sums have the exact exponent

$$\limsup_{N\to\infty}\frac{\log\Bigl(1+\sum_{n_0\le n\le N}\mathcal{A}(n^k)^+/n\Bigr)}{\log N}=k(\Theta-\tfrac12). \tag{82.8}$$

Proof. The mean value theorem gives $x_{n+1}-x_n\asymp_k x_n^{1-1/k}$, so (82.2) holds with $\eta=1/k$. Moreover

$$\lambda_n=k\log(1+1/n)\asymp_k 1/n.$$

Positive weighted partial sums are therefore comparable. Apply Theorem 82.2 and use $\log(n^k)=k\log n$. The one-cell shift between a sum ending at N and the full-cell mass at x_{N+1} has no effect on this comparison or on the logarithmic growth exponent. Indeed, monotonicity of $S_{\mathcal{X}}$ and $\log x_{n+1}/\log x_n\to 1$ identify its full-real-cutoff limsup with the limsup at the mesh endpoints. □

Corollary 82.5 (Pointwise sampled positive growth). *For every mesh satisfying* (82.2),

$$\limsup_{n\to\infty}\frac{\log(1+\mathcal{A}(x_n)^+)}{\log x_n}=\delta.$$

In particular, for every fixed $k>0$,

$$\limsup_{n\to\infty}\frac{\log(1+\mathcal{A}(n^k)^+)}{\log n}=k\delta.$$

Thus RH is also equivalent to boundedness above of $\mathcal{A}(n^k)$ on any one fixed polynomial mesh.

Proof. For $\delta>0$, integrating the derivative bound gives $\mathcal{A}(x)=O_\delta(x^\delta\log x)$, which proves the upper bound for the pointwise limsup. If that limsup were less than δ, there would be some $d<\delta$, $d\ge 0$, such that $\mathcal{A}(x_n)^+\ll x_n^d$ eventually (with a harmless arbitrarily small increase of d when necessary). Summing the positive rectangle weights as in the theorem gives $S_{\mathcal{X}}(X)\ll 1+X^d$ for $d>0$, or $O(\log X)$ for $d=0$, contradicting (82.6). The case $\delta=0$ follows from eventual negativity. □

Corollary 82.6 (A finite strip-improvement argument for nodal signs). *Assume $\mathcal{A}(x_n)\le 0$ eventually on a mesh satisfying* (82.2). *If all nontrivial zeros initially lie in $\Re s\le B$, $1/2\le B\le 1$, the same hypothesis improves that strip to*

$$\Re\rho\le\max\{1/2,B-\eta\}.$$

Starting from $B=1$, after at most $\lceil 1/(2\eta)\rceil$ such implications one obtains RH. For $x_n=n^k$ this number is at most $\lceil k/2\rceil$.

Proof. For $x_n \le x \le x_{n+1}$, the nodal sign hypothesis and Lemma 82.1 give

$$\mathcal{A}(x) \le \int_{x_n}^{x} |\mathcal{A}'(t)|\, dt \ll x^{B-1/2-\eta} \log x.$$

Set $\sigma = \max\{0, B - 1/2 - \eta\}$. This one-sided upper bound implies $I(c) < \infty$ for every $c > \sigma$. The positive-damping exclusion theorem already proved in the preceding section then excludes every zero with real part greater than $1/2 + \sigma$. If $\sigma = 0$, this and functional-equation symmetry yield RH. Iteration proves the count. This is an implication from an unproved eventual nodal-sign hypothesis, not an unconditional narrowing of the known zero strip. □

The same finite strip-improvement argument can be made from bounded sampled positive mass, using (82.5) with $B - 1/2$ in place of δ. It clarifies why an intermediate, weaker strip is not the end of the implication: improving the strip improves the next interpolation estimate.

Corollary 82.7 (Unconditional negative values on every such mesh)**.** *For every mesh satisfying* (82.2)*,*

$$\limsup_{n\to\infty} \frac{\log(1 + \mathcal{A}(x_n)^-)}{\log x_n} = \delta. \tag{82.9}$$

More particularly,

$$\liminf_{n\to\infty} \frac{\mathcal{A}(x_n)}{\log\log x_n} = \begin{cases} -1, & \text{RH } \textit{holds}, \\ -\infty, & \text{RH } \textit{fails}. \end{cases} \tag{82.10}$$

Thus $\liminf_n \mathcal{A}(n^k) = -\infty$ *for every fixed* $k > 0$*, unconditionally. In particular, infinitely many nodes on any one fixed polynomial mesh have negative values.*

Proof. We first record the lower-barrier counterpart of the strip iteration. Fix $d \ge 0$, and suppose $\mathcal{A}(x_n) \ge -Cx_n^d$ eventually. If the zeros currently lie in $\Re s \le B$, the cell derivative estimate gives

$$\mathcal{A}(x) \ge -C'x^d - C''x^{B-1/2-\eta} \log x.$$

The lower-bound version of the previously proved Landau argument excludes zeros with real part greater than

$$\max\{1/2 + d, B - \eta\}.$$

Indeed, for every $c > \max\{d, B - 1/2 - \eta\}$ the negative part has finite x^{-c-1}-integral, and the same positive Laplace argument with the two signs exchanged makes the positive part finite as well. The logarithmic zero singularity then gives the stated exclusion, on letting c decrease to that maximum. This argument also covers $d = 0$ by using arbitrary $c > 0$. Starting from $B = 1$, finitely many iterations therefore show $\Theta \le 1/2 + d$ whenever the eventual nodal lower bound holds and $d \le 1/2$.

If $\delta > 0$, this consequence implies that for every $0 < d < \delta$ there are infinitely many n with $\mathcal{A}(x_n) < -x_n^d$. Otherwise the eventual lower bound with $C = 1$ would imply $\delta \le d$. Together with the global upper bound $|\mathcal{A}(x)| \ll_\delta x^\delta \log x$, this proves (82.9) and the second line of (82.10). When $\delta = 0$, the already proved RH asymptotic $\mathcal{A}(x) = -\log\log x + C_* + O(1/\log x)$ proves both assertions. The two cases cover all possibilities, so the asserted negative divergence is unconditional. □

There is no assertion here that the absolute interpolation error is summable on every polynomial mesh. In particular, $\delta \ge 1/k$ permits the error in (82.5) to grow. What is proved is that its growth exponent is strictly smaller than the exponent δ of a hypothetical obstruction to RH, so it cannot conceal that obstruction. Each k is fixed before the limit is taken; the theorem neither permits $k = k(X) \to \infty$ nor reduces RH to finitely many computations. No historical-priority claim is made for this sampling argument.

83 A square-summed zero-band bound at the critical damping

This section improves the sufficient separation condition at an unattained positive supremum of zero real parts. It does not prove that the supremum is $1/2$, or that its separation hypothesis holds. Retain

$$\Theta = \sup_{\rho} \Re\rho, \qquad \delta = \Theta - \tfrac{1}{2} > 0, \qquad d_\rho = \Theta - \Re\rho \geq 0, \qquad c_\rho = \frac{m_\rho}{\rho(\rho - 1/2)}.$$

Here zeros are indexed by distinct locations and their multiplicities are carried by m_ρ. Define the real-valued series

$$S(u) = -\sum_{\rho} c_\rho e^{-d_\rho u + i\Im\rho u}, \qquad B_k(U) = \sum_{k \leq \Im\rho < k+1} |c_\rho| e^{-d_\rho U/4} \quad (k \in \mathbb{Z}).$$

Absolute summability of c_ρ makes all these expressions unambiguous for $u, U > 0$. Theorem 79.2 gives

$$e^{-\delta u}\mathcal{A}(e^u) = \frac{S(u)}{u} + \widetilde{R}(u), \qquad \int_{\log 2}^{\infty} |\widetilde{R}(u)|\, du = K_\delta < \infty. \tag{83.1}$$

In particular, Proposition 79.1 implies the exact integrability equivalence

$$I(\delta) < \infty \quad \Longleftrightarrow \quad \int_{\log 2}^{\infty} |S(u)| \frac{du}{u} < \infty. \tag{83.2}$$

The estimates below supply sufficient conditions for this norm; no necessity for the square-summed band condition is asserted.

Lemma 83.1 (Uniform almost orthogonality with variable damping)**.** *There is an absolute constant C such that, for every $U \geq 1$,*

$$\int_U^{2U} |S(u)|^2\, du \leq CU \sum_{k \in \mathbb{Z}} B_k(U)^2, \tag{83.3}$$

$$\int_U^{2U} |S(u)| \frac{du}{u} \leq C \left(\sum_{k \in \mathbb{Z}} B_k(U)^2 \right)^{1/2}. \tag{83.4}$$

The constant is independent of all $d_\rho \geq 0$, including damping parameters arbitrarily close to zero and coincident ordinates.

Proof. Choose once and for all a nonnegative $g \in C_c^\infty((1/2, 3))$ with $g \geq 1$ on $[1, 2]$. For $a \geq 0$ and $b \in \mathbb{R}$, the direct estimate and two integrations by parts give

$$\left| \int g(s) e^{-as+ibs}\, ds \right| \leq C \frac{(1+a)^2 e^{-a/2}}{1+b^2} \leq C' \frac{e^{-a/4}}{1+b^2}.$$

There are no boundary terms because g is compactly supported. Consequently, the kernel occurring after expanding $\int g(u/U)|S(u)|^2 du$ is bounded in absolute value by

$$\frac{CU e^{-(d_\rho + d_\omega)U/4}}{1 + U^2(\Im\rho - \Im\omega)^2}.$$

If $\Im\rho \in [k, k+1)$ and $\Im\omega \in [l, l+1)$, then for $|k - l| \geq 2$ their distance is at least $|k - l|/2$. For the three adjacent band differences use the trivial bound. Since $U \geq 1$, this bounds the expanded integral by

$$CU \sum_{k,l \in \mathbb{Z}} \frac{B_k(U) B_l(U)}{1 + (k-l)^2} \leq C' U \sum_k B_k(U)^2.$$

The last inequality is the convolution inequality for the summable sequence $(1+j^2)^{-1}$, or follows directly from $2B_kB_l \le B_k^2+B_l^2$. First applying the argument to finite sums and then using absolute convergence justifies the expansion. This proves (83.3); Cauchy–Schwarz and $\int_U^{2U} u^{-2}du = 1/(2U)$ prove (83.4). □

Theorem 83.2 (A square-summed sufficient condition)**.** *If*

$$\sum_{j=0}^{\infty}\left(\sum_{k\in\mathbb{Z}} B_k(2^j)^2\right)^{1/2} < \infty, \tag{83.5}$$

then $I(\delta) < \infty$. Quantitatively,

$$I(\delta) \le K_\delta + \int_{\log 2}^{1} |S(u)|\frac{du}{u} + C\sum_{j\ge 0}\|B(2^j)\|_{\ell^2}.$$

If a zero attains Θ, condition (83.5) cannot hold.

Proof. Sum (83.4) over the disjoint dyadic intervals. The interval $[\log 2, 1]$ contributes a finite amount. Taking positive parts in (83.1) proves the bound. An attained zero has $d_\rho = 0$, and its nonzero coefficient contributes a fixed positive lower bound to one $B_k(2^j)$ for every j. Thus the displayed series then diverges. □

For comparison, the earlier condition $\sum_\rho |c_\rho|\log^+(1/d_\rho) < \infty$, with no $d_\rho = 0$, implies (83.5): use $\|B(U)\|_2 \le \|B(U)\|_1$ and

$$\sum_{j\ge 0} e^{-d2^j/4} \ll 1 + \log^+(1/d) \qquad (d>0).$$

The band estimate retains cancellation across separated ordinates and gives the stronger implications below.

Corollary 83.3 (A separation range from the total zero count)**.** *Suppose $\Theta > 1/2$ is not attained. If for some $C > 0$ and $0 < \alpha < 3/2$ all sufficiently high zeros satisfy*

$$\Theta - \Re\rho \ge \exp\bigl(-C|\Im\rho|^\alpha\bigr), \tag{83.6}$$

then $I(\delta) < \infty$. Using only the total and local zero counts, the same conclusion holds under the borderline bound

$$\Theta - \Re\rho \ge \exp\left(-\frac{C|\Im\rho|^{3/2}}{(\log|\Im\rho|)^b}\right), \qquad b > 2.$$

At $b = 2$ it is enough to put the additional factor $(\log\log|\Im\rho|)^q$, $q > 1$, in that denominator.

Proof. The classical local zero count, also used in [45, Section 13], is

$$\sum_{k\le\Im\rho<k+1} m_\rho \ll \log(2+|k|).$$

Since $|c_\rho| \ll m_\rho/(1+|\Im\rho|^2)$, it gives

$$\left\|\left(\sum_{\substack{k\le\Im\rho<k+1\\ |\Im\rho|>T}} |c_\rho|\right)_{k\in\mathbb{Z}}\right\|_{\ell^2} \ll \frac{\log T}{T^{3/2}} \qquad (T\ge 3). \tag{83.7}$$

For (83.6), put $T = (\log U/(2C))^{1/\alpha}$. All zeros of height at most T then have $d_\rho U \geq \sqrt{U}$ once U is sufficiently large. Here the finitely many low zeros are harmless because their positive d_ρ have a positive minimum; this is where nonattainment is required. Splitting at T yields

$$\|B(U)\|_2 \ll e^{-\sqrt{U}/4} + \frac{\log\log U}{(\log U)^{3/(2\alpha)}}.$$

This is summable at $U = 2^j$ for $\alpha < 3/2$.

For the borderline cases choose T by $CT^{3/2}/(\log T)^b = (\log U)/2$, including the additional $(\log\log T)^q$ denominator when present. The second term in the preceding splitting is respectively comparable to

$$\frac{(\log\log U)^{1-b}}{\log U}, \qquad \frac{1}{\log U\ \log\log U\ (\log\log\log U)^q} \quad (b = 2).$$

At $U = 2^j$ the usual logarithmic series tests prove the claims. These borderline thresholds describe this sufficient estimate, not necessary thresholds for endpoint convergence. □

Theorem 83.4 (Density improves the separation threshold)**.** *Suppose $\Theta > 1/2$ is not attained, and fix $1/2 < \sigma < \Theta$. Suppose an established zero-density estimate has the form*

$$N(\sigma, T) := \sum_{\substack{\Re\rho \geq \sigma \\ |\Im\rho| \leq T}} m_\rho \ll_{\sigma,\varepsilon} T^{\nu(\sigma)+\varepsilon} \quad \text{for every } \varepsilon > 0.$$

If (83.6) holds for all sufficiently high zeros with $\Re\rho \geq \sigma$ and

$$0 < \alpha < 2 - \frac{\nu(\sigma)}{2}, \tag{83.8}$$

then $I(\delta) < \infty$.

One available choice, combining Ingham's estimate with Guth–Maynard's Theorem 1.2, is

$$\nu(\sigma) = \min\left\{\frac{3(1-\sigma)}{2-\sigma}, \frac{15(1-\sigma)}{3+5\sigma}\right\}. \tag{83.9}$$

Both estimates, with these exponents and the $o(1)$ convention, are recorded in [45, equation (1.2), Theorem 1.2]. In particular $\nu(\sigma) < 1$ for every $\sigma > 1/2$. Consequently the clean hypothesis

$$\Theta - \Re\rho \geq e^{-C|\Im\rho|^{3/2}} \quad \text{for all sufficiently high zeros} \tag{83.10}$$

already implies $I(\delta) < \infty$, for any fixed $C > 0$.

Proof. Zeros with $\Re\rho < \sigma$ have $d_\rho \geq \Theta - \sigma > 0$. Their absolute contribution to $\int |S(u)|du/u$ is finite by absolute summability of c_ρ. For the remaining zeros let $M_k = \sum_{k \leq \Im\rho < k+1,\ \Re\rho \geq \sigma} m_\rho$. The local zero count gives $M_k \ll \log(2+|k|)$, while the density hypothesis and partial summation give

$$\sum_{|k|>T}\left(\sum_{\substack{k\leq\Im\rho<k+1\\ \Re\rho\geq\sigma}} |c_\rho|\right)^2 \ll \sum_{|k|>T} \frac{M_k \log(2+|k|)}{(1+|k|)^4} \ll_{\sigma,\varepsilon} T^{\nu(\sigma)+\varepsilon-4}\log T.$$

Changing T by a bounded amount includes the at most two boundary bands. Thus the corresponding high-height ℓ^2 norm is

$$\ll_{\sigma,\varepsilon} T^{-2+\nu(\sigma)/2+\varepsilon}.$$

The same splitting as in the preceding proof, with $T = (\log U/(2C))^{1/\alpha}$, gives

$$\|B_{\geq\sigma}(U)\|_2 \ll e^{-\sqrt{U}/4} + (\log U)^{(-2+\nu(\sigma)/2+\varepsilon)/\alpha}.$$

By (83.8), choose $\varepsilon > 0$ small enough that the last exponent is less than -1. The dyadic series converges. Theorem 83.2, or its proof applied after discarding the integrable lower-real-part terms, proves $I(\delta) < \infty$. Finally, (83.9) is less than 1 whenever $\sigma > 1/2$, so $\alpha = 3/2$ satisfies the strict threshold. □

For the same density input $N(\sigma, T) \ll T^{\nu(\sigma)+o(1)}$, the earlier absolute coefficient condition yields only the range $\alpha < 2-\nu(\sigma)$ by partial summation of $\sum_{\Re\rho\geq\sigma} m_\rho|\Im\rho|^{\alpha-2}$. The range $\alpha < 2-\nu(\sigma)/2$ therefore records a genuine gain from the square-summed frequency bands, with the arithmetic density hypothesis held fixed.

Corollary 83.5 (What divergent critical mass would force)**.** *If $\Theta > 1/2$ is unattained and $I(\Theta - 1/2) = \infty$, then for every $C > 0$ and every $T_0 > 0$ there is a nontrivial zero $\rho = \beta + i\gamma$ such that*

$$|\gamma| > T_0, \qquad 0 < \Theta - \beta < e^{-C|\gamma|^{3/2}}.$$

More generally the exponent $3/2$ may be replaced by any $\alpha < 2 - \nu(\sigma)/2$, for fixed $1/2 < \sigma < \Theta$, and the zeros may be required to have $\beta \geq \sigma$.

Proof. If this conclusion failed for some C, T_0, the corresponding separation bound would hold for every zero beyond that height. The preceding theorem would give finite critical mass. Nonattainment makes every horizontal distance strictly positive. □

Scope of the improvement. The Fourier estimate and the conditional implications above are proved without RH. The separation premise itself is not established for the unknown Θ. When $\Theta = 1$, the known zero-free region near 1 already supplies much stronger separation, and the earlier unconditional $I(1/2) < 180$ is unchanged. Thus this section does not prove finiteness at any new unconditional damping value, and does not settle the original $I(0)$ target. It strengthens the description of the only possible divergent endpoint when a positive supremum is unattained; it does not replace the original problem by that endpoint question. No historical priority is claimed for the almost-orthogonality or zero-density ingredients.

84 A sparsity gap for the signs of polynomial samples

The sampled positive mass in Section 82 retains the full zero exponent. Combining that fact with an upper bound for a single sample removes the sample magnitudes from the criterion altogether. The resulting condition allows infinitely many positive samples, but requires a fixed power saving in their number. Its validity for the actual primes remains unproved. Integrated sign criteria and persistence of inequalities have substantial precedents, including Johnston [39] and Bhattacharya–Martin–Simpson [48]. The specific deductions for the present primitive are proved below; historical priority is not asserted.

Fix $k > 0$, choose n_0 with $n_0^k \geq 3$, and put

$$A_n = \mathcal{A}(n^k), \qquad M \geq 0,$$
$$E^{\pm}_{k,M}(N) = \#\{n_0 \leq n \leq N : \pm A_n > M\}.$$

All inequalities defining these counts are strict. The parameter M is fixed before N tends to infinity. Retain $\delta = \Theta - 1/2$.

Lemma 84.1 (Both truncated signs retain the zero exponent)**.** *Suppose $\delta > 0$, and write $\sigma = k\delta$. For every fixed $M \geq 0$,*

$$\limsup_{N\to\infty} \frac{\log\Big(1 + \sum_{n_0\leq n\leq N} (\pm A_n - M)^+/n\Big)}{\log N} = \sigma. \tag{84.1}$$

Moreover, for all sufficiently large n,

$$|A_n| \leq C_{k,\delta} \frac{n^\sigma}{\log n}. \tag{84.2}$$

Proof. Theorem 79.2 gives, for $u \to \infty$,

$$|\mathcal{A}(e^u)| \leq Be^{\delta u}/u + O_\delta(1 + \log u + e^{\delta u}/u^2) \ll_\delta e^{\delta u}/u.$$

Substitute $u = k \log n$ to obtain (84.2). The positive-sign statement in (84.1) for $M = 0$ is Corollary 82.4.

Here is the needed negative-sign extension. For every $c > 0$, the proof of Proposition 79.1, with the positive and negative parts interchanged, shows that

$$\int_2^\infty \mathcal{A}(x)^- x^{-c-1}\, dx < \infty \quad \Longleftrightarrow \quad I(c) < \infty.$$

Indeed, the two transforms differ by the same function H, holomorphic through every positive real point, so the Landau positivity argument and monotone convergence apply with either sign designated first. Thus the negative-part transform has the same abscissa $\delta > 0$. The proof of Corollary 79.6 consequently gives

$$\limsup_{X\to\infty} \frac{\log\Big(1 + \int_2^X \mathcal{A}(x)^-\, dx/x\Big)}{\log X} = \delta.$$

The map $y \mapsto y^-$ is also 1-Lipschitz. All full-cell and terminal-cell error bounds in Theorem 82.2 therefore hold with $\mathcal{A}^-$ in place of $\mathcal{A}^+$. Their exponent is strictly below δ, so the same comparison proves (84.1) for the negative sign and $M = 0$. Finally, for either sign,

$$0 \leq \sum_{n_0\leq n\leq N} \frac{(\pm A_n)^+ - (\pm A_n - M)^+}{n} \leq M \sum_{n_0\leq n\leq N} \frac{1}{n} = O_M(1 + \log N).$$

Since $\sigma > 0$, subtracting this term cannot change the exponent. □

Theorem 84.2 (The positive set is finite or has full count exponent)**.** *For every fixed $k > 0$ and $M \geq 0$, the following dichotomy holds. If RH holds, then $E^+_{k,M}(N)$ is eventually constant and $E^-_{k,M}(N) = N + O_{k,M}(1)$. If RH fails, then*

$$\limsup_{N\to\infty} \frac{\log(1 + E^+_{k,M}(N))}{\log N} = \limsup_{N\to\infty} \frac{\log(1 + E^-_{k,M}(N))}{\log N} = 1. \tag{84.3}$$

In particular, unconditionally,

$$\limsup_{N\to\infty} \frac{\log(1 + E^-_{k,M}(N))}{\log N} = 1. \tag{84.4}$$

There is also a blockwise quantitative bound. In the case $\delta > 0$, put

$$D^\pm_{k,M}(N) = \sum_{N<n\leq 2N} \frac{(\pm A_n - M)^+}{n}.$$

For sufficiently large N,

$$E^{\pm}_{k,M}(2N) - E^{\pm}_{k,M}(N) \geq c_{k,\delta}\, N^{1-k\delta} \log N\, D^{\pm}_{k,M}(N). \tag{84.5}$$

For each sign separately and every $\varepsilon > 0$, infinitely many dyadic N satisfy

$$E^{\pm}_{k,M}(2N) - E^{\pm}_{k,M}(N) \geq N^{1-\varepsilon}. \tag{84.6}$$

The two signs need not use the same dyadic subsequence.

Proof. Under RH the previously proved asymptotic

$$\mathcal{A}(x) = -\log\log x + C_* + O(1/\log x)$$

implies both assertions about the counts.

Suppose $\delta > 0$ and set $\sigma = k\delta > 0$. On the block $N < n \leq 2N$, (84.2) gives

$$\frac{(\pm A_n - M)^+}{n} \leq C_{k,\delta} \frac{N^{\sigma-1}}{\log N} \mathbf{1}_{\{\pm A_n > M\}}.$$

This proves (84.5). For either sign and every $0 < \eta < \sigma$, Lemma 84.1 implies

$$D^{\pm}_{k,M}(2^j) \geq (2^j)^{\sigma-\eta} \quad \text{for infinitely many } j.$$

Otherwise, summing the eventual upper bounds over dyadic blocks would give exponent at most $\sigma - \eta$ for the cumulative mass, contradicting (84.1). Given $\varepsilon > 0$, choose $0 < \eta < \min(\varepsilon, \sigma)$. On that infinite subsequence, (84.5) yields

$$E^{\pm}_{k,M}(2N) - E^{\pm}_{k,M}(N) \geq c_{k,\delta} N^{1-\eta} \log N \geq N^{1-\varepsilon}$$

for sufficiently large members. This proves (84.6) and the lower bounds in (84.3). The upper bounds follow from $E^{\pm}(N) \leq N$. The RH and non-RH cases together prove (84.4). □

Corollary 84.3 (An amplitude-free fixed-series criterion)**.** *Fix $k > 0$, $M \geq 0$, and $0 < s < 1$. Then*

$$\boxed{\mathsf{B} \iff \mathrm{RH} \iff \sum_{\substack{n \geq n_0 \\ \mathcal{A}(n^k) > M}} \frac{1}{n^s} < \infty.} \tag{84.7}$$

Equivalently, RH holds if and only if there exist $\varepsilon > 0$ and $C < \infty$ such that

$$E^{+}_{k,M}(N) \leq C N^{1-\varepsilon} \quad \textit{for all sufficiently large } N. \tag{84.8}$$

Thus the concrete target

$$\boxed{\sum_{\substack{n \geq 2 \\ \mathcal{A}(n^8) > 0}} \frac{1}{\sqrt{n}} < \infty} \tag{84.9}$$

is exactly equivalent to the unchanged target B*.*

The Dirichlet series

$$Z^{+}_{k,M}(s) = \sum_{n \geq n_0} \mathbf{1}_{\{A_n > M\}} n^{-s}$$

has convergence abscissa $-\infty$ under RH and exactly 1 under its failure. No assertion about its convergence at $s = 1$ is made in the unattained-supremum case.

Proof. RH makes the positive-index set finite. Conversely, if the series in (84.7) converges, then

$$E^{+}_{k,M}(N) \le N^{s} \sum_{\substack{n\ge n_0\\ A_n>M}} n^{-s} = O(N^{s}).$$

Since $s<1$, Theorem 84.2 forces RH. The power-saving count is equivalent for the same reason: RH gives a bounded count, while any such power saving contradicts (84.3) under failure of RH. For $\Re s>1$ the Dirichlet series converges absolutely by comparison with $\sum n^{-\Re s}$. Under failure of RH it diverges at every real $s<1$ by the count argument above (and trivially for $s\le 0$). Its coefficients are nonnegative, so these facts identify its convergence abscissa. A finite Dirichlet polynomial has abscissa $-\infty$ in the RH case. □

Remark 84.4 (The negative series is not an RH criterion). For each fixed $k>0$, $M\ge 0$, and $0<s<1$, the series

$$\sum_{\substack{n\ge n_0\\ \mathcal{A}(n^k)<-M}} n^{-s}$$

diverges unconditionally by (84.4). Its convergence abscissa is always 1. In particular, infinitely many negative values, or even the full count exponent of their index set, does not prove the positive-index criterion.

84.1 A stronger recurrence statement when the supremum is attained

The preceding count exponent does not assert positive natural density. A stronger conclusion follows in the attained case, using the almost-periodic profile already proved in Theorem 79.3. This separates that case from the unattained case without assuming independence of zero ordinates.

Theorem 84.5 (Both signs recur with positive lower density). *Suppose $\delta>0$ and at least one zero has real part Θ. With the notation of Theorem 79.3, set*

$$T(u) = -\sum_{\Re\rho=\Theta} c_\rho e^{i\Im\rho u}, \qquad B_* = \sum_{\Re\rho=\Theta} |c_\rho|, \qquad q_* = \sum_{\Re\rho=\Theta} |c_\rho|^2 > 0, \qquad \mu = \operatorname{mean}(T^{+}).$$

Then for every fixed $k>0$ and $M\ge 0$,

$$\liminf_{N\to\infty} \frac{E^{+}_{k,M}(N)}{N} > 0, \qquad \liminf_{N\to\infty} \frac{E^{-}_{k,M}(N)}{N} > 0. \tag{84.10}$$

The harmonic lower densities have the explicit bounds

$$\liminf_{N\to\infty} \frac{1}{\log N} \sum_{\substack{n_0\le n\le N\\ \pm A_n>M}} \frac{1}{n} \ge \frac{\mu}{B_*} \ge \frac{q_*}{2B_*^{2}} > 0. \tag{84.11}$$

In the continuous variable, both signs contain intervals of fixed relative width with bounded multiplicative gaps: there are $r,L,\eta>0$ such that each sufficiently late logarithmic interval $[U,U+L]$ contains a subinterval of length r on which

$$\mathcal{A}(e^{u}) \ge \eta e^{\delta u}/u,$$

and another subinterval of length r on which the reverse negative bound holds.

Proof. The already proved normalized formula is

$$F(u) := u e^{-\delta u}\mathcal{A}(e^{u}) = T(u) + o(1).$$

The real uniformly almost-periodic function T has mean zero, mean square $q_* > 0$, and $|T| \le B_*$. Therefore it has both a strictly positive interval and a strictly negative interval. Uniform continuity supplies a fixed positive height and width on each. Relatively dense almost periods replicate both intervals with uniformly bounded gaps, after decreasing their heights by a fixed factor. The uniform $o(1)$ error preserves these intervals for F at sufficiently large u. This proves the continuous recurrence assertion. For completeness, relative density follows by uniformly approximating T with a finite trigonometric sum: the orbit of its finitely many phases has compact closure, and finitely many translates of a small identity neighborhood cover that closure. The corresponding return times have bounded gaps. No rational independence of the phases is needed.

On such intervals $e^{\delta u}/u \to \infty$, so their positive and negative values eventually exceed any fixed level M. Apply the recurrence assertion to a logarithmic interval ending at $k \log N$. It supplies a positive subinterval of fixed length, bounded distance from that endpoint, and likewise a negative subinterval. Under $n = e^{u/k}$ each contains at least $c_k N + O(1)$ integers, for some $c_k > 0$ independent of N. This proves (84.10).

For the explicit harmonic bounds, the mesh $u_n = k \log n$ has maximum gap tending to zero, and

$$\log(1 + 1/n) = 1/n + O(1/n^2).$$

Uniform continuity of $T^{\pm}$ therefore gives

$$\frac{1}{\log N} \sum_{n_0 \le n \le N} \frac{T(u_n)^{\pm}}{n} \longrightarrow \mu.$$

One may check this by comparison with $k^{-1} \int_{u_{n_0}}^{u_N} T(u)^{\pm} du$: the summable weight error is bounded, while uniform continuity bounds the normalized quadrature error by an arbitrarily small constant after deleting finitely many cells. Since $F(u_n) - T(u_n) \to 0$, the same limit holds with $F(u_n)^{\pm}$.

If $\pm A_n \le M$, then $F(u_n)^{\pm} \le M u_n e^{-\delta u_n} \to 0$. On the remaining indices, $F(u_n)^{\pm} \le B_* + o(1)$. Thus, after averaging with $1/n$,

$$\mu \le B_* \liminf_{N \to \infty} \frac{1}{\log N} \sum_{\substack{n_0 \le n \le N \\ \pm A_n > M}} \frac{1}{n}.$$

The bound $\mu \ge q_*/(2B_*)$ was proved in Theorem 79.3, and yields (84.11). □

Exact scope. The full count exponent for negative fixed-level samples is an unconditional theorem. The positive count has a proved dichotomy: it is finite under RH and has exponent 1 otherwise. The fixed series (84.9) therefore supplies an exact amplitude-free replacement for the earlier weighted positive-mass criterion. Its convergence, the fixed power saving in (84.8), and eventual nonpositivity of the actual prime samples are all still unproved. The stronger lower density conclusion requires an attained positive supremum, and does not settle the unattained case. These arguments combine previously proved explicit-formula growth with counting and almost-periodic recurrence; no historical-priority claim is made.

85 A summable critical quadrature on stretched exponential meshes

The arbitrary polynomial meshes above preserve the undamped obstruction by a strict gap between growth exponents. A different gain is possible after retaining the actual critical damping. The nodes may then be stretched exponential in their index, while the entire quadrature error is still absolutely summable. This statement concerns $I(\delta)$, and must not be read as an extension of the undamped RH equivalence to these much sparser meshes.

Write $b(u) = \mathcal{A}(e^u)$, $u_0 = \log 2$, and retain $\delta = \Theta - 1/2$. Fix $0 < \alpha < 1$, and put

$$u_n = n^\alpha, \qquad x_n = e^{u_n}, \qquad \Delta_n = u_{n+1} - u_n, \qquad p = \frac{1-\alpha}{\alpha} > 0.$$

For a fixed sufficiently large n_0, define the exact weights

$$w_n(\delta) = \int_{u_n}^{u_{n+1}} e^{-\delta u}\, du = \begin{cases} (e^{-\delta u_n} - e^{-\delta u_{n+1}})/\delta, & \delta > 0, \\ \Delta_n, & \delta = 0. \end{cases}$$

Theorem 85.1 (Absolutely summable critical sampling error)**.** *For the actual value of δ, without assuming RH or attainment of Θ, set*

$$e_n = \int_{u_n}^{u_{n+1}} b(u)^+ e^{-\delta u}\, du - w_n(\delta) b(u_n)^+.$$

Then

$$\sum_{n \ge N} |e_n| \ll_{\alpha,\delta} u_N^{-p} = N^{-(1-\alpha)}. \tag{85.1}$$

In particular, for every $M > n_0$, the exact finite identity is

$$\mathcal{I}_\delta(x_M) - \mathcal{I}_\delta(x_{n_0}) = \sum_{n=n_0}^{M-1} w_n(\delta) \mathcal{A}(x_n)^+ + \sum_{n=n_0}^{M-1} e_n, \tag{85.2}$$

and

$$I(\delta) < \infty \quad \Longleftrightarrow \quad \sum_{n \ge n_0} w_n(\delta) \mathcal{A}(e^{n^\alpha})^+ < \infty. \tag{85.3}$$

The weights $w_n(\delta)$ may also be replaced by $\widehat{w}_n(\delta) = \Delta_n e^{-\delta u_n}$. For this replacement the total absolute correction has the same tail order as (85.1).

Proof. Changing variables in Lemma 81.1, and covering a fixed multiplicative interval by finitely many dyadic intervals, gives for every sufficiently large integer j

$$\int_j^{j+2} |b'(u)|^2 e^{-2\delta u}\, du \ll j^{-2}.$$

Indeed, $b'(u) = e^u \mathcal{A}'(e^u)$, so the left side equals $\int_{e^j}^{e^{j+2}} x^{1-2\delta} |\mathcal{A}'(x)|^2 dx$. Consequently Cauchy–Schwarz gives

$$\int_j^{j+2} |b'(u)| e^{-\delta u}\, du \ll j^{-1}. \tag{85.4}$$

The mean value theorem yields $\Delta_n \asymp_\alpha u_n^{-p}$, and hence $\Delta_n \le 1$ eventually. The positive-part map is 1-Lipschitz, so

$$\begin{aligned} |e_n| &\le \int_{u_n}^{u_{n+1}} e^{-\delta u} \int_{u_n}^{u} |b'(v)|\, dv\, du \\ &\le e^{\delta \Delta_n} \Delta_n \int_{u_n}^{u_{n+1}} |b'(v)| e^{-\delta v}\, dv. \end{aligned}$$

Group cells by their left endpoints $u_n \in [j, j+1)$. They have disjoint interiors and lie in $[j, j+2]$. Equation (85.4) therefore bounds the sum of their absolute errors by $O_{\alpha,\delta}(j^{-p-1})$. Summing $j \ge \lfloor u_N \rfloor$ proves (85.1), with the finitely many initial cells absorbed. Additivity proves the finite identity. Its absolutely convergent correction proves the equivalence.

If $\delta = 0$, the two weights coincide. Suppose $\delta > 0$. Theorem 79.2, or an integration of (85.4) over preceding unit intervals, gives

$$|b(u)| e^{-\delta u} \ll_\delta u^{-1}.$$

Since

$$0 \le \widehat{w}_n(\delta) - w_n(\delta) \le \frac{\delta}{2} e^{-\delta u_n} \Delta_n^2,$$

the additional absolute correction is $O_\delta(\Delta_n^2/u_n)$. On each unit interval its sum is $O_{\alpha,\delta}(j^{-p-1})$, because the cell lengths have bounded total and $\Delta_n \ll j^{-p}$. The same tail summation completes the proof. □

Corollary 85.2 (A sparse exact criterion for the critical integral)**.** *For $x_n = e^{\sqrt{n}}$,*

$$I(\delta) < \infty \quad \Longleftrightarrow \quad \sum_{n \ge n_0} \frac{e^{-\delta\sqrt{n}}}{\sqrt{n}} \mathcal{A}(e^{\sqrt{n}})^+ < \infty.$$

Using the exact weights in the theorem gives an absolutely summable quadrature error with tail $O_\delta(N^{-1/2})$, or $O_\delta(1/\log x_N)$. There are only $O((\log X)^2)$ such nodes below X.

Proof. Take $\alpha = 1/2$, so $p = 1$, and use $\Delta_n = 1/(\sqrt{n+1} + \sqrt{n}) \asymp n^{-1/2}$. The displayed simple weights are used only for convergence; the finite-error statement retains the theorem's exact weights. □

Corollary 85.3 (The attained-edge obstruction is still detected)**.** *Suppose $\delta > 0$ and at least one nontrivial zeta zero satisfies $\Re\rho = \Theta$. Then for every fixed $0 < \alpha < 1$ the sampled series in (85.3) diverges. In particular, $\mathcal{A}(e^{n^\alpha}) > 0$ for infinitely many n. Conversely, eventual nonpositivity on any one such mesh implies either RH, or the alternative*

$$\Theta > 1/2, \qquad \Re\rho < \Theta \text{ for every } \rho, \qquad I(\delta) < \infty.$$

The alternative has not been excluded.

Proof. Proposition 79.1 shows that finiteness of $I(\delta)$, when $\delta > 0$, excludes zeros on the boundary $\Re s = 1/2 + \delta = \Theta$. Thus an attained boundary forces $I(\delta) = \infty$; apply the theorem. If the sampled values are eventually nonpositive, the sampled positive series has finite support and hence $I(\delta) < \infty$. The same boundary exclusion gives the stated alternative when $\delta > 0$; $\delta = 0$ is RH. □

Scope. This result removes the polynomial-density restriction only for the critically damped integral, where the factor $e^{-\delta n^\alpha}$ remains explicit. For $\delta > 0$, dropping that factor would change the problem. The proof does not claim that eventual nonpositivity on a stretched exponential mesh is equivalent to RH when the zero supremum is unattained. The new feature is a complete, summable quadrature estimate on these meshes, including its quantitative tail, rather than an unproved extension of the earlier exponent-gap argument.

86 Fixed-modulus arithmetic does not remove the comparison obstruction

The previous staircase obstruction can be made arithmetically more restrictive. Every fixed modulus can be balanced, with bounded rather than merely asymptotic discrepancy, while every fixed small prime divisor is eventually excluded. These conditions still do not imply the zero-tilt conclusion. The constructed jump locations are deliberately composite; this is not a counterexample for the actual primes.

Proposition 86.1 (Residue balance leaves the common counting error)**.** *For any counting function N supported on positive integers, fix $q \ge 1$ and put*

$$D_q(x) = N(x) - \sum_{\substack{a \bmod q \\ (a,q)=1}} N(x; q, a), \qquad \phi = \phi(q), \qquad L = \mathrm{Li}_2(x).$$

Then the exact identity

$$\sum_{\substack{a \bmod q \\ (a,q)=1}} \left(N(x;q,a) - \frac{L}{\phi}\right)^2 = \sum_{\substack{a \bmod q \\ (a,q)=1}} \left(N(x;q,a) - \frac{N(x) - D_q(x)}{\phi}\right)^2 + \frac{(N(x) - D_q(x) - L)^2}{\phi} \tag{86.1}$$

separates the centered residue discrepancy from the common counting error. For the actual primes, $D_q(x) = \#\{p \le x : p \mid q\}$*; for the comparison model below,* $D_q(x)$ *is eventually constant. In that model the first term on the right is* $O_q(1)$*, while the second term retains the prescribed global oscillation.*

Proof. The centered quantities $N(x;q,a) - (N(x) - D_q(x))/\phi$ sum to zero over the reduced classes. Expand each square after adding the common quantity $(N(x) - D_q(x) - L)/\phi$; the cross term vanishes, and the sum of the common squares is the last term in (86.1). The assertions about D_q follow by counting the noncoprime support points; the model's bounded centered term follows from (86.2) below. □

This is elementary orthogonality of a constant vector and a vector whose coordinates sum to zero, not a new orthogonality principle. It identifies why centered balance among fixed residue classes does not, by itself, bound their shared error relative to $\mathrm{Li}_2(x)$.

Lemma 86.2 (Compatible ordered lists of reduced residue classes)**.** *Put* $M_j = j!$ *for integers* $j \ge 2$*. There are ordered lists* U_j *of the reduced residue classes modulo* M_j*, each class appearing once, such that the projection of* U_{j+1} *modulo* M_j *is a concatenation of identical copies of* U_j*. Consequently, for* $J \ge j$*, the projection of* U_J *modulo* M_j *is a concatenation of copies of* U_j*.*

Proof. For positive integers $m \mid n$, reduction maps $(\mathbb{Z}/n\mathbb{Z})^\times$ onto $(\mathbb{Z}/m\mathbb{Z})^\times$. Indeed, an integer coprime to m can be lifted to an integer avoiding each additional prime divisor of n, by the Chinese remainder theorem. Each fiber has the same cardinality $\phi(n)/\phi(m)$, since the map is a surjective group homomorphism. Start with any list U_2. Having chosen U_j, enumerate the $f_j = \phi(M_{j+1})/\phi(M_j)$ lifts of each of its entries. For each $1 \le r \le f_j$, list the rth lift of every entry in the order prescribed by U_j. Concatenating these f_j rounds gives U_{j+1} and the desired projection. Iteration proves the last claim. □

Theorem 86.3 (The tilted barrier survives every fixed modulus)**.** *Fix* $0 < \delta < 1/2$*,* $\gamma > 0$*, and* $\eta > 0$*. For every finite* X_0 *and every* $\varepsilon > 0$*, the comparison counting function in Theorem 80.1 can be chosen to have all the properties of that theorem and the following additional properties.*

1. *All jump locations after a finite cutoff are composite integers. Their least prime factors tend to infinity.*

2. *Write*
$$N(x;q,a) = \#\{v \le x : v \text{ is a jump location of } N,\ v \equiv a \pmod q\}.$$
For every fixed positive integer q *and every reduced residue class* $a \pmod q$*,*
$$N(x;q,a) = \frac{N(x)}{\phi(q)} + O_q(1) \qquad (x \longrightarrow \infty). \tag{86.2}$$
The constant may depend on the constructed model. For a nonreduced residue class, $N(x;q,a)$ *is eventually constant.*

3. *For every fixed* q*, reduced* a*, and fixed* $0 < a_0 < b_0 < 1$*, uniformly for* $x^{a_0} \le h \le x^{b_0}$*,*
$$N(x+h;q,a) - N(x;q,a) = \frac{h}{\phi(q)\log x} + O_q\Big(h\exp\big(-(\log x)^{1/4}\big)\Big). \tag{86.3}$$

More generally, for sufficiently large x and $1 \le h \le x$,

$$N(x+h;q,a) - N(x;q,a) \le \frac{3h}{2\phi(q)\log x} + O_q(1). \tag{86.4}$$

No estimate uniform in a modulus growing with x is asserted. In particular, despite these arithmetic properties,

$$I_N(c) < \infty \quad \Longleftrightarrow \quad c > \delta, \qquad \int_2^X \mathcal{A}_N(x)^+ x^{-\delta-1}\,dx = \frac{\eta}{\pi}\log\log X + O(1).$$

Proof. We retain the prescribed function and its differentiated error from the earlier comparison theorem:

$$g(x) = -\log\log x + \eta\frac{x^\delta\cos(\gamma\log x)}{\log x}, \qquad E(x) = x^{3/2}g'(x).$$

Choose a sufficiently large integer $T \ge X_0$, and put

$$F_T(x) = \pi(T) + \mathrm{Li}_2(x) - \mathrm{Li}_2(T) + E(x) - E(T), \qquad x \ge T.$$

The earlier derivative calculation ensures, after increasing T,

$$\frac{1}{2\log x} \le F_T'(x) \le \frac{3}{2\log x} \qquad (x \ge T). \tag{86.5}$$

Let $n_0 = \pi(T)$. For every integer $m > n_0$, define the unique target location $y_m > T$ by $F_T(y_m) = m$. For each $j \ge 2$, choose any prime $Q_j > j$, and put $L_j = Q_j M_j$ with $M_j = j!$.

Stages and complete cycles. We partition the target indices into finite consecutive stages $[m_j, m_{j+1})$, $j \ge 2$. Increase T initially so that $\log y_{n_0+1} \ge 100L_2$, and take $m_2 = n_0 + 1$. Given m_j, choose $m_{j+1} > m_j$ so that

$$m_{j+1} - m_j \quad \text{is a multiple of } \phi(M_j), \qquad \log y_{m_{j+1}} \ge 100L_{j+1}.$$

Such a finite index exists because $y_m \to \infty$. Therefore, for every m in stage j,

$$L_j \le \frac{1}{100}\log y_m. \tag{86.6}$$

Assign to consecutive indices in stage j the entries of U_j cyclically, beginning with its first entry. Denote the entry assigned to m by r_m.

Rounding to composite locations. For m in stage j, let v_m be the least integer not below y_m satisfying

$$v_m \equiv 0 \pmod{Q_j}, \qquad v_m \equiv r_m \pmod{M_j}.$$

Since Q_j is coprime to M_j, the Chinese remainder theorem gives one class modulo L_j, and hence

$$0 \le v_m - y_m < L_j \le \frac{1}{100}\log y_m. \tag{86.7}$$

On the other hand, (86.5) gives

$$1 = \int_{y_m}^{y_{m+1}} F_T'(t)\,dt \le \frac{3(y_{m+1}-y_m)}{2\log y_m}, \qquad y_{m+1} - y_m \ge \frac{2}{3}\log y_m.$$

Thus $v_m < y_{m+1} \le v_{m+1}$, including at a stage transition. The rounded locations are distinct and strictly increasing. Moreover $v_m > Q_j$, because $y_m \ge \exp(100Q_jM_j)$. Each v_m is consequently

composite, and $\gcd(v_m, j!) = 1$ makes its least prime factor exceed j. Since every stage is finite and the stages exhaust the target indices, these least prime factors tend to infinity.

Define N by keeping every actual prime at most T and then putting one jump at each v_m. No further jump is placed at an unlisted integer. Since $v_m \in [y_m, y_{m+1})$, at every $x \geq T$,

$$N(x) \in \{\lfloor F_T(x) \rfloor - 1, \lfloor F_T(x) \rfloor\}, \qquad -2 < N(x) - F_T(x) \leq 0, \tag{86.8}$$

with the evident initial interpretation before y_{n_0+1}. In particular, $N = \pi$ on $[2, T]$ and all new jumps have size one. For sufficiently large m, the lower bound in (86.5) also gives $y_{m+1} < 2y_m$ and $y_{m+1} - y_m \leq 2 \log y_{m+1}$. Together with (86.7), this proves $v_{m+1} - v_m = O(\log v_m)$.

Bounded discrepancy for every fixed modulus. Fix $q \geq 2$ and choose $j_0 \geq q$, so $q \mid M_{j_0}$. All locations in stages $j \geq j_0$ are coprime to q. By Lemma 86.2, the list in every such stage, reduced modulo M_{j_0}, is a repetition of U_{j_0}. Each period contains exactly $\phi(M_{j_0})/\phi(q)$ entries in each reduced residue class modulo q. Every completed stage contains an integral number of these periods. The incomplete period in the current stage contributes discrepancy at most $\phi(M_{j_0})$, independently of the number of stages passed. The prefix and the finitely many stages before j_0 contribute a fixed constant. This proves (86.2); the assertion for $q = 1$ is an identity. It also proves eventual constancy in each nonreduced class. Subtracting (86.2) at two endpoints gives the uniform-in-x, h identity

$$N(x + h; q, a) - N(x; q, a) = \frac{N(x + h) - N(x)}{\phi(q)} + O_q(1).$$

Equations (86.8) and (86.5) immediately imply (86.4).

Preservation of the analytic estimates. The crucial tracking estimate is exactly the bounded remainder used in the earlier staircase proof. Namely, for $x \geq T$,

$$N(x) - \mathrm{Li}_2(x) = E(x) + C_T + R_T(x), \qquad C_T = P(T) - E(T), \qquad -2 < R_T(x) \leq 0. \tag{86.9}$$

It gives $N(x) \sim \mathrm{Li}_2(x)$ and, for $1 \leq h \leq x$,

$$N(x + h) - N(x) - \frac{h}{\log x} \ll \frac{h^2}{x(\log x)^2} + \frac{hx^{\delta - 1/2}}{\log x} + 1.$$

Combining this with the fixed-modulus difference identity proves (86.3). All parameters are fixed before $x \to \infty$. The global local-count bound and the exact between-jump curvature from Theorem 80.1 also follow from the same bounded-remainder formula.

Integration of (86.9) gives

$$\mathcal{A}_N(x) = -\log\log x + K_T + \eta \frac{x^\delta \cos(\gamma \log x)}{\log x} + O_T(x^{-1/2}) \tag{86.10}$$

for a finite constant K_T. The integral of the bounded rounding remainder is absolutely convergent, with a tail $O(x^{-1/2})$. The positive-part argument with the periodic function $[\cos(\gamma u)]_+$, whose mean is $1/\pi$, now gives exactly the earlier convergence barrier, the critical truncated asymptotic, and

$$I_N(c) = \frac{\eta}{\pi} \log \frac{1}{c - \delta} + O(1) \qquad (c \downarrow \delta).$$

The estimate for $\int |a_N(u) - a(u)| e^{-u/2}\, du$ in the earlier proof used only (86.9) and the unchanged finite prefix. Its upper bound therefore tends to zero as $T \to \infty$ here too. Choose T large enough at the outset to make it less than ε, and, if necessary, less than one. The controlled Poisson budgets remain arbitrarily close to those of the actual primes, and the previously proved $I_N(1/2) < 180$ remains valid. Finally, (86.10) has the same $O_T(e^{-u/2})$ remainder after $x = e^u$. Its signed Laplace transform consequently has the same continuation through every positive real point as in the earlier comparison theorem. □

What this excludes, and what it does not. Fixed-modulus residue balance, eventual exclusion of each fixed small prime divisor, logarithmic gaps, the stated short-interval estimates, integer unit jumps, and the controlled tilted budgets can all coexist with an infinite zero-tilt mass. A proof using only these properties cannot establish the missing sign condition. The constants in (86.2) are not uniform as q grows. Indeed, all new jumps within stage j are divisible by the moving prime Q_j; this explicitly displays arithmetic information that the fixed-modulus statements fail to capture. The construction does not satisfy the assertion that its late jumps are the actual primes, and does not assign the Riemann zeta function or its functional equation to the comparison staircase. Its bounded residue discrepancy is an artificial balancing property: it does not reproduce actual prime-race biases or discrepancies, or the functional equations of Dirichlet L-functions.

Part VI
Additional synthesis theorems

The following sections combine the preceding geometric, arithmetic, operator, and scalar theories. Their proofs are given in full and were checked independently against the stated inputs. They are additional deductions in this manuscript, with no claim that their historical priority has been established. The total Toeplitz-leakage amplification was already present in the earlier spectral module; the new spectral result locates that leakage in one individual eigenvalue and bounds the remaining eigenvalues quantitatively.

87 A sharp synthesis: tangent geometry and prime quadrature

The earlier one-sided comparison between the prime-circle deficit and the optimized prime-counting variance can be made asymptotically exact in the divergent regime. The three inputs have different origins: local geometry of a concentrating measure, quadrature at the actual prime endpoints, and a lower bound forced by the rightmost zeta zeros. The result below is a deduction made in this synthesis; no assertion of historical priority is attached to it.

For clarity, the arithmetic notation used in this section is

$$E(x) = \sum_{p \le x} \frac{1}{p} - \log\log x - B_{\mathbb{P}}, \qquad E_n = E(p_n),$$

$$U_n = \exp\left(2\pi i \left(\sum_{p \le p_n} \frac{1}{p} - \log\log p_n\right)\right), \qquad h_n = p_{n+1} - p_n,$$

$$w_n = h_{n-1} + h_n = p_{n+1} - p_{n-1} \quad (n \ge 2), \qquad W_N = \sum_{n=2}^{N} w_n = p_N + p_{N+1} - 5,$$

$$D_N = W_N - \left|\sum_{n=2}^{N} w_n U_n\right|, \qquad V(X) = \min_{c \in \mathbb{R}} \int_2^X (E(t) - c)^2 \, dt.$$

The exact finite variance identity established earlier identifies the last quantity also as $\min_a \int_2^X (\pi(t) - \mathrm{Li}_2(t) - a)^2 t^{-2}\, dt$. Thus the normalization agrees with the original prime-counting target.

87.1 A geometric limit that retains the weights

Lemma 87.1 (Tangent covariance of a concentrating weighted sequence)**.** *Let $t_n \to 0$ be real, let $w_n > 0$, and suppose $W_N = \sum_{n \le N} w_n \to \infty$. Put*

$$Q_N = \min_{c \in \mathbb{R}} \sum_{n \le N} w_n (t_n - c)^2.$$

Suppose $F(t_n)$ is bounded in $\mathbb{R}^d$ and F is continuously differentiable in a neighborhood of zero. Write $v = F'(0)$ and $\overline{F}_N = W_N^{-1} \sum_{n \le N} w_n F(t_n)$. If $Q_N \to \infty$, then

$$\frac{1}{Q_N} \sum_{n \le N} w_n (F(t_n) - \overline{F}_N)(F(t_n) - \overline{F}_N)^T \longrightarrow v v^T.$$

Proof. The exact pair identities express the scalar variance and the covariance matrix, respectively, as

$$\frac{1}{2W_N} \sum_{i,j \le N} w_i w_j (t_i - t_j)^2, \quad \frac{1}{2W_N} \sum_{i,j \le N} w_i w_j [F(t_i) - F(t_j)][F(t_i) - F(t_j)]^T.$$

By continuous differentiability, the second pair integrand divided by $(t_i - t_j)^2$ tends uniformly to vv^T when both indices tend to infinity; coincident values contribute zero to both sums. Pairs involving a fixed finite prefix contribute $O(1)$ to either normalized pair sum, by boundedness of the sequences. Divide by Q_N, discard that prefix, and then let its length tend to infinity. □

Corollary 87.2 (Circular variance at a fixed frequency)**.** *In the setting of Lemma 87.1, set*

$$D_N(\lambda) = W_N - \left| \sum_{n \le N} w_n e^{i\lambda t_n} \right|.$$

For every fixed real $\lambda \neq 0$, if $Q_N \to \infty$, then

$$D_N(\lambda) \sim \frac{\lambda^2}{2} Q_N.$$

Proof. Apply the lemma to $F(t) = (\cos \lambda t, \sin \lambda t)$ and take traces. The resulting scatter equals $W_N - |\sum w_n e^{i\lambda t_n}|^2 / W_N$ and is asymptotic to $\lambda^2 Q_N$. It also equals $(1 + |m_N|) D_N(\lambda)$, where $m_N = W_N^{-1} \sum w_n e^{i\lambda t_n}$. Positive weighted averaging gives $m_N \to 1$. □

87.2 Arithmetic sampling with a controlled endpoint

The following two inputs are retained from the earlier proofs, with their arguments reproduced to make the new comparison checkable.

Lemma 87.3 (Summable quadrature error)**.** *For $X = p_{N+1}$ put*

$$T_N(c) = \sum_{n=1}^{N} h_n \{ (E_n - c)^2 + (E_{n+1} - c)^2 \}.$$

There is a finite absolute constant C such that, for every real c,

$$\left| T_N(c) - 2 \int_2^X (E(t) - c)^2 \, dt \right| \le C(1 + |c|) + 2|c| \log(X/2).$$

Proof. Use the quantitative prime number theorem to obtain $E(x) = O((\log x)^{-2})$, and use Peck's estimate as reproduced in [26, Theorem 3.1]:

$$\sum_{p_n \le Y} h_n^2 \ll_\eta Y^{5/4+\eta} \qquad (\eta > 0).$$

On a gap $[p, q]$, set $h = q - p$ and $F(t) = E(p) - c - \log\log t + \log\log p$. Except at q this equals $E(t) - c$, and $F(q) = E(q) - c - 1/q$. For $G = F^2$,

$$G''(t) = \frac{2}{t^2 \log^2 t} + \frac{2F(t)(\log t + 1)}{t^2 \log^2 t}.$$

The smooth trapezoidal error is consequently $O((1 + |c|)h^3/(p^2 \log p))$. Replacing the left-hand value at q by the actual prime value adds exactly $h\{2(E(q) - c)/q - 1/q^2\}$. All contributions not containing c are absolutely summable: the jump terms are bounded by constant multiples of

$$\sum_n \frac{h_n}{p_{n+1} \log^2 p_{n+1}} + \sum_n \frac{h_n}{p_{n+1}^2} < \infty.$$

For the smooth terms, the inequality between finite ℓ^r norms gives

$$\sum_{Y \le p_n < 2Y} h_n^3 \le \left(\sum_{p_n \le 2Y} h_n^2 \right)^{3/2} \ll_\eta Y^{15/8+3\eta/2}.$$

Choose $\eta < 1/12$ and sum dyadically to obtain $\sum h_n^3/(p_n^2 \log p_n) < \infty$. Finally, $\sum_{n=1}^{N} h_n/p_{n+1} \le \log(X/2)$ controls the only potentially divergent contribution, which is multiplied by $2|c|$. □

Lemma 87.4 (Lower growth uniform over bounded centers)**.** *If $\Theta > 1/2$, then for every $C > 0$ and every $\varepsilon > 0$,*

$$\int_2^X |E(t) - c|^2 \, dt \gg_{C,\varepsilon} X^{2\Theta-1-\varepsilon} \quad (|c| \le C)$$

for all sufficiently large X. In particular, $V(X) \gg_\varepsilon X^{2\Theta-1-\varepsilon}$.

Proof. Let $P(x) = \pi(x) - \mathrm{Li}_2(x)$ and $C_\pi = 2(\log\log 2 + B_{\mathbb{P}})$. Exact inverse partial summation gives

$$P(x) = x(E(x) - c) - \int_2^x (E(t) - c) \, dt + C_\pi + 2c.$$

Hence

$$\int_Y^{2Y} |P(t)| \, dt \le 3Y \int_2^{2Y} |E(t) - c| \, dt + Y(|C_\pi| + 2C).$$

Theorem 12, equation (15.2), of Stechkin–Popov [25] states that for every $\delta > 0$, $\int_Y^{2Y} |P(t)| \, dt \gg_\delta Y^{1+\Theta-\delta}$ for all sufficiently large Y. Their theorem includes $\Theta = 1$; changing the normalization of li adds only a constant and does not change this bound. Choose $\delta < \min(\Theta, \varepsilon/2)$. The $O_C(Y)$ term is absorbed, and Cauchy–Schwarz then gives the stated lower bound with $X = 2Y$. The minimizing center for V is a mean of the bounded function E, so it lies in a fixed bounded interval. □

We now return to the actual primes, retaining the original phases, weights, and deficit. Set

$$E_n = E(p_n), \qquad Q_N = \min_{c \in \mathbb{R}} \sum_{n=2}^{N} w_n (E_n - c)^2, \qquad X = p_{N+1}.$$

Theorem 87.5 (Sharp comparison in the divergent regime)**.** *If the Riemann hypothesis is false, then*

$$Q_N \sim 2V(p_{N+1}), \qquad \boxed{D_N \sim 4\pi^2 V(p_{N+1}).}$$

No fixed zero-free half-plane is assumed. In particular, the assertion includes the possible case $\Theta = 1$.

Proof. Write $h_n = p_{n+1} - p_n$ and let

$$c_d = \frac{\sum_{n=2}^{N} w_n E_n}{W_N}, \qquad c_X = \frac{1}{X - 2} \int_2^X E(t) \, dt.$$

The quantitative prime number theorem implies $E(x) = O((\log x)^{-2})$. Averaging this estimate, and using $W_N \sim 2X$, gives $c_d, c_X = O((\log X)^{-2})$. For the discrete mean one may split at $\sqrt{X}$; the total weight below that point is $O(\sqrt{X})$ by Bertrand's postulate.

Define

$$T_N(c) = \sum_{n=1}^{N} h_n \{(E_n - c)^2 + (E_{n+1} - c)^2\}, \qquad G_N(c) = \sum_{n=2}^{N} w_n (E_n - c)^2.$$

Lemma 87.3 gives, for $c = c_d$ and $c = c_X$,

$$T_N(c) = 2 \int_2^X (E(t) - c)^2 \, dt + O(1).$$

Its exact endpoint relation is

$$T_N(c) = G_N(c) + h_1(E_1 - c)^2 + h_N(E(X) - c)^2.$$

The first endpoint is bounded. Peck's estimate $\sum_{p_n \le Y} h_n^2 \ll_\eta Y^{5/4+\eta}$, with $\eta = 1/12$, implies $h_N \ll X^{2/3}$.

Suppose first that $1/2 < \Theta < 1$. The classical estimate $\pi(x) - \mathrm{li}(x) = O(x^\Theta \log x)$, recorded in [1, equation (23)], and partial summation give

$$E(x) = O_\Theta(x^{\Theta-1} \log x).$$

Integration gives the same estimate for c_X. Neighboring-gap summation gives it for c_d: compare each weighted value with the integral on its two adjacent prime gaps, using Bertrand's postulate, and sum. Hence the last endpoint term is

$$O_\Theta(X^{2\Theta-4/3}(\log X)^2).$$

Lemma 87.4 gives

$$V(X) \gg_\varepsilon X^{2\Theta-1-\varepsilon}.$$

Choosing $0 < \varepsilon < \min(2\Theta - 1, 1/3)$ makes the ratio of the endpoint term to $V(X)$ tend to zero.

If $\Theta = 1$, boundedness of E and of both centers instead gives an endpoint bound $O(X^{2/3})$, while $V(X) \gg_\varepsilon X^{1-\varepsilon}$. Again the ratio tends to zero, with $\varepsilon < 1/3$.

At the continuous minimizing center,

$$Q_N \le G_N(c_X) = 2V(X) + o(V(X)).$$

At the discrete minimizing center,

$$Q_N = G_N(c_d) = 2\int_2^X (E(t) - c_d)^2\, dt + o(V(X)) \ge 2V(X) + o(V(X)).$$

Therefore $Q_N \sim 2V(X) \to \infty$. A fixed rotation removes $e^{2\pi i B_{\mathbb{P}}}$ from the phases without changing the deficit. Corollary 87.2, with $\lambda = 2\pi$, then proves $D_N \sim 2\pi^2 Q_N \sim 4\pi^2 V(X)$. □

The finite-energy case. Under RH, D_N and $V(X)$ have finite limits. The universal factor $4\pi^2$ is not asserted for their ratio: finite early phases and the finite quadrature correction need not disappear. This distinction is essential; a divergent-energy asymptotic must not be extended to the convergent case without another argument.

87.3 A fixed harmonic and a strict limiting gap

Theorem 87.6 (A fixed-harmonic ratio criterion)**.** *For an integer $m \ge 2$, define*

$$D_N^{(m)} = W_N - \left|\sum_{n=2}^{N} w_n U_n^m\right|.$$

Then $D_N > 0$ and $0 \le D_N^{(m)} < m^2 D_N$ for $N \ge 3$. The limit $r_m = \lim_{N\to\infty} D_N^{(m)}/D_N$ exists, and

$$\mathrm{RH} \implies 0 \le r_m < m^2, \qquad \neg\mathrm{RH} \implies r_m = m^2.$$

Consequently, for any one fixed $m \ge 2$,

$$\boxed{\mathrm{RH} \iff r_m < m^2.}$$

Proof. For the finite inequality, let b be a unit minimizing direction for D_N, so that $2D_N = \sum w_n|U_n - b|^2$. For unit u, b,

$$|u^m - b^m| \le m|u - b|,$$

with strict inequality if $u \neq b$ and $m \ge 2$. This follows by factoring $u^m - b^m$ and applying the triangle inequality to the sum of m unit factors; equality would require $u = b$. Therefore

$$2D_N^{(m)} \le \sum w_n|U_n^m - b^m|^2 < m^2 \sum w_n|U_n - b|^2 = 2m^2 D_N$$

whenever the phases are not all identical. Their nonidentity for $N \ge 3$ is verified below.

If RH is false, apply Corollary 87.2 at frequencies $2\pi m$ and 2π, with the same divergent Q_N.

If RH holds, the limiting-center identity gives finite sums

$$D_\infty^{(m)} = \sum_{n\ge 2} w_n[1 - \cos(2\pi m E_n)], \qquad D_\infty = \sum_{n\ge 2} w_n[1 - \cos(2\pi E_n)].$$

For integer $m \ge 2$, $|\sin(mx)| < m|\sin x|$ whenever $\sin x \neq 0$; the inequality follows by applying the triangle inequality to the geometric-sum formula for $\sin(mx)/\sin x$, and equality of the unit summands would require $e^{2ix} = 1$. Thus the first summand is at most m^2 times the second, strictly so at any nonintegral E_n.

There is such an index without any numerical assumption on $B_{\mathbb{P}}$. Indeed

$$E(5) - E(3) = \frac{1}{5} - \log\frac{\log 5}{\log 3} \in (-1, 0).$$

For the sign, use $5^3 > 3^4$ and $\log(4/3) > 1/4 > 1/5$; for the lower bound use $5 < 9$ and $\log 2 < 1$. Therefore $E(3), E(5)$ cannot both be integers. Their positive weights ensure $D_\infty > 0$ and strictness after summing. They also imply $U_2 \neq U_3$, so $D_N > 0$ for $N \ge 3$. Taking ratios proves the result. □

The theorem produces a strict limiting gap, not an effective lower bound for that gap. It does not establish which alternative holds. Its value is structural: the same finite prime data connect a geometric frequency ratio to the original quadratic-energy target. In fact every finite ratio is already strictly below m^2 under either alternative. Only a strict gap remaining in the limit is equivalent to RH; finite verification of the displayed inequality is not evidence that this limit gap has been proved.

88 A universal second-eigenvalue law for collapsing circle measures

The fixed-degree Toeplitz comparison can be strengthened from two-sided inequalities to an exact asymptotic law whenever the deficit diverges. The new input is a second-order expansion transverse to the limiting rank-one direction. All statements in this section hold for general weighted circle measures; no prime estimate is assumed.

Let $t_n \in \mathbb{R}$ tend to zero, let $w_n > 0$, and suppose $W_N = \sum_{n\le N} w_n \to \infty$. Fix a real $\lambda \neq 0$ and an integer $m \ge 1$. Define the unnormalized Gram matrix

$$G_{m,N} = \left(\sum_{n\le N} w_n e^{i\lambda(j-k)t_n}\right)_{0\le j,k\le m}.$$

Its eigenvalues, in decreasing order, are $\Lambda_{1,N} \ge \cdots \ge \Lambda_{m+1,N} \ge 0$. Put

$$\bar t_N = W_N^{-1}\sum_{n\le N} w_n t_n, \quad S_{2,N} = \sum_{n\le N} w_n(t_n - \bar t_N)^2, \quad Q_{3,N} = \sum_{n\le N} w_n|t_n - \bar t_N|^3.$$

Theorem 88.1 (Finite transverse spectral certificate)**.** *Write $d = m+1$ and $c_m = m(m+1)(m+2)/12$. For one finite prefix set*

$$r_N = \frac{d|\lambda|^3 m^3}{6} Q_{3,N}, \qquad b_N = \frac{d\lambda^2 m^2}{2} S_{2,N} + r_N.$$

If $dW_N > 2b_N$, then

$$\lambda^2 c_m S_{2,N} - r_N - \frac{b_N^2}{dW_N - 2b_N} \le \Lambda_{2,N} \le \lambda^2 c_m S_{2,N} + r_N.$$

In addition, $0 \le \Lambda_{j,N} \le r_N$ for $3 \le j \le d$.

Proof. Diagonal unitary conjugation replaces every t_n by $x_n = t_n - \bar{t}_N$ without changing the spectrum. Let $e = (1,\dots,1)^T$, $u = e/\sqrt{d}$, $v = (0,1,\dots,m)^T$, $v^{[2]} = (0^2, 1^2, \dots, m^2)^T$, and $P = I - uu^*$. Taylor's formula, $\sum w_n x_n = 0$, and $|e^{iy} - 1 - iy + y^2/2| \le |y|^3/6$ give

$$G_{m,N} = dW_N uu^* + B_N,$$

$$B_N = \lambda^2 S_{2,N}\left(vv^* - \tfrac{1}{2}v^{[2]}e^* - \tfrac{1}{2}e(v^{[2]})^*\right) + R_N, \qquad \|R_N\| \le r_N, \quad \|B_N\| \le b_N.$$

The norm bounds follow by bounding each matrix entry and then using $\|A\| \le d \max_{j,k} |A_{jk}|$. The compression to $u^\perp$ is

$$C_N = PG_{m,N}P|_{u^\perp} = \lambda^2 S_{2,N}(Pv)(Pv)^* + PR_NP|_{u^\perp}.$$

Since $Pv = (j - m/2)_{j=0}^m$,

$$\|Pv\|^2 = \sum_{j=0}^{m} (j - m/2)^2 = c_m.$$

Writing c for the largest eigenvalue of C_N, Weyl's inequality gives $|c - \lambda^2 c_m S_{2,N}| \le r_N$, while all its other eigenvalues lie in $[0, r_N]$. Interlacing gives the upper bounds for $\Lambda_{2,N}$ and $\Lambda_{j,N}$, $j \ge 3$.

For the lower bound, take a unit top eigenvector $q \in u^\perp$ of C_N. The compression of $G_{m,N}$ to $\operatorname{span}(u,q)$ has matrix $\left(\begin{smallmatrix} a & h \\ \bar{h} & c \end{smallmatrix}\right)$, where $a \ge dW_N - b_N$, $c \le b_N$, and $|h| \le b_N$. Thus $a - c \ge dW_N - 2b_N > 0$. Its smaller eigenvalue is at least

$$c - \frac{|h|^2}{a-c} \ge c - \frac{b_N^2}{dW_N - 2b_N};$$

the first inequality follows from $\sqrt{A^2 + 4x} \le A + 2x/A$ for $A > 0$. The two-dimensional min–max principle bounds $\Lambda_{2,N}$ below by this smaller eigenvalue. This proves the theorem. □

Theorem 88.2 (One transverse eigenvalue carries the divergent defect)**.** *If $S_{2,N} \to \infty$, then*

$$\Lambda_{2,N} \sim \lambda^2 c_m S_{2,N}, \qquad \Lambda_{j,N} = o(S_{2,N}) \quad (3 \le j \le m+1),$$

$$\Lambda_{1,N} = (m+1)W_N - \lambda^2 c_m S_{2,N} + o(S_{2,N}).$$

Equivalently, if

$$D_N = W_N - \left|\sum_{n \le N} w_n e^{i\lambda t_n}\right| \longrightarrow \infty,$$

then

$$\boxed{\frac{\Lambda_{2,N}}{D_N} \longrightarrow \binom{m+2}{3}, \qquad \Lambda_{j,N} = o(D_N) \quad (j \ge 3).}$$

Proof. Weighted Cesaro convergence gives $\bar{t}_N \to 0$ and $S_{2,N}/W_N \to 0$. Also $Q_{3,N} = o(S_{2,N})$: for any $\varepsilon > 0$, choose a fixed tail on which $|t_n| < \varepsilon$; eventually $|t_n - \bar{t}_N| < 2\varepsilon$ there. The tail contribution to $Q_{3,N}$ is at most $2\varepsilon S_{2,N}$, whereas the fixed-prefix contribution is bounded independently of N and hence negligible compared with $S_{2,N}$. It follows that $r_N = o(S_{2,N})$, $b_N = O(S_{2,N}) = o(W_N)$, and $b_N^2/(dW_N - 2b_N) = o(S_{2,N})$. Apply Theorem 88.1. The assertion for $\Lambda_{1,N}$ follows from $\operatorname{tr} G_{m,N} = dW_N$.

For $m = 1$, the two eigenvalues are exactly $W_N \pm |\sum w_n e^{i\lambda t_n}|$, so $D_N \sim \lambda^2 S_{2,N}/2$ whenever $S_{2,N} \to \infty$. Conversely, the circle minimization identity gives

$$2D_N = \min_{|b|=1} \sum_{n \le N} w_n |e^{i\lambda t_n} - b|^2 \le \sum_{n \le N} w_n |e^{i\lambda t_n} - e^{i\lambda \bar{t}_N}|^2 \le \lambda^2 S_{2,N}.$$

Hence $D_N \to \infty$ implies $S_{2,N} \to \infty$. Dividing the general asymptotic by its $m = 1$ instance proves the displayed ratio. □

Remark 88.3 (Meaning for the prime circle). Take $t_n = E(p_n)$, $\lambda = 2\pi$, and the original adjacent-gap weights. Rotation by the prime Mertens constant does not affect the matrices. The previously established deficit criterion makes the divergent alternative exactly the failure of RH. The theorem identifies the asymptotic shape of that alternative; it does not exclude it. It is stronger than fixed comparison constants but does not supply an unconditional upper bound for D_N.

89 Constant-free Hardy coefficients and an invertible Toeplitz filter

The canonical Laguerre coefficients require exact centering by the prime Mertens constant. A common centering error creates an artificial interior pole. Multiplicative differencing removes that error exactly, and the Hardy-space representation shows that no true off-critical zero is erased in the process.

Fix $\lambda > 1$, $L = \log \lambda$, $q = \lambda^{-1/2}$, and $\alpha > 1/2$. Put

$$g_\lambda(x) = E(x) - E(\lambda x) = -\sum_{x < p \le \lambda x} \frac{1}{p} + \log \frac{\log(\lambda x)}{\log x}, \qquad x \ge 2,$$

$$h_\lambda(t) = \sqrt{2e^t}\, g_\lambda(2e^t), \quad f(t) = \sqrt{2e^t}\, E(2e^t), \qquad t \ge 0.$$

Then $h_\lambda = f - qR_L f$, where $R_L f(t) = f(t + L)$. Define, without assuming RH,

$$b_n^{(\lambda,\alpha)} = \sqrt{2\alpha} \int_0^\infty h_\lambda(t) e^{-\alpha t} L_n(2\alpha t)\, dt.$$

Every individual integral converges absolutely since E is bounded. Neither g_λ nor these coefficients require a value of the prime Mertens constant.

Theorem 89.1 (Sharp inverse and exact Hardy criterion). *For arbitrary $f \in L^2(0, \infty)$,*

$$(1 - q)\|f\|_2 \le \|(I - qR_L)f\|_2 \le (1 + q)\|f\|_2,$$

and both constants are sharp. The inverse is the norm-convergent series $\sum_{j \ge 0} q^j R_L^j$.

For the prime error,

$$\text{RH} \iff \sum_{n \ge 0} |b_n^{(\lambda,\alpha)}|^2 < \infty.$$

When finite, the sum equals $\int_2^\infty |g_\lambda(x)|^2\, dx$, and

$$(1 - q)^2 \int_2^\infty E(x)^2\, dx \le \sum_{n \ge 0} |b_n^{(\lambda,\alpha)}|^2 \le (1 + q)^2 \int_2^\infty E(x)^2\, dx.$$

Proof. The advance operator R_L is a contraction. The triangle inequality and its reverse give the norm bounds, and its Neumann series gives the inverse. The functions $f_\varepsilon(t) = e^{-\varepsilon t}$ are eigenvectors of R_L with eigenvalues $e^{-\varepsilon L}$; letting $\varepsilon \downarrow 0$ proves sharpness of the lower constant. Replacing these by $e^{-\varepsilon t}e^{i\pi t/L}$ gives eigenvalues $-e^{-\varepsilon L}$ and proves sharpness of the upper constant.

The standard Laguerre functions $\sqrt{2\alpha}e^{-\alpha t}L_n(2\alpha t)$ form an orthonormal basis of $L^2(0,\infty)$. Thus $h_\lambda \in L^2$ implies the coefficient identity. Conversely, if its coefficients are in ℓ^2, take the L^2 function with those coefficients. The difference has every moment against $e^{-\alpha t}t^k$ equal to zero. Its Laplace transform is analytic on $\Re w > 1/2$, all its derivatives at $w = \alpha$ vanish, and uniqueness gives equality almost everywhere. Therefore $h_\lambda \in L^2$.

One must also justify inversion before assuming the actual prime function f is in L^2. The identity $h_\lambda = f - qR_Lf$ telescopes to

$$\sum_{j=0}^{J} q^j R_L^j h_\lambda(t) = f(t) - \sqrt{2e^t}E(2e^t\lambda^{J+1}).$$

The last term tends pointwise to zero by Mertens' theorem. If $h_\lambda \in L^2$, the series on the left converges in L^2, so an almost-everywhere convergent subsequence identifies its limit with f. This proves $f \in L^2$. The established prime-error energy criterion $E \in L^2(2,\infty) \iff$ RH now proves the equivalence. The norm inequalities and the change of variables $x = 2e^t$ prove the remaining statements. □

Proposition 89.2 (The exact disk operator)**.** *Let $\mathcal{U}_\alpha : L^2(0,\infty) \to H^2(\mathbb{D})$ be the Laguerre unitary, so that*

$$(\mathcal{U}_\alpha f)(z) = \frac{\sqrt{2\alpha}}{1-z}\int_0^\infty f(t)\exp\left(-\alpha\frac{1+z}{1-z}t\right)dt.$$

Define the singular inner function

$$\Theta_{L,\alpha}(z) = \exp\left(-\alpha L\frac{1+z}{1-z}\right).$$

Multiplication by $\Theta_{L,\alpha}$ is an isometry, and on $H^2(\mathbb{D})$

$$\mathcal{U}_\alpha R_L \mathcal{U}_\alpha^{-1} = M^*_{\Theta_{L,\alpha}}.$$

When the prime functions f, h_λ belong to L^2 (equivalently, when RH holds), this gives

$$\mathcal{U}_\alpha h_\lambda = (I - qM^*_{\Theta_{L,\alpha}})\mathcal{U}_\alpha f.$$

Thus the constant-free coefficients are obtained from the original coefficients by a bounded invertible co-analytic Toeplitz filter. For $\lambda = 4$, its condition number is at most 3.

Proof. The delay $S_Lf(t) = \mathbf{1}_{t\ge L}f(t-L)$ has adjoint R_L. Its Laplace transform is multiplied by e^{-Lw}. Under the Cayley map $w = \alpha(1+z)/(1-z)$ this is multiplication by $\Theta_{L,\alpha}$. The latter is bounded by one on the disk and has modulus one almost everywhere on its boundary, so its multiplication operator is isometric. Taking adjoints gives the formula. The condition number bound is $(1+q)/(1-q)$; at $q = 1/2$ this equals 3. □

The operator is explicit on coefficients. If $\Theta_{L,\alpha}(z) = \sum_{k\ge0}\theta_k z^k$, then

$$\theta_0 = e^{-\alpha L}, \qquad \theta_k = e^{-\alpha L}\big(L_k(2\alpha L) - L_{k-1}(2\alpha L)\big), \quad k \ge 1,$$

and $(M^*_\Theta a)_n = \sum_{k\ge0}\overline{\theta_k}a_{n+k}$ for $a \in \ell^2$. The infinite tail is part of this formula: bounded invertibility does not make a finite coefficient list a certificate for all indices.

Proposition 89.3 (No zeta singularity is hidden by the filter)**.** *Let $F(s) = \int_2^\infty E(x)x^{-s}\,dx$ initially for $\Re s > 1$, and define F_λ by replacing E with g_λ. Then*

$$F_\lambda(s) = (1-\lambda^{s-1})F(s) + \lambda^{s-1}\int_2^{2\lambda} E(x)x^{-s}\,dx.$$

Every nonremovable zero-induced logarithmic singularity of F at ρ with $1/2 < \Re\rho < 1$ remains nonremovable for F_λ. The prime-square singularity at $s = 1/2$ remains as well. Consequently the coefficient generating function $\sum_{n\ge0} b_n^{(\lambda,\alpha)} z^n$ has the exact radius

$$\min\left\{1, \inf_{\Re\rho>1/2}\left|\frac{\rho-1/2-\alpha}{\rho-1/2+\alpha}\right|\right\}.$$

In particular, its coefficient root-growth criterion is also equivalent to RH.

Proof. The Mellin identity follows by substitution $y = \lambda x$. The finite-interval term is entire. At a zero in the stated strip, $|\lambda^{\rho-1}| < 1$, so the multiplier is nonzero; at $s = 1/2$ it equals $1 - q > 0$. The established singularity expansion of F therefore survives. Near the origin the generating function is

$$\frac{\sqrt{2\alpha}}{1-z}2^{w_\alpha(z)}F_\lambda(1/2 + w_\alpha(z)), \qquad w_\alpha(z) = \alpha\frac{1+z}{1-z}.$$

The additional exponential factor has no zeros or singularities in the disk. The exact radius follows by the same analytic continuation and nonremovability argument as for the canonical Mellin–Laguerre transform. The endpoint at $z = -1$ supplies radius one when there is no interior transformed zero. □

Corollary 89.4 (Convex dilation filters)**.** *Let $a_1, \dots, a_r \ge 0$, $\sum a_j = 1$, and $\lambda_j > 1$. The filter*

$$g(x) = E(x) - \sum_{j=1}^{r} a_j E(\lambda_j x)$$

is constant-free and has the same L^2 criterion and the same zero-induced singularities as E. Its inverse norm is at most $1/(1-q_*)$, where $q_* = \sum_j a_j \lambda_j^{-1/2} < 1$.

Proof. The corresponding Hilbert-space operator is $I - \sum a_j \lambda_j^{-1/2} R_{\log\lambda_j}$, a perturbation of the identity by an operator of norm at most q_*. Pointwise identification follows by iterating the finite dilation relation: the remaining term is a convex average of E at arguments at least $x(\min_j \lambda_j)^k$, and tends to zero. Its Mellin multiplier is $1 - \sum a_j \lambda_j^{s-1}$. For $\Re s < 1$, $|\sum a_j \lambda_j^{s-1}| \le \sum a_j \lambda_j^{\Re s - 1} < 1$, so no singularity in this half-plane is canceled. □

These theorems supply stable, constant-free coordinates and an exact noncancellation result. They do not establish the finiteness of the Hardy norm. That arithmetic estimate remains the original RH target.

90 Nonlinear moments, critical endpoints, and moving sign thresholds

This section recombines three ingredients developed in the preceding analysis: the integrated explicit formula, the positive-mass growth law, and sampling on polynomial meshes. The resulting statements concern nonlinear powers and increasing thresholds of the same prime statistic. Their proofs are included. They neither assume nor establish the missing arithmetic inequality, and no claim of historical priority is made.

Throughout this section, retain

$$P(x) = \pi(x) - \mathrm{Li}_2(x), \qquad \mathrm{Li}_2(x) = \int_2^x \frac{dt}{\log t}, \qquad \mathcal{A}(X) = \int_2^X \frac{P(x)}{x^{3/2}}\, dx.$$

For a real number y, write $y^+ = \max(y, 0)$. Set

$$\Theta = \sup_{\zeta(\rho)=0,\ 0<\Re\rho<1} \Re\rho, \qquad \delta = \Theta - \frac{1}{2} \in [0, 1/2], \qquad u_0 = \log 2.$$

The functional equation identifies $\delta = 0$ with RH. The value of δ is not supplied by any theorem below.

Here are the precise earlier inputs. Theorem 67.1 gives, under RH,

$$\mathcal{A}(X) = -\log\log X + C_* + O(1/\log X), \tag{90.1}$$

so that $\mathcal{A}^+$ vanishes outside a compact interval. In the case $\delta > 0$, Theorem 79.2 and Corollary 79.6 give

$$|\mathcal{A}(X)| \ll_\delta \frac{X^\delta}{\log X}, \tag{90.2}$$

$$\limsup_{X\to\infty} \frac{\log\Big(1 + \int_2^X \mathcal{A}(x)^+\, dx/x\Big)}{\log X} = \delta. \tag{90.3}$$

For every fixed $k > 0$, Corollary 82.4 gives

$$\limsup_{N\to\infty} \frac{\log\Big(1 + \sum_{n_0\le n\le N} \mathcal{A}(n^k)^+/n\Big)}{\log N} = k\delta, \qquad n_0^k \ge 3. \tag{90.4}$$

The proof of this last statement controls the interpolation error by a strictly smaller growth exponent. It does not require absolute summability of that error on every polynomial mesh.

90.1 The full spectrum of positive powers

For fixed $r, k > 0$ and real c, define the nonnegative quantities

$$\mathscr{M}_r(X) = \int_2^X \big(\mathcal{A}(x)^+\big)^r \frac{dx}{x}, \tag{90.5}$$

$$\mathscr{J}_r(c) = \int_2^\infty \big(\mathcal{A}(x)^+\big)^r x^{-c-1}\, dx, \tag{90.6}$$

$$\mathscr{S}_{r,k}(N) = \sum_{n_0\le n\le N} \frac{\big(\mathcal{A}(n^k)^+\big)^r}{n}. \tag{90.7}$$

The infinite integrals and sums are allowed to be infinite.

Theorem 90.1 (Nonlinear positive-mass spectrum)**.** *For every fixed $r, k > 0$, unconditionally,*

$$\limsup_{X\to\infty} \frac{\log(1 + \mathscr{M}_r(X))}{\log X} = r\delta, \tag{90.8}$$

$$\limsup_{N\to\infty} \frac{\log(1 + \mathscr{S}_{r,k}(N))}{\log N} = rk\delta. \tag{90.9}$$

Consequently, for every such pair (r, k),

$$\mathrm{RH} \iff \mathscr{J}_r(0) < \infty \iff \sum_{n\ge n_0} \frac{\big(\mathcal{A}(n^k)^+\big)^r}{n} < \infty. \tag{90.10}$$

If RH fails, $\mathscr{J}_r(c)$ converges for $c > r\delta$ and diverges for $c < r\delta$. Under RH it converges for every real c.

The Dirichlet series

$$\mathscr{D}_{r,k}(s) = \sum_{n \geq n_0} (\mathcal{A}(n^k)^+)^r n^{-s}$$

has convergence abscissa $1 + rk\delta$ if RH fails, and $-\infty$ under RH. These abscissa statements alone make no assertion at their boundary lines.

Proof. Under RH, (90.1) makes every positive part eventually zero. Both partial masses are then eventually constant, and all assertions in that case follow.

Suppose $\delta > 0$. Let

$$B_X = \sup_{2 \leq x \leq X} \mathcal{A}(x)^+, \qquad L_X = \int_2^X \frac{dx}{x} = \log(X/2).$$

Local continuity of $\mathcal{A}$ and (90.2) imply, after increasing a constant, $B_X \ll_\delta X^\delta / \log X$ for sufficiently large X. The inequality $\mathscr{M}_r(X) \leq L_X B_X^r$ proves the upper bound $r\delta$ in (90.8).

If $r \geq 1$, Hölder's inequality gives

$$\mathscr{M}_1(X) \leq \mathscr{M}_r(X)^{1/r} L_X^{1-1/r}, \qquad \mathscr{M}_r(X) \geq L_X^{1-r} \mathscr{M}_1(X)^r.$$

For every $\epsilon > 0$, the growth law (90.3) supplies an unbounded sequence of X on which $\mathscr{M}_1(X) \geq X^{\delta-\epsilon}$. On that sequence the last inequality gives lower exponent at least $r(\delta - \epsilon)$. Let ϵ decrease to zero.

If $0 < r < 1$, pointwise $\mathcal{A}(x)^+ \leq B_X^{1-r} (\mathcal{A}(x)^+)^r$ on $[2, X]$. Thus

$$\mathscr{M}_r(X) \geq B_X^{r-1} \mathscr{M}_1(X).$$

Using the same sequence and the upper bound for B_X gives lower exponent at least $\delta - \epsilon - (1-r)\delta = r\delta - \epsilon$. This proves (90.8) for every $r > 0$.

For the discrete statement, use the measure $\sum_{n_0 \leq n \leq N} n^{-1} \delta_n$, whose total mass is $O(\log N)$. Its first positive moment has exponent $k\delta$ by (90.4), and its largest sample is $O_{k,\delta}(N^{k\delta}/\log N)$ by (90.2). The identical two inequalities, with integrals replaced by sums, prove (90.9).

The positive exponents force both undamped masses to diverge when $\delta > 0$, proving (90.10). The pointwise amplitude bound makes $\mathscr{J}_r(c)$ finite for $c > r\delta$. If it were finite at $0 < c < r\delta$, then $\mathscr{M}_r(X) \leq X^c \mathscr{J}_r(c)$ would contradict (90.8). For $c \leq 0$, convergence would imply convergence of the undamped mass, also impossible.

Finally, write the sample Dirichlet series as $\sum [(\mathcal{A}(n^k)^+)^r / n] n^{-(s-1)}$. The amplitude bound gives absolute convergence for $\Re s > 1 + rk\delta$. For real $1 < s < 1 + rk\delta$, convergence would imply $\mathscr{S}_{r,k}(N) = O(N^{s-1})$, contradicting its growth exponent. At $s \leq 1$ the series dominates the divergent undamped sample mass. Nonnegative coefficients now identify its convergence abscissa. Under RH the series is a finite Dirichlet polynomial, as asserted. □

90.2 A critical endpoint that closes for every power above one

The preceding abscissa does not determine convergence at $c = r\delta$. The integrated explicit formula provides enough information to analyze that boundary. In the rest of this subsection suppose $\delta > 0$, and use the profile from Theorem 79.3:

$$F(u) := u e^{-\delta u} \mathcal{A}(e^u) = T(u) + o(1), \qquad T(u) = -\sum_{\Re\rho = \Theta} \frac{m_\rho e^{i \Im \rho u}}{\rho(\rho - 1/2)}. \tag{90.11}$$

The sum runs over distinct zero locations, with multiplicities m_ρ. It is absolutely and uniformly convergent. If no zero attains Θ, it is the empty sum. The function T is real, uniformly almost periodic, and has mean zero. For $r > 0$, put

$$\mu_r = \operatorname{mean}\big((T^+)^r\big). \tag{90.12}$$

The mean exists: uniformly approximate the continuous function $y \mapsto (y^+)^r$ on the bounded range of T by polynomials, and approximate T uniformly by trigonometric polynomials.

Theorem 90.2 (Critical endpoint hierarchy)**.** *For every fixed $r > 1$, unconditionally,*

$$\mathscr{J}_r(r\delta) < \infty, \qquad \sum_{n \ge n_0} \frac{\big(\mathcal{A}(n^k)^+\big)^r}{n^{1+rk\delta}} < \infty. \tag{90.13}$$

In the case $\delta > 0$, set $U = \log X$. Then

$$\int_2^X \big(\mathcal{A}(x)^+\big)^r x^{-r\delta-1} dx = \frac{\mu_r}{1-r} U^{1-r} + o(U^{1-r}) \qquad (0 < r < 1), \tag{90.14}$$

$$\int_2^X \mathcal{A}(x)^+ x^{-\delta-1} dx = \mu_1 \log U + o(\log U), \tag{90.15}$$

$$\int_X^\infty \big(\mathcal{A}(x)^+\big)^r x^{-r\delta-1} dx = \frac{\mu_r}{r-1} U^{1-r} + o(U^{1-r}) \qquad (r > 1). \tag{90.16}$$

Moreover, $\mu_r > 0$ if and only if at least one zero has real part exactly Θ. If the supremum is not attained, the formulas give the corresponding little-o estimates. In that case they do not decide convergence at $0 < r \le 1$.

Proof. If RH holds, both quantities in (90.13) have only a compact or finite contribution. Suppose $\delta > 0$. The amplitude bound yields

$$\big(\mathcal{A}(x)^+\big)^r x^{-r\delta-1} \ll_{r,\delta} \frac{1}{x \log^r x}, \qquad \frac{\big(\mathcal{A}(n^k)^+\big)^r}{n^{1+rk\delta}} \ll_{r,k,\delta} \frac{1}{n \log^r n}.$$

These are integrable and summable, respectively, for $r > 1$.

To obtain the asymptotics, change variables $x = e^u$. The critical integrand becomes

$$\big(\mathcal{A}(e^u)^+\big)^r e^{-r\delta u} du = \big(F(u)^+\big)^r u^{-r} du. \tag{90.17}$$

The function F is bounded on $[u_0, \infty)$, by local continuity and (90.11). Uniform continuity of positive powers on a bounded interval implies

$$\big(F(u)^+\big)^r = (T(u)^+)^r + o(1).$$

Let $h(u) = (T(u)^+)^r$. Its primitive satisfies

$$\int_{u_0}^U h(u) du = \mu_r U + o(U).$$

For $0 < r < 1$, integration by parts against u^{-r} gives

$$\int_{u_0}^U h(u) u^{-r} du = \frac{\mu_r}{1-r} U^{1-r} + o(U^{1-r}).$$

For $r = 1$ the same calculation gives $\mu_1 \log U + o(\log U)$. For $r > 1$, integrate by parts on $[U, V]$ and send V to infinity; boundedness of h and $r > 1$ justify the limit and give

$$\int_U^\infty h(u) u^{-r} du = \frac{\mu_r}{r-1} U^{1-r} + o(U^{1-r}).$$

In each case the replacement error $o(1)$ contributes the indicated little-o weighted integral. For example, on a tail it is bounded by an arbitrary small constant times $\int_U^\infty u^{-r}du$; the initial segment is harmless in the other two cases. This proves (90.14)–(90.16).

If the edge contains no zero, $T = 0$ and all its means vanish. If it contains a zero, the coefficient at its frequency is nonzero; uniqueness of Fourier coefficients for an absolutely convergent trigonometric series makes T nonzero. A nonzero real uniformly almost-periodic function of mean zero has a strictly positive interval. Relatively dense almost periods repeat a smaller positive interval with fixed height and bounded gaps. Hence $(T^+)^r$ has positive mean for every $r > 0$. This proves the last assertion without an independence assumption on zero ordinates. □

Remark 90.3 (The first-power endpoint remains different). Theorem 90.2 closes the critical $r > 1$ endpoints even in the case $\delta > 0$. Therefore their finiteness cannot exclude zeros on the supremal line. When that line is attained, the first-power integral diverges logarithmically in $\log X$, whereas every higher-power critical integral converges. The original undamped target is still $\mathscr{J}_1(0) < \infty$, not any of these automatically finite damped endpoints.

90.3 Recovering the edge through a limit of nonlinear norms

There is nevertheless a precise relation between the convergent $r > 1$ endpoints and the first-power obstruction. The parameter limit must be taken with its normalization, rather than by exchanging an unbounded integral with a pointwise limit.

Theorem 90.4 (Renormalized critical-moment limit). *Suppose $\delta > 0$. With μ_1 defined by (90.12),*

$$\lim_{r\downarrow 1}(r-1)\mathscr{J}_r(r\delta) = \mu_1. \tag{90.18}$$

The limit is positive exactly when the supremal zero line is attained. Under RH the same normalized integral tends to zero, but the positive-δ profile (90.11) is not applicable in that case.

Proof. Assume first $\delta > 0$ and put $h(u) = F(u)^+$. This is a bounded nonnegative function, and (90.11) shows that it has Cesàro mean μ_1. Choose a finite bound B for h. For $0 < \epsilon \le 1/2$, the mean value theorem in the exponent gives

$$\sup_{0\le y\le B} |y^{1+\epsilon} - y| \le C_B\epsilon. \tag{90.19}$$

Indeed, $y^q \log y$, with value zero at $y = 0$, is bounded uniformly for $0 \le y \le B$ and $1 \le q \le 3/2$.

Using (90.17) with $r = 1 + \epsilon$,

$$\epsilon\mathscr{J}_{1+\epsilon}((1+\epsilon)\delta) = \epsilon\int_{u_0}^\infty h(u)^{1+\epsilon}u^{-1-\epsilon}du.$$

By (90.19), replacing $h^{1+\epsilon}$ by h changes the right side by at most

$$C_B\epsilon^2\int_{u_0}^\infty u^{-1-\epsilon}du = C_B\epsilon u_0^{-\epsilon} = O_B(\epsilon).$$

This estimate is uniform as ϵ decreases to zero.

Write $H_0(U) = \int_{u_0}^U h(u)du = \mu_1 U + o(U)$. Integration by parts is justified by boundedness of h and gives

$$\epsilon\int_{u_0}^\infty h(u)u^{-1-\epsilon}du = \epsilon(1+\epsilon)\int_{u_0}^\infty H_0(u)u^{-2-\epsilon}du.$$

The contribution of $\mu_1 u$ tends to μ_1. For the error, given $\eta > 0$, choose U_1 so large that $|H_0(u) - \mu_1 u| \le \eta u$ for $u \ge U_1$. Its initial contribution tends to zero, and its tail has absolute value at most

$$\eta\epsilon(1+\epsilon)\int_{U_1}^\infty u^{-1-\epsilon}du = \eta(1+\epsilon)U_1^{-\epsilon}.$$

Letting ϵ decrease to zero and then η to zero proves (90.18). The positivity characterization follows from Theorem 90.2.

Under RH, $\mathcal{A}^+$ is bounded and compactly supported. The integrals $\mathscr{J}_r(0)$ are uniformly bounded for $1 \le r \le 3/2$, by an elementary bound on that compact interval. Multiplication by $r-1$ therefore gives zero. This separate argument avoids using the positive-δ asymptotic in the RH case. □

90.4 Increasing thresholds and an exact hierarchy of zero strips

The fixed-level sign count can be sharpened to increasing thresholds. Fix $k > 0$, a real number $\lambda \ge 0$, and an integer n_0 with $n_0^k \ge 3$. Define

$$\mathscr{E}_{k,\lambda}(N) = \#\{n_0 \le n \le N : \mathcal{A}(n^k) > n^\lambda\}, \qquad \sigma = k\delta. \tag{90.20}$$

Every parameter is fixed before N tends to infinity.

Theorem 90.5 (Sharp transition for growing-threshold counts)**.** *If $\lambda \ge \sigma$, the count $\mathscr{E}_{k,\lambda}(N)$ is eventually constant. If $0 \le \lambda < \sigma$, then*

$$\limsup_{N\to\infty} \frac{\log(1+\mathscr{E}_{k,\lambda}(N))}{\log N} = 1. \tag{90.21}$$

In the second case, for every $\epsilon > 0$, infinitely many dyadic values N satisfy

$$\mathscr{E}_{k,\lambda}(2N) - \mathscr{E}_{k,\lambda}(N) \ge N^{1-\epsilon}. \tag{90.22}$$

The finite-count conclusion includes the boundary $\lambda = \sigma$.

Proof. Under RH all these counts are eventually constant by eventual negativity of $\mathcal{A}$. Suppose $\delta > 0$. From (90.2),

$$|\mathcal{A}(n^k)| \le C_{k,\delta} \frac{n^\sigma}{\log n} \tag{90.23}$$

for sufficiently large n. If $\lambda \ge \sigma$, its right side is strictly smaller than n^λ once $\log n > C_{k,\delta}$. This proves the finite-count conclusion, including equality of the exponents.

Suppose $0 \le \lambda < \sigma$, and put $b_n = (\mathcal{A}(n^k) - n^\lambda)^+$. Then

$$0 \le \sum_{n_0\le n\le N} \frac{\mathcal{A}(n^k)^+ - b_n}{n} \le \sum_{n_0\le n\le N} n^{\lambda-1} = \begin{cases} O(\log N), & \lambda = 0, \\ O_\lambda(N^\lambda), & \lambda > 0. \end{cases}$$

The subtracted term has strictly smaller growth exponent than σ. Equation (90.4) consequently gives

$$\limsup_{N\to\infty} \frac{\log\Big(1 + \sum_{n_0\le n\le N} b_n/n\Big)}{\log N} = \sigma. \tag{90.24}$$

For clarity, the upper bound follows from $b_n \le \mathcal{A}(n^k)^+$. For the lower bound, along a sequence where the original mass is at least $N^{\sigma-\eta}$, choose $\eta < \sigma - \lambda$; the error is smaller by a positive power and cannot change its limsup exponent.

For every $0 < \eta < \sigma$, infinitely many dyadic N satisfy

$$\sum_{N<n\le 2N} \frac{b_n}{n} \ge N^{\sigma-\eta}.$$

Otherwise summing the eventual strict upper bounds over dyadic blocks would give cumulative exponent at most $\sigma - \eta$, contradicting (90.24). On each such block, (90.23) gives

$$\frac{b_n}{n} \le C'_{k,\delta} \frac{N^{\sigma-1}}{\log N} \mathbf{1}_{\{\mathcal{A}(n^k) > n^\lambda\}}.$$

Hence the block count is at least $c_{k,\delta} N^{1-\eta} \log N$. Choosing $0 < \eta < \min(\epsilon, \sigma)$ proves (90.22) for sufficiently large members of the subsequence. These bounds imply the lower bound 1 in (90.21); the reverse bound is immediate from $\mathscr{E}_{k,\lambda}(N) \le N$. □

Corollary 90.6 (A quantitative strip criterion using only threshold signs)**.** *Fix $k > 0$, $\lambda \ge 0$, and a real s with $0 < s < 1$. The following assertions are equivalent:*

1. *Every nontrivial zeta zero satisfies $\Re\rho \le 1/2 + \lambda/k$.*
2. *Only finitely many n satisfy $\mathcal{A}(n^k) > n^\lambda$.*
3. *For some $\epsilon > 0$ and $C < \infty$, $\mathscr{E}_{k,\lambda}(N) \le C N^{1-\epsilon}$ for every sufficiently large N.*
4. *The nonnegative series*

$$\sum_{\substack{n \ge n_0 \\ \mathcal{A}(n^k) > n^\lambda}} \frac{1}{n^s}$$

converges.

For $\lambda = 0$, this is an RH criterion at the fixed level 1. For $\lambda > 0$, it gives a hierarchy of exact zero-strip criteria.

Proof. Assertion 1 says exactly $\sigma = k\delta \le \lambda$. Theorem 90.5 identifies this with assertion 2. Assertion 2 implies both assertions 3 and 4. If assertion 3 held with $\lambda < \sigma$, it would contradict the full count exponent in (90.21); hence 3 implies 1. Finally, convergence in assertion 4 gives

$$\mathscr{E}_{k,\lambda}(N) \le N^s \sum_{\substack{n \ge n_0 \\ \mathcal{A}(n^k) > n^\lambda}} n^{-s} = O(N^s).$$

Since $s < 1$, assertion 3 follows. This proves all equivalences. □

The precise additional conclusion. The nonlinear moment exponents are now determined, every critical power $r > 1$ has finite mass, the normalized $r \downarrow 1$ limit recovers the attained-edge mean, and increasing sign thresholds give a sharp hierarchy of zero-strip tests. Each is a theorem deduced from established inputs in this work. No fixed power saving for the actual positive-index count has been proved here. In particular, the equivalences in Corollary 90.6 do not themselves improve the unconditional zero-free region.

Part VII

Extremal polynomials, finite certificates, and theta obstructions

Dependency convention for this part. Every application below of a quoted fixed strip $\Theta < 7/8$ is conditional on the explicit input (H_b) in (1.7). In particular this applies to automatic finiteness of the eighth phase moment, the actual-prime degree-four limit, and the cubic-coefficient target. Finite linear-algebra identities, abstract theorems with a stated moment hypothesis, the Li-coordinate identity, and the RH-conditional tail laws do not use that input. Statements about reported 2026 preprints document the dependency; they are not an independently certified zero-free theorem of this manuscript.

91 Consequences of a fixed zero-free half-plane

91.1 A theorem with an explicit external input

Let b be any fixed number with $1/2 < b < 1$, and assume the zero-boundary estimate $\Theta \le b$. This subsection first proves a general implication, so that the external analytic input is visible and can be replaced independently of the geometry. Set

$$\alpha_b = 2b - 1, \qquad q_b = \frac{1}{1-b}, \qquad v_n = w_n p_n^{-\alpha_b},$$

and define a modified deficit by

$$D_N^{[b]} = \sum_{n=2}^{N} v_n - \left| \sum_{n=2}^{N} v_n U_n \right|. \tag{91.1}$$

The unit vectors are exactly those in Section 7; only their weights have changed.

Theorem 91.1 (Closure supplied by a fixed strip)**.** *If $\Theta \le b$, where $1/2 < b < 1$, then*

$$E(x) = O_b\left(\frac{x^{b-1}}{\log x} \right), \tag{91.2}$$

$$D_N \ll_b \frac{p_N^{\alpha_b}}{(\log p_N)^2}, \tag{91.3}$$

$$\int_2^\infty |E(x)|^{q_b}\, \mathrm{d}x < \infty, \qquad \int_2^\infty x^{-\alpha_b} E(x)^2\, \mathrm{d}x < \infty, \tag{91.4}$$

$$\sup_N D_N^{[b]} < \infty. \tag{91.5}$$

The same L^q conclusion holds for every finite $q \ge q_b$. An admissible finite upper bound in (91.5) is

$$K_b = 2\pi^2 \sum_{n \ge 2} w_n p_n^{-\alpha_b} E(p_n)^2. \tag{91.6}$$

No numerical value or effective constant for K_b is asserted.

Proof. If $\Theta > 1/2$, Grosswald's estimate (2.7) and partial summation give (91.2). If $\Theta = 1/2$, the classical RH bound $E(x) = O(x^{-1/2} \log x)$ is absorbed by the right side of (91.2) because $b > 1/2$.

For any positive weights t_n, using $u_\infty = e^{2\pi \mathrm{i} B_{\mathbb{P}}}$ and $1 - \cos t \le t^2/2$,

$$\sum t_n - \left| \sum t_n U_n \right| \le \sum t_n \{1 - \Re(U_n \overline{u_\infty})\} \le 2\pi^2 \sum t_n E(p_n)^2. \tag{91.7}$$

The neighboring prime ratios are bounded by Bertrand's postulate. Comparison on the adjacent prime gaps consequently gives

$$\sum_{n=2}^{N} \frac{w_n p_n^{2b-2}}{(\log p_n)^2} \ll_b 1 + \int_3^{p_{N+1}} \frac{x^{2b-2}}{(\log x)^2}\,\mathrm{d}x \ll_b \frac{p_N^{2b-1}}{(\log p_N)^2},$$

and

$$\sum_{n\geq 2} \frac{w_n}{p_n(\log p_n)^2} < \infty. \tag{91.8}$$

For completeness, on each adjacent gap the ratios of the endpoint values of either positive comparison function are bounded; the gap intervals have overlap at most two. This proves the stated sum-integral comparisons. Apply (91.7) first with $t_n = w_n$ and then with $t_n = v_n$. The second application is bounded by (91.8), since $-\alpha_b + 2b - 2 = -1$.

At the two integral endpoints in (91.4), (91.2) gives respectively the integrable majorants $1/[x(\log x)^{q_b}]$ and $1/[x(\log x)^2]$. Boundedness of E then includes every larger finite q. □

91.2 The reported 2026 input and its exact consequences

OpenAI's September 2026 preprint and accompanying formalization state a uniform zero-free region $\Re s > 7/8$ for zeta and the specified Dirichlet and Hecke families [49]. Liu's subsequent manuscript states the sharper endpoint bound $\Theta \leq b_*$, where

$$b_* = \frac{18\xi_* + 7}{18\xi_* + 9} = 0.87495701942009894612860385056145\ldots, \tag{91.9}$$

and $\xi_* \in (0.38858, 0.38859)$ is the root of

$$1188\xi_*^3 - 2160\xi_*^2 + 171\xi_* + 190 = 0.$$

This is Theorem 27.1 of [13]. The zero-free region is open; nonvanishing on its boundary is not claimed.

Verification recorded in the source note. The earlier source note records inspection of the full primary manuscript, public formalization scope, and original exported Lean theorem signatures on 8 November 2026. The present consolidation does not independently rebuild those developments and retains their applications under (H_b). The sources are preprints with supplied formalizations; journal peer review was not established here. The conclusions below are explicit deductions from their stated zero-free theorem, not an independent proof of that theorem. In particular, they use no RH assumption once that input is admitted.

Applying Theorem 91.1 gives

$$\alpha_* := 2b_* - 1 = 0.74991403884019789225720770112291\ldots, \tag{91.10}$$

$$q_* := \frac{1}{1-b_*} = 7.99725018839431200117685191958404\ldots, \tag{91.11}$$

$$D_N \ll \frac{p_N^{\alpha_*}}{(\log p_N)^2}, \qquad \sup_N D_N^{[b_*]} < \infty. \tag{91.12}$$

In particular $E \in L^8$. For the centered circle $Z(x) = e^{2\pi \mathrm{i} E(x)}$, this proves

$$\int_2^{\infty} |(Z(x)-1)^4|^2\,\mathrm{d}x < \infty. \tag{91.13}$$

Its tail is

$$O\left(\frac{X^{-\delta_8}}{(\log X)^8}\right), \qquad \delta_8 = 7 - 8b_* = 0.00034384463920843097\ldots > 0.$$

Target	Necessary and sufficient boundary	Consequence of $\Theta \le b_*$
Original D_N, ordinary circle or fixed ellipse	$\Theta \le 1/2$	Uniform bound unresolved
Energy $\int \lvert Z-1\rvert^4$	$\Theta \le 3/4$	Not implied
Energy $\int \lvert Z-1\rvert^6$	$\Theta \le 5/6$	Not implied
Energy $\int \lvert Z-1\rvert^8$	$\Theta \le 7/8$	Finite
Energy $\int x^{-\alpha_*}E^2$	$\Theta \le b_*$	Finite, including endpoint
Modified deficit $D_N^{[b_*]}$	Weights specified in (91.1)	Uniformly bounded

Table 3: What the fixed zero-free strip closes. The first five boundary tests follow from the exact spectra proved earlier. The modified deficit is bounded by Theorem 91.1.

This fourth-power transformation changes the local contact order and therefore changes the criterion. It does not establish finite ordinary circle energy, or finite energy for a nondegenerate ellipse.

The cited manuscript reports, in Theorem 34.4 and Corollary 34.5, optimality of b_* within the explicitly specified moment, length, zero-detector, and contour comparison system [13]. According to that reported result, further tuning or iteration within that same system cannot reach $1/2$. This is a limitation of that collection of estimates, not a universal impossibility theorem for arithmetic methods. A stronger analytic estimate is still needed to bound the original D_N by an absolute constant.

92 The original target and the inputs retained

The target is the original, unmodified inequality

$$\text{there exists a finite } K \text{ such that } D_N \le K \text{ for every } N.$$

It concerns the original prime phases and original gap weights below. The three supplied manuscripts are denoted by [50], [51], and [52]. The prior expanded note is [53]. This module develops the additional steps that their combination permits. It distinguishes unconditional functional-analytic statements, consequences of a reported fixed-strip theorem, and results conditional on RH itself.

92.1 Definitions and exact geometric identities

Write p_n for the n-th prime, with $p_1 = 2$, and put

$$S(x) = \sum_{p \le x} \frac{1}{p}, \quad E(x) = S(x) - \log\log x - B_{\mathbb{P}}, \quad B_{\mathbb{P}} = \gamma - \sum_{p} \sum_{k \ge 2} \frac{1}{kp^k}.$$

Mertens' theorem and the prime number theorem give $E(x) \to 0$. The vectors and weights, indexed from $n = 2$, are

$$U_n = e^{2\pi i(S(p_n) - \log\log p_n)}, \quad a = e^{2\pi i B_{\mathbb{P}}}, \quad w_n = p_{n+1} - p_{n-1}, \quad W_N = p_N + p_{N+1} - 5.$$

Thus $U_n \to a$, $|U_n| = |a| = 1$, and no value of $B_{\mathbb{P}}$ is needed to compute the U_n. Define

$$M_N = \sum_{n=2}^{N} w_n U_n, \qquad m_N = M_N / W_N,$$
$$D_N = W_N - |M_N|, \qquad V_N = W_N - |M_N|^2 / W_N,$$
$$J_{k,N} = \sum_{n=2}^{N} w_n |U_n - a|^{2k}.$$

The letter a here denotes a limiting direction; in the finite-field elliptic-curve subsection it denotes the usual Frobenius trace instead.

Lemma 92.1 (Variance and the limiting center)**.** *For any positive weighted unit vectors $u_n \to a$, with $W_N \to \infty$, the quantities defined above satisfy*

$$V_N = \min_{c \in \mathbb{C}} \sum_{n \le N} w_n |u_n - c|^2, \quad D_N \le V_N \le 2D_N, \quad J_{1,N} = V_N + W_N |m_N - a|^2.$$

Both D_N and V_N are nondecreasing, and in the extended real sense

$$\lim_N V_N = J_{1,\infty}, \qquad \lim_N D_N = \frac{1}{2} J_{1,\infty}.$$

If these limits are finite, then

$$2(D_\infty - D_N) = J_{1,\infty} - J_{1,N} + W_N |m_N - a|^2 - D_N^2 / W_N.$$

Proof. Complete the square about m_N. Since $|u_n| = 1$, $V_N = D_N(1 + |m_N|)$ and $V_N = 2D_N - D_N^2/W_N$. Adding another point increases the minimum squared error; the triangle inequality proves monotonicity of D_N. Weighted convergence gives $m_N \to a$. For any fixed prefix L, its error about m_N tends to $J_{1,L}$, and that error is bounded above by V_N for $N \ge L$. Therefore $J_{1,\infty} \le \lim_N V_N$. The reverse inequality follows from $V_N \le J_{1,N}$. Divide by $1 + |m_N| \to 2$ and then substitute the finite limiting identity to obtain the final formula. □

92.2 The analytic baseline, including endpoints

Let

$$\Theta = \sup\{\Re\rho : \zeta(\rho) = 0,\ 0 < \Re\rho < 1\}.$$

The functional equation gives $\Theta \ge 1/2$. We recall the precise parts of the supplied energy theory that are needed here, together with their proof mechanism.

Theorem 92.2 (The original energy criterion)**.** *For the original prime phases and weights,*

$$\text{RH} \iff \int_2^\infty E(x)^2 \, \mathrm{d}x < \infty \iff J_{1,\infty} < \infty \iff \sup_N D_N < \infty.$$

Theorem 92.3 (The moment and sampling input)**.** *For each real $q > 1$,*

$$\int_2^\infty |E(x)|^q \, \mathrm{d}x < \infty \iff \sum_{n \ge 2} w_n |E(p_n)|^q < \infty \iff \Theta \le 1 - 1/q. \tag{92.1}$$

In particular no moment with $1 < q < 2$ is finite. For each integer $k \ge 1$,

$$J_{k,\infty} < \infty \iff \Theta \le 1 - \frac{1}{2k}, \tag{92.2}$$

including equality at the endpoint. For a general convergent unit-vector sequence the pair energy

$$C_{k,N} = \frac{1}{W_N} \sum_{2 \le j < l \le N} w_j w_l |u_j - u_l|^{2k}$$

has the extended-real limit $J_{k,\infty}$.

Proof of the analytic and sampling assertions. For $\Re s > 1$, put $P(s) = \sum_p p^{-s}$ and $F(s) = \int_2^\infty E(x)x^{-s}\,\mathrm{d}x$. Euler products and partial summation give

$$F(s) = \frac{\log((s-1)\zeta(s)) - R_0(s) - J_0(s)}{s-1}, \tag{92.3}$$

where

$$R_0(s) = \sum_{k\ge 2} \frac{P(ks)}{k},$$

$$J_0(s) = (\log\log 2 + B_{\mathbb{P}})2^{1-s} - \gamma - \log\log 2 - \sum_{j\ge 1} \frac{[-(s-1)\log 2]^j}{j\,j!}.$$

R_0 is holomorphic in $\Re s > 1/2$, J_0 is entire, and $s = 1$ is removable in (92.3). An off-critical zero ρ with multiplicity m creates the local term $m\log(s-\rho)/(s-1)$, which is not holomorphic. At $s = 1/2$, the prime-square term produces $F(1/2+\varepsilon) = -\log\varepsilon + O(1)$.

If $E \in L^q$, Hölder's inequality makes its Mellin integral holomorphic for $\Re s > 1 - 1/q$, including differentiation on compact subsets. For $q \ge 2$, this excludes all zeros to the right of $1 - 1/q$. For $1 < q < 2$, it would also remove the prime-square singularity, which is impossible. This proves necessity.

For sufficiency at $q = 2$, the conditional Cramér bound [54, eq. (1)], applied after partial summation, gives

$$\int_X^{2X} E(x)^2\,\mathrm{d}x \ll (\log X)^{-2}.$$

A direct derivation of this local estimate is included in Lemma 96.4; summing dyadic intervals proves finiteness. Under RH, the pointwise bound $E(x) = O(x^{-1/2}\log x)$ also gives every $q > 2$. If $1/2 < \Theta < 1$, Grosswald's estimate, as stated precisely in Zhao [55, eq. (24)], gives

$$\pi(x) - \mathrm{li}(x) = O_\Theta(x^\Theta/\log x), \qquad E(x) = O_\Theta(x^{\Theta-1}/\log x).$$

The second estimate follows from the exact identity

$$E(x) = \frac{\pi(x) - \mathrm{li}(x)}{x} - \int_x^\infty \frac{\pi(t) - \mathrm{li}(t)}{t^2}\,\mathrm{d}t.$$

If $q(1-\Theta) \ge 1$, it is integrable to power q; at equality the integral is $\int dx/(x\log^q x)$, finite because $q > 1$. The case $\Theta = 1$ satisfies neither side of (92.1).

For sampling, let $p < r$ be consecutive primes, $h = r - p$, $e = E(p)$, and $b = E(r) - 1/r = E(r^-)$. On this gap the only variation is $-\log\log x$. Set $v = \log\log x$; the Jacobian $x\log x$ varies by a factor at most four because $r < 2p$. The integral of the absolute q-th power of a real linear function on a unit interval is at least the sum of its two endpoint powers divided by $2(q+1)$. Therefore

$$\frac{h}{8(q+1)}(|e|^q + |b|^q) \le \int_p^r |E(x)|^q\,\mathrm{d}x \le h(|e|^q + |b|^q).$$

The endpoint jump changes the summed comparison by a finite term: $\sum h/r^q \le \int_2^\infty x^{-q}\,\mathrm{d}x < \infty$. Adding the two adjacent gap contributions gives exactly the weights w_n. Finally $|U_n - a| = 2|\sin(\pi E(p_n))| \sim 2\pi|E(p_n)|$. This proves the claimed sampling and even-moment equivalences. The variance lemma supplies the last part of Theorem 92.2. □

Proof of the pair-energy limit. For $p = 2k$ put $d_n = u_n - a$. The inequality $|d_j - d_l|^p \le 2^{p-1}(|d_j|^p + |d_l|^p)$ gives $C_{k,N} \le 2^{p-1}J_{k,N}$. Fix a finite prefix L. For each $j \le L$, the weighted mean of $|u_j - u_l|^p$ over $L < l \le N$ tends to $|u_j - a|^p$; hence the prefix–tail pairs tend to $J_{k,L}$. This proves $\liminf C_{k,N} \ge J_{k,L}$, and then the required divergence if $J_{k,\infty} = \infty$. When it is finite, the pairs with both indices larger than L are at most $2^{p-1}(J_{k,\infty} - J_{k,L})$; the finite prefix–prefix pairs divided by W_N vanish. Let first N and then L tend to infinity to obtain the stated limit. □

92.3 The separately identified fixed-strip input

Some conclusions below require the existence of a finite fourth even moment. A sufficient input is $\Theta < 7/8$. The reported 2026 theorem of Liu [13, Theorem 27.1] states

$$\Theta \le b_* = \frac{18\xi + 7}{18\xi + 9} = 0.874957019420098946\ldots,$$

where $\xi \in (0.38858, 0.38859)$ solves $1188\xi^3 - 2160\xi^2 + 171\xi + 190 = 0$. The source is a preprint extending the reported quasi-RH argument. The theorem statement and scope have been checked against its primary text; the underlying proof and formal verification have not been independently rebuilt here. We explicitly admit this as an external input when applying the degree-four results to the actual primes. The general extremal-polynomial theorem, endpoint Li identity, finite spectral identities, and RH-conditional tail theorem do not depend on that 2026 input.

Under this input, $J_{4,\infty} < \infty$ and $H_{4,N} \le J_{4,\infty}$, where $H_{4,N}$ is the least squared norm of a monic degree-four polynomial on the prime measure. What does not follow is $J_{1,\infty} < \infty$. The next section identifies the missing condition within the limiting minimizing polynomial itself.

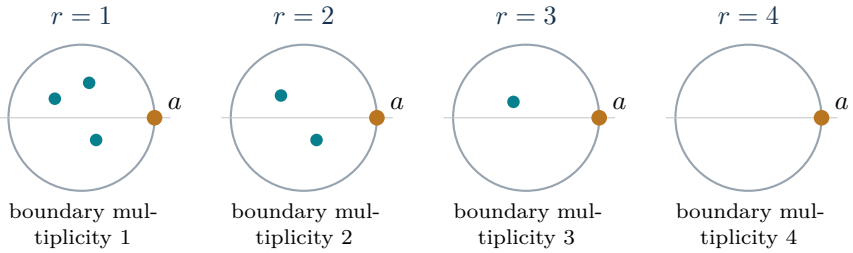


Figure 4: The possible multiplicities of the unique boundary root of the degree-four limiting extremizer. Gold marks the limiting direction; teal marks the remaining interior roots. This is a schematic of the theorem, not a numerical plot of the prime polynomial. Under the stated fixed-strip input, RH is exactly the first case.

93 The limiting extremal polynomial and its boundary multiplicity

The boundedness hierarchy for monic polynomial norms can be sharpened to a theorem about the actual minimizing polynomials. This gives an exact location for the remaining obstruction inside a degree-four polynomial whose norm is already bounded under the stated fixed-strip input. It also gives an explicit inverse bound for the original deficit if that obstruction is removed.

We first state the result for a general sequence. Let $u_n \to a$ with $|u_n| = |a| = 1$, let $w_n > 0$, and suppose $W_N = \sum_{n=2}^{N} w_n \to \infty$. Write

$$J_{k,N} = \sum_{n=2}^{N} w_n |u_n - a|^{2k}, \qquad J_{k,\infty} = \sum_{n\ge 2} w_n |u_n - a|^{2k},$$

and, for monic polynomials of degree m, put

$$H_{m,N} = \min_{P(z)=z^m+\sum_{j<m} c_j z^j} \sum_{n=2}^{N} w_n |P(u_n)|^2.$$

Once at least m distinct sampled points have occurred, the minimizer is unique; denote it by $P_{m,N}$.

Theorem 93.1 (Actual extremizer and exact boundary multiplicity)**.** *Assume that the set of sampled points is infinite and that $J_{m,\infty} < \infty$. Define*

$$r = \min\{k \in \{1, \dots, m\} : J_{k,\infty} < \infty\}, \qquad \nu_r = \sum_{n\geq 2} w_n |u_n - a|^{2r} \delta_{u_n}.$$

Then ν_r is a finite positive measure with infinite support. The polynomials $P_{m,N}$ converge coefficient by coefficient to the unique polynomial

$$P_{m,\infty}(z) = (z-a)^r Q_{m-r}(z), \tag{93.1}$$

where Q_{m-r} is the monic orthogonal polynomial of degree $m-r$ for ν_r; when $m = r$, set $Q_0 = 1$. Moreover,

$$H_{m,N} \uparrow H_{m,\infty} = \sum_{n\geq 2} w_n |P_{m,\infty}(u_n)|^2 = \min_{Q \text{ monic, } \deg Q = m-r} \int |Q|^2 \, d\nu_r > 0. \tag{93.2}$$

Every zero of Q_{m-r} lies strictly inside the unit disk. Consequently a is the only zero of $P_{m,\infty}$ on the unit circle, and its multiplicity is exactly r.

Proof. The minimum is nondecreasing in N, and the trial polynomial $(z-a)^m$ bounds it by $J_{m,\infty}$. Choose m distinct sampled points at fixed indices. Evaluation of the lower coefficients at these points is an invertible Vandermonde map. The bound on the objective bounds the value of each minimizing polynomial at each of these fixed points, and hence bounds all its lower coefficients.

Every subsequence therefore has a further coefficient-convergent subsequence, with a monic degree-m limit P. If $H_{m,N} \uparrow H$, passage to the limit on each fixed finite prefix gives

$$\sum_{n\geq 2} w_n |P(u_n)|^2 \leq H.$$

Conversely, for every monic degree-m polynomial R with finite full energy, one has

$$H_{m,N} \leq \sum_{n\leq N} w_n |R(u_n)|^2, \qquad H \leq \sum_{n\geq 2} w_n |R(u_n)|^2.$$

Thus every subsequential limit is a minimizer of the full energy and its energy is exactly H.

A nonzero polynomial R with $R(a) \neq 0$ has infinite energy, because its modulus is bounded below on a tail while $W_N \to \infty$. More generally, if $R(z) = (z-a)^s T(z)$ with $T(a) \neq 0$, its energy is finite precisely when $J_{s,\infty} < \infty$: on a tail $|T(u_n)|$ is bounded above and below by positive constants. Since $|u_n - a| \leq 2$, finite moments form an upper interval of integer orders. The monic degree-m polynomials of finite energy are therefore exactly

$$(z-a)^r Q(z), \qquad Q \text{ monic of degree } m-r.$$

Their full energy is $\int |Q|^2 d\nu_r$.

Deleting the possible atom at a leaves infinitely many distinct sampled points, so ν_r has infinite support. Its Gram matrix on any fixed finite-dimensional polynomial space is positive definite: a nonzero polynomial cannot vanish at infinitely many distinct points. The preceding minimization consequently has a unique minimizer, characterized by orthogonality to the lower-degree polynomials. This also makes its minimum strictly positive. All subsequential limits are this same polynomial, proving convergence of the entire sequence.

It remains to locate the zeros of $Q = Q_{m-r}$. If a zero ξ satisfies $|\xi| > 1$, replacing the factor $z - \xi$ by $z - 1/\overline{\xi}$ gives another monic polynomial whose modulus on the unit circle is smaller by the constant factor $|\xi|$. Its norm would be smaller, a contradiction. If $|\xi| = 1$, write $Q(z) = (z-\xi)R(z)$. Orthogonality to R gives

$$0 = \int (z - \xi) |R(z)|^2 \, d\nu_r(z).$$

Thus ξ would be the average of unit vectors with positive weights proportional to $|R|^2 d\nu_r$. An average has unit modulus only when all its mass is at that same unit vector; equivalently, integrating $|z-\xi|^2$ gives zero. This is impossible for a nonzero polynomial R and a measure with infinite support. All the zeros are therefore strictly interior, in particular $Q(a) \neq 0$. □

Proposition 93.2 (Exact remaining-energy identity)**.** *Under the hypotheses of Theorem 93.1, for every N at which the finite minimizer is unique,*

$$H_{m,\infty} - H_{m,N} = \sum_{n>N} w_n |P_{m,\infty}(u_n)|^2 + \sum_{n=2}^{N} w_n |P_{m,\infty}(u_n) - P_{m,N}(u_n)|^2. \tag{93.3}$$

Fix m distinct sampled points and let $\lambda_0 > 0$ be the smallest eigenvalue of their weighted Gram matrix on $1, z, \dots, z^{m-1}$. If $c_{m,N}$ and $c_{m,\infty}$ are the vectors of lower coefficients, then, after these points have occurred,

$$\|c_{m,N} - c_{m,\infty}\|_2^2 \leq \frac{H_{m,\infty} - H_{m,N}}{\lambda_0}. \tag{93.4}$$

Proof. The difference between two monic degree-m polynomials has degree at most $m-1$. Finite least-squares orthogonality therefore gives

$$\sum_{n\leq N} w_n |P_{m,\infty}(u_n)|^2 = H_{m,N} + \sum_{n\leq N} w_n |P_{m,\infty}(u_n) - P_{m,N}(u_n)|^2.$$

Add the tail and use (93.2). Keeping the fixed m points in the final sum bounds it below by λ_0 times the squared coefficient distance. □

The identity separates two quantities that a rate argument must control: the tail of the limiting polynomial and the finite projection error. Smallness of the former alone does not bound the latter.

93.1 What a degree-four boundary root would settle

Return to the original prime vectors and weights,

$$u_n = U_n = e^{2\pi i (S(p_n) - \log\log p_n)}, \quad a = e^{2\pi i B_{\mathbb{P}}}, \quad w_n = p_{n+1} - p_{n-1}.$$

Their support is infinite. Indeed, finite support together with $U_n \to a$ would imply $U_n = a$ eventually. Since $E(p_n) \to 0$, this would imply $E(p_n) = 0$ eventually, making the sampled $3/2$-moment finite. The sampling and integrability spectrum in Theorem 92.3 exclude this: no L^q energy with $0 < q < 2$ is finite.

Corollary 93.3 (A boundary multiplicity that encodes the zero strip)**.** *Whenever the prime $2m$-moment is finite, its limiting extremizer has boundary multiplicity*

$$r = \left\lceil \frac{1}{2(1-\Theta)} \right\rceil. \tag{93.5}$$

In particular, using the separately supplied zero-free input $\Theta \leq b_* < 7/8$, the degree-four extremizers converge to a well-defined polynomial $P_{4,\infty}$. Its only unit-circle zero is a, and the possibilities are exactly

multiplicity of a	*zero-boundary condition*
1	$\Theta = 1/2$ (RH)
2	$1/2 < \Theta \leq 3/4$
3	$3/4 < \Theta \leq 5/6$
4	$5/6 < \Theta \leq b_*$

Consequently,

$$\mathrm{RH} \iff P'_{4,\infty}(a) \neq 0. \tag{93.6}$$

The finite minimizing polynomials use no supplied value of $B_{\mathbb{P}}$; the boundary point appears as a root of their limit.

Proof. The input spectrum states $J_{k,\infty} < \infty$ exactly when $\Theta \leq 1 - 1/(2k)$, including the endpoints. The smallest integer satisfying this condition is (93.5). Apply Theorem 93.1, and use that the functional equation gives $\Theta \geq 1/2$. □

Corollary 93.4 (Exact equality and strict improvement of the limit)**.** *Let $C_{m,N}$ be the original normalized pair energy of order $2m$. If $J_{m,\infty} < \infty$, then*

$$H_{m,\infty} = \lim_N C_{m,N} = J_{m,\infty} \iff r = m.$$

For $r < m$, one has the strict inequality $H_{m,\infty} < \lim_N C_{m,N}$. Thus under the stated fixed-strip input,

$$H_{4,\infty} < \lim_N C_{4,N} \iff \Theta \leq 5/6. \tag{93.7}$$

Proof. The pair-energy limit is $J_{m,\infty}$. If $r = m$, the only admissible monic polynomial of finite energy is $(z-a)^m$. If $r < m$, the trial quotient $(z-a)^{m-r}$ has a boundary zero and therefore differs from the unique minimizing quotient, whose zeros are all strictly interior. Positive definiteness of the finite-measure Gram matrix makes the inequality of their squared norms strict. □

A finite positive difference between the two diagnostics is not by itself a certificate for (93.7): that statement compares their infinite limits. The projection term in (93.3) is among the quantities that must be controlled to turn it into a certified finite computation.

93.2 An inverse bound for the original deficit

Proposition 93.5 (Interior-root gaps control the original budget)**.** *In Theorem 93.1, suppose $r = 1$, and write*

$$P_{m,\infty}(z) = (z-a)\prod_{j=1}^{m-1}(z-\xi_j), \qquad g = \prod_{j=1}^{m-1}(1-|\xi_j|) > 0.$$

Empty products equal one. For the original deficit $D_N = W_N - |\sum_{n\leq N} w_n u_n|$, one has

$$\sup_N D_N = D_\infty = \frac{J_{1,\infty}}{2} \leq \frac{H_{m,\infty}}{2g^2}. \tag{93.8}$$

There is also a bound using the boundary derivative and a finite prefix. Suppose $m \geq 2$, put $d = |P'_{m,\infty}(a)| > 0$ and $s = m-1$, and choose L so that

$$|u_n - a| \leq \frac{d}{s2^s} \qquad (n > L).$$

Then

$$D_\infty \leq \frac{J_{1,L}}{2} + \frac{2H_{m,\infty}}{d^2} \leq 2W_L + \frac{2H_{m,\infty}}{d^2}. \tag{93.9}$$

For degree four the required tail radius is $d/24$.

Proof. For $|z| = 1$, the reverse triangle inequality gives $|\prod(z - \xi_j)| \geq g$. Hence $H_{m,\infty} \geq g^2 J_{1,\infty}$. The usual variance identity, or the limiting pair-energy identity at order two, gives $V_N \to J_{1,\infty}$. Since $V_N = D_N(1 + |\sum w_n u_n|/W_N)$ and the normalized resultant tends to a, it follows that $D_N \to J_{1,\infty}/2$. The triangle inequality makes D_N nondecreasing, which proves (93.8).

Let $Q(z) = \prod_{j=1}^{s}(z - \xi_j)$. On the closed unit disk, $|Q'(z)| \leq s2^{s-1}$. The line segment joining a and any u_n lies in that disk, so

$$|Q(u_n) - Q(a)| \leq s2^{s-1}|u_n - a|.$$

Because $|Q(a)| = d$, the stated tail radius ensures $|Q(u_n)| \geq d/2$ for $n > L$. Therefore

$$\sum_{n>L} w_n|u_n - a|^2 \leq \frac{4}{d^2}\sum_{n>L} w_n|P_{m,\infty}(u_n)|^2 \leq \frac{4H_{m,\infty}}{d^2}.$$

Add the prefix and divide by two; finally use $J_{1,L} \leq 4W_L$. □

The simple-root condition, or a positive lower bound for its derivative, is the remaining arithmetic assertion. Positivity of the finite Gram matrices proves neither. The inverse bounds show precisely how such an assertion would supply the original uniform constant, without changing the weights or replacing the objective by a higher moment.

Proposition 93.6 (A finite radial-gap certificate without centering)**.** *For a finite positive weighted unit-circle configuration, let $P_{m,N}(z) = \prod_{j=1}^{m}(z - \xi_{j,N})$ be its monic minimizing polynomial, and put $M_N = \sum_{n\leq N} w_n u_n$ and $m_N = M_N/W_N$. Choose one root $\xi_{1,N}$. If the other roots satisfy $|\xi_{j,N}| < 1$, define*

$$g_N = \prod_{j=2}^{m}(1 - |\xi_{j,N}|) > 0.$$

Then the original variance and deficit satisfy

$$V_N \leq \frac{H_{m,N}}{g_N^2}, \qquad D_N \leq \frac{H_{m,N}}{g_N^2(1 + |m_N|)}. \tag{93.10}$$

In particular, if those $m - 1$ roots lie in $|z| \leq 1 - \eta$, then

$$D_N \leq \frac{H_{m,N}}{\eta^{2m-2}(1 + |m_N|)}.$$

Every term in these finite bounds is obtained from the original vectors and weights; no limiting direction or centering constant is supplied.

Proof. For $|u_n| = 1$, factorization and the reverse triangle inequality give

$$|P_{m,N}(u_n)|^2 \geq g_N^2|u_n - \xi_{1,N}|^2.$$

The weighted variance minimizes the last squared-distance sum over all complex centers, so

$$H_{m,N} \geq g_N^2 \sum_{n\leq N} w_n|u_n - \xi_{1,N}|^2 \geq g_N^2 V_N.$$

Finally $V_N = D_N(1 + |m_N|)$. □

Corollary 93.7 (The second largest root is the remaining condition)**.** *For the original prime sequence, use the stated fixed-strip input that bounds $H_{4,N}$. Order the moduli of the four roots of $P_{4,N}$ as $\rho_{1,N} \geq \rho_{2,N} \geq \rho_{3,N} \geq \rho_{4,N}$. Then*

$$\text{RH} \iff \limsup_{N\to\infty} \rho_{2,N} < 1. \tag{93.11}$$

More concretely, if $H_{4,N} \leq B$ and $\rho_{2,N} \leq 1 - \eta$ for all sufficiently large N, then

$$\sup_N D_N \leq \frac{B}{2\eta^6}. \tag{93.12}$$

Proof. Coefficient convergence of monic polynomials implies convergence of their multisets of roots, including multiplicities, and hence of their ordered root moduli. By Corollary 93.3, under RH exactly one limiting root is on the unit circle; without RH at least two are. This proves (93.11).

Alternatively the sufficient implication follows immediately from Proposition 93.6, the bound on $H_{4,N}$, and the original deficit criterion. For its explicit constant, use $|m_N| \to 1$ in the finite bound and take the limit. Monotonicity of D_N makes its limit its supremum, so the eventual root-gap bound also controls every earlier prefix. □

Proposition 93.8 (A fourfold boundary limit with divergent deficit)**.** *Consider the independent sequence*

$$w_n = 1, \qquad u_n = \exp(in^{-1/7}) \quad (n \geq 1), \qquad \mathcal{D}_N = N - \left|\sum_{n=1}^{N} u_n\right|.$$

Its fourth extremal norms are bounded, but their limiting polynomial is

$$P_{4,\infty}(z) = (z-1)^4,$$

and

$$\mathcal{D}_N = \frac{7}{360} N^{5/7} + O(N^{3/7}). \tag{93.13}$$

Thus a uniform bound for the fourth extremal norm, positivity of the finite Gram matrices, and convergence of the phase vectors do not alone force a simple boundary root.

Proof. The support is infinite and $u_n \to 1$. Since $|u_n - 1| \sim n^{-1/7}$, the moment $\sum_n |u_n - 1|^{2k}$ is finite precisely when $2k > 7$. Its smallest finite integer order is therefore $r = 4$, so Theorem 93.1 gives the asserted limiting polynomial.

Elementary sums of powers and the Taylor remainders for sine and cosine give

$$\Re \sum_{n=1}^{N} u_n = N - \frac{7}{10} N^{5/7} + O(N^{3/7}),$$
$$\Im \sum_{n=1}^{N} u_n = \frac{7}{6} N^{6/7} + O(N^{4/7}).$$

Squaring and taking the square root yields

$$\left|\sum_{n=1}^{N} u_n\right| = N + \frac{1}{2}\left(-\frac{7}{5} + \frac{49}{36}\right) N^{5/7} + O(N^{3/7}) = N - \frac{7}{360} N^{5/7} + O(N^{3/7}).$$

This proves (93.13). □

This example is not the prime construction and is not a counterexample to RH. It isolates the additional arithmetic content required by the radial-gap criterion: in this example all four extremal roots approach the boundary point, so the required uniform gap fails.

93.3 A quantitative fourth-pair-energy rate under (H_b)

The pair energy has a rate that follows directly from a strict fixed-strip input; it does not require the simple-root condition.

Proposition 93.9 (Rate from a fixed strip)**.** *Suppose the original prime sequence satisfies*

$$|U_n - a| \ll \frac{p_n^{-\eta}}{\log p_n}, \qquad \frac{1}{8} < \eta < \frac{1}{7}.$$

Then, with $X = p_N$ and $\delta = 8\eta - 1 > 0$,

$$C_{4,N} = J_{4,\infty} + O\left(\frac{X^{-\delta}}{(\log X)^8}\right). \tag{93.14}$$

In particular the separately supplied bound $E(x) = O(x^{b_*-1}/\log x)$ gives

$$b_* = 0.87495701942009894612860385056145\ldots,$$
$$\delta = 7 - 8b_* = 0.00034384463920843097116919550 84\ldots.$$

The implicit constants include those in the analytic fixed-strip estimate; no certified numerical error bar follows without them.

Proof. Put $d_n = U_n - a$ and $S_{r,s,N} = \sum_{n\le N} w_n d_n^r \overline{d_n}^{\,s}$, with $S_{0,0,N} = W_N$. Expanding the two fourth powers gives the exact finite identity

$$C_{4,N} = \frac{1}{2W_N} \sum_{r,s=0}^{4} (-1)^{r+s} \binom{4}{r}\binom{4}{s} S_{4-r,4-s,N} S_{r,s,N}. \tag{93.15}$$

The terms $(r,s) = (0,0), (4,4)$ sum to $J_{4,N}$. For every other term the two total orders are integers ℓ and $8-\ell$, with $1 \le \ell \le 7$.

Bertrand's postulate bounds neighboring prime ratios. Comparison over the adjacent prime gaps, whose overlap is at most two, gives for $1 \le \ell \le 7$

$$\sum_{n\le N} w_n |d_n|^\ell \ll 1 + \int_3^{2X} \frac{x^{-\ell\eta}}{(\log x)^\ell}\,dx \ll \frac{X^{1-\ell\eta}}{(\log X)^\ell}.$$

Here $\ell\eta < 1$, and $W_N \asymp X$. Every non-endpoint term in (93.15) is consequently $O(X^{1-8\eta}/(\log X)^8)$. The same gap comparison on the tail gives

$$J_{4,\infty} - J_{4,N} \ll \int_X^\infty \frac{x^{-8\eta}}{(\log x)^8}\,dx \ll \frac{X^{1-8\eta}}{(\log X)^8}.$$

Combining these estimates proves the proposition. □

At the weaker boundary value $\eta = 1/8$, the same proof instead gives $C_{4,N} = J_{4,\infty} + O((\log X)^{-7})$: the non-endpoint terms are $O((\log X)^{-8})$, while the tail integral is of order $(\log X)^{-7}$. Neither rate is a bound for the degree-one deficit. The limiting-polynomial criterion above states the additional condition that would yield that bound.

94 The exactly centered Laguerre endpoint and Li's chart

The Laguerre scale $\alpha = 1/2$ places the expansion center at $s = 1$ and makes the disk coordinate exactly Li's coordinate. This endpoint is admissible for the exactly centered prime error. Its justification requires both absolute convergence of the individual coefficients and a genuine holomorphic germ at $s = 1$. We retain

$$E(x) = \sum_{p\le x} \frac{1}{p} - \log\log x - B_{\mathbb{P}}, \qquad F(s) = \int_2^\infty E(x) x^{-s}\,\mathrm{d}x \quad (\Re s > 1),$$

and use the ordinary Laguerre polynomials

$$L_n(t) = \sum_{j=0}^{n} (-1)^j \binom{n}{j} \frac{t^j}{j!}.$$

Lemma 94.1 (Absolute endpoint moments and a holomorphic germ)**.** *For every nonnegative integer j,*

$$\int_2^\infty |E(x)|x^{-1}(\log x)^j \,\mathrm{d}x < \infty.$$

The initially defined function F extends holomorphically across $s = 1$, and its derivatives there are

$$F^{(j)}(1) = \int_2^\infty E(x)x^{-1}(-\log x)^j \,\mathrm{d}x. \tag{94.1}$$

Proof. Write $\Delta(x) = \pi(x) - \mathrm{li}(x)$. The unconditional estimate of Johnston and Yang [27, Corollary 1.3] gives

$$|\Delta(x)| \le 9.59x(\log x)^{0.515}e^{-0.8274\sqrt{\log x}} \qquad (x \ge 2).$$

Partial summation, with the constant fixed by $E(x) \to 0$, yields

$$E(x) = \frac{\Delta(x)}{x} - \int_x^\infty \frac{\Delta(y)}{y^2}\,\mathrm{d}y.$$

After decreasing the positive exponential constant, this implies

$$E(e^t) = O(e^{-c_0\sqrt{t}}) \qquad (t \to \infty) \tag{94.2}$$

for some $c_0 > 0$. It proves the asserted absolute moments.

To establish holomorphic continuation, define

$$P(s) = \sum_p p^{-s} \quad (\Re s > 1), \qquad R_0(s) = \sum_{k\ge2} \frac{P(ks)}{k}. \tag{94.3}$$

$$\begin{aligned} c &= \log\log 2 + B_{\mathbb{P}}, \\ J_0(s) &= c\,2^{1-s} - \gamma - \log\log 2 - \sum_{j\ge1} \frac{[-(s-1)\log 2]^j}{j\,j!}. \end{aligned} \tag{94.4}$$

The series for R_0 converges normally on $\Re s > 1/2$, and J_0 is entire. For $\Re s > 1$, integration of the prime sum and of $\log\log x$ gives

$$(s-1)F(s) = P(s) - c\,2^{1-s} - \int_2^\infty \frac{x^{-s}}{\log x}\,\mathrm{d}x.$$

Using the Euler product and the exponential-integral expansion gives

$$F(s) = \frac{\log((s-1)\zeta(s)) - R_0(s) - J_0(s)}{s-1}. \tag{94.5}$$

The logarithm is initially the branch agreeing with real values for real $s > 1$.

The function $H(s) = (s-1)\zeta(s)$ is holomorphic near 1, with $H(1) = 1$, so $\log H$ has a holomorphic branch there with value zero at 1. The exact constant identity $B_{\mathbb{P}} = \gamma - R_0(1)$ gives $J_0(1) = B_{\mathbb{P}} - \gamma = -R_0(1)$. The numerator of (94.5) therefore vanishes at 1, making its apparent pole removable. Finally, differentiate the integral at real $\sigma > 1$ and let $\sigma \downarrow 1$. The absolute moments already proved give dominated convergence, and continuity of the holomorphic derivatives gives (94.1). □

Remark 94.2. The decay (94.2) alone supplies all one-sided derivatives at the original boundary; it does not by itself supply a holomorphic extension. Formula (94.5) is the additional step that makes the endpoint argument valid.

Theorem 94.3 (The endpoint Laguerre criterion)**.** *The numbers*

$$b_n = \int_2^\infty E(x)x^{-1}L_n(\log x)\,\mathrm{d}x, \qquad n \geq 0, \tag{94.6}$$

are unconditionally absolutely convergent. The holomorphic germ

$$G_*(z) = \frac{1}{1-z}F\Big(\frac{1}{1-z}\Big) \tag{94.7}$$

has Taylor coefficients b_n. Moreover,

$$\mathrm{RH} \iff (b_n)_{n\geq 0} \in \ell^2 \iff G_* \text{ extends to } H^2(\mathbb{D}). \tag{94.8}$$

When these conditions hold,

$$\sum_{n\geq 0}|b_n|^2 = \|G_*\|_{H^2}^2 = \int_2^\infty E(x)^2\,\mathrm{d}x. \tag{94.9}$$

The Hardy norm uses normalized angular measure on the unit circle.

Proof. Absolute convergence follows from Lemma 94.1 and the finite degree of L_n. Expanding that polynomial gives

$$b_n = \sum_{j=0}^{n}\binom{n}{j}\frac{F^{(j)}(1)}{j!}. \tag{94.10}$$

Since F is an actual holomorphic germ at 1, ordinary Taylor algebra applied to $s-1 = z/(1-z)$ shows that the coefficient of z^n in (94.7) is exactly (94.10). No exchange of an infinite Laguerre series with the endpoint integral is needed.

Put

$$f(t) = e^{t/2}E(e^t)\mathbf{1}_{\{t\geq\log 2\}}, \qquad \ell_n(t) = e^{-t/2}L_n(t), \qquad t \geq 0.$$

The functions ℓ_n are orthonormal in $L^2(0,\infty)$ by the Laguerre orthogonality relation [56, Table 18.3.1]. For completeness, if $h \in L^2(0,\infty)$ is orthogonal to all ℓ_n, triangularity of the polynomials makes every derivative of $\mathcal{L}h(w) = \int_0^\infty h(t)e^{-wt}\,\mathrm{d}t$ vanish at $w = 1/2$. This Laplace transform is holomorphic on $\Re w > 0$, so the identity theorem and Laplace uniqueness give $h = 0$. Thus the system is an orthonormal basis. Also $b_n = \int_0^\infty f(t)\ell_n(t)\,\mathrm{d}t$, and

$$\int_0^\infty |f(t)|^2\,\mathrm{d}t = \int_2^\infty E(x)^2\,\mathrm{d}x.$$

Theorem 92.2 and Parseval prove the forward implication and the norm identity.

Conversely, suppose $(b_n) \in \ell^2$. The orthonormal basis produces $h \in L^2(0,\infty)$ with precisely those coefficients. For $g = f - h$, polynomial triangularity gives

$$\int_0^\infty g(t)e^{-t/2}t^j\,\mathrm{d}t = 0 \qquad (j \geq 0). \tag{94.11}$$

All these integrals converge absolutely, by (94.2) for f and by Cauchy–Schwarz for h. Initially on $\Re w > 1/2$,

$$\mathcal{L}g(w) = F(w+1/2) - \mathcal{L}h(w).$$

Lemma 94.1 and the holomorphy of $\mathcal{L}h$ on $\Re w > 0$ extend this difference across $w = 1/2$. Its derivatives at that point are zero by (94.11), where the derivative identities follow by the same dominated limits as in Lemma 94.1. The identity theorem first makes the difference zero locally, and then on $\Re w > 1/2$. For any real $\sigma > 1/2$, the function $e^{-\sigma t}g(t)$ is in $L^1(0,\infty)$, and its Fourier transform is zero. Fourier uniqueness gives $g = 0$ almost everywhere. Consequently $f = h \in L^2$, and Theorem 92.2 gives RH. Finally, the usual coefficient characterization of $H^2(\mathbb{D})$ proves the other equivalence and its norm equality. □

94.1 An explicit corrected-Li square-summability condition

Use the conventional completed zeta function and Li coefficients [42, §1]:

$$\xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \qquad \xi(1) = \tfrac{1}{2}, \tag{94.12}$$

$$\lambda_n = \frac{1}{(n-1)!}\frac{\mathrm{d}^n}{\mathrm{d}s^n}\{s^{n-1}\log\xi(s)\}\Big|_{s=1}, \qquad n \geq 1. \tag{94.13}$$

Set

$$s(z) = \frac{1}{1-z}, \qquad A(s) = -\log s + \frac{s}{2}\log\pi - \log\Gamma(s/2), \tag{94.14}$$

$$C(z) = A(s(z)) - R_0(s(z)) - J_0(s(z)), \qquad c_n = [z^n]C(z). \tag{94.15}$$

The logarithms in A are the holomorphic branches on $\Re s > 1/2$ that are real on the positive real axis. In particular, $A(1) = 0$.

Proposition 94.4 (Exact bridge to Li's coefficients)**.** *The function C is holomorphic throughout $\mathbb{D}$, unconditionally, and $C(0) = 0$. The following identities hold as holomorphic germs at $z = 0$:*

$$zG_*(z) = \log(2\xi(s(z))) + C(z), \tag{94.16}$$

$$b_{n-1} = \frac{\lambda_n}{n} + c_n \qquad (n \geq 1). \tag{94.17}$$

Consequently,

$$\mathrm{RH} \iff \sum_{n\geq 1}\left|\frac{\lambda_n}{n} + c_n\right|^2 < \infty. \tag{94.18}$$

If finite, this sum is exactly $\int_2^\infty E(x)^2\,\mathrm{d}x$.

Proof. The map $s(z)$ sends $\mathbb{D}$ into $\Re s > 1/2$. The asserted holomorphy therefore follows from the definitions of A, R_0, J_0; their values at 1 give $C(0) = 0$. Near $s = 1$, the definition of ξ gives

$$\log((s-1)\zeta(s)) = \log(2\xi(s)) + A(s).$$

Insert this into (94.5) and use $s(z) - 1 = z/(1-z)$ to obtain (94.16). If $\Phi(s) = \log(2\xi(s))$, finite Taylor algebra and the Leibniz rule in (94.13) give, for $n \geq 1$,

$$[z^n]\Phi(s(z)) = \sum_{j=1}^{n}\binom{n-1}{j-1}\frac{\Phi^{(j)}(1)}{j!} = \frac{\lambda_n}{n}.$$

This proves (94.17); Theorem 94.3 then proves (94.18) and the norm assertion. □

Corollary 94.5 (The prime-power endpoint has no prime-zeta correction)**.** *Define*

$$E^{\mathrm{pp}}(x) = \sum_{p^k \leq x}\frac{1}{kp^k} - \log\log x - \gamma, \qquad b_n^{\mathrm{pp}} = \int_2^\infty E^{\mathrm{pp}}(x)x^{-1}L_n(\log x)\,\mathrm{d}x,$$

and put $J_{\mathrm{pp}}(s) = J_0(s) + (\gamma - B_{\mathbb{P}})2^{1-s}$. All these endpoint coefficients converge absolutely, and

$$b_{n-1}^{\mathrm{pp}} = \frac{\lambda_n}{n} + [z^n]\{A(s(z)) - J_{\mathrm{pp}}(s(z))\} \qquad (n \geq 1). \tag{94.19}$$

Moreover, $(b_n^{\mathrm{pp}}) \in \ell^2$ if and only if RH*; when finite, its squared norm is $\int_2^\infty |E^{\mathrm{pp}}(x)|^2\,\mathrm{d}x$.*

Proof. The bias is

$$E(x) - E^{\mathrm{pp}}(x) = \sum_{\substack{k\geq 2\\ p^k>x}} \frac{1}{kp^k} = O\Big(\frac{1}{\sqrt{x}\log x}\Big).$$

For $k = 2$, this bound follows by partial summation from $\pi(y) \ll y/\log y$; splitting $k \geq 3$ at $\lfloor \log x/\log 2\rfloor$ gives $O(x^{-2/3}\log x)$ for the remaining terms. The bias is therefore in $L^2(2,\infty)$ and has every endpoint logarithmic moment. Initially for $\Re s > 1$, the Euler product gives

$$F^{\mathrm{pp}}(s) := \int_2^\infty E^{\mathrm{pp}}(x)x^{-s}\,\mathrm{d}x = \frac{\log((s-1)\zeta(s)) - J_{\mathrm{pp}}(s)}{s-1}.$$

Since $J_{\mathrm{pp}}(1) = 0$, its singularity at 1 is removable. The proof of Theorem 94.3 now applies unchanged, and the L^2 bias transfers the RH equivalence. The identity with ξ gives (94.19). □

Corollary 94.6 (The endpoint radius and its singularities)**.** *The Taylor radius of G_* is*

$$R_* = \min\left\{1,\ \inf_{\substack{\zeta(\rho)=0\\ \Re\rho>1/2}} \Big|1-\frac{1}{\rho}\Big|\right\}, \tag{94.20}$$

where the infimum of the empty set is $+\infty$. Thus $R_* = 1$ if and only if RH*; equivalently,* $\limsup_{n\to\infty}|b_n|^{1/n} = 1$ *if and only if* RH*.*

Proof. Within $\Re s > 1/2$, the only possible obstructions in (94.5), apart from the already removed point 1, are zeros of ζ. At a zero ρ of multiplicity m_ρ, on a slit neighborhood,

$$F(s) = \frac{m_\rho}{s-1}\log(s-\rho) + H_\rho(s),$$

where H_ρ is holomorphic. Since $z_\rho = 1 - 1/\rho \neq 0$ and the change of variable has nonzero derivative there, G_* has a nonremovable logarithmic branch at z_ρ. A disk about zero containing no such image admits the normalized holomorphic logarithm in (94.5).

There is also an unconditional boundary obstruction at $z = -1$. For real $s = 1/2 + \varepsilon$,

$$R_0(s) = -\tfrac{1}{2}\log\varepsilon + O(1), \qquad F(s) = -\log\varepsilon + O(1) \quad (\varepsilon \downarrow 0).$$

The first relation comes from the term $P(2s)/2$; all terms with $k \geq 3$ remain bounded. The other terms in the numerator of (94.5) are regular on this real approach. Hence $G_*(z) = -\frac{1}{2}\log(1+z) + O(1)$ as real $z \downarrow -1$ from within the disk, preventing a radius larger than one. This proves (94.20). Finally, $|1 - 1/\rho| < 1$ exactly when $\Re\rho > 1/2$, and the functional equation supplies such a zero whenever RH fails. Cauchy–Hadamard gives the coefficient statement. □

Remark 94.7 (Exact centering is essential at this endpoint)*.* If E is replaced by $E_\delta = E - \delta$ with $\delta \neq 0$, every integral in (94.6) fails to converge: the added term is the integral of a nonzero constant times the polynomial $L_n(t)$ over an infinite interval. Analytically,

$$F_\delta(s) = F(s) - \delta\frac{2^{1-s}}{s-1}, \qquad G_{*,\delta}(z) = G_*(z) - \frac{\delta}{z}2^{1-1/(1-z)}.$$

Thus the miscentered expression has a simple pole at the expansion center, with residue $-\delta$. The endpoint bridge uses the exact constant $B_{\mathbb{P}}$; it does not turn an approximate centering constant into valid endpoint coefficients. The corrected-Li criterion is a proved equivalence, not an estimate of its infinite sum.

95 Exact spectral mass limits for the circle measures

The normalized Toeplitz matrices converge to rank one. The following results identify exactly what remains after multiplication by the diverging total mass. They apply to arbitrary converging unit phases, including finite-support examples, and specialize to the prime phases $u_n = U_n$, weights w_n, and limit $a = u_\infty$.

Fix $r \geq 1$ and write $L = r + 1$. Throughout this section assume

$$|u_n| = |a| = 1, \qquad u_n \longrightarrow a, \qquad w_n > 0, \qquad W_N = \sum_{n=2}^{N} w_n \longrightarrow \infty. \tag{95.1}$$

Define

$$v(u) = (1, u, \dots, u^r)^{\mathsf{T}}, \qquad e = \frac{v(a)}{\sqrt{L}}, \qquad \Pi = I - ee^*, \tag{95.2}$$

$$A_N = \sum_{n=2}^{N} w_n v(u_n) v(u_n)^*, \qquad T_{r,N} = \frac{A_N}{W_N}, \tag{95.3}$$

$$m_{k,N} = \frac{1}{W_N} \sum_{n=2}^{N} w_n u_n^k, \qquad V_N^{(k)} = W_N(1 - |m_{k,N}|^2). \tag{95.4}$$

Thus $(T_{r,N})_{jk} = m_{j-k,N}$ for $0 \leq j, k \leq r$. Let $\ell_{1,N} \geq \cdots \geq \ell_{L,N} \geq 0$ be the eigenvalues of A_N, and put

$$\mathcal{L}_{r,N} = LW_N - \ell_{1,N} = \sum_{j=2}^{L} \ell_{j,N}, \tag{95.5}$$

$$\mathcal{T}_{r,N} = W_N e_2(T_{r,N}), \qquad e_2(T) = \sum_{i<j} \lambda_i(T)\lambda_j(T). \tag{95.6}$$

Theorem 95.1 (Monotone leakage and its exact limiting energy)**.** *Under* (95.1)*, $\mathcal{L}_{r,N}$ is nondecreasing, and*

$$\mathcal{L}_{r,\infty} := \lim_{N\to\infty} \mathcal{L}_{r,N} = \sum_{n=2}^{\infty} w_n \|\Pi v(u_n)\|^2 \tag{95.7}$$

$$= \frac{1}{L} \sum_{n=2}^{\infty} w_n \sum_{k=1}^{r} (L-k)|u_n^k - a^k|^2. \tag{95.8}$$

The equalities hold in $[0, \infty]$; no summability assumption is needed. In particular, this limit is finite exactly when $\sum_n w_n|u_n - a|^2 < \infty$. For the prime phases and weights, its finiteness is equivalent to RH*.*

Proof. The Rayleigh principle gives the exact finite-dimensional identity

$$\mathcal{L}_{r,N} = \min_{\|b\|=1} Q_N(b), \qquad Q_N(b) = \sum_{n=2}^{N} w_n\{L - |b^* v(u_n)|^2\}. \tag{95.9}$$

Each summand is nonnegative by Cauchy–Schwarz, since $\|v(u_n)\|^2 = L$. Hence Q_N increases pointwise with N, and its minima are nondecreasing.

Suppose first that their limit q is finite. Choose a minimizer b_N on the compact unit sphere and take a convergent subsequence $b_N \to b_*$. For each fixed M, $Q_M(b_N) \leq \mathcal{L}_{r,N}$ when $N \geq M$. Continuity and then monotone convergence give $Q_\infty(b_*) \leq q$. If $L - |b_*^* v(a)|^2$ were positive, convergence $u_n \to a$ would give a positive lower bound on that factor on a tail, and $W_N \to \infty$

would force $Q_\infty(b_*)=\infty$. Equality in Cauchy–Schwarz therefore implies that b_* is a unit scalar multiple of e. Thus $Q_\infty(e)\le q$. Conversely, testing $b=e$ in (95.9) gives $\mathcal{L}_{r,N}\le Q_N(e)$, so $q=Q_\infty(e)$. If $q=\infty$, the same test shows that $Q_\infty(e)$ cannot be finite. This proves (95.7) in both cases.

For $u\in\mathbb{T}$, expand the squared geometric sum to obtain

$$\|\Pi v(u)\|^2 = L-\frac{1}{L}\left|\sum_{j=0}^{r}(u\overline{a})^j\right|^2 = \frac{1}{L}\sum_{k=1}^{r}(L-k)|u^k-a^k|^2.$$

This gives (95.8). Its $k=1$ term gives the lower bound $r|u-a|^2/L$, while $|u^k-a^k|\le k|u-a|$ gives a fixed multiple of $|u-a|^2$ as an upper bound. The finiteness equivalence follows. The specialization to primes uses Theorem 92.2. □

Corollary 95.2 (The original deficit is the first spectral case)**.** *Let $D_N=W_N-|\sum_{n=2}^{N}w_nu_n|$. Then $\mathcal{L}_{1,N}=D_N$, and*

$$\lim_{N\to\infty}D_N=\frac{1}{2}\sum_{n=2}^{\infty}w_n|u_n-a|^2 \quad \text{in } [0,\infty]. \tag{95.10}$$

For every fixed r, if these limits are finite, then

$$\frac{2r}{r+1}D_\infty\le\mathcal{L}_{r,\infty}\le\binom{r+2}{3}D_\infty. \tag{95.11}$$

Proof. For $r=1$, the eigenvalues of $T_{1,N}$ are $1\pm|m_{1,N}|$, proving the first assertion and then (95.10). The comparison follows from the bounds used in Theorem 95.1 and

$$\sum_{k=1}^{r}(r+1-k)k^2=\frac{r(r+1)^2(r+2)}{12}.$$

□

Theorem 95.3 (Every small eigenvalue has an exact limit)**.** *Assume (95.1) and $\mathcal{L}_{r,\infty}<\infty$. On the r-dimensional space $e^\perp$, define*

$$B_\infty=\sum_{n=2}^{\infty}w_n\Pi v(u_n)v(u_n)^*\Pi\Big|_{e^\perp}. \tag{95.12}$$

This positive matrix series converges in trace norm. Write its eigenvalues as $\beta_1\ge\cdots\ge\beta_r\ge0$. Then, with multiplicities and including possible zeros,

$$\ell_{j+1,N}\longrightarrow\beta_j \qquad (1\le j\le r), \tag{95.13}$$

$$\ell_{1,N}=LW_N-\operatorname{tr}B_\infty+o(1), \tag{95.14}$$

$$W_N^r\det T_{r,N}\longrightarrow L\det B_\infty. \tag{95.15}$$

Proof. The trace of each positive summand in (95.12) is $w_n\|\Pi v(u_n)\|^2$. Theorem 95.1 makes their total finite. Thus the partial sums B_N converge in trace norm, and hence in operator norm, to B_∞. In the orthogonal decomposition $\mathbb{C}e\oplus e^\perp$, write

$$A_N=\begin{pmatrix}d_N & q_N^*\\ q_N & B_N\end{pmatrix}, \qquad d_N=LW_N-\operatorname{tr}B_N\sim LW_N.$$

The off-diagonal vector is $q_N=\sum_{n=2}^{N}w_n\Pi v(u_n)(v(u_n)^*e)$. Split it at a fixed index K. The prefix has fixed norm, while Cauchy–Schwarz gives

$$\left\|\sum_{n>K}^{N}w_n\Pi v(u_n)(v(u_n)^*e)\right\|^2\le LW_N\sum_{n>K}w_n\|\Pi v(u_n)\|^2.$$

First let $N \to \infty$, then $K \to \infty$. The finite tail energy proves

$$\|q_N\| = o(\sqrt{W_N}), \qquad \frac{\|q_N\|^2}{d_N} \to 0. \tag{95.16}$$

The min–max principle applied to $e^{\perp}$ gives $\ell_{2,N} \le \|B_N\|$, so all r small eigenvalues are bounded. Take any subsequence along which these eigenvalues and an orthonormal choice of their eigenvectors converge. Write such an eigenvector as $(x_{j,N}, y_{j,N})$. The first row of its eigenvalue equation gives

$$x_{j,N} = -\frac{q_N^* y_{j,N}}{d_N - \ell_{j+1,N}} \to 0.$$

The second row, together with (95.16), gives

$$(B_N - \ell_{j+1,N} I) y_{j,N} = -q_N x_{j,N} \longrightarrow 0.$$

The limiting y-vectors form an orthonormal basis of $e^{\perp}$; their limiting eigenvalues therefore form the full spectrum of B_∞. Ordering forces those limits to be $\beta_1, \dots, \beta_r$. Since this applies to every convergent subsequence, it proves (95.13) for the full sequence. The trace identity gives (95.14). Finally,

$$W_N^r \det T_{r,N} = \frac{\ell_{1,N}}{W_N} \prod_{j=2}^{L} \ell_{j,N} \longrightarrow L \prod_{j=1}^{r} \beta_j,$$

which is (95.15). □

Remark 95.4 (Support assumptions and determinant degeneracy). Theorems 95.1 and 95.3 do not assume infinite phase support. The matrix B_∞ is positive definite exactly when the vectors $\Pi v(u_n)$ span $e^{\perp}$. Infinite distinct phase support is sufficient: an annihilating vector would produce a polynomial of degree at most r vanishing at infinitely many points. For smaller finite supports, B_∞ can be singular, and the determinant limit can be zero. Positivity of a determinant by itself supplies no upper bound. A converse for boundedness of a properly scaled determinant becomes available once a finite prefix has full rank; this is proved below. The hypotheses $u_n \to a$ and $W_N \to \infty$ are what force the variational direction to $v(a)$; that forcing argument is unavailable without them.

Lemma 95.5 (The prime phases have infinite support)**.** *The set of prime phases*

$$\left\{ U_n = \exp\left(2\pi \mathrm{i} \left[\sum_{p \le p_n} \frac{1}{p} - \log\log p_n \right] \right) : n \ge 2 \right\}$$

is infinite. In particular, for every fixed r some finite prefix of the prime moment matrix A_N is positive definite.

Proof. Suppose the phase set were finite. Since $U_n \to a = e^{2\pi \mathrm{i} B_{\mathbb{P}}}$, the sequence would eventually equal a. Then $E(p_n)$ would eventually be an integer; since $E(p_n) \to 0$, it would eventually equal zero. For consecutive sufficiently large primes $p < q$, subtraction would give

$$\log\left(\frac{\log q}{\log p}\right) = \frac{1}{q}, \qquad \log p = e^{-1/q} \log q.$$

Consequently,

$$\log(q/p) = (1 - e^{-1/q}) \log q = \frac{\log q}{q} + O\left(\frac{\log q}{q^2}\right),$$

and expansion of $p = q \exp(-\log(q/p))$ gives

$$q - p = \log q + O\left(\frac{(\log q)^2}{q}\right). \tag{95.17}$$

Both primes are odd, so $q - p \in 2\mathbb{Z}$. Equation (95.17) would force $\operatorname{dist}(\log q, 2\mathbb{Z}) \to 0$ along all primes.

For integers $j \to \infty$, let $x_j = \lceil e^{2j+1} \rceil$. Bertrand's postulate [57] supplies a prime q_j with $x_j < q_j < 2x_j$. Thus

$$1 + o(1) < \log q_j - 2j < 1 + \log 2 + o(1),$$

so its distance from $2\mathbb{Z}$ is at least $1 - \log 2 - o(1) > 0$. This contradicts (95.17). The phase support is therefore infinite.

Finally, any $L = r + 1$ distinct phases give an invertible Vandermonde matrix with columns $v(u)$. Because their weights are positive, the sum of their rank-one matrices is positive definite. A prefix containing these phases is positive definite as well. □

Theorem 95.6 (A monotone determinant criterion after a full-rank prefix)**.** *Assume* (95.1) *and that the phase support contains at least $L = r + 1$ distinct points. Define*

$$\mathcal{Z}_{r,N} = W_N^r \det T_{r,N} = \frac{\det A_N}{W_N}. \tag{95.18}$$

Then $\mathcal{Z}_{r,N}$ is nondecreasing, including its possibly singular initial segment, and

$$\sup_N \mathcal{Z}_{r,N} < \infty \quad \Longleftrightarrow \quad \sup_N D_N < \infty. \tag{95.19}$$

Choose any N_0 for which A_{N_0} is positive definite and put $c = \lambda_{\min}(A_{N_0}) > 0$. For $N \geq N_0$,

$$D_N \leq c^{1-r} \mathcal{Z}_{r,N}. \tag{95.20}$$

If the equivalent conditions in (95.19) *hold, the limits satisfy the stronger bound*

$$D_\infty \leq \frac{\mathcal{Z}_{r,\infty}}{(r+1)c^{r-1}}. \tag{95.21}$$

For the prime phases and weights, for every fixed $r \geq 1$,

$$\mathrm{RH} \quad \Longleftrightarrow \quad \sup_N W_N^r \det T_{r,N} < \infty. \tag{95.22}$$

Proof. The Vandermonde argument in Lemma 95.5 shows that the support hypothesis supplies a positive-definite prefix. Whenever A_N is positive definite, the rank-one determinant identity gives, writing $w = w_{N+1}$ and $v = v(u_{N+1})$,

$$\frac{\mathcal{Z}_{r,N+1}}{\mathcal{Z}_{r,N}} = \frac{1 + w\, v^* A_N^{-1} v}{1 + w/W_N}.$$

Since $\|v\|^2 = L$ and $\lambda_{\max}(A_N) \leq \operatorname{tr} A_N = L W_N$,

$$v^* A_N^{-1} v \geq \frac{\|v\|^2}{\lambda_{\max}(A_N)} \geq \frac{1}{W_N}.$$

The displayed ratio is at least one. Before the first positive-definite prefix, the determinants are zero, so monotonicity holds for the entire sequence.

For $N \geq N_0$, the positive increments give $A_N \geq A_{N_0} \geq cI$, hence $\ell_{j,N} \geq c$ for all j. The leading 2×2 principal submatrix of A_N has eigenvalues $W_N \pm |\sum_{n=2}^N w_n u_n|$. Eigenvalue interlacing therefore gives $\ell_{2,N} \geq D_N$. Also $\ell_{1,N} \geq \operatorname{tr}(A_N)/L = W_N$. It follows that

$$\mathcal{Z}_{r,N} = \frac{\ell_{1,N}}{W_N} \prod_{j=2}^{L} \ell_{j,N} \geq c^{r-1} \ell_{2,N} \geq c^{r-1} D_N,$$

which proves (95.20) and one implication in (95.19).

Conversely, bounded D_N makes $\sum_n w_n|u_n - a|^2$ finite by (95.10). Theorem 95.1 then makes $\mathcal{L}_{r,\infty}$ finite, and Theorem 95.3 gives the finite limit

$$\mathcal{Z}_{r,\infty} = L \prod_{j=1}^{r} \beta_j.$$

Here $\beta_j \geq c$, and $\beta_1 \geq D_\infty$ by $\ell_{2,N} \geq D_N$. Thus $\mathcal{Z}_{r,\infty} \geq Lc^{r-1}D_\infty$, proving (95.21). For the prime sequence, Lemma 95.5 supplies the support hypothesis for every r, and Theorem 92.2 completes (95.22). □

Corollary 95.7 (Factorization into the monic extremal energies)**.** *Let $H_{j,N}$ be the monic extremal energies of Section 93, and set $H_{0,N} = W_N$. For every N,*

$$\mathcal{Z}_{r,N} = \prod_{j=1}^{r} H_{j,N}. \tag{95.23}$$

Under the hypotheses of Theorem 95.6, choose a positive-definite prefix A_{N_0}. If $\mathcal{Z}_{r,\infty} < \infty$, then

$$D_\infty \leq \frac{\mathcal{Z}_{r,\infty}}{2\prod_{j=2}^{r} H_{j,N_0}}, \tag{95.24}$$

where the empty product for $r = 1$ equals one.

Proof. With the convention $\langle P, Q\rangle_N = \sum_{n=2}^{N} w_n P(u_n)\overline{Q(u_n)}$, linear in its first argument, the monomial Gram matrix is exactly A_N. In the positive-definite case, monic Gram–Schmidt gives a unit lower triangular coefficient matrix C with $CA_NC^* = \operatorname{diag}(H_{0,N}, \dots, H_{r,N})$. Taking determinants and dividing by W_N proves (95.23). In the singular case there are $q \leq r$ distinct phases; a monic polynomial vanishing on them gives $H_{q,N} = 0$, and both sides are zero. Each $H_{j,N}$ is nondecreasing, and the positive-definite prefix makes $H_{j,N_0} > 0$ for $j \leq r$. Finally, $H_{1,N} = V_N^{(1)} = D_N(1 + |m_{1,N}|) \to 2D_\infty$. Using $H_{j,N} \geq H_{j,N_0}$ in (95.23) and taking limits proves (95.24). The determinant thus packages the existing extremal energies; positivity of finite Gram matrices supplies no bound on their product. □

Remark 95.8. The full-rank prefix supplies the positive constant c in the quantitative bounds. With fewer than L distinct phases, the determinant is identically zero and cannot quantify the lower-dimensional energy. In the convergent-phase setting (95.1), such finite support forces $u_n = a$ eventually, so the deficit is automatically bounded; outside that setting, a zero determinant need not imply bounded deficit. Under the support assumption, the theorem concerns boundedness of $W_N^r \det T_{r,N}$, not positivity of $\det T_{r,N}$. For $r = 1$ it reduces to $\mathcal{Z}_{1,N} = V_N^{(1)}$ and $D_\infty = \mathcal{Z}_{1,\infty}/2$.

Proposition 95.9 (A sharp finite relation between the spectral tests)**.** *For every finite positive circle measure in the notation above,*

$$\mathcal{T}_{r,N} = L\mathcal{L}_{r,N} - \frac{\mathcal{L}_{r,N}^2 + \sum_{j=2}^{L} \ell_{j,N}^2}{2W_N}. \tag{95.25}$$

Consequently,

$$L\mathcal{L}_{r,N} - \frac{\mathcal{L}_{r,N}^2}{W_N} \leq \mathcal{T}_{r,N} \leq L\mathcal{L}_{r,N} - \frac{L\mathcal{L}_{r,N}^2}{2rW_N}. \tag{95.26}$$

Both bounds are sharp among finite positive circle measures. Under (95.1), if the leakage is eventually positive,

$$\frac{\mathcal{T}_{r,N}}{\mathcal{L}_{r,N}} \longrightarrow L. \tag{95.27}$$

Proof. Since $\ell_{1,N} = LW_N - \mathcal{L}_{r,N}$, separating this eigenvalue from the others in $e_2(A_N)/W_N$ gives (95.25). For r nonnegative numbers with sum $\mathcal{L}_{r,N}$, their sum of squares lies between $\mathcal{L}_{r,N}^2/r$ and $\mathcal{L}_{r,N}^2$. This proves (95.26). The lower bound is attained by measures supported at two distinct points, whose moment matrices have rank at most two. The upper bound is attained by the uniform measure on the L-th roots of unity, whose normalized moment matrix is I. Under (95.1), weighted Cesaro convergence gives $T_{r,N} \to v(a)v(a)^*$, and hence $\mathcal{L}_{r,N}/W_N \to 0$. Divide (95.26) by $\mathcal{L}_{r,N} > 0$ to get (95.27). □

Proposition 95.10 (Exact amplification if the original deficit diverges)**.** *Under* (95.1)*, suppose* $D_N \to \infty$*. For each fixed* $r \geq 1$*,*

$$\frac{\mathcal{L}_{r,N}}{D_N} \longrightarrow \binom{r+2}{3}, \qquad \frac{\mathcal{T}_{r,N}}{D_N} \longrightarrow (r+1)\binom{r+2}{3}. \tag{95.28}$$

Proof. For a fixed positive integer k, the pairwise identity gives

$$V_N^{(k)} = \frac{1}{2W_N} \sum_{j,l=2}^{N} w_j w_l |u_j^k - u_l^k|^2.$$

The polynomial quotient

$$u^k - v^k = (u-v) \sum_{h=0}^{k-1} u^{k-1-h} v^h$$

has squared modulus tending uniformly to k^2 as $u, v \to a$, including the diagonal by its polynomial extension. Thus, given $\varepsilon > 0$, the part of the pair sum with both indices sufficiently large differs from k^2 times the first-harmonic pair sum by at most $\varepsilon V_N^{(1)}$. Pairs involving the fixed prefix contribute a bounded quantity, because the prefix has fixed total mass and the normalization is $1/(2W_N)$. Therefore

$$|V_N^{(k)} - k^2 V_N^{(1)}| \leq \varepsilon V_N^{(1)} + C_{k,\varepsilon}.$$

Since $m_{1,N} \to a$, $V_N^{(1)} = D_N(1 + |m_{1,N}|) \sim 2D_N \to \infty$. It follows that $V_N^{(k)}/V_N^{(1)} \to k^2$. The identity

$$\mathcal{T}_{r,N} = \frac{W_N}{2}\{L^2 - \operatorname{tr}(T_{r,N}^2)\} = \sum_{k=1}^{r} (L-k) V_N^{(k)}$$

and $\sum_{k=1}^{r} (L-k)k^2 = r(r+1)^2(r+2)/12$ now give the second limit in (95.28). Equation (95.27) gives the first. □

Remark 95.11. For the prime sequence, these results close the limiting variational problem and identify the complete finite spectrum that remains after the leading eigenvalue diverges. They also determine the exact amplification factors if the deficit diverges and give a monotone determinant criterion after a finite full-rank prefix. Their finite limiting energies remain equivalent to RH. None of these identities proves $\sup_N D_N < \infty$.

96 Closing the conditional tail law for the original polygon

The addendum to the first supplied manuscript proposes a sharp tail law for the original prime polygon, but leaves the passage from a logarithmic mean square to the sampled polygon as a sketch. This section supplies that passage. The conclusion is conditional on RH, and does not supply the unconditional boundedness sought in the original question. Neither simplicity of the zeros nor linear independence of their ordinates is assumed.

Write

$$E(x)=\sum_{p\le x}\frac{1}{p}-\log\log x-B_{\mathbb{P}},\qquad E^{*}(x)=\sum_{p^{k}\le x}\frac{1}{kp^{k}}-\log\log x-\gamma.$$

The letters t_n, F will denote either the prime sequence and E, or the increasing sequence of distinct prime powers and E^{*}. Put

$$d_n=t_{n+1}-t_n,\quad w_n=d_{n-1}+d_n\quad(n\ge 2),\qquad u_n=\exp(2\pi\mathrm{i}F(t_n)).$$

These u_n rotate the original vectors by a fixed unit scalar, which leaves the deficit unchanged. Thus

$$W_N=\sum_{n=2}^{N}w_n=t_N+t_{N+1}-5,\quad M_N=\sum_{n=2}^{N}w_nu_n,\quad D_N=W_N-|M_N|,\quad J_N=\sum_{n=2}^{N}w_n|u_n-1|^2.$$

For the prime-power choice, superscripts $*$ distinguish the resulting quantities. The equivalence with finite scalar energy is the imported criterion in Theorem 92.2.

Theorem 96.1 (Exact conditional deficit tails)**.** *Assume* RH*. For each distinct ordinate γ of a nontrivial zero, let m_γ be its multiplicity and set*

$$\kappa=\sum_{\gamma\ mdistinct}\frac{m_\gamma^{2}}{1/4+\gamma^{2}}.$$

The sum includes both signs of the ordinate and converges. Then

$$\lim_{Y\to\infty}\frac{1}{\log Y}\int_{2}^{Y}(\log x)^{2}E^{*}(x)^{2}\,\mathrm{d}x=\kappa,\tag{96.1}$$

$$\lim_{Y\to\infty}\frac{1}{\log Y}\int_{2}^{Y}(\log x)^{2}E(x)^{2}\,\mathrm{d}x=1+\kappa.\tag{96.2}$$

Consequently,

$$\int_{X}^{\infty}E^{*}(x)^{2}\,\mathrm{d}x\sim\frac{\kappa}{\log X},\qquad\int_{X}^{\infty}E(x)^{2}\,\mathrm{d}x\sim\frac{1+\kappa}{\log X},\tag{96.3}$$

$$D_{\infty}^{*}-D_{N}^{*}\sim\frac{4\pi^{2}\kappa}{\log q_N},\qquad D_{\infty}-D_N\sim\frac{4\pi^{2}(1+\kappa)}{\log p_N}.\tag{96.4}$$

Here q_N is the N-th distinct prime power. Every quantity and weight in the second formula of (96.4) belongs to the original prime polygon.

96.1 The spectral statement and its measure

Set $y=\log x$ and

$$H(y)=e^{-y/2}(\psi(e^{y})-e^{y}),\qquad G^{*}(y)=ye^{y/2}E^{*}(e^{y}).$$

For a locally square-integrable function define its Besicovitch seminorm by

$$\|h\|_{B^{2}}^{2}=\limsup_{Y\to\infty}\frac{1}{Y}\int_{\log 2}^{Y}|h(y)|^{2}\,\mathrm{d}y.$$

The mean-square explicit-formula theorem of Akbary–Ng–Shahabi [58, Theorem 1.2, Corollary 1.3, Theorem 1.14 and Corollary 1.15] gives, under RH, the B^2 expansion

$$H(y)=-\sum_{\gamma\ mdistinct}\frac{m_\gamma}{\rho}\,e^{\mathrm{i}\gamma y},\qquad\rho=\tfrac{1}{2}+\mathrm{i}\gamma.\tag{96.5}$$

This is convergence in the stated seminorm, not pointwise absolute convergence. Grouping equal ordinates is essential. Their hypotheses follow from the truncated explicit formula and the elementary zero-count bound $N(T+1)-N(T)=O(\log T)$; in particular, the sum of the absolute coefficients in a unit frequency interval is $O(\log T/T)$. There is no zero with ordinate zero. The multiplicity bound $m_\gamma = O(\log(2+|\gamma|))$ and $N(T)=O(T\log T)$ also show directly that $\sum m_\gamma^2/(1/4+\gamma^2)<\infty$.

A second input is the conditional Cramér estimate, in the form proved and studied by Brent–Platt–Trudgian [54, eq. (1), Theorem 1]:

$$\int_X^{2X} (\psi(x)-x)^2 \,\mathrm{d}x \ll X^2, \qquad \int_y^{y+\log 2} |H(v)|^2 \,\mathrm{d}v \ll 1. \tag{96.6}$$

The latter follows by substitution and is uniform for $y \geq \log 2$. It is stronger than the sole assertion that the long mean is bounded, and makes the following operator step legitimate.

Lemma 96.2 (The Hardy operator after normalization)**.** *Assume* RH*, and let*

$$(\mathcal{T}h)(y) = \int_0^\infty e^{-v/2} h(y+v)\,\mathrm{d}v.$$

Then

$$G^*(y) = H(y) - \mathcal{T}H(y) + O(1/y). \tag{96.7}$$

On the B^2 closure of the trigonometric approximants to H, $\mathcal{T}$ has norm at most 2. Therefore

$$G^*(y) = \sum_{\gamma \ mdistinct} \frac{m_\gamma}{1-\rho}\, e^{\mathrm{i}\gamma y} \quad \text{in } B^2. \tag{96.8}$$

Its mean is zero and its mean square is κ.

Proof. The exact partial-summation identity is

$$E^*(x) = f^*(x) - \int_x^\infty \frac{f^*(t)}{t}\left(1+\frac{1}{\log t}\right)\mathrm{d}t, \qquad f^*(x) = \frac{\psi(x)-x}{x\log x}.$$

Consequently

$$G^*(y) = H(y) - \int_0^\infty e^{-v/2} H(y+v)\left\{\frac{y}{y+v} + \frac{y}{(y+v)^2}\right\}\mathrm{d}v.$$

The expression in braces differs from 1 by at most $(v+1)/y$. Divide the v-axis into intervals of length $\log 2$ and apply Cauchy–Schwarz on each. The uniform bound in (96.6) gives

$$\int_0^\infty e^{-v/2}(1+v)|H(y+v)|\,\mathrm{d}v \ll 1.$$

This proves (96.7); its error has zero B^2 seminorm.

For completeness, the operator may be passed through the B^2 limit without an unproved interchange of an infinite zero sum. If P is a trigonometric polynomial, $H-P$ has a uniform bound for its square integral on intervals of length $\log 2$. Thus fixed translations preserve its B^2 seminorm; the finitely long boundary intervals contribute $O(1/Y)$. Minkowski's inequality, followed by the dominated limsup inequality (reverse Fatou) using that same local bound, gives

$$\|\mathcal{T}(H-P)\|_{B^2} \leq \int_0^\infty e^{-v/2}\|H-P\|_{B^2}\,\mathrm{d}v = 2\|H-P\|_{B^2}.$$

For a single frequency, $\mathcal{T}e^{\mathrm{i}\gamma y} = (1/2-\mathrm{i}\gamma)^{-1}e^{\mathrm{i}\gamma y}$. Its coefficient in $H-\mathcal{T}H$ is therefore

$$-\frac{m_\gamma}{\rho}\left(1-\frac{1}{1-\rho}\right) = \frac{m_\gamma}{1-\rho}.$$

Applying this calculation first to finite sums and then to their B^2 limit proves (96.8). Orthogonality of distinct frequencies and B^2 Parseval give mean zero and mean square $\sum m_\gamma^2/|1-\rho|^2 = \kappa$. Pairwise distinctness suffices; no linear-independence hypothesis enters a second moment. □

The difference between the two errors is exact and nonnegative:

$$B(x) := E(x) - E^*(x) = \sum_{\substack{p^k > x \\ k \geq 2}} \frac{1}{kp^k} \sim \frac{1}{\sqrt{x}\log x}. \tag{96.9}$$

For the asymptotic, the $k = 2$ terms contribute $\frac{1}{2}\sum_{p>\sqrt{x}} p^{-2} \sim 1/(\sqrt{x}\log x)$ by partial summation and the prime number theorem; the terms $k \geq 3$ are $O(x^{-2/3}\log x)$, which is smaller. Hence $ye^{y/2}E(e^y) = G^*(y) + 1 + o(1)$. Lemma 96.2 proves (96.1)–(96.2), since

$$\int_2^Y (\log x)^2 F(x)^2 \, \mathrm{d}x = \int_{\log 2}^{\log Y} \left(ye^{y/2}F(e^y)\right)^2 \, \mathrm{d}y.$$

Lemma 96.3 (From a long mean to a scalar tail)**.** *Suppose G is locally square-integrable and $A(U) := \int_{\log 2}^{U} |G(y)|^2 \, \mathrm{d}y = CU + o(U)$, with $C > 0$. Then*

$$\int_U^\infty \frac{|G(y)|^2}{y^2} \, \mathrm{d}y \sim \frac{C}{U}.$$

More precisely, if $\eta(U) = \sup_{v \geq U} |A(v) - Cv|/v$, then the absolute error is at most $3\eta(U)/U$.

Proof. Integration by parts yields

$$\int_U^\infty y^{-2} \, \mathrm{d}A(y) = -\frac{A(U)}{U^2} + 2\int_U^\infty \frac{A(y)}{y^3} \, \mathrm{d}y.$$

Substitute $A(y) = Cy + (A(y) - Cy)$. The main terms sum to C/U, and the two error terms total at most $3\eta(U)/U$. □

This proves (96.3). In particular, existence of the limiting mean is enough; a pointwise or a fixed-width dyadic asymptotic has not been assumed. The lemma gives no numerical error at a specified finite U unless $\eta(U)$ is separately bounded.

96.2 Prime sampling, boundary terms and the moving centroid

We now supply the additional estimate which a mere two-sided sampling comparison does not provide. We use the classical mean-square-gap bound of Peck, reproved by Maynard [26]:

$$\sum_{p_n \leq X} (p_{n+1} - p_n)^2 \ll_\varepsilon X^{5/4+\varepsilon} \qquad (\varepsilon > 0). \tag{96.10}$$

This input is unconditional. Its exponent need only be strictly below $3/2$ for the argument below. The finer partition by prime powers also satisfies (96.10): within a prime gap, the sum of the squares of the subgap lengths does not exceed the square of the entire gap. A boundary subgap lies in a prime gap counted with left endpoint at most X.

Lemma 96.4 (Asymptotically exact tail sampling)**.** *Assume* RH*. For either choice of (t_n, F), and every fixed $0 < \varepsilon < 1/4$,*

$$\sum_{n>N} w_n F(t_n)^2 = 2\int_{t_N}^\infty F(x)^2 \, \mathrm{d}x + O_\varepsilon(t_N^{-1/4+\varepsilon}). \tag{96.11}$$

Moreover

$$\sum_{n=2}^{N} w_n |F(t_n)| \ll \frac{\sqrt{t_N}}{\log t_N}, \qquad \frac{|M_N - W_N|^2}{W_N} \ll \frac{1}{\log^2 t_N}. \tag{96.12}$$

Proof. The usual pointwise RH bound and partial summation give $F(x) = O(x^{-1/2}\log x)$. On $(a,b) = (t_n, t_{n+1})$, $F'(x) = -1/(x\log x)$, and the jump at b has size at most $1/b$. Both sequences satisfy $b < 2a$. With $h = b - a$,

$$|F(a)^2 - F(x)^2| \ll \frac{x-a}{a^{3/2}},$$
$$|F(b)^2 - F(x)^2| \ll \frac{b-x+\log a}{a^{3/2}}.$$

Thus the error in the two-endpoint trapezoid satisfies

$$\left| h\{F(a)^2 + F(b)^2\} - 2\int_a^b F(x)^2 \, \mathrm{d}x \right| \ll \frac{h^2 + h\log a}{a^{3/2}}. \tag{96.13}$$

Summing the first term over dyadic intervals above X, using (96.10), gives $O_\varepsilon(X^{-1/4+\varepsilon})$. The second gives $O(X^{-1/2}\log X)$, because the gap lengths telescope.

There is one boundary term, which must not be silently omitted:

$$\sum_{n>N} w_n F(t_n)^2 = \sum_{n\ge N} d_n\{F(t_n)^2 + F(t_{n+1})^2\} - d_N F(t_N)^2.$$

The bound (96.10) gives $d_N \ll_\varepsilon t_N^{5/8+\varepsilon/2}$, so the subtracted term is $O_\varepsilon(t_N^{-3/8+\varepsilon/2}\log^2 t_N)$. This is smaller than the error in (96.11). The argument applies without change to the prime-power partition, with its smaller jumps and its inherited square-gap bound.

To bound the centroid we require local energy bounds, not an assertion that every dyadic block has a limiting density. Put $f(x) = (\vartheta(x) - x)/(x\log x)$ for $F = E$, and use f^* above for $F = E^*$. Cramér's bound, and $\psi(x) - \vartheta(x) = O(\sqrt{x})$, give $\int_X^{2X} |f|^2 \ll 1/\log^2 X$. The exact Hardy identity and Cauchy–Schwarz on successive dyadic intervals imply

$$\left| \int_x^\infty \frac{f(t)}{t}\left(1 + \frac{1}{\log t}\right) \mathrm{d}t \right| \ll \sum_{j\ge 0} \frac{1}{\sqrt{2^j x}\log(2^j x)} \ll \frac{1}{\sqrt{x}\log x}.$$

Consequently $\int_X^{2X} F^2 \ll 1/\log^2 X$. The local version of (96.13), or the existing finite sampling inequality, now gives

$$\sum_{X\le t_n<2X} w_n F(t_n)^2 \ll \frac{1}{\log^2 X}, \qquad \sum_{X\le t_n<2X} w_n \ll X.$$

The possible two outer gaps only enlarge the interval by a bounded factor; their endpoint terms obey the same estimates above. Apply Cauchy–Schwarz to each dyadic block and sum the resulting geometric series. This proves the first part of (96.12). Finally $|u_n - 1| \le 2\pi|F(t_n)|$ and $W_N \asymp t_N$ imply its second part. □

Completion of Theorem 96.1. The scalar energy criterion gives bounded D_N and $J_\infty = 2D_\infty$. The chord expansion and the pointwise RH bound show

$$J_\infty - J_N = 4\pi^2 \sum_{n>N} w_n F(t_n)^2 + O\left(\sum_{n>N} w_n F(t_n)^4\right).$$

The last sum is $O((\log^2 t_N)/t_N)$ times the quadratic tail, hence is $o(1/\log t_N)$. Lemma 96.4 and (96.3) therefore give

$$J_\infty - J_N \sim \frac{8\pi^2 C}{\log t_N}, \qquad C = \kappa \text{ or } 1+\kappa.$$

The exact identity, retaining the moving-centroid correction, is

$$2(D_\infty - D_N) = J_\infty - J_N + \frac{|M_N - W_N|^2}{W_N} - \frac{D_N^2}{W_N}. \tag{96.14}$$

Indeed $J_N = V_N + |M_N - W_N|^2/W_N$ and $V_N = 2D_N - D_N^2/W_N$. By (96.12), the second term in (96.14) is $O(1/\log^2 t_N)$; the third is $O(1/t_N)$. They are both smaller than the stated main term, proving (96.4). □

Corollary 96.5 (The unique constant bias removal)**.** *Assume* RH. *For real a, put $F_a = E - a(E - E^*)$. Then*

$$\lim_{Y\to\infty} \frac{1}{\log Y} \int_2^Y (\log x)^2 F_a(x)^2 \, \mathrm{d}x = \kappa + (1-a)^2, \qquad \int_X^\infty F_a(x)^2 \, \mathrm{d}x \sim \frac{\kappa + (1-a)^2}{\log X}.$$

Among these constant multiples of the exact prime-power bias, $a = 1$ is the unique minimizer. Its remaining coefficient $\kappa > 0$ comes from the zeta zeros and cannot be removed by a constant bias.

Proof. The normalized error is $G^*(y) + (1-a) + o(1)$. Use its zero mean, its mean square κ, and Lemma 96.3. □

96.3 A corrected, certified multiplicity allowance

Li's first coefficient is $\lambda_1 = 1 + \gamma/2 - \frac{1}{2}\log(4\pi)$ [59]. Under RH, the absolutely convergent sum gives

$$\kappa - 2\lambda_1 = \sum_{\gamma \ m\text{distinct}} \frac{m_\gamma(m_\gamma - 1)}{1/4 + \gamma^2} \geq 0. \tag{96.15}$$

Equality is equivalent to simplicity of every nontrivial zero. One must not replace the square of the multiplicity by the multiplicity when applying Parseval.

Proposition 96.6 (Explicit allowance without a simplicity assumption)**.** *Assume* RH. *Then*

$$2 + \gamma - \log(4\pi) \ \leq \ \kappa < 2 + \gamma - \log(4\pi) + 4.637 \times 10^{-11}. \tag{96.16}$$

The first term is approximately 0.04619141793224206763*.*

Proof. Platt–Trudgian's interval-arithmetic verification accounts for all zeros up to $T = 3 \cdot 10^{12}$ by distinct sign changes together with Turing's total count [8]. These zeros are therefore simple. For larger heights we use the independently checked bound [54, Corollary 1]

$$N(t) = \mathcal{M}(t) + R(t), \qquad \mathcal{M}(t) = \frac{t}{2\pi}\log\frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8}, \qquad |R(t)| \leq A\log t, \quad A = 0.28.$$

A jump at γ satisfies $m_\gamma \leq 2A\log\gamma$. Let $a = 2A$, $L = \log T$, $c = \log(2\pi)$ and $g(t) = (a\log t - 1)/t^2$. This function is positive and decreasing for $t \geq T$. With N counting positive zeros with multiplicity, (96.15) and Stieltjes integration by parts give

$$\kappa - 2\lambda_1 \leq 2\int_T^\infty g(t) \, \mathrm{d}N(t) \leq 2(I_1 + I_2 + I_3),$$

where the factor 2 includes negative ordinates and

$$I_1 = \int_T^\infty g(t)\mathcal{M}'(t) \, \mathrm{d}t = \frac{a\{L^2 + (2-c)L + 2 - c\} - L - 1 + c}{2\pi T},$$

$$I_2 = Ag(T)\log T = \frac{AL(aL-1)}{T^2},$$

$$I_3 = A\int_T^\infty (-g'(t))\log t \, \mathrm{d}t = \frac{A\{aL^2 + (a/2-1)L + a/4 - 1/2\}}{T^2}.$$

There is no uncontrolled numerical quadrature in these expressions. For example, use the outward rational bounds

$$L < 28.729635, \qquad c > 1.837877, \qquad \pi > 3.1415926.$$

The expression is increasing in L, decreasing in c, and its positive first summand is decreasing in the denominator π on this range. Exact rational substitution gives $2(I_1 + I_2 + I_3) < 4.637 \cdot 10^{-11}$. The certificate below verifies those elementary bounds and the last inequality with rational arithmetic alone. □

Remark 96.7 (Why the supplied smaller allowance was replaced). The supplied manuscript used the constants $0.112 \log t + 0.278 \log\log t + 2.510 + 0.2/t$ from a 2014 zero-count estimate. Brent–Platt–Trudgian explicitly note a problem with an input lemma behind those sharper constants [54, p. 743, footnote 1]; they instead use independently checked constants. Thus the supplied $3.73 \cdot 10^{-11}$ allowance is not certified by the citation given there. Proposition 96.6 uses their safe Corollary 1. It states an interval, rather than claiming a shared number of significant digits for its endpoints.

```
# Exact rational certificate for kappa - 2*lambda_1 < 4.637e-11.
# Only Python's standard library is required.
from fractions import Fraction as F

def atan_interval(x, terms):
    s = sum(((-1)**k * x**(2*k+1) / (2*k+1)
             for k in range(terms)), F(0))
    e = x**(2*terms+1) / (2*terms+1)
    return (s, s+e) if terms % 2 == 0 else (s-e, s)

def log_interval(x, terms=120):
    # For x>1: log(x)=2*atanh((x-1)/(x+1)).
    assert x > 1
    z = (x-1)/(x+1)
    s = 2*sum((z**(2*k+1)/(2*k+1)
               for k in range(terms)), F(0))
    e = 2*z**(2*terms+1)/((2*terms+1)*(1-z*z))
    return s, s+e

# Machin's identity: pi=16*atan(1/5)-4*atan(1/239).
xlo, xhi = atan_interval(F(1, 5), 20)
ylo, yhi = atan_interval(F(1, 239), 6)
pi_lo, pi_hi = 16*xlo-4*yhi, 16*xhi-4*ylo
assert pi_lo > F('3.1415926')

l3lo, l3hi = log_interval(F(3))
l10lo, l10hi = log_interval(F(10))
assert l3hi + 12*l10hi < F('28.729635')
c_lo, _ = log_interval(2*pi_lo)
assert c_lo > F('1.837877')

T = 3*10**12
A = F('0.28')
a = 2*A
L, c, pi_lower = F('28.729635'), F('1.837877'), F('3.1415926')
I1 = (a*(L*L+(2-c)*L+2-c)-L-1+c)/(2*pi_lower*T)
I2 = A*L*(a*L-1)/T**2
I3 = A*(a*L*L+(a/2-1)*L+a/4-F(1, 2))/T**2
assert 2*(I1+I2+I3) < F('4.637e-11')
print('Certified: kappa - 2*lambda_1 < 4.637e-11, under RH.')
```

Remark 96.8 (What the conditional closure does not certify). Equations (96.3) and (96.4) are asymptotic statements. They do not turn the supplied predictions $D_\infty \approx 40.14$ and $D_\infty^* \approx 3.3412$ into rigorous upper bounds, even after assuming RH and simplicity. A finite certificate would additionally need an explicit bound for the mean-square remainder, such as the function η in Lemma 96.3, and certified numerical integration errors. Knowing κ in a narrow interval does not bound those remainders.

97 An exact finite-field tail, including resonant Frobenius roots

The supplied note *Closing What Can Be Closed* proves an explicit uniform bound for the curve analogue and the rate $O(1/N)$. The coefficient of $1/N$ can also be determined. The calculation distinguishes repeated Frobenius roots from distinct roots, and shows when the prime-square contribution interacts with the spectrum.

Let $C/\mathbb{F}_q$ be a smooth projective geometrically irreducible curve of genus g. Write

$$N_d = \#C(\mathbb{F}_{q^d}) = q^d + 1 - \sum_{i=1}^{2g} \alpha_i^d, \qquad N_d = \sum_{e|d} e a_e.$$

Here a_e counts closed points of degree e. The Hasse–Weil theorem gives $|\alpha_i| = \sqrt{q}$; these standard facts, including their cohomological formulation, are reviewed in [60, Sections 26–27]. Put $r = q^{-1/2}$ and $z_i = \alpha_i/\sqrt{q}$. For a distinct unit complex number z, let m_z be the multiplicity of z among the z_i, and take $m_z = 0$ if it does not occur. All sums over z below are over the finite set consisting of these points and $1, -1$.

Define the prime and prime-power errors by

$$S_C(n) = \sum_{d \le n} a_d q^{-d}, \qquad E_C(n) = S_C(n) - H_n - c_C,$$

$$S_C^*(n) = \sum_{dk \le n} \frac{a_d}{k q^{dk}} = \sum_{j \le n} \frac{N_j}{j q^j}, \qquad E_C^*(n) = S_C^*(n) - H_n - c_C^*,$$

where the constants are the respective limits of $S - H_n$. Both constants exist by the estimates proved below. Use weights $w_n = q^{n+1} - q^{n-1}$, $n \ge 1$, and form the deficits $D_N^C, D_N^{C,*}$ from the phases $\exp(2\pi i(S_C(n) - H_n))$ and $\exp(2\pi i(S_C^*(n) - H_n))$, respectively. Rotating all phases by their limiting direction has no effect on either deficit.

Theorem 97.1 (Exact curve tail coefficients)**.** *Set*

$$c_z = m_z + \frac{1}{2}\mathbf{1}_{z=1} + \frac{1}{2}\mathbf{1}_{z=-1},$$

and define

$$\mathcal{K}_C = 2\pi^2(1 - q^{-2}) \sum_z \frac{c_z^2}{|1 - rz|^2}, \tag{97.1}$$

$$\mathcal{K}_C^* = 2\pi^2(1 - q^{-2}) \sum_z \frac{m_z^2}{|1 - rz|^2}. \tag{97.2}$$

Then, with constants in the remainder allowed to depend on the curve,

$$D_\infty^C - D_N^C = \frac{\mathcal{K}_C}{N} + O_C(N^{-2}), \tag{97.3}$$

$$D_\infty^{C,*} - D_N^{C,*} = \frac{\mathcal{K}_C^*}{N} + O_C(N^{-2}). \tag{97.4}$$

The second leading coefficient is zero for genus zero; in that case the sharper formula is

$$D_\infty^{C,*} - D_N^{C,*} = \frac{2\pi^2(q+1)}{q^2}\, q^{-N}(\log N)^2 + O_q(q^{-N}\log N). \tag{97.5}$$

If neither $\sqrt{q}$ nor $-\sqrt{q}$ is a Frobenius root, then

$$\mathcal{K}_C - \mathcal{K}_C^* = \frac{\pi^2(q+1)^2}{q(q-1)}. \tag{97.6}$$

In general the right side must be increased by

$$2\pi^2(1-q^{-2})\left\{\frac{m_1}{(1-r)^2}+\frac{m_{-1}}{(1+r)^2}\right\}. \tag{97.7}$$

Proof. Möbius inversion gives

$$d\left(a_dq^{-d}-\frac{1}{d}\right)=q^{-d}-\sum_i(\alpha_i/q)^d+\sum_{\substack{k\mid d\\k\ge2}}\mu(k)q^{-d}N_{d/k}.$$

The $q^{d/2}$ term at $k=2$ contributes $-\mathbf{1}_{2|d}q^{-d/2}$. All remaining terms with $k\ge3$ have main factor at most $q^{-2d/3}$ after division by q^d. The non-main parts at $k=2$ are at most $(1+2g)q^{-3d/4}$ up to a constant depending on q. Bounding the number of divisors by d is enough for

$$a_dq^{-d}-\frac{1}{d}=-\frac{1}{d}\left\{\sum_i(rz_i)^d+\mathbf{1}_{2|d}r^d\right\}+O_{q,g}(q^{-2d/3}). \tag{97.8}$$

This estimate also proves convergence of the defining constant c_C. Since $\mathbf{1}_{2|d}r^d=(r^d+(-r)^d)/2$, the negative tail of (97.8) yields

$$E_C(n)=\sum_z c_zT_n(rz)+O_{q,g}(q^{-2n/3}),\qquad T_n(v)=\sum_{d>n}\frac{v^d}{d}.$$

For fixed $|v|<1$, elementary summation gives

$$T_n(v)=\frac{v^{n+1}}{n(1-v)}+O_v(|v|^n/n^2).$$

Consequently

$$E_C(n)=\frac{r^n}{n}F_n+O_{q,g}(r^n/n^2+q^{-2n/3}),\qquad F_n=\sum_z c_z\frac{rz}{1-rz}z^n. \tag{97.9}$$

The conjugate pairing of the Frobenius roots makes F_n real. The prime-power calculation is shorter: its exact error is

$$E_C^*(n)=\sum_z m_zT_n(rz)-T_n(q^{-1}).$$

Thus (97.9) holds with c_z replaced by m_z and remainder $O_C(r^n/n^2+q^{-n}/n)$. The arithmetic part of the remainder is exponentially smaller; the r^n/n^2 tail-expansion correction remains.

For any finite sum $F_n=\sum_z b_zz^n$ with distinct $|z|=1$,

$$\sum_{n>N}\frac{|F_n|^2}{n^2}=\frac{1}{N}\sum_z|b_z|^2+O_F(N^{-2}). \tag{97.10}$$

Indeed the diagonal terms use $\sum_{n>N}n^{-2}=N^{-1}+O(N^{-2})$. For $z\ne z'$, partial sums of $(z\overline{z'})^n$ are bounded; summation by parts therefore makes each off-diagonal tail $O(N^{-2})$. Repeated roots must first be grouped, which is why the squares c_z^2 and m_z^2 occur. Multiplying (97.9) by the weights gives

$$\sum_{n>N}w_nE_C(n)^2=\frac{1-q^{-2}}{N}\sum_z\frac{c_z^2}{|1-rz|^2}+O_C(N^{-2}).$$

The same formula holds for E_C^* with m_z.

For completeness, pass from these fixed-center scalar energies to the actual moving-center deficit. For either error e_n put

$$J_N = \sum_{n\le N} w_n|e^{2\pi i e_n} - 1|^2, \quad A_N = \sum_{n\le N} w_n(e^{2\pi i e_n} - 1), \quad \delta_N = |A_N|^2/W_N.$$

The variance identities in Lemma 92.1 imply

$$2(D_\infty - D_N) = J_\infty - J_N + \delta_N - D_N^2/W_N. \tag{97.11}$$

Here $w_n e_n^4 \ll q^{-n}/n^4$ and $|A_N| \ll q^{N/2}/N$, whereas $W_N \asymp q^N$. Hence $\delta_N = O(N^{-2})$, and $D_N^2/W_N = O(q^{-N})$. Finally $|e^{2\pi i e_n} - 1|^2 = 4\pi^2 e_n^2 + O(e_n^4)$. This proves both tail formulas. In genus zero, $E_C^*(n) = -q^{-n}/((q-1)n) + O_q(q^{-n}/n^2)$ and $W_N = (q+1)(q^N - 1)$. Therefore

$$A_N = -2\pi i(1 + q^{-1})\log N + O_q(1), \qquad J_\infty - J_N = O_q(q^{-N}/N^2).$$

Substitution in (97.11) proves (97.5). Here the moving-center correction, rather than the fixed-center energy tail, supplies the leading term.

To subtract the constants, only $z = 1, -1$ remain. At either point $(m_z + 1/2)^2 - m_z^2 = m_z + 1/4$. The two $1/4$ terms simplify to

$$\frac{\pi^2}{2}(1 - q^{-2})\left\{(1 - r)^{-2} + (1 + r)^{-2}\right\} = \frac{\pi^2(q+1)^2}{q(q-1)}.$$

The other two terms are exactly (97.7). □

Corollary 97.2 (A closed expression for a nonextremal elliptic curve)**.** *For genus one write the Frobenius polynomial as $X^2 - aX + q$. If $a^2 < 4q$, then*

$$\mathcal{K}_C^* = \frac{4\pi^2(q^2-1)}{q(q+1-a)}, \qquad \mathcal{K}_C = \frac{4\pi^2(q^2-1)}{q(q+1-a)} + \frac{\pi^2(q+1)^2}{q(q-1)}.$$

At $a = \pm 2\sqrt{q}$, use Theorem 97.1 instead: the two roots coincide, and replacing m_z^2 by m_z would lose a factor of two in the spectral contribution as well as the resonance term.

Proof. The two roots are distinct conjugates, and for each of them $|1 - \alpha_i/q|^2 = (q + 1 - a)/q$. □

Scope of the hypothetical converse. The supplied note's converse for a "zeta function of the same shape" needs hypotheses. A safe version assumes a finite conjugation-stable multiset of roots, real resulting N_d, and $\sqrt{q} < R := \max_i |\alpha_i| < q$. These ensure that the centered errors exist and tend to zero. The terms of largest modulus then give

$$E_C(n) = \frac{(R/q)^n}{n}\left(\sum_j b_j z_j^n + o(1)\right), \qquad |z_j| = 1,$$

with a nonzero finite exponential sum. Its positive mean square forces $\sum_n w_n E_C(n)^2 = \infty$, so the variance lemma proves $D_N \to \infty$. If $R \ge q$, the centering constant need not exist; the original formulation cannot simply be used. This converse concerns formal spectral data, not a curve violating Hasse–Weil.

What is gained. The $O(1/N)$ statement is now an exact leading term, with an $O_C(N^{-2})$ remainder. The prime and prime-power constructions have an explicit difference, including exceptional resonance. The calculation uses the already proved Hasse–Weil theorem; it supplies no new zero-location estimate for the number-field zeta function.

98 What the family average closes, and what it does not

The large-sieve transfer in the supplied companion survives the audit as a finite-window family estimate. Its interpretation needs care. For a primitive character $\chi \bmod q$ put

$$B_\chi(y) = \sum_{y<p\le 2y} \frac{\chi(p)}{p}, \qquad \Delta_\chi(x;y) = \sum_{y<p\le x} \frac{\chi(p)}{p}.$$

The multiplicative large sieve of Montgomery–Vaughan [28, 61] gives

$$\sum_{q\le Q} \frac{q}{\varphi(q)} \sideset{}{^*}\sum_{\chi \bmod q} |B_\chi(y)|^2 \le (y+Q^2) \sum_{y<p\le 2y} p^{-2}. \tag{98.1}$$

The sequence is supported on an interval of at most $y+1$ integers, so the usual $N-1+Q^2$ constant is at most $y+Q^2$. The short-interval Brun–Titchmarsh bound $\pi(2y)-\pi(y) \le 2y/\log y$ gives, when $y = Q^2 \ge 4$,

$$\sum_{2\le q\le Q} \frac{q}{\varphi(q)} \sideset{}{^*}\sum_{\chi \bmod q} |B_\chi(y)|^2 \le \frac{4}{\log y},$$

and application to each truncated sequence, followed by integration, gives

$$\sum_{2\le q\le Q} \frac{q}{\varphi(q)} \sideset{}{^*}\sum_{\chi \bmod q} \int_y^{2y} |\Delta_\chi(x;y)|^2 \,\mathrm{d}x \le \frac{4y}{\log y}.$$

For example, at most $4y/A$ members of this family can have local energy larger than $A/\log y$, since $q/\varphi(q) \ge 1$. This is a direct exceptional-count consequence of the positive sum.

The estimate averages over conductors and changes the family as the window grows. It neither bounds the infinite tail of one fixed character nor selects the untwisted prime error E. The local displacement Δ_χ is also different from its globally centered tail error. In particular, the claimed fixed-dyadic asymptotic "energy $\asymp \kappa_\chi/\log^2 y$" does not follow merely from a long logarithmic mean-square formula. This module uses the large-sieve inequalities themselves and does not use that interpretation as a theorem.

98.1 Other corrections retained in the combined argument

1. **A zero-count allowance requires a valid input bound.** The replacement of $3.73\cdot 10^{-11}$ by the certified allowance $4.637\cdot 10^{-11}$ is justified in Proposition 96.6. Even a narrow interval must be reported as an interval, rather than as a fixed string of decimal digits shared by all its possible values.

2. **The exact circle is not its linearization.** Zhao's theorem concerns $\int_2^X E(x)\,\mathrm{d}x$ [55]. An assertion about the sign of $\sum w_n \sin(2\pi E(p_n))$ needs additional control of the cubic remainder and the quadrature error. The supplied linearized-centroid discussion is not used as an exact all-N equivalence here.

3. **The Li endpoint is admissible only after exact centering.** Section 94 removes an earlier restriction $\alpha > 1/2$, while proving why an approximate centering constant cannot be substituted at $\alpha = 1/2$.

4. **A formal curve converse needs a defined limiting phase.** Section 97 adds the condition $\max_i |\alpha_i| < q$ and reality assumptions to the hypothetical off-circle model. Actual curves already satisfy Hasse–Weil.

5. **A predicted limit is not a certified upper bound.** The supplied extrapolations for D_∞ and D_∞^* are not promoted to numerical values of the desired K. The exact asymptotic coefficient and the finite-cutoff remainder are different quantities.

99 The fixed target and the retained analytic input

Write p_n for the nth prime, $p_1 = 2$, and set

$$S(x) = \sum_{p \le x} \frac{1}{p}, \qquad E(x) = S(x) - \log\log x - B_{\mathbb{P}}, \qquad B_{\mathbb{P}} = \gamma - \sum_{p} \sum_{k \ge 2} \frac{1}{kp^k}.$$

The phases and weights remain those of the preceding note [62]:

$$U_n = e^{2\pi i (S(p_n) - \log\log p_n)}, \quad a = e^{2\pi i B_{\mathbb{P}}}, \quad w_n = p_{n+1} - p_{n-1}, \quad n \ge 2.$$

Thus $U_n \to a$, and the finite U_n require no numerical input for $B_{\mathbb{P}}$. For monic polynomials put

$$H_{m,N} = \min_{\deg P = m} \sum_{n=2}^{N} w_n |P(U_n)|^2, \qquad J_{m,N} = \sum_{n=2}^{N} w_n |U_n - a|^{2m}.$$

The minimizing polynomial is denoted $P_{m,N}$. Work after a prefix with at least five distinct phases, so all norms up to degree four are positive. The original deficit remains

$$D_N = \sum_{n=2}^{N} w_n - \left| \sum_{n=2}^{N} w_n U_n \right|;$$

the present target is the weaker, specifically selected coefficient condition, not an asserted bound for D_N.

We separately retain a fixed-strip input $\Theta \le b < 7/8$, where

$$\Theta = \sup\{\Re\rho : \zeta(\rho) = 0,\ 0 < \Re\rho < 1\}.$$

The previous note used Liu's reported $b_* = 0.874957019420098946\ldots$ [13]. The more recent Xu–Sang value 0.8748867 is discussed below with its explicitly imported analytic assumptions [63]. Neither source's full underlying analytic proof is independently rebuilt here. All finite algebraic results below are independent of those inputs; the application of the degree-four limiting framework to the actual primes uses the retained strip input.

Proposition 99.1 (Why the coefficient target has its stated meaning)**.** *Under the retained input, $J_{4,\infty} < \infty$ and $P_{4,N}$ converges coefficientwise to*

$$P_{4,\infty}(z) = (z-a)^r Q_{4-r}(z), \qquad r = \min\{j \in \{1,2,3,4\} : J_{j,\infty} < \infty\}.$$

Every zero of Q_{4-r} is strictly inside the unit disk. Consequently the following conditions are equivalent:

$$\exists \varepsilon > 0 :\ |c_{3,N}| \le 4 - \varepsilon \quad \textit{eventually};$$
$$|c_{3,\infty}| < 4; \qquad r \le 3; \qquad J_{3,\infty} < \infty; \qquad \sup_N H_{3,N} < \infty; \qquad \Theta \le 5/6.$$

Proof. We give the essentials to keep the target explicit. For $q = 6, 8$, the Mellin identity for E and Hölder's inequality show that $E \in L^q$ excludes zeros in $\Re s > 1 - 1/q$: a zero there creates a nonremovable logarithmic singularity. For sufficiency, Grosswald's estimate as stated in [55, eq. (24)] gives $E(x) = O_\Theta(x^{\Theta - 1}/\log x)$ for $1/2 < \Theta < 1$. At the endpoint the integrable majorant is $x^{-1}(\log x)^{-q}$. Under RH the usual $E(x) = O(x^{-1/2}\log x)$ suffices for these two exponents. On a prime gap, E varies only by $-\log\log x$ and then jumps by $1/p$. Changing variables to $\log\log x$ compares the gap integral of $|E|^q$ with its endpoint powers; Bertrand's bound keeps the Jacobian ratio bounded, and the jump corrections sum to a finite amount. Thus the original adjacent-gap weights give exactly the same finiteness condition. Since $|U_n - a| \sim 2\pi |E(p_n)|$, this proves the required moment claims.

For completeness, bounded $H_{4,N}$ bounds all minimizing coefficients by evaluation at four fixed distinct phases and Vandermonde inversion. A subsequential limit has finite full energy by passage to each finite prefix. A polynomial with a zero of exact order j at a has finite full energy precisely when J_j is finite. The admissible monic quartics are therefore exactly $(z-a)^r Q$, and minimizing their energy is ordinary orthogonal projection for the finite measure $\sum_n w_n|U_n-a|^{2r}\delta_{U_n}$. Its infinite support gives a unique minimizer. The usual reflection argument excludes zeros outside the disk; orthogonality excludes boundary zeros. This proves convergence and factorization. The same compactness argument at degree three proves $\sup H_3<\infty \iff J_3<\infty$.

Finally Viète gives $c_{3,\infty}=-ra-\sum_{j=1}^{4-r}\xi_j$ with $|\xi_j|<1$. Its modulus is four exactly when $r=4$; otherwise it is strictly less than four. Coefficient convergence turns a strict limiting inequality into an eventual uniform margin. □

The infinite-support assertion used here does not assume RH. If the prime phases had finite support, convergence would force $E(p_n)=0$ eventually. For consecutive sufficiently large odd primes $p<q$ this gives $\log(\log q/\log p)=1/q$, hence $q-p=\log q+O(\log^2 q/q)$. Parity would imply $\operatorname{dist}(\log q, 2\mathbb{Z})\to 0$. Bertrand's postulate applied near e^{2j+1} supplies primes whose logarithms stay a distance at least $1-\log 2-o(1)$ from $2\mathbb{Z}$, a contradiction.

100 The compression and its one-dimensional defect

Let $\mu_N=\sum_{n=2}^{N} w_n\delta_{u_n}$, where $w_n>0$ and $|u_n|=1$. Work after a prefix containing at least five distinct phases. Let $P_{j,N}$ be the monic orthogonal polynomial of degree j, and let $H_{j,N}=\|P_{j,N}\|_{\mu_N}^2>0$, for $0\le j\le 4$; thus $P_{0,N}=1$ and $H_{0,N}=W_N$. Our inner product is linear in its first argument:

$$\langle f,g\rangle_N=\int f\overline{g}\,d\mu_N.$$

Put $\varphi_{j,N}=P_{j,N}/\sqrt{H_{j,N}}$. Let V_N be the space of polynomials of degree at most three, with its induced Hilbert norm, and let $C_N=\Pi_{V_N}M_z|_{V_N}$, where $M_zf(z)=zf(z)$.

Theorem 100.1 (Exact compression and trace bound)**.** *The characteristic polynomial of C_N is $P_{4,N}$, and $\operatorname{tr} C_N=-c_{3,N}$. In the ordered orthonormal basis $\varphi_{0,N},\dots,\varphi_{3,N}$,*

$$I-C_N^*C_N=\frac{H_{4,N}}{H_{3,N}}e_3e_3^*. \tag{100.1}$$

Consequently,

$$|c_{3,N}|\le 3+\sqrt{1-\frac{H_{4,N}}{H_{3,N}}},\qquad 4-|c_{3,N}|\ge 1-\sqrt{1-\frac{H_{4,N}}{H_{3,N}}}\ge\frac{H_{4,N}}{2H_{3,N}}. \tag{100.2}$$

Every ratio here belongs to $(0,1]$.

Proof. In the monomial basis, $C_Nz^j=z^{j+1}$ for $j<3$, while

$$C_Nz^3=z^4-P_{4,N}(z)=-c_{0,N}-c_{1,N}z-c_{2,N}z^2-c_{3,N}z^3.$$

Its matrix is therefore the companion matrix of $P_{4,N}$. This proves the characteristic-polynomial and trace assertions; the monomial basis need not be orthonormal for either assertion.

Multiplication by z is unitary on $L^2(\mu_N)$. For $j<3$, $z\varphi_{j,N}$ remains in V_N. For $j=3$, its component orthogonal to V_N is $P_{4,N}/\sqrt{H_{3,N}}$, whose squared norm is $H_{4,N}/H_{3,N}$. Thus the squared norm lost by compression is exactly this ratio times the squared last coordinate, proving (100.1). The ratio is at most one also directly from the monic trial polynomial $zP_{3,N}$.

The singular values of C_N are consequently $1, 1, 1, \sqrt{1 - H_{4,N}/H_{3,N}}$. The inequality $|\operatorname{tr} C| \le \sum_j s_j(C)$ follows from a singular-value decomposition: after cyclically moving the unitary factors inside the trace, every diagonal entry of the remaining unitary matrix has modulus at most one. This proves (100.2); its final inequality is $1 - \sqrt{1-x} \ge x/2$. □

If $H_{3,N} \le B_3$ eventually and $N \ge N_0$, this yields the explicit margin

$$\varepsilon = 1 - \sqrt{1 - H_{4,N_0}/B_3} > 0.$$

The numerator has a positive finite-prefix lower bound. Bounding the denominator is the remaining arithmetic assertion: $\sup_N H_{3,N} < \infty$ is itself equivalent to $\Theta \le 5/6$ for the prime sequence. The trace inequality is sharp for the underlying rank-one-defect contraction problem: $C = \operatorname{diag}(a, a, a, ta)$, $|a| = 1$, $0 \le t < 1$, has equality. This algebraic extremizer is not asserted to be a finite prime-circle compression.

101 Exact coefficient increments and a tail radius

For a measure μ, suppress its index and put

$$K_k(z, v) = \sum_{j=0}^{k} \frac{P_j(z)\overline{P_j(v)}}{H_j}.$$

With our inner-product convention, $\langle Q, K_k(\,\cdot\,, v)\rangle_\mu = Q(v)$ for every polynomial Q of degree at most k.

Proposition 101.1 (One-atom update and total variation)**.** *When $\mu_{N+1} = \mu_N + w\delta_v$,*

$$c_{3,N+1} - c_{3,N} = -\frac{wP_{4,N}(v)\overline{P_{3,N}(v)}}{H_{3,N}(1 + wK_{3,N}(v,v))}, \tag{101.1}$$

$$|c_{3,N+1} - c_{3,N}|^2 = (H_{4,N+1} - H_{4,N})\left(\frac{1}{H_{3,N}} - \frac{1}{H_{3,N+1}}\right). \tag{101.2}$$

In particular, for $M > N$,

$$\sum_{k=N}^{M-1} |c_{3,k+1} - c_{3,k}| \le \sqrt{(H_{4,M} - H_{4,N})\left(\frac{1}{H_{3,N}} - \frac{1}{H_{3,M}}\right)}. \tag{101.3}$$

If $H_{4,N} \uparrow H_{4,\infty} < \infty$, the coefficient has finite total variation. With $1/H_{3,\infty} = 0$ when $H_{3,N} \to \infty$,

$$|c_{3,\infty} - c_{3,N}| \le \sqrt{(H_{4,\infty} - H_{4,N})\left(\frac{1}{H_{3,N}} - \frac{1}{H_{3,\infty}}\right)} \le \sqrt{\frac{H_{4,\infty} - H_{4,N}}{H_{3,N}}}. \tag{101.4}$$

Proof. Reproduction and orthogonality show directly that

$$P_m^{\mathrm{new}}(z) = P_m(z) - \frac{wP_m(v)}{1 + wK_{m-1}(v,v)} K_{m-1}(z, v).$$

Indeed, its inner product for $\mu + w\delta_v$ with any lower-degree Q is

$$[-t + w\{P_m(v) - tK_{m-1}(v,v)\}]\,\overline{Q(v)} = 0, \qquad t = \frac{wP_m(v)}{1 + wK_{m-1}(v,v)}.$$

The norm update is

$$H_m^{\mathrm{new}} - H_m = \frac{w|P_m(v)|^2}{1 + wK_{m-1}(v,v)};$$

it also follows by taking the new inner product of P_m^{new} with P_m. The leading coefficient of $K_3(z,v)$, as a polynomial in z, is $\overline{P_3(v)}/H_3$, giving (101.1). Use

$$K_3(v,v) = K_2(v,v) + \frac{|P_3(v)|^2}{H_3}$$

and the norm update at degree three to obtain

$$\frac{w|P_3(v)|^2}{H_3^2(1 + wK_3(v,v))} = \frac{1}{H_3} - \frac{1}{H_3^{\mathrm{new}}}.$$

Multiplication by the degree-four norm increment proves (101.2). Cauchy–Schwarz and telescoping give (101.3). Passage to $M \to \infty$ proves absolute convergence and (101.4). □

Remark 101.2 (There is no one-sided sign supplied by positivity). The two factors in (101.2) are nonnegative, but they control the squared magnitude, not the direction, of the complex increment. For unit weights on the fifth roots of unity, $P_3 = z^3$, $P_4 = z^4$, and adding $w\delta_v$ gives

$$\Delta c_3 = -\frac{wv}{5 + 4w}.$$

For a fixed unit a, the sign of $\Re(\overline{a}\,\Delta c_3)$ changes between $v = a$ and $v = -a$. Thus positive atomic updates alone cannot supply a one-sided radial estimate for the prime coefficient.

102 What a finite coefficient certificate needs

Corollary 102.1 (A directional finite-prefix certificate)**.** *Suppose B_4 is a rigorously established upper bound for $H_{4,\infty}$. Define*

$$R_N(B_4) = \sqrt{\frac{B_4 - H_{4,N}}{H_{3,N}}}.$$

If, at one finite index,

$$|c_{3,N}| + R_N(B_4) < 4, \tag{102.1}$$

then $|c_{3,\infty}| < 4$. More precisely, if $\delta = 4 - |c_{3,N}| - R_N(B_4) > 0$, then $|c_{3,M}| \le 4 - \delta$ for every $M \ge N$, and the same bound holds for $|c_{3,\infty}|$.

Proof. For $M > N$, use (101.3), $H_{4,M} \le B_4$, and $1/H_{3,N} - 1/H_{3,M} \le 1/H_{3,N}$ to obtain $|c_{3,M} - c_{3,N}| \le R_N(B_4)$. The triangle inequality gives the asserted bound for every $M \ge N$; coefficient convergence gives the limiting bound. Every finite quantity is obtained from the original uncentered phases and weights. The certified bound B_4, rather than a numerical estimate of it, is an essential input. □

The usual bound $H_{4,\infty} \le J_{4,\infty}$ cannot make this certificate succeed. The obstruction is exact.

Proposition 102.2 (The centered-quartic bound always includes the bad coefficient)**.** *Assume $u_n \to a$, $|a| = 1$, and*

$$J_{4,\infty} = \sum_{n \ge 2} w_n |u_n - a|^8 < \infty.$$

For every admissible finite prefix,

$$J_{4,N} - H_{4,N} \ge H_{3,N}|-4a - c_{3,N}|^2. \tag{102.2}$$

Consequently, if $B_4 \ge J_{4,N}$, then

$$R_N(B_4) \ge |-4a - c_{3,N}|, \qquad |c_{3,N}| + R_N(B_4) \ge 4.$$

In particular, $B_4 = J_{4,\infty}$, or any larger bound, cannot verify (102.1).

Proof. Let $R(z) = (z-a)^4$. The difference $R - P_{4,N}$ has degree at most three, so orthogonality gives

$$J_{4,N} = H_{4,N} + \|R - P_{4,N}\|_{\mu_N}^2.$$

Its cubic coefficient is $-4a - c_{3,N}$. By the defining extremal property of $H_{3,N}$, its norm squared is at least $H_{3,N}|-4a-c_{3,N}|^2$, also when that coefficient vanishes. This proves (102.2); the remaining statements follow from the triangle inequality and $|4a| = 4$. □

There is a strict finite improvement at every nontrivial prefix, but it need not persist. For real t, the monic trial polynomial $R_t(z) = (z-a)^3(z-(1-t)a)$ has the exact energy

$$\|R_t\|_{\mu_N}^2 = (1-t)J_{4,N} + t^2 J_{3,N}.$$

Thus

$$H_{4,N} \le J_{4,N} - \frac{J_{4,N}^2}{4J_{3,N}} < J_{4,N}.$$

In the permitted fourfold-limit case, $H_{4,\infty} = J_{4,\infty}$ and $J_{3,N} \to \infty$; indeed $J_{4,N} - H_{4,N} \to 0$. This rules out inferring $H_{4,\infty} \le H_{4,N} + J_{4,\infty} - J_{4,N}$ from a finite optimization and the centered-quartic tail alone.

The remaining target. The compression supplies a quantitative margin once $H_{4,N}/H_{3,N}$ has a positive eventual lower bound. The coefficient-ball certificate instead needs a sufficiently sharp global norm bound; any successful instance necessarily has $B_4 < J_{4,N}$, hence $H_{4,\infty} < J_{4,\infty}$. By the established limit-factorization theorem, that strict global improvement is exactly the exclusion of the fourfold boundary root. The finite identities are proved. The additional prime-specific uniform inequality remains to be established.

103 What the reciprocal recurrence and small gaps do not prove

The following construction concerns an increasing sequence of integers, not the actual primes. Its purpose is to test precisely which inputs can force the proposed uniform coefficient gap. It preserves an arbitrary finite observed prime prefix, the reciprocal-step recurrence, odd-integer parity, much smaller gaps than currently needed in the argument, oscillation of the centered error, and the admitted power bound for that error. It nevertheless has the forbidden fourfold boundary limit. No claim about a counterexample for the actual primes is made.

Theorem 103.1 (A prime-shaped integer continuation with fourfold collapse)**.** *Fix $b \in (5/6, 7/8)$, a nonzero real c, and*

$$\max\{1/8, 1-b\} < \alpha < 1/6.$$

For every fixed $M \ge 2$, there is a strictly increasing integer sequence (q_n), beginning with $q_n = p_n$ for $1 \le n \le M$, whose subsequent terms are odd, such that the following statements hold. Let $A(x) = \#\{n : q_n \le x\}$ and $\mathrm{Li}_2(x) = \int_2^x dt/\log t$. Then

$$A(x) = \mathrm{Li}_2(x) + c\frac{x^{1-\alpha}}{\log x}\sin(\log x) + O(1), \qquad q_{n+1} - q_n = \log q_n + O(1). \tag{103.1}$$

The limit

$$B_A = \lim_{x\to\infty}\left(\sum_{q_n\le x}\frac{1}{q_n} - \log\log x\right)$$

exists. Its centered error satisfies

$$\begin{aligned} E_A(x) &= \sum_{q_n\le x}\frac{1}{q_n} - \log\log x - B_A \\ &= c\frac{x^{-\alpha}}{\log x}\operatorname{Im}\left(\left(1-\frac{1}{\alpha-i}\right)e^{i\log x}\right) + O\left(\frac{x^{-\alpha}}{\log^2 x}\right) + O(x^{-1}). \end{aligned} \tag{103.2}$$

In particular the error changes sign infinitely often, and $E_A(x) = O(x^{b-1}/\log x)$. At every successive pair it obeys the exact prime-form recurrence

$$E_A(q_{n+1}) - E_A(q_n) = \frac{1}{q_{n+1}} - \log\left(\frac{\log q_{n+1}}{\log q_n}\right).$$

Define the corresponding phases and the same adjacent-gap weights by

$$U_n^A = \exp\left(2\pi i\left[\sum_{j\le n}\frac{1}{q_j} - \log\log q_n\right]\right), \quad a_A = e^{2\pi i B_A}, \quad w_n^A = q_{n+1} - q_{n-1},$$

and let $P_{4,N}^A$ be the monic least-squares minimizer for this weighted configuration. Then

$$\sum_{n\ge 2} w_n^A |U_n^A - a_A|^8 < \infty, \qquad \sum_{n\ge 2} w_n^A |U_n^A - a_A|^6 = \infty,$$

and consequently

$$P_{4,N}^A(z) \longrightarrow (z-a_A)^4, \qquad c_{3,N}^A \longrightarrow -4a_A, \qquad |c_{3,N}^A| \longrightarrow 4. \tag{103.3}$$

More quantitatively, with $X = q_N$,

$$\sum_{n=2}^{N} w_n^A |U_n^A - a_A|^6 \asymp \frac{X^{1-6\alpha}}{\log^6 X}, \tag{103.4}$$

$$\sum_{n>N} w_n^A |U_n^A - a_A|^8 \asymp \frac{X^{1-8\alpha}}{\log^8 X}. \tag{103.5}$$

Thus the eighth-energy tail can even decay faster than the bound deduced from the currently admitted exponent b, while the requested uniform coefficient gap fails.

Proof. For sufficiently large x, put

$$F(x) = \operatorname{Li}_2(x) + c\frac{x^{1-\alpha}}{\log x}\sin(\log x).$$

Direct differentiation gives

$$F'(x) = \frac{1}{\log x}\left[1 + cx^{-\alpha}\left((1-\alpha)\sin\log x + \cos\log x - \frac{\sin\log x}{\log x}\right)\right].$$

Hence $F' > 0$ eventually. If $x_k = F^{-1}(k)$, the mean value theorem gives

$$x_{k+1} - x_k = \log x_k + o(1).$$

Choose an integer K so large that these gaps exceed four for $k \ge K$, and $x_{K+1} > p_M + 2$. Continue the prescribed prime prefix with the largest odd integer not exceeding x_{K+j}, $j = 1, 2, \ldots$. Each rounding changes the quantile by less than two, so the new terms are strictly increasing and their gaps are $\log q_n + O(1)$.

Counting rounded quantiles at x differs from counting real quantiles at x by at most the number in an interval of length two, which is bounded. The finite prefix and the starting index contribute only a constant. This proves (103.1). In particular $q_{n+1} < 2q_n$ for all sufficiently large n.

Set $\Delta_A(x) = A(x) - \mathrm{Li}_2(x)$. Partial summation shows that

$$B_A = -\log\log 2 + \int_2^\infty \frac{\Delta_A(t)}{t^2}\,dt, \qquad E_A(x) = \frac{\Delta_A(x)}{x} - \int_x^\infty \frac{\Delta_A(t)}{t^2}\,dt.$$

The first integral converges absolutely since $\alpha > 0$. To evaluate the tail, write the sine as an imaginary part and use

$$\int_x^\infty \frac{t^{-1-\alpha+i}}{\log t}\,dt = \frac{x^{-\alpha+i}}{(\alpha - i)\log x} + O\!\left(\frac{x^{-\alpha}}{\log^2 x}\right),$$

which follows by integration by parts after $t = e^y$. The $O(1)$ counting error contributes $O(x^{-1})$. This proves (103.2). Its oscillation has nonzero amplitude

$$|c|\left|1 - \frac{1}{\alpha - i}\right| = |c|\sqrt{\frac{(1-\alpha)^2 + 1}{\alpha^2 + 1}} > 0.$$

The bound with exponent $b - 1$ follows from $\alpha > 1 - b$.

Equation (103.2) implies

$$\int_2^X |E_A(x)|^6\,dx \asymp X^{1-6\alpha}\log^{-6} X, \qquad \int_X^\infty |E_A(x)|^8\,dx \asymp X^{1-8\alpha}\log^{-8} X.$$

For the lower bounds, any logarithmic interval of fixed length 2π contains an interval of a fixed positive length on which the leading sinusoid is bounded away from zero. The error in (103.2) is smaller there for sufficiently large x. The resulting intervals in the original variable have length comparable to their endpoints. The upper bounds follow directly from $|E_A(x)| \ll x^{-\alpha}/\log x$.

On every gap (q_n, q_{n+1}), the centered error has the exact form $E_A(q_n) - \log(\log x/\log q_n)$, and at the next point it jumps by $1/q_{n+1}$. The same endpoint sampling comparison as for primes therefore applies: the Jacobian of $v = \log\log x$ varies by a bounded factor, and the summed endpoint-jump corrections are finite. For tail comparisons their order is $O(X^{1-s})$ at exponent $s > 1$, which is negligible here. The one boundary endpoint omitted or included at q_N contributes at most $O(X^{-s\alpha}\log^{1-s} X)$, also negligible compared with $X^{1-s\alpha}\log^{-s} X$. Finally

$$|U_n^A - a_A| = 2|\sin(\pi E_A(q_n))| \sim 2\pi|E_A(q_n)|.$$

This proves the two quantitative moment estimates. The phases have infinite support, since their centered errors tend to zero while taking nonzero values on infinitely many logarithmic intervals. The general extremal-polynomial theorem now gives (103.3): the first finite integer moment is exactly the fourth. □

Remark 103.2 (Concrete parameters and preservation of finite experiments). For the admitted $b_* = 0.874957\ldots$, the simple choice $\alpha = 1/7$ is valid. In that case the leading bracket in (103.2) is

$$\frac{43\sin(\log x) - 49\cos(\log x)}{50},$$

whose squared amplitude is $17/10$. The sixth energy grows like $X^{1/7}/\log^6 X$, while the eighth-energy tail is comparable to $X^{-1/7}/\log^8 X$. Choosing the preserved prefix through p_{N_*+1} keeps

every original phase and gap weight, hence every computed $P_{4,N}$, unchanged for $N \le N_*$. This is a statement about the insufficiency of those finite data and stated analytic hypotheses for a larger class of sequences. The constructed continuation is not asserted to consist of primes and does not preserve the Euler product or functional equation of the ordinary Riemann zeta function.

A sign input that would be circular. For the actual primes, Zhao's Theorem 1 in *On the mean values of the error terms in Mertens' theorems* (Research in Number Theory 11 (2025), article 62, https://doi.org/10.1007/s40993-025-00640-y) states that $\int_2^X E(x)\,dx > 0$ for every $X > 2$ is equivalent to RH. It cannot be used as an unconditional sign input in a proposed proof of the weaker target $\Theta \le 5/6$.

The finite variation has a nonzero cross term but a divergent cost. Write $\delta_n = u_n - a$, $|u_n| = |a| = 1$. The identity $\mathrm{Re}(\bar{a}\delta_n) = -|\delta_n|^2/2$ gives

$$\mathrm{Re}\left(\bar{a}\sum_{n\le N} w_n \delta_n^4 \overline{\delta_n^3}\right) = -\frac{1}{2}J_{4,N}.$$

Consequently the monic trial polynomial $R_t(z) = (z-a)^3(z-(1-t)a)$, with real t, has exact energy

$$\sum_{n\le N} w_n |R_t(u_n)|^2 = (1-t)J_{4,N} + t^2 J_{3,N}.$$

Its optimal finite gain is $J_{4,N}^2/(4J_{3,N})$. In the constructed fourfold-collapse example this gain tends to zero, and every fixed nonzero t has infinite full energy. Thus the geometrically nonzero cross term does not justify varying the infinite energy before proving the needed sixth-moment control.

104 The exact analytic cost of the cubic-coefficient gap

Put

$$E(x) = \sum_{p\le x}\frac{1}{p} - \log\log x - B_{\mathbb{P}}, \qquad \Theta = \sup\{\Re\rho : \zeta(\rho) = 0,\ 0 < \Re\rho < 1\},$$

and define two different accumulated quantities,

$$A_6(X) = \int_2^X |E(x)|^6\,dx, \qquad G(X) = \int_2^X E(x)\,dx.$$

The first is nonnegative and nondecreasing; the second retains signs. The target remains the same coefficient gap throughout this section. When the separately stated fixed-strip input supplies the existence and factorization of the degree-four extremizer, the previous argument gives

$$|c_{3,\infty}| < 4 \quad\Longleftrightarrow\quad \Theta \le \frac{5}{6}.$$

The results below give exact analytic versions of this condition. They do not establish the condition for the primes.

104.1 Only subpower growth of the sixth energy would suffice

Two standard prime-distribution inputs are sufficient for the argument. For $1/2 < \Theta < 1$, Grosswald's estimate, as recorded in [55, eq. (24)], and partial summation give

$$E(x) = O_\Theta\left(\frac{x^{\Theta-1}}{\log x}\right). \tag{104.1}$$

Under RH, the usual pointwise bound instead gives $E(x) = O(x^{-1/2}\log x)$. For every $\Theta > 1/2$ and every $\varepsilon > 0$, Zhao proves the two-sided oscillation [55, eq. (20)]

$$G(X) = \Omega_{\pm}(X^{\Theta-\varepsilon}). \tag{104.2}$$

No linear-independence or simplicity hypothesis is used here.

Theorem 104.1 (Exact growth exponent of the sixth energy)**.** *For the original prime-reciprocal error,*

$$\limsup_{X\to\infty} \frac{\log(1 + A_6(X))}{\log X} = \max\{0, 6\Theta - 5\}. \tag{104.3}$$

In particular, the following conditions are equivalent:

$$\Theta \le 5/6; \tag{104.4}$$

$$A_6(\infty) < \infty; \tag{104.5}$$

$$\text{for every } \varepsilon > 0, \quad A_6(X) = O_\varepsilon(X^\varepsilon). \tag{104.6}$$

Proof. First consider the upper bound. If $\Theta = 1/2$, the RH pointwise estimate gives $|E(x)|^6 \ll x^{-3}(\log x)^6$, so $A_6(\infty)$ is finite. If $1/2 < \Theta \le 5/6$, (104.1) gives

$$|E(x)|^6 \ll_\Theta \frac{x^{6\Theta-6}}{(\log x)^6}.$$

This is integrable, including the endpoint $\Theta = 5/6$: the endpoint majorant is $1/(x(\log x)^6)$. Thus the left side of (104.3) is zero throughout $1/2 \le \Theta \le 5/6$.

If $5/6 < \Theta < 1$, integration of the same estimate gives

$$A_6(X) \ll_\Theta \frac{X^{6\Theta-5}}{(\log X)^6} \qquad (X \text{ sufficiently large}). \tag{104.7}$$

The implicit constant may depend on Θ and is not claimed to be numerically certified. Finally, if $\Theta = 1$, the unconditional boundedness of E gives $A_6(X) \ll X$, sufficient for the asserted upper exponent one. No fixed zero-free half-plane is assumed in this last case.

For the lower bound when $\Theta > 5/6$, Hölder's inequality gives

$$|G(X)|^6 \le (X - 2)^5 A_6(X).$$

For each fixed $\varepsilon > 0$, (104.2) supplies an unbounded sequence X_j on which $|G(X_j)| \ge c_\varepsilon X_j^{\Theta-\varepsilon}$. Consequently

$$A_6(X_j) \ge c_\varepsilon^6 X_j^{6\Theta-5-6\varepsilon}.$$

Taking the logarithmic limsup and then letting ε decrease to zero proves the lower exponent $6\Theta - 5$, also when $\Theta = 1$. For the remaining cases the lower bound zero is automatic because $\log(1 + A_6(X)) \ge 0$. This proves (104.3). It also proves that (104.6) forces $\Theta \le 5/6$; the upper estimates, with their endpoint logarithm, then give (104.5). The other implications are immediate. □

This reduces the analytic quantifiers: a direct proof of convergence of the sixth energy is unnecessary if every positive power of growth can be excluded. It does not eliminate the arithmetic difficulty. For example, a single estimate $A_6(X) \ll X^\delta$ with $\delta > 0$ gives only $\Theta \le (5 + \delta)/6$, not $\Theta \le 5/6$. Finite samples cannot certify the universal quantifier in (104.6).

104.2 A one-sided mean condition is already equivalent

Proposition 104.2 (One-sided accumulated-error criterion)**.** *Fix $b \in (1/2, 1)$. Each of the following is equivalent to $\Theta \le b$:*

1. *there exist $C > 0$ and X_0 such that $G(X) \ge -CX^b$ for every $X \ge X_0$;*
2. *there exist $C > 0$ and X_0 such that $G(X) \le CX^b$ for every $X \ge X_0$.*

Thus at $b = 5/6$, either single-sided bound is an exact alternative to the cubic-coefficient gap, within the degree-four framework.

Proof. If $1/2 < \Theta \le b$, integration of (104.1) gives $G(X) = O_\Theta(X^b/\log X)$. If $\Theta = 1/2$, the RH pointwise estimate gives $G(X) = O(X^{1/2}\log X) = O_b(X^b)$. Hence both assertions follow whenever $\Theta \le b$.

Conversely, suppose $\Theta > b$ and choose $0 < \varepsilon < \Theta - b$. The negative excursions in (104.2) violate the first assertion, since $X^{\Theta-\varepsilon}/X^b \to \infty$. Its positive excursions violate the second assertion in exactly the same manner. □

In particular, no unjustified conversion of a signed estimate into an absolute sixth moment is needed: the implication passes through the independently established oscillation theorem. The estimate $G(X) \ge -CX^{5/6}$ itself is still unproved in this investigation.

104.3 What the available fixed-strip inputs actually supply

The two external inputs should not be conflated. Liu's [13, Theorem 27.1] supplies the stated boundary

$$B_{\mathrm{L}} = 0.874957019420098946\ldots.$$

Xu–Sang [63, Theorem 1.1] state the smaller boundary

$$B_{\mathrm{XS}} = 8748867/10000000 = 0.8748867,$$

explicitly assuming the analytic inputs of the OpenAI quasi-RH preprint. Their Section 9 says those input proofs were not independently verified there. These are separately admitted preprint inputs, not proofs rebuilt in this note.

Upon admitting either bound $\Theta \le B$, the sixth-energy exponent in (104.3) is at most $6B - 5$. Numerically,

$$6B_{\mathrm{L}} - 5 = 0.249742116520593676\ldots, \qquad 6B_{\mathrm{XS}} - 5 = 0.2493202.$$

Both remain positive. They therefore establish neither (104.6) nor the coefficient gap. The fact that they already imply a finite eighth moment does not interpolate down to a finite sixth moment on an infinite-measure space.

There are also precise limitations on a parameter-only attack. Liu proves optimality at B_{L} inside his explicitly specified comparison system, including successive admissible iterations [13, Theorem 34.4 and Corollary 34.5]. Xu–Sang change an analytic numerator estimate, so their improvement does not contradict that restricted optimality statement.

Even the latter argument cannot simply be evaluated at $5/6$. Its stated admissibility includes $B(\ell) > 87/100$, where

$$B(\ell) = 11/12 - \ell/4.$$

Furthermore, the particular small-class estimate in their equations (28)–(29) requires

$$C_0 = -1/4 + b/6 + 5\ell/4 < 0.$$

Keeping just this weaker necessary condition already gives

$$\ell < 1/5 - 2b/15, \qquad B(\ell) > 13/15 + b/30 > 13/15.$$

The desired 5/6 would require $\ell = 1/3$ in this balanced normalization; then $C_0 = 1/6 + b/6 > 0$. Thus preserving that small-class treatment cannot reach the target. This is a limitation of the specified estimates, not an impossibility theorem for a new prime argument or a different probe.

Status of the targeted assertion. The exact energy-growth exponent and the one-sided equivalences are proved above. No estimate reducing the exponent to zero, no proof of the one-sided $X^{5/6}$ bound, and no eventual uniform positive margin in $4 - |c_{3,N}|$ has been obtained here.

105 Interpolation pinpoints the missing power saving

Let $I_q(X) = \int_X^{2X} |E(x)|^q \, dx$. Suppose an eighth-moment tail estimate has the form

$$I_8(X) \le C \frac{X^{-\delta}}{(\log X)^8}.$$

Hölder's inequality gives the exact implication

$$I_6(X) \le X^{1/4} I_8(X)^{3/4} \le C^{3/4} \frac{X^{(1-3\delta)/4}}{(\log X)^6}.$$

Therefore $\delta \ge 1/3$ suffices for the target. At the endpoint, dyadic summation is bounded by a constant times $\sum_k k^{-6}$. The retained pointwise strip bound supplies only

$$\delta = 7 - 8b.$$

The Liu and Xu–Sang inputs give respectively $0.000343844639208432\ldots$ and 0.0009064. Neither reaches 1/3. In Theorem 103.1, $\alpha = 1/7$ even supplies the stronger eighth-energy exponent $\delta = 1/7$ with the same logarithmic factor, while the sixth energy still diverges. Thus the existence of some positive power saving is not sufficient.

106 Finite checks on the actual prime coefficients

The following computations use the original phases and weights, 75 decimal digits, and the prime-index cutoff N. The sieve includes p_{N+1}, so the final weight is exact. Normal equations are solved at degrees three and four. Independently, multiplication by the circle coordinate is compressed in the orthonormal basis, and its defect and trace are checked. These are numerical algebra checks, not interval certificates or proofs of limiting behavior.

N	p_N	$\lvert c_{3,N}\rvert$	$H_{3,N}$	$H_{4,N}$	Bound
100	541	1.83566520	18.65576619	4.91220962	3.85830773
1000	7919	1.74325964	26.07364725	8.79299688	3.81410261
10000	104729	1.71895081	29.36053158	10.74112505	3.79634445
100000	1299709	1.71010218	30.95815086	11.72273944	3.78824865

The last column is the proved finite bound $3 + \sqrt{1 - H_{4,N}/H_{3,N}}$, evaluated numerically. At the last sample the actual coefficient remains well inside the radius-four circle, but no bound for

future $H_{3,N}$ follows. The smallest displayed finite margin from the compression bound is about 0.14169; it is not asserted to persist beyond the computed indices. The independent trace and rank-one-defect residuals are below 10^{-50} in every run.

At $N = 100000$, a hypothetical certified bound $H_{4,\infty} \le B_4$ would make the coefficient-ball certificate succeed if

$$B_4 < H_{4,N} + H_{3,N}(4 - |c_{3,N}|)^2 \approx 174.05589117.$$

This is a required upper bound, not one we have established. The centered-quartic bound is already too large on this finite prefix. Its implied radius is at least 5.7193734, while the distance to the bad coefficient is about 3.2024066; its modulus certificate is at least $7.4294756 > 4$ before adding any positive tail. Proposition 102.2 proves the failure for every prefix, without relying on these decimals.

106.1 A rigorous finite certificate with one unproved global hypothesis

An independent computation uses 256-bit MPFR outward interval arithmetic, including the prime reciprocal sum, logarithms, phases, moments, and linear-system residuals. It certifies the following strict rational bounds at $N_* = 100000$:

$$|c_{3,N_*}| < 1.710103, \qquad H_{3,N_*} > 30.958150, \qquad H_{4,N_*} > 11.722738. \tag{106.1}$$

Here the finite decimals denote exact rational numbers, not rounded confidence limits. The exact sieve gives $p_{N_*} = 1299709$, $p_{N_*+1} = 1299721$, and total weight 2599425.

Theorem 106.1 (Conditional computer-assisted certificate)**.** *For the original prime phases and weights, suppose that*

$$H_{4,\infty} := \sup_M H_{4,M} \le 174. \tag{106.2}$$

Then every $M \ge 100000$ satisfies

$$|c_{3,M}| < \frac{39999}{10000} = 4 - 10^{-4}.$$

The hypothesis (106.2) is not proved in this note.

Proof. The finite-variation certificate gives, for every $M \ge N_*$,

$$|c_{3,M}| \le |c_{3,N_*}| + \sqrt{\frac{174 - H_{4,N_*}}{H_{3,N_*}}}.$$

Using (106.1), the right side is strictly less than $1.710103 + \sqrt{(174 - 11.722738)/30.958150}$. The desired inequality now follows entirely by rational arithmetic:

$$(3.9999 - 1.710103)^2\, 30.958150 - (174 - 11.722738) = \frac{831813207468067}{20000000000000000} > 0.$$

All quantities being compared before squaring are positive. □

What was verified by the interval computation. Let $C_k = \sum_{n=2}^{N_*} w_n U_n^k$ and $C_{-k} = \overline{C_k}$. At degree m, the coefficient vector solves $Gc = -b$ with $G_{jk} = C_{k-j}$ and $b_j = C_{m-j}$, $0 \le j, k < m$. The program first chooses binary rational candidate coefficients c_0 and a preconditioner R. With

$$\|v\|_{\infty,1} = \max_j (|\Re v_j| + |\Im v_j|), \qquad \|A\|_{\infty,1} \le \max_j \sum_k (|\Re A_{jk}| + |\Im A_{jk}|),$$

outward intervals certify $q = \|I - RG\|_{\infty,1} < 10^{-40}$ and

$$\|c - c_0\|_{\infty,1} \le \frac{\|R(Gc_0 + b)\|_{\infty,1}}{1 - q} < 10^{-40}.$$

This is a Neumann-series residual proof; the accuracy of the initial numerical inverse is not an assumption. Orthogonality gives $H_{m,N_*} = C_0 + \sum_{k<m} c_k C_{k-m}$, which is enclosed by the same interval calculation. The implementation uses correctly rounded MPFR operations followed by outward neighboring floating-point numbers. Sine and cosine are enclosed from a midpoint value and their global Lipschitz constant one; see the MPFR specification [64]. Appendix 124 contains the complete executable certificate.

This result verifies the finite part of a sufficient condition. It does not bound the uncomputed tail. The numerical value 174 is chosen below the finite threshold above; it is not claimed to be a necessary condition or an equivalent reformulation of the original target. This conditional implication itself needs no external fixed-strip theorem: it uses only the original finite prime data, the update identity, and the explicitly stated global energy hypothesis.

107 The unchanged target and the role of the new auxiliary problem

Let p_n be the nth prime and define

$$S(x) = \sum_{p \le x} \frac{1}{p}, \qquad U_n = \exp\bigl(2\pi i[S(p_n) - \log\log p_n]\bigr), \qquad w_n = p_{n+1} - p_{n-1} \quad (n \ge 2).$$

The classical prime Mertens theorem [65] gives

$$S(x) - \log\log x \longrightarrow B_{\mathbb{P}}, \qquad U_n \longrightarrow a = e^{2\pi i B_{\mathbb{P}}}.$$

The uncentered circle advances by $1/p$ turns at prime p; U_n removes its main drift. The weights and the phases in this note are the original ones. They are not optimized, replaced by random points, or rescaled to improve a numerical threshold.

For the finite measure $\mu_N = \sum_{n=2}^{N} w_n \delta_{U_n}$, let

$$H_{m,N} = \min_{\substack{P \text{ monic} \\ \deg P = m}} \int |P|^2 \, \mathrm{d}\mu_N, \qquad H_{m,\infty} = \sup_N H_{m,N}.$$

For a prefix with at least $m+1$ distinct points the minimizer is unique. Write its quartic form as

$$P_{4,N}(z) = z^4 + c_{3,N} z^3 + c_{2,N} z^2 + c_{1,N} z + c_{0,N}.$$

The target retained from the preceding note is exactly

$$\exists \epsilon > 0 \; \exists N_0 \; \forall N \ge N_0 : |c_{3,N}| \le 4 - \epsilon. \tag{107.1}$$

This is a criterion developed inside the prime-circle programme, not itself a historically named conjecture. The earlier relation to prime-error moments is background motivation; none of the proofs of the finite certificates below assumes the Riemann hypothesis or a new zero-free strip.

Define also

$$J_{m,N}(a) = \int |z - a|^{2m} \, \mathrm{d}\mu_N, \qquad T_m(\mu_N) = \min_{|b|=1} \int |z - b|^{2m} \, \mathrm{d}\mu_N.$$

The difference between these quantities matters. $J_{m,N}(a)$ fixes the actual limiting point, whereas T_m minimizes over every possible boundary direction. The latter is used only as an auxiliary bound; it does not change the original coefficient target.

107.1 What improves over the preceding certificate

The preceding note [66] controlled the movement of $c_{3,N}$ by a disk whose radius involved the unspent energy $H_{4,\infty} - H_{4,N}$. It certified a gap 10^{-4} under the global hypothesis $H_{4,\infty} \le 174$. The new argument controls the whole quartic near its boundary family. At the same hypothesis it certifies the larger gap $1/2000$. More substantially, the allowable hypothesis becomes $H_{4,\infty} \le 800$, with an explicit smaller gap, or $H_{4,\infty} \le 1000$, with a nonexplicit eventual cutoff.

All three bounds on $H_{4,\infty}$ are additional global hypotheses. The first 100000 primes determine finite energies and give lower bounds on $H_{4,\infty}$; they do not give an upper bound on the unseen tail. The distinction is preserved in every theorem and in the certificate output.

107.2 Provenance and reproducibility

The corpus snapshot used here is commit `c30d40119134a59eda6fa5cc6331f27c8e67d034` of the repository specified by the user. Its index and selected relevant proof sections were examined; this is not a verification of every proof in its 130 numbered TeX manuscripts. Appendix 122 contains the complete finite certificate, including its exact integer sieve, interval arithmetic, global angular cover, and rational final tests. The PDF and its single TeX source are self-contained: no external figure, bibliography, or included source file is required to compile the paper.

108 A boundary-energy problem and its complete finite solution

Fix a finite positive measure $\mu = \sum_{j=1}^{L} w_j \delta_{u_j}$ on the unit circle, and write $W = \sum_j w_j$. Define

$$T_m(\mu) = \min_{|b|=1} \int |z-b|^{2m}\, d\mu(z).$$

This is an auxiliary extremal problem, not a replacement for the original prime assertion. Its purpose is to separate all boundary polynomials $(z-b)^m$ from a given energy sublevel set. The minimum exists by continuity and compactness.

Lemma 108.1 (Stability near the boundary coefficient)**.** *Let $m \ge 2$ and let $P(z) = \prod_{j=1}^{m}(z - r_j)$, where $|r_j| \le 1$. Put*

$$\epsilon = m - \left| \sum_{j=1}^{m} r_j \right|.$$

If $0 \le \epsilon \le 1$, set $b = (\sum_j r_j)/|\sum_j r_j|$. Then

$$\sup_{|z|=1} |P(z) - (z-b)^m| \le C_m \epsilon,$$

where

$$C_m = 2^{m-1} + \frac{3}{2}\, 2^{m-2} + \sum_{k=3}^{m} 2^{m-k} \binom{m}{k} \left(\frac{2}{m}\right)^{k/2}.$$

In particular one may use $C_4 = 18$ for quartics.

Proof. The sum defining b is nonzero because its modulus is at least $m - 1 \ge 1$. Put $d_j = r_j - b$, and let e_k be the elementary symmetric functions of the d_j. The exact first-order cancellation is

$$e_1 = \sum_j d_j = -\epsilon b.$$

Also

$$\sum_j |d_j|^2 = \sum_j |r_j|^2 + m - 2\Re\left(\overline{b}\sum_j r_j\right) \le 2\epsilon.$$

Thus

$$|e_2| = \frac{1}{2}\left|e_1^2 - \sum_j d_j^2\right| \le \epsilon + \frac{\epsilon^2}{2} \le \frac{3}{2}\epsilon.$$

Maclaurin's inequality applied to $|d_j|$, followed by Cauchy–Schwarz, gives for $k \ge 3$

$$|e_k| \le \binom{m}{k}\left(\frac{\sum_j |d_j|}{m}\right)^k \le \binom{m}{k}\left(\frac{2\epsilon}{m}\right)^{k/2}.$$

Expand $P(z) = \sum_{k=0}^m (-1)^k e_k (z-b)^{m-k}$ and use $|z-b| \le 2$ and $\epsilon^{k/2} \le \epsilon$ for $k \ge 2$. This proves the stated bound. At $m = 4$, the constant is $57/4 + 2\sqrt{2} < 18$. □

Theorem 108.2 (Finite boundary-energy certificate)**.** *Suppose P is monic of degree m, all its zeros lie in the closed unit disk, and $\|P\|_\mu^2 \le B < T_m(\mu)$. If c_{m-1} is its next-to-leading coefficient, then*

$$m - |c_{m-1}| \ge \min\left\{1, \frac{\sqrt{T_m(\mu)} - \sqrt{B}}{C_m\sqrt{W}}\right\}.$$

For quartics, C_4 may be replaced by 18*.*

Proof. If $\epsilon = m - |c_{m-1}| > 1$, the conclusion is immediate. Otherwise use Lemma 108.1 and the triangle inequality:

$$\sqrt{T_m(\mu)} \le \|(z-b)^m\|_\mu \le \|P\|_\mu + C_m\epsilon\sqrt{W} \le \sqrt{B} + C_m\epsilon\sqrt{W}.$$

□

Apply the theorem to an original finite prime prefix μ_N and $P = P_{4,M}$ for any $M \ge N$. These minimizing polynomials have all zeros strictly inside the unit disk, and $\|P_{4,M}\|_{\mu_N}^2 \le H_{4,M}$. For completeness, a zero r of modulus greater than one can be replaced by $1/\overline{r}$; on the unit circle this strictly reduces the norm by the factor $|r|^{-2}$. If $|r| = 1$ and $P = (z-r)Q$, orthogonality with Q would make r the weighted average of the support points under weights $w_j|Q(u_j)|^2$. A prefix with at least $m+1$ distinct points gives at least two positive such weights, so this average has modulus less than one, a contradiction.

Consequently any independently proved bound $H_{4,\infty} \le B < T_4(\mu_N)$ yields a uniform coefficient gap for every $M \ge N$. This argument uses no asymptotic zero-free strip and no numerical value of the prime Mertens constant.

108.1 Certified boundary minimum for the original primes

Write $C_k = \sum_{n=2}^N w_n U_n^k$ and $W = C_0$. For $b = e^{i\theta}$, the boundary energy is exactly

$$f(\theta) = 70W + 2\Re\left(-56C_1e^{-i\theta} + 28C_2e^{-2i\theta} - 8C_3e^{-3i\theta} + C_4e^{-4i\theta}\right).$$

At $N = 100000$, outward-rounded 256-bit MPFR arithmetic certifies

$$W = 2599425, \qquad T_4(\mu_N) > \frac{1601}{2} = 800.5. \tag{108.1}$$

The same computation verifies five distinct original phase points.

The global minimum certificate uses an exact dyadic cover of $[0,7] \supset [0,2\pi]$ by 1792 intervals of width $1/256$. At the midpoint of each interval it encloses $f, f', \dots, f^{(4)}$. Since $|C_k| \le W$,

$$\sup_\theta |f^{(5)}(\theta)| \le 2W \sum_{k=1}^{4} |a_k| k^5 = 7840W, \qquad (a_1,a_2,a_3,a_4) = (-56, 28, -8, 1).$$

Writing $h = 1/512$, the interval lower bound is

$$f(\theta_0) - \sum_{j=1}^{4} \frac{|f^{(j)}(\theta_0)| h^j}{j!} - \frac{7840 W h^5}{120},$$

with every arithmetic bound taken outward. Every such bound is strictly above 800.5. The smallest computed lower endpoint is $800.514800821199094\ldots$; this decimal is descriptive, while (108.1) is the exact rational claim checked by the program. No samples between grid points are assumed safe: the fifth-derivative remainder covers their whole intervals.

Corollary 108.3 (A stronger conditional prime certificate)**.** *If the original prime sequence satisfies the additional, presently unproved hypothesis $H_{4,\infty} \le 800$, then*

$$|c_{3,M}| < 4 - 10^{-7} \qquad (M \ge 100000).$$

In fact $3 \cdot 10^{-7}$ can replace 10^{-7}.

Proof. For a proposed gap $\delta \le 1$, Theorem 108.2 and (108.1) apply whenever

$$\sqrt{B} + 18\delta\sqrt{W} < \sqrt{1601/2}.$$

Put $A = 1601/2 - B - 324\delta^2 W$. The last assertion follows by exact rational arithmetic from $A > 0$ and $A^2 - 1296\delta^2 BW > 0$. At $B = 800$, $\delta = 10^{-7}$, this last quantity is

$$\frac{22304073953393239164 6769}{10^{24}} > 0.$$

At $\delta = 3 \cdot 10^{-7}$ it is $736666091252372338 8289/10^{24} > 0$. □

Other certified choices from the same finite prefix are:

Assumed global bound B	Certified gap δ	Valid indices
174	$1/2000$	every $M \ge 100000$
300	$3/10000$	every $M \ge 100000$
500	$1/5000$	every $M \ge 100000$
800	$3/10^7$	every $M \ge 100000$

Each row assumes its displayed global bound; none proves that bound. Compared with the earlier coefficient-disk certificate, the same bound 174 now gives a five-times larger explicit gap. The allowed energy hypothesis can also be weakened to 800, at the cost of a smaller explicit gap.

109 Sharpness of the finite boundary-energy problem

Theorem 109.1 (Arbitrarily inexpensive collapsing completions)**.** *Fix a finite positive circle measure μ_0 with at least $m+1$ distinct support points, a point b on the unit circle, and $\eta_0 > 0$. There is an infinite positive atomic continuation whose atoms converge to b, whose total mass is infinite, and for which the degree-m monic minimizers satisfy*

$$P_{m,N}(z) \longrightarrow (z-b)^m,$$

while

$$H_{m,\infty} < \int |z-b|^{2m}\, d\mu_0(z) + \eta_0.$$

Thus, for every $B > T_m(\mu_0)$, some such continuation has $H_{m,\infty} < B$ but no eventual positive gap in $m - |c_{m-1,N}|$.

Proof. Set $\alpha = 1/(2m-1)$ and append unit-mass atoms at

$$u_j = b\exp(i\eta j^{-\alpha}), \qquad j = 1, 2, \ldots,$$

where $0 < \eta < 1$ is small enough. Since $|e^{it} - 1| \le |t|$,

$$\sum_{j\ge 1} |u_j - b|^{2m} \le \eta^{2m} \sum_{j\ge 1} j^{-2m/(2m-1)} \le 2m\eta^{2m} < \eta_0.$$

Here the final series bound follows by the integral test. For every $r < m$, the sum of $|u_j - b|^{2r}$ diverges, because $|e^{it} - 1| \sim |t|$ and $2r/(2m-1) < 1$. Consequently the only monic polynomial of degree m having finite full energy is $(z-b)^m$: every lower vanishing order at b would give infinite energy.

Finite minimizing polynomials have all their zeros in the unit disk, so their coefficients lie in a fixed compact set. Any subsequential limit has finite full energy, by first passing to each finite prefix and then increasing the prefix. It must therefore be $(z-b)^m$. This proves coefficient convergence. The same finite-prefix argument and the trial polynomial $(z-b)^m$ show

$$H_{m,\infty} = \int |z-b|^{2m}\, d\mu_0(z) + \sum_j |u_j - b|^{2m},$$

which proves the claimed estimate. Finally choose b attaining $T_m(\mu_0)$ and take $\eta_0 < B - T_m(\mu_0)$. □

This sharpness statement concerns the class of positive circle measures. The appended atoms are not claimed to arise from the actual primes or from their reciprocal recurrence. It proves that a universal certificate using only the finite prefix, positive circle atoms, and a global energy budget cannot cross the threshold $T_m(\mu_0)$. It is not a counterexample to the prime target and is not a general impossibility theorem for methods using additional arithmetic information.

Proposition 109.2 (Exhaustion of the boundary energy)**.** *Suppose $u_n \to a$, $|a| = 1$, $w_n > 0$, and $\sum_n w_n = \infty$. Then*

$$T_m(\mu_N) \uparrow \sum_{n\ge 1} w_n |u_n - a|^{2m},$$

where both sides may be infinite.

Proof. The minima are nondecreasing, and choosing $b = a$ gives the upper bound. If their limit L is finite, choose minimizers b_N. Any cluster point b has finite full centered energy at most L, by passage to each finite prefix. Infinite total mass and $u_n \to a$ force $b = a$. Thus $b_N \to a$. For every fixed K, continuity gives

$$\sum_{n\le K} w_n |u_n - a|^{2m} = \lim_{N\to\infty} \sum_{n\le K} w_n |u_n - b_N|^{2m} \le L.$$

Increasing K proves equality and also excludes a finite L when the full centered energy is infinite. □

110 Using the known limiting point without an explicit cutoff

The boundary family can be reduced to $(z-a)^4$ when passing to subsequential limits. This permits a still weaker global energy hypothesis, but the argument below supplies only an eventual index, not the explicit cutoff 100000.

Lemma 110.1 (Anchored quartic estimate)**.** *Let $P(z) = (z-a)\prod_{j=1}^{3}(z-r_j)$ with $|a| = 1$ and $|r_j| \le 1$. Put $r = |a + \sum_j r_j|$, and $D^2 = \sum_j |r_j - a|^2$. Then $D^2 \le 16 - r^2$, and, with $J_k = \int |z-a|^{2k}\, d\mu$,*

$$\|P - (z-a)^4\|_\mu \le \sqrt{3}D\sqrt{J_3} + D^2\sqrt{J_2} + \frac{D^3}{3\sqrt{3}}\sqrt{J_1}.$$

Proof. Write $S = a + \sum_j r_j$. Since $|S - a| \le 3$, $2\Re(\overline{a}S) \ge |S|^2 - 8$. Hence

$$D^2 \le 8 - 2\Re(\overline{a}S) \le 16 - |S|^2.$$

For the three shifts $r_j - a$, Cauchy–Schwarz and Maclaurin give bounds $\sqrt{3}D, D^2, D^3/(3\sqrt{3})$ for their first three elementary symmetric functions. Expansion around a and the triangle inequality in $L^2(\mu)$ finish the proof. □

Corollary 110.2 (Conditional bound 1000 with an eventual cutoff)**.** *For the original prime circle, if $H_{4,\infty} \le 1000$, then*

$$|c_{3,M}| < 4 - 10^{-5} \qquad \textit{for all sufficiently large } M.$$

The global bound 1000 *is not proved here, and the cutoff is not asserted to equal* 100000*.*

Proof. At the fixed prefix $N = 100000$, outward interval arithmetic certifies

$$J_{1,N} < 75, \quad J_{2,N} < 108, \quad J_{3,N} < 310, \quad J_{4,N} > 1024.$$

These are centered at the actual point $a = e^{2\pi i B_{\mathbb{P}}}$. The computation uses the identity

$$B_{\mathbb{P}} = \gamma + \sum_{k\ge 2} \frac{\mu(k)}{k} \log\zeta(k).$$

The identity follows by absolutely convergent rearrangement of the Euler product, using $\sum_{d|n,\ d\ge 2} \mu(d) = -1$ for $n \ge 2$. Its tail after $k = 360$ has absolute value less than 2^{-360}; the program uses the looser outward bound 2^{-358}. Indeed $\log\zeta(k) \le 3\cdot 2^{-k}$ for $k \ge 2$, so the absolute tail after $K \ge 3$ is at most $3\cdot 2^{-K}/(K+1) < 2^{-K}$.

Suppose infinitely many M violated the conclusion. Compactness of the roots supplies a subsequential monic quartic limit P with $|c_3(P)| \ge 4 - 10^{-5}$ and full energy at most 1000. Since the original phases tend to a and the total weight is infinite, $P(a) = 0$. All its roots remain in the closed unit disk. Let $\epsilon = 4 - |c_3(P)| \le 10^{-5}$. In Lemma 110.1, $D^2 \le 8\epsilon - \epsilon^2 < (9/1000)^2$. Therefore

$$\|P - (z-a)^4\|_{\mu_N} < 31\frac{9}{1000} + 11\left(\frac{9}{1000}\right)^2 + 2\left(\frac{9}{1000}\right)^3 < \frac{7}{25}.$$

Here $\sqrt{3\cdot 310} < 31$, $\sqrt{108} < 11$, and $\sqrt{75}/(3\sqrt{3}) < 2$. But

$$32 < \sqrt{J_{4,N}} \le \|P\|_{\mu_N} + \|P - (z-a)^4\|_{\mu_N} < \sqrt{1000} + \frac{7}{25} < 32,$$

where the final inequality follows from $(32 - 7/25)^2 > 1000$. This contradiction proves the eventual assertion. □

The conditional conclusions in this section and Corollary 108.3 do not establish a new zero-free half-plane. They improve the finite-to-global transfer once an independent prime-specific energy bound is supplied. Establishing such a bound remains the unresolved arithmetic part of the original problem.

111 Completeness of boundary-energy certification

Theorem 111.1 (Exact abstract completeness)**.** *Suppose $u_n \to a$ on the unit circle, $w_n > 0$, $\sum_n w_n = \infty$, the support contains at least five distinct points, and*

$$J_{4,\infty} = \sum_n w_n |u_n - a|^8 < \infty.$$

Let $P_{4,N}$ be the monic degree-four minimizers, with cubic coefficient $c_{3,N}$, and put $T_N = \min_{|b|=1} \sum_{n \le N} w_n |u_n - b|^8$. The following statements are equivalent:

1. *There exist $\epsilon > 0$ and N_0 such that $|c_{3,N}| \le 4 - \epsilon$ for every $N \ge N_0$.*
2. *$J_{3,\infty} = \sum_n w_n |u_n - a|^6 < \infty$.*
3. *$H_{4,\infty} < J_{4,\infty}$.*
4. *Some finite N satisfies $T_N > H_{4,\infty}$.*
5. *There exist a finite N and rational numbers B, τ with $H_{4,\infty} \le B < \tau < T_N$.*

Whenever the last condition is proved, Theorem 108.2 gives an explicit positive coefficient gap for every $M \ge N$.

Proof. The trial polynomial $(z-a)^4$ bounds $H_{4,N}$ uniformly. Finite minimizing polynomials have roots in the unit disk, so their coefficients lie in a fixed compact set. Let

$$r = \min\{j \in \{1,2,3,4\} : J_{j,\infty} < \infty\}.$$

A polynomial has finite full energy exactly when its vanishing order at a is at least r. Indeed, infinite total mass first forces a zero at a; after factoring its exact vanishing order, the remaining factor is bounded above and away from zero near a. Finitely many earlier terms do not affect convergence.

Thus the admissible monic quartics are exactly $(z-a)^r Q(z)$ with Q monic of degree $4-r$. Their minimization is ordinary orthogonal projection for the finite positive measure

$$\nu_r = \sum_n w_n |u_n - a|^{2r} \delta_{u_n}.$$

This measure has at least four distinct support points: only a can be removed from the original support. Therefore its Gram matrix through degree $4 - r \le 3$ is positive definite and its monic minimizer Q_* is unique. Every zero of Q_* lies strictly inside the unit disk, by the reflection and weighted average arguments already given above.

Any subsequential limit of $P_{4,N}$ has full energy at most $H_{4,\infty}$, by passing first to each finite prefix. Conversely, for any admissible full-energy quartic P, one has $H_{4,N} \le \|P\|^2_{\mu_N}$, so $H_{4,\infty} \le \|P\|^2_\mu$. Consequently every subsequential limit is the unique full-energy minimizer

$$P_* = (z-a)^r Q_*,$$

and $P_{4,N} \to P_*$ coefficientwise. Viète gives $|c_3(P_*)| < 4$ when $r \le 3$, and $c_3(P_*) = -4a$ when $r = 4$. Coefficient convergence proves the equivalence of (1) and (2).

If $J_{3,\infty} = \infty$, the sole admissible monic quartic is $(z-a)^4$, so $H_{4,\infty} = J_{4,\infty}$. If instead $J_{3,\infty} < \infty$, the full-energy trial

$$R_t(z) = (z-a)^3 (z - (1-t)a)$$

has norm squared $(1-t)J_{4,\infty}+t^2J_{3,\infty}$. Taking $t=J_{4,\infty}/(2J_{3,\infty})$ makes this

$$J_{4,\infty}-\frac{J_{4,\infty}^2}{4J_{3,\infty}}<J_{4,\infty}.$$

Both moments here are positive because the support is nondegenerate. This proves (2)⇔(3). Proposition 109.2 gives $T_N\uparrow J_{4,\infty}$, proving (3)⇔(4). Density of the rationals proves (4)⇔(5), and Theorem 108.2 supplies the claimed explicit margin from a verified instance of (5). □

The completeness statement does not provide a method for proving $H_{4,\infty}\le B$. It says that this family of boundary-energy certificates can express a successful proof whenever the original coefficient gap is true. Neither the equivalence nor the finite computations establish the remaining global prime estimate.

112 A theta-cusp obstruction with infinitely many strict failures

This section concerns the spatial theta density

$$\Phi(t)=\sum_{n\ge1}\left(2\pi^2n^4e^{9t}-3\pi n^2e^{5t}\right)e^{-\pi n^2e^{4t}}. \tag{112.1}$$

It is not the Fourier transform of this density and must not be identified with the Riemann Ξ function. The all-derivative inequality in Csordas's Open Problem 4.13, restating Coffey–Csordas Conjecture 2.5, asks whether

$$J_n(t):=\left(\Phi^{(n)}(t)\right)^2-\Phi^{(n-1)}(t)\Phi^{(n+1)}(t)>0\qquad(n\ge1,\ t\in\mathbb{R}). \tag{112.2}$$

The normalization in (112.1) is exactly the one in [67, Theorem 4.1 and Open Problem 4.13].

Theorem 112.1 (Cusp flatness, nonvanishing, and strict sign failures)**.** *Put $R=\pi/8$ and $a_k=(-1)^k\Phi^{(2k)}(0)$. Then:*

1. *For every fixed integer $j\ge0$,*
$$\lim_{y\uparrow R}\frac{d^j}{dy^j}\Phi(iy)=0.$$
More precisely, as $\epsilon\downarrow0$,
$$\Phi\big(i(R-\epsilon)\big)=-\frac{\pi^2}{4096}\epsilon^{-9/2}\exp\left(-\frac{\pi}{16\epsilon}\right)(1+O(\epsilon)). \tag{112.3}$$
2. *Every even derivative $\Phi^{(2k)}(0)$ is nonzero.*
3. *The sequence (a_k) has infinitely many positive terms and infinitely many negative terms. In particular,*
$$J_{2k+1}(0)<0\quad\text{for infinitely many integers } k\ge0. \tag{112.4}$$

The proof of the second assertion uses the established algebraic independence of π and $\Gamma(1/4)$. The first assertion and the existence of infinitely many nonpositive $J_{2k+1}(0)$ do not require that transcendence input. We give the full argument to distinguish these two levels of conclusion.

112.1 The exact cusp and all-order flatness

Write

$$\vartheta(x) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 x}, \qquad \Re x > 0, \qquad F(t) = e^t \vartheta(e^{4t}).$$

Termwise differentiation, justified by locally uniform convergence, gives

$$\Phi(t) = e^{9t}\vartheta''(e^{4t}) + \frac{3}{2}e^{5t}\vartheta'(e^{4t}), \tag{112.5}$$

$$16\Phi(t) = F''(t) - F(t). \tag{112.6}$$

Poisson summation gives $\vartheta(x) = x^{-1/2}\vartheta(1/x)$ in $\Re x > 0$, with the principal square root. Since $|\Im t| < R$ implies $\Re(e^{4t}) > 0$, this proves that F and Φ are analytic and even in that strip. They are real on the real axis. Moreover, $\Phi(0) > 0$, since every summand in (112.1) at zero is positive.

Lemma 112.2 (Flatness at the first imaginary cusp)**.** *For each fixed $j \geq 0$ there are $c > 0$, $C_j > 0$ and $\epsilon_j > 0$ such that*

$$\left|\Phi^{(j)}(i(R-\epsilon))\right| \leq C_j \epsilon^{-2j-9/2} e^{-c/\epsilon} \qquad (0 < \epsilon < \epsilon_j).$$

The leading asymptotic is (112.3)*.*

Proof. For $t = i(R-\epsilon)$ put $x = e^{4t} = i + \delta$. The exact identities

$$\delta = 2\sin(2\epsilon)e^{-2i\epsilon}, \qquad \delta^{-1} = \tfrac{1}{2}\cot(2\epsilon) + \tfrac{i}{2} \tag{112.7}$$

hold. In particular, $\Re\delta > 0$ for sufficiently small positive ϵ, $|\delta| \asymp \epsilon$, and $\Re(\delta^{-1}) \asymp \epsilon^{-1}$.

The alternating Gaussian Poisson identity is

$$\vartheta(i+\delta) = \sum_{n\in\mathbb{Z}} (-1)^n e^{-\pi n^2 \delta} = \delta^{-1/2} \sum_{m\in\mathbb{Z}} e^{-\pi(m+1/2)^2/\delta}. \tag{112.8}$$

For positive real δ this is the Fourier transform of a Gaussian evaluated on the shifted dual lattice. Analytic continuation proves it throughout $\Re\delta > 0$; both series converge locally uniformly there. It is also the classical theta transformation in [68, §20.7(viii)].

Let $A = \pi/4$ and $A_m = \pi(m+1/2)^2$. For every fixed $r \geq 0$, direct differentiation gives

$$\frac{d^r}{d\delta^r}\left(\delta^{-1/2}e^{-A_m/\delta}\right) = \delta^{-r-1/2}e^{-A_m/\delta}P_r(A_m/\delta),$$

where P_r is a polynomial of degree r. Consequently, on the curve (112.7), the absolute value of this expression is at most

$$C_r|\delta|^{-2r-1/2}(1+A_m)^r e^{-A_m\Re(1/\delta)}.$$

After factoring out $e^{-A\Re(1/\delta)}$, the sum of the remaining bounds over $m \in \mathbb{Z}$ is uniformly bounded once $\Re(1/\delta) \geq 1$. Differentiating the locally uniformly convergent series in (112.8) therefore proves

$$|\vartheta^{(r)}(i+\delta)| \leq C_r'|\delta|^{-2r-1/2}e^{-A\Re(1/\delta)}. \tag{112.9}$$

Every fixed derivative of (112.5) is a finite sum of bounded exponential factors times $\vartheta^{(r)}(e^{4t})$, with $r \leq j+2$. This proves the claimed flatness bound. Multiplication by i^j gives the same conclusion for the real derivatives of $y \mapsto \Phi(iy)$.

For the leading term, the terms $m = 0, -1$ in (112.8) both have $A_m = A$. All other terms have $A_m \geq 9A$, and the same differentiated estimates show that their relative contribution is exponentially small. In particular,

$$\vartheta''(i+\delta) = 2e^{-A/\delta}\left(A^2\delta^{-9/2} - 3A\delta^{-7/2} + \tfrac{3}{4}\delta^{-5/2}\right) + O\left(|\delta|^{-9/2}e^{-9A\Re(1/\delta)}\right).$$

Similarly, the leading order of $\vartheta'(i+\delta)$ is $2Ae^{-A/\delta}\delta^{-5/2}$. It follows that

$$\Phi(t) = 2A^2 e^{9t}\delta^{-9/2}e^{-A/\delta}\bigl(1+O(\epsilon)\bigr).$$

The phase of the leading expression is exactly

$$9(R-\epsilon)+9\epsilon - A/2 = \pi.$$

Its magnitude is

$$2A^2\bigl(2\sin(2\epsilon)\bigr)^{-9/2}\exp\left(-\tfrac{\pi}{8}\cot(2\epsilon)\right).$$

Using $2\sin(2\epsilon) = 4\epsilon(1+O(\epsilon^2))$ and $\cot(2\epsilon) = (2\epsilon)^{-1}+O(\epsilon)$ gives (112.3). □

Since Φ is even and real on the real axis, $\Phi(iy)$ is real for $|y| < R$. Its positive value at zero and its negative values near R also prove that the spatial density has a nonreal zero inside its analytic strip. This is a statement about Φ, not a statement about the zeros of its Fourier transform.

112.2 Flatness forces infinitely many Taylor sign changes

Lemma 112.3 (Flat boundary and Taylor coefficients)**.** *Let g be a nonzero even function analytic on $|z| < R$, real on $(-R,R)$, and suppose*

$$g^{(j)}(y) \longrightarrow 0 \quad (y \uparrow R) \qquad \text{for every fixed } j \geq 0.$$

If $g(z) = \sum_{k\geq 0} a_k z^{2k}/(2k)!$, then (a_k) contains infinitely many positive terms and infinitely many negative terms.

Proof. Suppose first that $a_k \geq 0$ for every $k \geq K$. Termwise differentiation in $0 < y < R$ gives

$$g^{(2K)}(y) = \sum_{k\geq K} \frac{a_k y^{2k-2K}}{(2k-2K)!}.$$

If at least one coefficient in this tail is positive, this sum is bounded below by a positive constant for all y sufficiently close to R, contradicting flatness. If the tail vanishes identically, g is a polynomial; flatness of all its derivatives at R then forces that polynomial to be zero, again a contradiction. Thus the coefficients are not eventually nonnegative. Applying the same argument to $-g$ shows that they are not eventually nonpositive. □

Apply this lemma to $g(y) = \Phi(iy)$, using Lemma 112.2. Its Taylor coefficients are precisely $a_k = (-1)^k\Phi^{(2k)}(0)$. Since all odd derivatives of Φ vanish at zero,

$$J_{2k+1}(0) = -\Phi^{(2k)}(0)\Phi^{(2k+2)}(0) = a_k a_{k+1}. \tag{112.10}$$

If these products were eventually positive, the coefficients would be nonzero and have constant sign from some point onward. This contradicts Lemma 112.3. We have therefore already proved, without transcendence theory,

$$J_{2k+1}(0) \leq 0 \quad \text{for infinitely many } k.$$

To make the inequalities strictly negative, it remains to exclude zero even derivatives. The next argument closes exactly that gap.

112.3 An algebraic-independence argument excludes zero derivatives

Let $\theta_j(\tau)$ denote the classical theta constants; thus $\theta_3(ix) = \vartheta(x)$. Put

$$q_* = \pi\theta_3(i)^4 = \frac{\Gamma(1/4)^4}{4\pi^2}.$$

The displayed evaluation follows from $K(1/\sqrt{2}) = \frac{\pi}{2}\theta_3(i)^2 = \Gamma(1/4)^2/(4\sqrt{\pi})$, the classical lemniscatic period formula; see [68, §20.9(i)] and [69, p. 1321]. The algebraic independence of π and $\Gamma(1/4)$ implies that q_* is transcendental. This independence was proved by Chudnovsky; it is also contained in Nesterenko's stronger Corollary 5. Nesterenko explicitly records the earlier attribution on p. 1322 of [69].

Proposition 112.4 (Nonzero polynomials for the even derivatives)**.** *For every $k \geq 0$ there is a nonzero polynomial $R_k \in \mathbb{Q}[X]$ such that*

$$\Phi^{(2k)}(0) = \theta_3(i)R_k(q_*).$$

Consequently, $\Phi^{(2k)}(0) \neq 0$ for every $k \geq 0$.

Proof. Let $x = e^{4t}$ and introduce the scaled functions

$$E = \pi x E_2(ix), \qquad A = \pi x\theta_2(ix)^4, \qquad B = \pi x\theta_3(ix)^4, \qquad C = \pi x\theta_4(ix)^4.$$

Here E_2 is the normalized weight-two Eisenstein series. The classical theta differential identities, with the chain rule, give the polynomial differential system

$$\begin{aligned} E' &= 4E - \tfrac{2}{3}E^2 + \tfrac{1}{3}(A^2 + B^2 + C^2), \\ A' &= 4A - \tfrac{4}{3}A(E + B + C), \\ B' &= 4B - \tfrac{4}{3}B(E + A - C), \\ C' &= 4C - \tfrac{4}{3}C(E - A - B), \\ F' &= \Big(1 - \tfrac{1}{3}(E + A - C)\Big)\,F. \end{aligned} \tag{112.11}$$

For a primary statement of the differential identities, see [70, Proposition 1, equation (8)]; their Weierstrass quantity is $\eta = \pi^2 E_2/12$. This substitution yields exactly (112.11).

At $t = 0$, the modular transformation at $\tau = i$ and Jacobi's quartic identity give

$$(E, A, B, C) = (3, q_*/2, q_*, q_*/2). \tag{112.12}$$

Indeed, $E_2(i) = 3/\pi$ and $\theta_2(i)^4 = \theta_4(i)^4 = \theta_3(i)^4/2$. Let $\mathcal{D}$ be the polynomial vector field consisting of the first four equations of (112.11), and put $L = 1 - (E + A - C)/3$. The recursion

$$Q_0 = 1, \qquad Q_{r+1} = \mathcal{D}Q_r + LQ_r$$

defines polynomials $Q_r \in \mathbb{Q}[E, A, B, C]$ and proves by induction that $F^{(r)} = Q_r(E, A, B, C)F$. Evaluation at (112.12), followed by (112.6), proves the claimed polynomial representation, with

$$R_k(X) = \frac{1}{16}(Q_{2k+2} - Q_{2k})(3, X/2, X, X/2).$$

It remains essential to prove that R_k is not the zero polynomial. Specialize the polynomial initial data formally to $X = 0$. The solution of (112.11) through $(E, A, B, C) = (3, 0, 0, 0)$ satisfies

$$A = B = C = 0, \qquad E(t) = \frac{6e^{4t}}{1 + e^{4t}}.$$

The last equation then becomes $F'/F = -\tanh(2t)$. With normalization $F(0) = 1$ its solution is

$$f(t) = (\cosh(2t))^{-1/2}.$$

This specialization is an auxiliary solution of the polynomial system, used to evaluate a polynomial's constant term; it is not an assertion that theta constants vanish at $\tau = i$.

Write $h(y) = f(iy) = (\sec(2y))^{1/2}$ near zero and $b_k = h^{(2k)}(0)$. Every b_k is strictly positive. One elementary proof uses $T(y) = \tan(2y)$: the differential equation $T' = 2(1 + T^2)$ and $T(0) = 0$ imply that all its odd Taylor coefficients are positive. Since $h' = Th$ and $h(0) = 1$, coefficient comparison then proves inductively that every even Taylor coefficient of h is positive as well. Therefore

$$f^{(2k)}(0) = (-1)^k b_k, \qquad R_k(0) = \frac{(-1)^{k+1}}{16}(b_{k+1} + b_k) \neq 0.$$

The polynomials are nonzero. Their evaluation at the transcendental number q_* cannot vanish, proving the proposition. □

Completion of Theorem 112.1. The flatness and asymptotic statement is Lemma 112.2. The nonvanishing is Proposition 112.4. Lemma 112.3 shows that the nonzero real sequence (a_k) takes both signs infinitely often. It must therefore change sign between adjacent terms infinitely often. Equation (112.10) then gives (112.4). □

112.4 What this closes, and its prior-art status

The theorem rules out the strict all-order positivity claim (112.2), and shows that its failure at the origin occurs at arbitrarily high derivative orders. The cusp argument is a concrete obstruction to transferring a global positivity premise from the spatial theta density to all of its derivatives.

It is not a new resolution of a previously unresolved named problem. Theorem 5.2 of the public preprint [71], dated 30 August 2026, already gives an interval-certified counterexample at $n = 9$, $t = 1/50$. The cusp mechanism also appears in the 18 October 2026 public manuscript [72, Section 3.1]. Thus the proposed first-resolution claim fails G3 and G6. The proof above is supplied as a complete mathematical argument, including the additional nonvanishing step needed for infinitely many *strict* failures. No priority claim for that strengthening is made; the literature audit does not establish one.

The statements in this section are logically independent of the conditional prime-circle boundary-energy certificates. They do not bound $H_{4,\infty}$, do not prove the cubic coefficient gap, and do not supply a zero-free half-plane for the Riemann zeta function. In particular, zeros or derivative sign failures of Φ cannot be substituted for assertions about zeros of Ξ.

113 Corpus transfers and the missing arithmetic input

We inspected selected relevant statements and proof mechanisms in the *Verifiable Mathematical Reasoning Corpus*, at commit `c30d40119134a59eda6fa5cc6331f27c8e67d034` (accessed 8 November 2026). These are research manuscripts; this section does not certify all proofs in the repository. In particular, an implication in a source is not an unconditional proof of its premise for the prime sequence used here.

Source	Transferable mechanism	Input still required here
[004]	Prime-phase hulls, local Jensen counts, and recovery from divisor and boundary data in $\Re s > 1$.	A prime-error estimate across the relevant analytic boundary. The manuscript explicitly limits its claims to the absolute-convergence half-plane.
[006]	Polynomial-coefficient recurrences impose a finite singularity budget; critical values can establish lower bounds for recurrence order.	An actual finite recurrence for the object under study. The integral Franel lattice is not inherited by an arbitrary weighted prime sequence.
[028]	Spectral surgery and a quantitative squared-distance remainder away from extremal spectra.	Uniform separation from the extremal set, backed by arithmetic information. Positive defect at each finite stage is insufficient.
[103]	A nodal-tangency criterion converts strict shifted Laguerre positivity into real-rootedness.	The strict positivity premise for the actual Riemann function; the criterion does not establish that premise.
[126]	Rationally weighted cancellation controls heat-flow collisions and prevents collision mass escaping to spatial infinity.	A global Laguerre sign for the actual Riemann heat family. The source explicitly retains an unresolved interior region.
[097]	Metric completeness for a class containing the Fisher metric of Dirichlet probability distributions.	This is not a theorem about Dirichlet series or Hardy-space completeness. No direct transfer to the present prime-circle problem has been established.

113.1 Why the spectral stability result is relevant

The finite-dimensional count model of corpus manuscript [028] writes $G = P + Q$, where

$$P \succeq 0, \qquad \operatorname{rank} P \le s, \qquad \operatorname{tr} P \le s, \qquad n_+(Q) \le b, \qquad s + 2b \le N.$$

It sets

$$L_0 = 4 \operatorname{tr} G - \operatorname{tr} G^2 - 2N, \qquad D_0 = \operatorname{tr}\Big([G(2I - G)]^2\Big)$$

and proves, for $0 < \theta < 1$,

$$s \ge \frac{2L_0}{\theta} - \frac{D_0}{\theta^2} + c_\theta \sum_j \operatorname{dist}(\lambda_j, \mathcal{S}_\theta)^2, \tag{113.1}$$

where

$$\mathcal{S}_\theta = \{0, 2, 1 - \sqrt{1-\theta}, 1 + \sqrt{1-\theta}\}, \qquad c_\theta = \min\{2/\theta, (1-\theta)/\theta^2\}.$$

The proof supplies an exact decomposition into nonnegative rank, trace, and polynomial defects. The manuscript also proves strict finite nonattainment for a sinc evaluation frame, but expressly does not infer a uniform normalized defect from it. Its zeta application retains unproved third- and fourth-moment asymptotics as well as the required positive limiting spectral-distance bound.

This is the useful conceptual transfer to the coefficient problem: identify the extremal configuration, write an explicit distance defect, and then obtain a quantitative lower bound for that defect. The new energy-barrier estimates in the present work concern a different extremal set, namely fourth powers $(z - b)^4$ with $|b| = 1$. They make the required finite-prefix separation quantitative. They still require an independent bound on the energy of the actual infinite prime sequence. Neither the old spectral certificate nor the new polynomial separation turns finite nonattainment alone into that arithmetic bound.

113.2 The positivity converses preserve their premises

For $L[g] = (g')^2 - gg''$ and $f_\mu(z) = f(z + i\mu) + f(z - i\mu)$, manuscript [103] claims

$$L[f_\mu](x) > 0 \quad (x \in \mathbb{R},\ \mu \neq 0) \quad \Longrightarrow \quad f \in \mathcal{LP}$$

in its real entire strip class. Its proof excludes horizontal nodal tangencies by means of a Jensen count and a descent argument. This does not establish the strict inequalities for a specified f. Moreover, the distinction between strict and weak inequalities is essential: for $p(z) = z(z^2 + 1)$,

$$L[p_\mu](x) = 4\big(3x^4 + (1 - 3\mu^2)^2\big) \geq 0,$$

although p has nonreal zeros.

Manuscript [126] applies a collision theorem to the detector

$$\Delta_f(t, x) = L(F_t)(x)L(F_t')(x).$$

For the Riemann family, its global sign condition is another criterion; the source expressly leaves the range $0 < t < 1/5$, $|x| < \exp(C/t) + 2$ unresolved. Thus these manuscripts do not provide automatic Li or Weil positivity, a new zeta zero-free region, or the missing prime-circle energy estimate.

113.3 A direct limit on the finite-recurrence transfer

Proposition 113.1. *The prime zeta function $P(s) = \sum_p p^{-s}$ is not D-finite over $\mathbb{C}(s)$.*

Proof. In $\Re s > 1$, Mobius inversion of the logarithmic Euler product gives

$$P(s) = \sum_{k \geq 1} \frac{\mu(k)}{k} \log \zeta(ks), \qquad P'(s) = \sum_{k \geq 1} \mu(k) \frac{\zeta'(ks)}{\zeta(ks)}.$$

The second series continues meromorphically to $\Re s > 0$: on each compact subset its tail converges normally, since its terms decrease exponentially in k. If m is squarefree, then $s = 1/m$ is a pole of the $k = m$ term with residue $-\mu(m)/m$. Every other term is analytic near that point. Here one uses that zeta has no positive real zero and has its sole positive real pole at 1.

Thus P' has infinitely many finite singularities. A solution of a polynomial-coefficient linear differential equation can have finite singularities only at the finitely many zeros of its leading coefficient, and differentiation preserves D-finiteness. This proves the assertion. □

This is a structural consequence of the classical prime-zeta expansion, not a novelty claim. It blocks a direct finite-holonomic argument for P itself. It does not rule out auxiliary recurrences and does not, without an additional argument, establish non-holonomicity of the separate prime-circle coefficient sequence.

113.4 Prior art for the theta-derivative candidate

Csordas's Open Problem 4.13 in https://arxiv.org/html/1309.0055v2, restating Coffey–Csordas Conjecture 2.5, asks whether

$$J_n(t) = (\Phi^{(n)}(t))^2 - \Phi^{(n-1)}(t)\Phi^{(n+1)}(t) > 0 \quad (n \in \mathbb{N},\ t \in \mathbb{R}),$$

for

$$\Phi(t) = \sum_{m \geq 1} \pi m^2 (2\pi m^2 e^{4t} - 3) e^{5t - \pi m^2 e^{4t}}.$$

This is not an eligible newly solved open problem in the present work. Theorem 5.2 of *GPT, the Counterexample Machine*, https://arxiv.org/html/2608.29595v1, publicly dated 30 August 2026, already gives $J_9(1/50) < 0$ by interval arithmetic with an explicit tail bound. The theorem's original hypotheses and the normalization above agree. We checked this prior statement; we do not claim to have rerun its certificate.

The modular boundary mechanism also has a public antecedent. Section 3.1 of the 18 October 2026 manuscript *A Fabry obstruction to the separated meromorphic Pick construction of the Riemann xi function*, https://rstation.ai/problems/riemann_hypothesis/papers/20/v1, uses the normalization $P(x) = 2\Phi(x/2)$ and proves all-orders flatness at the theta cusp by Poisson summation. It applies that fact to a different obstruction involving Fabry's theorem. This is a public research manuscript, not evidence here of conventional journal review.

The signed-even-derivative consequence developed in the present paper should therefore be described as an independently derived analytic consequence, with novelty unresolved. The focused search did not establish an exhaustive priority result. In particular, neither this consequence nor the preceding converse criteria establish the required unconditional arithmetic estimate for the prime circle.

Part VIII
Analogues, computation, and the complete transfer ledger

114 Addendum: an attempt to close the remaining step

This section was added after an independent check of Sections 2–123. It records what can be closed, what cannot, and why. No proof of RH is claimed.

114.1 Independent verification

Every proof in Sections 2–10 was rechecked line by line, including the constants in Lemma 6.1, the update (7.13), the generating function (9.3), the asymptotics (9.13)–(9.14), the bound (9.16) and the miscentering law (9.17). No error was found. The cited inputs were checked against the source: (2.7) is exactly [1, eq. (24)], whose proof uses Ingham's zero-density estimate, and the forward half of Theorem 4.1 is the case $\Theta = 1/2$ of [1, Lemma 9]. The embedded program reproduces Tables 11 and 12 and the seven Laguerre coefficients to every printed digit.

The caption of Table 12 uses an inequality that is not proved in the text. It holds for every N:

$$\mathcal{J}_N - 2D_N = W_N\big(|m_N - u_\infty|^2 - (1 - |m_N|)^2\big) \;\geq\; 0, \tag{114.1}$$

by (7.5), (7.7) and $|m_N - u_\infty| \geq 1 - |m_N|$.

114.2 Why a change of representation cannot close the gap

Proposition 114.1 (Representation invariance)**.** *Except for the root criterion, each criterion collected in Section 123 is equivalent to $\int_2^\infty E(x)^2\,\mathrm{d}x < \infty$ through isometries, two-sided bounded distortions, or exact identities. These are the circle and the fixed ellipse (2.6), (4.3); the prime and integer samplings (6.6) and Corollary 6.3; the deficit identities of Lemma 7.1; and the Laguerre isometry with its Cayley image (9.5). The root criterion of Theorem 9.2 and the zero geometry of Section 10 are bijective relabellings of $\mathcal{Z}$ that preserve the property $\Re\rho = 1/2$.*

The proofs are the cited equations. Consequently no statement about these maps alone can decide RH. The missing ingredient is an a priori arithmetic estimate.

The same applies to the classical criteria built on the coordinate $u = 1 - 1/s$ of Section 10. One is Li's positivity criterion [73, 74]. Another is the criterion of Balazard, Saias and Yor [75], which in this coordinate is Jensen's formula on the unit disk. Indeed, put $f(u) = (s-1)\zeta(s)/s$ with $s = 1/(1-u)$. Then $f(0) = 1$ and $|\,\mathrm{d}u| = |\,\mathrm{d}s|/|s|^2$. Moreover $|s-1| = |s|$ on the critical line, and $|u_\rho| < 1$ exactly when $\Re\rho > 1/2$. Letting the radius tend to 1 in Jensen's formula (justified in [75]) gives

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{\log|\zeta(\frac{1}{2}+\mathrm{i}t)|}{\frac{1}{4}+t^2}\,\mathrm{d}t = \sum_{\Re\rho>1/2} \log\left|\frac{\rho}{1-\rho}\right|,$$

so RH is equivalent to the vanishing of the left side. It is a further exact identity, and again it supplies no bound.

114.3 What is unconditionally true

Theorem 114.2 (Unconditional deficit bound)**.** *There is an absolute constant $c > 0$ such that, for $N \geq 2$,*

$$D_N \leq V_N \leq \mathcal{J}_N \ll p_N \exp\Big(-c\,(\log p_N)^{3/5}(\log\log p_N)^{-1/5}\Big).$$

In particular $D_N = o(W_N)$.

Proof. The Vinogradov–Korobov zero-free region gives [76, Ch. 6]

$$\pi(x) - \mathrm{li}(x) \ll x \exp\big(-c_0(\log x)^{3/5}(\log\log x)^{-1/5}\big).$$

Inserting this in (5.1) and integrating the tail gives, for some $0 < c_1 < c_0$,

$$E(x) \ll \exp\big(-c_1(\log x)^{3/5}(\log\log x)^{-1/5}\big).$$

Since $|e^{\mathrm{i}\phi} - 1| \leq |\phi|$, one has $\mathcal{J}_N \leq 4\pi^2 A_N$. Then (6.6) gives $A_N \ll 1 + I(p_{N+1})$. Lemma 7.1 gives the first two inequalities. □

Theorem 114.3 (Dictionary for zero-free half-planes)**.** *For $1/2 \leq \sigma_0 < 1$ the following are equivalent:*

(i) $\zeta(s) \neq 0$ *for* $\Re s > \sigma_0$*, that is,* $\Theta \leq \sigma_0$*;*

(ii) $D_N \ll_\varepsilon p_N^{2\sigma_0 - 1 + \varepsilon}$ *for every* $\varepsilon > 0$*;*

(iii) $\int_2^\infty x^{1-2\sigma_0} E(x)^2\,\mathrm{d}x < \infty$*;*

(iv) when $\sigma_0 > 1/2$*:* $\int_2^\infty |E(x)|^q\,\mathrm{d}x < \infty$ *with* $q = 1/(1-\sigma_0)$*.*

Proof. Use Theorem 8.5 for (ii), Theorem 5.2 for (iii), and Theorem 5.1 for (iv). □

Corollary 114.4 (The 2026 quasi-Riemann preprints)**.** *A preprint released in October 2026 and attributed to an OpenAI model claims $\zeta(s) \neq 0$ for $\Re s > 7/8$, with a Lean formalization [77]. Liu claims the slight improvement $\sigma_0 \approx 0.874957$ [13]. Neither had been refereed when this addendum was written.*

By Theorem 114.3, the first claim is equivalent *to $E \in L^8(2,\infty)$. Equivalently, it is finite energy for every curve of contact order $k = 4$ (Corollary 5.3). It implies $D_N \ll_\varepsilon p_N^{3/4+\varepsilon}$.*

Contact order k	Finite energy $\Longleftrightarrow$	Status (November 2026)
1	$\Theta = 1/2$ (RH)	open
2	$\Theta \leq 3/4$	open
3	$\Theta \leq 5/6$	open
4	$\Theta \leq 7/8$	claimed in [77], unrefereed
≥ 5	$\Theta \leq 1 - 1/(2k)$	follows from $k = 4$

Table 5: The contact-order hierarchy of Corollary 5.3 against current knowledge. Without the 2026 claim, only $\Theta \leq 1$ is known, and no level is closed.

114.4 A positivity form of the same question

Zhao proved that RH is equivalent to $\int_2^X E(x)\,\mathrm{d}x > 0$ for all $X > 2$ [1, Theorem 1]. This is a one-sided, or positivity, statement. It is of the type that closes the function-field case below, and the note's geometry displays it.

Write $A_N = M_N - W_N u_\infty = \sum_{n\le N} w_n(U_n - u_\infty)$, so that $m_N = u_\infty + A_N/W_N$. Then

$$\Im(\overline{u_\infty} A_N) = \sum_{n=2}^{N} w_n \sin(2\pi E(p_n)) = 2\pi \sum_{n=2}^{N} w_n E(p_n) + O\Big(\sum_{n\le N} w_n |E(p_n)|^3\Big),$$

and $\sum_n w_n E(p_n)$ is a two-sided Riemann sum for $2\int_2^{p_N} E$. After linearization, Zhao's criterion says that the centroid of the prime polygon never crosses the radius through its limit point.

Under RH the margin is large. The prime-square tail $\sum_{p^k>x,\,k\ge2} 1/(kp^k) \sim 1/(\sqrt{x}\log x)$ contributes $\sim 2\sqrt{X}/\log X$ to $\int_2^X E$. The zeros contribute oscillations bounded by $(\sum_\gamma(\frac14+\gamma^2)^{-1} + o(1))\sqrt{X}/\log X$, where the sum is $0.0461\ldots$ (Zhao's constant B_1 [1, (16)]).

In the computed range the criterion holds trivially. Indeed $E(p_n) > 0$ for every prime $3 \le p_n \le 10^9$, with minimum $4.35\cdot10^{-7}$ at $p_n = 905\,055\,691$, so $\Im(\overline{u_\infty}A_N) > 0$ throughout. Proving the positivity for all X is again equivalent to RH.

114.5 A better-conditioned variant: prime powers and Euler's constant

Let $q_1 = 2 < q_2 = 3 < q_3 = 4 < q_4 = 5 < q_5 = 7 < q_6 = 8 < \cdots$ be the prime powers, and put

$$S^*(x) = \sum_{p^k\le x} \frac{1}{kp^k} = \sum_{n\le x} \frac{\Lambda(n)}{n\log n}, \qquad E^*(x) = S^*(x) - \log\log x - \gamma. \tag{114.2}$$

Mertens' theorem gives $E^*(x) \to 0$. Define U_n^*, D_N^*, V_N^* and $\mathcal{J}_N^*$ as in (7.1)–(7.4). In these definitions replace p_n by q_n, w_n by $w_n^* = q_{n+1} - q_{n-1}$, and S by S^*. The construction is again free of an input constant; its limit point is $e^{2\pi\mathrm{i}\gamma}$.

Theorem 114.5 (Prime-power criterion)**.**

(a) For $\Re s > 1$, $F^*(s) = \int_2^\infty E^*(x)x^{-s}\,\mathrm{d}x = \big[\log((s-1)\zeta(s)) - J^*(s)\big]/(s-1)$, where

$$J^*(s) = (\gamma + \log\log 2)\,2^{1-s} - \gamma - \log\log 2 - \sum_{j\ge1} \frac{[-(s-1)\log 2]^j}{j\,j!}$$

is entire and $J^*(1) = 0$. The singularities of F^* in $\Re s > -2$ are exactly the nontrivial zeros of ζ. The prime-square singularity of Lemma 3.2 is absent.

(b) $\mathrm{RH} \iff E^* \in L^2(2,\infty) \iff \sup_N D_N^* < \infty$.

(c) Theorems 5.1, 5.2 and 8.5 hold verbatim for E^* and D_N^*, with p_N replaced by q_N.

Proof. (a) Since $\int_{2^-}^\infty x^{1-s}\,\mathrm{d}S^*(x) = \sum_n \Lambda(n)n^{-s}/\log n = \log\zeta(s)$, the computation leading to (3.3) and (3.6) applies with $B_{\mathbb{P}}$ replaced by γ and with $R_0 \equiv 0$. The expansion of E_1 gives the formula. At $s = 1$ both $\log((s-1)\zeta(s))$ and J^* vanish.

(b) If $E^* \in L^2$, then F^* is holomorphic in $\Re s > 1/2$, hence so is $\log((s-1)\zeta(s))$, and ζ has no zeros there. Conversely, partial summation gives the exact analogue of (4.2):

$$E^*(x) = f^*(x) - \int_x^\infty \frac{f^*(t)}{t}\Big(1 + \frac{1}{\log t}\Big)\,\mathrm{d}t, \qquad f^*(x) = \frac{\psi(x) - x}{x\log x}.$$

Then (4.1) and the Hardy bound give $E^* \in L^2$. Lemma 6.1 and (6.2) hold for consecutive prime powers: $q_{n+1} < 2q_n$ by Bertrand's postulate, and the jump at $q_{n+1} = p^k$ is $1/(kq_{n+1}) \le 1/q_{n+1}$. Then Lemma 7.1 applies.

(c) The proofs use only Lemma 3.1, a barrier at $\Re s = 1/2$, and pointwise bounds. Here the barrier is supplied by any zero on the critical line, for example $\rho = \frac{1}{2} + 14.1347\ldots\mathrm{i}$, where F^* has a logarithmic singularity by (a). The pointwise bounds follow from $\psi(x) - x = O(x^{\Theta}\log^2 x)$ and from (2.7). □

The variant matters numerically. In E, the prime-square tail $E - E^*$ is a positive bias of normalized size near 1. The oscillation coming from the zeros has normalized mean square only 0.046. The variant removes this bias, as quantified next.

114.6 The exact energy density under RH, and the constant in condition (a)

Proposition 114.6 (Energy density; sketch of proof)**.** *Assume RH, and let* $\kappa = \sum_\gamma m_\gamma^2/(\frac{1}{4} + \gamma^2)$, *summed over distinct ordinates with multiplicities* m_γ. *Then, as* $Y \to \infty$,

$$\frac{1}{\log Y}\int_2^Y (\log x)^2 E^*(x)^2\,\mathrm{d}x \to \kappa, \qquad \frac{1}{\log Y}\int_2^Y (\log x)^2 E(x)^2\,\mathrm{d}x \to 1 + \kappa. \tag{114.3}$$

Consequently $\int_X^\infty E^{*2} \sim \kappa/\log X$, $\int_X^\infty E^2 \sim (1+\kappa)/\log X$, *and*

$$D_\infty^* - D_N^* \sim \frac{4\pi^2\kappa}{\log q_N}, \qquad D_\infty - D_N \sim \frac{4\pi^2(1+\kappa)}{\log p_N}. \tag{114.4}$$

If all zeros are simple, then

$$\kappa = \sum_\rho \frac{1}{\rho(1-\rho)} = 2\lambda_1 = 2 + \gamma - \log 4\pi = 0.0461914179\ldots.$$

This is twice the first Li coefficient [73], and it is also Zhao's B_1 *[1, (16)].*

Sketch. Under RH, the explicit formula gives $x^{-1/2}(\psi(x) - x) = -\sum_\gamma m_\gamma x^{\mathrm{i}\gamma}/\rho + o(1)$ in the Besicovitch mean-square sense with respect to $\mathrm{d}x/x$. This is the mean-value principle behind Cramér's theorem (cf. [78]).

In the identity of Theorem 114.5(b), the operator $f^* \mapsto \int_x^\infty f^*(t)t^{-1}\,\mathrm{d}t$ multiplies $x^{\rho-1}/\log x$ by $1/(1-\rho)$ to leading order, since $\int_x^\infty t^{\rho-2}\,\mathrm{d}t = x^{\rho-1}/(1-\rho)$. Hence

$$\sqrt{x}\,\log x\; E^*(x) = \sum_\gamma \frac{m_\gamma x^{\mathrm{i}\gamma}}{1-\rho} + o(1)$$

in the same sense. Parseval's identity for Besicovitch almost periodic functions gives the mean square $\sum_\gamma m_\gamma^2/|1-\rho|^2 = \kappa$.

For E, add $E - E^* = \sum_{p^k > x,\, k \ge 2} 1/(kp^k) = (1 + o(1))/(\sqrt{x}\log x)$; its cross term with the oscillation has mean zero. Partial integration against $\mathrm{d}x/(x\log^2 x)$ gives the tails.

Finally, the following facts give (114.4): $2D_\infty = \mathcal{J}_\infty$ (Lemma 7.1); $|e^{\mathrm{i}\phi} - 1|^2 = \phi^2(1 + O(\phi^2))$; $\sum_{n>N} w_n E(p_n)^2 = (2 + o(1))\int_{p_N}^\infty E^2$; and $\mathcal{J}_N - V_N = W_N|m_N - u_\infty|^2 = O(1/\log^2 p_N)$. □

Computation to 10^9. The construction was streamed with a segmented sieve through $X = 10^9$: 50 847 534 primes, plus the prime powers. Every gap was integrated by Gauss–Legendre quadrature, with 32 nodes below 10^5 and 4 nodes above. The calculation used extended precision. Up to 10^6 it reproduces Table 12 to all digits. D_N was evaluated both by direct subtraction and through the stable identity $D_N = V_N/(1 + |m_N|)$, $V_N = \mathcal{J}_N - |A_N|^2/W_N$. The two values agree to better than $5 \cdot 10^{-9}$.

Decade	$[10^5, 10^6)$	$[10^6, 10^7)$	$[10^7, 10^8)$	$[10^8, 10^9)$
Log-mean of $(\sqrt{x}\log x\, E^*)^2$	0.045211	0.046274	0.046252	0.046114
Log-mean of $(\sqrt{x}\log x\, E)^2$	0.927024	0.913109	0.899210	0.913860
Log-mean of $\sqrt{x}\log x\,(E - E^*)$	0.93744	0.93119	0.93212	0.93249

Table 6: Measured energy densities. The E^* row should tend to $\kappa = 2\lambda_1 = 0.0461914$ under RH with simple zeros; on the last three decades it agrees to 0.2%. The E row equals the square of the prime-square bias plus κ, up to cross terms. The bias tends to 1 only like $1 - 2/\log x$.

Extrapolated limits. By Lemma 7.1 and Proposition 114.6,

$$2D_\infty = \mathcal{J}_N + 8\pi^2 \int_{\log X}^{\infty} \frac{\varrho(u)}{u^2}\,\mathrm{d}u\,(1 + o(1)). \tag{114.5}$$

Here $\varrho = \kappa$ for E^*, and $\varrho(u) = b(u)^2 + \kappa$ for E. The bias $b(u)$ is modelled by the logarithmic-integral density of prime powers, $b(u) = \sum_{k\ge 2}(u/k)e^{u/2}\mathrm{E}_1((k-1)u/k)$. This model agrees with the measured mean 0.93249 on $[10^8, 10^9)$ to 0.4%.

X	D_N	$\mathcal{J}_N$	limit from (114.5)	D_N^*	limit from (114.5)
10^6	36.2562939	74.8491148	40.1568	3.2073175	3.34106
10^7	37.0363059	75.5998205	40.1482	3.2277388	3.34118
10^8	37.5420619	76.1524252	40.1409	3.2422226	3.34122
10^9	37.8925538	76.5892248	40.1397	3.2532151	3.34122

Table 7: Deficits at $p_N \le X < p_{N+1}$ and the RH-conditional extrapolations. Thirteen checkpoints between 10^6 and 10^9 give extrapolated values in $[40.1397, 40.1568]$ for D and in $[3.3404, 3.3420]$ for D^*.

Remark 114.7 (Uncertified numerical extrapolations). Assume RH and simple zeros. Then the computations above give the following numerical predictions. The displayed error bars are heuristic extrapolation ranges, not rigorously certified enclosures:

$$\sup_N D_N = D_\infty = 40.14 \pm 0.02, \qquad \sup_N D_N^* = D_\infty^* = 3.3412 \pm 0.0002,$$

$$\int_2^\infty E^2 = 0.8566 \pm 0.0005, \qquad \int_2^\infty E^{*2} = 0.05387 \pm 0.00001.$$

By (9.5), the third value is also $\sum_n |a_n^{(1)}|^2$ under RH. The coefficients with $n \ge 7$ therefore carry about 0.147 of the total, beyond the partial sum 0.7092 of Section 117.

The approach to the limits is only logarithmic. At $p_N \approx 10^9$ the extrapolation places the deficit about 2.25 below the proposed limit. In that model, a correction of size 0.1 occurs near $\log p_N \approx 400$. No finite computation can replace condition (a), even with the constant known.

114.7 The function-field analogue: here the closure is a theorem

Let $C/\mathbb{F}_q$ be a smooth projective geometrically irreducible curve of genus g. Then $Z_C(T) = \prod_{i=1}^{2g}(1 - \alpha_i T)/((1-T)(1-qT))$ and $N_n = \#C(\mathbb{F}_{q^n}) = q^n + 1 - \sum_i \alpha_i^n$. Let a_d be the number of closed points of degree d, so that $N_n = \sum_{d\mid n} d a_d$. The analogue of p^{-1} is $q^{-\deg P}$, and the analogue of $\log\log x$ is the harmonic number H_n. Put

$$S_C(n) = \sum_{d\le n} a_d q^{-d}, \qquad E_C(n) = S_C(n) - H_n - c_C, \qquad c_C = \lim_{n\to\infty}(S_C(n) - H_n),$$

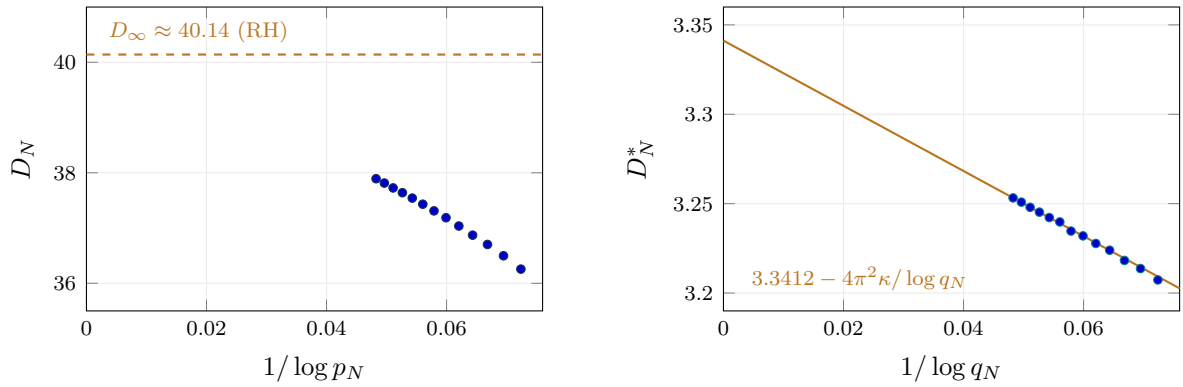


Figure 5: Deficits at thirteen checkpoints $10^6 \le X \le 10^9$, plotted against $1/\log X$. Right: the prediction $D_\infty^* - 4\pi^2\kappa/\log q_N$ with $\kappa = 2\lambda_1$. It has no fitted slope; only the intercept is taken from Table 7.

and, with the x-measure weights $w_n = q^{n+1} - q^{n-1}$,

$$D_N^C = \sum_{n=1}^{N} w_n - \Big|\sum_{n=1}^{N} w_n\, e^{2\pi \mathrm{i}(S_C(n) - H_n)}\Big|.$$

Again no constant is supplied.

Theorem 114.8 (Function-field closure)**.**

(a) $|\alpha_i| = \sqrt{q}$ *for all* i *if and only if* $\sup_N D_N^C < \infty$.

(b) The left side holds: Hasse proved it for $g = 1$ *[79], and Weil in general [80]. Hence* $\sup_N D_N^C < \infty$ *unconditionally, and* $D_\infty^C - D_N^C = O_C(1/N)$.

Proof. Möbius inversion gives $na_n = N_n + \sum_{d|n,\, d<n} \mu(n/d) N_d$, and the inner sum is $O_g(q^{n/2})$. Hence

$$a_n q^{-n} - \frac{1}{n} = -\frac{1}{n}\sum_i \Big(\frac{\alpha_i}{q}\Big)^n + O_g\Big(\frac{q^{-n/2}}{n}\Big).$$

If $|\alpha_i| = \sqrt{q}$, then $|E_C(n)| \le \sum_{d>n} |a_d q^{-d} - 1/d| \ll_g q^{-n/2}/n$. Therefore $\sum_n w_n |e^{2\pi \mathrm{i} E_C(n)} - 1|^2 \ll \sum_n n^{-2} < \infty$, with tail $O(1/N)$, and Lemma 7.1 applies.

Conversely, suppose $r = \max_i |\alpha_i|/q > q^{-1/2}$. The terms with $|\alpha_i| = rq$ dominate and give $E_C(n) = n^{-1} r^n \big(\sum_i c_i z_i^n + o(1)\big)$, with distinct $|z_i| = 1$ and $c_i \ne 0$. A nondegenerate exponential sum $\sum_i c_i z_i^n$ has positive mean square. Hence $w_n E_C(n)^2 \gg (r^2 q)^n / n^2$ on a set of n of positive density, and $\mathcal{J} = \infty$. □

For $g = 1$, the input in (b) is Hasse's inequality $|a| \le 2\sqrt{q}$, where $a = q + 1 - \#C(\mathbb{F}_q)$. Its standard proof shows that the degree form $\deg(m - n\pi) = m^2 - a\,mn + q\,n^2 \ge 0$ on $\mathrm{End}(C)$ is positive semidefinite. It is positive definite, with elliptic level curves, when $a^2 < 4q$, and degenerate when $a^2 = 4q$. The discriminant condition $a^2 \le 4q$ then places the Frobenius eigenvalues on the circle $|\alpha| = \sqrt{q}$. In this analogue, therefore, the closure missing from Section 123 is supplied by the nonnegative degree form, including its degenerate equality case. Both cases give the required spectral modulus. Weil's proof for general g uses the positivity of the Castelnuovo–Severi, or Hodge index, form. The number-field counterpart of that positivity is Weil's explicit-formula positivity, which is equivalent to RH [74].

N	5	10	20	40	60	140
D_N^C, $y^2 = x^3 + x + 1$ over $\mathbb{F}_5$	27.738	33.164	35.216	36.191	36.536	36.887
D_N, fake $\lvert\alpha\rvert = 5^{0.7}$	108.5	808.5	$1.28 \cdot 10^5$	$1.18 \cdot 10^{10}$	$2.03 \cdot 10^{15}$	–

Table 8: The curve has $a = -3$ and $a^2 - 4q = -11 < 0$, so its degree form is an ellipse; $|q^{N/2} N E_C(N)| \le 1.22$ for $5 \le N \le 140$, and D_N^C converges. Replacing α by $5^{0.7} e^{1.1\mathrm{i}}$ violates the analogue of RH, and the deficit grows geometrically. Exact integer point counts were used, with 260-digit arithmetic.

114.8 What remains

The attempt ends where Section 123 ended, with a sharper description of the gap.

- Every representation used here is equivalent to RH and supplies no bound by itself (Proposition 114.1). This includes the circle, the fixed and spectral ellipses, the Cayley/Hardy disk, Li's coordinate and Jensen's formula.
- Unconditionally, $D_N = o(p_N)$ with the Vinogradov–Korobov saving (Theorem 114.2). A power saving $D_N \ll p_N^{1-\delta}$ is equivalent to a zero-free half-plane (Theorem 114.3). The unrefereed 2026 claims would give the exponent 3/4.
- Under the RH and simplicity model, the extrapolations are near 40.14 for D_∞ and 3.3412 for D_∞^*. These are not upper-bound certificates. The conditional tail theorem describes a logarithmic approach (Proposition 114.6 and Remark 114.7).
- In the function-field analogue the same deficit is bounded, and the bound comes from the nonnegative degree form and the Hasse–Weil theorem (Theorem 114.8). For $\mathbb{Q}$ the corresponding positivity statements are Weil's criterion and Zhao's one-sided integral criterion. Both are equivalent to RH.

No unconditional bound of the required strength is supplied here.

115 Independent interval evaluation of finite deficits

The original prime-endpoint deficit was recomputed by a direct vector sum using outward-rounded `mpmath.iv` intervals at 40 decimal places. This uses no centering constant, no quadrature, and a different summation identity from the manuscript's stable nonnegative-update calculation. A deterministic Eratosthenes sieve includes the next prime needed for the final weight. For display, the raw enclosures have been widened to the following shorter intervals:

X	p_N	W_N	Enclosure for D_N
10	7	13	[1.969759465990, 1.969759465991]
10^2	97	193	[16.710860675974, 16.710860675975]
10^3	997	2001	[26.969965365289, 26.969965365290]
10^4	9973	19975	[32.298046003525, 32.298046003526]
10^5	99991	199989	[34.913607775328, 34.913607775329]
10^6	999983	1999981	[36.256293902575, 36.256293902576]
10^7	9999991	20000005	[37.036305899751, 37.036305899752]

The first six reproduce the original checkpoints, while the last extends the cutoff by a factor of ten. These are reproducible interval-library enclosures, not a formally verified proof of the software's

implementation. The complete interval source is embedded in Appendix 127; the intervals above are widened from its output.

Monotonicity gives the finite lower bound $\sup_N D_N > 37.036305899751473531 38$, subject to the computational certificate just described. If RH is assumed, the supremum is D_∞, and the manuscript's identity gives the corresponding weighted circle energy greater than 74.072611799502947062 76. Neither statement is an upper bound, and neither verifies RH.

A separate non-certified long-double experiment evaluated the new integer-window criterion through $m = 4{,}194{,}303$. It gave $D_{21}^* \approx 15.703975172$, $L_{21} \approx 5.127737034$, and $d_{20} \approx 0.011272267$. These values concern integer sampling, not the differently weighted prime-endpoint deficit; their numerical scales must not be conflated. Agreement with the finite inequalities is a consistency test, not evidence controlling all subsequent windows.

A rigorously certified obstruction to local monotonicity. Although the cumulative global deficit is nondecreasing, the new local deficits are not nonincreasing. An independent calculation using only exact Python integers, with no floating-point operations or external numerical packages, proves

$$d_5 \in [0.174263205633101609 79, 0.174263205633101609 80],$$
$$d_6 \in [0.181163953074081467 71, 0.181163953074081467 72].$$

In particular $d_6 > d_5$. The exact-integer program in Appendix 128 represents every interval as $[a/10^{70}, b/10^{70}]$, rounding each integer operation outwards. For logarithms, it first scales the argument into $[1, 2)$ and uses

$$\log y = 2\sum_{j=0}^{99} \frac{t^{2j+1}}{2j+1} + R, \qquad t = \frac{y-1}{y+1}, \qquad |R| \le \frac{9}{4 \cdot 201 \cdot 3^{201}},$$

with the appropriate integer multiple of $\log 2$ restored. Monotonicity of the logarithm handles interval endpoints. Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ is combined with 100-term alternating series and their next-term bounds. The sine and cosine Taylor polynomials have degrees 161 and 160; the code verifies $|\text{phase}| \le 6$ and bounds the respective remainders by $6^{162}/162!$ and $6^{161}/161!$. Square-root bounds use integer square root. The output records exact integer numerators and verifies that the upper numerator for d_5 is strictly less than the lower numerator for d_6. This is a fully specified finite rational certificate; it is not an estimate of subsequent deficits and does not settle their boundedness.

The larger interval run used Python 3.12.14, NumPy 2.3.5, and mpmath 1.3.0, with `iv.dps=40` decimal interval precision.

116 Finite checks and the role of computation

The new finite calculations use 90 decimal digits in `mpmath`. They are checks of algebra and normalization, not interval certificates of an infinite assertion. The distinct exception is the exact rational certificate following Proposition 96.6. All code needed for the tables is embedded in this TeX file.

116.1 The actual degree-four prime polynomials

The prime index N is used, not a cutoff x. The sieve includes p_{N+1} so that the final original weight is correct. The normal equations for $P_{4,N}$ are solved directly, and its norm is independently evaluated as a positive weighted sum. The deficit is computed from a positive variance sum to avoid subtracting two nearly equal large numbers. Neither computation supplies $B_{\mathbb{P}}$.

N	p_N	D_N	$H_{4,N}$	$\rho_{1,N}$	$\rho_{2,N}$
100	541	24.716380	4.912210	0.99763801	0.96186498
1000	7919	31.902292	8.792997	0.99960317	0.95471782
10000	104729	34.944886	10.741125	0.99995002	0.95455190

Table 9: Finite values for the original prime weights and phases. The final two columns are the largest and second largest root moduli, not zeta-zero locations.

Every displayed finite root lies inside the disk, as expected from the finite OPUC theory. What is unproved is a gap for the second root that persists for all sufficiently large N. At the largest sample, the bound in Proposition 93.6 is about $4.92 \cdot 10^7$, whereas D_N is about 34.94. Thus the general root-gap inequality is valid but loose in this numerical example; it is not presented as a good numerical estimate for D_∞.

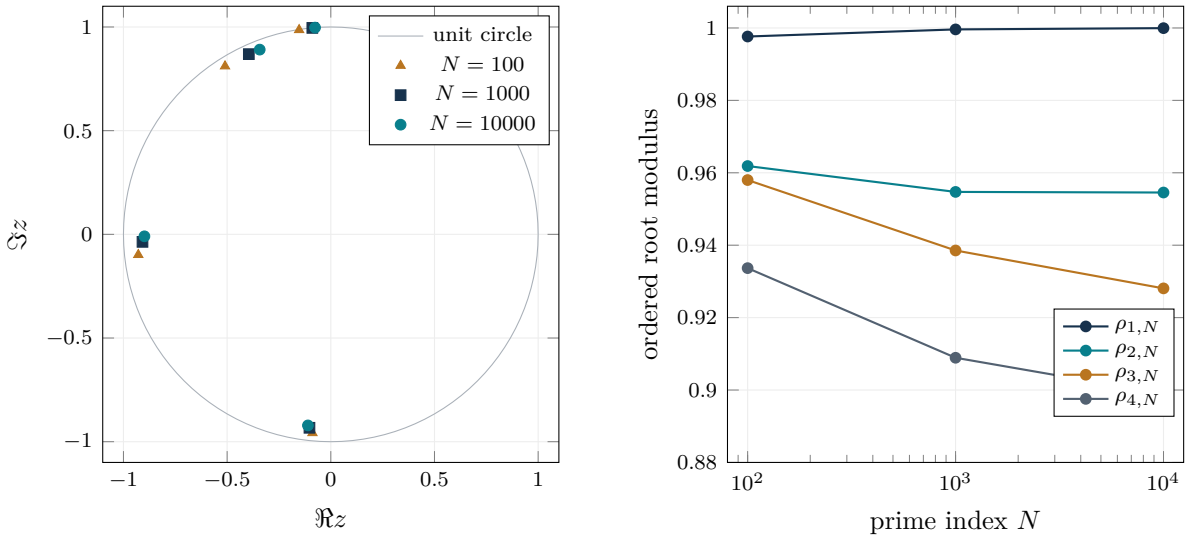


Figure 6: Left: roots of the computed degree-four prime extremizers. Right: their ordered moduli at three finite indices. The lines join observations only; they are not estimates of future behavior or certified limiting root gaps.

The relative least-squares orthogonality residual is below $7 \cdot 10^{-88}$ in these runs. Evaluation of the computed roots in the polynomial gives residuals below $4 \cdot 10^{-91}$. For the $r = 3$ spectral matrices, the finite leakage identity and its independent harmonic expression agree to absolute error below $1.2 \cdot 10^{-84}$. The ratios $\mathcal{T}_{3,N}/\mathcal{L}_{3,N}$ are respectively 3.89775697, 3.98992652, and 3.99912027; their convergence to four is predicted by an unconditional theorem for convergent phase sequences, so it does not constitute evidence resolving RH.

116.2 Curve-tail coefficients

We evaluate closed-point counts using integer recurrences and Möbius inversion, then compute the errors by backward summation of their tails. The sum is carried 160 degrees past the largest displayed cutoff. Stable phase differences use `expm1`, and the deficit is recovered through the variance identity. The quantity

$$2N\{D_{2N}^C - D_N^C\}$$

must tend to $\mathcal{K}_C$, by Theorem 97.1; this comparison requires no extrapolated value of D_∞^C.

The $q = 9$, trace-six spectral data test the repeated-root and prime-square resonance case. The genus-zero row tests the nonzero prime-square coefficient even when there are no Frobenius

Data	$\mathcal{K}_C$ (exact formula, decimal)	$2N(D_{2N} - D_N)$ at $N = 800$	$\mathcal{K}_C^*$ (exact formula, decimal)
$q = 5$, $a = -3$	38.820444	38.845534	21.055156
$q = 3$, $a = 0$	52.637890	52.564379	26.318945
$q = 9$, $a = 6$	276.897235	276.842347	175.459634
$q = 5$, genus zero	17.765288	17.759789	0.000000

Table 10: The curve-tail constant compared with finite differences. An $O(1/N)$ error in the middle column follows from the proved $O(1/N^2)$ remainder of the deficit tail.

roots. The prime-power variant has zero $1/N$ coefficient in genus zero, and its finer behavior is the separate formula (97.5).

117 Finite experiments and what they verify

The computations in this section are finite numerical experiments. They are not interval-certified, and they do not establish any infinite convergence claim. The equations proved above are exact; the displayed decimals here are approximations.

117.1 Continuous energies

We sieved primes through 10^6 and evaluated each prime gap separately, so no quadrature panel crossed a discontinuity of S. For each cutoff, Gauss–Legendre quadrature of orders 16 and 32 was compared. The largest absolute difference among the reported energy values was below $6 \cdot 10^{-14}$. That agreement is a numerical consistency check, not a rigorous quadrature-error bound.

Let $J_{\mathrm{circ}}(X) = \int_2^X |Z(x) - 1|^2 \, \mathrm{d}x$ and let $J_{\mathrm{ell}}(X)$ use axes $a = 2, b = 1/2$ in Theorem 4.1.

X	$I(X)$	$J_{\mathrm{circ}}(X)$	$J_{\mathrm{ell}}(X)$
10	0.4528982208	12.3501439535	28.9243950387
10^2	0.6453466953	19.7963685485	32.3999690652
10^3	0.7183439651	22.6756655349	33.1480535456
10^4	0.7541530540	24.0893056345	33.5019817574
10^5	0.7745490579	24.8945062699	33.7032968338
10^6	0.7879412220	25.4232076629	33.8354727627

Table 11: Finite continuous energies. Distinct initial behavior is compatible with equivalent tail-convergence criteria.

117.2 The constant-free deficit

The deficit was accumulated using the nonnegative update (7.13), with extended-precision floating-point arithmetic. The next prime after each cutoff is needed for the last weight w_N. Direct subtraction and the variance identity were used as independent numerical checks.

117.3 Initial Laguerre coefficients

For $\alpha = 1$, a high-precision evaluation of (9.7)–(9.6), checked against derivatives of the generating function, gives: Their squared sum is approximately

Cutoff X	Last prime p_N	W_N	D_N	$\mathcal{J}_N$
10	7	13	1.9697594660	34.1413094763
10^2	97	193	16.7108606760	60.8759929534
10^3	997	2,001	26.9699653653	68.7831808384
10^4	9,973	19,975	32.2980460035	72.0512604990
10^5	99,991	199,989	34.9136077753	73.7657173538
10^6	999,983	1,999,981	36.2562939026	74.8491148279

Table 12: The deficit does not use $B_{\mathbb{P}}$. The comparison column $\mathcal{J}_N = \sum_{n=2}^{N} w_n |U_n - e^{2\pi i B_{\mathbb{P}}}|^2$ does use it. At finite N, $2D_N \le \mathcal{J}_N$; equality is a limit statement under RH.

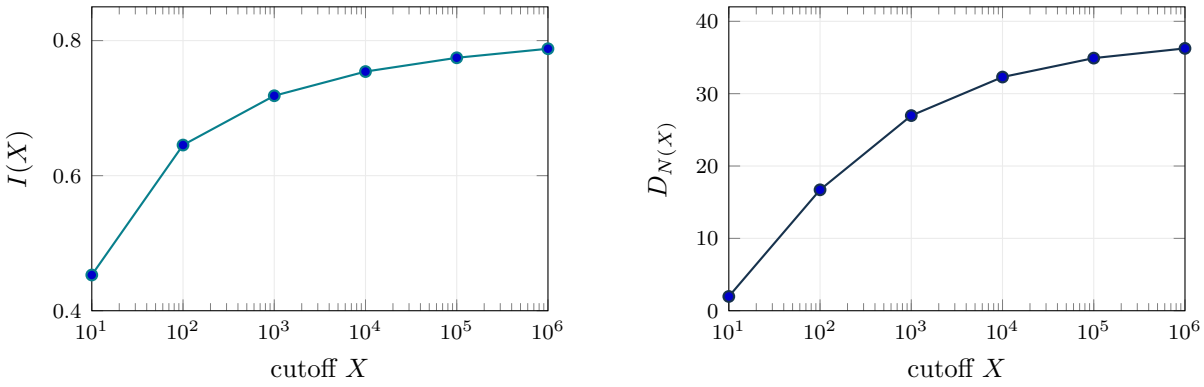


Figure 7: Reported checkpoints joined for readability. The visible flattening over this finite range does not prove boundedness. Here $N(X) = \pi(X)$.

n	$a_n^{(1)}$
0	0.346934737830795621
1	−0.661616469769648581
2	0.225894182779432296
3	−0.140258082155750076
4	0.228257943870596736
5	−0.040192481906854658
6	0.163353841105629616

Table 13: Approximate initial coefficients, not certified intervals.

$$\sum_{n=0}^{6} |a_n^{(1)}|^2 = 0.709202178753600527.$$

These are global transform coefficients, so their finite partial sum is not an integral truncated at $x = 10^6$. Under RH, all coefficient squares sum to $I(\infty)$, but neither the printed coefficients nor Table 11 controls the remaining tail. Appendix 119 supplies explicit bounds that can be used to certify individual finite calculations.

117.4 Finite checks of the harmonic and determinant transfers

The following values use the original prime vectors and weights. The pair definition of $C_{4,N}$ was evaluated independently of its alternating harmonic formula. The least-squares value $H_{4,N}$ was also compared with the Gram determinant ratio. Calculations used 90 decimal digits; they are floating-point checks, not interval certificates. No infinite bound is inferred from the table.

N	p_N	D_N	$C_{4,N}$	$H_{4,N}$
8	19	7.16815104	225.62037355	0.14477688
16	53	13.36793576	510.93018613	0.76241711
32	131	18.13936274	700.80614780	2.04459049
64	311	22.31007541	828.12192867	3.75468037
128	719	25.82085230	909.14157506	5.47180933

Table 14: Finite checks of the original deficit, fourth pair moment, and degree-four monic extremal norm.

Across all tested prefixes and moment orders $1 \le m \le 4$, the maximum absolute discrepancy in the harmonic identity was less than 10^{-86}. The determinant-ratio discrepancy was also less than 10^{-86}. The finite Hölder bridge was satisfied at every tested prefix. Its values grow much faster than the observed deficit; it is a valid upper bound, not an effective numerical estimate of the remaining constant. The nonmonotonicity example gives exactly $9/2$ followed by $11/3$, as stated in Remark 15.10. The complete calculation is embedded in Appendix 120.2.

118 Eighty methods: complete transfer and source audit

Every numbered example supplied for this extension appears below. An application is counted as a proved transfer only when its hypotheses have been checked in the present construction. Otherwise the row states the additional mathematical input required. The original target remains $\sup_N D_N < \infty$. A different finite-moment threshold, a finite computation, or an existence theorem is not substituted for that target.

The links identify the supplied source or a primary replacement. “Access” distinguishes a full text, an abstract or author survey, indexed source text, and an unavailable original. This is a targeted mechanism audit, not an independent reproduction of eighty original proofs. Several entries share one proof technology; the eighty examples do not provide eighty independent estimates for the prime error.

The connections and missing-hypothesis statements in the right column are deductions in this note. They are not claims made by the authors of the linked papers. Complete proofs of the successful transfers are in Sections 15–20.

118.1 Fourier analysis, circle integrals, and determinants: 1–20

Method and source	Mechanism, attempted transfer, and outcome
1. Ternary Goldbach Source *Access:* Primary author overview opened; the whole multi-paper proof and computation were not independently reproduced.	**Mechanism.** Helfgott combines major-arc estimates, minor-arc bounds, the large sieve, and finite verification to prove the three-prime theorem for odd integers greater than five. **Transfer.** Apply Fourier orthogonality to the dyadic polynomials A_j whose coefficients are E(n). Both arc regions would need a uniform or summable quadratic budget. **Outcome.** An exact Fourier criterion is obtained in the main text. Published threefold-convolution estimates do not provide the required bound for these error coefficients.
2. Integer partitions Source *Access:* User's circle-method overview was read. Rademacher's 1937 primary record was retrieved; full PMC access hit a bot check.	**Mechanism.** Cauchy coefficient extraction and the modular transformation of the partition generating function yield asymptotics and Rademacher's convergent exact series. **Transfer.** A comparable transformation law for the relevant prime-error generating function would be needed, with rigorous uniform control of the transformed remainder. **Outcome.** No such modular law follows from circle integration alone. The finite Laguerre-polynomial construction instead gives an exact Szego determinant identity, with its uniform constant still unknown.
3. Waring representation counts Source *Access:* Vaughan's author-hosted Waring notes were read; the user's Encyclopedia overview was also checked.	**Mechanism.** The circle method extracts representations by powers and bounds minor arcs; singular series and singular integrals supply the main term under the relevant hypotheses. **Transfer.** A prime-energy violation would need to force an impossible representation count, or the relevant exponential-sum bounds would have to control the dyadic error norms uniformly. **Outcome.** The coefficient-extraction step transfers exactly, but no such arithmetic implication or bound has been established. Hilbert's prior existence proof is a separate method.
4. Sixteen fourth powers Source *Access:* Supplied author lecture notes opened. The number is G(4), the eventual threshold, not the all-integer Waring number g(4).	**Mechanism.** The exact eventual threshold G(4)=16 combines analytic representation estimates with congruence obstructions that show optimality. **Transfer.** The analogue would be a sharp minor-arc bound for the original quadratic error, with local arithmetic conditions strong enough to prevent remaining exceptional concentration. **Outcome.** No threshold from additive fourth powers bounds our second moment. The four-harmonic statistic bounded under (H_b) has eighth-order contact and a different exact zero-location threshold.
5. Five prime squares Source *Access:* Primary paper opened. The linked proof is a transference proof; it should not be described as merely the original circle-method proof.	**Mechanism.** The linked paper gives an L-function-free transference proof of Hua's theorem for sufficiently large N congruent to 5 modulo 24, using a sieve majorant and pseudorandomness estimates. **Transfer.** A dense-model or transference argument would need a verified majorant and pseudorandom estimates for the actual centered-error observable, together with a quantitative inverse comparison. **Outcome.** The theorem's admissible congruence class and its pseudorandomness hypotheses do not automatically control D_N. Higher moments can remain finite while its quadratic energy diverges.
6. Roth's three-term progressions Source *Access:* Original 1953 publisher metadata retrieved; direct full text blocked. The supplied 2018 article is a Roth obituary, not the original proof.	**Mechanism.** Fourier analysis detects a large coefficient when a dense set has too few progressions; a density increment then forces a contradiction after iteration. **Transfer.** One would need a density or bounded potential attached to prime-error blocks, and a quantitatively controlled increment caused by excessive variance. **Outcome.** The block merger identity is exact, but no bounded density parameter or exhausting increment argument has been constructed for the original prime sequence.

Continued on next page

Method and source	Mechanism, attempted transfer, and outcome
7. Square differences in dense sets Source *Access:* Original Sarkozy reference identified, full DOI access unavailable. Scope corroborated by primary difference-set research; the supplied MathWorld entry is secondary.	**Mechanism.** Fourier estimates for square exponential sums and density increment, or ergodic recurrence, force a square difference in a set of positive upper density. **Transfer.** A high-error subset would need positive density with a quantitatively controlled square-difference obstruction derived from the prime-reciprocal recurrence. **Outcome.** No such forbidden-difference implication is present. An energy can diverge on a set of density zero, so the qualitative density theorem alone cannot settle the required norm.
8. Prime-minus-one differences Source *Access:* Primary Ruzsa-Sanders abstract opened. The user's Oxford link is a research seminar announcement; it was also read.	**Mechanism.** Shifted-prime exponential sums, Fourier concentration and density increment control sets avoiding differences p-1; Ruzsa and Sanders give a quantitative bound. **Transfer.** Large prime-circle errors would need to create a sufficiently dense difference-avoiding set, or the shifted-prime estimates would need to control the actual error coefficients directly. **Outcome.** The required connection to a forbidden difference has not been proved. A theorem about existence of one additive configuration is not a uniform upper bound for all weighted error blocks.
9. Birch's systems of forms Source *Access:* Original DOI/full text unavailable; bibliographic and abstract text retrieved. Related author research on forms was opened. No full fresh reading of Birch's proof is claimed.	**Mechanism.** A torus integral, Weyl differencing and local densities give asymptotic solution counts when the singular locus has sufficiently large codimension and the required local solubility holds. **Transfer.** An explicit system of fixed-degree forms would have to encode a prime-energy violation while satisfying the rank and local hypotheses uniformly in the index. **Outcome.** A multitorus coordinate change supplies no such system. The original analytic target has no verified reduction to these finite-dimensional polynomial constraints.
10. Cubic forms in many variables Source *Access:* The supplied SciXplorer page had a bot wall. Dietmann-Wooley's primary paper was opened and explicitly records Davenport's sixteen-variable result.	**Mechanism.** Davenport's circle-method estimates, with arithmetic and geometric reductions, show that sixteen variables suffice for a rational cubic form to have a nontrivial rational zero. **Transfer.** One needs a faithful polynomial encoding in which the existence or impossibility of a cubic zero gives a uniform prime-error estimate. **Outcome.** Adding auxiliary variables can make a new equation soluble without retaining any bound for the original observable. No faithful encoding with the required quantitative comparison is available.
11. Longest increasing subsequences Source *Access:* Primary arXiv record and the relevant method description opened. No identity with the random permutation model is assumed.	**Mechanism.** Baik-Deift-Johansson use a Toeplitz determinant representation, orthogonal polynomials and Riemann-Hilbert asymptotics to determine the limiting distribution. **Transfer.** Form the Toeplitz moment matrix of the original positive atomic prime measure and study its monic extremal polynomial norms. **Outcome.** This produces the proved hierarchy H_m: boundedness is exactly the 2m-moment threshold. H_4 is bounded using the stated zero-free input; H_1 remains the RH-equivalent variance.
12. Fisher-Hartwig asymptotics Source *Access:* Primary Annals theorem record opened; its scope is retained rather than calling every Fisher-Hartwig regime solved.	**Mechanism.** The cited theorem analyzes Toeplitz and related determinants with controlled root and jump singularities, using orthogonal-polynomial and Riemann-Hilbert methods within stated parameter ranges. **Transfer.** The prime-dependent symbols would need a justified singularity model and estimates uniform as the number or location of singularities varies. **Outcome.** Finite determinant identities transfer, but the cited fixed-symbol theorem does not furnish a uniform bound over all prime prefixes. The exact double-index criterion isolates that missing uniformity.

Continued on next page

Method and source	Mechanism, attempted transfer, and outcome
13. Dyson's large-gap constant Source *Access:* Primary arXiv paper record opened.	**Mechanism.** Krasovsky analyzes orthogonal polynomials on an arc and Toeplitz determinants to compute the constant in the large-gap asymptotic. **Transfer.** An arc-supported prime measure would need uniformly controlled weights and a matching asymptotic regime; the constant must then dominate the original quadratic deficit. **Outcome.** The atomic prime measure and its increasing total mass are not the fixed ensemble in that theorem. Our degree-four determinant ratio closes a different observable; no original constant follows.
14. One-dimensional Bose occupations Source *Access:* Claeys-Krasovsky primary record, revised version, opened. The application concerns the specified model and conjecture.	**Mechanism.** Uniform Toeplitz asymptotics for two merging Fisher-Hartwig singularities resolve the cited occupation-number conjecture for the impenetrable Bose gas. **Transfer.** A proven correspondence between the arithmetic symbol and the physical symbol, including uniform control of its merging parameter, is required. **Outcome.** No physical-model identity has been established. The useful imported operation is finite positive Toeplitz optimization, whose exact arithmetic threshold is proved separately.
15. A random-matrix second moment Source *Access:* Same primary paper as item 14; this is a separate application, not a second independent proof technology.	**Mechanism.** The same merging-singularity analysis evaluates a particular averaged characteristic-polynomial moment and resolves the corresponding conjectured case. **Transfer.** A random model would need a rigorous comparison controlling this deterministic prime sequence, not only its ensemble mean or typical behavior. **Outcome.** No deterministic comparison of that strength is supplied. Positive atomic Gram matrices are available, but their positivity alone is compatible with an unbounded quadratic deficit.
16. Strong Szego theorem Source *Access:* Simon's primary proof and sharp Theorem 1.1 were read. This replaces reliance on the supplied encyclopedia exposition.	**Mechanism.** The normalized Toeplitz determinant limit is the exponential of the weighted square sum of the Fourier coefficients of the log symbol. **Transfer.** Use the smooth finite symbol L_M=2 Re g_M built from the unconditional Laguerre coefficients; then take a supremum over both polynomial degree and determinant size. **Outcome.** An exact identity is proved: the double supremum of log determinants equals the full coefficient energy. Finiteness is RH-equivalent, and a bound C converts to an explicit original K(C).
17. Optimal prediction of time series Source *Access:* Bingham's author survey was opened; it is a primary author synthesis of prediction theory, not a new proof of every cited historical result.	**Mechanism.** Spectral measures, outer functions and orthogonal polynomials on the circle express prediction errors as extremal polynomial norms and determinant ratios. **Transfer.** Use monic least squares for the atomic prime measure. A prediction interpretation is legitimate only after its precise spectral measure and normalization are specified. **Outcome.** The extremal norm and Schur-complement identities are exact. They yield H_1=V and the full H_m hierarchy, but existence of an optimal predictor does not bound its unnormalized error uniformly.
18. Ising correlations and magnetization Source *Access:* Primary author/publisher abstract opened. The website's later metadata date is not treated as the original publication date.	**Mechanism.** Correlation formulas reduce the model to determinants with explicit circle symbols; their asymptotics give an alternative derivation of spontaneous magnetization. **Transfer.** An explicit arithmetic symbol with an equally controlled determinant identity and limiting regime would have to be found. **Outcome.** The double-index Szego construction gives a rigorous symbol, but its limiting constant is precisely the unknown coefficient energy. The Ising model's integrability does not transfer by analogy alone.

Continued on next page

Method and source	Mechanism, attempted transfer, and outcome
19. Fourier proof of isoperimetry Source *Access:* The author page was search-indexed but direct access returned 403. The transfer result here has an independent elementary proof in the main text.	**Mechanism.** For an arc-length parametrized closed curve, Fourier orthogonality and a spectral inequality compare area with squared perimeter and identify the circle equality case. **Transfer.** One needs a coercive inequality that turns an actually bounded derivative or curvature norm into the unbounded-domain displacement norm of the prime phase. **Outcome.** Finite bending of a natural prime-phase interpolation is proved, but it is insufficient: a function may tend to zero and have square-integrable derivative while its own square integral diverges.
20. Weyl equidistribution Source *Access:* Weyl's 1916 primary bibliographic record located. The supplied Hamenstadt paper discusses linear irrational rotation in its introduction, not the full polynomial theorem; its scope was corrected.	**Mechanism.** Fourier characters detect the uniform distribution of polynomial sequences with an irrational nonconstant coefficient; differencing controls their exponential sums. **Transfer.** Determine the actual limiting measure of the centered prime points before applying an equidistribution criterion. **Outcome.** Their weighted empirical measures converge to a point mass, and all normalized harmonic moments converge to powers of that point. This gives a sublinear deficit, not a uniform deficit bound.

118.2 Elliptic curves and Diophantine equations: 21–40

Method and source	Mechanism, attempted transfer, and outcome
21. Fermat's Last Theorem via Frey curves and modularity Source *Access:* Publisher record checked; detailed operative mechanism checked in the primary Darmon-Merel and BCDT papers.	**Mechanism.** A hypothetical solution produces a Frey curve; modularity, an irreducible residual representation, and level lowering force an impossible residual level. **Transfer.** A large prime-circle error must yield similarly constrained Galois data, including removable new conductor primes. **Outcome.** The explicit curve Y^2=X(X-A_n)(X+Q_n) retains discriminant valuation 2 at every new odd denominator prime. The standard odd-prime removal mechanism fails for this encoding, with the stated irreducibility qualification.
22. Perfect powers in the Fibonacci sequence Source *Access:* Primary publisher abstract checked; both the theorem and the combined methods are explicitly stated.	**Mechanism.** Bugeaud, Mignotte and Siksek combine logarithmic lower bounds, Thue equations, and modular methods; the perfect-power values are 0, 1, 8 and 144. **Transfer.** The actual centered prime phases would need an exact fixed algebraic recurrence or a controlled Thue-equation encoding, with the required height estimates. **Outcome.** No such recurrence is supplied. The direct expression contains log log p, which is not already a logarithm of an algebraic argument of the required kind.
23. Perfect powers in the Lucas sequence Source *Access:* Primary publisher abstract checked; this is a separate application within the same paper.	**Mechanism.** The same classical and modular program uses the special recurrence structure to prove that the Lucas perfect-power values are 1 and 4. **Transfer.** An exact Lucas-type or Thue identity must encode the original reciprocal recurrence and logarithmic centering, rather than only a point on a circle. **Outcome.** No required identity has been constructed. The explicit comparison sequences preserve primorial order, height zero and convergence while having opposite quadratic-deficit behavior, so those data alone cannot replace the recurrence.

Continued on next page

Method and source	Mechanism, attempted transfer, and outcome
24. Three distinct positive nth powers in arithmetic progression Source *Access:* Primary main theorem, Frey invariants and residual-conductor argument checked directly.	**Mechanism.** Darmon-Merel treat x^n+y^n=2z^n using Frey curves, level lowering and winding quotients; nontrivial primitive solutions are excluded for n at least 3. **Transfer.** A violation of the prime-circle bound must force this exact primitive identity or an equally restrictive modular representation. **Outcome.** Prime-reciprocal rationality supplies no such identity. In the natural numerator-primorial Frey curve, the new discriminant valuations are 2 rather than multiples of an odd residual exponent.
25. The generalized Fermat signature x^n+y^n=z^2 Source *Access:* Primary main theorem and reduction to prime exponents checked directly.	**Mechanism.** The Darmon-Merel theorem excludes nontrivial primitive solutions for n at least 4, combining small-exponent descent with modular arguments for larger primes. **Transfer.** A faithful encoding of prime errors must produce the square identity and preserve primitive conditions uniformly over all indices. **Outcome.** The reduced pair A_n,Q_n is coprime, but it satisfies no established square identity of this form. Attaching an elliptic curve does not manufacture the special equation needed for the theorem.
26. The generalized Fermat signature x^n+y^n=z^3 Source *Access:* Primary conditional statement checked; BCDT's primary Theorem A separately checked.	**Mechanism.** Darmon-Merel exclude nontrivial primitive solutions for n at least 3 under full modularity. BCDT later proved that modularity hypothesis for elliptic curves over Q. **Transfer.** Prime-circle errors must produce the corresponding Frey representation, with its restrictive local behavior, including the prime 3. **Outcome.** The historical modularity condition is now satisfied, but no transfer from the actual centered errors to those special Galois data is established.
27. Primitive solutions of x^2+y^3=z^7 Source *Access:* Primary abstract, solution theorem, and Chabauty and Mordell-Weil sieve section checked.	**Mechanism.** Nonabelian descent, modularity and level lowering leave ten genus-3 curves; Chabauty and the Mordell-Weil sieve determine their rational points. **Transfer.** One needs a finite specified family of arithmetic covers receiving hypothetical prime-error violations, together with certified ranks and finite-index information. **Outcome.** No such cover is constructed. The natural periodic lift instead produces elliptic torsion of unbounded order, which cannot remain over number fields of uniformly bounded degree.
28. Primitive positive solutions of x^2+y^8=z^3 Source *Access:* Primary publisher full text checked; the original DOI landing request failed.	**Mechanism.** Genus-2 covers, rank-zero elliptic quotients and effective Chabauty leave the positive primitive solution (1549034,33,15613). **Transfer.** An actual prime-error encoding must satisfy the covering equations and the rank condition making the relevant p-adic intersection finite. **Outcome.** Those hypotheses have not been supplied. The raw reciprocal sums are not Cauchy in any fixed Q_l, so direct local summation cannot stand in for a constructed Chabauty map.
29. Primitive solutions of x^2+y^4=z^5 Source *Access:* Primary Theorem 1.1 and the covering framework checked directly.	**Mechanism.** Bruin's elliptic Chabauty methods use higher-genus covers and arithmetic over number fields; every primitive solution of this equation has xyz equal to zero. **Transfer.** A fixed arithmetic cover must encode the centered prime phase while preserving the weighted error needed for the original deficit. **Outcome.** A rational coordinate change from the circle cannot supply the requested elliptic point: every rational map from a smooth conic to an elliptic curve in characteristic zero is constant.

Continued on next page

Method and source	Mechanism, attempted transfer, and outcome
30. Primitive positive solutions of x^2-y^4=z^5 Source *Access:* Primary Theorem 1.2 and local logarithmic framework checked directly.	**Mechanism.** Elliptic covers over number fields and Chabauty determine the primitive solutions; the positive primitive triple is (122,11,3). **Transfer.** The prime sequence would require a fixed number field, a covering condition, certified Mordell-Weil information and the appropriate rank restriction. **Outcome.** The natural group-preserving lift of the raw rational phase has exact primorial torsion order. Merel's theorem forces its necessary field degrees to become unbounded, so this lift does not enter the fixed-field calculation.
31. Primitive positive solutions of x^8+y^3=z^2 Source *Access:* Primary Theorem 1.3 and the stated genus-5 and genus-17 covering program checked.	**Mechanism.** Higher-genus elliptic-cover Chabauty gives positive primitive solutions (1,2,3) and (43,96222,30042907). **Transfer.** A controlled arithmetic covering construction must constrain the prime-indexed points, and its estimates must transfer to the weighted archimedean energy. **Outcome.** The fixed-cover classification does not bound an unconstrained sequence of complex torus points. Explicit countermodels preserve exact primorial order and zero height while all higher moments converge and the quadratic deficit diverges.
32. Primitive solutions of x^2+y^3=z^9 Source *Access:* Primary full text, main solution theorem and covering method checked.	**Mechanism.** The source treats a^3+b^9=c^2 using genus-10 covers and Chabauty on Weil restrictions. The substitution (a,b,c)=(-y,z,x) connects the equations. **Transfer.** A concrete Weil-restriction model must encode actual prime errors and satisfy the required arithmetic and rank conditions. **Outcome.** The positive primitive solution is (13,7,2). The signed substitution is essential; the linked title is not literally the requested equation. No analogous prime-error lift has been constructed.
33. Three consecutive positive cubes with square sum Source *Access:* Primary introduction checked; it states the completed square classification and the elliptic-logarithm method.	**Mechanism.** For fixed length, genus-1 integral-point methods and elliptic logarithms give the positive starting values 1 and 23. **Transfer.** The actual prime-error data must become integral points on a controlled curve, with a height estimate that implies the required weighted energy bound. **Outcome.** Rationality of the reciprocal sum is insufficient: its reduced denominator contains every prime up to the cutoff. No fixed integral-point equation for the logarithmically centered errors is present.
34. Five consecutive positive cubes with square sum Source *Access:* Primary introduction and table checked; positive and nonnegative starting conventions are distinguished.	**Mechanism.** The positive starting values are 1,25,96,118; allowing zero adds start 0. Fixed-length genus-1 integral-point methods complete the square case. **Transfer.** A fixed polynomial identity, or genuinely uniform height estimates for the varying prime-dependent parameters, must control the actual error sequence. **Outcome.** The finite-length classifications provide no automatic uniform bound over changing arithmetic complexity. They cannot be substituted for the original weighted quadratic estimate.
35. Common tetrahedral and square-pyramidal values Source *Access:* Image-only primary PDF read by OCR, including the theorem and explicit proof-dependency disclaimer.	**Mechanism.** The only positive common value is 1. The completing proof uses a small cubic-form inequality, continued fractions and Easton's irrationality bound. **Transfer.** A large prime error would need to imply a forbidden approximation to a fixed algebraic target with a controlled irrationality measure. **Outcome.** The moving logarithmic reference is not the fixed algebraic target used here. The authors expressly state that their elliptic-curve calculation is not used in the completing proof.

Continued on next page

Method and source	Mechanism, attempted transfer, and outcome
36. Lucas' cannonball or square-pyramid problem Source *Access:* Primary solution theorem, quartic reduction and congruent-number correspondence checked.	**Mechanism.** The positive solutions of 1^2+...+n^2=t^2 are (1,1) and (24,70). Bennett's treatment uses quartic Pell equations and a congruent-number correspondence. **Transfer.** The prime data must satisfy a fixed quartic identity or lie in a controlled Pell-unit orbit whose estimates bound the original error. **Outcome.** No such orbit is established. The expanding denominator support obstructs treating the raw reciprocal sums as a sequence in one fixed ring of S-integers.
37. An explicit rational right triangle of area 157 Source *Access:* Direct page requests timed out; indexed author-book side lengths were independently verified by exact integer arithmetic.	**Mechanism.** A rational triangle corresponds to a non-torsion rational point on Y^2=X^3-157^2 X, certifying that 157 is a congruent number. **Transfer.** A rational elliptic point must encode the actual centered phase, and elliptic-logarithm estimates must control the weighted distance to its limit. **Outcome.** This existence certificate supplies no quadratic-energy estimate. Analytic transport preserves the energy question without proving rationality of the transported phases.
38. Four rational squares in nonconstant arithmetic progression Source *Access:* Primary exposition checked, including its express qualification about the numerical L-value.	**Mechanism.** A birational reduction gives an elliptic curve with eight rational torsion points; rank zero excludes additional rational points. **Transfer.** One needs an exact map from prime-error violations to additional rational points, plus a certified rank-zero conclusion. **Outcome.** The classical nonexistence theorem is unconditional. The supplied exposition invokes Kolyvagin with an explicitly uncertified numerical L-value; that illustration cannot be promoted to a certificate. A direct rational circle-to-elliptic map is constant.
39. The Lebesgue-Nagell equation x^2+D=y^n Source *Access:* Primary abstract and stated range checked directly; no extension to arbitrary D is asserted.	**Mechanism.** The linked work completes the range 1 through 100 for D and n at least 3, using Thue equations, logarithmic bounds and modular techniques. **Transfer.** A prime-error violation must force a solution within this fixed parameter range, or new estimates uniform in the prime-dependent parameters must be proved. **Outcome.** No such bounded-parameter encoding is established. The finite-D classification cannot automatically become a theorem about arbitrary denominator primes or the moving logarithmic reference.
40. Elliptic binomial equations with pairs (2,3) and (2,4) Source *Access:* Primary abstract, parameter scope and elliptic-logarithm and LLL reduction steps checked.	**Mechanism.** Elliptic logarithms, explicit height bounds, lattice reduction and finite checks determine solutions for eight specified binomial-index pairs, including the two requested pairs. **Transfer.** Uniform polynomial identities and arithmetic height bounds must encode the actual prime errors in controlled elliptic models. **Outcome.** No such identity is supplied. Finite interpolation of sampled phases would not provide a uniform arithmetic equation, and the source does not solve all repeated entries of Pascal's triangle.

118.3 Geometric transformations and inequalities: 41–60

Method and source	Mechanism, attempted transfer, and outcome
41. Circle inversion and straight-line linkages Source *Access:* Supplied Cornell teaching notes were opened; their inversion and linkage sections were inspected. They are not an original research proof.	**Mechanism.** A linkage enforces an exact distance-product identity realizing inversion. Circles through the inversion center become straight lines. **Transfer.** Rotate the limiting prime direction to one and apply a Cayley map. On a sufficiently short tail arc its squared displacement is comparable to the original squared chord distance. Closing D_N therefore requires a uniform sum of the transformed squared displacements, with the original weights. **Outcome.** A rigorous equivalent scalar model follows; inversion supplies no new upper budget. A pole at the limiting point invalidates bounded distortion.
42. Tarski circle squaring Source *Access:* The supplied MathWorld URL did not resolve. The alternative primary Marks–Unger manuscript and its graph-flow description were retrieved.	**Mechanism.** Discrepancy estimates and graph flows permit finite translation equidecompositions. The constructive Borel version preserves measurability while matching sets of equal measure. **Transfer.** Treat weighted prime directions as mass to transport toward their limit. The required certificate is a bounded unnormalized squared transport cost. A finite number of pieces with bounded translation lengths only gives a cost proportional to the growing total weight. **Outcome.** A transport formulation is available, but neither equidecomposability nor equality of mass proves the required finite cost.
43. Kakeya needle compression Source *Access:* MathWorld opened. Davies's primary research-paper extract confirmed the needle result; the original Besicovitch DOI was access-blocked.	**Mechanism.** Overlapping thin configurations permit all needle directions inside arbitrarily small area. The area counts occupied locations rather than repeated visits or transport multiplicity. **Transfer.** Compressing a phase curve must retain its energy measure. A legitimate transfer needs a measure-preserving rearrangement, or a Jacobian bounded below together with a uniformly bounded transformed energy. Otherwise a small image area can hide arbitrarily large weighted multiplicity. **Outcome.** Measure-preserving rearrangement preserves the energy; area compression alone does not bound D_N. No suitable Jacobian and budget estimate was obtained.
44. Dehn invariant and Hilbert's third problem Source *Access:* The supplied encyclopedia page repeatedly returned 502. Definitions and original references were checked in Gaifullin–Ignashchenko's primary paper.	**Mechanism.** An additive dissection invariant distinguishes polyhedra whose equal volumes do not imply scissors congruence. Tracking an invariant prevents an invalid geometric replacement. **Transfer.** For prime blocks, the exact variance merger identity has an additional nonnegative term: the product of their weights divided by the combined weight, times the squared difference of their mean directions. A multiscale proof must budget every such merger term. **Outcome.** Exact bookkeeping transfers, but an unrelated additive invariant cannot eliminate the interblock costs. Their total summability remains unproved.
45. Flexible closed polyhedra Source *Access:* The supplied original 1977 paper was opened; its introduction and construction were inspected.	**Mechanism.** Connelly constructs an embedded closed polyhedral sphere that flexes while its faces remain congruent. Local metric constraints therefore need not determine the whole configuration. **Transfer.** Represent the weighted prime vectors as a linkage. Fixing each segment length does not determine the resultant endpoint and hence does not determine D_N. A useful rigidity transfer would require additional global constraints forcing a quantitatively bounded endpoint deficit. **Outcome.** The test identifies missing global constraints. It does not prove that every geometric approach to the prime phases must fail.

Continued on next page

Method and source	Mechanism, attempted transfer, and outcome
46. Bellows volume invariance Source *Access:* The supplied source is Sabitov's own survey; its record and abstract were retrieved, rather than a new full proof audit.	**Mechanism.** A volume polynomial ties volume to fixed metric and combinatorial data. During a continuous flex, volume cannot move among finitely many roots of that fixed polynomial. **Transfer.** A sufficient analogue would be a monic equation for D_N whose coefficients are uniformly bounded across N. Cauchy's root bound would then give a uniform constant. The available quadratic relation involving W_N and the resultant has coefficients growing with N. **Outcome.** This yields a concrete sufficient algebraic certificate. Finite algebraicity separately at each cutoff does not establish uniform boundedness.
47. Countable circle-domain uniformization Source *Access:* The supplied Annals record and this full primary-paper copy were opened; Theorem 0.1 and the proof architecture were read.	**Mechanism.** He–Schramm combine fixed-point indices, normal families and controlled limiting maps to uniformize domains with countably many boundary components by circle domains. **Transfer.** A conformal change can round boundaries, but the prime-circle weighted one-dimensional energy is not automatically conformally invariant. Transfer requires bounded pullback and inverse-pullback operators, uniform throughout the exhaustion, together with a uniform energy bound on the transformed side. **Outcome.** Existence of the circle-domain map supplies geometric organization, not the required norm estimate. No uniform energy budget was established.
48. Nitsche's harmonic annulus theorem Source *Access:* The supplied primary arXiv record and full manuscript were opened. The endpoint-inclusive radius statement was checked.	**Mechanism.** Harmonic homeomorphisms between circular annuli obey the sharp target-radius restriction expressed by the hyperbolic cosine of the source modulus. **Transfer.** An annular estimate could control the existing Hardy representation only after an appropriate harmonic homeomorphism and uniform boundary data are constructed. A sufficient conclusion would be a uniform bound on all radial boundary square norms of the generating function. **Outcome.** Neither the required homeomorphism nor its norm budget is known. The radius inequality alone does not bound the prime-circle energy.
49. Poincaré–Birkhoff annulus twist Source *Access:* The supplied AMS PDF returned 403. The matching original-paper record and indexed text were inspected; the full PDF was not freshly read.	**Mechanism.** Area preservation together with opposite rotations of the two boundary components forces fixed points. These hypotheses concern a single annulus transformation. **Transfer.** The prime construction instead follows a nonautonomous sequence of rotations. To control its entire orbit, one could seek a bounded nonnegative potential whose decrease dominates each original deficit increment. Fixed-point existence for an auxiliary map would not provide that inequality. **Outcome.** No autonomous twist representation or decreasing potential was obtained. An auxiliary periodic orbit would not bound the full prime sequence.
50. Alexandrov reflection and moving planes Source *Access:* The supplied Magnanini article was opened. It explicitly describes itself as a survey without rigorous proofs; its hypotheses and method description were checked.	**Mechanism.** Constant mean curvature, embeddedness and a strong comparison principle force spherical symmetry. The differential equation supplies the decisive rigidity, not visual roundness. **Transfer.** Seek an equation or defect identity with a uniform coercivity constant: its residual must dominate the prime variance and remain uniformly bounded. The prime phase has jumps and has not been shown to solve the relevant elliptic equation. **Outcome.** A precise coercive certificate is identified but unproved. Zeta's functional-equation symmetry alone allows paired zeros away from the critical line.

Continued on next page

Method and source	Mechanism, attempted transfer, and outcome
51. Lawson conjecture and the Clifford torus Source *Access:* The supplied primary record and full paper were opened; the main theorem and explicit embeddedness qualification were read.	**Mechanism.** Brendle's two-point maximum-principle argument exploits an embedded minimal torus in the three-sphere. Embeddedness and the minimal-surface equation are essential inputs. **Transfer.** The prime analogue starts from pairwise chord distances. A valid maximum-principle argument needs a discrete elliptic operator, uniformly controlled boundary terms and a bounded forcing term that dominates those distances. A positive pair kernel by itself is insufficient. **Outcome.** The two-point formulation transfers, but the required differential inequality has not been established. No uniform D_N bound follows.
52. The double bubble theorem Source *Access:* The supplied primary Annals record and its exact minimization statement were retrieved; no complete new proof audit was attempted.	**Mechanism.** A sharp constrained minimization theorem identifies the least-area enclosure and separation of two regions of prescribed volumes. **Transfer.** To use this architecture, construct a variational problem whose minimizer is the actual prime configuration, whose excess energy dominates a fixed positive multiple of D_N, and whose explicit competitor has uniformly bounded excess. All three requirements must hold with constants independent of cutoff. **Outcome.** No such minimizer identification or competitor budget was obtained. A geometric lower bound alone does not give an arithmetic upper bound.
53. The honeycomb theorem Source *Access:* The supplied primary Hales record was opened, including the qualification permitting disconnected cells and gaps.	**Mechanism.** Local geometric inequalities and global accounting establish a perimeter-minimizing equal-area partition. The proof controls contributions across an entire partition rather than checking isolated cells. **Transfer.** Partition the prime range into blocks and apply the exact variance merger formula. A sufficient certificate needs summable local defects, including every interblock mean displacement. A positive fixed allowance per block would accumulate without bound over infinitely many blocks. **Outcome.** The method gives a valid design for a block certificate. The necessary summable arithmetic budget has not been proved.
54. The Kepler packing theorem Source *Access:* The supplied Annals record and full original paper were opened. The table records the proof architecture, not an independent revalidation of all computations.	**Mechanism.** Geometric reduction and rigorously checked local inequalities cover every admissible packing configuration, converting an infinite packing problem into exhaustive controlled cases. **Transfer.** A comparable prime argument needs a uniform finite description of every admissible phase-block configuration and a summable error or telescoping potential across scales. Merely verifying many prime cutoffs does not demonstrate that all possible later blocks lie in the checked cases. **Outcome.** Finite certification remains viable in principle. The exhaustive arithmetic reduction needed for the original D_N is presently missing.
55. The dodecahedral volume theorem Source *Access:* The supplied DOI returned 403; the matching primary Hales–McLaughlin arXiv manuscript record was retrieved.	**Mechanism.** Voronoi cells in a separated sphere packing satisfy a uniform geometric lower-volume bound. Local cells connect the packing constraints to a global density estimate. **Transfer.** The original prime weights have an exact Voronoi meaning: the cell between neighboring prime midpoints has length w_n divided by two. A unit-speed polygon with direction U_n on that cell has D_N equal to twice its length excess over endpoint distance. **Outcome.** This gives an exact realization of the original target. A uniform excess-length estimate is still required.

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Method and source	Mechanism, attempted transfer, and outcome
56. The strong dodecahedral surface theorem Source *Access:* The supplied paper was opened; it explicitly sketches results and points elsewhere for complete proofs.	**Mechanism.** A sharper local surface-area inequality controls bounded Voronoi cells in a separated sphere packing. The cited construction uses computer-assisted geometric estimates. **Transfer.** Seek local surface or bending defects controlling the excess length of each prime block. The transfer needs both a uniform lower coercivity constant and a finite total sum of these defects. Sharpening each local positive inequality does not itself make the infinite sum converge. **Outcome.** A natural phase interpolation has finite bending energy, proved separately, but this supplies no uniform displacement-energy bound on an unbounded interval.
57. The four-dimensional kissing number Source *Access:* The supplied primary Annals record and full Musin paper were opened.	**Mechanism.** Musin modifies Delsarte's method, combining positive polynomial kernels with angular separation constraints to bound a spherical code and prove the kissing number is twenty-four. **Transfer.** Positive-definite circle kernels yield positive combinations of harmonic variances. A proved comparison shows these retain the original quadratic criterion. Closure requires their weighted arithmetic sum to be bounded above by a constant; nonnegativity alone is not sufficient. **Outcome.** The harmonic-kernel comparison theorem is proved. Packing separation is unavailable unchanged because the prime directions converge and cluster.
58. Eight-dimensional sphere packing Source *Access:* The primary paper was opened; the auxiliary-function sign conditions, normalization and Poisson argument were inspected.	**Mechanism.** Viazovska constructs an auxiliary function with nonnegative Fourier transform and a spatial sign restriction. Poisson summation and packing separation turn these into matching bounds. **Transfer.** An arithmetic auxiliary function needs both a positivity identity and a uniform signed-sum budget for the actual prime configuration. The logarithms of neighboring primes have arbitrarily small spacing, so the spatial exclusion hypothesis cannot be imported unchanged. **Outcome.** The auxiliary-function architecture yields exact pair-moment formulas and finite bounds for D_N. No uniformly bounded quadratic certificate has been found.
59. Twenty-four-dimensional sphere packing Source *Access:* The supplied primary record and full paper were opened.	**Mechanism.** An optimal interpolating auxiliary function matches the Leech lattice's structure and saturates the linear-programming bound. Its prescribed zeros are tied to the problem's geometry. **Transfer.** Cancel successive low-order phase terms using harmonic interpolation. This produces the nonnegative fourth pair moment, exactly 56 V1 minus 28 V2 plus 8 V3 minus V4. The admitted zero-free input below seven eighths proves this combination bounded. **Outcome.** Under (H_b), the constant-free four-harmonic statistic is bounded with the original weights. Its finite Holder bridge bounds D_N by order W_N to the three-quarters power, not by a constant.
60. Poncelet closure by elliptic-curve translation Source *Access:* The supplied Griffiths–Harris paper was read through its planar Poncelet argument and the explicit translation formula.	**Mechanism.** Two involutions lift to a fixed translation on an elliptic curve. Polygon closure becomes a torsion condition independent of the starting point. **Transfer.** A direct transfer would require a homeomorphic conjugacy between the centered prime sequence and a fixed translation orbit. A nonconstant convergent orbit cannot arise from a nonzero translation on a compact group, whereas the centered prime directions converge and are nonconstant. **Outcome.** This specific fixed-translation construction is obstructed. Nonautonomous or noninjective alternatives remain possible, but none presently supplies the required energy bound.

118.4 Disks, spheres, tori, and topology: 61–80

Method and source	Mechanism, attempted transfer, and outcome
61. Lovasz and the Kneser conjecture Source *Access:* The original journal abstract was verified through primary-source search. Direct opening returned 403; the complete paper was not inspected.	**Mechanism.** Connectivity and equivariant topology of a graph's neighborhood complex obstruct colorings and give the required chromatic lower bound. **Transfer.** One would need a graph construction whose topological index controls the arithmetic weighted error, with comparison constants independent of the number of primes. **Outcome.** No such encoding has been established. A coloring obstruction records combinatorial structure, whereas the target requires an upper bound on the size of a specific arithmetic vector.
62. Ham-sandwich bisection Source *Access:* The supplied DOI could not be opened. The 1942 bibliography was verified, and a primary 2006 generalization was checked; the original proof was not inspected.	**Mechanism.** An odd continuous map on a sphere and the Borsuk-Ulam theorem produce a hyperplane bisecting finitely many measures simultaneously. **Transfer.** Moment balancing would have to imply a uniform bound for the original positive-weight squared error, rather than merely make selected signed integrals vanish. **Outcome.** Finite balancing is available in an appropriate nonatomic model, but it does not control total error energy. The explicit Hobby-Rice application below divides the deficit equally and preserves its sum.
63. Hobby-Rice simultaneous cancellation Source *Access:* JSTOR supplied no article body. Pinkus's author-hosted three-page PDF was read instead; fractional splitting of original weights is explicitly allowed.	**Mechanism.** For finitely many real integrable functions on a nonatomic interval, a sign function with a controlled number of changes annihilates every prescribed integral. **Transfer.** The cancellation must reduce the original deficit or furnish an independent uniform energy estimate; cancellation alone is insufficient. **Outcome.** Apply the theorem to mass and the real and imaginary phase components. At most three cuts give two parts with half the mass and half the complex moment, hence exactly half the deficit each. Total deficit is conserved.
64. Necklace splitting Source *Access:* The supplied scan was accessible without searchable text. The exact cut-count statement was verified from Alon's original journal abstract.	**Mechanism.** Equivariant topology gives an equitable division of a necklace with t colors among k recipients using at most (k-1)t cuts. **Transfer.** Prime phases would need a controlled conversion to color measures, followed by a summable estimate for the arithmetic energy across all scales. **Outcome.** Equitable subdivision can balance finitely many quantities. Recombining the parts restores the original mass and complex moment, and therefore the original deficit. The cut-count theorem supplies no global upper bound.
65. Fundamental theorem of algebra through winding Source *Access:* Gowers's Cambridge exposition was opened and read. It is an explanatory proof sketch, not an original research paper on this arithmetic problem.	**Mechanism.** A nonzero winding number of a polynomial's image of a large boundary circle forces a zero inside by homotopy invariance. **Transfer.** A zero-exclusion argument would require certified boundary control on every relevant region, not only the existence of a winding-number calculation. **Outcome.** The Mellin transform has nontrivial logarithmic monodromy around a hypothetical off-line zero. Winding detects this obstruction but does not rule it out. The monodromy also survives finite ramified changes of variable.
66. Brouwer fixed point and no retraction Source *Access:* Direct opening failed. Indexed text from the same official Duke teaching note supplied the theorem and no-retraction argument.	**Mechanism.** The impossibility of retracting a closed ball onto its boundary forces a continuous self-map of that ball to have a fixed point. **Transfer.** An arithmetic fixed-point construction would need an invariant ball or coercive estimate whose radius does not increase with the number of primes. **Outcome.** Finite-prefix minimizers and compact fixed-point problems can be constructed, but existence does not bound their objective values. Establishing an invariant ball of uniform radius would require the missing arithmetic estimate.

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Method and source	Mechanism, attempted transfer, and outcome
67. Nash equilibrium existence Source *Access:* The PDF of Nash's original 1951 paper was opened, including its finite-game definitions and equilibrium-existence statement.	**Mechanism.** A fixed-point construction on a finite product of mixed-strategy simplices establishes equilibrium for every finite noncooperative game. **Transfer.** The proposed game would need payoffs and equilibrium values that quantitatively bound the prime-error norm with constants uniform across finite prefixes. **Outcome.** Finite optimization or equilibrium existence supplies no bound independent of game size or payoff scale. Without an additional arithmetic comparison, it does not close the required uniform deficit estimate.
68. Riemann mapping and Dirichlet's principle Source *Access:* The open-access Greene-Kim research article was read, including its rigorous account of Riemann's Dirichlet-problem approach.	**Mechanism.** Solving an appropriate Dirichlet problem, or using normal families, constructs a conformal equivalence between a proper simply connected plane domain and the disk. **Transfer.** The arithmetic boundary trace must be retained and its Dirichlet energy bounded uniformly; existence of a conformal coordinate alone is insufficient. **Outcome.** The Laguerre coefficients transfer exactly to disk Dirichlet energy and boundary H^(1/2) energy. Conformal changes of domain preserve this energy, so uniformization does not reduce the missing upper bound.
69. Circle-packing realization Source *Access:* Rodin-Sullivan's original paper was opened. It states packing realization as background; the earlier original realization proof was not separately inspected.	**Mechanism.** A triangulated sphere admits a realization by tangent circles, with the normalized configuration determined up to the appropriate Mobius freedom. **Transfer.** Arithmetic gaps and sampling weights must become quantitatively controlled radii or conductances, and the corresponding geometric energy must dominate the original deficit. **Outcome.** A packing realizes incidence information. It does not automatically preserve the arithmetic weights or provide an energy bound independent of the number of vertices. Those additional comparisons remain unproved.
70. Rodin-Sullivan convergence of circle packings Source *Access:* The original paper's proof structure and compact convergence were checked. It explicitly notes that its argument gives no definite convergence rate.	**Mechanism.** Ring bounds, length-area estimates, approximate rigidity, and compactness show that refined circle-packing maps converge to the normalized conformal map. **Transfer.** Arithmetic functions transported through the refinements would need uniform energy or boundary-tail control, beyond convergence on compact subsets. **Outcome.** Compact-open convergence alone cannot bound Dirichlet energy: the polynomials z to the power m converge locally to zero while their normalized Dirichlet energies equal m. The missing uniform energy estimate remains explicit.
71. Planar and sphere-packing separators Source *Access:* The original Miller-Teng-Thurston-Vavasis paper was opened and its packing and separator statements were checked.	**Mechanism.** Packing geometry produces a sphere meeting few balls while separating the remaining system into balanced parts, yielding graph-separator theorems. **Transfer.** Small separator cardinality would have to control weighted arithmetic interactions and cumulative error across every scale, with constants independent of the prefix. **Outcome.** The ordered prime samples already form a path, which has a one-vertex separator. That combinatorial fact imposes no bound on cumulative weighted variance. Planarity alone does not supply the desired arithmetic upper estimate.

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Method and source	Mechanism, attempted transfer, and outcome
72. Fenchel total-curvature inequality Source *Access:* The supplied MIT document is a two-page teaching summary, not Milnor's original paper or Fenchel's original proof. That distinction is retained.	**Mechanism.** The tangent indicatrix or directional-extrema argument gives a lower bound of two pi for the total curvature of a closed space curve. **Transfer.** One would need an arithmetic curve and a quantitative comparison converting curvature information into an upper bound for the weighted deficit. **Outcome.** The supporting unit circle has fixed total curvature while its sampling mass can be arbitrarily rescaled. The resulting deficit can grow without changing the curve. Geometry that forgets the weights cannot provide the required bound.
73. Fary-Milnor knot-curvature inequality Source *Access:* The supplied two-page MIT teaching summary was read. It summarizes Milnor's argument; the original 1950 research paper was not inspected in full.	**Mechanism.** A nontrivial knot requires additional directional extrema, forcing total curvature strictly greater than four pi. **Transfer.** The prime-error data would need a canonical nontrivial spatial knot together with a quantitative inequality relating knot geometry to the original sampling weights. **Outcome.** The current phase data are planar. An arbitrary lift into three dimensions introduces knotting that is not determined by the arithmetic deficit. The curvature lower bound therefore supplies no upper bound for that deficit.
74. Rectangles inscribed in smooth Jordan curves Source *Access:* The original publisher abstract confirmed both the theorem and the Lagrangian Klein-bottle mechanism. The full proof was not opened.	**Mechanism.** A symplectic obstruction to a smooth Lagrangian Klein-bottle embedding forces a rectangle of each prescribed aspect ratio on every smooth Jordan curve. **Transfer.** A smooth Jordan completion of the prime data would need to preserve their weights and make the resulting rectangle constraint control arithmetic energy quantitatively. **Outcome.** The existence conclusion is shared by all smooth Jordan curves. A chosen geometric completion may satisfy it without constraining the weighted prime error, so it gives no uniform deficit estimate.
75. Willmore conjecture and min-max theory Source *Access:* The original Annals abstract verified the theorem and method. The full hundred-page proof was not inspected.	**Mechanism.** Min-max theory of minimal surfaces, combined with conformal geometry and topology, proves the sharp Willmore-energy lower bound for immersed tori. **Transfer.** An arithmetic surface and a quantitative energy comparison must convert the geometric inequality into an upper bound for the prime-circle deficit. **Outcome.** No such surface or comparison has been supplied. A lower bound for curvature energy is a different assertion from the required upper bound on weighted arithmetic error. The shared use of an energy functional does not establish a transfer.
76. Mobius-invariant knot energy Source *Access:* The original arXiv abstract was checked for invariance and minimizer statements. The complete proof was not inspected.	**Mechanism.** Conformal invariance and renormalized self-repulsion characterize round-circle minimizers and impose quantitative restrictions on knot types. **Transfer.** The geometric construction must retain the prime sampling weights and provide a bound on the arithmetic deficit from the knot energy. **Outcome.** An unparametrized curve invariant ignores rescaling of its sampling measure. On the same circle, uniform mass L gives zero complex moment and deficit L, although the geometric image is unchanged. Weight control is indispensable.
77. Min-max theory and link energy Source *Access:* The original arXiv PDF was opened and its Main Theorem checked, including the nonsplit hypothesis and sharp two-pi-squared constant.	**Mechanism.** Linking constraints and min-max minimal-surface theory establish the sharp cross-energy lower bound, attained by the conformal Hopf-link model. **Transfer.** An arithmetic two-component link must be constructed, and a comparison must turn its geometric energy information into an upper bound for the weighted prime error. **Outcome.** The theorem concerns nonsplit two-component links and supplies a different lower bound. Without the missing arithmetic construction and comparison, it does not bound the prime-circle deficit.

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Method and source	Mechanism, attempted transfer, and outcome
78. Ricci flow with surgery and finite extinction Source *Access:* Perelman's original preprint abstract was checked. This is not a full inspection of all Ricci-surgery papers, and the tested heat flow is not the Xi deformation.	**Mechanism.** Nonlinear geometric estimates, controlled surgery, and a minimal-disk argument give finite extinction for the specified class of three-manifolds. **Transfer.** One would need an arithmetic metric and flow whose estimates retain the prime information through surgery and control the original deficit uniformly. **Outcome.** No such nonlinear identification is established. A separate rigorous linear test shows that every fixed positive time of Gaussian heat smoothing of the logarithmic prime error preserves RH equivalence and the exact energy-growth exponent.
79. Smale's sphere eversion Source *Access:* Smale's original 1959 paper was opened and its classification and three-dimensional corollary were checked.	**Mechanism.** Regular-homotopy classification through Stiefel-manifold obstruction groups shows that immersions of the two-sphere in three-space can be deformed into one another. **Transfer.** An arithmetic realization would need to preserve the sampling information and provide quantitative energy bounds throughout the regular homotopy. **Outcome.** Existence of a deformation, even one reversing the sphere, does not bound its geometric energy or the arithmetic variance. Self-intersections are permitted, and no uniform deficit estimate follows from the classification alone.
80. Inverse Galois theory over C(t) and branched covers Source *Access:* Direct opening returned 403. Indexed primary text from the same author-hosted document was recovered; the complete long paper was not inspected.	**Mechanism.** Fundamental-group monodromy, Riemann existence, and patching construct finite branched covers realizing prescribed finite Galois groups. **Transfer.** A proposed covering must remove the offending arithmetic singularity or furnish a uniform norm estimate on the original domain. **Outcome.** A logarithmic branch pulls back under finite ramification to a nonzero integer multiple of a logarithm. Its infinite additive monodromy survives. Finite covers therefore cannot remove this particular obstruction; universal covering does not itself restore the required energy bound.

119 Explicit error bounds for finite coefficient calculations

The following bounds explain how one could turn finite coefficient calculations into rigorous intervals. They are not a claim that the decimals in Section 117 were computed by interval arithmetic.

119.1 Direct prime tails

For real $s_0 > 1$ and integer $j \geq 0$,

$$P^{(j)}(s_0) = (-1)^j \sum_p (\log p)^j p^{-s_0}.$$

If the integer $N \geq e^{j/s_0}$, the summand as a function of its real argument is decreasing after N. Thus

$$\sum_{p>N} \frac{(\log p)^j}{p^{s_0}} \leq \frac{\Gamma(j+1, (s_0-1)\log N)}{(s_0-1)^{j+1}}. \tag{119.1}$$

For integer j, the bound is

$$N^{1-s_0} \sum_{r=0}^{j} \frac{j!}{(j-r)!} \frac{(\log N)^{j-r}}{(s_0-1)^{r+1}}.$$

The proof is the integral test after bounding the prime tail by the integer tail. Equations (9.7) and (9.6) propagate these bounds to any fixed coefficient, provided the elementary functions and $B_{\mathbb{P}}$ are also enclosed rigorously.

119.2 A geometric tail through Möbius inversion

For $\Re s \geq \sigma_0 > 1$, truncate (3.2) after K. Its tail $\mathcal{T}_K(s)$ satisfies

$$|\mathcal{T}_K(s)| \leq \frac{1}{K+1}\left(1+\frac{2}{(K+1)\sigma_0-1}\right)\frac{2^{-(K+1)\sigma_0}}{1-2^{-\sigma_0}}. \tag{119.2}$$

Indeed,

$$|\log\zeta(ks)| \leq \log\zeta(k\sigma_0) \leq \zeta(k\sigma_0)-1 \leq 2^{-k\sigma_0}\left(1+\frac{2}{k\sigma_0-1}\right),$$

and the remaining sum is geometric after replacing the decreasing prefactors by their first values.

Choose

$$0<r<\frac{\alpha-\frac{1}{2}}{\alpha+\frac{1}{2}}, \qquad \sigma_r=\frac{1}{2}+\alpha\frac{1-r}{1+r}>1.$$

Let ε_K denote the right side of (119.2) with $\sigma_0=\sigma_r$. Cauchy's coefficient estimate gives

$$|\Delta a_n^{(\alpha)}| \leq \frac{\sqrt{2\alpha}}{(1-r)(\sigma_r-1)}r^{-n}\varepsilon_K. \tag{119.3}$$

This bounds only the prime-zeta truncation error. The centering error from $B_{\mathbb{P}}$, finite precision in ζ, and the numerical extraction of coefficients need their own bounds. Theorem 9.3 explains why the centering error cannot be ignored at high order.

120 Embedded reproduction code

The following Python program uses NumPy and mpmath. Its default part reproduces the finite energy and deficit experiments. The optional coefficient routine is slower and should be run with increasing precision as an empirical check; it does not implement the interval certification of Appendix 119. No random inputs are used.

```
import math
import numpy as np

LD = np.longdouble
CD = np.clongdouble
PI = LD("3.14159265358979323846264338327950288419716939937751")
B = LD("0.26149721284764278375542683860869585905156664826l2")
CUTS = [10, 100, 1000, 10000, 100000, 1000000]

def sieve(limit):
    a = np.ones(limit + 1, dtype=bool)
    a[:2] = False
    for p in range(2, math.isqrt(limit) + 1):
        if a[p]:
            a[p*p::p] = False
    return np.flatnonzero(a)

# Bertrand's postulate ensures enough primes for the next endpoint.
primes = sieve(2 * max(CUTS) + 64)
p = primes.astype(LD)
S = np.cumsum(1 / p, dtype=LD)

def energies(X, order):
    n = np.searchsorted(primes, X, side="left")
    left = p[:n]
    right = np.minimum(p[1:n+1], LD(X))
    nodes, weights = np.polynomial.legendre.leggauss(order)
    nodes, weights = nodes.astype(LD), weights.astype(LD)
    half = (right - left) / 2
    mid = (right + left) / 2
    x = mid[:, None] + half[:, None] * nodes[None, :]
    e = S[:n, None] - np.log(np.log(x)) - B
    jac = half[:, None] * weights[None, :]
    phase = 2 * PI * e
    scalar = np.sum(jac * e**2, dtype=LD)
    circle = np.sum(jac * 4*np.sin(PI*e)**2, dtype=LD)
    ellipse = np.sum(
        jac * (4*(np.cos(phase)-1)**2 + LD(".25")*np.sin(phase)**2),
        dtype=LD
    )
    return np.array([scalar, circle, ellipse], dtype=LD)

print("Continuous energies: X, scalar, circle, ellipse")
max_difference = LD(0)
for X in CUTS:
    a, b = energies(X, 16), energies(X, 32)
    max_difference = max(max_difference, np.max(np.abs(a-b)))
    print(X, *[f"{v:.12f}" for v in b])
print("Largest order-16/order-32 difference:", max_difference)

# U, D, and V use no B. Only the comparison J uses B.
checkpoints = {
    np.searchsorted(primes, X, side="right") - 1: X for X in CUTS
}
last = max(checkpoints)
W = LD(0)
M = CD(0)
D = LD(0)
V = LD(0)
mean = CD(0)
J = LD(0)
print("Deficit: X, p_N, W, D, V, fixed-center J")
for i in range(1, last + 1):  # i=1 corresponds to p_2=3
    w = p[i+1] - p[i-1]
    phase = 2*PI*(S[i] - np.log(np.log(p[i])))
```

```
    u = CD(np.cos(phase) + 1j*np.sin(phase))
    r = abs(M)
    newM = M + w*u
    if r > 0:
        D += w*r*abs(u-M/r)**2 / (r+w+abs(newM))
    if W > 0:
        V += W*w/(W+w) * abs(u-mean)**2
    mean += w/(W+w) * (u-mean)
    M = newM
    W += w
    e = S[i] - np.log(np.log(p[i])) - B
    J += w*4*np.sin(PI*e)**2
    if i in checkpoints:
        print(checkpoints[i], int(primes[i]), int(W),
              f"{D:.12f}", f"{V:.12f}", f"{J:.12f}")
        print("  subtraction check:", float((W-abs(M))-D))
        print("  variance check:", float(V-D*(1+abs(M)/W)))

def laguerre_coefficients(nmax=6, dps=70, alpha="1"):
    import mpmath as mp
    mp.mp.dps = dps
    aa = mp.mpf(alpha)
    if aa <= mp.mpf(".5"):
        raise ValueError("alpha must exceed 1/2")
    # Increase this constant's precision for larger n or larger dps.
    bb = mp.mpf(
        "0.261497212847642783755426838608695859051566648261199206"
        "192064213924924510897368209714142631434246651051617728876"
    )
    ell = mp.log(2)
    cc = mp.log(ell) + bb
    def F(s):
        return (mp.primezeta(s) - cc*mp.power(2, 1-s)
                - mp.e1((s-1)*ell)) / (s-1)
    # taylor returns F^(k)(s0)/k!, so no extra factorial is needed.
    deriv = mp.taylor(F, mp.mpf(".5")+aa, nmax)
    values = [
        mp.sqrt(2*aa) * sum(
            mp.binomial(n, k)*(2*aa)**k*deriv[k]
            for k in range(n+1)
        ) for n in range(nmax+1)
    ]
    for n, value in enumerate(values):
        print(n, mp.nstr(value, min(40, dps-15)))
    print("Squared partial sum:",
          mp.nstr(sum(v*v for v in values), min(40, dps-15)))
    return values

# Optional, slower transform calculation:
# laguerre_coefficients()
```

120.1 Interval certificate for the rectangular Epstein crossing

This executable Python certificate uses `mpmath.iv`. The signed polynomial bounds and positive series-tail bound are proved in Section 14. All numerical comparisons below are interval comparisons. The printed enclosures are asserted before printing, so their deliberately rounded endpoints remain rigorous.

```
from mpmath import iv

iv.dps = 75
k0 = 4 * iv.pi * iv.exp(-iv.euler)


def K0_polynomial_bound(x, last):
    # Odd last: lower bound. Even last: upper bound.
    term = iv.mpf(1)
    poly = term
    for j in range(1, last + 1):
        term *= -iv.mpf((2*j - 1)**2) / (8*j*x)
        poly += term
    return iv.exp(-x) * iv.sqrt(iv.pi / (2*x)) * poly


def central_bounds(k):
    q = iv.exp(-2 * iv.pi * k)
    main = 2 * iv.ln(k / k0)

    lower = main + 8 * (
        K0_polynomial_bound(2 * iv.pi * k, 13)
        + 2 * K0_polynomial_bound(4 * iv.pi * k, 13)
    )

    # Bounds the entire positive tail with r >= 3.
    tail = q**3 * (3 - 2*q) / (
        2 * iv.sqrt(k) * (1 - q)**2
    )
    upper = main + 8 * (
        K0_polynomial_bound(2 * iv.pi * k, 12)
        + 2 * K0_polynomial_bound(4 * iv.pi * k, 12)
        + tail
    )
    return lower, upper


# Certify the constants used in global monotonicity/existence.
q1 = iv.exp(-2 * iv.pi)
B = (
    8 * iv.pi * iv.sqrt(iv.pi / (2 * (2 * iv.pi - 1)))
    * q1 * (1 + q1) / (1 - q1)**3
)
F1_upper = -2 * iv.log(k0) + 4 * q1 / (1 - q1)**2
assert B < iv.mpf("0.026")
assert F1_upper < iv.mpf("-3.9001")

c_small = iv.mpf("2.96037427750280e-19")
c_large = iv.mpf("2.96037427750281e-19")
lo_small, hi_small = central_bounds(k0 - c_small)
lo_large, hi_large = central_bounds(k0 - c_large)

# Certify the outward-rounded printed intervals.
assert lo_small > iv.mpf("1.5398e-34")
assert hi_small < iv.mpf("1.5459e-34")
assert lo_large > iv.mpf("-1.2949e-34")
```

```
assert hi_large < iv.mpf("-1.2888e-34")
assert lo_small > 0
assert hi_large < 0

print("B < 0.026; F(1) < -3.9001")
print("F(k0-c_small) in [1.5398, 1.5459] * 1e-34")
print("F(k0-c_large) in [-1.2949, -1.2888] * 1e-34")
print("2.96037427750280e-19 < k0-k_rect")
print("k0-k_rect < 2.96037427750281e-19")
```

120.2 Finite checks for the eighty-method extension

This standalone script uses only `mpmath` and Python's standard library. It checks algebraic formulas at finite prefixes; it does not certify an infinite bound or replace any proof in the main text.

```
"""Finite algebra checks for the eighty-method extension.

These are high-precision floating-point checks, not interval certificates
and not evidence for a uniform bound over all primes.
"""
import json
from pathlib import Path
import mpmath as mp

mp.mp.dps = 90

def first_primes(count):
    primes = []
    candidate = 2
    while len(primes) < count:
        if all(candidate % p for p in primes if p*p <= candidate):
            primes.append(candidate)
        candidate += 1
    return primes

primes = first_primes(129)
partial = mp.mpf(0)
points = []
weights = []
for index, p in enumerate(primes[:-1]):
    partial += mp.mpf(1)/p
    if index:
        points.append(mp.exp(2j*mp.pi*(partial-mp.log(mp.log(p)))))
        weights.append(primes[index+1]-primes[index-1])

rows = []
max_harmonic_error = mp.mpf(0)
max_ratio_error = mp.mpf(0)
for N in [8, 16, 32, 64, 128]:
    u, w = points[:N-1], weights[:N-1]
    W = mp.mpf(sum(w))
    moment = [mp.fsum(wi*ui**k for wi,ui in zip(w,u))
              for k in range(5)]
    variance = [W-abs(moment[k])**2/W for k in range(1,5)]
    D = W-abs(moment[1])
    direct = []
    for m in range(1,5):
        pair = mp.fsum(w[j]*w[k]*abs(u[j]-u[k])**(2*m)
                       for j in range(len(u)) for k in range(j+1,len(u)))/W
        harmonic = mp.fsum((-1)**(k+1)*mp.binomial(2*m,m-k)*variance[k-1]
                           for k in range(1,m+1))
        max_harmonic_error = max(max_harmonic_error,abs(pair-harmonic))
        direct.append(pair)
    gram = mp.matrix([[mp.fsum(wi*ui**(k-j) for wi,ui in zip(w,u))
                       for k in range(5)] for j in range(5)])
    G4 = gram[:4,:4]
    coefficients = mp.lu_solve(G4,-gram[:4,4])
    H4 = mp.fsum(wi*abs(ui**4+mp.fsum(coefficients[k]*ui**k for k in range(4)))**2
```

```
                for wi,ui in zip(w,u))
    ratio = mp.det(gram)/mp.det(G4)
    max_ratio_error = max(max_ratio_error,abs(H4-ratio))
    bound = (W/2)**mp.mpf('0.75')*direct[3]**mp.mpf('0.25')
    assert D <= variance[0] <= 2*D
    assert D <= bound
    assert all(x >= 0 for x in direct)
    rows.append({"N":N,"p_N":primes[N-1],"D":mp.nstr(D,14),
                 "C4":mp.nstr(direct[3],14),"H4":mp.nstr(H4,14),
                 "holder_bound":mp.nstr(bound,14)})

assert max_harmonic_error < mp.mpf('1e-80')
assert max_ratio_error < mp.mpf('1e-75')
assert all(mp.mpf(rows[i]['H4']) <= mp.mpf(rows[i+1]['H4'])
           for i in range(len(rows)-1))

nonmono = [mp.exp(1j*mp.pi/3),mp.exp(-1j*mp.pi/3),mp.mpf(1)]
c22 = abs(nonmono[0]-nonmono[1])**4/2
c23 = mp.fsum(abs(nonmono[j]-nonmono[k])**4
              for j in range(3) for k in range(j+1,3))/3
assert abs(c22-mp.mpf(9)/2) < mp.mpf('1e-85')
assert abs(c23-mp.mpf(11)/3) < mp.mpf('1e-85')
result = {"decimal_precision":mp.mp.dps,"rows":rows,
          "max_harmonic_identity_abs_error":mp.nstr(max_harmonic_error,8),
          "max_gram_ratio_abs_error":mp.nstr(max_ratio_error,8),
          "nonmonotone_C2":[mp.nstr(c22,12),mp.nstr(c23,12)]}
print(json.dumps(result,indent=2))
Path(__file__).with_suffix('.json').write_text(json.dumps(result,indent=2))
```

121 Reproducible interval certificate

The following complete Python program uses `mpmath`'s interval context. It reproduces the four enclosures, checks the strict rational lower bounds and the logarithmic comparisons, and writes the enclosing intervals to a JSON file. It certifies Theorem 42.1 only; it does not test an infinite set of large primes. No auxiliary mathematical data file is required.

```
"""Finite interval certificates, not a proof of the infinite target.

Uses mpmath interval arithmetic and analytic composite-trapezoid remainders.
Only logarithms and exact prime enumeration occur; no zeta-zero data or RH.
"""
import json
from math import isqrt, ceil, sqrt
from pathlib import Path
from mpmath import iv, mp

iv.dps = 50
mp.dps = 50
END = 200000
CHECKPOINTS = {3, 5, 23, END}
sieve = bytearray(b'\x01') * (END + 1)
sieve[0:2] = b'\x00\x00'
for p in range(2, isqrt(END) + 1):
    if sieve[p]:
        sieve[p*p:END+1:p] = b'\x00' * (((END-p*p)//p)+1)
primes = [p for p in range(2, END+1) if sieve[p]]
knots = sorted(set(primes + [END]))
prime_set = set(primes)
S = iv.mpf(0)
T1 = iv.mpf(0)
T2 = iv.mpf(0)
err1 = iv.mpf(0)
err2 = iv.mpf(0)
quarter = iv.mpf(1)/4
results = {}
panels = 0

def interval_text(v):
    return str(v)

for a,b in zip(knots, knots[1:]):
    if a in prime_set:
        S += iv.mpf(1)/a
    # The scale is only a performance choice; all errors are bounded below.
    count = max(1, ceil(128/sqrt(a)))
    step = iv.mpf(b-a)/count
    la = iv.log(iv.mpf(a))
    fa = S-quarter-iv.log(la)
    fb = S-quarter-iv.log(iv.log(iv.mpf(b)))
    M = max(abs(fa).b, abs(fb).b)
    F2 = (la+1)/(iv.mpf(a)**2*la**2)
    G2 = 2/(iv.mpf(a)**2*la**2) + 2*M*F2
    err1 += (iv.mpf(b-a)**3/(12*count**2)*F2).b
    err2 += (iv.mpf(b-a)**3/(12*count**2)*G2).b
    local1 = (fa+fb)/2
    local2 = (fa**2+fb**2)/2
    for j in range(1,count):
        t = iv.mpf(a)+iv.mpf((b-a)*j)/count
        f = S-quarter-iv.log(iv.log(t))
        local1 += f
        local2 += f**2
    T1 += step*local1
    T2 += step*local2
    panels += count
    if b in CHECKPOINTS:
        # [-err,err] formed using exact interval endpoint tuples.
        delta1 = iv.mpf([-err1.b, err1.b])
        delta2 = iv.mpf([-err2.b, err2.b])
        I1 = T1+delta1
        I2 = T2+delta2
        V = I2-I1**2/(b-2)
```

```
        results[str(b)] = {
            'V': interval_text(V),
            'integral_F': interval_text(I1),
            'integral_F_squared': interval_text(I2),
            'trapezoid_error_F': interval_text(err1),
            'trapezoid_error_F_squared': interval_text(err2),
            'panels': panels,
        }
        thresholds = {3: iv.mpf(1)/60, 5: iv.mpf(1)/25,
                      23: iv.mpf(17)/100, END: iv.mpf(37)/50}
        assert V.a > thresholds[b].b, (b,V,thresholds[b])
        print(b, V, 'panels', panels, flush=True)

# These overlapping ranges exclude every prime p_N <= 10^25.
# V(3)>1/60 covers p_N <= 3.
# V(5)>1/25 covers 3 <= p_N <= 19.
# V(23)>17/100 covers 19 <= p_N <= 200000.
# V(200000)>37/50 covers 200000 <= p_N <= 10^25.
assert (iv.log(3)/(8*iv.pi**2)).b < (iv.mpf(1)/60).a
assert (iv.log(19)/(8*iv.pi**2)).b < (iv.mpf(1)/25).a
assert (iv.log(200000)/(8*iv.pi**2)).b < (iv.mpf(17)/100).a
assert (25*iv.log(10)/(8*iv.pi**2)).b < (iv.mpf(37)/50).a
results['certificate'] = {
    'excluded_range': 'all indices N with p_N <= 10^25',
    'prime_count_through_200000':len(primes),
    'precision_decimal_digits':iv.dps,
    'algorithm':'interval composite trapezoid with rigorous second derivative bounds',
    'target_proved_infinitely_often':False,
}
dest = Path(__file__).with_name('variance_certificates.json')
dest.write_text(json.dumps(results,indent=2)+'\n')
print('CERTIFICATES PASSED',dest,flush=True)
```

122 Complete finite certificate

The following Python program uses gmpy2; it has no input data files and reconstructs the primes itself. It was independently rerun for this note at 256-bit precision. All assertions passed, including the whole-angle lower bound, the four comparison rows, and the limiting-point moment bounds. The final JSON labels the global energy bound as unproved. The proof relies on correct MPFR rounding and on the displayed implementation; it is a computer-assisted certificate, not a claim of formal proof-assistant verification.

The trigonometric interval routine combines a correctly rounded value at a binary midpoint with the global Lipschitz constant 1 for sine and cosine. The exact dyadic cover, together with a fifth-derivative bound, covers every angle rather than only finitely many samples. The final inequalities use Python exact rational arithmetic.

```
#!/usr/bin/env python3
"""Rigorous boundary-quartic certificate, conditional on H4,infty <= 800.

Arithmetic: MPFR at 256 bits, nearest rounding, followed by next_below / next_above
for every operation used as a bound.  MPFR log, sin, cos, sqrt and pi are correctly
rounded.  Trigonometric interval enclosures use their global Lipschitz constant 1.
A fifth-order Taylor remainder certifies a trigonometric polynomial on every
interval of an exact dyadic cover of [0,7], which contains [0,2*pi].

This does NOT certify the missing global hypothesis H4,infty <= 800.
"""
from __future__ import annotations

import json
import math
import time
from fractions import Fraction
from pathlib import Path

import gmpy2 as g
```

```
PREC = 256
N = 100000
g.get_context().precision = PREC
g.get_context().round = g.RoundToNearest
ZERO, ONE, TWO = g.mpfr(0), g.mpfr(1), g.mpfr(2)


def down(x):
    return g.next_below(x)


def up(x):
    return g.next_above(x)


class I:
    """Closed real interval with MPFR endpoints, each an exact binary rational."""
    __slots__ = ('lo', 'hi')

    def __init__(self, lo=0, hi=None):
        if not isinstance(lo, (int, g.mpfr, g.mpz)):
            raise TypeError('Only exact integers or already chosen binary MPFR points')
        if not isinstance(lo, g.mpfr):
            assert abs(int(lo)).bit_length() <= PREC
        self.lo = lo if isinstance(lo, g.mpfr) else g.mpfr(lo)
        if hi is None:
            self.hi = self.lo
        else:
            if not isinstance(hi, (int, g.mpfr, g.mpz)):
                raise TypeError('Invalid endpoint')
            if not isinstance(hi, g.mpfr):
                assert abs(int(hi)).bit_length() <= PREC
            self.hi = hi if isinstance(hi, g.mpfr) else g.mpfr(hi)
        assert self.lo <= self.hi

    @staticmethod
    def cast(x):
        return x if isinstance(x, I) else I(x)

    @staticmethod
    def rational(num, den=1):
        return I(num) / I(den)

    def iszero(self):
        return self.lo == 0 and self.hi == 0

    def __add__(self, other):
        b = I.cast(other)
        if b.iszero():
            return self
        if self.iszero():
            return b
        return I(down(self.lo + b.lo), up(self.hi + b.hi))

    __radd__ = __add__

    def __neg__(self):
        # Negation changes only the sign bit and is exact.
        return I(-self.hi, -self.lo)

    def __sub__(self, other):
        return self + (-I.cast(other))

    def __rsub__(self, other):
        return I.cast(other) - self

    def __mul__(self, other):
        b = I.cast(other)
        if self.iszero() or b.iszero():
            return I(0)
        products = (self.lo*b.lo, self.lo*b.hi,
                    self.hi*b.lo, self.hi*b.hi)
        return I(down(min(products)), up(max(products)))
```

```
    __rmul__ = __mul__

    def reciprocal(self):
        assert not self.lo <= 0 <= self.hi
        return I(down(ONE/self.hi), up(ONE/self.lo))

    def __truediv__(self, other):
        return self * I.cast(other).reciprocal()

    def __rtruediv__(self, other):
        return I.cast(other) / self

    def log(self):
        assert self.lo > 0
        return I(down(g.log(self.lo)), up(g.log(self.hi)))

    def square(self):
        maxabs = max(abs(self.lo), abs(self.hi))
        if self.lo <= 0 <= self.hi:
            lower = ZERO
        else:
            minabs = min(abs(self.lo), abs(self.hi))
            lower = down(minabs*minabs)
        return I(lower, up(maxabs*maxabs))

    def sqrt(self):
        assert self.lo >= 0
        lower = ZERO if self.lo == 0 else down(g.sqrt(self.lo))
        return I(lower, up(g.sqrt(self.hi)))

    def midpoint(self):
        # This is only a chosen binary point, not used as a bound.
        return (self.lo+self.hi)/TWO

    def sincos(self):
        mid = self.midpoint()
        # A computed midpoint is allowed anywhere; this max bounds its distance
        # from every point of the interval, with outward rounding.
        rad = up(max(abs(mid-self.lo), abs(self.hi-mid)))
        sm, cm = g.sin(mid), g.cos(mid)
        sval = I(down(down(sm)-rad), up(up(sm)+rad))
        cval = I(down(down(cm)-rad), up(up(cm)+rad))
        return sval, cval

    def abs_upper(self):
        return max(abs(self.lo), abs(self.hi))

    def contains_zero(self):
        return self.lo <= 0 <= self.hi


class CI:
    """Rectangular complex interval."""
    __slots__ = ('re', 'im')

    def __init__(self, re=0, im=0):
        self.re, self.im = I.cast(re), I.cast(im)

    @staticmethod
    def cast(z):
        return z if isinstance(z, CI) else CI(z)

    def __add__(self, other):
        b = CI.cast(other)
        return CI(self.re+b.re, self.im+b.im)

    __radd__ = __add__

    def __neg__(self):
        return CI(-self.re, -self.im)

    def __sub__(self, other):
        return self + (-CI.cast(other))
```

```
    def __rsub__(self, other):
        return CI.cast(other) - self

    def __mul__(self, other):
        if not isinstance(other, CI):
            real = I.cast(other)
            return CI(self.re*real, self.im*real)
        return CI(self.re*other.re-self.im*other.im,
                  self.re*other.im+self.im*other.re)

    __rmul__ = __mul__

    def conjugate(self):
        return CI(self.re, -self.im)

    def abs_interval(self):
        return (self.re.square()+self.im.square()).sqrt()

    def l1_upper(self):
        return up(self.re.abs_upper()+self.im.abs_upper())


def primes_count(k):
    bound = max(100, int(k*(math.log(k)+math.log(math.log(k))))+128)
    while True:
        s = bytearray(b'\x01')*(bound+1)
        s[:2] = b'\x00\x00'
        for p in range(2, math.isqrt(bound)+1):
            if s[p]:
                s[p*p::p] = b'\x00'*((bound-p*p)//p+1)
        out = [i for i in range(2, bound+1) if s[i]]
        if len(out) >= k:
            return out[:k]
        bound *= 2




def exact_binary(x):
    n, d = x.as_integer_ratio()
    return {'numerator': str(n), 'denominator': str(d)}


def rational_interval(q):
    return I.rational(q.numerator, q.denominator)


def mertens_interval(cap=360):
    # B_P = gamma + sum_{k>=2} mu(k) log(zeta(k))/k.
    # For k>=2, log(zeta(k)) <= zeta(k)-1 <= 3*2^(-k).
    # The absolute tail after cap is <2^(-cap) for cap>=3;
    # we pad by the looser exact binary bound 2^(2-cap).
    assert cap >= 3
    mu = [1]*(cap+1)
    for p in range(2,cap+1):
        if all(p % q for q in range(2,math.isqrt(p)+1)):
            for j in range(p,cap+1,p):
                mu[j] *= -1
            for j in range(p*p,cap+1,p*p):
                mu[j] = 0
    euler = g.const_euler()
    value = I(down(euler),up(euler))
    for k in range(2,cap+1):
        if mu[k]:
            zeta = g.zeta(g.mpfr(k))
            value = value+I(mu[k])*I(down(zeta),up(zeta)).log()/k
    tail = g.mpfr(2)**(2-cap)
    return value+I(-tail,tail)


def main():
    start = time.monotonic()
    primes = primes_count(N+1)
```

```
assert primes[N-1] == 1299709
pi = g.const_pi()
two_pi = I(2)*I(down(pi), up(pi))
assert two_pi.hi < 7
harmonic = I(0)
moments = [CI(0) for _ in range(5)]
initial_points = []
weight_sum = 0
for i, p in enumerate(primes[:-1]):
    harmonic = harmonic+I.rational(1, p)
    if i == 0:
        continue
    theta = two_pi*(harmonic-I(p).log().log())
    sin_i, cos_i = theta.sincos()
    u = CI(cos_i, sin_i)
    if len(initial_points) < 5:
        initial_points.append(u)
    w = primes[i+1]-primes[i-1]
    weight_sum += w
    power = CI(1)
    for j in range(1,5):
        power = power*u
        moments[j] = moments[j]+power*w
assert weight_sum == primes[N-1]+primes[N]-5 == 2599425
# Five disjoint rectangles certify five distinct circle points, hence
# positive degree-four norm and unique finite minimizers.
for i in range(5):
    for j in range(i):
        x,y=initial_points[i],initial_points[j]
        assert (x.re.lo>y.re.hi or y.re.lo>x.re.hi or
                x.im.lo>y.im.hi or y.im.lo>x.im.hi)
moments[0] = CI(weight_sum)
print('Certified prime moments:', round(time.monotonic()-start, 3),
      'seconds', flush=True)

# f(theta)=sum_n w_n |u_n-exp(i theta)|^8.
# Fourier coefficients are a[0]=70, a[1]=-56, a[2]=28,
# a[3]=-8, a[4]=1.  Since |C_k|<=W,
# |f^(5)| <= 2 W sum_{k=1}^4 |a[k]| k^5 = 7840 W.
a = (70,-56,28,-8,1)
assert 2*sum(abs(a[k])*k**5 for k in range(1,5)) == 7840
derivative5_bound = I(7840*weight_sum)
division = 256
h = I.rational(1, 2*division)
hp = [I(1)]
for j in range(1,6):
    hp.append(hp[-1]*h)
factorial = (1,1,2,6,24,120)
threshold = Fraction(1601,2)  # 800.5 exactly
threshold_interval = rational_interval(threshold)
minimum_lower = None
minimum_index = None

# Intervals [j/256,(j+1)/256], j=0,...,1791, cover [0,7].
# Exact binary centers avoid any grid coverage ambiguity.
for j in range(7*division):
    center = g.mpfr(2*j+1)/g.mpfr(2*division)
    assert center.as_integer_ratio() == g.mpq(2*j+1,2*division).as_integer_ratio()
    derivatives = [I(70*weight_sum)]+[I(0) for _ in range(4)]
    for k in range(1,5):
        s, c = (I(-k)*I(center)).sincos()
        term = moments[k]*CI(c,s)
        step = CI(0,-k)
        for order in range(5):
            derivatives[order] = derivatives[order]+(2*a[k])*term.re
            term = term*step
    lower = derivatives[0]
    for order in range(1,5):
        lower = lower-I(derivatives[order].abs_upper())*hp[order]/factorial[order]
    lower = lower-derivative5_bound*hp[5]/factorial[5]
    # Every actual f(theta) on this closed interval is >= lower.lo.
    assert lower.lo > threshold_interval.hi
    if minimum_lower is None or lower.lo < minimum_lower:
        minimum_lower, minimum_index = lower.lo, j
```

```
# For a monic quartic with all roots in the closed disk and
# epsilon=4-|sum roots|<=1, ||P-(z-b)^4|| <= 18 epsilon sqrt(W),
# where b is the direction of the sum.  Hence the following entirely
# rational check certifies the claimed coefficient gap under B4<=800.
B4 = Fraction(800)
epsilon = Fraction(1,10**7)
crossfree = threshold-B4-324*epsilon**2*weight_sum
rational_margin = crossfree**2-1296*epsilon**2*B4*weight_sum
assert crossfree > 0 and rational_margin > 0

comparisons = []
for bound,gap in [(174,Fraction(1,2000)),
                  (300,Fraction(3,10000)),
                  (500,Fraction(1,5000)),
                  (800,Fraction(3,10**7))]:
    left = threshold-bound-324*gap**2*weight_sum
    margin = left**2-1296*gap**2*bound*weight_sum
    assert 0<gap<=1 and left>0 and margin>0
    comparisons.append({'B4_unproved_global_hypothesis':str(bound),
                        'gap':str(gap),
                        'strict_rational_margin':str(margin)})

# Anchored limiting-point version.  These finite centered moments
# certify the separate implication H4,infty<=1000 => eventual gap10^-5.
# That implication has no explicit cutoff N0=100000.
bp = mertens_interval()
sa,ca=(two_pi*bp).sincos()
abar=CI(ca,-sa)
powers=[CI(1)]
for k in range(1,5):
    powers.append(powers[-1]*abar)
centered={}
for m in range(1,5):
    value=I(math.comb(2*m,m)*weight_sum)
    for k in range(1,m+1):
        value=value+(2*(-1)**k*math.comb(2*m,m+k))*(moments[k]*powers[k]).re
    centered[m]=value
assert centered[1].hi<75 and centered[2].hi<108
assert centered[3].hi<310 and centered[4].lo>1024
# For e<=10^-5: D^2<=8e-e^2<(9/1000)^2 and
# ||P-(z-a)^4|| <31D+11D^2+2D^3<7/25.
# Also sqrt(1000)+7/25<32, exactly checked by squaring.
anchored_gap=Fraction(1,10**5)
d=Fraction(9,1000)
assert 8*anchored_gap-anchored_gap**2<d*d
assert 31*d+11*d*d+2*d*d*d<Fraction(7,25)
assert (Fraction(32)-Fraction(7,25))**2>1000
ctx = g.get_context()
assert not ctx.overflow and not ctx.invalid and not ctx.divzero
result = {
    'scope': 'Finite boundary energy certified. Global coefficient conclusion CONDITIONAL on
 H4_infinity <= 800.',
    'conditional_conclusion': 'For every M >= 100000, abs(c3_M) < 4-1/10^7, provided H4_infinity <=
 800.',
    'N': N, 'p_N': primes[N-1], 'p_N_plus_1': primes[N],
    'W_N': weight_sum, 'atom_count': N-1,
    'precision_bits': PREC, 'gmpy2':g.version(), 'MPFR':g.mpfr_version(),
    'grid': {'interval_count':7*division,'range':'[0,7]',
             'width':'1/256','half_width':'1/512',
             'remainder_derivative_order':5,
             'derivative_bound':7840*weight_sum},
    'T_N_strict_lower_rational':str(threshold),
    'minimum_Taylor_lower_display':str(minimum_lower),
    'minimum_Taylor_lower_exact_binary':exact_binary(minimum_lower),
    'minimum_interval_index':minimum_index,
    'B4_unproved_global_hypothesis':str(B4),
    'epsilon':str(epsilon),
    'rational_crossfree':str(crossfree),
    'strict_rational_margin':str(rational_margin),
    'comparison_certificates':comparisons,
    'centered_moments':{str(m):{'lower_display':str(v.lo),
                                'upper_display':str(v.hi),
                                'lower_exact_binary':exact_binary(v.lo),
```

```
                                    'upper_exact_binary':exact_binary(v.hi)}
                                for m,v in centered.items()},
            'anchored_conditional_conclusion':
                'If H4_infinity<=1000, then abs(c3_M)<4-10^-5 eventually; cutoff not specified.',
            'all_checks_passed':True,
            'elapsed_seconds':round(time.monotonic()-start,3)
        }
        out=Path(__file__).with_suffix('.json')
        out.write_text(json.dumps(result,indent=2)+'\n')
        print('Boundary minimum >', threshold, 'certified; Taylor lower:', minimum_lower, flush=True)
        print('Exact positive final margin:',rational_margin,flush=True)
        print('Saved',out,'seconds',result['elapsed_seconds'],flush=True)


if __name__ == '__main__':
    main()
```

Part IX
Supporting proof dependencies and finite certificates

The following supporting sections are retained so that no cited certificate or proof dependency is external to the single source.

123 Baseline conclusions and their retained dependencies

This section records the baseline results and the dependencies of the earlier source stage. Table 1 and Section 131 give the additional results and the final consolidated status. The baseline constructions give a connected family of exact statements:

$$\begin{aligned}
\mathrm{RH} &\iff E \in L^2(2,\infty)\\
&\iff \text{finite circle or fixed-ellipse energy}\\
&\iff \sum_{n\geq 2} w_n E(p_n)^2 < \infty\\
&\iff \sup_N D_N < \infty\\
&\iff (a_n^{(\alpha)}) \in \ell^2\\
&\iff G_\alpha \in H^2(\mathbb{D})\\
&\iff \limsup_n |a_n^{(\alpha)}|^{1/n} = 1.
\end{aligned}$$

Beyond equivalence to RH, the deficit growth exponent recovers Θ exactly, and the full L^q, weighted-energy, contact-order, and shrinking-ellipse classifications are closed. The Taylor-radius formula and the centering instability are also proved, not heuristic. Fixed compact or half-line-supported logarithmic filters, and finite-time Gaussian smoothing, preserve both the RH equivalence and the exact energy exponent.

With the stated 2026 zero-free theorem as an external input, Section 91 closes the modified weighted deficit and the eighth-moment circle energy, and bounds the original deficit by $p_N^{0.749914\cdots}/(\log p_N)^2$. These statements concern a larger exponent than zero or altered weights/-contact order. They do not imply boundedness of the original deficit. Independently, Section 14 certifies a distinct Epstein threshold calculation; it does not replace the arithmetic estimate below.

The eighty-method audit also closes the following exact statements:

- Pair energies have exact finite harmonic formulas, extended-real limits, and rightmost-zero growth exponents. In particular the original weights support the bounded statistic $56V_N^{(1)} - 28V_N^{(2)} + 8V_N^{(3)} - V_N^{(4)}$, using the explicit fixed-strip hypothesis (H_b).
- The monic polynomial norms $H_{m,N}$ form a monotone hierarchy with the same even-moment thresholds. Degree four is bounded under (H_b); degree one is the original variance.
- The double supremum of the finite Szegő log determinants is exactly the Laguerre coefficient energy. An upper bound C would give the original explicit constant $K(C)$ in (16.10); no such uniform determinant bound is proved.
- Finite bending of a natural phase interpolation is unconditional. Finite covers retain the Mellin logarithmic branch; Hobby–Rice balancing splits a deficit exactly without reducing its total.

- Explicit comparison sequences retain the actual prime-gap weights, primorial torsion orders and zero heights, with every moment above two finite but the quadratic deficit divergent. A natural Frey curve retains new conductor primes. Both results specify missing transfer hypotheses, without refuting the actual prime-circle assertion.

The remaining task is to establish an unconditional bound strong enough to force one of these equivalent conditions. For example, any of the following would suffice:

(a) A uniform bound $D_N \le K$ for every N.

(b) For every $\varepsilon > 0$, a bound $D_N \ll_\varepsilon p_N^{\varepsilon}$ valid for all N.

(c) A uniform bound on $Q_m = \int_{2^m}^{2^{m+1}} E(x)^2 \, \mathrm{d}x$.

(d) For one fixed $\alpha > 1/2$, a proof that $\sum_n |a_n^{(\alpha)}|^2 < \infty$, or that its root limsup is 1.

(e) A common upper bound for $\log\det T_L(e^{L_M})$ over all finite M, L, as in Theorem 16.4.

(f) A positive-harmonic certificate as in Remark 15.12, valid for every prime prefix.

Condition (c) gives $I(X) = O(\log X)$, and Theorem 8.2 then forces $\Theta = 1/2$. Even $Q_m = O_\varepsilon(2^{\varepsilon m})$ for every $\varepsilon > 0$ would suffice by the same argument. No such unconditional bound is supplied by this note.

There is also a precise endpoint-approach criterion. For a real $q > 2$, put

$$\mathcal{P}_{q,N} = \frac{1}{W_N} \sum_{j<k} w_j w_k |U_j - U_k|^q.$$

The proof of Theorem 15.2 uses only $q \ge 1$, not its integrality; Proposition 15.4 also applies. Consequently, for the actual prime sequence,

$$\mathrm{RH} \quad \Longleftrightarrow \quad \text{there exists } q_j \downarrow 2 \text{ with } \sup_N \mathcal{P}_{q_j,N} < \infty \text{ for every } j.$$

The bounds need not be uniform in j: each one gives $\Theta \le 1 - 1/q_j$, and taking $j \to \infty$ gives $\Theta \le 1/2$. The converse follows from the moment criterion. This is an arithmetic implication using the Mellin obstruction. The comparison sequences in Theorem 18.4 do not satisfy the prime-error identities and therefore do not contradict it.

There is a precise obstacle to a simple geometric proof: a sum of nonnegative deficit increments need not remain bounded. The triangle inequality proves monotonicity, but supplies no finite total budget. Likewise, re-expressing a function in an orthogonal basis or mapping a line to a circle does not itself control its norm or its singularities. The unresolved arithmetic estimate survives every exact change of representation.

The constant-free deficit is a concrete candidate for further work because it has positive stable increments and exact variance identities. The weighted-energy threshold is useful for measuring how much a proposed estimate would actually prove. A prospective argument should therefore state its uniform bound explicitly, show that it does not assume the desired zero-free region, and keep the real-x measure or the corresponding prime-gap weights. Finite calculations can test an estimate, but cannot replace that last step.

124 Outward-rounded finite certificate

This standalone program requires `gmpy2` and `mpmath`. The verified run used `gmpy2 2.3.2` and MPFR 4.2.2. The program asserts (106.1) and the exact rational margin in Theorem 106.1; it

writes its full interval endpoints as exact binary fractions in JSON. Its `mpmath` calculations only propose candidate binary points. All claimed bounds are separately validated by MPFR intervals. The global hypothesis (106.2) is not an output of this program.

```
#!/usr/bin/env python3
"""Rigorous finite certificate for the c3 target, conditional on H4,infty <= 174.

Arithmetic: MPFR at 256 bits, nearest rounding, followed by next_below / next_above
for every operation used as a bound.  MPFR log, sin, cos, sqrt and pi are correctly
rounded.  Trigonometric interval enclosures use their global Lipschitz constant 1.
The mpmath solve ONLY proposes exact-binary centers and inverses; a rigorous
interval Neumann residual bound verifies every solution enclosure independently.

This does NOT certify the missing global hypothesis H4,infty <= 174.
"""
from __future__ import annotations

import json
import math
import time
from fractions import Fraction
from pathlib import Path

import gmpy2 as g
import mpmath as mp

PREC = 256
N = 100000
g.get_context().precision = PREC
g.get_context().round = g.RoundToNearest
mp.mp.dps = 105
ZERO, ONE, TWO = g.mpfr(0), g.mpfr(1), g.mpfr(2)


def down(x):
    return g.next_below(x)


def up(x):
    return g.next_above(x)


class I:
    """Closed real interval with MPFR endpoints, each an exact binary rational."""
    __slots__ = ('lo', 'hi')

    def __init__(self, lo=0, hi=None):
        if not isinstance(lo, (int, g.mpfr, g.mpz)):
            raise TypeError('Only exact integers or already chosen binary MPFR points')
        if not isinstance(lo, g.mpfr):
            assert abs(int(lo)).bit_length() <= PREC
        self.lo = lo if isinstance(lo, g.mpfr) else g.mpfr(lo)
        if hi is None:
            self.hi = self.lo
        else:
            if not isinstance(hi, (int, g.mpfr, g.mpz)):
                raise TypeError('Invalid endpoint')
            if not isinstance(hi, g.mpfr):
                assert abs(int(hi)).bit_length() <= PREC
            self.hi = hi if isinstance(hi, g.mpfr) else g.mpfr(hi)
        assert self.lo <= self.hi

    @staticmethod
    def cast(x):
        return x if isinstance(x, I) else I(x)

    @staticmethod
    def rational(num, den=1):
        return I(num) / I(den)

    def iszero(self):
        return self.lo == 0 and self.hi == 0
```

```
    def __add__(self, other):
        b = I.cast(other)
        if b.iszero():
            return self
        if self.iszero():
            return b
        return I(down(self.lo + b.lo), up(self.hi + b.hi))

    __radd__ = __add__

    def __neg__(self):
        # Negation changes only the sign bit and is exact.
        return I(-self.hi, -self.lo)

    def __sub__(self, other):
        return self + (-I.cast(other))

    def __rsub__(self, other):
        return I.cast(other) - self

    def __mul__(self, other):
        b = I.cast(other)
        if self.iszero() or b.iszero():
            return I(0)
        products = (self.lo*b.lo, self.lo*b.hi,
                    self.hi*b.lo, self.hi*b.hi)
        return I(down(min(products)), up(max(products)))

    __rmul__ = __mul__

    def reciprocal(self):
        assert not self.lo <= 0 <= self.hi
        return I(down(ONE/self.hi), up(ONE/self.lo))

    def __truediv__(self, other):
        return self * I.cast(other).reciprocal()

    def __rtruediv__(self, other):
        return I.cast(other) / self

    def log(self):
        assert self.lo > 0
        return I(down(g.log(self.lo)), up(g.log(self.hi)))

    def square(self):
        maxabs = max(abs(self.lo), abs(self.hi))
        if self.lo <= 0 <= self.hi:
            lower = ZERO
        else:
            minabs = min(abs(self.lo), abs(self.hi))
            lower = down(minabs*minabs)
        return I(lower, up(maxabs*maxabs))

    def sqrt(self):
        assert self.lo >= 0
        lower = ZERO if self.lo == 0 else down(g.sqrt(self.lo))
        return I(lower, up(g.sqrt(self.hi)))

    def midpoint(self):
        # This is only a chosen binary point, not used as a bound.
        return (self.lo+self.hi)/TWO

    def sincos(self):
        mid = self.midpoint()
        # A computed midpoint is allowed anywhere; this max bounds its distance
        # from every point of the interval, with outward rounding.
        rad = up(max(abs(mid-self.lo), abs(self.hi-mid)))
        sm, cm = g.sin(mid), g.cos(mid)
        sval = I(down(down(sm)-rad), up(up(sm)+rad))
        cval = I(down(down(cm)-rad), up(up(cm)+rad))
        return sval, cval

    def abs_upper(self):
        return max(abs(self.lo), abs(self.hi))
```

```
    def contains_zero(self):
        return self.lo <= 0 <= self.hi


class CI:
    """Rectangular complex interval."""
    __slots__ = ('re', 'im')

    def __init__(self, re=0, im=0):
        self.re, self.im = I.cast(re), I.cast(im)

    @staticmethod
    def cast(z):
        return z if isinstance(z, CI) else CI(z)

    def __add__(self, other):
        b = CI.cast(other)
        return CI(self.re+b.re, self.im+b.im)

    __radd__ = __add__

    def __neg__(self):
        return CI(-self.re, -self.im)

    def __sub__(self, other):
        return self + (-CI.cast(other))

    def __rsub__(self, other):
        return CI.cast(other) - self

    def __mul__(self, other):
        if not isinstance(other, CI):
            real = I.cast(other)
            return CI(self.re*real, self.im*real)
        return CI(self.re*other.re-self.im*other.im,
                  self.re*other.im+self.im*other.re)

    __rmul__ = __mul__

    def conjugate(self):
        return CI(self.re, -self.im)

    def abs_interval(self):
        return (self.re.square()+self.im.square()).sqrt()

    def l1_upper(self):
        return up(self.re.abs_upper()+self.im.abs_upper())


def primes_count(k):
    bound = max(100, int(k*(math.log(k)+math.log(math.log(k))))+128)
    while True:
        s = bytearray(b'\x01')*(bound+1)
        s[:2] = b'\x00\x00'
        for p in range(2, math.isqrt(bound)+1):
            if s[p]:
                s[p*p::p] = b'\x00'*((bound-p*p)//p+1)
        out = [i for i in range(2, bound+1) if s[i]]
        if len(out) >= k:
            return out[:k]
        bound *= 2


def exact_mp_to_mpmath(x):
    num, den = x.as_integer_ratio()
    return mp.mpf(int(num))/int(den)


def approximate_point(z):
    # This chooses an exact-binary point.  No accuracy assertion is attached
    # to the conversion; the interval residual will verify the choice.
    return CI(g.mpfr(mp.nstr(mp.re(z), 110)),
              g.mpfr(mp.nstr(mp.im(z), 110)))
```

```
def dot(a, b):
    out = CI(0)
    for x, y in zip(a, b):
        out = out+x*y
    return out


def matrix_product(A, B):
    cols = list(zip(*B))
    return [[dot(row, col) for col in cols] for row in A]


def matrix_norm_upper(A):
    rows = []
    for row in A:
        total = ZERO
        for z in row:
            total = up(total+z.l1_upper())
        rows.append(total)
    return max(rows)


def verified_solve(G, b):
    """Enclose solution of G*c=-b using an interval Neumann residual proof."""
    m = len(b)
    G0 = mp.matrix([[mp.mpc(exact_mp_to_mpmath(z.re.midpoint()),
                            exact_mp_to_mpmath(z.im.midpoint()))
                     for z in row] for row in G])
    b0 = mp.matrix([mp.mpc(exact_mp_to_mpmath(z.re.midpoint()),
                           exact_mp_to_mpmath(z.im.midpoint())) for z in b])
    c_mp = -mp.lu_solve(G0, b0)
    R_mp = G0**-1
    c0 = [approximate_point(c_mp[j]) for j in range(m)]
    R = [[approximate_point(R_mp[i,j]) for j in range(m)] for i in range(m)]
    RG = matrix_product(R, G)
    Q = [[CI(1 if i == j else 0)-RG[i][j] for j in range(m)]
         for i in range(m)]
    q_upper = matrix_norm_upper(Q)
    assert q_upper < ONE
    residual = [dot(G[i], c0)+b[i] for i in range(m)]
    transformed = [dot(R[i], residual) for i in range(m)]
    residual_upper = max(z.l1_upper() for z in transformed)
    denominator_lower = down(ONE-q_upper)
    assert denominator_lower > 0
    radius_upper = up(residual_upper/denominator_lower)
    radius = I(-radius_upper, radius_upper)
    enclosure = [CI(z.re+radius, z.im+radius) for z in c0]
    return enclosure, dict(q_upper=q_upper,
                           transformed_residual_upper=residual_upper,
                           coefficient_l1_radius_upper=radius_upper)


def exact_binary(x):
    n, d = x.as_integer_ratio()
    return {'numerator': str(n), 'denominator': str(d)}


def real_serial(x):
    return {'lower_display': str(x.lo), 'upper_display': str(x.hi),
            'lower_exact_binary': exact_binary(x.lo),
            'upper_exact_binary': exact_binary(x.hi)}


def scalar_serial(x):
    return {'display': str(x), 'exact_binary': exact_binary(x)}


def rational_interval(q):
    return I.rational(q.numerator, q.denominator)


def main():
```

```
start = time.monotonic()
primes = primes_count(N+1)
assert primes[N-1] == 1299709
pi = g.const_pi()
two_pi = I(2)*I(down(pi), up(pi))
harmonic = I(0)
moments = [CI(0) for _ in range(5)]
weight_sum = 0
for i, p in enumerate(primes[:-1]):
    harmonic = harmonic+I.rational(1, p)
    if i == 0:
        continue
    theta = two_pi*(harmonic-I(p).log().log())
    sin_i, cos_i = theta.sincos()
    u = CI(cos_i, sin_i)
    w = primes[i+1]-primes[i-1]
    weight_sum += w
    power = CI(1)
    for j in range(1,5):
        power = power*u
        moments[j] = moments[j]+power*w
    if (i+1) % 20000 == 0:
        print('Certified moments through prime index', i+1,
              'seconds', round(time.monotonic()-start, 2), flush=True)
assert weight_sum == primes[N-1]+primes[N]-5
moments[0] = CI(weight_sum)

def C(k):
    return moments[k] if k >= 0 else moments[-k].conjugate()

solutions, norms, verification = {}, {}, {}
for m in (3,4):
    G = [[C(k-j) for k in range(m)] for j in range(m)]
    b = [C(m-j) for j in range(m)]
    coefficients, info = verified_solve(G, b)
    norm = C(0)
    for k in range(m):
        norm = norm+coefficients[k]*C(k-m)
    assert norm.im.contains_zero()
    assert norm.re.lo > 0
    solutions[m], norms[m], verification[m] = coefficients, norm.re, info

cabs = solutions[4][3].abs_interval()
# Coarse rational claims are the mathematical output.  The decimal endpoint
# strings in JSON are for display; exact binary endpoints are also retained.
c_upper = Fraction('1.710103')
h3_lower = Fraction('30.958150')
h4_lower = Fraction('11.722738')
assert cabs.hi < rational_interval(c_upper).lo
assert norms[3].lo > rational_interval(h3_lower).hi
assert norms[4].lo > rational_interval(h4_lower).hi
for info in verification.values():
    assert info['q_upper'] < rational_interval(Fraction(1, 10**40)).lo
    assert info['coefficient_l1_radius_upper'] < rational_interval(Fraction(1, 10**40)).lo

B4 = Fraction(174)
target_modulus = Fraction('3.9999')
rational_margin = ((target_modulus-c_upper)**2*h3_lower
                   -(B4-h4_lower))
assert target_modulus > c_upper and B4 > h4_lower
assert rational_margin > 0
# No overflow/invalid/division-by-zero is allowed.  Extreme subnormal
# outward neighbors of exact zero may set MPFR underflow; outward padding
# still contains the exact result and is covered by the endpoint proof.
ctx = g.get_context()
assert not ctx.overflow and not ctx.invalid and not ctx.divzero
result = {
    'scope': 'Rigorously certified finite prefix. All-N conclusion is CONDITIONAL on H4_infinity <=
 174.',
    'conditional_conclusion': 'For every M >= 100000, abs(c3_M) < 3.9999, provided H4_infinity <= 174.
',
    'arithmetic': {
        'gmpy2': g.version(), 'mpfr': g.mpfr_version(), 'precision_bits': PREC,
        'rounding': 'Nearest MPFR operation with outward next_below / next_above padding.',
```

```
            'trigonometry': 'Correctly rounded midpoint sin/cos, enlarged by global Lipschitz radius.',
            'linear_system_proof': 'q=||I-RG||_infty,11<1; radius=||R(G*c0+b)||/(1-q). R and c0 are exact
     binary points.',
            'mpmath_role': 'Approximate candidate generation only; no unverified mpmath value is used as
    a bound.'
        },
        'N': N, 'p_N': primes[N-1], 'p_N_plus_1': primes[N],
        'weight_sum': weight_sum, 'atom_count': N-1,
        'c3_real': real_serial(solutions[4][3].re),
        'c3_imag': real_serial(solutions[4][3].im),
        'c3_abs': real_serial(cabs),
        'H3': real_serial(norms[3]), 'H4': real_serial(norms[4]),
        'verified_solve': {str(m): {key: scalar_serial(value)
                                    for key,value in info.items()}
                           for m,info in verification.items()},
        'coarse_rational_enclosures': {
            'c3_abs_strict_upper': str(c_upper),
            'H3_strict_lower': str(h3_lower), 'H4_strict_lower': str(h4_lower),
            'B4_hypothesis': str(B4), 'target_modulus': str(target_modulus),
            'strict_rational_margin': str(rational_margin),
            'proof': '(target-c_upper)^2*H3_lower > 174-H4_lower'
        },
        'all_checks_passed': True,
        'elapsed_seconds': round(time.monotonic()-start, 3)
    }
    out = Path(__file__).with_suffix('.json')
    out.write_text(json.dumps(result, indent=2)+'\n')
    print('Coarse rational bounds certified:', result['coarse_rational_enclosures'], flush=True)
    print('Saved', out, 'seconds', result['elapsed_seconds'], flush=True)


if __name__ == '__main__':
    main()
```

125 The exact variance and the strength of a sparse logarithmic bound

Throughout this appendix, p_n denotes the ordinary primes and

$$P(x) = \pi(x) - \mathrm{Li}_2(x), \qquad \mathrm{Li}_2(x) = \int_2^x \frac{dt}{\log t}, \qquad E(x) = \sum_{p \le x} \frac{1}{p} - \log\log x - B_{\mathbb{P}}.$$

Here $B_{\mathbb{P}}$ is the prime Mertens constant, so the classical unconditional Mertens theorem states $E(x) \to 0$. The functions are taken right-continuously at primes.

Lemma 125.1 (Exact equality of the two centered variances)**.** *For every real $X > 2$,*

$$V(X) := \min_{a \in \mathbb{R}} \int_2^X \frac{(P(t)-a)^2}{t^2}\, dt = \min_{c \in \mathbb{R}} \int_2^X (E(t)-c)^2\, dt. \tag{125.1}$$

Put

$$Q(X) = \int_2^X \frac{P(t)^2}{t^2}\, dt, \qquad A(X) = \int_2^X \frac{P(t)}{t^2}\, dt, \qquad b_0 = \log\log 2 + B_{\mathbb{P}}.$$

Then

$$Q(X) - V(X) = \frac{2X}{X-2} A(X)^2, \qquad A(X) = b_0 + E(X) - \frac{P(X)}{X} \longrightarrow b_0. \tag{125.2}$$

In particular $Q(X) = V(X) + O(1)$, uniformly for $X \ge 3$.

Proof. Let $J(t) = \int_2^t E(u)\, du$. Stieltjes partial summation gives

$$P(t) = tE(t) - J(t) + 2b_0. \tag{125.3}$$

Indeed, both sides have Stieltjes differential $t\,dE(t)$ and agree at $t = 2$, where $P(2) = 1$ and $E(2) = 1/2 - b_0$. For a real c, set $J_c(t) = J(t) - c(t-2)$. This function is absolutely continuous, $J_c(2) = 0$, and

$$\frac{P(t) - 2b_0 - 2c}{t} = J_c'(t) - \frac{J_c(t)}{t} \quad \text{almost everywhere.}$$

The identity $(J_c' - J_c/t)^2 = (J_c')^2 - (J_c^2/t)'$ therefore proves

$$\int_2^X (E-c)^2 dt = \int_2^X \frac{(P - 2b_0 - 2c)^2}{t^2}\, dt + \frac{(J(X) - c(X-2))^2}{X}.$$

The left side is minimized at $c_X = J(X)/(X-2)$, where the last term vanishes. Integrating (125.3) after division by t^2 gives

$$A(X) = 2b_0 \left(\frac{1}{2} - \frac{1}{X}\right) + \frac{J(X)}{X}.$$

Thus the corresponding $a_X = 2b_0 + 2c_X$ is also the weighted mean of P for the measure $t^{-2}dt$, and minimizes the other integral. This proves (125.1).

Expanding that weighted variance, whose total mass is $(X-2)/(2X)$, gives the first identity in (125.2). Combining the displayed formula for $A(X)$ with (125.3) gives its second identity. Mertens' theorem gives $E(X) \to 0$, and Chebyshev's estimate $\pi(X) = O(X/\log X)$, together with the elementary analogous bound for $\mathrm{Li}_2(X)$, gives $P(X)/X \to 0$. The bounded-correction conclusion follows. □

Theorem 125.2 (A sparse logarithmic variance bound implies RH)**.** *If there is any unbounded sequence of real numbers X_j such that $V(X_j) = O(\log X_j)$, then RH holds. In particular, the original statement*

$$V(p_{N+1}) \le \frac{\log p_N}{8\pi^2} \quad \textit{for infinitely many } N \tag{125.4}$$

would imply RH. No bound on the gaps between the selected endpoints is assumed.

Proof. Pass to a sequence tending to infinity and put $T_j = \log X_j$. Lemma 125.1 gives $Q(X_j) = O(T_j)$. Define the entire functions

$$F_j(s) = \int_2^{X_j} P(x) x^{-s-1} dx, \qquad G_j(s) = (sF_j(s))' = \int_2^{X_j} P(x) x^{-s-1}(1 - s\log x) dx.$$

For every compact set $K \subset \{\Re s > 1/2\}$, Cauchy–Schwarz gives

$$\sup_{s\in K} |G_j(s)| \le \sqrt{Q(X_j)} \sup_{s\in K} \left(\int_2^\infty x^{-2\Re s} |1 - s\log x|^2 dx\right)^{1/2} \ll_K \sqrt{T_j}. \tag{125.5}$$

The integral is uniformly finite because $\min_{s\in K} \Re s > 1/2$.

Initially for $\Re s > 1$, set

$$F(s) = \int_2^\infty P(x) x^{-s-1} dx, \qquad \mathcal{P}(s) = \sum_p p^{-s}.$$

Absolutely convergent termwise integration yields $sF(s) = \mathcal{P}(s) - \int_2^\infty x^{-s}/\log x\, dx$. Differentiation and the Euler product give

$$\begin{aligned} G(s) := (sF(s))' &= \mathcal{P}'(s) + \frac{2^{1-s}}{s-1} \\ &= \frac{\zeta'(s)}{\zeta(s)} - \sum_{k\ge 2} \mathcal{P}'(ks) + \frac{2^{1-s}}{s-1}. \end{aligned} \tag{125.6}$$

The chain factor k cancels $1/k$ when differentiating $\log\zeta(s) = \sum_{k\geq 1}\mathcal{P}(ks)/k$. Only the Euler-product logarithm in $\Re s > 1$ is used; no logarithm is continued through a zero.

The prime-power sum is normally convergent in $\Re s > 1/2$, as follows from, for every $\sigma_0 > 1/2$,

$$\sum_{k\geq 2}\sum_{p}(\log p)p^{-k\sigma_0} = \sum_{p}\frac{(\log p)p^{-2\sigma_0}}{1-p^{-\sigma_0}} < \infty.$$

At $s = 1$, the respective principal parts of ζ'/ζ and $2^{1-s}/(s-1)$ are $-1/(s-1)$ and $1/(s-1)$; they cancel. Thus (125.6) is meromorphic throughout $\Re s > 1/2$, with a simple pole at each zeta zero there, whose residue is that zero's positive integer multiplicity.

On any fixed bounded segment of $\Re s = b > 1$, the bound $P(x) = O(x)$ implies uniformly

$$G_j(s) - G(s) = -\int_{X_j}^{\infty} P(x)x^{-s-1}(1 - s\log x)dx = O\Big(T_j e^{-(b-1)T_j}\Big). \tag{125.7}$$

Suppose now that $\zeta(\rho) = 0$ with $\Re\rho > 1/2$. Choose a finite rectangle $\mathcal{R}$ containing ρ, with left edge $\Re s = \alpha$, where $1/2 < \alpha < \Re\rho$, and right edge $\Re s = b > \max(1, \Re\rho)$. Choose both horizontal heights and the left edge to avoid all zeta zeros. Discreteness of the zeros permits these choices. Let $\rho_1, \ldots, \rho_m$ be all its distinct interior zeros and put

$$\mathcal{C}(s) = \prod_{r=1}^{m}(s - \rho_r), \qquad H_j(s) = \mathcal{C}(s)(G_j(s) - G(s)).$$

Each distinct zero is included once: a zero of any multiplicity produces only a simple logarithmic-derivative pole. Consequently H_j is holomorphic on a neighborhood of the closed rectangle. By (125.5), it is bounded by $C\sqrt{T_j}$ on the boundary, while (125.7) improves the right-edge bound to $CT_j e^{-(b-1)T_j}$. The constants are fixed independently of j.

Let $\omega(s)$ be the harmonic measure of the right edge in this rectangle: its boundary values are one on that open edge and zero on the other open edges. The maximum principle gives $0 < \omega(s) < 1$ at every interior point. Applying the subharmonic maximum principle to $\log|H_j|$ gives

$$|H_j(s)| \leq \Big(CT_j e^{-(b-1)T_j}\Big)^{\omega(s)}\Big(C\sqrt{T_j}\Big)^{1-\omega(s)}.$$

The four corners have harmonic measure zero and do not change this two-constants estimate. In particular $H_j(\rho) \to 0$.

But if $\rho = \rho_r$ has multiplicity m_ρ, the removable value at that point is exactly

$$H_j(\rho) = -m_\rho \prod_{\substack{1\leq\ell\leq m\\ \ell\neq r}}(\rho - \rho_\ell) \neq 0,$$

independently of j. This contradiction rules out every zero with real part greater than $1/2$. The functional equation's symmetry $\rho \mapsto 1 - \rho$ for nontrivial zeros then gives RH. Finally, (125.4) supplies the hypothesis with $X_j = p_{N_j+1}$, since $\log p_{N_j} \leq \log X_j$. □

Scope. This proves only the implication from the stated sparse upper bound to RH. It neither proves that such endpoints exist nor uses the desired upper bound as an unconditional premise. The proof imports no every-scale oscillation theorem or assumed zero-free region beyond the absolutely convergent Euler product.

126 The original sparse target and its direct upper bound

Let p_n denote the nth ordinary prime and set

$$S(x) = \sum_{p \le x} \frac{1}{p}, \qquad E(x) = S(x) - \log\log x - B_{\mathbb{P}}, \tag{126.1}$$

$$U_n = e^{2\pi i (S(p_n) - \log\log p_n)}, \qquad w_n = p_{n+1} - p_{n-1} \quad (n \ge 2). \tag{126.2}$$

Here $B_{\mathbb{P}}$ is the prime Mertens constant. Define

$$W_N = \sum_{n=2}^{N} w_n = p_N + p_{N+1} - 5, \qquad M_N = \sum_{n=2}^{N} w_n U_n, \qquad D_N = W_N - |M_N|. \tag{126.3}$$

The earlier direct-upper-bound stage singled out the equivalent sparse assertion

$$\boxed{\forall N_0 \ge 2 \ \exists N \ge N_0 : \quad D_N \le \log p_N.} \tag{126.4}$$

No different sequence, normalization, or threshold replaces (126.4). The main proved estimate is

$$D_N \le 4\pi^2 V(p_{N+1}) + C, \qquad V(X) = \min_{a \in \mathbb{R}} \int_2^X \frac{(P(t) - a)^2}{t^2}\, dt, \tag{126.5}$$

where $P(x) = \pi(x) - \mathrm{Li}_2(x)$ and $\mathrm{Li}_2(x) = \int_2^x dt/\log t$. The constant C is finite and independent of N. This controls the original deficit from above, but a sufficiently small value of the right-hand side at infinitely many indices is not proved here.

All integrals use Lebesgue measure. The functions E and P are right-continuous at primes. The symbols $\ll$ and O permit fixed constants; a parameter subscript indicates allowed dependence.

127 Embedded interval reproduction code

The following standalone Python program uses NumPy and mpmath. It runs the deterministic sieve and interval vector sum, then writes the raw interval endpoints as JSON. The output path has been made local to the working directory; this does not change the arithmetic. The supplied cutoff includes a prime after 10^7, and an assertion checks that requirement.

```
"""Independent outward interval calculation of constant-free prime circle deficit.
No input Meissel-Mertens constant and no quadrature.
Uses mpmath.iv elementary intervals; implementation is not formally verified.
"""
import time, json, math
import numpy as np
from mpmath import iv
iv.dps=40
limit=10000100
sieve=np.ones(limit+1,dtype=np.bool_); sieve[:2]=False
for p in range(2,math.isqrt(limit)+1):
    if sieve[p]:sieve[p*p::p]=False
ps=[int(p) for p in np.flatnonzero(sieve)]
assert ps[-1] > 10000000  # next prime for the final weight
S=iv.mpf(0); re=iv.mpf(0); im=iv.mpf(0); W=0
cuts=[10,100,1000,10000,100000,1000000,10000000]
cutindices={sum(p<=X for p in ps)-1:X for X in cuts}
rows=[]
t0=time.time()
for i,p in enumerate(ps):
    if p>10000000: break
    S+=iv.mpf(1)/p
    if i==0:continue
    w=ps[i+1]-ps[i-1]
    phase=2*iv.pi*(S-iv.ln(iv.ln(iv.mpf(p))))
```

```
    re+=w*iv.cos(phase)
    im+=w*iv.sin(phase)
    W+=w
    if i in cutindices:
        D=W-iv.sqrt(re*re+im*im)
        row={'cutoff':cutindices[i], 'last_prime':p,'W':W,'D_interval':str(D)}
        rows.append(row)
        print(row,'seconds',round(time.time()-t0,1),flush=True)
with open('certified_deficit_intervals_1e7.json', 'w') as f:
    json.dump(rows, f, indent=2)
```

128 Exact rational local comparison code

This program uses only the Python standard library and exact integers. Its final assertion verifies that the upper bound for d_5 is smaller than the lower bound for d_6.

```
"""Exact integer fixed-point interval proof for local deficits d_5 < d_6.
No external packages and no floating-point arithmetic.
Every represented interval is [lo/S, hi/S], S=10**70.
Logarithms use atanh series with geometric tail; pi uses Machin's
formula and alternating arctan series; sin/cos use Taylor's theorem.
"""
import math, json
SCALE=10**70
N=100

def ceildiv(a,b): return -((-a)//b)
class I:
 def __init__(self,lo,hi=None): self.lo=lo; self.hi=lo if hi is None else hi
 @classmethod
 def rat(cls,n,d=1):
  assert d>0
  return cls(n*SCALE//d,ceildiv(n*SCALE,d))
 def __add__(self,b): return I(self.lo+b.lo,self.hi+b.hi)
 def __neg__(self): return I(-self.hi,-self.lo)
 def __sub__(self,b): return self+-b
 def __mul__(self,b):
  p=[self.lo*b.lo,self.lo*b.hi,self.hi*b.lo,self.hi*b.hi]
  return I(min(p)//SCALE,ceildiv(max(p),SCALE))
 def divint(self,n):
  assert n>0
  return I(self.lo//n,ceildiv(self.hi,n))
 def scaleint(self,n):
  return I(self.lo*n,self.hi*n) if n>=0 else I(self.hi*n,self.lo*n)
 def enlarge(self,r): return I(self.lo-r,self.hi+r)
 def sqrt(self):
  assert self.lo>=0
  a=math.isqrt(self.lo*SCALE); b=math.isqrt(self.hi*SCALE)
  if b*b<self.hi*SCALE: b+=1
  return I(a,b)
 def text(self):
  def dec(n):
   sign='-' if n<0 else ''; n=abs(n)
   return sign+str(n//SCALE)+'.'+str(n%SCALE).zfill(70)
  return '['+dec(self.lo)+', '+dec(self.hi)+']'
ZERO=I.rat(0);ONE=I.rat(1)

def atanh_twice(t):
 z=t*t; power=t; ans=ZERO
 for j in range(N):
  ans=ans+power.divint(2*j+1)
  power=power*z
 # Analytic tail for the underlying |t| <= 1/3.
 tail=ceildiv(9*SCALE,4*(2*N+1)*3**(2*N+1))
 return ans.scaleint(2).enlarge(tail)
LN2=atanh_twice(I.rat(1,3))

def logpoint(n):
 assert n>0
```

```
 d=SCALE; e=0
 while n>=2*d: d*=2; e+=1
 while n<d: n*=2; e-=1
 # Now 1 <= n/d < 2, so 0 <= t < 1/3.
 return LN2.scaleint(e)+atanh_twice(I.rat(n-d,n+d))

def log_interval(a):
 return I(logpoint(a.lo).lo,logpoint(a.hi).hi)

def atan_rational(q):
 t=I.rat(1,q); z=t*t; power=t; ans=ZERO
 for j in range(N):
  term=power.divint(2*j+1)
  ans=ans+term if j%2==0 else ans-term
  power=power*z
 tail=ceildiv(SCALE,(2*N+1)*q**(2*N+1))
 return ans.enlarge(tail)
PI=atan_rational(5).scaleint(16)-atan_rational(239).scaleint(4)

def sincos(a):
 z=a*a; cs=ONE; ct=ONE; ss=a; st=a
 for j in range(1,81):
  ct=-(ct*z).divint((2*j-1)*(2*j)); cs=cs+ct
  st=-(st*z).divint((2*j)*(2*j+1)); ss=ss+st
 # Taylor remainder: max |a|^(degree+1)/(degree+1)!.
 # Here we enforce |a| <= 6 and use the exact rational bound.
 assert max(abs(a.lo),abs(a.hi))<=6*SCALE
 rc=ceildiv(SCALE*6**161,math.factorial(161))
 rs=ceildiv(SCALE*6**162,math.factorial(162))
 return ss.enlarge(rs),cs.enlarge(rc)

prime=[True]*256;prime[:2]=[False,False]
for p in range(2,16):
 if prime[p]:
  for m in range(p*p,256,p):prime[m]=False
recip=ZERO; values={}
for m in range(2,256):
 if prime[m]:recip=recip+I.rat(1,m)
 a=PI.scaleint(2)*(recip-log_interval(log_interval(I.rat(m))))
 sn,cs=sincos(a);values[m]=(cs,sn)
rows=[]
for k in range(1,7):
 re=ZERO;im=ZERO
 for m in range(2**k,2**(k+2)):
  re=re+values[m][0];im=im+values[m][1]
 deficit=I.rat(3*2**k)-(re*re+im*im).sqrt()
 rows.append({'k':k,'lower_numerator':deficit.lo,'upper_numerator':deficit.hi,
              'denominator':SCALE,'decimal_interval':deficit.text()})
assert rows[4]['upper_numerator']<rows[5]['lower_numerator']
print(json.dumps({'arithmetic':'exact Python integers','interval_scale':SCALE,'rows':rows,
                  'verified_statement':'d_5 < d_6'},indent=2))
```

129 An independent check that the sparse target implies RH

Write

$$P(x)=\pi(x)-\operatorname{Li}_2(x),\qquad \operatorname{Li}_2(x)=\int_2^x\frac{dt}{\log t},\qquad V(X)=\min_{a\in\mathbb{R}}\int_2^X\frac{(P(t)-a)^2}{t^2}\,dt.$$

The target under investigation is exactly

$$V(p_{N+1})\le\frac{\log p_N}{8\pi^2}\quad\text{for infinitely many } N. \tag{129.1}$$

The argument below independently proves that this statement would imply RH. It does not establish (129.1).

Theorem 129.1. *If (129.1) holds for the ordinary primes, then every nontrivial zero of ζ has real part $1/2$.*

Proof. Step 1: recover the uncentered positive energy. Put

$$Q(X)=\int_2^X \frac{P(t)^2}{t^2}\,dt, \qquad A(X)=\int_2^X \frac{P(t)}{t^2}\,dt, \qquad b_0=\log\log 2+B_{\mathbb{P}}.$$

Expanding the quadratic in a gives

$$Q(X)=V(X)+\frac{2X}{X-2}A(X)^2. \tag{129.2}$$

For the reciprocal-prime error $E(x)=\sum_{p\le x}p^{-1}-\log\log x-B_{\mathbb{P}}$, exact partial summation gives

$$A(X)=b_0+E(X)-\frac{P(X)}{X}.$$

The ordinary prime number theorem and Mertens' theorem imply $A(X)\to b_0$. Thus $Q(X)=V(X)+O(1)$ for $X\ge 3$. Selecting the indices in (129.1) and writing $X_j=p_{N_j+1}\to\infty$, $T_j=\log X_j$, we obtain

$$Q(X_j)=O(T_j). \tag{129.3}$$

No bound on the gaps between successive X_j is needed.

Step 2: entire truncations and a meromorphic continuation. For each j define the entire functions

$$F_j(s)=\int_2^{X_j} P(x)x^{-s-1}\,dx, \qquad G_j(s)=(sF_j(s))'=\int_2^{X_j} P(x)x^{-s-1}(1-s\log x)\,dx.$$

If K is any compact subset of $\Re s>1/2$, Cauchy–Schwarz and (129.3) imply

$$\sup_{s\in K}|G_j(s)|\le\sqrt{Q(X_j)}\sup_{s\in K}\left(\int_2^\infty x^{-2\Re s}|1-s\log x|^2\,dx\right)^{1/2}\ll_K\sqrt{T_j}. \tag{129.4}$$

The last integral is uniformly finite on K because $\min_{s\in K}\Re s>1/2$.

Initially for $\Re s>1$, let

$$F(s)=\int_2^\infty P(x)x^{-s-1}\,dx, \qquad \mathcal{P}(s)=\sum_p p^{-s}.$$

Termwise integration, justified by absolute convergence, gives

$$sF(s)=\mathcal{P}(s)-\int_2^\infty \frac{x^{-s}}{\log x}\,dx.$$

Differentiating and using the Euler product yields

$$G(s):=(sF(s))'=\mathcal{P}'(s)+\frac{2^{1-s}}{s-1} \tag{129.5}$$

$$=\frac{\zeta'(s)}{\zeta(s)}-\sum_{k\ge 2}\mathcal{P}'(ks)+\frac{2^{1-s}}{s-1}. \tag{129.6}$$

There is no missing chain factor: differentiating $\log\zeta(s)=\sum_{k\ge1}\mathcal{P}(ks)/k$ cancels the factor k against $1/k$.

The higher-prime-power series in (129.6) is normally convergent on $\Re s>1/2$. Indeed, for any fixed $\sigma_0>1/2$,

$$\sum_{k\ge2}\sum_p(\log p)p^{-k\sigma_0}=\sum_p\frac{(\log p)p^{-2\sigma_0}}{1-p^{-\sigma_0}}<\infty.$$

At $s = 1$, the principal parts of ζ'/ζ and $2^{1-s}/(s-1)$ are respectively $-1/(s-1)$ and $1/(s-1)$; they cancel. Thus G is meromorphic in $\Re s > 1/2$ and its only poles there are the zeros of ζ, each giving a simple pole whose residue is its positive integer multiplicity.

On a bounded segment of the vertical line $\Re s = b > 1$, the elementary estimate $P(x) = O(x)$ gives

$$G_j(s) - G(s) = -\int_{X_j}^{\infty} P(x)x^{-s-1}(1 - s\log x)\,dx = O_b\Big(T_j e^{-(b-1)T_j}\Big), \tag{129.7}$$

uniformly on that segment. Dependence on its fixed height bound is included in the implied constant.

Step 3: a finite rectangle rules out a pole. Suppose $\zeta(\rho) = 0$ with $\Re\rho > 1/2$. Choose a rectangle $\mathcal{R}$ whose left edge is $\Re s = \alpha$ with $1/2 < \alpha < \Re\rho$, whose right edge is $\Re s = b > 1$, and whose horizontal edges have finite heights strictly below and above $\Im\rho$. Choose its boundary to avoid all zeros of ζ. This is possible since the zeros are discrete. There are only finitely many distinct zeros $\rho_1, \dots, \rho_m$ inside the rectangle. Define

$$B(s) = \prod_{r=1}^{m}(s - \rho_r).$$

Each zero is included once, because a zero of any multiplicity produces only a simple pole of ζ'/ζ. The functions

$$H_j(s) = B(s)(G_j(s) - G(s))$$

are holomorphic on a neighborhood of the closed rectangle. By (129.4), their modulus on its boundary is at most $C\sqrt{T_j}$ after enlarging a fixed constant. On the right edge, (129.7) improves this to $CT_j e^{-(b-1)T_j}$.

Let $\omega(s)$ denote the harmonic measure, in $\mathcal{R}$, of its right edge. It is the harmonic function with boundary values 1 on the open right edge and 0 on the other open edges. For every interior point, $0 < \omega(s) < 1$ by the strong maximum principle. Applying the subharmonic maximum principle to $\log|H_j|$ (or its standard regularization) gives the two-constants estimate

$$|H_j(s)| \le \Big(CT_j e^{-(b-1)T_j}\Big)^{\omega(s)} \Big(C\sqrt{T_j}\Big)^{1-\omega(s)}.$$

The corner discontinuities of the boundary data have harmonic measure zero and do not affect this estimate. In particular, $H_j(\rho) \to 0$.

On the other hand, if $\rho = \rho_r$ has multiplicity m_ρ, then the removable value at that point is exactly

$$H_j(\rho) = -m_\rho \prod_{\substack{1 \le \ell \le m \\ \ell \ne r}} (\rho - \rho_\ell) \ne 0,$$

independently of j. This contradiction rules out all zeros in $\Re s > 1/2$. The functional-equation symmetry of the nontrivial zeros then gives RH. □

What the argument uses. The sparse bound supplies uniform control of the entire truncations on a compact rectangle. Absolute convergence on its right edge supplies exponential approximation. Meromorphic continuation and the two-constants theorem are essential; positivity of an accumulated energy alone does not turn a sparse prefix estimate into a global pointwise bound. This proof does not import the Stechkin–Popov every-scale lower theorem and does not prove that the required sparse prefix estimate holds.

130 Finite checks of the additional synthesis results

The checks in this section were run for this consolidation. The first uses integer interval arithmetic and is rigorous for its finite range. The other checks use floating-point arithmetic to detect constant, index, or normalization errors; they are not proofs of asymptotic claims or RH.

130.1 Three exact crossing certificates

For $B = 2^{120}$ and the first n primes, put $L_n = \sum_{j \le n} \lfloor B/p_j \rfloor$. Integer arithmetic gives $L_n/B \le S(p_n) < (L_n + n)/B$. For each entry below we verified $L_n \ge mB$ and $L_{n-1} + n - 1 < mB$. These inequalities certify the first crossing; the decimals only display the excess and do not supply the proof.

Turn m	Crossing prime	Prime index	Excess (displayed)
1	5	3	$3.333333333333E-2$
2	277	59	$2.350151450293E-3$
3	5,195,977	361,139	$3.092735082356E-8$

The interval width is at most $n/2^{120}$, below 3×10^{-31} for these three certificates. The first twelve exact rational sums also have denominator equal to the product of their primes, as predicted by the proof.

130.2 Actual prime deficits

To reduce cancellation in $W - |M|$, the program computes the centered complex scatter and divides it by $1 + |M|/W$. The upper cutoff means $p_N \le X$; the next prime is retained to form the original neighboring-gap weight.

X	p_N	D_N	$D_N^{(2)}/D_N$	$D_N^{(3)}/D_N$
1,000	997	26.969965365	2.216332848	3.474532260
10,000	9,973	32.298046004	2.374465621	3.929850143
100,000	99,991	34.913607775	2.465151301	4.201386517
1,000,000	999,983	36.256293902	2.514984160	4.354281691

Every finite ratio must be smaller than 4 or 9, respectively, under both truth values of RH. This table therefore cannot certify the strict limiting gap.

130.3 A convergent-phase model with divergent variance

Take $t_n = n^{-1/4}$, $w_n = 1$, and phases $e^{0.4it_n}$. Its scalar variance diverges, so it provides a model in which the synthesis asymptotics apply without making any hypothesis about zeta zeros. With $m = 3$, the predicted second-eigenvalue ratio is $\binom{5}{3} = 10$; the third ratio tends to zero.

N	D_N	λ_2/D_N	λ_3/D_N	$D_N^{(2)}/D_N$
1,000	0.473182864	9.893059164	0.017650880	3.977983821
10,000	1.675504753	9.924717024	0.014735490	3.984914937
100,000	5.510718478	9.957455210	0.009488540	3.991733261
1,000,000	17.660537669	9.979181743	0.005099988	3.996064255

A separate deterministic finite test (seed 20261109) checked 560 weighted circle configurations satisfying the finite spectral theorem's sufficient hypothesis. All inequalities held to floating-point rounding tolerance. 0 configurations did not satisfy that hypothesis and were not counted as tests of its conclusion. The Taylor and min–max proof, rather than these trials, establishes the theorem.

130.4 Embedded reproduction program

```
from pathlib import Path
import math,json
from decimal import Decimal, localcontext
from fractions import Fraction
import numpy as np
from scipy.linalg import toeplitz

ROOT=Path('.')
(ROOT/'tmp').mkdir(exist_ok=True)
result={}

def primes_up_to(n):
 a=np.ones(n+1,dtype=bool);a[:2]=False
 for p in range(2,math.isqrt(n)+1):
  if a[p]:a[p*p::p]=False
 return np.flatnonzero(a)

primes=primes_up_to(5_300_000)
scale=1<<120
total=0;crossings=[];turn=1
for idx,p0 in enumerate(primes,1):
 p=int(p0);before=total;total+=scale//p
 if total>=turn*scale:
  assert before+idx-1<turn*scale
  with localcontext() as ctx:
   ctx.prec=42
   lo=Decimal(total-turn*scale)/Decimal(scale)
   hi=Decimal(total+idx-turn*scale)/Decimal(scale)
  crossings.append({'turn':turn,'prime':p,'prime_index':idx,'excess_low':str(lo),'excess_high':str(hi)})
  turn+=1
  if turn==4:break
assert [x['prime'] for x in crossings]==[5,277,5195977]
result['certified_laps']=crossings

small=primes[:12]
sumq=Fraction(0);product=1
for p0 in small:
 p=int(p0);sumq+=Fraction(1,p);product*=p
 assert sumq.denominator==product
result['rational_denominator_examples']=len(small)

def circle_deficit(angle,weight,m):
 u=np.exp(1j*m*angle);W=float(weight.sum());mean=np.sum(weight*u)/W
 scatter=np.sum(weight*np.abs(u-mean)**2)
 return float(scatter/(1+abs(mean)))

actual=[]
for X in [1000,10000,100000,1000000]:
 stop=int(np.searchsorted(primes,X,side='right'))
 pn=primes[:stop+1].astype(float)
 S=np.cumsum(1/pn)
 E=S-np.log(np.log(pn))-0.261497212847642783755426838608695859 05
 weight=pn[2:]-pn[:-2];angle=2*np.pi*E[1:-1]
 ds=[circle_deficit(angle,weight,m) for m in [1,2,3]]
 assert 0<ds[1]<4*ds[0] and 0<ds[2]<9*ds[0]
 actual.append({'cutoff':X,'p_N':int(pn[-2]),'D1':ds[0],'D2_over_D1':ds[1]/ds[0],'D3_over_D1':ds[2]/ds
     [0]})
result['actual_prime_deficits']=actual

model=[]
for N in [1000,10000,100000,1000000]:
 n=np.arange(1,N+1,dtype=float)
 t=n**(-.25);angles=.4*t;weights=np.ones(N)
 D=circle_deficit(angles,weights,1)
```

```
 row=np.array([np.sum(np.exp(1j*j*angles)) for j in range(4)])
 G=toeplitz(row,row.conj())
 eig=np.linalg.eigvalsh(G)[::-1]
 model.append({'N':N,'D':D,'lambda2_over_D':float(eig[1]/D),'lambda3_over_D':float(eig[2]/D),'D2_over_D':
     circle_deficit(angles,weights,2)/D})
result['divergent_convergent_phase_model']=model

rng=np.random.default_rng(20261109)
accepted=0;not_applicable=0;max_violation=0.0
for m in range(1,8):
 for rep in range(80):
  t=rng.normal(0,.003+rep*.0004,size=40)
  w=rng.uniform(.2,2,size=t.size);W=float(w.sum());x=t-np.sum(w*t)/W
  lam=1.0;d=m+1;S2=float(np.sum(w*x*x));Q3=float(np.sum(w*np.abs(x)**3))
  r=d*abs(lam)**3*m**3*Q3/6;b=d*lam*lam*m*m*S2/2+r
  if d*W<=2*b:not_applicable+=1;continue
  row=np.array([np.sum(w*np.exp(1j*lam*j*x)) for j in range(m+1)])
  eig=np.linalg.eigvalsh(toeplitz(row,row.conj()))[::-1]
  c=m*(m+1)*(m+2)/12
  low=lam*lam*c*S2-r-b*b/(d*W-2*b);up=lam*lam*c*S2+r
  violation=max(0.0,low-eig[1],eig[1]-up,(-float(eig[-1])),float(eig[2]-r) if m>=2 else 0.0)
  max_violation=max(max_violation,violation)
  assert violation<1e-10*d*W
  accepted+=1
result['finite_spectral_certificate']={'seed':20261109,'accepted':accepted,'not_applicable':
     not_applicable,'max_float_violation':max_violation}

(ROOT/'tmp/rh_synthesis_checks.json').write_text(json.dumps(result,indent=2))
print(json.dumps(result,indent=2))
```

131 Final proof ledger and the unchanged open estimate

The original target has not been replaced by a sensor constant or a finite height calculation. In the notation fixed at the beginning, it is still

$$\exists K < \infty \; \forall N \geq 2 : \; D_N \leq K.$$

The proofs in this manuscript give several exact equivalents, including the strict limiting harmonic gap, the centered variance bound, the critical prime Sobolev norm, the compensated correlation bound, and the positive-mass integral of the signed primitive. The additional theorems quantify how these objects behave and how a failure would appear. None establishes their required finite upper bound for the actual infinite prime sequence.

Question	Status in this manuscript
Exact reciprocal laps and finite arithmetic	Settled as rational finite sums; prime-endpoint closure is impossible. This does not test primality.
Energy, disk, operator, and sign equivalences	Proved with the displayed domains, weights, and quantifiers.
New cross-structure estimates	Sharp divergent-regime comparison, single-mode spectral localization, invertible constant-free Hardy coordinates, nonlinear moment endpoints, and count-threshold criteria are proved.
Finite computations and boundary certificates	Finite identities or implications with an explicit global-budget premise; no universal extrapolation.
Degree-four applications using a fixed strip	Kept under (H_b) or the directly stated required moment hypothesis, separately from the unconditional synthesis.
Original RH-equivalent upper bound	Remains unproved.
Historical novelty and external review	Not certified by this manuscript; bounded searches and internal checks are not substitutes for a literature review and independent expert scrutiny.

One should therefore distinguish two accomplishments. Several auxiliary problems have complete answers, and several changes of representation now have sharp error bounds. The global arithmetic assertion has not acquired a proof merely because it has acquired more equivalent formulations. A future advance must supply a noncancellation or upper-bound estimate for the actual prime sequence that is absent from the present hypotheses.

132 Provenance, version scope, and compilation

This document consolidates the following eleven working sources from the 8–9 November 2026 investigation. Repeated foundations were merged where they were verbatim; distinct proof routes and finite certificates were retained. The sign-sparsity source is its seventh revision, and the main circle source is its second revision. The separate Mittag–Leffler paper is excluded.

1. *Prime Reciprocal Circles* (expanded source, revision 2).
2. *Prime Reciprocal Circles: Closure Addendum.*
3. *Prime Reciprocal Circles: Eighty Methods.*

4. *Prime Reciprocal Circles: Further Closures.*
5. *Prime Circle: Cubic Coefficient Target.*
6. *Boundary Energy and Theta Kernel Obstructions.*
7. *Prime Circle: Uniform Growth and Subsequence Reduction.*
8. *Prime Circle: Direct Upper Bounds.*
9. *Prime Variance: Logarithmic Target.*
10. *Prime Variance: Dilation and Obstruction.*
11. *Prime Variance: A Sign-Sparsity Criterion* (revision 7; formerly *Compact Frequency Criterion*).

The title, unified reading guide, additional synthesis sections, and new finite checks belong to this consolidation. The eightieth-method ledger retains its source-access qualifications. Older computations are preserved as reported finite computations with their original code; they are not all recomputed in this consolidation. The separate numerical section specifies the checks actually rerun for the additional theorems.

This is a single-file LaTeX source using standard packages. Compile with `pdflatex` until cross-references stabilize, or use `latexmk -pdf`. No external figures, bibliography file, or auxiliary source file is required; the diagrams are drawn by TikZ and PGFPlots. The build-generated auxiliary files are disposable. The mathematical status is the one in Section 131, independent of whether the file compiles or finite tests pass.

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